The Igusa Square Root

The Borcherds determinant of $\phi_{0,1}$, its denominator algebra, and the $K3 \times E$ realization problem

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Frontispiece

The Igusa cusp form Δ_5 has five readings. The genus-two automorphic section, the denominator of a generalized Borcherds–Kac–Moody superalgebra on $\Lambda^{2,1}_{II}$, and the singular theta lift of the weak Jacobi form $\phi_{0,1}$ are theorem-level readings. The protected Pfaffian is a retained-data reading of the chosen E_3-prefactorization datum

$$\mathfrak{A}_{K3\times E} \;\in\; E_3\text{-HolFA}(K3\times E),$$

and the mirror discriminant is a conjectural period-side reading on the rank-3 Mukai-invariant $K3\times E$ component.

0.0.0.1 The five views.

1. *Automorphic.* The Borcherds–Gritsenko–Nikulin section

$$\mathcal{D}_X \;=\; \Delta_5 \;\in\; H^0\big(\mathbf{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2,\; \mathcal{L}^{\otimes 5}\otimes\nu_{\Delta_5}\big)$$

of the weight-5 automorphic line on Siegel upper-half space, with multiplier ν_{Δ_5} of [8, 71].

2. *Borcherds–Kac–Moody denominator.* The denominator of the generalized BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$ on the rank-three Lorentzian root lattice $\Lambda^{2,1}_{II}\subset\Lambda^{2,1}=\Lambda^{(1,1)}\oplus\langle 2\rangle$ of signature $(2,1)$, the index-four sublattice with even $f_{\pm 2}$ coordinates, abstractly $U(4)\oplus\langle 2\rangle$ (discriminant -32, the index-four scaling of the hyperbolic block). The ambient lattice $\Lambda^{2,1}$ has basis (f_2, f_3, f_{-2}) with $(f_2, f_{-2}) = -1$ and $(f_3, f_3) = 2$; the sublattice $\Lambda^{2,1}_{II}$ is generated by $2f_2, f_3, 2f_{-2}$. The type-II Cartan matrix on simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ is, with $\mathbf{J}_3$ the 3×3 all-ones matrix, $(\delta_i, \delta_j) = 4\mathbf{I}_3 - 2\mathbf{J}_3 = \left(\begin{smallmatrix} 2 & -2 & -2\\ -2 & 2 & -2\\ -2 & -2 & 2\end{smallmatrix}\right)$,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\,\Delta_5(2Z),$$

with signed root supermultiplicities smult $\mathfrak{g}_{\Delta_5,\alpha(n,l,m)} = f(nm, l)$ on the active support $\Gamma_{\mathrm{act}} = \{(n,l,m)\in\Gamma_{\mathrm{eff}} \mid f(nm,l)\neq 0\}$, with $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ [8, 43–45].

3. *Compact Pfaffian.* The retained Pfaffian line of the $K3\times E$ chiral E_3-prefactorization input, presented at finite stage R by the Dirac–Igusa pro-object

$$\mathfrak{D}^{DI}_X \;=\; \varprojlim_R \big(\mathcal{A}^{E_3}_{X,R},\, F^{\mathrm{hyb}}_{X,R},\, \Gamma_{X,R},\, \Pi_X,\, \Phi_R,\, o_R,\, H_R,\, P^{\Pi}_R,\, C_{X,R},\, \Theta_{\mathrm{Kos},R},\, \mathcal{L}_{\mathrm{Pf},R},\, \mathrm{pf}_{X,R},\, \epsilon_{o,R},\, \mathrm{Rec}_R\big)$$

(the chosen E_3-prefactorization datum $\mathcal{A}^{E_3}$, the hybrid factorization carrier F^{hyb} of $\mathrm{Ran}^{\mathrm{hyb}}(E)$, the Mukai charge lattice Γ, the Gram projection Π_X, the vanishing-cycle coefficient sheaf Φ, the orientation o, the Hall datum H, the Gram-descended primitive P^{Π}, the source coalgebra C, the Koszul cobar Θ_{Kos}, the Pfaffian line $\mathcal{L}_{\mathrm{Pf}}$, the Pfaffian section pf, the orientation character ϵ_o, and the recognition map Rec; Chapter 5 specifies the finite-stage data and their residuals, with the E_3, wrapped, Koszul, and recognition lanes retained or conditional exactly as stated there). The obstruction $(D0)$ is reduced in Appendix 7 to the retained D0-HN datum; $(O1)$, $(O1)^+$, and

$(O2)$ are reduced there to retained quotient-orientation, Weyl transport, and type-II wall-atlas data. Together these data supply the Pfaffian and orientation part of the Pfaffian–Dirac theorem

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI}) \;=\; \Delta_5, \qquad \epsilon_o \;=\; \nu_{\Delta_5}.$$

Here ϵ_o is the orientation character of the Pfaffian line on $W^{(2)}(\Lambda_{II}^{2,1})$, and ν_{Δ_5} is the Maass multiplier of [71] on $\mathbf{Sp}_4(\mathbb{Z})$, pulled back along the exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ and the inclusion $\Lambda_{II}^{2,1} \hookrightarrow \Lambda^{3,2}$ to a character on $W^{(2)}(\Lambda_{II}^{2,1})$ taking value -1 on each of the three simple type-II reflections $s_{\hat{\partial}_i}$; see Theorem 4.1. The symbol $\mathcal{P}_X$ denotes the protected primitive Lie superalgebra $P_X^{\Pi,\mathrm{red}}$ of Chapter 5.6 (the H^0 of the protected primitives of the hybrid factorization sheaf $\mathcal{F}_X$ after Gram-projection pushforward $\overline{\Pi}^{\Theta}_{X,*}$ and Hopf-radical quotient $\overline{\mathrm{Rad}\langle\,,\,\rangle_X}$; the three notations $\mathcal{P}_X = P_X^{\Pi,\mathrm{red}} = P_X$ denote one object, abbreviated P_X where the construction data are not in play). The Lie-superalgebra recognition

$$\mathcal{P}_X \;\cong\; \mathfrak{g}_{\Delta_5}$$

is the separate primitive-recognition theorem of Chapter 7, conditional on the finite compact-source datum $(S1)$–$(S10)$.

④ *Mirror discriminant. Conjectural.* On the rank-3 Mukai-invariant $(K3 \times E)$-period component, the discriminant section is conjecturally Δ_5^{-2}; its polar divisor is twice the reduced mirror discriminant, and its reduced automorphic Heegner support is the Humbert divisor where Δ_5 vanishes (Chapter 8).

⑤ *Singular theta lift.* The theta decomposition $\phi_{0,1}^{\mathrm{Weil}}$ of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ has weight $-1/2$ for the Weil representation of $\Lambda^{3,2}$; its Borcherds regularized theta lift is an orthogonal automorphic form, and the exceptional isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ identifies that form with the genus-two Siegel cusp form

$$\Theta_{\mathrm{sing}}^{\mathrm{Bor}}(\phi_{0,1}^{\mathrm{Weil}}) \;=\; \Delta_5.$$

This is the Borcherds–Gritsenko–Nikulin theta identity of [8, 9, 43].

0.0.0.2 The cross-level identities. The three edges ① ↔ ②, ② ↔ ⑤, ① ↔ ⑤ are theorems (Borcherds 1995; Gritsenko–Nikulin 1998); they identify the automorphic, BKM, and regularized-theta descriptions. The edge ② ↔ ③ is the conditional Pfaffian–Dirac theorem after $(D0)$, $(O1)$, $(O1)^+$, and the retained type-II wall-atlas datum $(O2_{\mathrm{atlas}})$, together with the finite Pfaffian section datum (P_{fin}), on the $K3 \times E$ specialization in Appendix 7 (Theorems 7.6, 7.12, 7.13, 7.18), making ② ↔ ③ a theorem at the Pfaffian and orientation level on $K3 \times E$; primitive Lie-superalgebra recognition remains conditional on the cofinal compact-source recognition datum. This edge relates the BKM denominator algebra to the Dirac–Igusa Pfaffian. The edge ① ↔ ③ is the two-step composite $\mathcal{D}_X = \Delta_5 = \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$, with the orientation match $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$ supplying the multiplier. The protected primitive Lie superalgebra $\mathcal{P}_X$ recovers the BKM target precisely when the source representatives, relations, pairings, radical quotient, PBW comparison, and full parity dimensions of $(S1)$–$(S10)$ are supplied.

0.0.0.3 The level-n scalar shadow. On the closed cobordism $K3 \times E \mapsto \mathrm{pt}$,

$$Z_{\mathrm{BPS}}^{K3\times E} \;=\; \left(\Delta_{10}\right)^{-1} \;=\; \Delta_5^{-2}, \qquad \Delta_{10} \;=\; \Delta_5^2 \;\in\; M_{10}\big(\mathbf{Sp}_4(\mathbb{Z})\big).$$

Here $\Delta_{10} = \Delta_5^2$ is the theta-normalized weight-ten cusp form on $\mathbf{Sp}_4(\mathbb{Z})$, equivalently $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$ in the unscaled Igusa convention ([48]; the multiplier ν_{Δ_5} squares to trivial [71]); the Oberdieck–Pixton-monic normalization $\chi_{10}^{\mathrm{OP}} := D_5^2 = 4096^{-1}\Delta_5^2$ (with $D_5 := 64^{-1}\Delta_5$ the Borcherds-monic form so that $[q^{1/2}r^{1/2}s^{1/2}]D_5 = 1$) gives the equivalent form $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$, where $4096 = 64^2$ is the squared theta-leading constant $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ of the Igusa form and the minus sign is the OP scalar branch convention [85, 86]. $Z_{\mathrm{BPS}}^{K3\times E}$ is the level-n protected trace of the retained $K3\times E$ data. A three-dimensional gravity-line interpretation with internal compact factor $K3\times E$ requires the Hall–Borcherds residual of Vol II and is not asserted here.

0.0.0.4 The κ-tuple of $K3\times E$. The retained $K3\times E$ data carry the five-coordinate κ-tuple

$$\kappa_{K3\times E} \;=\; \big(\kappa_{\mathrm{cat}},\ \kappa_{\mathrm{ch}}^{\mathrm{Hodge}},\ \kappa_{\mathrm{ch}}^{\mathrm{Heis}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}}\big) \;=\; (0,\ 0,\ 3,\ 5,\ 24).$$

The categorical Euler is $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3})\,\chi(\mathcal{O}_E) = 2\cdot 0 = 0$. The Hodge supertrace $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}}(K3\times E) = \sum_q(-1)^q h^{0,q}(K3\times E) = 0$. The Heisenberg rank $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$ is the rank of the type-II Heisenberg–Cartan obtained after the normal-ordered Gram projection Π_X to $\Lambda_{II}^{2,1}$. It is not the rank of the algebraic Mukai lattice $H^0(S,\mathbb{Z})\oplus \mathrm{NS}(S)\oplus H^4(S,\mathbb{Z})$; for the $U\oplus\langle -2\rangle$ polarisation that lattice has rank five, while the $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$-coordinate counts only the three Gram coordinates which enter the type-II BKM Cartan. The BKM coordinate

$$\kappa_{\mathrm{BKM}}(\Phi_N) \;=\; c_N(0)/2$$

(Borcherds 1995; Gritsenko 1999) reads, at the paramodular input $N = 1$, $c_1(0) = 10$, giving $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$. The fibre coordinate $\kappa_{\mathrm{fiber}} = \chi_{\mathrm{top}}(K3) = 24$. The additive identity $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ fails at every $N\in\{1,2,3,4,6\}$: the Lefschetz Euler $\chi_{\mathrm{top}}(K3) = 24$ and the holomorphic Euler $\chi(\mathcal{O}_{K3}) = 2$ are distinct invariants of the K3 fibre, and κ_{BKM} sees the $c_N(0)/2$ data of the Borcherds input rather than either of them. Each subscript on κ records a distinct invariant; the unsubscripted handle leaves the categorical dimension unspecified.

Chapter 1

Pentadic Δ_5

1 The retained E_3-object and the five readings of Δ_5

The five readings attached to Δ_5 have different logical status. The automorphic, BKM, and theta-lift readings are theorems; the Pfaffian reading is relative to retained $K3 \times E$ data; the mirror reading is conjectural. The trivial-canonical threefold $K3 \times E$ supplies the ambient CY-3 input for a retained holomorphic prefactorization datum in the Kontsevich–Tamarkin and Costello–Gwilliam–Li lineage [21, 22, 58]. The notation

$$\mathfrak{A}_{K3\times E} \;\in\; E_3\text{-HolFA}(K3 \times E)$$

denotes this chosen finite-stage E_3-prefactorization datum; the unconditional functor from CY-3 dg-categories to holomorphic E_3-factorization algebras is not asserted [21, 22, 58, 60, 93]. The retained protected sector supplies a Pfaffian line on the Dirac–Igusa pro-object. The Pfaffian–Dirac theorem compares that line with the automorphic section $\mathcal{D}_X = \Delta_5$ only under the named orientation and wall data. The five readings are the automorphic section, the BKM denominator, the retained Pfaffian section, the conjectural period discriminant, and the singular theta lift.

Definition 1.1 *(Pentadic Δ_5).* The five-tuple

$$\big(\mathcal{D}_X,\ \mathrm{den}(\mathfrak{g}_{\Delta_5}),\ \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI}),\ \mathrm{Disc}(\mathcal{P} \to \mathbb{D}),\ \Theta^{\mathrm{Bor}}_{\mathrm{sing}}(\phi^{\mathrm{Weil}}_{0,1})\big)$$

of automorphic, BKM, retained-Pfaffian, period, and theta-lift readings is the *pentadic* Δ_5. The reading of Δ_5 is fixed by the level: at level ①, Δ_5 is $\mathcal{D}_X$; at level ②, Δ_5 is the function whose value at $2Z$ is $64\ \mathrm{den}(\mathfrak{g}_{\Delta_5})$; at level ③, Δ_5 is the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$ under $(D0)$, $(O1)$, $(O1)^+$, and the retained $(O2_{\mathrm{atlas}})$ datum; at level ④, Δ_5^{-2} is the mirror discriminant section, conjecturally; at level ⑤, Δ_5 is the Borcherds theta lift of $\phi^{\mathrm{Weil}}_{0,1}$.

2 The ① ↔ ② edge: Borcherds–Gritsenko–Nikulin

The Borcherds product expansion of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ is the genus-two cusp form

$$\Delta_5(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \big(1 - q^n r^l s^m\big)^{f(nm,l)},$$

where $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right)$, $q = e^{2\pi i z_1}$, $r = e^{2\pi i z_2}$, $s = e^{2\pi i z_3}$, and $f(n,l)$ are the Fourier coefficients of $\phi_{0,1}$ (the weak Jacobi form whose discriminant decomposition records $f(0,0) = 10$, $f(0,\pm1) = 1$, $f(1,0) = 108$, $f(1,\pm1) = -64$, $f(1,\pm2) = 10, \ldots$) [8, 32, 43]. The leading constant $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64 = 2^6$ is the theta-leading coefficient (Igusa); the chamber $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ fixes the active semigroup Γ_{eff} of cusp-completed exponents.

The Gritsenko–Nikulin denominator identity converts the same product into the Weyl–Kac form on the type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1} = \Lambda^{(1,1)} \oplus \langle 2\rangle$:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; \sum_{w\in W^{(2)}(\Lambda^{2,1}_{II})} \mathrm{sgn}(w)\; w\big(e^{\rho}\big) \;=\; 64^{-1}\,\Delta_5(2Z),$$

with Weyl vector

$$\rho \;=\; \tfrac{1}{2}\big(\delta_1+\delta_2+\delta_3\big) \;=\; f_2 - \tfrac{1}{2}f_3 + f_{-2}$$

on the three simple type-II reflections $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ (Cartan matrix $(\delta_i,\delta_j) = 4\mathbf{I}_3 - 2\mathbf{J}_3 = \left(\begin{smallmatrix} 2 & -2 & -2\\ -2 & 2 & -2\\ -2 & -2 & 2\end{smallmatrix}\right)$, $\mathbf{J}_3$ the all-ones matrix); type-II reflection subgroup $W^{(2)}(\Lambda^{2,1}_{II})$ generated by these three reflections [43–45]. The signed root supermultiplicities recover the Fourier coefficients of $\phi_{0,1}$ under the doubling $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$:

$$\mathrm{smult}\;\mathfrak{g}_{\Delta_5,\alpha(n,l,m)} \;=\; f(nm,l) \qquad \text{on } (n,l,m)\in\Gamma_{\mathrm{act}} = \{(n,l,m)\in\Gamma_{\mathrm{eff}} \mid f(nm,l)\neq 0\}.$$

The edge ① ↔ ② is the joint statement: the automorphic section $\mathcal{D}_X = \Delta_5$ of $\mathcal{L}^{\otimes 5}\otimes \nu_{\Delta_5}$ on Siegel space *equals* the BKM denominator (after the constant 64^{-1} and the variable doubling $Z\mapsto 2Z$). This is Borcherds 1995 together with the Gritsenko–Nikulin identification of the underlying lattice with $\Lambda^{2,1}_{II}$ and the Weyl-group action with $W^{(2)}(\Lambda^{2,1}_{II})$ [8, 43].

3 THE ① ↔ ⑤ AND ② ↔ ⑤ EDGES: THE SINGULAR THETA LIFT

The Borcherds product expansion of the previous section converts $\phi_{0,1}$ into Δ_5 by an arithmetic identity in Γ_{eff}; the question of *where* this identity comes from has a single answer. The exceptional isomorphism for $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ identifies the genus-two Siegel space with the type-IV symmetric domain attached to the signature-$(3,2)$ lattice $\Lambda^{3,2} = \Lambda^{(1,1)}\oplus\Lambda^{(1,1)}\oplus\langle 2\rangle$, and it identifies the arithmetic quotient $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathcal{H}_2$ with $\Gamma_{3,2}\backslash\mathbb{H}^{\mathbf{IV}}_+$, where $\Gamma_{3,2} = \wedge^2\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$. Borcherds's regularisation pairs the $\mathbf{Mp}_2(\mathbb{Z})$ Weil-representation form ϕ^{Weil} with the Siegel theta kernel of $\Lambda^{3,2}$, and produces a meromorphic Siegel modular form whose divisor is prescribed by the principal part:

$$\Theta^{\mathrm{Bor}}_{\mathrm{sing}}(\phi) \;=\; \int^{\mathrm{reg}}_{\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}} \big\langle \phi(\tau),\; \Theta_{\Lambda^{3,2}}(\tau,Z)\big\rangle\,\frac{du\,dv}{v^2}, \qquad \tau = u+iv,$$

on the Weil-representation pairing of the vector-valued weakly-holomorphic input ϕ (the theta decomposition of $\phi_{0,1}$ into the discriminant form of $\Lambda^{3,2}$) against the Siegel theta series of the signature-$(3,2)$ lattice $\Lambda^{3,2}$ — not the rank-three target lattice $\Lambda^{2,1}_{II}$, which receives the BKM root structure under doubling — together with the antiholomorphic dependence required for absolute convergence after regularisation; the explicit construction is Chapter 5, Definition 1.9 and Theorem 1.10 [8, 9]. At the theta-decomposed input $\phi^{\mathrm{Weil}}_{0,1}$ the regularised integral computes the composite

$$\Theta^{\mathrm{Bor}}_{\mathrm{sing}}\colon J^{\mathrm{wk}}_{0,1} \;\longrightarrow\; M^{!}_{-1/2}(\rho_{\Lambda^{3,2}}) \;\longrightarrow\; M_5\big(\mathbf{Sp}_4(\mathbb{Z}),\;\nu_{\Delta_5}\big), \qquad \phi_{0,1} \;\longmapsto\; \phi^{\mathrm{Weil}}_{0,1} \;\longmapsto\; \Delta_5,$$

giving the edge ① ↔ ⑤. Reading the same lift through the Gritsenko–Nikulin denominator identity, $\phi_{0,1}$ is the character of the index-one Heisenberg sector and the multiplicative lift over $\Lambda^{2,1}_{II}$ is exactly $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, giving the edge ② ↔ ⑤ [8, 43–45].

4 The ② ↔ ③ edge: the Pfaffian–Dirac theorem

The theorem edges proved by Borcherds and Gritsenko–Nikulin identify the automorphic, BKM-denominator, and singular-theta readings of Δ_5; the chiral E_3-prefactorization datum of §1 is the retained operator-level source from which the View ③ datum is obtained. The structural question has two terms. The Pfaffian term asks whether the protected determinant of the $K3 \times E$ operator datum is the automorphic section Δ_5. The primitive term asks whether the compact Hall primitive sector is the BKM algebra $\mathfrak{g}_{\Delta_5}$. The first term is the Pfaffian–Dirac theorem after $(D0)$, $(O1)$, $(O1)^+$, and $(O2_{\text{atlas}})$, together with (P_{fin}), in Appendix 7. The second term is the finite compact-source recognition problem $(S1)$–$(S10)$.

Theorem 4.1 *(Pfaffian–Dirac). Conditional on* $(D0)$*,* $(O1)$*,* $(O1)^+$*,* $(O2_{\text{atlas}})$*, and* (P_{fin})*; ambient: cosection-twisted reduced d-critical chart on* $K3 \times E$ *at finite cusp-truncation height* R*, with chain-level pro-object, Mittag–Leffler vanishing of the inverse system in* R*, and Pfaffian orientation* o *with Weyl character* ϵ_o*.* The protected Pfaffian of the Dirac–Igusa pro-object $\mathfrak{D}^{DI}_X = \varprojlim_R \mathfrak{D}^{DI}_{X,R}$ satisfies

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{DI}_X) \;=\; \Delta_5, \qquad \epsilon_o \;=\; \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II}),$$

The primitive Lie-superalgebra assertion is a separate recognition statement. If the finite compact-source datum $(S1)$–$(S10)$ is supplied on a cofinal system of relation-closed degree windows, then

$$\mathcal{P}_X := H^0 \;\mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big)/\overline{\mathrm{Rad}\langle\,,\rangle_X} \;\cong\; \mathfrak{g}_{\Delta_5}.$$

The four obstructions and the finite Pfaffian section datum name the exact conditioning. $(D0)$ is the $D0$-degeneration limit of the cosection-reduced d-critical orientation theory: the limiting orientation in the $D0$-degenerate fibre is constructed and matches the orientation of the generic fibre under the cosection-twist [11, 51, 52]. $(O1)$ is the strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle: the square root of the determinant line of the cotangent complex extends across the cusp boundary as part of the retained quotient-orientation datum [15, 51, 52]. $(O1)^+$ is the Weyl-equivariant transport of the orientation along reflections in the type-II reflection subgroup $W^{(2)}(\Lambda^{2,1}_{II})$: the orientation cocycle $c_o \in Z^2(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2)$ trivializes in H^2, and the resulting character $\epsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \longrightarrow \{\pm 1\}$ is well-defined on generators. $(O2)$ is the local Pfaffian wall normal form on type-II walls: at each simple type-II wall s_{δ_i} the rank-one normal Pfaffian crossing of Definition 0.22 is part of the retained wall-atlas datum and gives the local sign $(-1)^{N^{\mathrm{Pf}}_{\delta_i}} = -1$, and the Hall-side sign sum on the half-Hilbert orbit of the self-Ext stack of size $2^{12} = 4096$ is compared with the squared theta-leading $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ only after scalarization. The wall transport gives the character value $\epsilon_o(s_{\delta_i}) = \nu_{\Delta_5}(s_{\delta_i})$ on each generator (Appendix E). The finite Pfaffian section datum gives support, parity, active-support coverage, leading coefficient, and Pfaffian-line Mittag–Leffler compatibility for the cofinal HN product.

Remark 4.2 *(The conditioning is named precisely).* The four obstruction classes $(D0)$, $(O1)$, $(O1)^+$, $(O2)$ and the finite Pfaffian section datum are the exact Pfaffian and orientation obligations; Appendix 7 reduces $(D0)$ to the retained D0-HN datum, names the retained quotient-orientation and Weyl transport data, and names $(O2_{\text{atlas}})$ and (P_{fin}). At the operator level ③, $\mathcal{D}_X = \Delta_5$ is compared with the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{DI}_X)$. After the level-$n$ protected trace, the scalar partition function is Δ_5^{-2}. The primitive algebra $\mathcal{P}_X \simeq \mathfrak{g}_{\Delta_5}$ is stronger; it requires the finite compact-source recognition datum $(S1)$–$(S10)$. The operator-level datum carries the orientation character ϵ_o that the scalar shadow has squared away, by view ③, and recovers it as ν_{Δ_5} on $W^{(2)}(\Lambda^{2,1}_{II})$ by $(O1)^+$ and $(O2)$.

5 View ④: mirror discriminant

Conjectural (ambient: variation of Hodge structure on the rank-3 Mukai-invariant $(K3 \times E)$-period component). On $\mathcal{P}^{(3)}_{K3\times E} \simeq \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z})$, the mirror discriminant section is conjecturally Δ_5^{-2}. Its polar divisor is $2\mathfrak{D}_{\mathrm{mir}}$, where the reduced support of $\mathfrak{D}_{\mathrm{mir}}$ is the Humbert–Heegner divisor $\mathfrak{H}_{\mathrm{II}}$ fixed by the Borcherds product of Δ_5. This mirror-discriminant identification is conjectural and is not a hypothesis of any theorem; see Conjecture 1.1.

6 The level-n scalar shadow

The scalar partition function on the closed cobordism $K3 \times E \mapsto \mathrm{pt}$ is the Borcherds determinant of the protected K3 index, computed by Oberdieck–Pixton 2018 and Oberdieck–Pandharipande 2018 [85, 86]: the OP partition function $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is the virtual count of Hilbert schemes of $K3 \times E$ in the $(-p_{\mathrm{DT}})^n$ DT-chamber, and the BPS normalization $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ rescales by -4096^{-1}. Two normalizations of the same weight-ten cusp form on $\mathbf{Sp}_4(\mathbb{Z})$ are in use, and naming them is necessary because both appear in the formulas:

- *Theta-normalized.* $\Delta_{10}(Z) := \Delta_5(Z)^2 \in M_{10}(\mathbf{Sp}_4(\mathbb{Z}))$ (the multiplier ν_{Δ_5} squares to trivial; the leading coefficient $[q\,r\,s]\Delta_{10} = 64^2 = 4096$).
- *Oberdieck–Pixton-monic.* $\chi^{\mathrm{OP}}_{10}(Z) := D_5(Z)^2 = 64^{-2}\Delta_5(Z)^2 = 4096^{-1}\Delta_{10}(Z)$ (so $[q\,r\,s]\chi^{\mathrm{OP}}_{10} = 1$).

Oberdieck–Pixton and Oberdieck–Pandharipande prove [85, 86]

$$Z^X_{\mathrm{OP}} = -\left(\chi^{\mathrm{OP}}_{10}\right)^{-1} = -4096\,\Delta_5^{-2},$$

and the level-n protected scalar shadow is

$$Z^{K3\times E}_{\mathrm{BPS}} = \left(\Delta_{10}\right)^{-1} = \Delta_5^{-2} = -\tfrac{1}{4096}\,Z^X_{\mathrm{OP}}.$$

The factor $4096 = 64^2$ is the squared theta-leading constant of Δ_5; the OP minus sign is a scalar branch convention absorbed into the OP $(-p_{\mathrm{DT}})^n$ chamber and is independent of the orientation character ϵ_o of view ③.

$Z^{K3\times E}_{\mathrm{BPS}}$ is the level-n protected trace of the retained $K3 \times E$ data: a charge-graded virtual partition function. A three-dimensional gravity-line interpretation of the scalar shadow with internal compact factor $K3 \times E$ requires the Hall–Borcherds residual of Vol II. The universal-stage chain names the required objects: primitive factorization dg-category $\mathcal{C} \longrightarrow$ boundary chart $A_b = \mathrm{End}_{\mathcal{C}}(b) \longrightarrow$ bar twisting $\mathrm{Bar}(A_b) \longrightarrow$ derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \longrightarrow$ ambient operator algebra $\mathcal{A}$ at the boundary vacuum b of the open factorization dg-category on $(K3 \times E,\ D,\ \tau)$; this chain is not constructed here. The local witness is the formal-Darboux stalk of the mixed-holomorphic-topological strings theory [23, 24, 58]. The class $\mathrm{ob}^{\mathrm{hol}}_{\mathrm{dR}}(K3 \times E)$ is the holomorphic de Rham obstruction to gluing this local BV factorization model over $K3 \times E$; its vanishing is a necessary global condition, not the gravity-line operator algebra.

Remark 6.1 *(Primitive order).* The five views are ordered: primitive objects (the retained chiral E_3-prefactorization datum on $K3 \times E$, its four obstructions $(D0)$, $(O1)$, $(O1)^+$, $(O2_{\mathrm{atlas}})$, and its finite Pfaffian section datum; the Cartan datum of $\mathfrak{g}_{\Delta_5}$) first; cross-level identities ((②↔③) Pfaffian–Dirac, $(\epsilon_o = \nu_{\Delta_5})$ orientation match, $(\mathcal{P}_X \cong \mathfrak{g}_{\Delta_5})$ primitive recognition) second; scalar modular forms last (the section $\mathcal{D}_X$, the BKM denominator $64^{-1}\Delta_5(2Z)$, the partition function Δ_5^{-2}). A statement is not primitive if it is true only after passing to a trace, choosing a chamber, completing a category, or installing descent data. On each first appearance the scalar reading is named: the theta-normalized form Δ_5 with leading $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$; the BKM denominator $64^{-1}\Delta_5(2Z)$ with the doubling $2Z$ on the type-II sublattice; the OP-monic form $D_5 = 64^{-1}\Delta_5$ with $\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$ the Oberdieck–Pixton normalization of the squared determinant section.

7 Placement in the Beilinson Chain

The pentadic Δ_5 is one identification at one specific level of the Beilinson chain. Under the chiral Koszul source datum of Definition 0.26 and the Koszul Mittag–Leffler exactness of Proposition 0.29, the source coalgebra C_X maps by cobar comparison to the target chiral shadow $\mathsf{Den}_{\Delta_5,E}$. Vol I's Theorem H supplies the chain-level identification $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) = \mathrm{ChirHoch}^\bullet(A_b, A_b)$ only on the retained Koszul locus, and Vol II's chiral-Hochschild Beilinson stratification (i)+(ii) supplies the CY-orientation trace pairing of degree $-\dim_{\mathbb{C}}(K3 \times E) = -3$. Relative to the retained trace datum, these inputs give the level-Z trace identity $\mathrm{Tr}^{\mathrm{bulk}}_n((-1)^F \cdot \mathrm{id})_{Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)} = \Delta_5^{-2}$ of Appendix 7, Theorem 7.22, relative to (Z_{HB}); the OP chamber scalar is -4096 times this trace. The level-A gravity-line operator algebra acting on the boundary is the open gravity-line Hall–Borcherds residual of Vol II. The Vol III κ-stratification places $\mathfrak{g}_{\Delta_5}$ at the $(d = 3,\ \text{rank } 3)$ chamber through the identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$, where $c_N(0)$ is the constant Fourier coefficient of the input weak Jacobi or vector-valued modular form. Applied at the paramodular row $(t, N; k) = (1, 1; 5)$ of the Gritsenko–Cléry catalogue [41, 43], with Jacobi seed $\phi_{0,1}$ and Borcherds product Δ_5, the identity reads $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5$. The companion identity at input Φ_{12}, the Fake-Monster Borcherds product on the Leech-extension lattice $\mathrm{II}_{25,1}$ [7, 8], reads $\kappa_{\mathrm{BKM}}(\Phi_{12}) = c_{\Phi_{12}}(0)/2 = 12$; the two values are different rows of one universal $c_N(0)/2$ rule, indexed by different input forms and lattices, not in contradiction. The mixed holomorphic-topological strings companion identifies the formal-Darboux stalk $A^{\mathrm{cl}}_{\partial,N} = C^\bullet_{\mathrm{CE}}(\mathfrak{gl}_N,\ \mathrm{Kosz}([\phi_1, \phi_2]))$ of the chiral Hochschild cochain at the Dirac brane vacuum. The obstruction to globalizing this local factorization model over $K3 \times E$ is the holomorphic de Rham class $\mathrm{ob}^{\mathrm{hol}}_{\mathrm{dR}}(K3 \times E)$. A three-dimensional gravitational path-integral interpretation of the level-n scalar shadow $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ requires this obstruction to vanish and separately requires the Vol II gravity-line Hall–Borcherds operator algebra.

Contents

Part I

Automorphic and BKM Data

Chapter 2

K3×E geometry, Mukai dictionary, lattice polarization, normal-ordered Gram cocycle, hybrid $\mathrm{Ran}^{\mathrm{hyb}}(E)$ carrier

The arithmetic source contains three named weight-five readings, and their normalizations are different. The automorphic chain produces the section

$$\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}),\ \nu_{\Delta_5}) \qquad \text{(View ①; proved [8, 43]).}$$

The Borcherds–Kac–Moody (BKM) chain produces the denominator

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\,\Delta_5(2Z) \qquad \text{(View ②; proved [43]).}$$

The compact Pfaffian chain produces, granted the retained Do-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and the retained type-II wall-atlas datum of Theorem 7.4 of Appendix 7, the protected Pfaffian

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5 \qquad \text{(View ③; relative to the retained orientation data at the Pfaffian level).}$$

The handle "Δ_5" with no superscript is reserved for working formulas; each occurrence names which of $\mathcal{D}_X$, $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, or $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ is meant. The cross-level identity ②↔③ is a theorem at the Pfaffian and orientation level on $K3 \times E$, granted (D0), the retained quotient-orientation datum for (O1), the chosen Weyl lifts for (O1$^+$), and the retained datum (O2$_{\mathrm{atlas}}$) in Theorems 7.6, 7.12, 7.13, 7.18. The primitive Lie-superalgebra recognition $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source datum (S1)–(S10) of Chapter 7.

0.0.0.1 The Mukai–Gram dictionary. The Mukai–Gram dictionary on $K3 \times E$ begins with the Mukai lattice $\widetilde{H}(S,\mathbb{Z})$ of the K3 surface S, of rank 24 and signature (4, 20) [77]. The even Kunneth sector

$$\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z}) \otimes H^{\mathrm{even}}(E,\mathbb{Z}) \cong \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$$

is the formal $K3 \times E$ charge lattice used below; the two summands are written (Q, P). For the Gram map this underlying Kunneth group is read with the two componentwise K3 Mukai pairings; the tensor Kunneth cross-pairing on $H^{\mathrm{even}}(X,\mathbb{Z})$ is a different bilinear form. The Gram-index lattice $\Gamma_{\mathrm{gram}} = \mathbb{Z}^3$ receives the quadratic shadow

$$\Pi_X(Q,P) = \big((Q,Q)/2,\ (Q,P),\ (P,P)/2\big).$$

This map is not additive; its additivity defect is the symmetric polarization $B = -\delta\Pi_X$. The additive lattice map is the normal-ordered homomorphism

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}.$$

After this normal ordering, Γ_{gram} embeds into $\Lambda^{2,1}_{II}$ and then into the type-IV ambient lattice $\Lambda^{3,2}$ of Chapter 5. Detailed lattice computations are in Appendix 3; the dictionary fixes the $K3 \times E$ charge convention.

The geometric and lattice data feed the three established Δ_5 constructions. The compact target is fixed at $X = S \times E$ with S a smooth complex projective K3 surface and E a smooth elliptic curve over $\mathbb{C}$; this is a trivial-canonical threefold ($\dim_{\mathbb{C}} X = 3$, $K_X \cong \mathcal{O}_X$; elementary from the product formula), so $n = \chi(\mathcal{O}_Y)$ appears directly. The $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ tuple of [65] (the K3×E row of the Gritsenko–Cléry catalogue) is the categorical-dimensional anchor; every use of κ carries one of these five subscripts.

The datum has a geometric part and an arithmetic part. The geometric part is the formal dyonic charge lattice $\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S$ on the Mukai sector, together with $\Gamma^{\mathrm{eff}}_{X,\sigma} \subset \Gamma_X^{\mathrm{alg}} \subset \Gamma_X^{\mathrm{form}}$. The arithmetic part is the quadratic Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}} = \mathbb{Z}^3$, its symmetric bilinear polarization $B = -\delta\Pi_X$, the normal-ordered extension $\widehat{\Gamma}_X = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}$, and the additive map $\overline{\Pi}_X(c, T) = \Pi_X(c) - T$. The fixed-lift raw-pushforward obstruction theorem forces the extension, and the projection-locality obstruction forces the wrapped stratum in $\mathrm{Ran}^{\mathrm{hyb}}(E) = \mathrm{Ran}(E)_{b=0} \sqcup \mathrm{Ran}^{\mathrm{wr}}(E)_{b>0}$.

1 Smooth K3×E threefold and Mukai lattice

1.1 The compact target

Definition 1.1 *(K3×E threefold; primitive geometric datum).* A *K3×E threefold* is a triple (X, S, E) in which S is a smooth complex projective K3 surface (so $K_S \cong \mathcal{O}_S$, $h^{1,0}(S) = 0$, $h^{2,0}(S) = 1$; [77]), E is a smooth elliptic curve over $\mathbb{C}$ with marked origin $0_E \in E$, and

$$X := S \times E.$$

The canonical bundle is trivial, $K_X \cong p_S^* K_S \otimes p_E^* K_E \cong \mathcal{O}_X$, and Hodge invariants are

$$\dim_{\mathbb{C}} X = 3, \qquad h^{1,0}(X) = h^{1,0}(E) = 1, \qquad \chi_{\mathrm{top}}(X) = \chi_{\mathrm{top}}(S) \cdot \chi_{\mathrm{top}}(E) = 24 \cdot 0 = 0, \qquad \chi(\mathcal{O}_X) = \chi(\mathcal{O}_S) \cdot \chi(\mathcal{O}_E) =$$

The threefold X is Calabi–Yau in the trivial-canonical sense used by Donaldson–Thomas theory and shifted symplectic geometry; the construction uses the chosen trivialization of K_X.

Remark 1.2 *(Distinguishing $\chi_{\mathrm{top}}(K3) = 24$ and $\chi(\mathcal{O}_S) = 2$).* The two integers enter the construction by different routes. The Borcherds weight is $f(0,0)/2 = 5$, where $f(0,0) = 10$ is the constant Fourier coefficient of the weight-zero index-one weak Jacobi form $\phi_{0,1} = r^{-1} + 10 + r + O(q)$ in the Eichler–Zagier normalization (computed, [32]). The topological Euler number $\chi_{\mathrm{top}}(K3) = 24$ enters one step removed, through the elliptic-genus specialization: at $z = 0$ the $\phi_{0,1}$ support $4n - l^2 \geq -1$ leaves only $l \in \{-1, 0, 1\}$ at $n = 0$, so the constant term is $\sum_l f(0, l) = f(0, -1) + f(0, 0) + f(0, 1) = 12$, and $Z_{K3}(\tau, 0) = 2\,\phi_{0,1}(\tau, 0)$ has constant term $2 \cdot 12 = 24 = \chi_{\mathrm{top}}(K3)$. Thus $\chi_{\mathrm{top}}(K3) = 2 \sum_l f(0, l)$, not $f(0,0)$; the Borcherds weight 5 sees only the single coefficient $f(0,0) = 10$. The integer $\chi(\mathcal{O}_S) = 2$ enters by a third route, the categorical Calabi–Yau dimension and the holomorphic part of the Mukai pairing. The two integers belong to different entries: the κ-tuple of [65] records $\kappa_{\mathrm{fiber}} = 24 = \chi_{\mathrm{top}}(K3)$, not $2 = \chi(\mathcal{O}_S)$, and the universal Borcherds identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ at $\Phi_1 = \Delta_5$ gives $c_1(0) = 10$ and $\kappa_{\mathrm{BKM}} = 5$. The expression $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} + \chi(\mathcal{O}_{K3}) = 0 + 2 = 2$ does not give the Borcherds weight, while $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} + \chi(\mathcal{O}_{K3}) = 3 + 2 = 5$ is only an accidental equality at the $N = 1$ row. Across the CHL frame the Borcherds weights are $c_N(0)/2 = (5, 4, 3, 2, 1)$, not a fixed sum of a chiral rank and $\chi(\mathcal{O}_{K3})$ (proved,

[65] counterexample). The coordinate $\kappa_{\mathrm{BKM}} = 5$ is the cusp-form weight; the protected scalar shadow is written at level $n = \chi(\mathcal{O}_Y)$ for the one-dimensional subscheme $Y \subset K3 \times E$. The target $X = K3 \times E$ is a trivial-canonical threefold.

1.2 Mukai lattice on the K3 factor

Definition 1.3 *(Mukai lattice and Mukai pairing).* Let S be a smooth complex K3 surface. The *Mukai lattice* is the total even cohomology

$$\Lambda_S := \widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z}),$$

identified with the K-theoretic charge lattice $K_0(D^b(\mathrm{Coh}\, S))_{\mathrm{tor.\,free}}$ via the Mukai vector

$$v(\mathcal{F}) := \mathrm{ch}(\mathcal{F})\sqrt{\mathrm{Td}(S)} = \big(r(\mathcal{F}),\ c_1(\mathcal{F}),\ r(\mathcal{F}) + \mathrm{ch}_2(\mathcal{F})\big),$$

where r is the rank, c_1 is the first Chern class, and the half-shift r in the four-form term comes from $\mathrm{Td}(S) = 1 + 2[\mathrm{pt}]$ (proved, [77, 78]). The *Mukai pairing* is

$$\big\langle (r, c, s),\ (r', c', s') \big\rangle_{\mathrm{Muk}} := c \cdot c' - rs' - r's,$$

where $c \cdot c' \in \mathbb{Z}$ is the cup product on $H^2(S, \mathbb{Z})$. It is even, unimodular, of signature $(4, 20)$, and rank 24:

$$\Lambda_S \cong II_{4,20} \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2},$$

where U denotes the rank-two hyperbolic plane $U \cong \mathbb{Z}e \oplus \mathbb{Z}f$ with $e^2 = f^2 = 0$, $e \cdot f = 1$, and $E_8(-1)$ denotes the negative-definite E_8 root lattice (proved, [78, 82]).

Remark 1.4 *(Sign convention).* The signature $(4, 20)$ uses the convention that $H^0 \oplus H^4$ contributes a hyperbolic plane U via $(1, 0, 0)$ and $(0, 0, 1)$ and that $H^2(S, \mathbb{Z}) \cong II_{3,19} \cong U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$, which has positive part of dimension 3. Together, $\Lambda_S = U \oplus II_{3,19} \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2}$ with positive part of dimension 4 (proved, [78]). The signature is $(4, 20)$, not $(20, 4)$: the positive direction is the smaller side. A reflection r_v across $v \in \Lambda_S$ acts by $r_v(x) = x - 2(x, v)v/(v, v)$, taking the $\mathbb{R}$-linear extension when $(v, v) \neq 0$, without implicit sign reversal.

1.3 Künneth and the formal even Mukai sector on $X = S \times E$

Definition 1.5 *(Formal even Mukai sector on $X = S \times E$).* Pick a basepoint $0_E \in E$ and the standard generators $1_E \in H^0(E, \mathbb{Z})$ and $\omega_E \in H^2(E, \mathbb{Z})$ with $\int_E \omega_E = 1$. Every even cohomology class on $X = S \times E$ decomposes as

$$v_X = P \otimes 1_E + Q \otimes \omega_E, \qquad P, Q \in \Lambda_S$$

(proved, Künneth on simply connected $\widetilde{H}$ tensor E-cohomology). The pair $(Q, P) \in \Lambda_S \oplus \Lambda_S$ is the *formal even charge*. The component $P \otimes 1_E$ is invariant under translation along E, while $Q \otimes \omega_E$ records the E-degree.

The *formal full-Mukai dyonic lattice* is the additive abelian group

$$\Gamma_X^{\mathrm{form}} := \Lambda_S \oplus \Lambda_S, \qquad c = (Q, P),$$

with electric component Q and magnetic component P. The symbol Γ_X^{phys} is defined here by the equality

$$\Gamma_X^{\mathrm{phys}} := \Gamma_X^{\mathrm{form}}.$$

This is not the full four-dimensional $\mathcal{N} = 4$ electric–magnetic charge lattice and not compact Hall support. It is the D6–D2–D0/OP even branch on which the Mukai Gram map is tested before imposing algebraicity, effectivity, stability, or non-emptiness of compact Hall moduli.

Definition 1.6 *(Algebraic/effective Hall support).* Fix a polarised K3 surface (S, H), a chosen Mukai vector lattice $\Lambda_{S,\mathrm{alg}} \subset \Lambda_S$ (the image of the algebraic K-class lattice generated by sheaves whose Mukai vector lies in $H^0 \oplus \mathrm{NS}(S) \oplus H^4$), and a stability datum σ on $D^b(\mathrm{Coh}\, S \times E)$ for which the retained sector has Harder–Narasimhan filtrations and wall-crossing. The datum is part of the compact Hall input; no unconditional Bridgeland stability theorem on the threefold $S \times E$ is used. Set

$$\Gamma_X^{\mathrm{alg}} := \Lambda_{S,\mathrm{alg}} \oplus \Lambda_{S,\mathrm{alg}}, \qquad \Gamma_{X,\sigma}^{\mathrm{eff}} \subset \Gamma_X^{\mathrm{alg}} \subset \Gamma_X^{\mathrm{form}},$$

where the *σ-effective semigroup* $\Gamma_{X,\sigma}^{\mathrm{eff}}$ is the additive sub-semigroup of charges supported by objects semistable for this datum and of nonzero Mukai vector. Membership in $\Gamma_{X,\sigma}^{\mathrm{eff}}$ is not implied by membership in Γ_X^{alg} or by formal lattice arithmetic (folklore).

Remark 1.7 *(Three-tier inclusion).* The hierarchy $\Gamma_{X,\sigma}^{\mathrm{eff}} \subset \Gamma_X^{\mathrm{alg}} \subset \Gamma_X^{\mathrm{form}}$ distinguishes three discipline registers. Γ_X^{form} is the formal additive group on which Π_X is defined and the polarization identity $B = -\delta\Pi_X$ holds. Γ_X^{alg} is the K-theoretic algebraic sublattice; its rank-one extension closure inside Γ_X^{form} records only the algebraic Mukai branch (computed, [59, 78]). $\Gamma_{X,\sigma}^{\mathrm{eff}}$ is the Hall support selected by σ; effectivity and semistability are not lattice-theoretic properties and are not fibre-functorial in the K3 type. The sectorial Hall truncation of §4 works at this third level after imposing finite Harder–Narasimhan height bounds.

Lemma 1.8 *(Two orthogonal hyperbolic planes inside Λ_S).* The Mukai lattice contains two orthogonal hyperbolic planes:

$$U \oplus U \hookrightarrow \Lambda_S, \qquad U_1 = \mathbb{Z}e_1 \oplus \mathbb{Z}f_1, \quad U_2 = \mathbb{Z}e_2 \oplus \mathbb{Z}f_2,$$

with $e_i^2 = f_i^2 = 0$, $e_i \cdot f_i = 1$, and $U_1 \perp U_2$ inside the Mukai pairing (proved, [82]).

Proof. Direct from $\Lambda_S \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2}$. □

2 Quadratic Gram map and lattice polarization

2.1 Igusa variables and the Gram-index lattice

For a Siegel period

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \in \mathcal{H}_2, \qquad q = e^{2\pi i z_1}, \quad r = e^{2\pi i z_2}, \quad s = e^{2\pi i z_3},$$

and

$$T(n,l,m) = \begin{pmatrix} n & l/2 \\ l/2 & m \end{pmatrix}, \qquad n, l, m \in \mathbb{Z},$$

the trace pairing

$$\big(T(n,l,m), Z\big)_{\mathrm{tr}} := \mathrm{tr}\big(T(n,l,m)\, Z\big) = nz_1 + lz_2 + mz_3$$

gives the Borcherds monomial $x^{(n,l,m)} = q^n r^l s^m$.

Definition 2.1 *(Gram-index lattice and effective semigroup).* The *Gram-index lattice* is

$$\Gamma_{\text{gram}} := \mathbb{Z}^3, \qquad \gamma = (n, l, m).$$

The *cusp-positive semigroup* at the standard cusp $\mathfrak{c}_\infty$ with chamber convention $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ is

$$\Gamma_{\text{eff}} = \{(n, l, m) : m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \cup \{(n, l, 0) : n > 0,\ l \in \mathbb{Z}\} \cup \{(0, l, 0) : l < 0\}.$$

The *Jacobi-active sublocus* $\Gamma_{\text{act}} \subset \Gamma_{\text{eff}}$ is the locus of nonvanishing $f(nm, l)$ in the weak Jacobi form $\phi_{0,1}$ (computed, [32]).

Proposition 2.2 *(Gram matrix convention for protected charges).* For a formal even charge $c = (Q, P) \in \Gamma_X^{\text{form}}$ in the unimodular lattice Λ_S, set

$$n = \frac{(Q, Q)}{2}, \qquad l = (Q, P), \qquad m = \frac{(P, P)}{2}.$$

Then $n, l, m \in \mathbb{Z}$ (since Λ_S is even unimodular, proved),

$$\begin{pmatrix} (Q, Q) & (Q, P) \\ (P, Q) & (P, P) \end{pmatrix} = 2\,T(n, l, m), \qquad \det\bigl(2T(n, l, m)\bigr) = 4nm - l^2,$$

and $x^{(n,l,m)} = q^n r^l s^m$ is the Fourier character of $\bigl(T(n, l, m), Z\bigr)_{\text{tr}}$.

Proof. The Mukai pairing is integer-valued and even on $\Lambda_S \oplus \Lambda_S$ by Definition 1.3. The matrix identity is the direct expansion

$$\begin{pmatrix} (Q, Q) & (Q, P) \\ (P, Q) & (P, P) \end{pmatrix} = \begin{pmatrix} 2n & l \\ l & 2m \end{pmatrix} = 2\,T(n, l, m).$$

The determinant follows. The Fourier character is $\exp(2\pi i\ \mathrm{tr}(TZ)) = q^n r^l s^m$. □

2.2 The Gram map and its bilinear cocycle

Definition 2.3 *(Quadratic Gram map and bilinear cocycle).* The *Gram map* is the quadratic projection

$$\Pi_X : \Gamma_X^{\text{form}} \longrightarrow \Gamma_{\text{gram}}, \qquad \Pi_X(Q, P) = \left(\frac{(Q, Q)}{2},\ (Q, P),\ \frac{(P, P)}{2}\right).$$

The *symmetric bilinear cocycle* B associated to Π_X is

$$B : \Gamma_X^{\text{form}} \times \Gamma_X^{\text{form}} \longrightarrow \Gamma_{\text{gram}}, \qquad B\bigl((Q, P), (Q', P')\bigr) = \bigl((Q, Q'),\ (Q, P') + (Q', P),\ (P, P')\bigr).$$

Lemma 2.4 *(Gram additivity defect).* For $c, c' \in \Gamma_X^{\text{form}}$,

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

The defect B is symmetric, $\mathbb{Z}$-bilinear, integer-valued, and satisfies polarization: $B(c, c) = 2\,\Pi_X(c)$. As an ordinary additive cochain identity,

$$B = -\delta\Pi_X, \qquad (\delta q)(c, c') = q(c) + q(c') - q(c + c').$$

(proved.)

Remark 2.5 *(Quadratic primitive, no linear primitive).* The equality $B = -\delta\Pi_X$ is an equality in the ordinary inhomogeneous cochain complex, where the primitive Π_X is quadratic. There is no additive homomorphism $L : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ with $B = -\delta L$. Thus the normal-ordering defect is removed by the quadratic Gram coordinate, not by a linear change of charge lattice.

Proof. Write $c = (Q, P)$, $c' = (Q', P')$; then

$$\frac{(Q+Q')^2}{2} = \frac{Q^2}{2} + \frac{Q'^2}{2} + Q \cdot Q',$$
$$(Q+Q') \cdot (P+P') = Q \cdot P + Q' \cdot P' + Q \cdot P' + Q' \cdot P,$$
$$\frac{(P+P')^2}{2} = \frac{P^2}{2} + \frac{P'^2}{2} + P \cdot P'.$$

The three cross terms are the three components of $B(c, c')$. Symmetry, bilinearity, and integrality follow from the corresponding properties of the Mukai pairing. Polarization is the case $c' = c$. The cochain identity rewrites the additivity defect. □

Lemma 2.6 *(Bilinear cocycle identity).* B is normalized, $B(c, 0) = B(0, c) = 0$, and the additive coboundary

$$(\delta B)(a, b, c) := B(b, c) - B(a + b, c) + B(a, b + c) - B(a, b)$$

vanishes identically on Γ_X^{form} (proved).

Proof. By bilinearity in each slot,

$$B(a + b, c) = B(a, c) + B(b, c), \qquad B(a, b + c) = B(a, b) + B(a, c).$$

Substituting,

$$(\delta B)(a, b, c) = B(b, c) - B(a, c) - B(b, c) + B(a, b) + B(a, c) - B(a, b) = 0.$$

The polarization identity $B(c, c) = 2\,\Pi_X(c)$ is the only place where B takes a non-vanishing value on the diagonal; this is structural, not a Bott-element framing. □

Remark 2.7 *(Polarization, not Bott element).* The identity $B(c, c) = 2\,\Pi_X(c)$ is the polarization identity for the symmetric bilinear B associated to the quadratic Π_X. It is forced by the homogeneous-of-degree-two structure of the Gram map. No external Bott-element framing or characteristic-class coincidence is involved. The full structural content is that Π_X is a homogeneous quadratic on the additive group Γ_X^{form} valued in Γ_{gram}, with associated symmetric bilinear B given by the Mukai pairing of components on each direction.

2.3 Formal lift of every Gram triple

Lemma 2.8 *(Formal primitive torsion-one lift of every Gram triple).* Suppose Λ_S contains two orthogonal hyperbolic planes $U_1 \oplus U_2$ with bases (e_1, f_1, e_2, f_2) as in Lemma 1.8. Then every Gram triple $(n, l, m) \in \mathbb{Z}^3$ admits a primitive saturated formal full-Mukai lift $c = (Q, P) \in \Gamma_X^{\mathrm{form}}$ with $\Pi_X(Q, P) = (n, l, m)$. One may take

$$Q = e_1 + nf_1, \qquad P = lf_1 + e_2 + mf_2.$$

For this lift, the wedge torsion invariant

$$I(Q,P) := \gcd\big\{\text{coordinates of } Q \wedge P \text{ in a } \mathbb{Z}\text{-basis of } \textstyle\bigwedge^2 \Lambda_S\big\}$$

equals 1. (proved.)

Proof. The hyperbolic-plane pairings give

$$Q^2 = 2n, \qquad P^2 = 2m, \qquad Q \cdot P = l,$$

hence $\Pi_X(Q,P) = (n,l,m)$. The submodule $\mathbb{Z}Q + \mathbb{Z}P \subset \Lambda_S$ contains the unimodular minor $\det\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$ in the rows indexed by e_1, e_2 of the coefficient matrix in (e_1, f_1, e_2, f_2), so the lift is primitive saturated. The wedge expansion

$$Q \wedge P = l\, e_1 \wedge f_1 + e_1 \wedge e_2 + m\, e_1 \wedge f_2 + n\, f_1 \wedge e_2 + nm\, f_1 \wedge f_2$$

contains the basis element $e_1 \wedge e_2$ with coefficient 1, so the gcd is 1 and $I(Q,P) = 1$. □

Remark 2.9 *(Pointwise existence is not Hall support).* Lemma 2.8 produces a formal charge plane realizing every prescribed Igusa degree. It does not prove algebraicity of (Q,P), effectivity, non-empty compact Hall moduli, duality-orbit descent, source-index descent, or any chain-level Hall strictification. The compact Hall obstruction is functorial compatibility of pointwise existence with Hall addition and the primitive bracket; only the latter is the working content of Chapter 6.

3 Normal-ordered Gram extension

3.1 *Central extension and the additive normal-ordered Gram homomorphism*

Definition 3.1 *(Normal-ordered extension and Gram homomorphism).* Set

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{form}} \oplus \Gamma_{\mathrm{gram}},$$

with operation

$$(c,T) \star (c',T') := \big(c + c',\ T + T' + B(c,c')\big).$$

Define the *normal-ordered Gram map*

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) = \Pi_X(c) - T.$$

The *raw placement* and the *split section* are

$$i_0 : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \quad i_0(c) = (c, 0), \qquad s : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \quad s(c) = (c, \Pi_X(c)).$$

Theorem 3.2 *(Group law of the normal-ordered extension; additivity of $\overline{\Pi}_X$).* Let $\widehat{\Gamma}_X, \star, \overline{\Pi}_X$, i_0, s be as in Definition 3.1. Then:

(i) $(\widehat{\Gamma}_X, \star)$ is an abelian group with identity $(0,0)$ and inverse $(c,T)^{-1} = (-c,\ -T + B(c,c)) = (-c,\ -T + 2\,\Pi_X(c))$.

(ii) The sequence

$$0 \to \Gamma_{\mathrm{gram}} \xrightarrow{T \mapsto (0,T)} \widehat{\Gamma}_X \xrightarrow{\mathrm{pr}_1} \Gamma_X^{\mathrm{form}} \to 0$$

is a central extension of abelian groups with cocycle B.

(iii) $\overline{\Pi}_X$ is a group homomorphism from $(\widehat{\Gamma}_X, \star)$ to the additive group Γ_{gram}.

(iv) The split section $s(c) = (c, \Pi_X(c))$ is additive, and

$$\overline{\Pi}_X(i_0(c)) = \Pi_X(c), \qquad \overline{\Pi}_X(s(c)) = 0.$$

(v) The change of variables $(c, T) \mapsto (c, T - \Pi_X(c))$ carries $\star$ to componentwise addition; thus $\widehat{\Gamma}_X \cong \Gamma_X^{\mathrm{form}} \oplus \Gamma_{\mathrm{gram}}$ as abstract abelian groups, and $[B] = 0$ in additive cohomology.

(proved.)

Proof. (i) Associativity follows from the cocycle identity $\delta B = 0$ (Lemma 2.6); the associator $B(a, b) + B(a + b, c) - B(a, b + c) - B(b, c)$ vanishes by bilinearity. Commutativity is symmetry of B. The unit calculation $(c, T) \star (0, 0) = (c, T + B(c, 0)) = (c, T)$ and inverse $(c, T) \star (-c, -T + B(c, c)) = (0, B(c, c) - B(c, c)) = (0, 0)$ (using $B(c, -c) = -B(c, c)$ by bilinearity) close the abelian-group structure.

(ii) Definition.

(iii) Apply Lemma 2.4:

$$\overline{\Pi}_X((c, T) \star (c', T')) = \Pi_X(c + c') - T - T' - B(c, c') = \Pi_X(c) + \Pi_X(c') - T - T' = \overline{\Pi}_X(c, T) + \overline{\Pi}_X(c', T').$$

(iv) For the section,

$$s(c) \star s(c') = (c + c',\ \Pi_X(c) + \Pi_X(c') + B(c, c')) = (c + c',\ \Pi_X(c + c')) = s(c + c').$$

The identities $\overline{\Pi}_X(i_0(c)) = \Pi_X(c) - 0 = \Pi_X(c)$ and $\overline{\Pi}_X(s(c)) = \Pi_X(c) - \Pi_X(c) = 0$ are immediate.

(v) The map $(c, T) \mapsto (c, T - \Pi_X(c))$ is a bijection with inverse $(c, T) \mapsto (c, T + \Pi_X(c))$. Direct calculation gives

$$(c, T - \Pi_X(c)) + (c', T' - \Pi_X(c')) = (c + c',\ T + T' - \Pi_X(c) - \Pi_X(c')) = (c + c',\ (T + T' + B(c, c')) - \Pi_X(c + c')),$$

matching the change of variables applied to $(c, T) \star (c', T')$. □

3.2 The fixed-lift raw-pushforward obstruction theorem

Definition 3.3 *(Raw homogeneous and fixed-lift Π_X-pushforwards).* Let $P = \bigoplus_{c \in \Gamma_X^{\mathrm{form}}} P_c$ be a Γ_X^{form}-graded primitive object. The *raw homogeneous Π_X-pushforward* is

$$\left(\Pi_{X,*}^{\mathrm{raw}} P\right)_\gamma := \bigoplus_{\Pi_X(c) = \gamma} P_c.$$

The *strict fixed-lift raw Π_X-pushforward* is the narrower operation obtained by choosing one lift $L(\gamma) \in \Gamma_X^{\mathrm{form}}$ for every retained γ and setting

$$\left(\Pi_{X,*}^{\mathrm{raw},L} P\right)_\gamma := P_{L(\gamma)}.$$

Theorem 3.4 *(Raw homogeneous fixed-lift Π_X-pushforward cannot realize the type-II real-root strings).* Let $P = \bigoplus_{c\in\Gamma_X^{\mathrm{form}}} P_c$ be a Γ_X^{form}-graded Lie superalgebra with $[P_c, P_{c'}] \subset P_{c+c'}$. Suppose a strict fixed-lift raw pushforward of P along the quadratic map Π_X were a Γ_{gram}-graded Lie superalgebra structure on the displayed bracket channels of chosen charge lifts. Then every nonzero bracket channel $[P_c, P_{c'}]$ contributing to that strict raw pushforward must satisfy $B(c, c') = 0$.

For the type-II real simple roots $(\delta_1, \delta_2, \delta_3)$ of the Gritsenko–Nikulin algebra $\mathfrak{g}_{\Delta_5}$ with Cartan matrix

$$A_{\mathrm{II}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

this condition is incompatible with the nonzero second real-root strings $2\delta_i + \delta_j$, $i \neq j$. The strict fixed-lift raw Π_X-descent therefore cannot carry these brackets; they require the normal-ordered extension $\widehat{\Gamma}_X$ and the additive homomorphism $\overline{\Pi}_X$. (proved, [43].)

Proof. If the raw pushforward is Γ_{gram}-graded, a nonzero bracket of classes of raw Gram degrees $\Pi_X(c)$ and $\Pi_X(c')$ must land in raw Gram degree $\Pi_X(c) + \Pi_X(c')$. But the upstairs bracket lands in physical degree $c + c'$, whose raw Gram shadow is

$$\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c').$$

Hence $B(c, c') = 0$ on every nonzero bracket channel.

Take physical lifts c_i, c_j of distinct type-II real simple root degrees γ_i, γ_j, $\Pi_X(c_i) = \gamma_i$ with $\gamma_i \neq 0$. The Gritsenko–Nikulin Cartan matrix A_{II} has off-diagonal entries -2, so distinct simple roots satisfy $\langle \delta_i, \delta_j \rangle = -2$, and the real-root string through δ_j in the δ_i-direction includes the levels $\delta_i + \delta_j$ and $2\delta_i + \delta_j$ (proved, [43, §2–3]). Both correspond to nonzero brackets, hence raw pushforward forces

$$B(c_i, c_j) = 0, \qquad B(c_i, c_i + c_j) = 0.$$

Bilinearity and polarization give

$$B(c_i, c_i + c_j) = B(c_i, c_i) + B(c_i, c_j) = 2\,\Pi_X(c_i) + 0 = 2\,\gamma_i,$$

which is nonzero in the torsion-free lattice Γ_{gram}. Contradiction. □

Remark 3.5 *(Scope of the no-go).* Theorem 3.4 excludes the strict *homogeneous fixed-lift* raw pushforward. It does not exclude isolated B-isotropic bracket channels and does not decide the mixed-lift fibre-summed construction in which several lifts of a single Gram degree are summed before bracketing; the latter remains an open mixed-lift question. The normal-ordered extension removes exactly the displayed obstruction because $\overline{\Pi}_X$ is additive on $\widehat{\Gamma}_X$. Chain-level Hall strictification (A_∞-compatibility, cyclic equivariance, Frobenius, multiplicative orientation) is a separate input system, named in §6 and developed in Chapter 6.

Remark 3.6 *(Quadratic obstruction; no linear trivialisation).* Suppose $B = \delta L$ for some additive linear $L : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$. Setting $c' = c$, $B(c, c) = 2L(c) - L(2c)$; additivity gives $L(2c) = 2L(c)$, so $B(c, c) = 0$, contradicting $B(c, c) = 2\,\Pi_X(c) \neq 0$ for any c with $(Q, Q), (Q, P), (P, P)$ not all zero. Hence no additive linear trivialisation exists. The quadratic primitive Π_X is necessary; trivializing the additive cocycle requires a quadratic cochain, not a linear one (proved).

3.3 Normal-ordering flag formula

Lemma 3.7 *(Multiplication of raw placements).* For $c_1, \dots, c_k \in \Gamma_X^{\mathrm{form}}$,

$$i_0(c_1) \star \cdots \star i_0(c_k) = \Big(\sum_i c_i,\ \sum_{i<j} B(c_i, c_j)\Big),$$

and consequently

$$\overline{\Pi}_X(i_0(c_1) \star \cdots \star i_0(c_k)) = \sum_i \Pi_X(c_i).$$

(proved.)

Proof. Induct on k. The case $k = 2$ is the definition of $\star$. At step $k \to k+1$,

$$B\Big(\sum_{i\le k} c_i,\ c_{k+1}\Big) = \sum_{i\le k} B(c_i,\ c_{k+1})$$

by bilinearity. Applying $\overline{\Pi}_X$ subtracts the accumulated pairwise B-term from $\Pi_X(\sum_i c_i) = \sum_i \Pi_X(c_i) + \sum_{i<j} B(c_i, c_j)$. □

Remark 3.8 *(Normal-ordered primitive recognition target).* The interface with the BKM denominator algebra is the normal-ordered charge route

$$\Gamma_X^{\mathrm{form}} \hookrightarrow \widehat{\Gamma}_X \xrightarrow{\overline{\Pi}_X} \Gamma_{\mathrm{gram}} \xrightarrow{\alpha} \Lambda_{II}^{2,1},$$

where $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$ is the type-IV embedding into the BKM root lattice (proved, [43]). At chain level this becomes the descent $\overline{\Pi}^{\Theta}_{X,*}$ of Chapter 6. The primitive recognition problem is to construct a protected primitive Hall algebra whose normal-ordered descended bracket is the Borcherds–Kac bracket of $\mathfrak{g}_{\Delta_5}$. Raw Gram pushforward is not an admissible recognition target.

4 The hybrid $\mathrm{Ran}^{\mathrm{hyb}}(E)$ factorization carrier

After integration along the K3 direction, the chiral object is supported on the elliptic factor E. The ordinary finite-support carrier is the Ran space $\mathrm{Ran}(E)$ of finite subsets, but a one-dimensional subscheme with positive elliptic degree projects onto all of E. The hybrid carrier $\mathrm{Ran}^{\mathrm{hyb}}(E)$ is the disjoint union of the projection-finite local stratum $\mathrm{Ran}(E)_{b=0}$ and the wrapped stratum $\mathrm{Ran}^{\mathrm{wr}}(E)_{b>0}$, with mixed local/wrapped correspondences supplied as additional finite-stage data.

4.1 Projection-locality obstruction

Lemma 4.1 *(Projection-locality obstruction).* Let $C \subset X = S \times E$ be a one-dimensional closed subscheme (reduced and irreducible, for definiteness) with

$$\deg_E(C) := \deg\big(p_E\restriction_C : C \to E\big) > 0.$$

Then $p_E(C) = E$, hence $p_E(C)$ is not a finite subset of E and C is not local on $\mathrm{Ran}(E)$. (proved.)

Proof. The pushforward $p_{E,*}[C]$ is a positive multiple of the fundamental class $[E]$, so the image $p_E(C)$ is a closed one-dimensional subset of the irreducible curve E. An irreducible curve has only itself as a one-dimensional closed subset, so $p_E(C) = E$. □

Remark 4.2 *(Geometric versus Borcherds elliptic degree).* The lemma fixes the necessity of a wrapped stratum. The geometric elliptic-degree map b_R^{geom}, on a finite Hall stage R, records the integer $\deg_E$ of a chosen retained class. The trace exponent $m_R^{\mathrm{Bch}} := \mathrm{pr}_3 \overline{\Pi}_X$, on the same stage, records the third coordinate of the normal-ordered Gram pushforward. The two are not equal as functions on the formal lattice; the Borcherds exponent m records the magnetic Mukai pairing $(P,P)/2$, which sees Λ_S-square data, while b_R^{geom} records the E-degree of the geometric support, which sees $H^0(S) \oplus H^4(S)$ data through r and ch_2 of the K3 factor. The wrapped stratum carries $b_R^{\mathrm{geom}} > 0$ branes; the local stratum carries $b_R^{\mathrm{geom}} = 0$ branes whose magnetic component lies in the projection-finite kernel. This is the $\mathrm{Ran}(E)$-support-locality obstruction in the K3-integrated chiral shadow.

4.2 The hybrid carrier

Definition 4.3 *(Hybrid Ran carrier).* The *hybrid Ran carrier* on E is the disjoint union

$$\mathrm{Ran}^{\mathrm{hyb}}(E) := \mathrm{Ran}(E)_{b=0} \sqcup \mathrm{Ran}^{\mathrm{wr}}(E)_{b>0},$$

with the projection-finite local stratum $\mathrm{Ran}(E)_{b=0}$ and the wrapped stratum $\mathrm{Ran}^{\mathrm{wr}}(E)_{b>0}$ defined as follows.

(i) $\mathrm{Ran}(E)_{b=0}$ is the standard Ran space of finite (unordered) subsets $T = \{x_1, \dots, x_k\} \subset E$ ([6, 33]); its functor of points sends a test scheme S' to the set of finite closed subschemes $T' \subset E \times S'$ whose projection to S' is finite étale of constant fibre size $\leq R$ for some chosen finite Hall height R. This is the $b_R^{\mathrm{geom}} = 0$ stratum: branes whose K3 factor has zero elliptic degree.

(ii) $\mathrm{Ran}^{\mathrm{wr}}(E)_{b>0}$ is the *wrapped prequotient stratum*: a colim of pre-quotient stacks $\mathfrak{W}_{R,b}$ parameterising rigidified compactly supported B-branes $\mathcal{F} \in D^b(\mathrm{Coh}\, X)$ of bounded retained Liu– Hilbert class $\Xi = (\gamma, [a,b], (P_i), N)$ with positive elliptic degree $b = \deg_E(\mathrm{ch}_2(\mathcal{F})) > 0$, modulo E-translation up to the chosen anchor (§4.4).

The two strata are exchanged only by the mixed local/wrapped correspondences of §4.3; they are not identified.

Remark 4.4 *(Two strata, two registers).* On the $b = 0$ stratum the standard Ran/factorization formalism applies and supplies the local chiral factorization input in $\mathrm{Fact}(\mathrm{Ran}(E))$ (folklore, [6, 33]). On the $b > 0$ stratum the support of $\mathrm{Supp}\,\mathcal{F}$ projects onto E, so the brane is global; its data live on the wrapped prequotient and after E-quotient become E-equivariant data on the elliptic curve. The hybrid carrier is the smallest carrier reflecting both strata; it is not a single Ran space.

4.3 Mixed local/wrapped correspondences

Definition 4.5 *(Mixed local/wrapped correspondences (finite Hall stage R)).* Fix a finite Hall stage R and retained Liu–Hilbert classes Ξ_A, Ξ_B, Ξ_C with $\Xi_C = \Xi_A + \Xi_B$ in the closure rules of [59]. The *retained extension correspondence*

$$\mathfrak{E}^{\mathrm{ret}}_{\Xi_A,\Xi_B;\Xi_C} \xrightarrow{p} \mathfrak{M}^{\mathrm{ss}}_{\Xi_A} \times \mathfrak{M}^{\mathrm{ss}}_{\Xi_B}, \qquad \mathfrak{E}^{\mathrm{ret}}_{\Xi_A,\Xi_B;\Xi_C} \xrightarrow{q} \mathfrak{M}^{\mathrm{ss}}_{\Xi_C}$$

is the closed derived flag-Quot/filtration stack over $\mathfrak{M}^{\mathrm{ss}}_{\Xi_C}$, with q required at this finite stage to be proper or $\mathrm{Coh}^{\mathrm{prop}}$-admissible in the compactified model (folklore). Three colours of correspondence appear, indexed by the location of Ξ_A, Ξ_B in $\mathrm{Ran}^{\mathrm{hyb}}(E)$:

(LL) *local-local*: $b_A = b_B = 0$;

(LW) *ordered local/wrapped*: $b_A = 0, b_B > 0$;
(WW) *wrapped-wrapped*: $b_A, b_B > 0$.

The arity-three shadow is the eight-word two-step binary flag atlas

$$\{L, W\}^3 = \{LLL, LLW, LWL, LWW, WLL, WLW, WWL, WWW\},$$

which records binary associativity only; higher-coloured tree configurations live on a separate coloured tree category $\mathrm{Tree}_R^{\mathrm{col}}$, supplied as additional input (folklore, [65]).

Remark 4.6 *(Eight binary words versus colored trees).* The eight binary words of Definition 4.5 exhaust the arity-three two-step compositions $\{L, W\}^3$, but they do not catalogue higher-coloured tree configurations, units, symmetry/order conventions, refinement maps, descent, or overlaps. Those data are additional finite Hall input named at §6; asserting them prematurely confuses binary associativity with the full factorization-coherence structure ([65], correspondence formalism).

4.4 Wrapped anchor and translation linearization

Definition 4.7 *(Determinant anchor on the wrapped stratum).* For a retained E-stable family $\mathcal{F}$ on the wrapped stratum, the *determinant anchor line bundle* on E is

$$\lambda(\mathcal{F}) := \det Rp_{E,*}\mathcal{F} \otimes \mathcal{O}_E\big(-\chi(\mathcal{F})\, 0_E\big).$$

This is a degree-zero line bundle on E, hence a point of $\mathrm{Pic}^0(E) \cong E$. The translation law

$$\lambda(t \cdot \mathcal{F}) = \lambda(\mathcal{F}) + \chi(\mathcal{F})\, t, \qquad t \in E,$$

holds in $\mathrm{Pic}^0(E)$ (proved, [78]).

Lemma 4.8 *(Connected E-descent: separated cohomological classes).* For the connected component E° of the identity in the E-translation action,

$$H^*(BE; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2], \qquad |u_i| = 2.$$

Hence

$$H^1(BE; \mathbb{F}_2) = 0, \qquad H^2(BE; \mathbb{F}_2) = \mathbb{F}_2 u_1 \oplus \mathbb{F}_2 u_2.$$

There is no degree-one connected character; there is a degree-two connected gerbe class

$$\alpha^{E,\mathrm{free}} = a_1 u_1 + a_2 u_2 \in H^2(BE; \mathbb{F}_2).$$

(proved, [82].)

Proof. $E^{\mathrm{top}} \simeq T^2$ has classifying space $BE \simeq BT^2$ and $H^*(BT^2; \mathbb{F}_2) = \mathbb{F}_2[u_1, u_2]$ with both u_i of degree 2 (mod-2 Chern classes of the universal line bundles). H^1 vanishes; H^2 is the displayed two-dimensional $\mathbb{F}_2$ vector space. Vanishing of the connected gerbe class is the equation $a_1 = a_2 = 0$ on the actual Borel/edge class of the equivariant reduced determinant complex, not a consequence of $H^1(BE) = 0$ or ordinary translation invariance. □

Lemma 4.9 *(Finite stabilizer $E[N]$-descent: gerbe and linearization classes).* For the order-two finite subgroup $E[2] \cong (\mathbb{Z}/2)^2$,

$$H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1,x_2], \qquad |x_i| = 1.$$

The degree-two gerbe class is

$$\beta^{E,E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \qquad (b_{20},b_{11},b_{02}) = (r_1,\ r_1+r_2+r_3,\ r_2),$$

detected on cyclic subgroup restrictions by joint reduction; the $r_i \in \mathbb{F}_2$ are the three nonzero point-stabilizer parities. After $\beta^{E,E[2]} = 0$ (i.e. $r_1 = r_2 = r_3 = 0$), a separate degree-one linearization class remains:

$$\lambda^{E,E[2]} = \lambda_1x_1 + \lambda_2x_2 \in H^1(BE[2];\mathbb{F}_2), \qquad \rho_1 = \lambda_1,\ \rho_2 = \lambda_2,\ \rho_3 = \lambda_1+\lambda_2.$$

The r_i are gerbe bits in H^2; the ρ_i are linearization bits in H^1; the two live in distinct cohomological degrees and remain independent. For even $N \geq 4$ with $2^a \parallel N,\ a \geq 2$,

$$H^*\big(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2\big) = \Lambda(x_1,x_2)\otimes\mathbb{F}_2[y_1,y_2],$$

$$\beta^{E,N} = A_1^{(N)}y_1 + A_{12}^{(N)}x_1x_2 + A_2^{(N)}y_2, \qquad \lambda^{E,N} = \lambda_1^{(N)}x_1 + \lambda_2^{(N)}x_2.$$

The mixed term $A_{12}^{(N)}$ is invisible to cyclic order-two restrictions; even $N \geq 4$ tests must be checked on the full two-primary generators. For odd N, the mod-2 orientation obstruction vanishes by transfer. (proved.)

Proof. $E[2] \cong (\mathbb{Z}/2)^2$ has $H^*(B(\mathbb{Z}/2)^2;\mathbb{F}_2) = \mathbb{F}_2[x_1,x_2]$ with $|x_i| = 1$. The class $\beta^{E,E[2]}$ is the finite-stabilizer Cech–Borel edge class, not the restriction of the connected BE-class $\alpha^{E,\mathrm{free}}$. Its restriction to the three non-zero cyclic subgroups gives the three parities r_i; inversion of this three-by-three system gives $(b_{20},b_{11},b_{02}) = (r_1, r_1+r_2+r_3, r_2)$. The degree-one linearization class $\lambda^{E,E[2]}$ is the residual character of the underlying line bundle after $\beta = 0$ is imposed; it lives in H^1 and is distinct from the gerbe $\beta \in H^2$ in degree. Even $N \geq 4$ introduces the polynomial generators $y_i \in H^2(B(\mathbb{Z}/2^a))$ and the mixed term x_1x_2, which on cyclic restrictions vanishes; full two-primary detection is needed. Odd N contributes only $\mathbb{F}_p$-cohomology with p odd, on which the mod-2 orientation has zero restriction by the transfer. □

Remark 4.10 *(Translation linearization criterion, not vanishing).* For a stabilizer $H \subset E[N]$ and a determinant anchor line bundle $\lambda(\mathcal{F})$, ordinary translation invariance $g^*\lambda \simeq \lambda$ in $\mathrm{Pic}^0(E)$ does not choose coherent isomorphisms $\phi_g : \lambda \to g^*\lambda$. After the square-root gerbe is killed, the choices form a character torsor in $H^1(BH;\mathbb{F}_2)$, and quotient orientation requires the linearization class

$$\lambda^H \in H^1(BH;\mathbb{F}_2)$$

to vanish. This is a *criterion* for descent on the wrapped stratum; it is not implied by $g^*[\lambda] = [\lambda]$ in $\mathrm{Pic}(E)$. The translation law $\lambda(t\cdot\mathcal{F}) = \lambda(\mathcal{F}) + \chi(\mathcal{F})t$ of Definition 4.7 has weight $\chi(\mathcal{F})$. It becomes unit weight only after the retained finite-cover normalisation when $\chi(\mathcal{F}) \neq 0$; for $\chi(\mathcal{F}) = 0$ it is unweighted and descends only after the linearization class vanishes. (proved as a criterion.)

Remark 4.11 *(Mod-2 character does not control odd-order anchor characters).* The mod-2 orientation character does not by itself control odd-order anchor characters. For anchor descent on the wrapped stratum, also compute the actual stabilizer character on the chosen anchor trivialization, valued in $\widehat{H} = \mathrm{Hom}(H,\mathbb{C}^\times)$, or define the retained stabilizer as preserving that trivialization. (unverified; supplied as part of the finite Hall datum.)

4.5 Cusp polarization and chamber transport

Definition 4.12 *(Cusp polarization on Γ_{eff}).* The chamber convention $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ at the standard cusp $\mathfrak{c}_\infty$ determines the cusp-positive semigroup Γ_{eff} of Definition 2.1. The order on triples is lexicographic: $m > 0$ first because s is smallest; if $m = 0$ then $n > 0$ because q is next; if $m = n = 0$ then $l < 0$ because $|r|^{-1} \ll 1$. This is a chamber convention for the completed product, not a microscopic charge positivity statement.

Remark 4.13 *(Chamber transport).* The pairing $\langle \gamma, \gamma' \rangle_{\mathrm{ind}} = 2(nm' + n'm) - ll'$ is the Gram pairing on triples; the order on Γ_{eff} is a Fock polarization at one cusp. An element of the duality group $\mathbf{Sp}_4(\mathbb{Z})$ transports both the chamber and the expansion to the corresponding product chamber at the transported cusp. The arithmetic group $\mathbf{Sp}_4(\mathbb{Z})$ is the symmetry of the genus-two chemical potential Z and of the invariant Gram plane on $\Lambda^{3,2}$; it is not asserted to be the full microscopic T-/U-duality group of the K3×T^2 compactification, which is larger (folklore, [29, 46]).

Remark 4.14 *(Chamber dependence of the wrapped stratum).* The cusp chamber chooses a positive direction on Γ_{eff}. At $\mathfrak{c}_\infty$ the wrapped stratum carries $\deg_E > 0$ branes; at a transported cusp the wrapped direction is the image of $\deg_E$ under the corresponding chamber transport. Mixed Hall correspondences are chamber-equivariant when the finite Hall stage R is compatible with the duality element; otherwise the correspondence stack is the original closed flag-Quot stack restricted to the new chamber.

Definition 4.15 *(Chamber-compatible Hall descent datum).* Let σ and σ' be two chosen stability data, and let $g \in \mathbf{Sp}_4(\mathbb{Z})$ transport the standard cusp chamber to another product chamber. A chamber-compatible Hall descent datum is a Kontsevich–Soibelman wall-crossing system

$$\mathsf{WC}_{g,R} : C_{\sigma,S,\leq R} \longrightarrow C_{\sigma',S,\leq R'}$$

on the retained finite Hall stages, with transition maps in R, which preserves the normal-ordered Gram map $\overline{\Pi}_X$, transports the local/wrapped split, the quotient-after-correspondence datum, the orientation lines, and the protected integration maps. The identity $\overline{\Pi}_X(\widehat{c} \star \widehat{c}') = \overline{\Pi}_X(\widehat{c}) + \overline{\Pi}_X(\widehat{c}')$ on $\widehat{\Gamma}_X$ is a chamber-independent lattice fact; the compact Hall category and its primitive bracket are not chamber-independent without $\mathsf{WC}_{g,R}$ (folklore/β, [59]).

5 Local protected observable algebra (sectorial source datum)

The compact realization route of Chapter 6 takes as supplied input a chain-level local observable system on opens $U \subset X = S \times E$. The system is graded by the algebraic/effective Hall support, factorizes on disjoint opens, and restricts on E-projection-finite supports to a chiral object on $\mathrm{Ran}(E)_{b=0}$. The data are external assumptions named in the literature and form the source datum for the compact realization.

Definition 5.1 *(Local protected observable algebra).* Let $U \subset X = S \times E$ be open. Let $\Gamma^{\mathrm{eff}}_{X,\sigma}(U) \subset \Gamma^{\mathrm{alg}}_X \subset \Gamma^{\mathrm{form}}_X$ be the supplied algebraic/effective charge support. For $c \in \Gamma^{\mathrm{eff}}_{X,\sigma}(U)$, let $\mathfrak{M}_c(U)$ be the supplied derived moduli stack of compactly supported B-branes on U of Mukai charge c. The PTVV theorem supplies the ambient (-1)-shifted symplectic structure on the CY_3 perfect-complex moduli problem (proved, [89]). Let Φ_c denote the BBDJS perverse sheaf $P^\bullet_{\mathfrak{M}_c,s_c}$ on an oriented d-critical truncation, when such a d-critical structure and square-root orientation are supplied (proved, [11, Theorem 6.9]). Let o_c be the orientation line, a square root of the determinant line of the cotangent complex of $\mathfrak{M}_c(U)$, supplied as part of the Hall datum.

The *local protected observable algebra of X on U* is the source object

$$\mathcal{A}_X(U) := \bigoplus_{c\in\Gamma^{\mathrm{eff}}_{X,\sigma}(U)} H^{\mathrm{BM}}_*\big(\mathfrak{M}_c(U),\ \Phi_c\otimes o_c\big),$$

graded by the supplied algebraic/effective support, not by the full formal lattice Γ^{form}_X. For pairwise disjoint $U_1,\dots,U_k\subset X$, the proposed factorization map is

$$\mathcal{A}_X(U_1)\otimes\cdots\otimes\mathcal{A}_X(U_k)\longrightarrow\mathcal{A}_X(U_1\sqcup\cdots\sqcup U_k),$$

defined chain-level by Massey–Saito Thom–Sebastiani for vanishing cycles

$$\Phi_{F_1\oplus\cdots\oplus F_k}\simeq\Phi_{F_1}\boxtimes\cdots\boxtimes\Phi_{F_k}$$

on disjoint d-critical charts (proved, [73, 91]), together with the Ext-vanishing $\mathrm{Ext}^*(\mathcal{F},\mathcal{G})=0$ for branes $\mathcal{F},\mathcal{G}$ with disjoint supports (folklore).

Remark 5.2 *(External source datum).* The components are external assumptions: PTVV (-1)-shifted symplectic structure on derived moduli of B-branes (proved, [89]); BBDJS vanishing-cycle perverse sheaf and its Thom–Sebastiani isomorphism (proved, [11, 73, 91]); orientation lines and Joyce–Upmeier-compatible multiplicative orientation data (ambient technology proved in [52, Theorem 4.1]; the $K3\times E$ quotient orientation is supplied separately). The factorization map relies on disjoint-supports Ext-vanishing for compactly supported branes (folklore; absent for non-compact supports, where one recovers chiral algebra structure, not commutative factorization). Compactness on $X=S\times E$ is open because the relevant moduli of branes are typically non-quasicompact in Λ_S-charge; the sectorial truncation of §4 is the finite-stage carrier, and the compact-source recognition datum $(S1)$–$(S10)$ is the remaining finite recognition problem.

6 The four obstructions and the comparison target

The hybrid carrier and the source datum are supplied as a finite holomorphic E_3-prefactorization datum and a $K3$-to-E specialization datum on $\mathrm{Ran}^{\mathrm{hyb}}(E)$. The cross-level comparison $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)=\Delta_5$ holds on $K3\times E$ relative to four retained data, with $(O2)$ represented by the retained wall-atlas datum of Appendix 7 (Theorems 7.6, 7.12, 7.13, 7.18). They are the orientation and Pfaffian data feeding Chapter 6.

(D0) D0-degeneration limit. The retained D0-HN datum supplies the cosection-reduced d-critical orientation theory in the D0-degeneration pro-limit and the orientation line on the local stratum (criterion Theorem 7.6).

(O1) Strong reduced orientation. The K3×E-quotient self-Ext frame bundle carries a strong reduced orientation on the wrapped stratum, descending the BBDJS orientation line o_c across the E-quotient relative to $(O1_{\mathrm{quot}})$ (Theorem 7.12).

(O1$^+$) Weyl-equivariant transport. The orientation transports $W^{(2)}(\Lambda^{2,1}_{II})$-equivariantly along reflections relative to the chosen Weyl lifts (Theorem 7.13); after the type-II wall atlas supplies the rank-one local signs, the transported character agrees with ν_{Δ_5} on the BKM Weyl group.

(O2) Local Pfaffian wall normal form. On type-II walls of $\Lambda^{2,1}_{II}$, the local Pfaffian wall normal form is the rank-one crossing $[\mathbb{R}\xrightarrow{u}\mathbb{R}]$; the retained $(O2_{\mathrm{atlas}})$ datum transports these local signs over the half-Hilbert orbit, whose sign support has cardinality $4096=2^{12}$ (Theorem 7.18).

The protected scalar shadow on the closed K3×E⟶point cobordism is the OP/DT identity

$$Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}}(P_{\mathrm{OP}}, Q_{\mathrm{OP}}, T_{\mathrm{OP}}) = -4096\,\Delta_5(Z)^{-2} \qquad \text{(proved, [85, 86, 94]).}$$

Equivalently $\chi_{10}^{\mathrm{OP}} = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2$ and $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1}$; the factor $4096 = 64^2$ is a theta-leading normalization conversion and the minus sign is the OP scalar convention. The orientation-forgetful shadow

$$Z^{K3\times E}_{\mathrm{BPS}} = (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$$

is the level-n protected trace of the retained $K3 \times E$ data on the closed cobordism in the unnormalized convention $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$; the inverse-square scalar cannot recover the orientation character that distinguishes Δ_5 from $-\Delta_5$. A $3d$ gravitational path-integral realization requires the Hall–Borcherds residual of [68] and is not claimed. The conditional cross-level identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5$ supplies the orientation/Pfaffian datum under (D0), (O1), (O1$^+$), (O2).

Remark 6.1 *(Compatibility of the κ-tuple).* The κ-tuple $(0, 0, 3, 5, 24)$ is consistent across [65] (Vol III), the Hall–Borcherds residual of [68] (Vol II), and the Beilinson–Drinfeld factorization framework of [6] via Vol I [67]. The local model $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ + brane completion of [66] contributes the holomorphic de Rham obstruction class on the global target; the identity $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ does not promote to the global target via formal Hamiltonian BF, and the Hall–Borcherds residual is the remaining chain-level bridge.

7 Notation and fixed data

$X = S \times E$. Smooth complex projective trivial-canonical threefold; S is a K3 surface, E a smooth elliptic curve. $\dim_{\mathbb{C}} X = 3$ and $h^{1,0}(X) = 1$.

Λ_S. Mukai lattice; even unimodular signature $(4, 20)$. $\Lambda_S \cong U^{\oplus 4} \oplus E_8(-1)^{\oplus 2} \cong II_{4,20}$.

$\Gamma_X^{\mathrm{form}}, \Gamma_X^{\mathrm{phys}}$. Formal full-Mukai dyonic lattice $\Lambda_S \oplus \Lambda_S$. The superscript "phys" denotes this formal even branch, not the $\mathcal{N} = 4$ charge lattice.

$\Gamma_X^{\mathrm{alg}}, \Gamma_{X,\sigma}^{\mathrm{eff}}$. Algebraic K-class lattice $\subset \Gamma_X^{\mathrm{form}}$; chosen σ-effective semigroup $\subset \Gamma_X^{\mathrm{alg}}$.

$\Gamma_{\mathrm{gram}} = \mathbb{Z}^3$. Gram-index lattice, (n, l, m).

Π_X. Quadratic Gram map, $\Pi_X(Q, P) = ((Q, Q)/2,\ (Q, P),\ (P, P)/2)$.

B. Symmetric bilinear cocycle, $B(c, c') = (Q \cdot Q',\ Q \cdot P' + Q' \cdot P,\ P \cdot P')$; satisfies $B = -\delta\Pi_X$ and $B(c, c) = 2\Pi_X(c)$.

$\widehat{\Gamma}_X$. Normal-ordered extension $\Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}$; abelian under $\star$.

$\overline{\Pi}_X$. Additive normal-ordered Gram homomorphism $\overline{\Pi}_X(c, T) = \Pi_X(c) - T$.

$\mathrm{Ran}^{\mathrm{hyb}}(E)$. Hybrid carrier $\mathrm{Ran}(E)_{b=0} \sqcup \mathrm{Ran}^{\mathrm{wr}}(E)_{b>0}$ on the elliptic factor.

$\mathcal{D}_X = \Delta_5$, $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$. Three named weight-five objects: automorphic section, BKM denominator, compact Pfaffian (View ①, ②, ③).

The fixed data are: (a) the formal Mukai sector $\Gamma_X^{\mathrm{form}} = \Lambda_S \oplus \Lambda_S$ and its three-tier support hierarchy; (b) the bilinear polarization $B = -\delta\Pi_X$ as the full structural content of the Gram defect; (c) the normal-ordered extension $\widehat{\Gamma}_X$ with additive homomorphism $\overline{\Pi}_X$, fixed by the fixed-lift raw-pushforward obstruction theorem; (d) the hybrid Ran carrier $\mathrm{Ran}^{\mathrm{hyb}}(E)$, fixed by the projection-locality obstruction; (e) the connected/finite split of the E-descent classes, with the linearization criterion separated from the gerbe class; (f) the local protected observable algebra source datum, taken as an external assumption; (g) the four obstructions (D0), (O1), (O1$^+$), (O2) conditioning the cross-level identity ②↔③.

Chapter 6 constructs from these fixed data the Dirac–Igusa pro-object $\mathfrak{D}_X^{\mathrm{DI}} = \lim_R(\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P$

Chapter 3

View I: the Borcherds–Gritsenko–Nikulin section

1 The pentadic level and the Beilinson cut

The automorphic reading of Δ_5 is the first theorem-level reading. It is a holomorphic section of the weight-5 automorphic line on the genus-two Siegel modular space $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$: the Borcherds–Gritsenko–Nikulin lift of the weight-zero index-one weak Jacobi form $\phi_{0,1}$, normalized at the isotropic cusp $\mathfrak{c}_\infty$ by the theta-constant leading coefficient 64. The $\mathbf{Sp}_4(\mathbb{Z})$-equivariant data of View I are the multiplier system ν_{Δ_5}, the Borcherds chamber, the divisor support, and the weight.

1.0.0.1 Three names. The handle "Δ_5" is shared among three structurally distinct objects and must be tagged on first use:

- *the Borcherds–Gritsenko–Nikulin section* $\mathcal{D}_X = \Delta_5$ of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$, View I;
- *the BKM denominator* $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, View II, the Borcherds–Kac–Moody denominator algebra;
- *the protected Pfaffian* $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$, View III, constructed in the Dirac–Igusa chapter as a section on the Pfaffian line of the cosection-reduced compact Hall datum after the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ wall datum, and finite Pfaffian section datum of Appendix 7.

The bare label Δ_5 never substitutes for any of the three: it names the automorphic section unless a reading is specified, while $\mathcal{D}_X$, $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, and $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ live at distinct categorical-dimensional levels.

1.0.0.2 Beilinson cut: scope of View I. View I is the scalar modular shadow. In the reconstitution order "primitive objects first, cross-level identities second, scalar shadows last" View I is a single holomorphic section of an automorphic line on a Shimura variety, with its multiplier system and its weight. No operator-level datum is asserted at View I. In particular, the equality $\mathcal{D}_X = \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of View I and View III is the Pfaffian–Dirac theorem of Chapter 7, relative to the retained data recorded in Appendix 7; it is not asserted in this chapter. The shadows are Δ_5, χ_{10}, the level-n BPS partition function $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$, and the Oberdieck–Pixton scalar branch $Z_{\mathrm{OP}}^{X} = -4096\,\Delta_5^{-2}$; none of them is a Hilbert space, a Hall pairing, an orientation line, or an operator product.

2 The genus-two Siegel space and the Borcherds chamber

Let $\mathbb{H}_2$ be the genus-two Siegel upper half space. An element $Z \in \mathbb{H}_2$ is a complex symmetric 2×2 matrix with positive-definite imaginary part,

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix}, \qquad \Im Z > 0.$$

Set Fourier variables
$$q = e^{2\pi i z_1}, \qquad r = e^{2\pi i z_2}, \qquad s = e^{2\pi i z_3}.$$
The fixed cusp polarization at the rational isotropic boundary $\mathfrak{c}_\infty$ of $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ — the standard type-II cusp polarization — is the ordered limit
$$0 < |s| \ll |q| \ll |r|^{-1} \ll 1.$$
Equivalently, $\Im z_3$ goes to $+\infty$ faster than $\Im z_1$ goes to $+\infty$, and $\Im z_2$ is bounded. This polarization selects a Weyl chamber on the $O(3,2)$ symmetric domain through the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ bridge of Chapter 5, and hence selects a positive cone, the effective semigroup
$$\Gamma_{\text{eff}} = \{(n,l,m) \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \cup \{(n,l,0) \mid n > 0,\ l \in \mathbb{Z}\} \cup \{(0,l,0) \mid l < 0\} \subset \mathbb{Z}^3,$$
and a corresponding completed Fourier expansion ring. In this completion every Fourier coefficient of every formal product receives only finitely many contributions: the chamber walls are the inequalities defining the ordering, and the chamber-completed product is the only product considered.

2.0.0.1 Choices fixed once and for all. The cusp $\mathfrak{c}_\infty$, the Fourier variables (q,r,s), the polarization $(0 < |s| \ll |q| \ll |r|^{-1} \ll 1)$, and the effective semigroup $\Gamma_{\text{eff}} \subset \mathbb{Z}^3$ are fixed. Every View-I product expansion is the product expansion in the cusp-completed semigroup algebra of Γ_{eff}. The cusp expansion at any other rational isotropic cusp is obtained by an $\mathbf{Sp}_4(\mathbb{Z})$-translate of these data.

3 The weak Jacobi input and the K3 elliptic genus

The Borcherds input is the *weight-zero index-one weak Jacobi form* $\phi_{0,1} \in J^{\text{weak}}_{0,1}$ of Eichler–Zagier [32, §2.2], normalized so that the K3 elliptic genus is
$$Z_{K3}(\tau,z) = 2\,\phi_{0,1}(\tau,z).$$
Write the Fourier expansion as
$$\phi_{0,1}(\tau,z) = \sum_{n\geq 0,\ l\in\mathbb{Z}} f(n,l)\,q^n r^l, \qquad q = e^{2\pi i\tau}, \quad r = e^{2\pi i z}.$$
The first Fourier coefficients are
$$\begin{aligned}
&f(0,-1) = 1, \quad f(0,0) = 10, \quad f(0,1) = 1,\\
&f(1,-2) = 10, \quad f(1,-1) = -64, \quad f(1,0) = 108, \quad f(1,1) = -64, \quad f(1,2) = 10,\\
&f(2,-3) = 1, \quad f(2,-2) = 108, \quad f(2,-1) = -513, \quad f(2,0) = 808,
\end{aligned} \tag{3.1}$$
extending in the standard way by hyperbolic symmetry $f(n,l) = f(n,-l)$ and Eichler–Zagier support $f(n,l) = 0$ for $4n - l^2 < -1$ [32, §2.2]. The values in (3.1) are obtained by the exact rational expansion of the standard theta-quotient expression $\phi_{0,1} = 4\sum_{i=2,3,4} \theta_i(\tau,z)^2/\theta_i(\tau,0)^2$ [*computed*; cross-checked against [32, Theorem 9.2]].

3.0.0.1 Borcherds-weight handle. For a fixed Borcherds input Φ, the weight identity $\kappa_{\mathrm{BKM}}(\Phi) = c_\Phi(0)/2$ (Borcherds 1995 [8, Theorem 10.1]; Gritsenko 2018 [40]) gives, on the row $(t, N) = (1, 1)$ of the Cléry–Gritsenko catalogue [41, Theorem 1.4],

$$c_1(0) := f(0,0) = 10, \qquad \kappa_{\mathrm{BKM}}(\Delta_5) = \frac{c_1(0)}{2} = 5,$$

to be matched in Proposition 6.1 below with the geometric weight of the Borcherds lift. The notation κ_{BKM} is the BKM cusp-form-weight entry of the κ-tuple $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ of the $K3 \times E$ target. The bare symbol κ is not a scalar invariant; every entry carries its semantic subscript; the additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(O_{\mathrm{fiber}})$ does not define the BKM coordinate. The BKM coordinate is $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5$.

4 The theta-constant normalization

The theta-constant Igusa form Δ_5 is fixed by its leading Fourier monomial at $\mathfrak{c}_\infty$. Writing the theta-constant representation of Igusa [48, §5], [49, §2]

$$\Delta_5(Z) = \prod_{\substack{a,b \in (\mathbb{Z}/2\mathbb{Z})^2 \\ {}^t ab \equiv 0 \bmod 2}} \nu_{a,b}(Z), \qquad \nu_{a,b}(Z) = \sum_{l \in \mathbb{Z}^2} \exp\!\Big(\pi i\big(z[l + \tfrac{1}{2}a] + {}^t bl\big)\Big),$$

the four characteristics with $a = (0, 0)$ start with the constant 1, the two factors with $a = (1, 0)$ start with $2q^{1/8}$, the two factors with $a = (0, 1)$ start with $2s^{1/8}$, and the two factors with $a = (1, 1)$ start with $2q^{1/8}r^{1/4}s^{1/8}$; multiplying gives

Proposition 4.1 *(Theta-constant leading coefficient).*

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 \;=\; 2^6 \;=\; 64.$$

[*proved*; Igusa 1962 [48, Theorem 3], Gritsenko 1999 [43, Theorem 4.1]; the rational theta-product expansion gives the same leading coefficient.]

Definition 4.2 *(Monic Borcherds–Gritsenko form).* The *monic genus-two Borcherds–Gritsenko form* is

$$D_5(Z) \;:=\; 64^{-1}\Delta_5(Z).$$

By Proposition 4.1, $D_5 = q^{1/2}r^{1/2}s^{1/2} + \dots$ at $\mathfrak{c}_\infty$.

This is the normalization used by Gritsenko–Nikulin [45, Theorem 2.1, (2.15)–(2.16)]: their Borcherds product is the monic form D_5; the theta-constant convention restores the prefactor 64.

5 The Borcherds–Gritsenko–Nikulin section

5.1 The product expansion

Borcherds's automorphic product theorem [8, Theorem 10.1], specialized to the genus-two Siegel modular variety $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ by Gritsenko–Nikulin [45, Theorem 2.1, Example 2.4] via the exceptional exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3, 2)_+$ of Chapter 5, lifts the weak Jacobi form $\phi_{0,1}$ to a meromorphic Siegel modular form of weight $f(0, 0)/2 = 5$ and multiplier system ν_{Δ_5} with infinite product expansion in the cusp chamber.

THEOREM 5.1 *(Borcherds–Gritsenko–Nikulin lift, monic form).* In the cusp chamber $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$,

$$D_5(Z) \;=\; q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

the product converging absolutely on a neighborhood of $\mathfrak{c}_\infty$ and continuing meromorphically to all of $\mathbb{H}_2$ as a holomorphic Siegel modular form of weight 5 on $\mathbf{Sp}_4(\mathbb{Z})$ with character ν_{Δ_5} of order two.

[*proved*; Borcherds 1995 [8, Theorem 10.1], Gritsenko–Nikulin 1998 [45, Theorem 2.1, (2.15)–(2.16)], Gritsenko 1999 [43, Theorem 4.1].]

COROLLARY 5.2 *(Theta-normalized section).*

$$\Delta_5(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

The exponent $f(nm, l)$ of the (n, l, m)-factor depends only on the Hecke–Verlinde–DMVV product nm and on the Jacobi index l, reflecting the second-quantized structure of the Borcherds lift, whose combinatorial origin is the Dijkgraaf–Moore–Verlinde–Verlinde elliptic genus of symmetric products [28].

5.2 The View I datum

Definition 5.3 *(The Borcherds–Gritsenko–Nikulin section).* The *Borcherds–Gritsenko–Nikulin section* of the weight-5 automorphic line is the View I representative

$$\mathcal{D}_X \;:=\; \Delta_5$$

on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$, with Borcherds product expansion at $\mathfrak{c}_\infty$ given by Corollary 5.2.

Remark 5.4 *($\mathcal{D}_X$ is the View I name).* The handle $\mathcal{D}_X$ is the View I name of the section. It coincides as a holomorphic Siegel modular form with the theta-constant Igusa cusp form Δ_5; the renaming records the role: View I scalar shadow, the determinant of the virtual K3 index in $K_0(\mathrm{sVect_{fd}})$ (the K_0-determinant identity proved at Proposition 7.2; the operator Pfaffian datum belongs to View III), and the input section of the Pfaffian–Dirac theorem of Chapter 7. The compact operator-level datum $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of View III is asserted equal to $\mathcal{D}_X$ under (D_0), (O_1), $(O_1)^+$, (O_2); without these obstruction trivializations, only $\mathcal{D}_X$ is rigorously available.

6 WEIGHT, MULTIPLIER, AND DIVISOR

6.1 The Borcherds weight

PROPOSITION 6.1 *(Borcherds–Gritsenko weight identity at $N = 1$).*

$$\mathrm{wt}(\mathcal{D}_X) \;=\; \frac{f(0,0)}{2} \;=\; 5 \;=\; \frac{c_1(0)}{2} \;=\; \kappa_{\mathrm{BKM}}(\Delta_5).$$

The weight is integral; no metaplectic cover enters the weight factor.

[*proved*; Borcherds 1995 [8, Theorem 10.1, weight formula]; Gritsenko–Nikulin 1998 [45, Example 2.4]; Gritsenko 1999 [43, Theorem 4.1, weight]; the Borcherds weight identity $\kappa_{\mathrm{BKM}}(\Phi) = c_\Phi(0)/2$ for a fixed input form and lattice; Cléry–Gritsenko [41], Vol III.]

The row $(t, N; \mathrm{wt}, c) = (1, 1; 5, 10)$ is the leading F_j-entry of the eight-row Cléry–Gritsenko catalogue [41, Theorem 1.4]. In the F_j-order the remaining rows carry weights $(2, 1, \frac{1}{2}, 3, 2, \frac{3}{2}, 1)$ and constants $(4, 2, 1, 6, 4, 3, 2)$; in the input-pair order $(2, 1), (1, 2), (3, 1), (1, 3), (4, 1), (1, 4), (2, 2)$ the same data read $(2, 3, 1, 2, \frac{1}{2}, \frac{3}{2}, 1)$ and $(4, 6, 2, 4, 1, 3, 2)$. Each row satisfies $\kappa_{\mathrm{BKM}}(F_j) = c_j(0)/2$. The half-integer rows are realized by $v_\eta^3 \times v_H$ and $\chi_4^{(4)} \times v_H$, respectively. The Vol III universal Borcherds-weight identity replaces the additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$: the κ-tuple records separate invariants, and the $K3 \times E$ entry is precisely $\kappa_{\mathrm{BKM}} = 5$.

6.2 The Maass multiplier system

The character $\nu_{\Delta_5} \colon \mathbf{Sp}_4(\mathbb{Z}) \to \mathbb{C}^\times$ of Maass [71] has order two, $\nu_{\Delta_5}^2 = 1$, and is determined on the standard generators of $\mathbf{Sp}_4(\mathbb{Z})$ by

$$\nu_{\Delta_5}\begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix} = 1, \qquad \nu_{\Delta_5}\begin{pmatrix} \mathbf{I}_2 & B \\ 0 & \mathbf{I}_2 \end{pmatrix} = (-1)^{b_1+b_2+b_3},$$

$$\nu_{\Delta_5}\begin{pmatrix} {}^tA^{-1} & 0 \\ 0 & A \end{pmatrix} = (-1)^{(1+a_1+a_4)(1+a_2+a_3)+a_1a_4},$$

with $B = \left(\begin{smallmatrix} b_1 & b_2 \\ b_2 & b_3 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} a_1 & a_2 \\ a_3 & a_4 \end{smallmatrix}\right)$ [71]. The square $\Delta_5^2 = \Delta_{10}$ is scalar. The multiplier records the obstruction to identifying Δ_5 with an ordinary modular form on $\mathbf{Sp}_4(\mathbb{Z})$: Δ_5 is a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$, where $\mathcal{L}$ is the Hodge automorphic line and ν_{Δ_5} is the order-two character; the section Δ_5 does not descend to a section of the bare Hodge line $\mathcal{L}^5$, but its square Δ_5^2 is a scalar section of $\mathcal{L}^{10}$.

6.3 The diagonal divisor

Let

$$\mathcal{H}_{\mathrm{diag}} = \left\{ \left(\begin{smallmatrix} a & 0 \\ 0 & b \end{smallmatrix}\right) \,\middle|\, a, b \in \mathbb{H}_1 \right\} \subset \mathbb{H}_2$$

be the locus of diagonal genus-two periods. The Cléry–Gritsenko classification [41] records

$$\operatorname{supp}\operatorname{div}(\Delta_5) = \mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}, \qquad \operatorname{mult}_{\mathcal{H}_{\mathrm{diag}}}(\Delta_5) = 1,$$

so the divisor of Δ_5 is the simple paramodular Humbert divisor. In the cusp-chamber language, the type-II reflective walls $\mathcal{H}_{\delta_i}$, $i = 1, 2, 3$, of the lattice $\Lambda_{II}^{2,1}$ (the type-II sublattice of the Borcherds even lattice $\Lambda^{2,1}$) are the components of $\mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}$ that meet the chosen fundamental Weyl chamber $\mathcal{P}_{II}$ (Chapter 4); the corresponding simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ have Gram pairings $(\delta_i, \delta_i) = 2$ and $(\delta_i, \delta_j) = -2$ for $i \neq j$ [45, Example 2.4], and the corresponding Borcherds divisor orders are

$$d_{\delta_1}^{\mathrm{aut}} = f(0, 1) = 1, \qquad d_{\delta_2}^{\mathrm{aut}} = f(0, 1) = 1, \qquad d_{\delta_3}^{\mathrm{aut}} = f(0, -1) = 1.$$

[*proved*; Cléry–Gritsenko 2011 [41, Theorem 1.4]; Gritsenko–Nikulin 1998 [45, Example 2.4]; the Gram computation is the one displayed in Appendix 1.2.]

6.4 Maass-character values on the type-II walls

Lemma 6.2 *(Maass-character values on the type-II reflections).* With the simple type-II roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ and chamber-permuting type-I roots $\alpha_1 = f_{-2} - f_2$, $\alpha_2 = f_2 - f_3$, $\alpha_3 = f_{-2} - f_3$,

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad \nu_{\Delta_5}(s_{\alpha_i}) = +1, \qquad i = 1, 2, 3.$$

Equivalently, for chosen $\mathbf{Sp}_4$-lifts and the weight-five slash action without character,

$$(\Delta_5|_5 s_{\delta_i})(Z) = -\Delta_5(Z), \qquad (\Delta_5|_5 s_{\alpha_i})(Z) = +\Delta_5(Z).$$

[*proved*; Maass [71]; Gritsenko–Nikulin [43, §3]; Gritsenko [42]. The type-II values follow from $d^{\mathrm{aut}}_{\delta_i} = 1$ and the chamber-wall cocycle calculation of Lemma 7.2, which fixes $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det$ on the three type-II generators; the type-I values follow from $d^{\mathrm{aut}}_{\alpha_i} = 0$ (vanishing of $f(0, \pm 2)$ by the support of $\phi_{0,1}$ at $q = 0$).]

The chamber-permuting reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$ act trivially on Δ_5 because the corresponding type-I roots α_i are not $\mathbf{Sp}_4$-images of timelike short roots in the cusp chamber: type-I square-2 vectors are outside the type-II sublattice $\Lambda^{2,1}_{II}$ on which the Borcherds product expansion of Δ_5 is supported, so their walls do not appear in $\mathrm{div}(\Delta_5)$ at all [32, §2.2].

7 IDENTIFICATION WITH THE VIRTUAL K_0-DETERMINANT

The Fock superdeterminant of the Borcherds-graded super vector space $\mathcal{V} = \bigoplus_{(n,l,m)\in\Gamma_{\mathrm{eff}}} \mathbb{U}_{nm,l}$ — the K_0-class of the second-quantized Hecke–DMVV half of the K3 elliptic genus, with the cusp/Weyl boundary factors at $m = 0$ carrying the input multiplicities $f(0, l)$ and the bulk factors at $m > 0$ carrying $f(nm, l)$ on Γ_{eff} (Definition 2.1) — is

LEMMA 7.1 *(K_0-Fock superdeterminant).*

$$\mathrm{sdet}_{\mathcal{V}}(1 - x) \;=\; \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}, \qquad x^{(n,l,m)} = q^n r^l s^m.$$

[*proved*; the Fock-trace lemma is the three-way comparison $\mathrm{sdet}_{\mathcal{V}}(1 - x) = \prod(1 - x^\gamma)^{\mathrm{sdim}\,\mathcal{V}_\gamma}$ under the Borcherds-graded second-quantization; the automorphic–orthogonal–BKM identification sits in Theorem 1.20 of Chapter 5.]

PROPOSITION 7.2 *(BGN identification of the virtual determinant).* With $v_0(Z) := 64\, q^{1/2} r^{1/2} s^{1/2}$ the cusp/Weyl monomial of Chapter 2, the normalized K_0-determinant agrees with the View I section,

$$\mathcal{D}_X(Z) \;:=\; v_0(Z)\ \mathrm{sdet}_{\mathcal{V}}(1 - x) \;=\; \Delta_5(Z),$$

in the cusp-chamber expansion at $\mathfrak{c}_\infty$ and hence as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

Proof. Combine Theorem 5.1, the theta-constant prefactor Proposition 4.1, Lemma 7.1, and Cohen's q-expansion principle for genus-two Siegel modular forms with half-integral character: a meromorphic Siegel form is determined by its expansion at one isotropic cusp. Both sides have the same Borcherds product expansion at $\mathfrak{c}_\infty$ by direct comparison of exponents, and $\mathcal{D}_X$ is holomorphic of weight 5 by Theorem 5.1. □

[*proved*; Borcherds 1995 [8, Theorem 10.1], Gritsenko–Nikulin 1998 [45, Theorem 2.1], Igusa 1962/1964 [48, 49].]

Remark 7.3 *(Scope of the BGN identification).* Proposition 7.2 is an equality of holomorphic sections of an automorphic line bundle. The carrier $\mathcal{V}$ is a class in $K_0(\Gamma_{\mathrm{eff}}$-graded super vector spaces), with no vertex, chiral, or holomorphic E_3-factorization structure. The passage from the K_0-Fock determinant to a section of an oriented Pfaffian line over the cosection-reduced compact Hall datum, and the identification of that section with $\mathcal{D}_X$, is the Pfaffian–Dirac theorem of Chapter 7, relative to the retained data recorded in Appendix 7. View I supplies only the automorphic determinant section.

8 The squared determinant section and OP normalization

The Maass character has order two, so the square $\Delta_5^2 = \Delta_{10}$ is a scalar weight-10 Igusa form on $\mathbf{Sp}_4(\mathbb{Z})$, with no multiplier system.

Proposition 8.1 *(The square is the Oberdieck–Pandharipande χ_{10}).*

$$\Delta_5^2 \;=\; \Delta_{10} \;=\; \Phi_{10}^{\mathrm{un}} \;=\; \chi_{10}.$$

[*proved*; Igusa [48, 49]; the symbol Φ_{10}^{un} is Borcherds's name for the squared theta-constant Igusa form, χ_{10} is the Oberdieck–Pandharipande [85, 86] name; both equal Δ_5^2. The naming is $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$, with the Oberdieck–Pixton OP normalization $\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$ the monic-on-D_5 variant.]

8.0.0.1 The OP scalar branch. With the OP normalization $\chi_{10}^{\mathrm{OP}} = D_5^2$, the Oberdieck–Pixton scalar partition function on $K3 \times E$ is [86, (0.4) and Theorem 1]

$$Z_{\mathrm{OP}}^{X} \;=\; -\,(\chi_{10}^{\mathrm{OP}})^{-1} \;=\; -\,4096\,\Delta_5^{-2}.$$

The factor $-4096 = -(64)^2$ records the squared theta-constant leading coefficient and the OP scalar sign convention. At View I this is the closed protected trace Δ_5^{-2}; the orientation character which distinguishes the Pfaffian section Δ_5 from its negative is lost under squaring. The operator datum entering that trace is defined at the Pfaffian and orientation level relative to the retained data of Appendix 7.

8.0.0.2 Three names for the squared form, distinguished. The notation is fixed as follows: $\Delta_5^2 = \Delta_{10}$ is the Igusa convention (theta-constant square); $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ is the Borcherds–Eisenstein convention; $\chi_{10} = \Delta_5^2$ is the Oberdieck–Pandharipande convention; and $\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2 = D_5^2$ is the Oberdieck–Pixton variant in which D_5 is the monic Borcherds form (Definition 4.2).

9 The View II shadow: the $2Z$ denominator

The Gritsenko–Nikulin denominator algebra of the Borcherds–Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ arises as the View II shadow of Δ_5 under the doubling $Z \mapsto 2Z$:

Proposition 9.1 *(Gritsenko–Nikulin doubled denominator).*

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; D_5(2Z) \;=\; 64^{-1}\Delta_5(2Z).$$

[*proved*; Gritsenko–Nikulin 1998 [45, Theorem 2.1, (2.15)–(2.16); §5]; Gritsenko 1999 [43, Theorem 4.1].] The doubling records the embedding $\Lambda_{II}^{2,1} = \alpha(\Gamma_{\mathrm{ind}}) \subset \Lambda^{2,1}$ of index 4. Its type-II walls are primitive roots of square 2, and the substitution $Z \mapsto 2Z$ replaces $\exp(-\pi i(\rho, Z))$ by $\exp(-2\pi i(\rho, Z))$ in the Gritsenko–Nikulin denominator normalization [45, Theorem 2.1, (2.15)]. At View I this is a pointer; the Cartan, Chevalley, real-and-imaginary roots, and Weyl–Kac–Borcherds denominator identity of $\mathfrak{g}_{\Delta_5}$ sit in Chapter 4.

10 The $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ bridge to View V

The Borcherds construction is naturally written on the orthogonal side: Δ_5 is the regularized theta lift of $\phi_{0,1}^{\mathrm{Weil}}$ on $\Lambda^{3,2}$, pulled back to $\mathbf{Sp}_4$ through the exterior-square isogeny

$$\mathbf{PSp}_4(\mathbb{R}) \;\simeq\; \mathbf{SO}(3,2)_+.$$

The corresponding spin cover is the real group with Lie algebra $\mathfrak{sp}_4(\mathbb{R})$. At the symbolic level View I (Δ_5 on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$) and View V ($\Delta_5$ as the singular theta lift on $\Gamma_{3,2}\backslash\mathbb{H}_+^{\mathbf{IV}}$) are the two sides of this bridge: the weight-5 automorphic line on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ is the pullback of the weight-5 orthogonal modular line on $\Gamma_{3,2}\backslash\mathbb{H}_+^{\mathbf{IV}}$, and the multiplier ν_{Δ_5} restricts to the orthogonal Maass character. The full theta-lift construction — Borcherds' *singular theta correspondence* [8], [9] — produces the orthogonal modular form; the automorphic section in View I is its pullback to the $\mathbf{Sp}_4$-side of the bridge.

11 Summary of View I

THEOREM 11.1 *(View I datum).* The genus-two Igusa cusp form Δ_5 is the Borcherds–Gritsenko–Nikulin section

$$\mathcal{D}_X \;:=\; \Delta_5$$

of the weight-5 automorphic line $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ over $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$:

(i) At the rational isotropic cusp $\mathfrak{c}_\infty$ with cusp polarization $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$, $\mathcal{D}_X$ admits the Borcherds product expansion

$$\mathcal{D}_X(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

with leading theta-constant coefficient $[q^{1/2}r^{1/2}s^{1/2}]\mathcal{D}_X \;=\; 64$, and exponents $f(nm,l)$ the Fourier coefficients of the K3 weight-zero index-one weak Jacobi form $\phi_{0,1} \;=\; \frac{1}{2}Z_{K3}$ (Eichler–Zagier [32, §2.2]).

(ii) $\mathcal{D}_X$ has weight $\mathrm{wt}(\mathcal{D}_X) \;=\; f(0,0)/2 \;=\; 5 \;=\; c_1(0)/2 \;=\; \kappa_{\mathrm{BKM}}(\Delta_5)$, the leading entry $(t, N; \mathrm{wt}, c) = (1, 1; 5, 10)$ of the Cléry–Gritsenko catalogue [41, Theorem 1.4]. The multiplier is the Maass character ν_{Δ_5} of order two.

(iii) The divisor of $\mathcal{D}_X$ is the simple paramodular Humbert divisor $\mathbf{Sp}_4(\mathbb{Z}) \cdot \mathcal{H}_{\mathrm{diag}}$; the type-II reflective walls $\mathcal{H}_{\delta_i}$ $(i = 1, 2, 3)$ carry order $d^{\mathrm{aut}}_{\delta_i} = 1$, and $\nu_{\Delta_5}(s_{\delta_i}) = -1 = (-1)^{d^{\mathrm{aut}}_{\delta_i}}$, while the chamber-permuting roots α_i carry order 0 and $\nu_{\Delta_5}(s_{\alpha_i}) = +1$ (Lemma 6.2).

(iv) The square $\mathcal{D}_X^2 = \Delta_{10} = \Phi_{10}^{\mathrm{un}} = \chi_{10}$ is a scalar weight-10 Igusa form, and the Oberdieck–Pixton scalar branch is $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$.

(v) The View II shadow is the doubled denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; D_5(2Z) \;=\; 64^{-1}\Delta_5(2Z)$ (Proposition 9.1); the View III operator-level datum $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ is the subject of Chapter 6 and Appendix 7.

[*proved*; Borcherds 1995 [8, Theorem 10.1], Borcherds 1998 [9], Gritsenko–Nikulin 1998 [45, Theorem 2.1, Example 2.4], Gritsenko 1999 [43, Theorem 4.1], Igusa 1962 [48, Theorem 3], Igusa 1964 [49, §2], Cléry–Gritsenko 2011 [41, Theorem 1.4], Maass 1964 [71], Eichler–Zagier 1985 [32, §2.2], Oberdieck–Pixton 2018 [86, (0.4) and Theorem 1], Oberdieck–Pandharipande 2016 [85]; the lattice and theta-product computations are displayed in Appendices 1.2 and 2.]

11.0.0.1 What View I does not assert. View I does not assert any operator-level datum on $K3 \times E$, does not construct the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$ (Chapter 4), does not construct the protected first-order operator $\mathfrak{D}_X^{\mathrm{DI}}$ or the Pfaffian line $\mathscr{L}_{\mathrm{Pf},X}$ (Chapter 6), and does not commit to the $K3 \times E$ realization of the chosen E_3-prefactorization datum $\mathfrak{A}_{K3\times E}$ at any level above the scalar shadow. In the language of the Beilinson cut, View I is the bottom of the reconstitution order: a holomorphic Siegel cusp form, with its weight, multiplier, divisor, and Borcherds product, and nothing more.

Chapter 4

The Borcherds–Kac–Moody denominator algebra $\mathfrak{g}_{\Delta_5}$ on $\Lambda^{2,1}_{II}$

The Igusa cusp form Δ_5 admits a Borcherds–Kac–Moody description as a denominator. The denominator algebra is the generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ on the index-four type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$, abstractly $U(4) \oplus \langle 2\rangle$, with three even real simple roots $\delta_1, \delta_2, \delta_3$, an infinite imaginary simple-root datum read from the K3 half-index $\phi_{0,1}$, and Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ generated by the three real reflections. Its Weyl–Kac–Borcherds denominator is the weight-five Borcherds–Kac–Moody denominator

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z),$$

the doubled Igusa form normalized by the theta-constant constant $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$. The Borcherds-weight characteristic of the denominator is $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 10/2 = 5$, the weight of the Gritsenko paramodular form $\Delta_5 = \Phi_1$ at level 1 — equivalently $(\Phi_{10}^{\mathrm{un}})^{1/2}$ in Igusa's normalization, where $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$. The handle Φ_1 denotes Gritsenko's paramodular Φ_t at $t = 1$, distinct from Igusa's weight-ten Φ_{10}.

The denominator algebra is target-side: it is the chiral shadow of an open compact realization. No compact Hall bracket, protected Pfaffian, or microscopic state space is claimed here. The protected Pfaffian is treated in Chapter 6 relative to the retained data of Appendix 7; the compact Hall primitive recognition remains conditional on $(S1)$–$(S10)$.

The three named instances of Δ_5 are tagged distinctly. In View II the canonical handle is the Borcherds–Kac–Moody denominator,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z).$$

The automorphic section $\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ of Chapter 3 is View I; the compact protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of Chapter 6 is View III, relative to retained data. The bare handle Δ_5 is the working symbol for the Siegel cusp form of weight five and never substitutes for any of the three.

1 The hyperbolic lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ and its type-II sublattice

Fix the rank-three Lorentzian lattice

$$\Lambda^{2,1} \;=\; \Lambda^{(1,1)} \oplus [2] \;=\; \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2},$$

where $\Lambda^{(1,1)}$ is the hyperbolic plane and $[2]$ is the rank-one lattice generated by a single vector of square $+2$. The bilinear form on the chosen basis is

$$(f_2, f_{-2}) \;=\; -1, \qquad (f_2, f_2) \;=\; (f_{-2}, f_{-2}) \;=\; 0, \qquad (f_3, f_3) \;=\; 2, \tag{4.1}$$

with all other pairings between basis vectors zero. The lattice is even, signature $(2,1)$, and primitively embedded in the rank-five Lorentzian lattice $\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ of Section 1.4.

Definition 1.1 *(Type-II sublattice).* The type-II sublattice of $\Lambda^{2,1}$ is

$$\Lambda^{2,1}_{II} \ = \ \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod 2\}.$$

Its dual lattice is

$$(\Lambda^{2,1}_{II})^* \ = \ \mathbb{Z}\tfrac{1}{2}f_2 \oplus \mathbb{Z}\tfrac{1}{2}f_3 \oplus \mathbb{Z}\tfrac{1}{2}f_{-2} \ = \ \tfrac{1}{2}\Lambda^{2,1}.$$

The inclusion $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$ has index four, and $\Lambda^{2,1} \subset (\Lambda^{2,1}_{II})^* = \frac{1}{2}\Lambda^{2,1}$ has index eight, so the discriminant group $(\Lambda^{2,1}_{II})^*/\Lambda^{2,1}_{II}$ has order 32. Abstractly the type-II sublattice is $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2 \rangle$ of discriminant -32.

Remark 1.2 *(The two divisibility classes of square-2 vectors).* A primitive element $\delta \in \Lambda^{2,1}$ with $(\delta, \delta) = 2$ has divisibility either $(\delta, \Lambda^{2,1}) = \mathbb{Z}$ (type I) or $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$ (type II). The type-I roots lie in the sublattice

$$\Lambda^{2,1}_{I} \ = \ \{mf_2 + lf_3 + nf_{-2} \in \Lambda^{2,1} \mid m + l + n \equiv 0 \pmod 2\}$$

and the type-II roots lie in $\Lambda^{2,1}_{II}$. The split is recorded in Gritsenko–Nikulin [43, Section 1]; the underlying classification of square-2 vectors on $\Lambda^{(1,1)} \oplus [2]$ is Nikulin's [82].

LEMMA 1.3 *(Reflection-group decomposition).* Let $W^{(2)}(\Lambda^{2,1}) \subset \mathbf{O}(\Lambda^{2,1})_+$ be the subgroup generated by reflections in primitive square-2 vectors. Then

$$\mathbf{O}(\Lambda^{2,1})_+ \ = \ W^{(2)}(\Lambda^{2,1}),$$

and there are two normal subgroups corresponding to the two divisibility classes,

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{I})] \ = \ 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{II})] \ = \ 6. \tag{4.2}$$

Proof. The first equality is Nikulin's [82] reflection-group theorem for $\Lambda^{(1,1)} \oplus [2]$. The two indices are the two cosets of the divisibility classes in $\mathbf{O}(\Lambda^{2,1})_+$; type-II reflections generate a subgroup of index six because the three orbit representatives $\{f_{-2} - f_2,\ f_2 - f_3,\ f_{-2} - f_3\}$ of the chamber automorphism S_3 are type-I and lie outside $W^{(2)}(\Lambda^{2,1}_{II})$. □

The change of polarization between Gram-index parametrization and the denominator-side Lorentzian root grading is the linear map

$$\alpha\colon \Gamma_{\text{ind}} \longrightarrow \Lambda^{2,1}_{II}, \qquad \alpha(n,l,m) \ = \ 2nf_2 - lf_3 + 2mf_{-2}.$$

Two pairings live on $\Gamma_{\text{ind}} = \mathbb{Z}^3$: the Gram-index pairing

$$\langle (n,l,m), (n',l',m') \rangle_{\text{ind}} \ = \ 2(nm' + n'm) - ll',$$

and the Lorentzian pairing on $\Lambda^{2,1}_{II}$ pulled back through α,

$$(\alpha(\gamma), \alpha(\gamma')) \ = \ -2\langle \gamma, \gamma' \rangle_{\text{ind}}, \qquad (\alpha(\gamma), \alpha(\gamma)) \ = \ -2(4nm - l^2). \tag{4.3}$$

The sign convention is fixed by the Borcherds product expansion and by the BKM denominator: positive root norm gives the real wall set, non-positive root norm gives the imaginary roots.

2 Real walls, the type-II Weyl chamber, and chamber automorphism

Reflections on a Lorentzian lattice

For a primitive vector $\alpha \in \Lambda^{2,1}$ with $(\alpha, \alpha) > 0$ and $(\alpha, \alpha) \mid 2(\alpha, \Lambda^{2,1})$, the Kac reflection [54, Chapter 3] is

$$s_\alpha : x \longmapsto x - \frac{2(x, \alpha)}{(\alpha, \alpha)}\alpha, \qquad s_\alpha(\alpha) = -\alpha, \qquad s_\alpha|_{\alpha^\perp} = \mathbf{I}.$$

For $(\alpha, \alpha) = 2$ this simplifies to

$$s_\alpha : x \longmapsto x - (x, \alpha)\alpha.$$

The simple type-II roots

Set

$$\delta_1 \;=\; 2f_2 - f_3, \qquad \delta_2 \;=\; 2f_{-2} - f_3, \qquad \delta_3 \;=\; f_3, \tag{4.4}$$

so $\delta_i \in \Lambda^{2,1}_{II}$, $(\delta_i, \delta_i) = 2$, and $(\delta_i, \Lambda^{2,1}) = 2\mathbb{Z}$ for $i = 1, 2, 3$. These are the Gritsenko–Nikulin type-II simple roots [43, Theorem 2.1, Section 4]; the same choice appears as case $(t = 1, II, 0)$ in [45, Section 5.1.1].

Definition 2.1 *(Type-II Weyl chamber).* For a positive-norm vector α set

$$\mathbb{H}_{\alpha,+} \;=\; \left\{\mathbb{R}_{>0}x \in C(\Lambda^{2,1})_+/\mathbb{R}_{>0} \;\middle|\; (x, \alpha) \le 0\right\}.$$

The type-II Weyl chamber is

$$\mathcal{P}_{II} \;=\; \bigcap_{i=1}^{3} \mathbb{H}_{\delta_i,+}.$$

Its primitive walls are

$$\mathcal{P}_{II,prim} \;=\; \{\delta_1, \delta_2, \delta_3\}.$$

The chamber decomposition is

$$W^{(2)}(\Lambda^{2,1}) \;\simeq\; W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}), \qquad \mathrm{Aut}(\mathcal{P}_{II}) \;\simeq\; S_3, \tag{4.5}$$

with $\mathrm{Aut}(\mathcal{P}_{II})$ generated by the three type-I reflections $s_{f_{-2}-f_2}$, $s_{f_2-f_3}$, $s_{f_{-2}-f_3}$. Compare (4.2).

The Cartan matrix is the symmetric Lorentzian Gram matrix

Lemma 2.2 *(Cartan matrix).* The Gram matrix of $(\delta_1, \delta_2, \delta_3)$ is

$$A \;=\; \big((\delta_i, \delta_j)\big)_{1\le i,j\le 3} \;=\; \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}. \tag{4.6}$$

Proof. Direct computation from (4.1) and (4.4): the diagonal entries are $(\delta_i, \delta_i) = 2$ for $i = 1, 2, 3$ as in the type-II classification, and the off-diagonal entries are

$$(\delta_1, \delta_2) \;=\; (2f_2 - f_3, 2f_{-2} - f_3) \;=\; -4 + 2 \;=\; -2,$$

and analogously $(\delta_1, \delta_3) = (\delta_2, \delta_3) = -2$. □

The matrix A is a symmetric generalized Cartan matrix in the sense of Kac [54, Chapter 1]: the diagonal entries are positive even integers, the off-diagonal entries are non-positive integers, and $A_{ij} = 0$ if and only if $A_{ji} = 0$. The associated Kac–Moody algebra $\mathfrak{g}(A)$ is hyperbolic; it is the rank-three hyperbolic algebra $H(2, 2, 2)$ of Feingold–Frenkel.

The chamber automorphism is S_3

LEMMA 2.3 *(S_3 chamber action).* The group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ acts on the simple roots by permutation: a reflection in $f_{-2} - f_2$ exchanges δ_1 and δ_2 and fixes δ_3; a reflection in $f_2 - f_3$ exchanges δ_1 and δ_3 and fixes δ_2; a reflection in $f_{-2} - f_3$ exchanges δ_2 and δ_3 and fixes δ_1. The Cartan matrix (4.6) is S_3-symmetric.

Proof. Each chamber-permuting reflection is the type-I reflection s_α in the primitive direction $\alpha \propto \delta_i - \delta_j$ that swaps the two type-II walls. For $(\delta_1\, \delta_2)$, the difference $\delta_1 - \delta_2 = 2(f_2 - f_{-2})$ has primitive direction $f_{-2} - f_2$ of square two; direct computation gives $s_{f_{-2}-f_2}(f_2) = f_{-2}$, $s_{f_{-2}-f_2}(f_3) = f_3$, hence $s_{f_{-2}-f_2}(\delta_1) = 2f_{-2} - f_3 = \delta_2$, $s_{f_{-2}-f_2}(\delta_2) = \delta_1$, $s_{f_{-2}-f_2}(\delta_3) = \delta_3$. For $(\delta_1\, \delta_3)$, $\delta_1 - \delta_3 = 2(f_2 - f_3)$ with primitive direction $f_2 - f_3$ of square two; one checks $s_{f_2-f_3}(\delta_1) = \delta_3$, $s_{f_2-f_3}(\delta_3) = \delta_1$, $s_{f_2-f_3}(\delta_2) = \delta_2$. For $(\delta_2\, \delta_3)$, $\delta_2 - \delta_3 = 2(f_{-2} - f_3)$ with primitive direction $f_{-2} - f_3$ of square two; one checks $s_{f_{-2}-f_3}(\delta_2) = \delta_3$, $s_{f_{-2}-f_3}(\delta_3) = \delta_2$, $s_{f_{-2}-f_3}(\delta_1) = \delta_1$. The three transpositions generate $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$; the Cartan matrix (4.6) is invariant under every permutation of $\{1, 2, 3\}$, so the S_3-action extends from the simple roots to all of $\overline{\mathcal{P}_{II}}$. □

This S_3-symmetry is part of the Gritsenko–Nikulin chamber stabilizer, it fixes the Weyl vector ρ, and it permutes the three primitive isotropic rays of $\overline{\mathcal{P}_{II}}$ (compare Lemma 6.3 below).

The Weyl vector

PROPOSITION 2.4 *(Weyl vector).* The Weyl vector $\rho \in \Lambda^{2,1} \otimes \mathbb{Q}$ defined by

$$(\rho, \delta_i) \;=\; -\frac{(\delta_i, \delta_i)}{2} \;=\; -1, \qquad i = 1, 2, 3,$$

is

$$\rho \;=\; \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) \;=\; f_2 - \tfrac{1}{2}f_3 + f_{-2},$$

and lies in the open positive cone $C(\Lambda^{2,1})_+ = C(\Lambda^{2,1}_{II})_+$.

Proof. The Cartan matrix $A = 4\mathbf{I}_3 - 2\mathbf{J}_3$ acts on $(1, 1, 1)^T$ as $A(1, 1, 1)^T = (-2, -2, -2)^T$ — the all-ones vector is an eigenvector with eigenvalue -2. The orthogonal complement of $(1, 1, 1)$ is the two-plane $\sum c_i = 0$, on which $\mathbf{J}_3$ vanishes and $A = 4\mathbf{I}_3$ acts by 4. Hence the spectrum of A is $(-2, 4, 4)$ and

$$A^{-1}(1, 1, 1)^T \;=\; -\tfrac{1}{2}(1, 1, 1)^T,$$

so the unique solution of $Ac = (-1, -1, -1)^T$ is $c = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Substituting $\rho = \sum c_i \delta_i$ and (4.4):

$$\tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) \;=\; \tfrac{1}{2}(2f_2 - f_3 + 2f_{-2} - f_3 + f_3) \;=\; f_2 - \tfrac{1}{2}f_3 + f_{-2}.$$

Its norm is

$$(\rho, \rho) \;=\; -2 + \tfrac{1}{2} \;=\; -\tfrac{3}{2} \;<\; 0,$$

so $\rho \in C(\Lambda^{2,1})_+$. Since $\rho \in \frac{1}{2}\Lambda^{2,1}_{II}$, ρ also lies in $C(\Lambda^{2,1}_{II})_+$. □

Remark 2.5 *(Weyl vector under the S_3 chamber action).* The expression $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$ is manifestly S_3-symmetric on the simple-root basis; consequently $g \cdot \rho = \rho$ for every $g \in \mathrm{Aut}(\mathcal{P}_{II})$. The Weyl vector is the only fixed line of the S_3-action up to scaling.

The dual cone and chamber containment

Lemma 2.6 *(Chamber and dual cone).* With

$$\Delta(\Lambda^{2,1}_{II})_+ = \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3,$$

$$\Delta(\Lambda^{2,1}_{II})^*_+ = \{x \in \Lambda^{2,1} \otimes \mathbb{R} \mid (x, \delta_i) \leq 0,\ i = 1, 2, 3\},$$

one has

$$\Delta(\Lambda^{2,1}_{II})^*_+ \subset \overline{C(\Lambda^{2,1})_+} \subset \Delta(\Lambda^{2,1}_{II})_+.$$

Proof. Both inclusions are convex-geometric statements about the simplicial chamber $\mathcal{P}_{II}$. The first follows from $(\rho, \delta_i) = -1 < 0$, which places ρ in $\Delta(\Lambda^{2,1}_{II})^*_+$, and the convexity of the cone. The second follows from the explicit decomposition of any $x \in \overline{C(\Lambda^{2,1})_+}$ as a non-negative linear combination of the three simple roots after passing to the chamber $\mathcal{P}_{II}$. □

3 Real and imaginary roots; the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$

Real roots and their multiplicities

Definition 3.1 *(Real-root system).* The real simple-root set in $\mathcal{P}_{II}$ is

$$\Delta^{re}_{\text{simp}} = \{\delta_1, \delta_2, \delta_3\}.$$

The full real-root set is the orbit of the simple roots under the type-II Weyl group,

$$\Delta^{re} = W^{(2)}(\Lambda^{2,1}_{II})\, \Delta^{re}_{\text{simp}}.$$

The real roots have norm $(\alpha, \alpha) = 2$ on every Weyl translate because reflections preserve the bilinear form, and the multiplicity of a real root in $\mathfrak{g}_{\Delta_5}$ is one in even parity (compare Lemma 4.6 below). The real roots index the perturbative wall set

$$\mathcal{W}_{\text{pert}} = \bigcup_{\delta \in \Delta^{re}_{\text{simp}}} \mathbb{H}_\delta;$$

together with the chamber-automorphism walls in $\mathrm{Aut}(\mathcal{P}_{II})$, the perturbative walls of the type-I cover are $\bigcup_{\alpha \in \Delta^{re}} \mathbb{H}_\alpha$.

Imaginary roots: the K3 half-index data

The imaginary simple-root data of $\mathfrak{g}_{\Delta_5}$ are read from the K3 half-index $\phi_{0,1}$ via the Borcherds product. Recall that the weak Jacobi form $\phi_{0,1} \in J^{\text{wk}}_{0,1}$ has Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0,\ l \in \mathbb{Z}} f(n, l)\, q^n r^l, \tag{4.7}$$

and, because the index is one, $f(n, l)$ is a function of the discriminant $4n - l^2$.

Definition 3.2 *(Imaginary simple roots).* For $a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$ set

$$\mu_{\bar{0}}(a) = \begin{cases} m(a), & (a, a) < 0 \text{ and } m(a) > 0, \\ \tau(a), & (a, a) = 0 \text{ and } \tau(a) > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\mu_{\bar{1}}(a) = \begin{cases} -m(a), & (a,a) < 0 \text{ and } m(a) < 0, \\ -\tau(a), & (a,a) = 0 \text{ and } \tau(a) < 0, \\ 0, & \text{otherwise.} \end{cases}$$

The functions $m(a)$ are the additive correction coefficients of Gritsenko–Nikulin [43, Theorem 2.1], read from the Weyl-alternating sum in Section 5.1 below; the functions $\tau(ta)$ on isotropic rays are the eta-product expansion exponents of Lemma 6.4 below.

The imaginary simple-root set is the multiset

$$\Delta^{im} = \Delta^{im}_{\bar{0}} \cup \Delta^{im}_{\bar{1}},$$

where $a \in \Delta^{im}_{\bar{p}}$ with multiplicity $\mu_{\bar{p}}(a)$.

Remark 3.3 *(Why the parity split appears).* The Gritsenko–Nikulin product expansion for $64^{-1}\Delta_5$ has positive and negative integer exponents. A negative exponent in a Borcherds product corresponds to an odd root in the BKM algebra; a positive exponent corresponds to an even root. The sign split of $m(a)$ and $\tau(a)$ across the imaginary support records this parity on the simple-generator fibre. The signed root supermultiplicity is the difference

$$\mathrm{smult}(\alpha) = \dim \mathfrak{g}_{\alpha,\bar{0}} - \dim \mathfrak{g}_{\alpha,\bar{1}}$$

for the full root space. The simple-generator signed dimension $\mu_{\bar{0}}(a) - \mu_{\bar{1}}(a)$ need not equal $\mathrm{smult}(a)$ when the degree a is reached by brackets of lower positive degrees. On the isotropic ray $a_{ij} = \delta_i + \delta_j$, the Gritsenko–Nikulin simple fibre has signed dimension 9, while the full root space has signed multiplicity 10, the missing direction being the real bracket $[e_i, e_j]$. The equality $\mathrm{smult}(\alpha(n,l,m)) = f(nm,l)$ is therefore a denominator identity, not a statement about simple fibres.

The full root datum

Definition 3.4 *(Real and imaginary root sets).* The real-root multiset is

$$\Delta^{re}_{\bar{0}} = \Delta^{re}, \qquad \Delta^{re}_{\bar{1}} = \varnothing.$$

The imaginary-root multiset is $\Delta^{im} = \Delta^{im}_{\bar{0}} \cup \Delta^{im}_{\bar{1}}$ with multiplicities $\mu_{\bar{p}}(a)$ on the imaginary simple support and Weyl-translated copies on the non-simple imaginary support.

LEMMA 3.5 *(Borcherds–Cartan inequalities on the simple support).* Let

$$\Pi = \{\delta_1, \delta_2, \delta_3\} \cup \Delta^{im}_{\mathrm{simp}}$$

be the simple support of the Borcherds–Cartan datum. For every $\alpha \in \Pi$ and every distinct pair $\alpha, \alpha' \in \Pi$,

$$(\alpha,\alpha) > 0 \text{ for } \alpha \in \{\delta_1,\delta_2,\delta_3\}, \qquad (\alpha,\alpha) \le 0 \text{ for } \alpha \in \Delta^{im}_{\mathrm{simp}}, \qquad (\alpha,\alpha') \le 0.$$

For a real simple root $\alpha = \delta_i$,

$$\frac{2(\alpha,\alpha')}{(\alpha,\alpha)} \in \mathbb{Z}.$$

The last two assertions are statements about the simple Borcherds–Cartan datum; arbitrary Weyl translates of real roots are not constrained by an off-diagonal Cartan inequality.

Proof. These are the Borcherds–Kac defining inequalities of the simple Cartan datum [54, Chapter 1], [7]. For the three real simple roots they are the entries of the Cartan matrix (4.6). If $a \in \Delta^{im}_{\text{simp}} \subset \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$, then $(a, \delta_i) \leq 0$ by the chamber inequalities. Two imaginary simple roots lie in the same closed Lorentzian cone and have non-positive norm; their pairing is therefore non-positive. The integrality condition follows from $(\delta_i, \delta_i) = 2$ and the integral pairing on $\Lambda^{2,1}_{II} = \mathbb{Z}\delta_1 + \mathbb{Z}\delta_2 + \mathbb{Z}\delta_3$. $\square$

4 Generators and relations: Cartan, Chevalley, Serre, Borcherds orthogonality, super sign

The Borcherds presentation conventions of [7] and the Kac sign conventions of [54, Chapter 1] are imported intact. In the type-II chamber these conventions give the denominator identity of Section 5.

Definition 4.1 *(Borcherds–Cartan superdatum $\mathscr{B}_{\Delta_5}$).* Let I be an index set with parity map $p\colon I \to \mathbb{Z}/2\mathbb{Z}$ and root map $\mathrm{ch}\colon I \to \Delta \subset \Lambda^{2,1}_{II}$, $i \mapsto \alpha_i$. The fibres over an imaginary root a have cardinality

$$\#\{i \in I \mid \alpha_i = a,\ p(i) = p\} \;=\; \mu_{\overline{p}}(a)$$

in each parity, and the three real simple roots $\delta_1, \delta_2, \delta_3$ each have a single index with parity $p = 0$.

Definition 4.2 *(Cartan, Chevalley, Serre, orthogonality, super sign).* The denominator algebra $\mathfrak{g}_{\Delta_5}$ is generated by the even Cartan $\Lambda^{2,1}_{II} \otimes \mathbb{C}$ and by parity-homogeneous generators

$$e_i,\ f_i \qquad (i \in I), \qquad |e_i| = |f_i| = p(i), \qquad \deg e_i = \alpha_i,\ \deg f_i = -\alpha_i,$$

together with the Cartan element $h_i \in \Lambda^{2,1}_{II} \otimes \mathbb{C}$ sending $i \mapsto \alpha_i$ under the linear identification of the Cartan with the lattice. The defining relations are:
Cartan relations. For $h, h' \in \Lambda^{2,1}_{II} \otimes \mathbb{C}$ and $i, j \in I$,

$$[h, h'] \;=\; 0, \qquad [h, e_j] \;=\; (h, \alpha_j)e_j, \qquad [h, f_j] \;=\; -(h, \alpha_j)f_j.$$

Chevalley relations.

$$[e_i, f_j] \;=\; \delta_{ij}h_i.$$

Real Serre relations. For real α_i and $i \neq j$,

$$(\operatorname{ad} e_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)}e_j \;=\; (\operatorname{ad} f_i)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)}f_j \;=\; 0. \tag{4.8}$$

Borcherds orthogonality. If $(\alpha_i, \alpha_j) = 0$, then

$$[e_i, e_j] \;=\; [f_i, f_j] \;=\; 0.$$

Super sign rule. For homogeneous elements X, Y of parities $|X|, |Y|$,

$$[X, Y] \;=\; -(-1)^{|X||Y|}[Y, X].$$

Remark 4.3 *(The real Serre exponent is three for $i \neq j$).* For the three type-II real simple roots, $(\delta_i, \delta_j) = -2$ and $(\delta_i, \delta_i) = 2$, so

$$1 - \frac{2(\delta_i, \delta_j)}{(\delta_i, \delta_i)} \;=\; 1 - \frac{-4}{2} \;=\; 3.$$

The real Serre relations (4.8) read

$$(\operatorname{ad} e_i)^3 e_j \;=\; (\operatorname{ad} f_i)^3 f_j \;=\; 0 \qquad (i \neq j).$$

The first commutators $[e_i, e_j]$ for $i \neq j$ are *not* forced to vanish; they generate the rank-three real-root sub-Kac–Moody algebra in the standard hyperbolic-Kac–Moody fashion.

The triangular decomposition

LEMMA 4.4 *(Triangular decomposition).* With

$$Q_+ \;=\; \sum_{\alpha\in\Delta} \mathbb{Z}_{\geq 0}\alpha \;\subset\; \Lambda^{2,1}_{II},$$

the algebra $\mathfrak{g}_{\Delta_5}$ has the standard triangular decomposition

$$\mathfrak{g}_{\Delta_5} \;=\; \Bigg(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_\alpha\Bigg) \oplus (\Lambda^{2,1}_{II}\otimes\mathbb{C}) \oplus \Bigg(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_{-\alpha}\Bigg),$$

with $\mathfrak{g}_0 = \Lambda^{2,1}_{II}\otimes\mathbb{C}$ and $[\mathfrak{g}_\alpha,\mathfrak{g}_\beta]\subset\mathfrak{g}_{\alpha+\beta}$. The positive root set is

$$\mathcal{R}_+ \;=\; \{\alpha\in Q_+\setminus\{0\} \mid \mathfrak{g}_\alpha\neq 0\}, \qquad \mathcal{R}_- \;=\; -\mathcal{R}_+.$$

Proof. This is the standard structure of a Borcherds–Kac–Moody Lie superalgebra associated to the superdatum (4.1) and the relations of Definition 4.2; see [7, 54]. □

The signed superdimension function

Definition 4.5 *(Signed root supermultiplicity).* For $\alpha\in\Lambda^{2,1}_{II}$,

$$\operatorname{smult}(\alpha) \;=\; \dim\mathfrak{g}_{\alpha,\bar 0} - \dim\mathfrak{g}_{\alpha,\bar 1}.$$

LEMMA 4.6 *(Real-root multiplicity is one).* For every $\alpha\in\Delta^{re}$,

$$\dim\mathfrak{g}_{\alpha,\bar 0} \;=\; 1, \qquad \dim\mathfrak{g}_{\alpha,\bar 1} \;=\; 0, \qquad \operatorname{smult}(\alpha) \;=\; 1.$$

Proof. The simple real roots δ_i are even by construction, with single generators e_i, f_i, so $\dim\mathfrak{g}_{\delta_i,\bar 0} = 1$ and $\dim\mathfrak{g}_{\delta_i,\bar 1} = 0$. By [54, Lemma 3.8 and Proposition 3.7], Weyl operators are automorphisms of $\mathfrak{g}_{\Delta_5}$. The $\mathfrak{g}_{\Delta_5}$ adaptation [43, Section 6, Statement 6.4] shows that, in absence of odd real simple roots, these Weyl operators are even. Hence $\mathfrak{g}_{\alpha,\overline{p}}\simeq\mathfrak{g}_{w\alpha,\overline{p}}$ for every $w\in W^{(2)}(\Lambda^{2,1}_{II})$ and every parity $\overline{p}$, and the equalities transfer from the simple roots to the full real-root orbit. □

LEMMA 4.7 *(Imaginary simple-fibre signed dimension).* For an imaginary simple degree $a\in\Delta^{im}_{\text{simp}}$, let

$$E_a^{\text{simp}} \subset \mathfrak{g}_a$$

be the span of the Chevalley generators whose root is a. Then

$$\operatorname{sdim} E_a^{\text{simp}} \;=\; \mu_{\bar 0}(a) - \mu_{\bar 1}(a).$$

If a is decomposable in the positive root monoid, the full signed root multiplicity $\operatorname{smult}(a)$ also contains the images of brackets whose total degree is a.

Proof. By construction of $\mathscr{B}_{\Delta_5}$, the imaginary simple-root fibres at a have $\mu_{\bar 0}(a)$ generators of even parity and $\mu_{\bar 1}(a)$ of odd parity, all of degree a. Their span is E_a^{simp}, so its signed dimension is the stated difference. The root space $\mathfrak{g}_a$ is the degree-a part of the quotient of the free Lie superalgebra by the Borcherds–Kac relations, and brackets of lower positive degrees may survive in that quotient. □

5 The denominator: Weyl alternating sum and Borcherds product

The Weyl alternating sum

Theorem 5.1 *(Weyl alternating sum for $64^{-1}\Delta_5$).* Let $W^{(2)}(\Lambda^{2,1}_{II})$ act on $\Lambda^{2,1}\otimes\mathbb{C}$ by reflection in the type-II walls, and let $\rho = \frac{1}{2}(\delta_1+\delta_2+\delta_3) = f_2 - \frac{1}{2}f_3 + f_{-2}$ be the Weyl vector. Then

$$\frac{1}{64}\Delta_5(Z) \;=\; \sum_{w\in W^{(2)}(\Lambda^{2,1}_{II})} \det(w)\left(e^{-\pi i(w(\rho),z)} - \sum_{a\in\Lambda^{2,1}_{II}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\,e^{-\pi i(w(\rho+a),z)}\right).$$

Proof. Write

$$\tfrac{1}{64}\Delta_5(Z) \;=\; \sum_{\lambda} c_\lambda e^{-\pi i(\lambda,z)}, \qquad \lambda\in\Lambda^{2,1}_{II}\cup(\rho+\Lambda^{2,1}_{II}).$$

The leading term has $\lambda=\rho$ with $c_\rho = 1$ by the $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ theta-constant normalization (4.9) below. The remaining exponents λ lie in $\rho + a$ with $a\in\Lambda^{2,1}_{II}\cap\mathbb{R}_{>0}\mathcal{P}_{II}$. The Maass multiplier restriction [43, (1.3)–(1.5)], citing [71], says $\Delta_5\,|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det$, so the Weyl translates satisfy

$$c_{w\lambda} \;=\; \det(w)\,c_\lambda.$$

If λ lies on a simple wall $\mathbb{H}_{\delta_i}$, the reflection s_{δ_i} fixes λ and forces $c_\lambda = -c_\lambda = 0$. Summing the signed Weyl translates of the remaining exponents gives the displayed alternating sum, with $m(a)$ the additive correction coefficient [43, Theorem 2.1] read from the active support of $\phi_{0,1}$. □

Borcherds product expansion of $D_5 = 64^{-1}\Delta_5$

Lemma 5.2 *(Vacuum coefficient).*

$$[q^{1/2}r^{1/2}s^{1/2}]\,\Delta_5(Z) \;=\; 64. \tag{4.9}$$

Proof. The leading exponent of the Gritsenko additive lift of $\eta^9\vartheta(z)$ at ρ is computed in [43, Theorem 2.1 and Section 3], with the constant 64 arising from the ten-theta-constant Maass formula [43, (1.3)]; the same factor is the value of the leading Fourier coefficient $c_\Delta(1,1,1)$ in the Gritsenko cusp expansion. □

Theorem 5.3 *(Borcherds product for D_5).* With $D_5(Z) = 64^{-1}\Delta_5(Z)$, the effective support

$$\Gamma_{\text{eff}} \;=\; \{(n,l,m)\in\mathbb{Z}^3 \mid m>0,\ n\ge 0,\ l\in\mathbb{Z}\}\cup\{(n,l,0)\in\mathbb{Z}^3 \mid n>0,\ l\in\mathbb{Z}\}\cup\{(0,l,0)\in\mathbb{Z}^3 \mid l<0\},$$

and the active subset $\Gamma_{\text{act}} = \{(n,l,m)\in\Gamma_{\text{eff}} \mid f(nm,l)\neq 0\}$,

$$D_5(Z) \;=\; q^{1/2}r^{1/2}s^{1/2}\prod_{(n,l,m)\in\Gamma_{\text{eff}}}(1-q^n r^l s^m)^{f(nm,l)}. \tag{4.10}$$

The cusp polarization selects one of the two half-spaces under $(n,l,m)\mapsto(-n,-l,-m)$ on the type-II character lattice, namely the half-space on which the chamber $0<|s|\ll|q|\ll|r|^{-1}\ll 1$ makes $q^n r^l s^m \to 0$ ([45, Example 2.4]). The boundary triple $(0,-1,0)$ is present and gives the simple root $\delta_3 = f_3$; the inactive triples $\Gamma_{\text{eff}}\setminus\Gamma_{\text{act}}$ contribute trivial factors $(1-q^n r^l s^m)^0 = 1$.

Proof. This is the Borcherds product associated to the singular theta lift of the weight-zero index-one weak Jacobi form $\phi_{0,1}$, [9, Theorem 13.3] specialized to the lattice $\Lambda^{2,1}$ and the Jacobi-form input $\phi_{0,1}$, equivalently [45, Theorem 2.1 and (2.1)–(2.3)]. The exponents $f(nm,l)$ are the Fourier coefficients of $\phi_{0,1}$ (4.7), and the prefactor $q^{1/2}r^{1/2}s^{1/2}$ is the Weyl-vector vacuum $e^{-\pi i(\rho,z)}$ in the chosen polarization. The Borcherds exponent identification is (4.3). □

The denominator identity

Theorem 5.4 *(Denominator identity, doubled form).* Let $\mathfrak{g}_{\Delta_5}$ be the Gritsenko–Nikulin denominator algebra of Definitions 4.1 and 4.2, with signed root supermultiplicity smult. Then

$$\frac{1}{64}\Delta_5(2Z) \;=\; \sum_{w\in W^{(2)}(\Lambda_{II}^{2,1})} \det(w)\left(e^{-2\pi i(w(\rho),z)} - \sum_{a\in\Lambda_{II}^{2,1}\cap\mathbb{R}_{>0}\mathcal{P}_{II}} m(a)\, e^{-2\pi i(w(\rho+a),z)}\right) \tag{4.11}$$

on the Weyl alternating side, and on the Borcherds product side

$$\frac{1}{64}\Delta_5(2Z) \;=\; e^{-2\pi i(\rho,z)} \prod_{\alpha\in\mathcal{R}_+} (1-e^{-2\pi i(\alpha,z)})^{\mathrm{smult}(\alpha)}. \tag{4.12}$$

The two expressions are equal as the Weyl–Kac–Borcherds denominator of $\mathfrak{g}_{\Delta_5}$:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; \frac{1}{64}\Delta_5(2Z). \tag{4.13}$$

On the active support, the signed multiplicities are the K3 half-index Fourier coefficients,

$$\mathrm{smult}(\alpha(n,l,m)) \;=\; f(nm,l) \qquad ((n,l,m)\in\Gamma_{\mathrm{act}}). \tag{4.14}$$

Proof. Apply the substitution $Z\mapsto 2Z$, equivalently $z\mapsto 2z$, to Theorem 5.1: every exponent $-\pi i(w(\rho+a),z)$ becomes $-2\pi i(w(\rho+a),z)$, giving the displayed Weyl alternating sum (4.11).

The product expression (4.12) is the Borcherds product expansion of Theorem 5.3 under $Z\mapsto 2Z$ and the Lorentzian degree identification (4.3): the Gram-index exponent $q^n r^l s^m$ is $e^{-\pi i(\alpha(n,l,m),z)}$, hence $Z\mapsto 2Z$ sends it to $e^{-2\pi i(\alpha(n,l,m),z)}$. The exponents $f(nm,l)$ on the active support are the signed multiplicities $\mathrm{smult}(\alpha(n,l,m))$ of the BKM denominator, because the Borcherds product and the Weyl–Kac–Borcherds product are the same element of the completed positive-root monoid algebra. The imaginary simple-root data of Definition 3.2 determine the additive correction terms in the Weyl alternating side; the full root multiplicities are read from the denominator product.

The equality $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ identifies the Borcherds–Kac–Moody denominator of $\mathfrak{g}_{\Delta_5}$ with the doubled-Igusa product. The factor 64 is the theta-constant normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ of Lemma 5.2; it is not a separate convention choice.

The signed-multiplicity equality (4.14) on the active support is the comparison of (4.12) with the Borcherds product expansion of D_5 at $Z\mapsto 2Z$. □

Two normalizations and the doubling factor

Remark 5.5 *(The doubling $Z\mapsto 2Z$).* The denominator identity is written for $\Delta_5(2Z)$ rather than $\Delta_5(Z)$ because the BKM root grading is on $\Lambda_{II}^{2,1}\subset\Lambda^{2,1}$ (index 4), while the additive lift of $\phi_{0,1}$ produces Δ_5 on the ambient $\Lambda^{2,1}$ in its primitive form. The root attached to a product monomial is

$$\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2} \in \Lambda_{II}^{2,1}.$$

The doubling acts on the Borcherds-product variables by $q\mapsto q^2$, $r\mapsto r^2$, $s\mapsto s^2$, so $q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m),z))$ becomes the standard denominator character $\exp(-2\pi i(\alpha(n,l,m),z))$. No equality $\Lambda_{II}^{2,1} = 2\Lambda^{2,1}$ is used; the f_3-coordinate remains primitive. The theta-constant normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ is untouched by doubling.

Remark 5.6 *(Doubling factor as a power of two).* The constant $64 = 2^6$ is the value of the leading Fourier coefficient $c_\Delta(1,1,1)$ of Δ_5 in the chosen polarization. It is *not* the power 2^d associated with a doubling on a rank-d lattice or a half-Hilbert orbit count. The $2^{12} = 4096$ orbit count appearing in the type-II Pfaffian wall normal form (Chapter 6, $(O2)$) is a distinct sign-sum on the half-Hilbert orbit and is unrelated to the prefactor 64 of the denominator identity.

Proposition 5.7 *(Higher-order Fourier coefficient verification of the Borcherds product).* The Fourier coefficients $f(n,l)$ of $\phi_{0,1}$ entering the Borcherds product of Theorem 5.3 match the expansion of $D_5(Z)$ to order $\max(n) \le 4$ at the type-II cusp. The non-zero $f(n,l)$ for $0 \le n \le 4$ are:

n	$f(n,l)$ for $\lvert l\rvert \le n+1$
0	$f(0,0) = 10, \quad f(0,\pm 1) = 1$
1	$f(1,0) = 108, \quad f(1,\pm 1) = -64, \quad f(1,\pm 2) = 10$
2	$f(2,0) = 808, \quad f(2,\pm 1) = -513, \quad f(2,\pm 2) = 108, \quad f(2,\pm 3) = 1$
3	$f(3,0) = 4016, \quad f(3,\pm 1) = -2752, \quad f(3,\pm 2) = 808, \quad f(3,\pm 3) = -64$
4	$f(4,0) = 16524, \quad f(4,\pm 1) = -11775, \quad f(4,\pm 2) = 4016, \quad f(4,\pm 3) = -513, \quad f(4,\pm 4) = 10$

Proof. The values are read from $\phi_{0,1}(\tau,z) = h_0(\tau)\theta_{1,0}(\tau,z) + h_1(\tau)\theta_{1,1}(\tau,z)$ by the Eichler–Zagier theta decomposition (Lemma 1.12) and the standard generating-function identities for the K3 elliptic genus. All values are obtained by direct rational expansion of the theta quotient and checked against the closed-form $\eta^{-2}\theta_1^2$ expansion to depth $(n,l) = (4,4)$. Squaring the Borcherds product doubles every exponent,

$$\Phi_{10}^{\mathrm{un}} = \Delta_5^2 = 64^2\, qrs \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{2f(nm,l)},$$

so the same coefficients $f(nm,l)$ control the weight-ten Igusa form with multiplicities $2f(nm,l)$.

The verified coefficients match the cross-check identities of [32, Proposition 5.2]: for each fixed n, the row $(f(n,l))_{\lvert l\rvert \le n+1}$ is the Eichler–Zagier expansion of the n-th Fourier component of $\phi_{0,1}$ in elliptic theta-functions. At the trivial elliptic variable one has

$$Z_{\mathrm{K3}}(\tau,0) = 2\phi_{0,1}(\tau,0) = 24,$$

so the same normalization records the Euler characteristic of K3 before the Borcherds multiplicative lift is formed. □

Remark 5.8 *(The verified arithmetic anchors all five views).* Every arithmetic identity about Δ_5, $\mathfrak{g}_{\Delta_5}$, or the type-II Cartan matrix factors through the table of Proposition 5.7 together with the Cartan matrix, Weyl vector, and reflection computation of Appendix 1.2. The five views of the frontispiece are arithmetically reconciled: View 1 via the divisor identification $f(0,\pm 1) = 1$, $f(0,0) = 10$; View 2 via the Borcherds-product exponents $f(nm,l)$; View 3 via the signed-multiplicity residual $\operatorname{sdim} P^{\Pi}_{X,(n,l,m)} = f(nm,l)$; View 5 via the singular-theta lift coefficients.

6 The active support: Lorentzian cone and finite checks

Definition 6.1 *(Active Gram-index support).* The active Gram-index support is

$$\Gamma_{\mathrm{act}} \;=\; \{(n,l,m) \in \Gamma_{\mathrm{eff}} \mid f(nm,l) \neq 0\}.$$

The Borcherds cone is the real cone generated by the Lorentzian images of the active support,

$$C_{\mathrm{Borch}} \;=\; \mathbb{R}_{\ge 0}\,\alpha(\Gamma_{\mathrm{act}}) \;\subset\; \Lambda_{II}^{2,1} \otimes \mathbb{R}.$$

LEMMA 6.2 *(Active support gives the type-II chamber cone).* For $(n, l, m) \in \Gamma_{\text{act}}$,

$$\alpha(n, l, m) \;=\; n\delta_1 + m\delta_2 + (n + m - l)\delta_3, \tag{4.15}$$

with

$$n \;\geq\; 0, \qquad m \;\geq\; 0, \qquad n + m - l \;\geq\; 0.$$

Consequently

$$C_{\text{Borch}} \;=\; \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3.$$

Proof. Substitute (4.4) into (4.15):

$$n(2f_2 - f_3) + m(2f_{-2} - f_3) + (n + m - l)f_3 \;=\; 2nf_2 - lf_3 + 2mf_{-2}.$$

The effective chamber Γ_{eff} has $n, m \geq 0$. Since $\phi_{0,1}$ is weak of index one, $f(nm, l) \neq 0$ implies $4nm - l^2 \geq -1$. If $n = 0$ or $m = 0$, this gives $l^2 \leq 1$, and the boundary cases in Γ_{eff} give $l \leq n + m$. If $n, m > 0$ and $l \geq n + m + 1$, then $l^2 \geq (n + m + 1)^2 = 4nm + (n - m)^2 + 2(n + m) + 1 > 4nm + 1$, contradicting $l^2 \leq 4nm + 1$. Hence $n + m - l \geq 0$.

The reverse inclusion $C_{\text{Borch}} \supset \mathbb{R}_{\geq 0}\delta_1 + \mathbb{R}_{\geq 0}\delta_2 + \mathbb{R}_{\geq 0}\delta_3$ follows from the active triples $(1, 0, 0)$, $(0, 0, 1)$, $(0, -1, 0)$ giving $\alpha = \delta_1 + \delta_3, \delta_2 + \delta_3, \delta_3$, and $(1, 1, 0)$, $(0, 1, 1)$ giving δ_1, δ_2. □

LEMMA 6.3 *(Isotropic primitive rays, S_3-orbit).* The primitive isotropic rays of $\overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$ are generated by

$$a^{(1)} \;=\; 2f_2, \qquad a^{(2)} \;=\; 2f_{-2}, \qquad a^{(3)} \;=\; 2f_2 - 2f_3 + 2f_{-2}.$$

The chamber-automorphism group $\text{Aut}(\mathcal{P}_{II}) \simeq S_3$ is transitive on the three primitive isotropic generators.

Proof. Each $a^{(j)}$ is primitive in $\Lambda_{II}^{2,1}$ and has $(a^{(j)}, a^{(j)}) = 0$ by (4.1); e.g. $(2f_2, 2f_2) = 0$. The condition $(a^{(j)}, \delta_i) \leq 0$ for all i places $a^{(j)}$ in $\overline{\mathcal{P}_{II}}$. The S_3-action permutes the three vectors: $s_{f_{-2}-f_2}$ exchanges $a^{(1)}$ and $a^{(2)}$ and fixes $a^{(3)}$, and similarly the other generators permute the remaining pairs. □

LEMMA 6.4 *(Eta-product expansion on isotropic rays).* For $a_0 = 2f_2$ and $t \in \mathbb{Z}_{\geq 1}$,

$$1 + \frac{1}{64}\sum_{t\geq 1} c_\Delta(1 + 2t, 1, 1)q^t \;=\; \prod_{k\geq 1}(1 - q^k)^9.$$

In denominator notation,

$$1 - \sum_{t\geq 1} m(ta_0)\, q^t \;=\; \prod_{t\geq 1}(1 - q^t)^{\tau(ta_0)}, \qquad \tau(ta_0) \;=\; 9 \quad (t \geq 1).$$

By S_3-symmetry, $\tau(ta) = 9$ for every primitive isotropic ray $\mathbb{Z}_{>0}a \subset \overline{\mathcal{P}_{II}} \cap \Lambda_{II}^{2,1}$.

Proof. The product $\prod_{k\geq 1}(1 - q^k)^9$ is the Gritsenko–Nikulin isotropic boundary expansion of the cusp form along the ray $\mathbb{R}_{>0}a_0$, computed via the eta-product η^9 at the relevant cusp [45, Section 5.1.1], using $\Delta_5 = \text{Lift}(\eta^9\vartheta)$ in Gritsenko's additive lift. The S_3-symmetry from Lemma 6.3 transfers the value $\tau(ta) = 9$ to every primitive isotropic ray. □

Low-degree denominator brackets

Proposition 6.5 *(Low-degree real-root brackets).* The denominator algebra has the following target-side low-degree checks. With

$$\alpha(n,l,m) \;=\; n\delta_1 + m\delta_2 + (n+m-l)\delta_3,$$

the active positive rows through height three are

$$\begin{array}{c|c|c|c|c} \mathrm{ht}\,\beta & \beta & \gamma(\beta) & (\beta,\beta) & \mathrm{smult}(\beta) \\ \hline 1 & \delta_1 & (1,1,0) & 2 & 1 \\ 1 & \delta_2 & (0,1,1) & 2 & 1 \\ 1 & \delta_3 & (0,-1,0) & 2 & 1 \\ 2 & \delta_1+\delta_2 & (1,2,1) & 0 & 10 \\ 2 & \delta_1+\delta_3 & (1,0,0) & 0 & 10 \\ 2 & \delta_2+\delta_3 & (0,0,1) & 0 & 10 \\ 3 & 2\delta_1+\delta_2 & (2,3,1) & 2 & 1 \\ 3 & \delta_1+2\delta_2 & (1,3,2) & 2 & 1 \\ 3 & 2\delta_1+\delta_3 & (2,1,0) & 2 & 1 \\ 3 & \delta_1+2\delta_3 & (1,-1,0) & 2 & 1 \\ 3 & 2\delta_2+\delta_3 & (0,1,2) & 2 & 1 \\ 3 & \delta_2+2\delta_3 & (0,-1,1) & 2 & 1 \\ 3 & \delta_1+\delta_2+\delta_3 & (1,1,1) & -6 & -64 \end{array} \tag{4.16}$$

After choosing even Chevalley generators e_i, f_i, h_i for the real simple roots and setting

$$A_{ij} = (\delta_i, \delta_j), \qquad E_{ij} := [e_i, e_j] \quad (i \neq j),$$

so $E_{ji} = -E_{ij}$, one has

$$[h_i,h_j] \;=\; 0, \qquad [h_i,e_j] \;=\; A_{ij}e_j, \qquad [h_i,f_j] \;=\; -A_{ij}f_j, \qquad [e_i,f_j] \;=\; \delta_{ij}h_i,$$

and

$$(\mathrm{ad}\,e_i)^3 e_j \;=\; 0 \quad (i \neq j).$$

For $\{i,j,k\} = \{1,2,3\}$,

$$[h_i,E_{ij}] \;=\; [h_j,E_{ij}] \;=\; 0, \qquad [h_k,E_{ij}] \;=\; -4E_{ij},$$

$$[f_i,E_{ij}] \;=\; 2e_j, \qquad [f_j,E_{ij}] \;=\; -2e_i, \qquad [f_k,E_{ij}] \;=\; 0.$$

For ordered $i \neq j$, put $E_{iij} := [e_i, E_{ij}]$. Then

$$[e_i,E_{iij}] \;=\; 0, \qquad [f_i,E_{iij}] \;=\; 2E_{ij}, \qquad [f_j,E_{iij}] \;=\; 0.$$

For $i < j$, the isotropic primitive degree $a_{ij} := \delta_i + \delta_j$ has

$$\mathrm{smult}(a_{ij}) \;=\; 10 \;=\; 1+9:$$

one even real-real commutator direction E_{ij} and nine even isotropic simple directions $u_{ij,1}, \ldots, u_{ij,9} \in \mathfrak{g}_{a_{ij}}$, $\tau(ta_{ij}) = 9$ for $t \geq 1$. The orthogonality relations give

$$[e_i, u_{ij,r}] = [e_j, u_{ij,r}] = 0, \qquad [f_i, u_{ij,r}] = [f_j, u_{ij,r}] = 0.$$

In the timelike degree $\delta_{123} := \delta_1 + \delta_2 + \delta_3$, the real bracketings

$$T_1 = [e_1, E_{23}], \qquad T_2 = [e_2, E_{31}], \qquad T_3 = [e_3, E_{12}]$$

satisfy $T_1 + T_2 + T_3 = 0$ by the graded Jacobi identity. Since $(\delta_k, a_{ij}) = -4$, the Borcherds–Kac presentation imposes

$$(\operatorname{ad} e_k)^5 u_{ij,r} = 0,$$

and analogously for the negatives. The product exponent gives

$$\operatorname{smult}(\delta_{123}) = f(1,1) = -64.$$

Proof. The table is obtained by writing $\beta = c_1\delta_1 + c_2\delta_2 + c_3\delta_3$ and solving $\gamma(\beta) = (c_1, c_1 + c_2 - c_3, c_2)$. The norm is $(\beta, \beta) = -2(4nm - l^2)$, and the multiplicity is $\operatorname{smult}(\beta) = f(nm, l)$ from Theorem 5.4. The bracket constants are the normalized Chevalley, Jacobi and Serre constants for the symmetric Cartan matrix A. The signs $E_{ji} = -E_{ij}$, $[f_i, E_{ij}] = 2e_j$, $[f_j, E_{ij}] = -2e_i$ follow from the graded Jacobi identity and the normalization $[e_i, f_i] = h_i$. The relation $T_1 + T_2 + T_3 = 0$ is the graded Jacobi identity for the even real generators e_1, e_2, e_3.

For $a_{ij} = \delta_i + \delta_j$, $(\delta_k, a_{ij}) = -4$, hence the real Serre exponent is $1 - 2(\delta_k, a_{ij})/(\delta_k, \delta_k) = 5$. The equality $\tau(ta_{ij}) = 9$ on primitive isotropic rays is Lemma 6.4. □

Remark 6.6 *(The terminal stop on the affine string).* For $\alpha = \delta_k$ and $\beta = a_{ij}$ with $\{i, j, k\} = \{1, 2, 3\}$, the real-root string theorem [54, Proposition 5.1(c)] with the GN super adaptation [43, Section 6, Statement 6.4] gives $(\beta, \alpha^\vee) = -4$ and $\beta - \alpha \notin \Delta$. Hence the string $\beta + \mathbb{Z}\alpha$ stops before $a_{ij} + 5\delta_k$:

$$a_{ij},\ a_{ij} + \delta_k,\ a_{ij} + 2\delta_k,\ a_{ij} + 3\delta_k,\ a_{ij} + 4\delta_k,\ \mathfrak{g}_{a_{ij}+5\delta_k} = 0.$$

The submatrix on δ_i, δ_j is $\left(\begin{smallmatrix} 2 & -2 \\ -2 & 2 \end{smallmatrix}\right)$, so $a_{ij} = \delta_i + \delta_j$ is the affine null root of $A_1^{(1)}$. [54, Theorem 7.4 and Corollary 8.3] gives $\operatorname{mult}(ta_{ij}) = 1$ in the affine subalgebra, and the extra nine in $\operatorname{smult}(a_{ij}) = 10$ are the GN isotropic simple fibres of Lemma 6.4.

The first timelike presentation count

PROPOSITION 6.7 *(Timelike presentation count 29|93 at δ_{123}).* With $\delta_{123} = \delta_1 + \delta_2 + \delta_3 = 2f_2 - f_3 + 2f_{-2}$, the additive correction coefficient is

$$m(\delta_{123}) = -93,$$

and consequently

$$\operatorname{mult}_{\bar{0}}(\delta_{123}) = 29, \qquad \operatorname{mult}_{\bar{1}}(\delta_{123}) = 93, \qquad 29 - 93 = -64 = f(1,1).$$

The 29 even directions are the two-dimensional real-real-real Borcherds–Kac presentation span

$$T_1 = [e_1, [e_2, e_3]], \qquad T_2 = [e_2, [e_3, e_1]], \qquad T_3 = [e_3, [e_1, e_2]], \qquad T_1 + T_2 + T_3 = 0,$$

together with the 27 mixed real-isotropic words

$$[e_k, u_{ij,r}], \qquad \{i, j, k\} = \{1, 2, 3\},\ 1 \leq r \leq 9.$$

The 93 odd directions are the negative-norm imaginary simple generators in degree δ_{123}.

Proof. The root ∂_{123} corresponds in the Weyl-sum coefficient formula to $(3-1)f_2 - \frac{3-1}{2}f_3 + (3-1)f_{-2}$. Hence $m(\partial_{123}) = -64^{-1}c_\Delta(3,3,3)$. Removing the leading monomial from $D_5 = 64^{-1}\Delta_5$ and computing $[qrs]\prod(1-q^n r^l s^m)^{f(nm,l)} = 93$ gives the integer 93: the factor $(1-qrs)^{-64}$ contributes 64, and the remaining root factors contribute 29. The signed equality $29 - 93 = -64 = f(1,1)$ is the Borcherds product identity (4.14) at $(n,l,m) = (1,1,1)$. The Borcherds–Kac presentation supplies $29 = 2 + 27$ even directions: the rank-three real-real-real Jacobi words T_1, T_2, T_3 with one linear relation $T_1 + T_2 + T_3 = 0$ giving two independent words, and the $3 \cdot 9 = 27$ mixed real-isotropic brackets $[e_k, u_{ij,r}]$. The remaining 93 odd directions are the negative-norm imaginary simple generators of Definition 3.2. □

Remark 6.8 *(Determinant alone forgets the parity split).* The determinant identity $\mathrm{smult}(\partial_{123}) = -64$ records only the difference $\mathrm{mult}_{\bar 0} - \mathrm{mult}_{\bar 1} = 29 - 93$. The two integers 29 and 93 are presentation counts: 29 even directions from the rank-three Borcherds–Kac presentation, and 93 odd directions read from $m(\partial_{123})$. They are determined by the BKM presentation of $\mathfrak{g}_{\Delta_5}$, not by the Borcherds product.

Remark 6.9 *(Target rows in the first relation-closed degree window).* Let $C_{k,s} = a_{ij} + s\partial_k$ for $\{i,j,k\} = \{1,2,3\}$. The target presentation supplies

$$2a_{ij} : 10|0, \qquad C_{k,3} : 29|93, \qquad C_{k,4} : 10|0, \qquad C_{k,5} : 0|0.$$

The first row uses the $A_1^{(1)}$ subalgebra generated by ∂_i, ∂_j, together with the nine GN isotropic simple fibres on the ray $\mathbb{Z}_{>0}a_{ij}$. The rows $C_{k,3}$ and $C_{k,4}$ are the even real-Weyl transports of ∂_{123} and a_{ij}, respectively, and $C_{k,5}$ is the terminal zero row of the real-root string [54, Proposition 5.1(c), Theorem 7.4, Corollary 8.3] [43, Sections 3–4, Statement 6.4]. In contrast,

$$C_{k,2} : \mathrm{sdim} = 108, \quad m = 90, \qquad 2\partial_{123} : \mathrm{sdim} = 4016, \quad m = -540$$

are signed target rows only. They cannot be used as parity blocks, PBW rows, or source comparison codomains until the finite target presentation is computed in those degrees.

7 Maass multiplier on $W^{(2)}(\Lambda^{2,1})$

The transformation law of Δ_5 under $\mathbf{Sp}_4(\mathbb{Z})$ is

$$\Delta_5(g \cdot Z) \;=\; \nu_{\Delta_5}(g)\det(CZ+D)^5\Delta_5(Z), \qquad g = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \mathbf{Sp}_4(\mathbb{Z}),$$

with multiplier system ν_{Δ_5} of order two. The scalar square is invariant of order 10:

$$\Delta_5^2(g \cdot Z) \;=\; \det(CZ+D)^{10}\Delta_5^2(Z).$$

Under the exterior-square identification of $\mathbf{PSp}_4(\mathbb{Z})$ with its projective image in $\mathbf{O}(\Lambda^{3,2})_+$ and the embedding $\mathbf{O}(\Lambda^{2,1})_+ \hookrightarrow \mathbf{O}(\Lambda^{3,2})$, the multiplier restricts to $W^{(2)}(\Lambda^{2,1})$ as a character to $\{\pm 1\}$.

Lemma 7.1 *(Reflection-class values of ν_{Δ_5}).* For the three type-II simple reflections,

$$\nu_{\Delta_5}(s_{\partial_i}) \;=\; -1, \qquad i = 1,2,3.$$

For the three type-I chamber-automorphism reflections,

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) \;=\; \nu_{\Delta_5}(s_{f_2-f_3}) \;=\; \nu_{\Delta_5}(s_{f_{-2}-f_3}) \;=\; +1.$$

Proof. Maass's multiplier formula for the product of the ten even theta constants is [43, (1.3)–(1.5)], citing [71]. Under the exterior-square identification of Section 1.4, Gritsenko–Nikulin compute the reflection values at [43, Proposition 2.1] and the type-II chamber restriction at [43, Proposition 2.2]. The character is -1 on the three type-II simple reflections and $+1$ on the three $\mathrm{Aut}(\mathcal{P}_{II})$-reflections. □

LEMMA 7.2 *(Multiplier restriction).* Under the chamber decomposition (4.5),

$$\nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})} = \det, \qquad \nu_{\Delta_5}\big|_{\mathrm{Aut}(\mathcal{P}_{II})} = 1.$$

Equivalently, for $h = wg$ with $w \in W^{(2)}(\Lambda_{II}^{2,1})$ and $g \in \mathrm{Aut}(\mathcal{P}_{II})$, the weight-five slash action without character satisfies

$$(\Delta_5|_5 h)(Z) = \det(w)\Delta_5(Z), \qquad (\Delta_5|_5 h)(Z) := j(h,Z)^{-5}\Delta_5(hZ),$$

where $j(h,Z)$ is the automorphy factor of the chosen $\mathbf{Sp}_4$-lift of h. Thus the lemma is a statement about the multiplier character, not a raw equality of functions on Siegel space.

Proof. Lemma 7.1 provides the values on the six generators of the chamber decomposition. The first generator class generates $W^{(2)}(\Lambda_{II}^{2,1})$ as a Coxeter group on three generators of det-character -1, hence $\nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})} = \det$. The second generator class generates the chamber automorphism S_3 with character $+1$ on each generator, hence trivially. □

Remark 7.3 *(Why the parity restricts to determinant).* The character ν_{Δ_5} has order two and is determined by six generator values; the type-II Coxeter relations among the s_{δ_i} force the restriction $\nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}$ to be either trivial or the determinant. Since $\nu_{\Delta_5}(s_{\delta_i}) = -1 \neq 1$, the restriction is the determinant character. This is a character computation: it does not depend on a normalization of Δ_5 and is fixed by the group theory plus the six generator values.

8 THE BORCHERDS-WEIGHT CHARACTERISTIC $\varkappa_{\mathrm{BKM}}(\Delta_5) = 5$

The Borcherds-weight characteristic of the BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ is the weight of the paramodular form whose product expansion is the denominator. The paramodular form here is $\Delta_5 = \Phi_1$ at level 1, with weight 5.

THEOREM 8.1 *(Universal Borcherds-weight identity for $\mathfrak{g}_{\Delta_5}$).*

$$\varkappa_{\mathrm{BKM}}(\Delta_5) = \frac{c_1(0)}{2} = \frac{10}{2} = 5. \tag{4.17}$$

The identity is universal across the five CHL frame shapes $N \in \{1,2,3,4,6\}$ with $\varphi(N) \mid 2$: for each N in the frame, the Borcherds product Φ_N has weight $c_N(0)/2$, with $c_N(0)$ the constant Fourier coefficient of the K3 elliptic genus $\phi_{0,1}^{(N)}$ at the g_N-twined level, tabulated as follows (values verified against the CHL elliptic genus of [19, §4]):

N	1	2	3	4	6
$c_N(0)$	10	8	6	4	2
$\varkappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$	5	4	3	2	1

Proof. Borcherds weight formula. The Borcherds weight formula [9, Theorem 13.3] states that for a Borcherds product $\Theta(\phi)$ on the Grassmannian $\mathbb{H}_+^{\mathbf{IV}}$ associated to a signature-$(3,2)$ lattice $\Lambda^{3,2}$ and a weakly holomorphic vector-valued modular form ϕ of weight $-1/2$ on $\mathbf{Mp}_2(\mathbb{Z})$ with constant Fourier coefficient $c(0,0)$, the weight of $\Theta(\phi)$ on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ is $\operatorname{wt}\Theta(\phi) = c(0,0)/2$. Pulled back along the exterior-square map from $\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$ to the projective orthogonal group of $\Lambda^{3,2}$, this is the weight on $\mathbf{Sp}_4(\mathbb{Z})$.

The $N = 1$ case. The input form is $\phi_{0,1} \in J_{0,1}^{\mathrm{weak}}$, the weight-zero index-one weak Jacobi form, with constant Fourier coefficient $c_1(0) = f(0,0) = 10$. Hence $\operatorname{wt}\Delta_5 = c_1(0)/2 = 10/2 = 5$, recovering equation (4.17).

The CHL-twined values for $N \in \{2,3,4,6\}$. For $g_N \in M_{24}$ of order $N \in \{2,3,4,6\}$ the canonical cycle shapes on the 24-dimensional Niemeier representation are

$$1^8 2^8, \quad 1^6 3^6, \quad 1^4 2^2 4^4, \quad 1^2 2^2 3^2 6^2$$

([19, §2]; Conway–Norton); the twined K3 elliptic genus $\phi_{0,1}^{(g_N)} \in J_{0,1}^{\mathrm{weak}}(\Gamma_0(N), \chi_{g_N})$ of [19, §3] has constant Fourier coefficient $c_N(0) = \chi^{g_N}(K3)$, the g_N-Lefschetz number on K3 cohomology. For $N \in \{2,3,4,6\}$, $\chi^{g_N}(K3)$ equals the number of 1-cycles in the cycle shape, giving $(c_2(0), c_3(0), c_4(0), c_6(0)) = (8,6,4,2)$. At $N = 1$ the EZ-normalised input $\phi_{0,1}^{(1)} = \phi_{0,1}$ has $c_1(0) = f(0,0) = 10$ by the $N = 1$ computation, and the table $(c_1(0), c_2(0), c_3(0), c_4(0), c_6(0)) = (10,8,6,4,2)$ is the CHL-twined constant-coefficient column of [19, Table 2].

Universality of the Borcherds product construction. The Borcherds singular-theta lift $\Theta(\phi_{0,1}^{(N)})$ of the g_N-twined input is well-defined for each $N \in \{1,2,3,4,6\}$ by [9, Theorem 13.3] applied to the corresponding CHL lattice $\Lambda^{(N),3,2}$; the weight is $c_N(0)/2$ on $\widetilde{\mathbf{O}}(\Lambda^{(N),3,2})_+$, pulled back to a Siegel modular form of weight $\kappa_{\mathrm{BKM}}(\Phi_N)$. For each $N \in \{1,2,3,4,6\}$, the table values follow from the CHL-twined constant terms.

Separation from the eight-row Gritsenko–Cléry catalogue. The Gritsenko–Cléry catalogue [41, Theorem 1.4] is indexed by the eight pairs

$$(1,1), (2,1), (1,2), (3,1), (1,3), (4,1), (1,4), (2,2),$$

or equivalently by the F_j-rows of Appendix 2. Its weights and constants in the input-pair order are

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1), \qquad (10, 4, 6, 2, 4, 1, 3, 2),$$

and each row satisfies $\operatorname{wt}(F_j) = c_j(0)/2$. The CHL-twining table of the present theorem is indexed by automorphisms g_N with $N \in \{1,2,3,4,6\}$ and has constants $(10,8,6,4,2)$. The common entry used by the $K3 \times E$ host is the $(t, N) = (1,1)$ row with Jacobi seed $\phi_{0,1}$, Borcherds product Δ_5, and weight 5. □

Remark 8.2 *(The additive formula fails).* The additive expression

$$\kappa_{\mathrm{BKM}}(\Phi_N) \overset{?}{=} \kappa_{\mathrm{ch}}(\mathcal{A}_X) + \chi(\mathcal{O}_{\mathrm{fiber}})$$

has no invariant meaning on the κ-tuple. On the Hodge reading at $N = 1$,

$$\kappa_{\mathrm{BKM}}(\Phi_1) = 5 \neq \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}(\mathcal{A}_{K3\times E}) + \chi(\mathcal{O}_{K3}) = 0 + 2 = 2.$$

On the Heisenberg reading,

$$\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(\mathcal{A}_{K3\times E}) + \chi(\mathcal{O}_{K3}) = 3 + 2 = 5$$

matches the $N = 1$ number only accidentally: the same fixed right hand side cannot produce the CHL weights $c_N(0)/2 = (5, 4, 3, 2, 1)$ for $N = 1, 2, 3, 4, 6$. The universal rule is the rowwise Borcherds weight $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$. The data live in distinct entries of the κ-tuple

$$(\kappa_{\mathrm{cat}},\ \kappa_{\mathrm{ch}}^{\mathrm{Hodge}},\ \kappa_{\mathrm{ch}}^{\mathrm{Heis}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}}) \ = \ (0, 0, 3, 5, 24),$$

matching the Vol III κ-stratification of Calabi–Yau quantum groups; the Borcherds-weight identity (4.17) is the correct universal formula on the BKM entry, and the additive formula confuses $\chi_{\mathrm{top}}(K3) = 24$ with $\chi(\mathcal{O}_{K3}) = 2$. The universal failure across the five-CHL frame $N \in \{1, 2, 3, 4, 6\}$ is the level-by-level evaluation of the κ-tuple recorded in Vol III.

Remark 8.3 *(Two Borcherds inputs and one weight rule).* The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ of [9, Theorem 13.3] applies after the input form, lattice, and Weyl chamber are fixed. The paramodular Δ_5 row, with Jacobi input $\phi_{0,1}$ and Borcherds product Δ_5 on $\Lambda_{II}^{2,1}$, reads $\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 10/2 = 5$. The Vol I row records the Fake-Monster Borcherds product Φ_{12} on the Leech-extension lattice $\mathrm{II}_{25,1}$ [7, 8], reading $\kappa_{\mathrm{BKM}}(\Phi_{12}) = c_{\Phi_{12}}(0)/2 = 12$. The values 5 and 12 are different inputs to the same Borcherds-weight rule, on different lattices, not a value disagreement and not a shifted indexing of one input; each site must name its input form and lattice. The additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ is not a structural identity: the Hodge reading fails at $N = 1$, and the Heisenberg reading gives only the accidental equality $3 + 2 = 5$. The number 5 is fixed by $c_1(0)/2$; the chiral and fibre entries belong to different slots of the κ-tuple.

9 The denominator-algebra theorem and its scope

Theorem 9.1 *(Denominator-algebra identity).* The denominator algebra $\mathfrak{g}_{\Delta_5}$ associated to the Borcherds–Cartan superdatum $\mathscr{B}_{\Delta_5}$ has the following properties.

(i) *Triangular decomposition.*

$$\mathfrak{g}_{\Delta_5} \ = \ \Bigg(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_\alpha\Bigg) \oplus \mathfrak{g}_{\Delta_5\,0} \oplus \Bigg(\bigoplus_{\alpha\in Q_+\setminus\{0\}} \mathfrak{g}_{-\alpha}\Bigg),$$

where $\mathfrak{g}_{\Delta_5\,0} = \Lambda_{II}^{2,1} \otimes \mathbb{C}$ is the Cartan subalgebra. The chamber-interior and cusp-boundary summands decompose only the positive half:

$$\mathfrak{g}_{\Delta_5+} = \mathfrak{g}_{\Delta_5+}^{\ \mathrm{int}} \oplus \mathfrak{g}_{\Delta_5+}^{\ \mathrm{cusp}}, \qquad \mathfrak{g}_{\Delta_5+}^{\ \mathrm{int}} = \bigoplus_{\alpha\in\mathcal{R}_+,\, m(\alpha)>0} \mathfrak{g}_\alpha, \quad \mathfrak{g}_{\Delta_5+}^{\ \mathrm{cusp}} = \bigoplus_{\alpha\in\mathcal{R}_+,\, m(\alpha)=0} \mathfrak{g}_\alpha.$$

(ii) *Bracket compatibility.*

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \ \subset \ \mathfrak{g}_{\alpha+\beta}.$$

(iii) *Visible signed multiplicity.* On the active support

$$\mathrm{smult}(\alpha(n, l, m)) \ = \ f(nm, l) \qquad ((n, l, m) \in \Gamma_{\mathrm{act}}).$$

(iv) *Denominator identity.*

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \ = \ \frac{1}{64}\Delta_5(2Z).$$

(v) *Borcherds-weight identity.*

$$\kappa_{\mathrm{BKM}}(\Delta_5) \;=\; \frac{c_1(0)}{2} \;=\; 5.$$

The theorem concerns the Borcherds denominator algebra. A microscopic state space and its operator product require separate compact realization; the operator-level pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ is defined in Chapter 6, and its protected Pfaffian is identified with Δ_5 by the Pfaffian–Dirac theorem of Chapter 7.

Proof. Parts (i) and (ii) are the standard structure of a generalized Kac–Moody Lie superalgebra associated to the superdatum $\mathscr{B}_{\Delta_5}$ and the relations of Definition 4.2; see [43, Sections 3–4] for the construction and [7, 54] for the underlying presentation theory.

Part (iii) is (4.14), comparing the Borcherds product expansion of D_5 (Theorem 5.3) with the Weyl–Kac–Borcherds denominator product (Theorem 5.4). The product exponents on the active Gram-index support Γ_{act} are the Fourier coefficients $f(nm, l)$ of $\phi_{0,1}$, and the BKM signed multiplicities $\mathrm{smult}(\alpha)$ are read off equality of the two expansions in their common Lorentzian chamber.

Part (iv) is Theorem 5.4: the doubling $Z \mapsto 2Z$ identifies the type-II sublattice grading with the BKM root lattice grading, and the prefactor 64 is the theta-constant normalization (4.9).

Part (v) is Theorem 8.1, the Borcherds-weight characteristic at $N = 1$. □

Corollary 9.2 *(Real-root and isotropic multiplicities).* The signed root supermultiplicity smult takes values

$$\mathrm{smult}(\delta_i) \;=\; 1 \quad (i = 1, 2, 3), \qquad \mathrm{smult}(a_{ij}) \;=\; 10 \quad (i < j), \qquad \mathrm{smult}(\delta_{123}) \;=\; -64,$$

with the parity split at δ_{123} given by $\mathrm{mult}_{\bar{0}}(\delta_{123}) = 29$, $\mathrm{mult}_{\bar{1}}(\delta_{123}) = 93$ of Proposition 6.7.

Proof. The values follow from the active support table (4.16) and Proposition 6.7. □

10 Boundary with View I (automorphic) and View V (theta lift)

The denominator algebra is the second of the five views of Δ_5. Two cross-level identities to neighbouring views are already theorems and are recorded here as boundary statements; the third cross-level identity, View II ↔ View III, is the Pfaffian–Dirac theorem of Chapter 7, with primitive recognition separated as $(S1)$–$(S10)$.

Theorem 10.1 *(Cross-level identity* View I ↔ View II*, Borcherds 1998 / Gritsenko–Nikulin 1998).* The Borcherds denominator of the BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$ is the doubled Igusa cusp form, normalized by the theta-constant constant:

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z).$$

Proof. This is Theorem 5.4 in the present chapter, equivalently [43, Sections 3–4, Proposition 3.1], with explicit product expansion in [45, Theorem 2.1, (2.1)–(2.3)] and the universal weight characteristic in [9, Theorem 13.3]. □

Theorem 10.2 *(Cross-level identity* View II ↔ View V*, Borcherds 1998).* The Borcherds denominator of $\mathfrak{g}_{\Delta_5}$ is obtained from the singular theta lift of the Weil-representation theta component $\phi_{0,1}^{\mathrm{Weil}}$ of $\phi_{0,1} \in J_{0,1}^{\mathrm{wk}}$ on the lattice $\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$. The bridge between the two sides is the exceptional exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ of Section 1.4.

Proof. This is the singular theta lift theorem [9, Theorem 13.3] specialized to the lattice $\Lambda^{3,2}$ and to the Weil-representation form $\phi^{\mathrm{Weil}}_{0,1}$ obtained by theta-decomposing the Jacobi input $\phi_{0,1}$, equivalently the additive lift in Gritsenko–Nikulin [43, Section 1, Theorem 2.1] cast through the exceptional isogeny. The bridge $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ and the metaplectic Weil action on the input are the subject of Chapter 5. □

11 THE $\mathfrak{g}_{\Delta_5}$ DENOMINATOR IN THE κ-TUPLE

The denominator algebra $\mathfrak{g}_{\Delta_5}$ sits at the BKM entry of the κ-tuple

$$(\kappa_{\mathrm{cat}},\ \kappa^{\mathrm{Hodge}}_{\mathrm{ch}},\ \kappa^{\mathrm{Heis}}_{\mathrm{ch}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24),$$

of Vol III's CY_d-stratification. The five-tuple records: $\kappa_{\mathrm{cat}} = 0$ for the categorical-Hodge characteristic, $\kappa^{\mathrm{Hodge}}_{\mathrm{ch}} = 0$ for the chiral-Hodge refined supertrace, $\kappa^{\mathrm{Heis}}_{\mathrm{ch}} = 3$ for the Heisenberg rank, $\kappa_{\mathrm{BKM}} = 5$ for the cusp-form weight, and $\kappa_{\mathrm{fiber}} = \chi_{\mathrm{top}}(K3) = 24$.

The five entries are independently defined and not summed: the categorical-Hodge characteristic is the trace of the BV bracket on the K3 chiral algebra; the chiral-Hodge refined supertrace is read on the protected reduced category; the Heisenberg rank is the rank-three type-II Heisenberg–Cartan obtained from the normal-ordered Gram projection to $\Lambda^{2,1}_{II}$; the cusp-form weight is $\kappa_{\mathrm{BKM}} = c_1(0)/2 = 5$ from Theorem 8.1; the fibre Euler character $\chi_{\mathrm{top}}(K3) = 24$ records the topology of the K3 fibre.

Remark 11.1 *(Paramodular row versus Fake-Monster row).* The value $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$ is the Borcherds-weight $c_1(0)/2$ on the paramodular row with Jacobi input $\phi_{0,1}$ at $\Lambda^{2,1}_{II}$. The Fake-Monster value $\kappa_{\mathrm{BKM}}(\Phi_{12}) = 12$ is the Borcherds-weight $c_{\Phi_{12}}(0)/2$ on a different input, the Fake-Monster Borcherds product on $\mathrm{II}_{25,1}$ of [7, 8]; the two rows are independent evaluations of one universal rule (Remark 8.3). The BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ realizes the paramodular row.

12 STRUCTURAL PROPERTIES OF $\mathfrak{g}_{\Delta_5}$

Real-root sub-Kac–Moody algebra

LEMMA 12.1 *(Real rank-three sub-Kac–Moody algebra).* The subalgebra ${\mathfrak{g}_{\Delta_5}}^{re} \subset \mathfrak{g}_{\Delta_5}$ generated by $\{e_i, f_i, h_i\}_{i=1}^3$ together with the Cartan $\Lambda^{2,1}_{II} \otimes \mathbb{C}$ is the rank-three hyperbolic Kac–Moody algebra $\mathfrak{g}(A)$ associated to the symmetric Cartan matrix $A = ((\delta_i, \delta_j)) = \left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$.

Proof. The relations of Definition 4.2 restricted to real $\alpha_i \in \Delta^{re}_{\mathrm{simp}}$ are exactly the Kac–Moody relations of [54, Chapter 1] for the symmetric Cartan matrix A. The real Serre exponent $1 - 2(\delta_i, \delta_j)/(\delta_i, \delta_i) = 3$ is the standard exponent for $A_{ij} = -2$, $A_{ii} = 2$. □

Remark 12.2 *(Hyperbolic structure of ${\mathfrak{g}_{\Delta_5}}^{re}$).* The Kac–Moody algebra $\mathfrak{g}(A)$ is hyperbolic in the sense of [54, Chapter 4]: every proper connected rank-two principal submatrix is

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix},$$

the affine $A^{(1)}_1$ Cartan matrix with null root $\delta_i + \delta_j$, while the full rank-three matrix has Lorentzian signature. The full BKM algebra $\mathfrak{g}_{\Delta_5}$ is the automorphic correction of $\mathfrak{g}(A)$ in the sense of [43, Section 3], i.e. adjoining the imaginary simple roots $a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$ with multiplicities $\mu_{\overline{p}}(a)$ to obtain the algebra whose denominator identity matches $64^{-1}\Delta_5(2Z)$.

Bilinear pairing on $\mathfrak{g}_{\Delta_5}$

Lemma 12.3 *(Invariant bilinear form).* There is a unique non-degenerate invariant supersymmetric bilinear form

$$(\cdot\,,\,\cdot)_{\mathfrak{g}_{\Delta_5}} : \mathfrak{g}_{\Delta_5} \otimes \mathfrak{g}_{\Delta_5} \longrightarrow \mathbb{C}$$

extending the bilinear form on $\Lambda^{2,1}_{II} \otimes \mathbb{C} \subset \mathfrak{g}_{\Delta_{5_0}}$ and satisfying the ad-invariance

$$([X,Y],Z)_{\mathfrak{g}_{\Delta_5}} \;=\; (X,[Y,Z])_{\mathfrak{g}_{\Delta_5}}.$$

The pairing $(\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha})_{\mathfrak{g}_{\Delta_5}}$ is non-degenerate; pairings between non-opposite root spaces vanish.

Proof. Standard for Borcherds–Kac–Moody algebras; see [7, Section 1] and [54, Section 2.2]. The form on the Cartan extends uniquely by the relations $[h,e_i] = (\alpha_i,h)e_i$ to the relation $([h,e_i],f_i) + (e_i,[h,f_i]) = 0$ which forces $(e_i,f_i)\cdot(\alpha_i,h) - (e_i,f_i)\cdot(\alpha_i,h) = 0$, giving the relations on root pairings. □

Weyl-equivariance of root spaces

Lemma 12.4 *(Even Weyl transport of parity).* For every $w \in W^{(2)}(\Lambda^{2,1}_{II})$ and every root $\alpha \in \Delta$, the Weyl operator induces a parity-preserving isomorphism

$$\mathfrak{g}_{\alpha,\overline{p}} \;\simeq\; \mathfrak{g}_{w\alpha,\overline{p}}, \qquad \overline{p} \in \mathbb{Z}/2.$$

Proof. This is the Weyl-transport-of-parity statement of [54, Lemma 3.8 and Proposition 3.7], with the GN super-version of [43, Section 6, Statement 6.4]. Since the GN real simple generators are even, the reflection operators are even automorphisms of $\mathfrak{g}_{\Delta_5}$. Hence Weyl transport preserves the parity decomposition. □

13 Comparison with the chiral shadow lattice Γ_{ind}

The Gram-index parametrization $\Gamma_{\mathrm{ind}} = \mathbb{Z}^3$ of $\Lambda^{2,1}_{II}$ via $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$ provides the comparison between the BKM denominator algebra and the K3 half-index $\phi_{0,1}$. The Gram-index pairing on Γ_{ind} is

$$\langle (n,l,m),(n',l',m')\rangle_{\mathrm{ind}} \;=\; 2(nm' + n'm) - ll',$$

and the pullback to the Lorentzian pairing on $\Lambda^{2,1}_{II}$ is (4.3).

Remark 13.1 *(The denominator polarization vs. the cusp polarization).* The Gram-index lattice carries two polarizations:

- the cusp polarization, in which $\Gamma_{\mathrm{eff}} \subset \Gamma_{\mathrm{ind}}$ is the effective subset (the active support of the K3 half-index in the Borcherds product) and the order is the Fock polarization at one cusp;
- the denominator polarization, in which Γ_{ind} maps via α to $\Lambda^{2,1}_{II}$ and the order is the type-II chamber $\mathcal{P}_{II}$.

The two polarizations are the same lattice with two ordered charts; the denominator side is the Borcherds chamber with $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$ symmetry, the cusp side is the Fock determinant at the chosen cusp. A duality element transports between the two; cf. Section 1.4 of Chapter 2.

Visible coefficients

LEMMA 13.2 *(Active multiplicity equals $f(nm, l)$).* On the active support $\Gamma_{\mathrm{act}} \subset \Gamma_{\mathrm{eff}}$,

$$\mathrm{smult}(\alpha(n, l, m)) = f(nm, l).$$

For triples $(n, l, m) \in \Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$, the product exponent vanishes and no visible product factor is contributed; the BKM root space $\mathfrak{g}_{\alpha(n,l,m)}$ is not pinned by the Borcherds product alone.

Proof. The first equality is (4.14), comparing the Borcherds product expansion with the Weyl–Kac–Borcherds denominator. The second statement records that exponent zero on a triple (n, l, m) does not exclude the existence of a root space with smult = 0; the Borcherds product expansion does not see zero-superdimension root spaces. □

What the determinant determines

PROPOSITION 13.3 *(Information content of the denominator product).* The denominator identity

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$$

determines the signed integers smult(α) on the active support after the BKM root system $\mathcal{R}_+$ has been fixed. It does not determine:

- invisible zero-superdimension root spaces;
- the parity decomposition $\dim \mathfrak{g}_{\alpha,\bar{0}}$, $\dim \mathfrak{g}_{\alpha,\bar{1}}$ beyond their difference;
- the structure constants of the BKM bracket $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \to \mathfrak{g}_{\alpha+\beta}$;
- simple primitive representatives, no-extra-relations theorems, or completed PBW compatibility;
- the closed radical of the positive-negative invariant pairing.

Consequently the denominator product is a necessary input for primitive recognition; it is not by itself a recognition datum.

Proof. Write $X^\alpha = e^{-2\pi i(\alpha,z)}$. In the completed positive root monoid algebra,

$$\log \prod_{\alpha\in\mathcal{R}_+} (1 - X^\alpha)^{\mathrm{smult}(\alpha)} = -\sum_{\alpha\in\mathcal{R}_+} \sum_{k\geq 1} \frac{\mathrm{smult}(\alpha)}{k} X^{k\alpha}.$$

The exponents smult(α) are recovered recursively by height, equivalently by Mobius inversion on each primitive ray. The recovered exponent is only $\mathrm{smult}(\alpha) = \mathrm{mult}_{\bar{0}}(\alpha) - \mathrm{mult}_{\bar{1}}(\alpha)$. The difference $a - b$ does not determine the pair (a, b). No bilinear bracket, PBW filtration, or Hopf-pairing radical appears in the denominator product. □

What the determinant forgets

PROPOSITION 13.4 *(Grothendieck class as the determinant invariant).* For every Γ_{eff}-graded super vector space $\mathcal{B} = \bigoplus_\gamma \mathcal{B}_\gamma$ with $\mathrm{sdim}\, \mathcal{B}_{(n,l,m)} = f(nm, l)$, the Fock determinant is

$$64q^{1/2}r^{1/2}s^{1/2} \prod_\gamma (1 - x^\gamma)^{\mathrm{sdim}\, \mathcal{B}_\gamma} = \Delta_5(Z).$$

The determinant remembers only the class of $\mathcal{B}$ in

$$K_0(\Gamma_{\mathrm{eff}}\text{-graded finite super vector spaces}).$$

A bracket $[\mathcal{B}_\gamma, \mathcal{B}_{\gamma'}] \to \mathcal{B}_{\gamma+\gamma'}$ does not appear in the determinant formula, nor do simple primitive representatives, parity dimensions, or pairing radicals.

Proof. The Fock determinant lemma gives $\mathrm{sdet}_{\mathcal{B}}(1-x) = \prod_\gamma (1-x^\gamma)^{\mathrm{sdim}\,\mathcal{B}_\gamma}$. Multiplying by the vacuum factor and using the hypothesis $\mathrm{sdim}\,\mathcal{B}_{(n,l,m)} = f(nm,l)$ identifies this product with the Borcherds product for $64^{-1}\Delta_5$. Replacing $\mathcal{B}$ by any other graded super vector space with the same Grothendieck class preserves the determinant; the zero bracket on such a space gives the same determinant and a non-isomorphic Lie algebra; adjoining a cancelling pair $W_\gamma \oplus \Pi W_\gamma$ in any active degree leaves the determinant unchanged while changing the parity dimensions. Hence the determinant invariant is the K_0-class. □

14 Comparison: rank-three hyperbolic vs. Borcherds correction

The denominator algebra $\mathfrak{g}_{\Delta_5}$ is the BKM correction of the rank-three hyperbolic Kac–Moody algebra $\mathfrak{g}(A)$ on the same Cartan matrix A. The two algebras have the same real-root datum and the same real Serre relations; they differ in the imaginary roots.

Table 4.1: Real-root datum of $\mathfrak{g}_{\Delta_5}$ and the rank-three hyperbolic $\mathfrak{g}(A)$.

	$\mathfrak{g}(A)$, $A = \left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$	$\mathfrak{g}_{\Delta_5}$
real simple roots	$\delta_1, \delta_2, \delta_3$	$\delta_1, \delta_2, \delta_3$
real-root mult.	1 each	1 each
real Serre exponent	3	3
imaginary simple roots	none	$\Lambda^{2,1}_{II} \cap \mathbb{R}_{>0}\mathcal{P}_{II}$
$\mu_{\bar{0}}(a)$	0	$\max(m(a), 0)$ or $\max(\tau(a), 0)$
$\mu_{\bar{1}}(a)$	0	$\max(-m(a), 0)$ or $\max(-\tau(a), 0)$
denominator	no closed product	$64^{-1}\Delta_5(2Z)$

Remark 14.1 *(Borcherds correction adjoins the imaginary simple roots).* The Gritsenko–Nikulin construction [43, Sections 3–4, Proposition 3.1] adjoins to $\mathfrak{g}(A)$ the imaginary simple generators with multiplicities $\mu_{\bar{p}}(a)$ from the Borcherds-product expansion of the additive lift of $\eta^9\vartheta$. The resulting BKM Lie superalgebra has denominator $64^{-1}\Delta_5(2Z)$, while $\mathfrak{g}(A)$ has no closed denominator product as a Kac–Moody algebra alone (its denominator is a formal power series whose closed form is unknown). The correction algebra is the central object of View II.

15 The character of the chiral shadow

The denominator algebra $\mathfrak{g}_{\Delta_5}$ is the Beilinson cut's chiral shadow at View II. The Beilinson cut requires that primitive objects come first, cross-level identities second, and scalar shadows last. Inscribed at View II:

- *Primitive object* (View III, relative to retained data): the supplied finite E_3-prefactorization input of the K3×E formal target, presented at finite stage as the pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ of Chapter 6.
- *Cross-level identity* (View II): the Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, with generators and relations as recorded above.
- *Automorphic determinant* (View I, K_0-determinant): the automorphic section $\mathcal{D}_X = \Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$, with squared determinant section $\Phi^{\mathrm{un}}_{10} = \Delta_5^2$ at weight 10.

The denominator algebra is target-side: it imports the Gritsenko–Nikulin Cartan, the Chevalley–Borcherds–Kac relations, the Weyl vector, and the Borcherds product, and supplies no compact Hall correspondence, no orientation line, no Pfaffian section, and no microscopic state space. The Pfaffian and orientation part of the cross-level identity View II ↔ View III is the Pfaffian–Dirac theorem after $(D0)$, $(O1)$, $(O1)^+$, and the retained $(O2_{\mathrm{atlas}})$ datum, together with the finite Pfaffian section datum, of Chapter 6. The primitive identification $P_X \simeq \mathfrak{g}_{\Delta_5}$ has the additional finite compact-source recognition hypothesis $(S1)$–$(S10)$.

Remark 15.1 *(The chiral shadow is target, not bulk).* The Beilinson cut classifies $\mathfrak{g}_{\Delta_5}$ as a chiral shadow, not a chiral bulk. The bulk object lives on the source side: at a finite retained stage the chiral bulk is $Z^{\mathrm{der}}_{\mathrm{ch}}(C_{X,R}) \simeq \mathrm{ChirHoch}^\bullet(C_{X,R}, C_{X,R})$, identified on the Koszul locus by Vol I Theorem H [67] together with Vol II's chiral-Hochschild Beilinson stratification (i)+(ii) [68], attached to a primitive $C^{op} = C^{op}_{X,R}$ on the closed factorization datum (X, D, τ) at the boundary vacuum b. Here $X = K3 \times E$ is the formal target, D the chosen boundary divisor at infinity carrying the cusp polarization, and $\tau \in H^*(D, \mathbb{Q})$ the boundary trace class supplying the factorization gluing — the data of a closed chiral C^{op} in the sense of [68, §2]. The denominator algebra $\mathfrak{g}_{\Delta_5}$ is the chiral shadow of this bulk, not the bulk itself; the bulk object lives one categorical level above the chiral shadow, and the BKM denominator records only the shadow.

Remark 15.2 *(Beilinson chain on $K3 \times E$).* The Universal Holography master theorem of Vol II [68, § 9] identifies, on a closed factorisation datum (X, D, τ) at boundary vacuum b:

$$\text{boundary} = A_b, \qquad \text{bulk} = Z^{\mathrm{der}}_{\mathrm{ch}}(A_b), \qquad \text{interaction} = \mathrm{SC}^{\mathrm{ch,top}}\text{-brace}.$$

On $K3 \times E$, the compact-source side supplies the *boundary chart* $A_b = \mathrm{End}_{\mathcal{C}}(b)$ at the boundary vacuum b, with $\mathcal{C}$ the open factorisation dg-category of Chapter 6, the augmented bar coalgebra $\mathrm{Bar}(A_b)$ and its MC-twisting structure of [68, §6], and the chiral shadow $\mathfrak{g}_{\Delta_5} = \mathrm{shadow}(Z^{\mathrm{der}}_{\mathrm{ch}}(A_b))$ of the bulk algebra at the BKM level — but *not* the bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$ itself as a chiral algebra on $K3 \times E$ with explicit OPE, which is the level-Z object of Vol II. The chain-level trace identification of Theorem 7.22 supplies the level-Z criterion (Definition 7.19) relative to (Z_{HB}), Vol I Theorem H, and Vol II's chiral-Hochschild Beilinson stratification; the level-A gravity-line operator-algebra facet remains open in Vol II. Equivalently, the Beilinson cut $\mathsf{P} \longrightarrow \mathsf{C} \longrightarrow \mathsf{S} \longrightarrow \mathsf{Z} \longrightarrow \mathsf{A}$ is exhibited at P,C,S as chiral-shadow constructions of Chapters 4 and 6, at $\mathsf{S} \longrightarrow \mathsf{Z}$ as a retained trace identity on the chain-level chiral derived centre (Theorem 7.22), and at $\mathsf{Z} \longrightarrow \mathsf{A}$ as the open gravity-line Hall–Borcherds residual. The K3×E threefold realisation question of Chapter 9 is the level-A open problem.

16 THE DENOMINATOR ALGEBRA OF THE CHIRAL SOURCE-TARGET FRAME

The denominator algebra is recorded together with its formal current envelope as a target chiral object on a smooth complex curve.

Definition 16.1 *(Formal current envelope).* For a smooth complex curve C, set

$$\mathrm{Cur}_C(\mathfrak{g}_{\Delta_5}) \;:=\; \mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_C$$

with the level-zero current bracket

$$[a \otimes f, b \otimes g] \;=\; [a, b] \otimes fg.$$

The chiral enveloping algebra of Beilinson–Drinfeld [6] is

$$\mathsf{Den}_{\Delta_5,C} \;:=\; U_C^{\mathrm{ch}}\big(\mathrm{Cur}_C(\mathfrak{g}_{\Delta_5})\big).$$

Proposition 16.2 *(Formal current envelope realizes the denominator).* The chiral algebra $\mathsf{Den}_{\Delta_5,C}$ is the formal completed ind Beilinson–Drinfeld current envelope on C obtained from the denominator Lie superalgebra. Its constant-current Lie superalgebra is $\mathfrak{g}_{\Delta_5}$, and the denominator of that constant-current bracket is $64^{-1}\Delta_5(2Z)$.

For $C = E$, the elliptic curve in $X = K3 \times E$, the chiral algebra $\mathsf{Den}_{\Delta_5,E}$ is the target chiral object used in the $K3 \times E$ realization problem of Chapter 6. A compact E_3/Hall/chiral protected realization requires the retained Dirac–Igusa pro-object data, the Hall correspondences, the orientation line, and the Koszul comparison.

Proof. In the completed ind setting, the current construction sends a Lie superalgebra to a Lie-* algebra on a smooth curve, and the chiral enveloping construction sends a Lie-* algebra to a chiral algebra [6]. Equivalently, in vertex-algebra language, this is the universal current vertex algebra at level zero [53]. Constant sections of $\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E$ recover the original Lie superalgebra, with denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ from Theorem 9.1. □

17 The closing identity and its conditioning

The denominator identity is

$$\boxed{\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z), \qquad \kappa_{\mathrm{BKM}}(\Delta_5) \;=\; 5, \qquad \mathrm{smult}(\alpha(n,l,m)) \;=\; f(nm,l).}$$

These three are imported theorems: the first from Borcherds 1995 [8] and Gritsenko–Nikulin 1998 [45] [45, Theorem 2.1, (2.1)–(2.3)], with the algebra constructed in [43, Sections 3–4, Proposition 3.1]; the second from Borcherds 1998 [9, Theorem 13.3] with the Vol III specialization to the paramodular family; the third from the same Borcherds product expansion as the first, by comparison with the Weyl–Kac–Borcherds denominator. The conditioning of the closing identity is target-side. No compact Hall correspondence, Pfaffian section, or microscopic state space is claimed.

The cross-level identity View II ↔ View III, the Pfaffian–Dirac theorem of Chapter 7, asks for the protected Pfaffian

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) \;=\; \Delta_5,$$

the orientation character $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$, and the primitive recognition theorem $P_X \simeq \mathfrak{g}_{\Delta_5}$, conditional on the finite compact-source recognition datum $(S1)$–$(S10)$ in addition to the four orientation and Pfaffian obstructions. The BKM denominator identity is independent of those four obstructions and is imported from primary literature.

Cited identities

The denominator calculation uses only proved identities. The type-II Cartan matrix of Lemma 2.2 and the Weyl vector $\rho = f_2 - \frac{1}{2}f_3 + f_{-2}$ of Proposition 2.4 are direct lattice computations. The reflection-group decomposition is Lemma 1.3, with Nikulin's theorem as input [82]. Real-root multiplicity one, Lemma 4.6, follows from the Kac–Moody root-space theorem and the Gritsenko–Nikulin normalization [43, 54]. The imaginary simple-root parity split is Definition 3.2, read from [43, Section 3]. The Maass multiplier values and their restriction to the Weyl group are Lemmas 7.1 and 7.2, using [43, 71] and

[43, Prop. 2.1, 2.2]. The Borcherds product for D_5 is Theorem 5.3, from [9, Thm. 13.3] and [45, Thm. 2.1]. The Weyl alternating sum is Theorem 5.1, from [43, Thm. 2.1]. The denominator identity (4.13) is the Gritsenko–Nikulin denominator theorem [43, Sections 3–4], [45, Thm. 2.1]. The Borcherds-weight value $\kappa_{\mathrm{BKM}} = 5$ is Theorem 8.1, from Borcherds' weight formula [9, Thm. 13.3]. The 29|93 presentation count of Proposition 6.7 is the direct finite count against the Gritsenko–Nikulin simple-root lists. The triangular decomposition and invariant bilinear form are Lemmas 4.4 and 12.3, by the general Borcherds–Kac–Moody theory [7, 54].

The closing identity is therefore proved as recorded. Its Pfaffian realization is the theorem of Chapter 7; its primitive compact-source realization remains conditional on $(S1)$–$(S10)$.

18 Arithmetic consistency of the denominator identity

The denominator-algebra theorem is constrained by the following computations; each one is fixed by the recorded primary sources and the Borcherds-weight identity.

Sign of imaginary multiplicities

The Borcherds product exponent $f(nm, l)$ on the active support determines the signed BKM multiplicity. The invariant smult is defined as the difference $\mathrm{mult}_{\bar{0}} - \mathrm{mult}_{\bar{1}}$ (Definition 4.5), and the comparison of the Borcherds product (4.10) with the Weyl–Kac–Borcherds denominator product (4.12) gives $\mathrm{smult}(\alpha(n, l, m)) = f(nm, l)$ on Γ_{act}. The parity split of smult into $(\mathrm{mult}_{\bar{0}}, \mathrm{mult}_{\bar{1}})$ requires the Gritsenko–Nikulin imaginary simple-root data $\mu_{\overline{p}}(a)$, recorded in Definition 3.2, and the Borcherds–Kac bracket calculus in the degree under consideration. The data $m(a)$ and $\tau(a)$ give the signed simple fibres; lower brackets supply the remaining summands of the full root space. Compare [43, Section 3].

Doubled vs. undoubled Δ_5

The denominator identity uses $\Delta_5(2Z)$. The additive Gritsenko lift produces the monomial product for $D_5(Z)$, and the root map $\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$ lands in $\Lambda_{II}^{2,1}$. The substitution $Z \mapsto 2Z$ changes $\exp(-\pi i(\alpha, z))$ into the standard denominator character $\exp(-2\pi i(\alpha, z))$; no equality of the ambient and type-II lattices is asserted. The prefactor $64 = [q^{1/2}r^{1/2}s^{1/2}]\Delta_5$ is the theta-constant normalization and is untouched by doubling. See Remark 5.5.

Doubling factor as power of two

The factor 64 is the value of the leading Fourier coefficient $c_\Delta(1, 1, 1)$ of Δ_5, computed in the Maass formula for the product of ten even theta constants [43, (1.3)–(1.5)]. It is not a half-Hilbert orbit count. The orbit-count factor $2^{12} = 4096$ arises in the type-II Pfaffian wall normal form $(O2)$ of Chapter 6 as the size of the half-Hilbert orbit on the local sign atlas; it is unrelated to the denominator prefactor 64. See Remark 5.6.

Simple-root system on $\Lambda_{II}^{2,1}$

Nikulin's reflection-group theorem [82] fixes $W^{(2)}(\Lambda_{II}^{2,1})$ as a normal subgroup of index 6 in $\mathbf{O}(\Lambda^{2,1})_+$, and the Gritsenko–Nikulin chamber $\mathcal{P}_{II}$ has fixed walls δ_1, δ_2, δ_3 with Gram matrix (4.6). Any simple-root triple for the same type-II chamber is Weyl-conjugate to a permutation of $(\delta_1, \delta_2, \delta_3)$. The BKM algebra depends on the chamber Cartan datum, not on a labelling of its three walls.

Weyl vector ρ

The Weyl vector is fixed by the condition $(\rho, \delta_i) = -(\delta_i, \delta_i)/2 = -1$ for $i = 1, 2, 3$. The displayed value $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$ satisfies this and lies in $(\Lambda^{2,1}_{II})^* \cap C(\Lambda^{2,1})_+$; $\rho' = \delta_1 + \delta_2 + \delta_3$ would have $(\rho', \delta_i) = -2$, violating the condition. The factor $\frac{1}{2}$ is forced. See Proposition 2.4.

Real-root norm convention

The lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ is even by (4.1). The type-II sublattice $\Lambda^{2,1}_{II}$ is also even (a sublattice of an even lattice). Hence $(\delta, \delta) \in 2\mathbb{Z}$ for every $\delta \in \Lambda^{2,1}_{II}$. The choice $(\delta, \delta) = 2$ is forced for primitive square-norm-2 vectors. The convention $(\delta_i, \delta_i) = 1$ does not occur.

Second Serre bracket compatibility (Gram cocycle)

The Gram cocycle $B(c, c')$ is the source-side quadratic correction on the upstairs lattice; its compatibility with the BKM bracket on $\Lambda^{2,1}_{II}$ is mediated by the normal-ordered homomorphism $\overline{\Pi}_X : \widehat{\Gamma}_X \to \Gamma_{\mathrm{gram}}$ of Chapter 2, which corrects the raw quadratic shadow $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$ to the additive map. The target-side Lorentzian Cartan pairing on $\Lambda^{2,1}_{II}$ needs no Gram cocycle correction. The compatibility is precisely between $\overline{\Pi}_X$ on the source and the BKM grading on the target, as established in Chapter 2.

Alternative Cartan-matrix choice

The symmetrized matrix $\left(\begin{smallmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{smallmatrix}\right)$ is not the Cartan datum of the denominator algebra. The matrix A in (4.6) has integer entries with $A_{ii} = 2$, and is the symmetrized form fixed by the even-lattice convention. A rescaled matrix $\frac{1}{2}A$ with $A_{ii} = 1$ would not be a generalized Cartan matrix in the Kac–Moody sense for the real simple roots, where the standard normalization is $A_{ii} = 2$ [54, Chapter 1]. The symmetric form A is the normalization compatible with the even lattice $\Lambda^{2,1}_{II}$.

The two values of κ_{BKM}

The values $\kappa_{\mathrm{BKM}} = 5$ and $\kappa_{\mathrm{BKM}} = 12$ index different Borcherds inputs to the same universal rule $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ of [9, Theorem 13.3]. The Δ_5 row evaluates the rule at the paramodular row with Jacobi seed $\phi_{0,1}$ and Borcherds product Δ_5 on $\Lambda^{2,1}_{II}$, with $c_1(0) = 10$ and $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$. The Vol I row evaluates the rule at the Fake-Monster row Φ_{12} on $\mathrm{II}_{25,1}$ [7, 8], with $c_{\Phi_{12}}(0) = 24$ and $\kappa_{\mathrm{BKM}}(\Phi_{12}) = 12$. Each site must name its input form and lattice; the values 5 and 12 are independent evaluations, not a single shifted quantity. See Remark 8.3 and the Vol III universal-Borcherds-weight identity (Theorem 8.1).

The determinant is a Grothendieck-class invariant

A graded super vector space with the same signed multiplicities and zero bracket gives the same denominator product but is not a Lie superalgebra. The denominator identity fixes only the signed Grothendieck class (Proposition 13.4). The bracket, the parity split, the radical, the simple primitives, and the no-extra-relations theorem are required for a recognition theorem; all of these are read on the BKM presentation of Definition 4.2, not from the denominator product alone. The recognition target $P_X \simeq \mathfrak{g}_{\Delta_5}$ requires the compact-source data of Chapter 7.

19 $\mathfrak{g}_{\Delta_5}$ AS ONE OF FIVE VIEWS

The denominator algebra $\mathfrak{g}_{\Delta_5}$ is one of five views of Δ_5: View I (automorphic section), View II (Borcherds–Kac–Moody denominator), View III (compact protected Pfaffian, conditional), View IV (mirror discriminant, conjectural), View V (singular theta lift). The cross-level identities are:

- View I ↔ View II: theorem, [8, 45].
- View II ↔ View V: theorem, [9].
- View I ↔ View V: theorem, [9, 43].
- View II ↔ View III: Pfaffian and orientation theorem on $K3 \times E$, Chapter 7; primitive recognition conditional on $(S1)$–$(S10)$.
- View II ↔ View IV: *conjectural*, Chapter 8.

The denominator identity is

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; 64^{-1}\Delta_5(2Z),$$

the Borcherds-weight characteristic

$$\varkappa_{\mathrm{BKM}}(\Delta_5) \;=\; 5,$$

and the active multiplicity formula

$$\mathrm{smult}(\alpha(n,l,m)) \;=\; f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{act}}).$$

These three identities are the target-side data of View II.

Chapter 5

The singular theta lift

1 The singular theta lift of $\phi_{0,1}$

The automorphic, Borcherds–Kac–Moody, and singular-theta descriptions of Δ_5 are theorems of Borcherds, Gritsenko, and Nikulin. The Pfaffian description is relative to the retained Dirac–Igusa data, and the mirror-discriminant description is conjectural. The exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ identifies the genus-two Siegel upper half-space and the type-IV symmetric domain attached to the signature-$(3,2)$ lattice into one homogeneous space; the singular theta integral of [8, 9] sends the Weil-representation theta component $\phi_{0,1}^{\mathrm{Weil}}$ of $\phi_{0,1}$ to the weight-five Borcherds product Δ_5, and the Gritsenko–Nikulin denominator identity then reads the lift as the BKM denominator on $\Lambda_{II}^{2,1}$. These identifications are theorems of [8, 9, 43, 45]; the integral lattice, arithmetic subgroup, and multiplier convention are fixed by the following normalization.

1.1 *Notation and the canonical name*

Three Borcherds-product handles for the weight-five Igusa cusp form are distinguished. The *automorphic section* $\mathcal{D}_X = \Delta_5$ of the line $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$ is the View-I name. The *BKM denominator* $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ on $\Lambda_{II}^{2,1}$ is the View-II name. The *protected Pfaffian* $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$, defined on $K3 \times E$ relative to the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, $(O2_{\mathrm{atlas}})$, and the finite Pfaffian section datum, is the View-III name. The View-V canonical name is

$$\Delta_5 \;=\; \Theta\big(\phi_{0,1}^{\mathrm{Weil}}\big),$$

the *singular theta lift of the Weil component* $\phi_{0,1}^{\mathrm{Weil}}$ obtained from $\phi_{0,1}$ by theta decomposition. Throughout $\phi_{0,1} \in J_{0,1}^{\mathrm{weak}}$ is the unique up to scale weight-zero index-one weak Jacobi form on the Jacobi group $\Gamma_1^J = \mathbf{SL}_2(\mathbb{Z}) \ltimes \mathbb{Z}^2$ with constant Fourier coefficient fixed by the K3 elliptic genus normalization $Z_{\mathrm{K3}} = 2\phi_{0,1}$ of [32, §2.2]; $\phi_{0,1}^{\mathrm{Weil}}$ is its weight-$-1/2$ Weil-representation theta component, and Θ is the Borcherds singular-theta integral of [8, §6], [9, §6] with the Igusa cusp trivialization $C_{\phi_{0,1}} = 64$. Its monic Borcherds product is $D_5 = 64^{-1}\Delta_5$. The bare handle Δ_5 is reserved for cross-view identifications.

1.2 *The weak Jacobi form $\phi_{0,1}$*

Definition 1.1 *(Weak Jacobi form, weight zero, index one).* A weak Jacobi form of weight k and index m on the Jacobi group $\Gamma_1^J = \mathbf{SL}_2(\mathbb{Z}) \ltimes \mathbb{Z}^2$ is a holomorphic function $\phi : \mathbb{H} \times \mathbb{C} \to \mathbb{C}$ satisfying, for $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathbf{SL}_2(\mathbb{Z})$ and $(\lambda, \mu) \in \mathbb{Z}^2$,

$$\phi\!\left(\frac{a\tau+b}{c\tau+d},\, \frac{z}{c\tau+d}\right) = (c\tau+d)^k e^{2\pi i m c z^2/(c\tau+d)}\phi(\tau, z),$$
$$\phi(\tau,\, z+\lambda\tau+\mu) = e^{-2\pi i m(\lambda^2\tau+2\lambda z)}\phi(\tau, z),$$

together with the cusp expansion

$$\phi(\tau,z)=\sum_{n\geq 0,\,l\in\mathbb{Z}} f(n,l)\,q^n r^l,\qquad q=e^{2\pi i\tau},\ r=e^{2\pi i z},$$

in which the coefficient depends only on the pair $(4mn-l^2,\ l \bmod 2m)$ (Eichler–Zagier [32, Theorem 2.2]). For $m=1$ the residue class is fixed by the parity of $4n-l^2$. The *weight-zero index-one* weak Jacobi form $\phi_{0,1}$ is fixed by $k=0$, $m=1$, the discriminant condition $4n-l^2\geq -1$ and the normalization

$$\phi_{0,1}(\tau,z)=r^{-1}+10+r+O(q).$$

The space $J_{0,1}^{\mathrm{weak}}$ is one-dimensional over $\mathbb{C}$ (Eichler–Zagier [32, Theorem 9.4 and Table 9, §9]); $\phi_{0,1}$ generates it. The theta-quotient presentation is the input to the singular-theta integral below.

Lemma 1.2 *(Eichler–Zagier theta-quotient presentation).* The weak Jacobi form $\phi_{0,1}$ admits the closed form

$$\phi_{0,1}(\tau,z)=4\sum_{i=2}^{4}\frac{\theta_i(\tau,z)^2}{\theta_i(\tau,0)^2},$$

where $\theta_2,\theta_3,\theta_4$ are the three even Jacobi theta functions in the Mumford normalization (Eichler–Zagier [32, §9, equation (9.4)]); equivalently $\phi_{0,1}=Z_{\mathrm{K3}}/2$ is one half of the K3 elliptic genus [28]. The leading Fourier coefficients are

$$f(0,0)=10,\quad f(0,\pm1)=1,\quad f(1,0)=108,\quad f(1,\pm1)=-64,\quad f(1,\pm2)=10,$$
$$f(2,0)=808,\quad f(2,\pm1)=-513,\quad f(2,\pm2)=108,\quad f(2,\pm3)=1,$$

in agreement with the Mukai–Gram normalization of Appendix 3. The discriminant relation $f(n,l)=f(n',l')$ when $4n-l^2=4n'-(l')^2$ (Eichler–Zagier [32, Theorem 2.2]) is the index-one specialization of the $(4mn-l^2, l \bmod 2m)$ rule and organises these coefficients into the equivalence classes

$$\{f(1,0),f(2,\pm2)\}=108\,(\mathrm{disc}\ 4),\quad \{f(0,0),f(1,\pm2)\}=10\,(\mathrm{disc}\ 0),\quad \{f(0,\pm1),f(2,\pm3)\}=1\,(\mathrm{disc}\ -1),$$

each independently consistent with the verified expansion.

Proof. The closed form is standard in Eichler–Zagier [32, §§9–10]; the K3 elliptic-genus identity $Z_{\mathrm{K3}}=2\phi_{0,1}$ is [32, §2.2] together with the sigma-model right-moving Witten index of [28]. The constant term $f(0,0)=10$ is the right-moving-localized character of the K3 Ramond sector at $L_0=c/24$; the remaining values follow from the discriminant relation $f(n,l)=f(n',l')$ when $4n-l^2=4n'-(l')^2$ (Eichler–Zagier [32, Theorem 2.2]) together with explicit theta-quotient evaluation, which fixes the disc-3 class ($f(1,\pm1)=-64$) and the disc-4 class ($f(1,0)=f(2,\pm2)=108$) by direct computation from the theta-quotient of Lemma 1.2. The squared theta-leading coefficient $([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2=64^2=4096$ of the Borcherds-product expansion in §3 is the next arithmetic check on these coefficients via the cusp-chamber product $\Delta_5(Z)=64q^{1/2}r^{1/2}s^{1/2}\prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1-q^nr^ls^m)^{f(nm,l)}$. □

1.3 The signature-(3, 2) lattice and the type-IV domain

Definition 1.3 *(The signature-(3, 2) lattice).* The lattice $\Lambda^{3,2}$ is the orthogonal complement of the alternating Pfaffian element $q_0=e_1\wedge e_3+e_2\wedge e_4\in\wedge^2\Lambda^4$ inside the signature-(3, 3) lattice $\left(\wedge^2\Lambda^4,\ (\ ,\ )\right)\simeq \mathrm{II}_{3,3}$, where $\Lambda^4=\bigoplus_{i=1}^4\mathbb{Z}e_i$ and the bilinear form (u,v) is the Pfaffian pairing $u\wedge v=(u,v)\,e_1\wedge e_2\wedge e_3\wedge e_4$. The vector q_0 has square -2. In the BKM sign convention it is the even discriminant-two lattice

$$\Lambda^{3,2}\ \simeq\ \Lambda^{(1,1)}\oplus\Lambda^{(1,1)}\oplus[2],\qquad \Lambda^{(1,1)}=\left(\begin{smallmatrix}0&-1\\-1&0\end{smallmatrix}\right),$$

written in the basis $(f_1,f_2,f_3,f_{-2},f_{-1})$ of the exterior-square model section.

Remark 1.4 *(Signature conventions).* The lattice $\Lambda^{3,2}$ carries three positive and two negative directions in the (u,v)-pairing of Definition 1.3. The *primitive hyperbolic sublattice* $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ on which the BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ is graded has signature $(2,1)$ in the fixed convention: the $[2]$ factor and one hyperbolic direction are positive, and the other hyperbolic direction is negative. The decomposition $\Lambda^{3,2} = \Lambda^{2,1} \oplus \Lambda^{(1,1)}$ splits an extra hyperbolic plane on which the singular theta integral lives; this is the $\mathrm{II}_{2,2}$-fibre at the standard cusp under the exceptional isomorphism of §1.4. No $\Lambda^{2,1} \oplus \langle -2 \rangle$ presentation is invoked; the fixed convention is the $\Lambda^{(1,1)} \oplus [2]$ Nikulin form.

Let

$$\Gamma_{3,2} := \wedge^2 \mathbf{Sp}_4(\mathbb{Z})/\{\pm \mathbf{I}_4\} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+$$

be the projective exterior-square arithmetic group. The type-IV symmetric domain attached to $\Lambda^{3,2}$ is

$$\mathbb{H}^{\mathbf{IV}} = \left\{ Z \in \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C}) \;\middle|\; (Z,Z) = 0,\ (Z,\overline{Z}) < 0 \right\},$$

with positive component $\mathbb{H}^{\mathbf{IV}}_+$ singled out by $\mathrm{Im} z_1 > 0$ in the parametrization ${}^t(z_2^2 - z_1 z_3, z_3, z_2, z_1, 1) z_0$ of the exterior-square model section. Its arithmetic quotient $\Gamma_{3,2} \backslash \mathbb{H}^{\mathbf{IV}}_+$ is the moduli space on which the Borcherds singular-theta lift of [8] produces automorphic forms with prescribed singularities along rational quadratic divisors.

1.4 The exceptional isogeny $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$

THEOREM 1.5 *(Exterior-square exceptional isomorphism).* The exterior-square representation $\wedge^2 : \mathbf{SL}_4 \to \mathrm{Aut}(\wedge^2 \Lambda^4, (\ ,\ ))$ restricts to a homomorphism

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm \mathbf{I}_4\} \xrightarrow{\ \simeq\ } \Gamma_{3,2},$$

realizing the exceptional isomorphism of adjoint real Lie groups $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$, together with its spin lift on the covering groups. Under this identification the genus-two Siegel upper half-space $\mathbb{H}_2$ embeds onto the positive component $\mathbb{H}^{\mathbf{IV}}_+$ by

$$\left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right) \longmapsto {}^t(z_2^2 - z_1 z_3,\ z_3,\ z_2,\ z_1,\ 1) z_0,$$

intertwining the $\mathbf{Sp}_4(\mathbb{Z})$ and $\Gamma_{3,2}$ actions.

Proof. The Pfaffian form $u \wedge v = (u,v) e_1 \wedge e_2 \wedge e_3 \wedge e_4$ is the canonical $\mathbf{SL}_4$-invariant symmetric bilinear form on $\wedge^2 \Lambda^4$; the subgroup of $\mathbf{SL}_4$ fixing $q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$ is the symplectic group $\mathbf{Sp}_4$ of the alternating form B_{q_0}. The orthogonal complement $\Lambda^{3,2} = q_0^\perp$ is preserved, and the kernel of $\wedge^2|_{\mathbf{Sp}_4}$ on $q_0^\perp$ is $\{\pm \mathbf{I}_4\}$ (the exterior-square model section). This is the integral form of the classical exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ of [56, Chapter II.6]; the spin cover supplies the corresponding lift with Lie algebra $\mathfrak{sp}_4(\mathbb{R})$. This cover is distinct from the metaplectic cover on the $\mathbf{Mp}_2(\mathbb{Z})$ input. The displayed substitution into the type-IV domain produces the commutative square of the exterior-square model section. The integral-Igusa form of the identification $\mathbb{H}_2 \simeq \mathbb{H}^{\mathbf{IV}}_+$ is the content of [48, §2]; see also [34, Chapter II] and [96, §6]. □

Remark 1.6 *(Three guises of one homogeneous space).* The same homogeneous space carries three names: the genus-two Siegel domain $\mathbf{Sp}_4(\mathbb{R})/\mathbf{U}(2)$, the type-IV symmetric domain $\mathbf{SO}(3,2)_+/(\mathbf{SO}(3) \times \mathbf{SO}(2))$, and the spin double cover $\mathbf{Spin}(3,2)/(\mathbf{U}(2)\text{-cover})$. The regularized theta integral below uses the metaplectic cover $\mathbf{Mp}_2(\mathbb{Z})$ on the input variable τ through the Weil representation of the discriminant form of $\Lambda^{3,2}$. The integrality of $\mathrm{wt}(\Delta_5) = 5$ places the resulting weight on the ordinary automorphic line; the multiplier system ν_{Δ_5} records the residual reflection character on $W^{(2)}(\Lambda^{2,1}_{II})$ (Lemma 7.2).

1.5 The Borcherds theta correspondence and the exceptional isomorphism

Theorem 1.7 *(Regularized theta correspondence on* $\Lambda^{3,2}$*).* Let $L = \Lambda^{3,2}$. Borcherds's regularized theta integral pairs a weakly holomorphic form $\phi \in M^!_{-1/2}(\rho_L)$ for $\mathbf{Mp}_2(\mathbb{Z})$ with the non-holomorphic Siegel theta kernel of L, and produces an automorphic form on $\widetilde{\mathbf{O}}(L)_+$ of weight $c_\phi(0,0)/2$. Its rational-quadratic divisor is determined by the negative-index part of ϕ, and its Weyl-chamber product uses the Fourier coefficients of ϕ in all effective product degrees. For $\phi = \phi^{\mathrm{Weil}}_{0,1}$, the pullback along

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm \mathbf{I}_4\} \longrightarrow \Gamma_{3,2} \subset \mathbf{O}(\Lambda^{3,2})$$

is the Igusa cusp form Δ_5.

[proved; [8, 9, 43, 48]]

Remark 1.8 *(The two groups in the lift).* The metaplectic group in the construction is $\mathbf{Mp}_2(\mathbb{Z})$, acting on the input form through the Weil representation $\rho_{\Lambda^{3,2}}$. The genus-two group $\mathbf{Sp}_4$ enters after the orthogonal automorphic form has been constructed, through the exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ and the integral exterior-square map of Theorem 1.5. Thus the theta input, the orthogonal automorphic form, and the Siegel modular form are three successive incarnations of one Borcherds product.

1.6 The Borcherds singular-theta integral

Borcherds's correspondence assigns to a Weil-representation form obtained from a Jacobi input a meromorphic orthogonal modular form by integration against a vector-valued Siegel theta series. The integral is divergent at the lift's input weight; the Borcherds regularization produces a meromorphic Siegel modular form, after the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ pullback, with explicit divisor.

Definition 1.9 *(Siegel theta-series of* $\Lambda^{3,2}$*).* For the lattice $\Lambda^{3,2}$ of Definition 1.3, the Siegel theta-series at $Z \in \mathbb{H}^{\mathbf{IV}}_+$ and $\tau \in \mathbb{H}$ is

$$\Theta_{\Lambda^{3,2}}(\tau, Z) = \sum_{\gamma \in A_{\Lambda^{3,2}}} e_\gamma \sum_{\lambda \in \Lambda^{3,2}+\gamma} \exp\!\big(\pi i\, \overline{\tau}\, (\lambda,\lambda)_{Z^+} + \pi i\, \tau\, (\lambda,\lambda)_{Z^-}\big),$$

where $(\ ,\ )_{Z^\pm}$ are the positive and negative parts of the (u,v)-pairing under the orthogonal decomposition determined by Z (Borcherds [8, §4], [9, §4]). The theta-series is a non-holomorphic modular form of weight $(3/2, 1)$ on $\mathbf{Mp}_2(\mathbb{Z})$ valued in the Weil representation of $\Lambda^{3,2}$ on $\mathbb{C}[A_{\Lambda^{3,2}}]$. The scalar sum over $\Lambda^{3,2}$ is only the $\gamma = 0$ component.

Theorem 1.10 *(Singular theta lift* [8, Theorem 13.3], [9, Theorem 6.2]*).* Let ϕ be a weakly holomorphic vector-valued modular form for the dual Weil representation of $\Lambda^{3,2}$ on $\mathbf{Mp}_2(\mathbb{Z})$ of weight $-1/2$ with integral Fourier coefficients $c_\phi(n,\lambda) \in \mathbb{Z}$ at the cusp. The regularized integral

$$\Theta(\phi)(Z) = \int^{\mathrm{reg}}_{\mathbf{SL}_2(\mathbb{Z})\backslash\mathbb{H}} \big\langle \phi(\tau), \Theta_{\Lambda^{3,2}}(\tau, Z)\big\rangle\, v\, \frac{du\, dv}{v^2}, \qquad \tau = u + iv,$$

in the regularization sense of [9, §6] produces a meromorphic function on $\mathbb{H}^{\mathbf{IV}}_+$ with the following properties.

(1) *Modular invariance.* $\Theta(\phi)$ is an automorphic form of weight $c_\phi(0,0)/2$ on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ with multiplier system determined by the Weil action on $\Lambda^{3,2*}/\Lambda^{3,2}$.

(2) *Borcherds product.* On the cone of definition, $\Theta(\phi)$ admits the Borcherds product expansion

$$\Theta(\phi)(Z) = C_\phi e^{-2\pi i(\rho,Z)} \prod_{\substack{\alpha\in\Lambda^{3,2*}\\ (\alpha,W)>0}} \left(1 - e^{-2\pi i(\alpha,Z)}\right)^{c_\phi(-(\alpha,\alpha)/2,\, \alpha \bmod \Lambda^{3,2})},$$

with $C_\phi \in \mathbb{C}^\times$ fixed by the chosen cusp trivialization, Weyl vector ρ determined by ϕ, and a chamber W of the positive cone.

(3) *Divisor.* The divisor of $\Theta(\phi)$ is the linear combination of rational quadratic divisors $\lambda^\perp$ weighted by $c_\phi(-(\lambda,\lambda)/2, \lambda \bmod \Lambda^{3,2})$ over primitive $\lambda \in \Lambda^{3,2*}$ with negative discriminant.

The factor v is the invariant normalization for signature $(3,2)$: the input has holomorphic weight $-1/2$, the theta kernel has weight $(3/2, 1)$, and $v\langle\phi, \Theta_{\Lambda^{3,2}}\rangle$ has weight $(0,0)$.

[proved; [8, 9]]

Remark 1.11 *(Two regularization conventions).* The Borcherds regularization [9, §6] subtracts the divergent $v \to \infty$ contribution by analytic continuation in an auxiliary parameter, and is the convention used here. The Harvey–Moore regularization [46] differs from the Borcherds regularization by a finite renormalization of the Weyl vector and the constant term, with no effect on the product expansion's exponents or divisor. The two conventions agree on quotient divisors $\Theta(\phi_1)/\Theta(\phi_2)$, on the BKM denominator identity, and on the weight identity $\mathrm{wt}\,\Theta(\phi) = c_\phi(0,0)/2$.

1.7 From $\phi_{0,1}$ to a vector-valued Weil-representation form

The singular-theta integral takes a weakly holomorphic vector-valued form on $\mathbf{Mp}_2(\mathbb{Z})$; the input $\phi_{0,1}$ is a scalar weak Jacobi form of index one. Theta-decomposition reduces the latter to the former.

Lemma 1.12 *(Theta-decomposition of $\phi_{0,1}$).* Let $\phi_{0,1} \in J_{0,1}^{\text{weak}}$ and decompose along the elliptic theta-functions of index one,

$$\phi_{0,1}(\tau, z) = h_0(\tau)\theta_{1,0}(\tau, z) + h_1(\tau)\theta_{1,1}(\tau, z),$$

where $\theta_{1,\mu}(\tau, z) = \sum_{\ell\equiv\mu\,(2)} q^{\ell^2/4} r^\ell$ for $\mu \in \{0, 1\}$ (Eichler–Zagier [32, §5]). Then $h = (h_0, h_1)$ is a weakly holomorphic vector-valued modular form of weight $-1/2$ for the dual Weil representation $\rho_{[2]}^\vee$, equivalently for the Weil representation of the opposite discriminant form $[-2]$, on $\mathbf{Mp}_2(\mathbb{Z})$. Under the finite quadratic-module identification

$$A_{[2]} \;\simeq\; A_{\Lambda^{(1,1)}\oplus[2]} \;\simeq\; A_{\Lambda^{3,2}},$$

it pairs with the $\rho_{\Lambda^{3,2}}$-valued theta kernel of Definition 1.9. Its Fourier coefficients are exactly

$$[q^{D/4}]h_\mu(\tau) = c(D), \qquad c(4n - l^2) = f(n, l), \qquad D \equiv -\mu^2 \pmod 4,$$

or equivalently $[q^{n-l^2/4}]h_{l \bmod 2} = f(n, l)$. In the index-one normalization the first coefficients are $[q^0]h_0 = 10$ and $[q^{-1/4}]h_1 = 1$ reproducing $\phi_{0,1} = r^{-1} + 10 + r + O(q)$. Since $\theta_{1,1}(\tau + 1, z) = i\,\theta_{1,1}(\tau, z)$, invariance of $\phi_{0,1}$ under T forces $h_1(\tau + 1) = -i\,h_1(\tau)$, the dual eigenvalue. The vector h is the weakly holomorphic input $\phi_{0,1}^{\text{Weil}}$ of the weight and representation type required by Theorem 1.10.

[proved; [8, 32, 43]]

Proof. Theta-decomposition along the lattice $\Lambda_{\mathrm{ind}} = \mathbb{Z}$ inside the Jacobi-elliptic variable is Eichler–Zagier [32, Theorem 5.1]. The two-component vector h is weakly holomorphic of weight $-1/2$ on $\mathbf{Mp}_2(\mathbb{Z})$ with the Weil representation of the rank-one lattice $\langle 2\rangle$. Comparing the coefficient of r^ℓ gives $[q^{n-\ell^2/4}]h_{\ell \bmod 2} = f(n,\ell)$, hence the discriminant formula above. The hyperbolic plane $\Lambda^{(1,1)}$ is unimodular, so adjoining one or two copies of $\Lambda^{(1,1)}$ does not change the discriminant form. Thus

$$(\Lambda^{(1,1)} \oplus [2])^*/(\Lambda^{(1,1)} \oplus [2]) \simeq [2]^*/[2] \simeq (\Lambda^{3,2})^*/\Lambda^{3,2},$$

with the same finite quadratic form. This discriminant-form isometry, is the transport of h to the required weakly holomorphic vector-valued form for the Weil representation of $\Lambda^{3,2}$. The constant Fourier coefficient $c_{\phi_{0,1}^{\mathrm{Weil}}}(0,0) = f(0,0) = 10$ is preserved under this discriminant-form identification because the trivial coset $0 \in \Lambda^{3,2*}/\Lambda^{3,2}$ matches the trivial coset on the rank-one factor. Borcherds [8, §15] records the explicit computation for the $\mathbf{Sp}_4$-case in the equivalent paramodular language. $\square$

Remark 1.13 *(Index-one normalization of $c_1(0)$).* The constant Fourier coefficient $c_1(0) := f(0,0) = 10$ of $\phi_{0,1}$ is the Eichler–Zagier index-one normalization of the K3 elliptic genus. The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ of [42] names N as the paramodular level of the input denominator; at $N = 1$ the input form is $\phi_{0,1}$ and $c_1(0) = 10$, giving

$$\kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5.$$

The integer-versus-half-integer ambiguity for the metaplectic weight factor is killed by the integrality of $c_1(0)/2 = 5$, and the Eichler–Zagier normalization is the row F_1 normalization in the Cléry–Gritsenko catalogue [41]. On the other seven rows the symbol $c(0)$ is indexed by the pair (t, N), not by a CHL twining order alone. No $c_N(0) = 2k_N + 1$ half-integer convention enters at $N = 1$.

1.8 Theta lift identifies Δ_5

The lift produces an automorphic form on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$, pulled back along $\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \xrightarrow{\simeq} \Gamma_{3,2}$ of Theorem 1.5 to a Siegel modular form on $\mathbf{Sp}_4(\mathbb{Z})$.

THEOREM 1.14 *(Δ_5 as singular theta lift).* The singular-theta lift of the weak Jacobi form $\phi_{0,1}$ is the weight-five Igusa cusp form:

$$\Theta(\phi_{0,1}^{\mathrm{Weil}})\big|_{\mathbb{H}_2} = \Delta_5,$$

the right-hand side written in the convention $\mathcal{D}_X = \Delta_5$ of §1.1 and the left-hand side regularized in the Borcherds sense of [9, §6]. Equivalently, in the BKM normalization,

$$\Theta(\phi_{0,1}^{\mathrm{Weil}}) = 64 \cdot \mathrm{den}(\mathfrak{g}_{\Delta_5})\big|_{Z\mapsto Z/2},$$

i.e. the lift is Δ_5 on $\mathbb{H}_2$ and $64^{-1}\Delta_5(2Z)$ on $\Lambda_{II}^{2,1}$-coordinates of $\mathbb{H}_+^{\mathbf{IV}}$.

[proved; [8, 9, 43, 45]]

Proof. The Borcherds singular-theta lift in Theorem 1.10 produces a meromorphic automorphic form $\Theta(\phi_{0,1}^{\mathrm{Weil}})$ on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ of weight $c_1(0)/2 = 5$. Its divisor is the linear combination of rational quadratic divisors weighted by $c_{\phi_{0,1}^{\mathrm{Weil}}}(-(\lambda,\lambda)/2, \lambda \bmod \Lambda^{3,2})$. For $\phi_{0,1}$, the negative-discriminant coefficients $f(0,\pm1) = 1$ give the simple reflective Humbert divisor of Δ_5, while $f(0,0) = 10$ gives the weight 5. The Borcherds-product expansion of Theorem 1.10(2), specialized to $\phi_{0,1}^{\mathrm{Weil}}$, reads

$$\Theta(\phi_{0,1}^{\mathrm{Weil}}) = C_{\phi_{0,1}} q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}$$

in the cusp chamber, with Γ_{eff} the cusp-positive semigroup and Weyl vector $\rho_0 = (1/2, 1/2, 1/2)$. The monic Borcherds product is $D_5(Z)$ of §1.1; the Igusa theta-constant cusp trivialization fixes $C_{\phi_{0,1}} = 64$, hence $\Theta(\phi_{0,1}^{\text{Weil}}) = \Delta_5$. The pullback to $\mathbb{H}_2$ along $\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \to \Gamma_{3,2}$ of Theorem 1.5 identifies $\Theta(\phi_{0,1}^{\text{Weil}})\big|_{\mathbb{H}_2}$ with the $\mathbf{Sp}_4(\mathbb{Z})$-cusp form whose Borcherds product is exactly the Igusa cusp form Δ_5 of weight five. The BKM normalization $64^{-1}\Delta_5(2Z)$ on $\Lambda_{II}^{2,1}$ is Gritsenko–Nikulin [43, Theorem 2.1], [45, Theorem 2.1, Section 5.1.1]: the doubled lattice argument $Z \mapsto 2Z$ corresponds to the polarization change $\alpha : \Gamma_{\text{ind}} \to \Lambda_{II}^{2,1}$ of §1.9. Both sides have the same Borcherds-product expansion, and their leading coefficients agree by the theta-constant $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$. □

Corollary 1.15 *(Borcherds–Gritsenko–Nikulin pentadic identity).* The theorem-level automorphic, BKM-denominator, and theta-lift readings coincide:

$$\begin{aligned}(\text{View I})\ \mathcal{D}_X &= (\text{View II})\ 64 \cdot \operatorname{den}(\mathfrak{g}_{\Delta_5})\big|_{Z\mapsto Z/2} \\ &= (\text{View V})\ \Theta\big(\phi_{0,1}^{\text{Weil}}\big).\end{aligned}$$

The Pfaffian identification $\operatorname{Pf}_{\text{prot}}(\mathfrak{D}_X^{\text{DI}}) = \Delta_5$ of View III holds on $K3 \times E$ at the Pfaffian and orientation level, granted the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\text{atlas}})$ wall datum, and retained finite Pfaffian section datum of Theorem 7.4 of Appendix 7. The lift does not provide the level-A gravity-line operator algebra; a 3d-gravity path-integral realization requires the level-A Hall–Borcherds residual, while Definition 7.19 records only the level-Z trace comparison.

[proved for Views I, II, V; conditional for View III]

1.9 The multiplier system and the Weyl vector

Theorem 1.16 *(Multiplier system of the lift).* The lift $\Theta(\phi_{0,1}^{\text{Weil}})$ carries the $\mathbf{Sp}_4(\mathbb{Z})$-multiplier system ν_{Δ_5} of Maass [71], equivalently described as the character $v_\eta^{12} \times v_H$ on the Jacobi parabolic and as the Gritsenko character on the BKM side [43, Proposition 2.1], [45, §5.1.1, case $(t = 1, \text{II}, 0)$]. On the type-II generators of the Weyl group $W^{(2)}(\Lambda_{II}^{2,1})$,

$$\nu_{\Delta_5}(s_{\delta_i}) = -1, \qquad i = 1, 2, 3,$$

and on the chamber automorphism factor $\operatorname{Aut}(\mathcal{P}_{II}) \simeq S_3$ the multiplier is trivial:

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_{-2}-f_3}) = +1.$$

This is exactly the multiplier of Lemma 7.2.

[proved; [43, 45, 71]]

Remark 1.17 *(Singular-theta multiplier vs Maass multiplier).* The Borcherds singular-theta lift produces a multiplier system on $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$ determined by the Weil action on $\Lambda^{3,2*}/\Lambda^{3,2}$ (Borcherds [8, Theorem 13.3]). For the present $\Lambda^{3,2}$, the discriminant group has order two, generated by $f_3/2$ in the convention of Remark 1.37. The finite Weil module is the two-dimensional module attached to this discriminant form. The Borcherds multiplier determined by that module and by the Weyl vector restricts on the Borcherds chamber to the same reflection character. Pulled back along $\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \to \Gamma_{3,2}$ it matches Maass's multiplier formula for the product of the ten even theta constants [43, (1.3)–(1.5)], [71]: the -1-eigenvalues on the three type-II simple reflections and the $+1$-eigenvalues on the three chamber automorphisms. The two descriptions are not independent — they are the two sides of the same character — and their agreement is part of the regularized theta correspondence of Theorem 1.7.

THEOREM 1.18 *(Weyl vector of the lift).* The Weyl vector of the Borcherds product $\Theta(\phi_{0,1}^{\mathrm{Weil}})$ in the type-II chamber $\mathcal{P}_{II} \subset \Lambda_{II}^{2,1} \otimes \mathbb{R}$ is

$$\rho = \tfrac{1}{2}\delta_1 + \tfrac{1}{2}\delta_2 + \tfrac{1}{2}\delta_3 = f_2 - \tfrac{1}{2}f_3 + f_{-2},$$

satisfying $(\rho, \delta_i) = -1$ for $i = 1, 2, 3$, with simple-real-root Cartan matrix

$$A = \big((\delta_i, \delta_j)\big) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

This is the Weyl vector of $\mathfrak{g}_{\Delta_5}$ on $\Lambda_{II}^{2,1}$ of the Weyl-chamber section, identified with the Weyl vector of the Borcherds product by [43, Theorem 4.1].

[proved; [43, 45]]

Proof. The Weyl vector of this Borcherds product is determined by the q^0-row of ϕ at the cusp. For $\phi_{0,1} = r^{-1} + 10 + r + O(q)$, the negative-discriminant coefficients are $f(0, \pm 1) = 1$, and the constant coefficient is $f(0, 0) = 10$; the Weyl-vector formula of [8, Theorem 13.3, equation (13.10)] yields $\rho = (1/2, 1/2, 1/2)$ in cusp coordinates, transported to $\Lambda_{II}^{2,1}$ by the polarization change $\alpha : \Gamma_{\mathrm{ind}} \to \Lambda_{II}^{2,1}$ of the Weyl-chamber section. This recovers $\rho = f_2 - \frac{1}{2}f_3 + f_{-2}$; the Cartan matrix is the Gram matrix of $\{\delta_1, \delta_2, \delta_3\}$ computed in the Weyl-chamber section. $\square$

1.10 Compatibility with View II: lift's image is the BKM denominator

THEOREM 1.19 *(Lift–denominator compatibility* [43, Proposition 3.1]*).* The image of the singular-theta lift, $\Theta(\phi_{0,1}^{\mathrm{Weil}}) = \Delta_5$, satisfies the Weyl–Kac–Borcherds denominator identity for the Lorentzian BKM superalgebra $\mathfrak{g}_{\Delta_5}$ on $\Lambda_{II}^{2,1}$:

$$64^{-1}\Delta_5(2Z) = e^{-2\pi i(\rho, Z)} \prod_{\alpha \in \mathcal{R}_+} \big(1 - e^{-2\pi i(\alpha, Z)}\big)^{\mathrm{smult}(\alpha)},$$

with positive-root system $\mathcal{R}_+$, Weyl vector ρ of Theorem 1.18, and signed root supermultiplicities

$$\mathrm{smult}\big(\alpha(n, l, m)\big) = f(nm, l), \qquad (n, l, m) \in \Gamma_{\mathrm{act}}.$$

The simple imaginary-root data are the Gritsenko–Nikulin coefficients $m(a)$ on negative-norm rays and $\tau(a)$ on isotropic rays of the Weyl-chamber section.

[proved; [43, 45]]

Proof. The Borcherds-product expansion $\Delta_5 = 64\, q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ of Theorem 1.14 is the BKM denominator product for $\mathfrak{g}_{\Delta_5}$ after the polarization change $\alpha : \Gamma_{\mathrm{ind}} \to \Lambda_{II}^{2,1}$: the exponent $f(nm, l)$ is the signed multiplicity of the root $\alpha(n, l, m)$, and the doubled argument $2Z$ converts $\exp(-\pi i(\alpha, z))$ into the standard denominator character $\exp(-2\pi i(\alpha, z))$. The algebra $\mathfrak{g}_{\Delta_5}$ is the Gritsenko–Nikulin algebra of [43, §§3–4, Proposition 3.1]; the denominator identity is read off the product expansion as in Chapter 4. Equivalently, the imaginary-simple-root characters $m(a)$ and $\tau(a)$ are the additive data of [45, Theorem 2.1, equations (2.1)–(2.3)]. $\square$

1.11 Compatibility with View I and the cross-level identities

Theorem 1.20 *(Borcherds–Gritsenko–Nikulin three-way agreement).* The three identifications

$$\begin{aligned} \Theta(\phi_{0,1}^{\mathrm{Weil}}) &= \Delta_5 && \text{(Borcherds 1995, 1998)},\\ 64^{-1}\Delta_5(2Z) &= \mathrm{den}(\mathfrak{g}_{\Delta_5}) && \text{(Gritsenko–Nikulin 1998)},\\ \mathcal{D}_X &= \Delta_5 && \text{(Igusa 1962)} \end{aligned}$$

are mutually consistent: $\Theta(\phi_{0,1}^{\mathrm{Weil}}) = \mathcal{D}_X = 64 \cdot \mathrm{den}(\mathfrak{g}_{\Delta_5})\big|_{Z\mapsto Z/2}$ on $\mathbb{H}_+^{\mathbf{IV}}$, and the multiplier system ν_{Δ_5} of Theorem 1.16 is common.

[proved; [8, 9, 43, 45, 48]]

Proof. Theorem 1.14 is identification (1). Theorem 1.19 is identification (2). Identification (3) is the Igusa identification $\mathcal{D}_X = \Delta_5$ of [48, Theorem of §2], transported through the exterior-square model section. All three sides have the Borcherds-product expansion $\Delta_5 = 64\, q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ in the cusp chamber and pull back along the exceptional isomorphism of Theorem 1.5 to the same automorphic form on $\mathbf{Sp}_4(\mathbb{Z})$. The multiplier-system agreement is Theorem 1.16. □

1.12 The weight-five universal Borcherds identity at $N = 1$

Theorem 1.21 *(Universal Borcherds weight identity, $N = 1$).* At paramodular level $N = 1$ the Vol III universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ specializes to

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \frac{c_1(0)}{2} = \frac{f(0,0)}{2} = \frac{10}{2} = 5.$$

This is the row ($N = 1$, wt $= 5$, $c_1(0) = 10$) of the Cléry–Gritsenko catalogue [41], and is the value entered in the K3×E $\kappa_\bullet$-spectrum $\{\kappa_{\mathrm{cat}} = 0, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = 0, \kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3, \kappa_{\mathrm{BKM}} = 5, \kappa_{\mathrm{fiber}} = 24\}$ of Vol III.

[proved; [8, 41]; Vol III universal identity]

Proof. The lift's weight is $c_1(0)/2$ by Theorem 1.10(1), and $c_1(0) = f(0,0) = 10$ by Lemma 1.2 and Lemma 1.12. The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is [42, §§2–3], generalized in Vol III across the eight Cléry–Gritsenko rows; at $N = 1$ it is exactly the Borcherds weight formula of Theorem 1.10(1). □

Remark 1.22 *(Subscript discipline).* The bare symbol κ does not appear in any theorem-level identity. Every occurrence carries the subscript fixing its construction reading (κ_{cat}, $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}}$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$, κ_{BKM}, κ_{fiber}). The additive expression $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{K3})$ is a mixed-reading collision. With $\kappa_{\mathrm{ch}} = \kappa_{\mathrm{ch}}^{\mathrm{Heis}}$ it gives the accidental equality $3 + 2 = 5$ at $N = 1$, but it has no rowwise extension to $N \in \{2, 3, 4, 6\}$. The universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is the invariant statement; at $N = 1$ the right hand side is $c_1(0)/2 = 5$.

1.13 Beilinson cut: the lift is the structural content

The five views of §1 lie on the Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z} \to \mathsf{A}$ of Vol II. View V's primitive content is the singular-theta integral $\Theta : J_{0,1}^{\mathrm{weak,Weil}} \to M_*(\widetilde{\mathbf{O}}(\Lambda^{3,2})_+)$ of Theorem 1.10, sending Jacobi forms to Borcherds products. Its value at $\phi_{0,1}$ is the automorphic section Δ_5, the level-S image of the lift.

Remark 1.23 *(Beilinson level).* The integral operator Θ lives at level C of the universal chain: it is a chart-dependent passage from a primitive Jacobi-form input to its Borcherds product, dependent on the Weil representation of $\Lambda^{3,2}$ and on the regularization of [9, §6]. The scalar Δ_5 lives at level S: it is the automorphic value of the lift on $\phi_{0,1}$. The retained $\mathfrak{D}_X^{\mathrm{DI}}$ datum of View III is the conditional Pfaffian construction whose protected Pfaffian recovers Δ_5; the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ of View II lives at level S under the trace lane of Vol II. The primitive object in this passage is the integral operator Θ, and the automorphic, BKM, Pfaffian, and theta-lift names of Δ_5 are readings, not primitive objects.

1.14 Consistency of the lattice conventions

Remark 1.24 *($\Lambda^{3,2}$ versus $\Lambda^{2,1} \oplus \langle -2\rangle$).* The signature-(3, 2) lattice $\Lambda^{3,2}$ of Definition 1.3 decomposes as $\Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ on the integral level. The sublattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ is primitive in $\Lambda^{3,2}$; its orthogonal complement is the second hyperbolic plane $\Lambda^{(1,1)}$, not the rank-one form $\langle -2\rangle$ (which would have wrong signature). The polarization change $\alpha : \Gamma_{\mathrm{ind}} \to \Lambda^{2,1}_{II}$ sends $(n, l, m) \mapsto 2nf_2 - lf_3 + 2mf_{-2}$, so the BKM grading lattice is $\Lambda^{2,1}_{II}$, the index-four sublattice of $\Lambda^{2,1}$ with primitive f_3-coordinate. The doubled argument $Z \mapsto 2Z$ in $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ of Gritsenko–Nikulin changes the exponential convention from $\exp(-\pi i(\alpha, z))$ to $\exp(-2\pi i(\alpha, z))$. The $\mathrm{II}_{2,2}$ presentation of $\Lambda^{3,2}$ used in some references (Borcherds 1995 [8, §15], Borcherds 1998 [9, Example 16.4]) is the same lattice expressed via two hyperbolic planes plus the [2]-form, and produces the same Borcherds product Δ_5 under the same lift. No $\mathrm{II}_{3,2}$ versus $\mathrm{II}_{2,1} \oplus \langle -2\rangle$ ambiguity remains.

Remark 1.25 *(Multiplier and metaplectic accounting).* The Borcherds singular-theta lift produces an automorphic form on the integral form $\widetilde{\mathbf{O}}(\Lambda^{3,2})_+$, pulled back to $\mathbf{Sp}_4(\mathbb{Z})$ by Theorem 1.5. Since $c_1(0)/2 = 5$ is integral, the resulting automorphy factor is defined on $\mathbf{Sp}_4(\mathbb{Z})$. The metaplectic cover enters on the $\mathbf{Mp}_2(\mathbb{Z})$ input side through $\rho_{\Lambda^{3,2}}$; after restriction to integral weight its central sign does not alter the $\mathbf{Sp}_4(\mathbb{Z})$ automorphy factor. The residual reflection character is exactly the multiplier ν_{Δ_5} of Theorem 1.16. For the CHL constants $N \in \{1, 2, 3, 4, 6\}$ of Theorem 1.21, the numbers $c_N(0)/2 = (5, 4, 3, 2, 1)$ are integral. Thus the metaplectic cover belongs to the input Weil representation, while the resulting automorphic forms live on the corresponding integral orthogonal or Siegel arithmetic groups.

Remark 1.26 *(The eta-multiplier v_η^{12}).* The character of Δ_5 on the Jacobi parabolic of $\mathbf{Sp}_4(\mathbb{Z})$ is described in [42] as $v_\eta^{12} \times v_H$, where v_η is the eta-multiplier on $\mathbf{SL}_2(\mathbb{Z})$ and v_H is the Heisenberg multiplier. The first four Cléry–Gritsenko rows carry eta exponents 12, 6, 4, 3 for $N = 1, 2, 3, 4$, respectively [41, Theorem 1.4]; the present form is the $N = 1$ row. Equivalently, after the lift through the exceptional isomorphism, v_η^{12} remains the eta multiplier in the Jacobi–Heisenberg presentation; it is not a discriminant character of $\mathrm{II}_{2,2}$, since $\mathrm{II}_{2,2}$ is unimodular. No v_η^N inconsistency between [42], [43], and [41] arises at $N = 1$; the three computations agree.

1.15 Theta-lift boundary

The singular theta lift is one realization of Δ_5 among five. The lift's primitive content is the Borcherds regularized theta operator $\Theta : \phi^{\mathrm{Weil}} \mapsto \Theta(\phi^{\mathrm{Weil}})$ of Theorem 1.10; its value at $\phi^{\mathrm{Weil}}_{0,1}$ is the weight-five Igusa cusp form. The five-view identification $\mathcal{D}_X = 64 \cdot \mathrm{den}(\mathfrak{g}_{\Delta_5})\big|_{Z \mapsto Z/2} = \Theta(\phi^{\mathrm{Weil}}_{0,1})$ is the theorem-level automorphic–BKM–theta identity; its Pfaffian extension to View III is conditional on the retained $K3 \times E$ data of Appendix 7. The mirror-discriminant View IV is the remaining conjectural cross-level identity, recorded in the mirror-discriminant chapter (Chapter 8) and not used here.

Borcherds's theta-correspondence formulation makes two structural facts visible. First, the weight-five integrality is forced by $c_1(0) = 10$ and the integrality of $c_1(0)/2$; no metaplectic cover is needed at the weight level. Second, the BKM-denominator structure of View II is not an extra postulate but a theorem about the lift's image: the Gritsenko–Nikulin denominator identity reads the lift's Borcherds product as the denominator of $\mathfrak{g}_{\Delta_5}$ on $\Lambda_{II}^{2,1}$ (Theorem 1.19), and the Borcherds–Gritsenko–Nikulin three-way agreement (Theorem 1.20) crystallizes the lift's value in three convertible normalizations. The universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is, at $N = 1$, precisely the Borcherds-weight formula $\mathrm{wt}\,\Theta(\phi) = c_\phi(0,0)/2$ of Theorem 1.10(1) applied to $\phi = \phi_{0,1}$.

Part II

Dirac–Igusa Pro-Object

Chapter 6

The Dirac–Igusa pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ on $K3 \times E$ (View ③)

The Igusa cusp form Δ_5 of weight 5 on $\mathrm{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2$ has three established identities: the Borcherds–Gritsenko–Nikulin section (View ①), the denominator of a generalized Borcherds–Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ on the root lattice $\Lambda_{II}^{2,1}$ (View ②), and the singular theta lift of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ (View ⑤). The View ③ datum is the protected Pfaffian line and compact source comparison for a finite E_3-prefactorization algebra on the $K3 \times E$ formal target, presented at finite stage as the Dirac–Igusa pro-object

$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R \big(\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{o,R}, \mathrm{Rec}_R\big)$$

with eight finite constructions and four named obstructions.

Let S be a smooth complex K3 surface and let E be a smooth elliptic curve. Singular K3 limits, nodal or cuspidal elliptic degenerations, imprimitive curve-class contributions, the $s = 0$ compact Hall cusp, and multi-component charge closures belong to separate boundary data. The standing finite-stage hypotheses are [K3×E-fin] (finite-type semistable substacks at bounded HN height), [K3×E-atlas] (finite d-critical/cosection Hall atlas), and [ML] (Mittag–Leffler descent for the HN inverse system). Two further standing functorial inputs are [E_3-HolFA], a chosen finite holomorphic E_3-prefactorization model for $\mathcal{A}_{X,R}^{E_3}$, including the BV quantum master equation, anomaly cancellation, locality, and framing/formality data needed to make the E_3-operations act on the retained finite stage, and [$K3{\to}E$-sp], a chosen chain-level specialization from the $K3$-direction to the hybrid chiral carrier on E.

The bar coalgebra is MC twisting, not the level-Z bulk. The imported Borcherds–Kac–Moody current envelope $\mathsf{Den}_{\Delta_5,E}$ is the target chiral shadow, and the cobar comparison $\Theta_{\mathrm{Kos},R}$ is the source-to-target map on the hybrid source. Its invertibility is the Koszul source datum, not a target-internal counit.

Construction theorem for the Dirac–Igusa package

Theorem 0.1 *(Construction of the Dirac–Igusa pro-object on $K3 \times E$).* Granted the retained Do-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and the retained $(O2_{\mathrm{atlas}})$ wall datum of Theorem 7.4 and the standing finite-stage hypotheses [K3×E-fin], [K3×E-atlas], [ML], the standing hypotheses [E_3-HolFA] and [$K3{\to}E$-sp], the chiral Koszul source datum of Definition 0.26, its Mittag–Leffler exactness of Proposition 0.29, and the cofinal primitive-recognition datum of Definition 0.31, there exists a finite-stage Dirac–Igusa pro-object

$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_{R \in \mathcal{R}_{\mathrm{HN}}} \big(\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{o,R}, \mathrm{Rec}_R\big)$$

on the moduli stack $\mathfrak{M}_c^{\mathrm{red},+}(X)$ of effective dyonic-class objects in the chosen stability heart of Chapter 2, with the following typed structure:

(L1) (*Hybrid carrier*, §0.1.) $F_{X,R}^{\mathrm{hyb}}$ is the hybrid wrapped factorization carrier on $\mathrm{Ran}^{\mathrm{hyb}}(E)$ over the elliptic factor, with local/wrapped/mixed atlas matching the type-II wall structure $\{\gamma(\partial_1), \gamma(\partial_2), \gamma(\partial_3)\} = \{(1,1,0), (0,1,1), (0,-1,0)\}$ on $\Lambda_{II}^{2,1}$.

(L2) (*Reduced d-critical orientation*, §0.2.) o_R is the reduced orientation datum: a square root of the reduced determinant line, Thom–Sebastiani transport on retained extension strata, null-trivialisations of the reduced and equivariant degree-two orientation gerbes, and zero finite-stabilizer linearization characters, as in Definition 0.8.

(L3) (*Hall correspondences*, §0.3.) H_R is the finite protected Hall datum on $H_c^\bullet(\mathfrak{M}_{c,R}^{\mathrm{red},+}(X), \Phi_W \otimes o_R)$: product, collision coproduct, primitive projection, and Hopf pairing on the cosection-reduced heart. The comparison with the Drinfeld double $G(X) = D(Y^+(X))$ requires the Hall pairing, integral form, stable-envelope transport, completion, and primitive-recognition data.

(L4) (*First-order Dirac block and Pfaffian line*, §0.4.) $\mathscr{L}_{\mathrm{Pf},R}$ is the Pfaffian line bundle on $\mathfrak{Q}_R = (\mathfrak{M}_{c,R}^{\mathrm{red},+}(X)/E)^{\mathrm{red}}$ with finite Pfaffian sections $\mathrm{pf}_{X,R}$ carried into the inverse limit by Mittag–Leffler descent.

(L5) (*Chiral Koszul source coalgebra*, §0.5.) $C_{X,R}$ is the bar of the augmented binary/two-step hybrid correspondence algebra; it becomes the bar of a hybrid factorization algebra only after the higher-coloured residual vanishes. The chiral Koszul cobar map $\Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_X \longrightarrow \mathtt{Den}_{\Delta_5,E}$ is the source-to-target comparison and is a quasi-isomorphism under the chiral Koszul source datum and its Mittag–Leffler exactness.

(L6) (*Primitive recognition*, §4.) Rec_R is the cofinal-window primitive recognition criterion (S1)–(S10) of Definition 0.31. Under that finite compact-source datum, it identifies $P_X^{\Pi,\mathrm{red}}$ with $\mathfrak{g}_{\Delta_5}$.

(L7) (*Eight finite constructions*, §0.7.) The pro-object is the inverse limit of the eight finite constructions of Appendix 6, each carrying its obstruction class.

(L8) (*Obstruction classes*, §1.2.) The obstruction $(D0)$ is reduced on $K3 \times E$ to the retained D0-HN datum, $(O1)$ and $(O1)^+$ are reduced to retained orientation data, and $(O2)$ is reduced to $(O2_{\mathrm{atlas}})$, by Theorem 7.4.

Proof. The pro-object is the inverse limit of the finite-stage data on the HN system $\mathcal{R}_{\mathrm{HN}}$. The Pfaffian, orientation, Hall, and carrier entries are obtained from the finite constructions of Appendix 6, and the inverse limit exists by the fixed data [K3×E-fin], [K3×E-atlas], [ML] together with the retained D0-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and $(O2_{\mathrm{atlas}})$ of Theorem 7.4. The Koszul entry $\Theta_{\mathrm{Kos},R}$ is supplied by the chiral Koszul source datum and its exact inverse-system condition. The recognition entry Rec_R is supplied by the finite compact-source recognition datum. Thus (L1)–(L8) are finite components with separate source, orientation, Hall, Koszul, Pfaffian, and recognition hypotheses; no component is inferred from another. □

Entries (L1)–(L8) are the finite constituents of $\mathfrak{D}_X^{\mathrm{DI}}$. Each constituent has its finite-stage morphism and obstruction class. The finite formulas are in Appendix 6; the retained obstruction data are in Appendix 7.

0.1 The hybrid factorization carrier $\mathrm{Ran}^{\mathrm{hyb}}(E)$

The first entry of the Dirac–Igusa pro-object is the carrier on which chiral operations live. An ordinary Beilinson–Drinfeld factorization algebra on $\mathrm{Ran}(E)$ [6, §3.1] sees only a finite multiset of points of E and the operations attached to their collisions; this is the projection-finite stratum. The Igusa product

Δ_5 carries a Borcherds variable s whose positive powers record positive elliptic-degree wrapping, and a one-dimensional subscheme of $K3 \times E$ of positive elliptic degree projects *onto* E. Locality on $\mathrm{Ran}(E)$ alone has no mechanism for it.

Lemma 0.2 *(Projection-to-E support-locality obstruction; proved).* Let $C \subset K3 \times E$ be a one-dimensional subscheme whose curve class has positive elliptic degree, $(p_E)_*[C] = b[E]$ with $b > 0$. Then $p_E(C) = E$. In particular C is not localized at finitely many points of E, and is not contained in $p_E^{-1}(U)$ for any proper open $U \subsetneq E$.

Proof. If $p_E(C)$ were zero-dimensional, the pushforward of $[C]$ to E would vanish, contradicting $(p_E)_*[C] = b[E]$ with $b > 0$. Properness of p_E makes $p_E(C)$ closed, and the irreducibility of E forces $p_E(C) = E$. □

The hybrid carrier replaces the Ran space by a two-stratum object that records, simultaneously, finite local supports and wrapped elliptic legs. At every finite Harder–Narasimhan height R the datum is a finite-coloured correspondence base, not a Ran space with a scalar attached.

Definition 0.3 *(Hybrid Ran base).* Let $p_E : K3 \times E \to E$ be projection. A compact Hall or chiral sector is *projection-finite over E* if every object A in it satisfies $p_E(\mathrm{Supp}\, A) \subset F$ for some finite subset $F \subset E$. A one-dimensional sector is *wrapped over E* if its curve class has positive elliptic degree, in which case write

$$b(A) = \deg(p_E)_*[\mathrm{Supp}_1 A] \in \mathbb{Z}_{\geq 0}.$$

At finite height R, a point of the hybrid base $\mathrm{Ran}^{\mathrm{hyb}}(E)_{\leq R}$ consists of a finite projection-local support $F \subset E$, finitely many wrapped colours $\eta_j \in \Gamma_R^{\mathrm{wr}}$ (the height-R charges of positive elliptic degree $b > 0$, the wrapped window made precise at (6.1) below; the projection-local charges of degree $b = 0$ form the complementary window Γ_R^{loc}), and rigidified wrapped objects W_j carrying anchor values $\lambda_{\eta_j,R}(W_j) \in E$. Its arrows are generated by collisions of F, by extensions of local colours, by mixed local/wrapped correspondences, and by wrapped/wrapped correspondences. The underlying degree shadow is

$$\mathrm{Ran}^{\mathrm{hyb}}(E) = \bigsqcup_{b \geq 0} \mathrm{Ran}(E)_{[b]}, \qquad \mathrm{Ran}(E)_{[b]} := \mathrm{Ran}(E) \times \{b\}.$$

The stratum $b = 0$ is the ordinary Beilinson–Drinfeld finite-support Ran stratum on E [6, §3.4]. A stratum $b > 0$ records finite local collision points together with a global wrapped elliptic leg of degree b; it does *not* assert that the wrapped support is finite over E.

Corollary 0.4 *(Finite-support Ran locality misses the geometric elliptic-degree direction; proved).* A support-local factorization algebra on $\mathrm{Ran}(E)$, defined only by separating supports over disjoint opens of E, sees the projection-finite sector. It does not realize the sectors of positive geometric elliptic degree.

Proof. The obstruction uses the geometric degree $b_R^{\mathrm{geom}} = \deg_E$, not the Borcherds coordinate $m = \mathrm{pr}_3\, \overline{\Pi}_X$. On the rank-one Oberdieck–Pandharipande/Donaldson–Thomas branch one has the branch-local equality $m_{\mathrm{Bch}} = d_E - 1$ [84]; it is not a definition of the wrapped stratum. By Lemma 0.2, objects with $b_R^{\mathrm{geom}} > 0$ project onto all of E. Such objects cannot be separated by disjoint opens. Therefore ordinary Ran-locality has no locality mechanism for the wrapped sector. □

FINITE HYBRID LOCAL/WRAPPED FACTORIZATION DATUM

The full carrier at finite stage is the following supplied chain datum, with named obstruction classes that must be trivialized.

Definition 0.5 *(Finite hybrid local/wrapped factorization datum).* Fix a Bridgeland-style stability sector (σ, S) in the K3-surface lineage [4, 12] and a finite HN height R. A finite-stage hybrid datum is

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} = \big(C_{\sigma,S,\le R},\ b^{\mathrm{geom}}_R,\ \mathcal{F}^{\mathrm{loc}}_R,\ \mathcal{G}^{\mathrm{wr}}_R,\ \mathfrak{E}^{\mathrm{loc}}_R,\ \mathfrak{E}^{\mathrm{hyb}}_R,\ \mathfrak{E}^{\mathrm{wr}}_R,\ \mathfrak{Flag}^{\mathrm{hyb}}_R,\ \mathsf{Corr}^{E,\mathrm{hyb}}_R,\ \lambda_R,\ Q_{E,R},\ \theta^Q_R,\ o_R,\ I^{\mathrm{prot}}_R\big)$$

with the seven clauses below. Every stack, sheaf, operation, and coherence lives at the fixed height R; the inverse limit appears only after all finite stages and transition maps have been supplied.

(i) *Finite Hall stage and elliptic degree.* $C_{\sigma,S,\le R}$ is a supplied extension-closed charge-support HN quotient of the candidate compact Hall category under the finite-type hypothesis [K3×E-fin], with finite-type substacks $\mathfrak{M}_{\gamma,R}$ for $\gamma \in \Gamma^{\mathrm{HN}}_{\sigma,S}(R)$, vanishing-cycle coefficients $\Phi_{\gamma,R}$, and orientation lines $o_{\gamma,R}$. The geometric elliptic-degree component

$$b^{\mathrm{geom}}_R : \Gamma^{\mathrm{HN}}_{\sigma,S}(R) \longrightarrow \mathbb{Z}_{\ge 0}$$

splits the finite charge support into local and wrapped pieces:

$$\Gamma^{\mathrm{loc}}_R = \{b^{\mathrm{geom}}_R = 0\}, \qquad \Gamma^{\mathrm{wr}}_R = \{b^{\mathrm{geom}}_R > 0\}. \tag{6.1}$$

The map is additive on retained extensions; terms of height $> R$ vanish.

(ii) *Projection-finite local sector through closed configurations.* For a finite set I, set

$$Z_I = \bigcup_{i\in I} \{(e, (e_j)_j) \in E \times E^I \mid e = e_i\}.$$

For $\alpha \in \Gamma^{\mathrm{loc}}_R$ the stage supplies a closed-configuration incidence stack

$$\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} = \{(A, \mathbf{e}) \in \mathfrak{M}_{\alpha,R} \times E^I \mid p_E(\mathrm{Supp}\, A) \subset Z_I(\mathbf{e})\}$$

compatible with diagonals $E^J \to E^I$, symmetric-group descent, and the d-critical/vanishing-cycle structure. Open-support locality is recovered as a limit

$$\mathcal{M}^{\mathrm{loc}}_{\alpha,R}(U) \simeq \operatorname*{colim}_I \big(\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \times_{E^I} U^I\big),$$

and the configuration map $\pi_{\alpha,R,I}$ produces

$$\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I} = (\pi_{\alpha,R,I})_!(\Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I}).$$

For an open $U \subset E$,

$$\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U) = \bigoplus_{I/S_I} R\Gamma_c\Big(U^I, j^!_{U,I}\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I}\Big),$$

and $\mathcal{F}^{\mathrm{loc}}_R(U)$ is the direct sum over Γ^{loc}_R. Locality is !-restriction on closed configuration spaces; the open-support Borel–Moore formula is a theorem to be derived from this atlas, not the primary datum.

(iii) *Wrapped prequotient sector and rigidification.* For $\eta \in \Gamma_R^{\mathrm{wr}}$, a prequotient stack

$$\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \xrightarrow{\rho_{\eta,R}} \mathcal{M}_{\eta,R}^{\mathrm{wr}} \subset \mathfrak{M}_{\eta,R}$$

is supplied with a fixed origin $o_E \in E$ and an E-equivariant anchor $\lambda_{\eta,R} : \mathcal{M}_{\eta,R}^{\mathrm{wr,rig}} \to E$ satisfying $\lambda_{\eta,R}(t \cdot W) = t + \lambda_{\eta,R}(W)$. A natural candidate anchor is the normalized determinant of pushforward $\det Rp_{E,*}A \otimes \mathcal{O}_E(-\chi(A) \cdot o_E) \in \mathrm{Pic}^0(E) \simeq E$, with translation weight $\chi(A)$; on $\chi(A) = 0$ strata it does not by itself remember the relative E-position, and rank-one wrapped pairs of identical determinant and charge can have distinct wrapped self-Ext dimensions ($\dim \mathrm{Ext}^1_E$ on $\mathcal{O}_E^{\oplus 2}$ is 4, on $L \oplus L^{-1}$ with $L^2 \neq \mathcal{O}_E$ is 2). The anchor residual is therefore a typed finite datum

$$\mathfrak{o}_R^{\lambda} = \left(\mathfrak{o}_R^{\lambda,\mathrm{ex}},\ \mathfrak{o}_R^{\lambda,\mathrm{unit}},\ \mathfrak{o}_R^{\lambda,\mathrm{loss}},\ \mathfrak{o}_R^{\lambda,\mathrm{multi}},\ \mathfrak{o}_{R'R}^{\lambda,\mathrm{tr}}\right).$$

With the anchor in place, set

$$\mathcal{G}_{\eta,R}^{\mathrm{wr}} = H_*^{\mathrm{BM}}\Big(\mathcal{M}_{\eta,R}^{\mathrm{wr,rig}}, \Phi_{\eta,R}^{\mathrm{wr}} \otimes o_{\eta,R}^{\mathrm{wr}}\Big), \qquad \mathcal{G}_R^{\mathrm{wr}} = \bigoplus_{\eta} \mathcal{G}_{\eta,R}^{\mathrm{wr}}.$$

(iv) *Local, mixed, and wrapped extension correspondences.* For disjoint opens $(U_i) \subset E$, local charges α_i, and wrapped charges η, η_1, η_2, the finite stage contains the correspondence stacks

$$\mathfrak{E}^{\mathrm{loc}}_{\alpha_1,\ldots,\alpha_r,R}(U_1,\ldots,U_r), \quad \mathfrak{E}^{\mathrm{hyb}}_{\alpha_1,\ldots,\alpha_r;\eta,R}, \quad \mathfrak{E}^{\mathrm{wr}}_{\eta_1,\eta_2,R},$$

together with the ordered *two-sided* mixed stacks

$$\mathfrak{E}^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}((U_i),(V_j))$$

classifying ordered insertions of finite local supports on both sides of one wrapped input. The defects of closed-configuration finite-type properness, pull-push admissibility, and Thom–Sebastiani transport are recorded by

$$\mathfrak{o}_R^{\mathrm{prop,LL}},\ \mathfrak{o}_R^{\mathrm{prop,mix}},\ \mathfrak{o}_R^{\mathrm{prop,WW}},$$

and aggregate residuals $\mathfrak{o}_R^{\mathrm{conf}}, \mathfrak{o}_R^{\mathrm{conf-prop}}, \mathfrak{o}_R^{\mathrm{adm}}, \mathfrak{o}_R^{\mathrm{TS}}$. Mixed and wrapped stacks remember source and target anchors before quotienting,

$$\mathfrak{E}^{\mathrm{wr}}_{\eta_1,\eta_2,R} \to E \times E \times E, \qquad \xi \mapsto (\lambda_{\eta_1,R}(W_1), \lambda_{\eta_2,R}(W_2), \lambda_{\eta_1+\eta_2,R}(W));$$

replacing this memory by a scalar Fock or Hecke factor is not the hybrid datum. These stacks generate a finite E-equivariant correspondence category $\mathsf{Corr}_R^{E,\mathrm{hyb}}$ with operation maps $\mu_R^{\bullet} = q_! \circ \mathrm{TS}_R \circ p^*$.

(v) *Eight-word two-step binary flag atlas.* Write L for a projection-finite local input and W for a rigidified wrapped input. For every word $w = (\epsilon_1, \epsilon_2, \epsilon_3) \in \{\mathrm{L},\mathrm{W}\}^3$ and every compatible ordered triple (ξ_1, ξ_2, ξ_3) of charges and local opens, the datum supplies a two-step flag stack $\mathfrak{Flag}^{w}_{\xi_1,\xi_2,\xi_3,R}$ classifying filtrations with the prescribed associated graded pieces, with comparison maps to the iterated correspondence fibre products. The eight words

LLL, LLW, LWL, WLL, LWW, WLW, WWL, WWW

are all part of the datum; in mixed and wrapped words the flag stack retains initial, intermediate, and terminal anchors before E-quotient. The two parenthesizations give the same map in the finite quotient,

$$\mu_{\xi_1+\xi_2,\xi_3,R} \circ (\mu_{\xi_1,\xi_2,R} \otimes 1) = \mu_{\xi_1,\xi_2+\xi_3,R} \circ (1 \otimes \mu_{\xi_2,\xi_3,R}),$$

with defect classes $\mathfrak{o}^{\mathrm{assoc}}_{w,R}$ required to vanish for all eight words; the chosen 2-morphisms must satisfy the four-input pentagon on the four-step flag-of-flags stacks. The higher-coloured residual $\mathfrak{o}^{\mathrm{col}}_R$ records the remaining colored configuration data, units, symmetry conventions, refinement maps, descent, and overlap coherences for all finite HN windows. The eight-word atlas gives only binary associativity; full hybrid factorization requires the higher-coloured residual to vanish in addition.

(vi) *Quotient-after-correspondence functor and reduced E-quotient.* All stacks and correspondences above are E-equivariant before quotienting. The quotient datum supplies a pseudofunctor

$$\mathsf{Corr}_R^{E,\mathrm{hyb}} \longrightarrow \mathsf{Corr}_R^{\mathrm{red,hyb}}$$

with coherent 2-morphisms for composition, base-change, and Thom–Sebastiani transport; applying vanishing-cycle Borel–Moore chains gives the finite correspondence module categories $\mathsf{BM}_R^{E,\mathrm{hyb}}$ and $\mathsf{BM}_R^{\mathrm{red,hyb}}$, and the induced functor $Q_{E,R}$ between them. For each operation $\mu_R^{\bullet} = q_! \,\mathrm{TS}\, p^*$ the datum carries a comparison isomorphism

$$\theta^{Q}_{\mu,R} : Q_{E,R}\, q_! \,\mathrm{TS}\, p^* \xrightarrow{\sim} \overline{q}_! \overline{\mathrm{TS}}\, \overline{p}^* (Q_{E,R}^{\boxtimes k}),$$

compatible with the flag-stack coherences. The four quotient residuals $\mathfrak{o}_R^{Q,\mathrm{form}}$, $\mathfrak{o}_R^{Q,\mathrm{adm}}$, $\mathfrak{o}_R^{Q,\mathrm{flag}}$, $\mathfrak{o}_{R'R}^{Q,\mathrm{tr}}$ must vanish. A quotient-first construction (object-stack quotient followed by a new Hall product) is *not* this datum.

(vii) *Protected integration with Gram-trace degree.* The finite stage carries a protected integration map

$$I_R^{\mathrm{prot}} : Q_{E,R}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \longrightarrow \mathbb{C}[\Gamma_{\mathrm{eff}}]_{\le R}$$

compatible with reduced product, coproduct, primitive projection, and HN transition. For homogeneous x of charge γ with full Gram label $\overline{\Pi}_X(\gamma) = (n, l, m)$,

$$I_R^{\mathrm{prot}}(Q_{E,R}x) \in s^{m_R^{\mathrm{Bch}}(\gamma)}\mathbb{C}[q^{\pm 1}, r^{\pm 1}], \qquad m_R^{\mathrm{Bch}}(\gamma) = m,$$

in the OP/Borcherds variables of Proposition 1.1. The Gram trace m_R^{Bch} is additive under $\overline{\Pi}_X$, so $s^{m_R^{\mathrm{Bch}}(\gamma_1+\cdots+\gamma_k)} = \prod_i s^{m_R^{\mathrm{Bch}}(\gamma_i)}$ inside the finite quotient: the s-degree is coupled to the actual hybrid operation, not appended as an external Fock determinant. The integration defects

$$\mathfrak{o}_R^{I,\mathrm{ch}},\ \{\mathfrak{o}_{w,R}^{I,\mathrm{prod}}\}_w,\ \mathfrak{o}_R^{I,\mathrm{co}},\ \mathfrak{o}_R^{I,\mathrm{Prim}},\ \mathfrak{o}_{R'R}^{I,\mathrm{tr}}$$

record chain compatibility, multiplicativity on each word, comultiplicativity, primitive compatibility, and HN transition.

The aggregate finite obstruction tuple of the hybrid lane is

$$\mathfrak{O}_{\mathrm{hyb},R} = (\mathfrak{o}_R^{\mathrm{stage}}, \mathfrak{o}_R^{\lambda}, \mathfrak{o}_R^{\mathrm{corr,LL}}, \mathfrak{o}_R^{\mathrm{corr,mix}}, \mathfrak{o}_R^{\mathrm{corr,WW}}, \mathfrak{o}_R^{\mathrm{conf}}, \{\mathfrak{o}_{w,R}^{\mathrm{assoc}}\}_w, \mathfrak{o}_R^{\mathrm{col}}, \mathfrak{o}_R^{Q,\bullet}, \mathfrak{o}_R^{I,\bullet}, \mathfrak{o}^{\mathrm{tr}}_{\mathrm{hyb},R'R}).$$

Proposition 0.6 *(Finite consequences of the hybrid datum; conditional on $\mathfrak{O}_{\mathrm{hyb},R} = 0$).* Assume the data of Definition 0.5 are supplied for all R with compatible transition maps, and $\mathfrak{O}_{\mathrm{hyb},R} = 0$. Then:

(i) the $b = 0$ restriction is an ordinary support-local factorization algebra on $\mathrm{Ran}(E)$ in the sense of Beilinson–Drinfeld [6, §3.4], equivalently a Costello–Gwilliam factorization algebra over the smooth real curve E_{top} [22, Vol. II §10];

(ii) every operation with a wrapped input lands in a wrapped stratum;

(iii) the ordered mixed maps, including the two-sided $\mu^{\mathrm{hyb}}_{\alpha_-;\eta;\beta_+,R}$, are the only Hall actions of projection-finite insertions on wrapped sectors in $\mathsf{Corr}^{E,\mathrm{hyb}}_R$ before reduced E-quotient;

(iv) the eight-word atlas plus pentagon coherence yield an associative finite-stage reduced Hall product after vanishing cycles, orientations, HN quotienting, the quotient-after-correspondence pseudofunctor, and the $\theta^{Q}_{\mu,R}$;

(v) for homogeneous x of charge γ, $I^{\mathrm{prot}}_R(Q_{E,R}x) \in s^{m^{\mathrm{Bch}}_R(\gamma)}\mathbb{C}[q^{\pm 1}, r^{\pm 1}]$, and the local/wrapped type of x is still measured by $b^{\mathrm{geom}}_R(\gamma)$.

Proof. Items (i) and (ii) are part of the elliptic-degree stratification and Lemma 0.2. Item (iii) is the definition of the mixed operations: before reduced E-quotient they are the only displayed operations remembering relative E-position between finite local supports and a wrapped leg. Items (iv) and (v) are the eight-word coherence and the protected integration clause. □

Corollary 0.7 *(No quotient-first universal hybrid repair; proved).* A candidate finite correspondence category for the wrapped sector gives the source object $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ only if it contains the unreduced local, wrapped, and ordered mixed source stacks of Definition 0.5, their anchors, the eight-word flag atlas, the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$ after Borel–Moore chains, the comparison maps $\theta^{Q}_{\mu,R}$, and protected integration with $\mathfrak{O}_{\mathrm{hyb},R} = 0$. A quotient of object stacks formed *before* the mixed and wrapped correspondences, or a determinant/Fock category carrying only the trace degree $s^{m^{\mathrm{Bch}}_R}$, does not define the hybrid source.

Proof. By Lemma 0.2, a positive elliptic-degree sector is not a finite-support Ran sector; only the rigidified wrapped prequotient with anchor, together with the unreduced mixed and wrapped correspondences, retains the relative E-position used by the source data. A quotient-first construction loses these anchors. □

Comparison with the two-stage chiral construction

The hybrid carrier is the finite $K3 \times E$ specialization datum attached to the two-stage construction CY_d-Cat $\to E_d$-HolFA$(X) \to$ ChirAlg(C) of Vol III Calabi–Yau quantum groups, read here as a chosen specialization datum rather than a constructed functor. The projection-finite local stratum is the retained holomorphic E_3-prefactorization input restricted to the smooth elliptic factor, and the wrapped stratum is the chain-level $K3$-fusion datum that the local stratum cannot see. Beilinson and Drinfeld's Ran construction [6, §3.4–§3.5] provides the $b = 0$ part; the $b > 0$ part is the Hall–Borcherds residual. The eight-word two-step binary flag atlas $w \in \{\mathrm{L}, \mathrm{W}\}^3$ is the smallest closed coherence calculus that detects the local/wrapped split through binary associativity; it is independent of the higher-coloured residual $\mathfrak{o}^{\mathrm{col}}_R$ which closes higher symmetric arities, units, and overlaps.

THE $D0$ SPECIALIZATION

The finite-stage components of $\mathfrak{o}_R^{\mathrm{stage}}$ and $\mathfrak{o}_R^{\lambda}$ are constructed in Appendix 7.2. The wrapped-leg construction and the eight-word atlas at finite height remain conditional on the named hybrid obstruction tuple; the $b = 0$ stratum is the classical projection-finite Ran sector of [6]. The hybrid carrier is the projection-finite Ran stratum on the $b = 0$ component and the wrapped stratum relative to the named finite obstruction tuple $\mathfrak{O}_{\mathrm{hyb},R}$.

Chiral operations on $\mathrm{Ran}^{\mathrm{hyb}}(E)$ require an orientation line on the cosection-reduced d-critical heart. On $K3 \times E$ this orientation descends through the reduced E-quotient only after the retained quotient-orientation datum is supplied; on rank-2 E-strata the Klein-four calculation gives the edge criterion and leaves the degree-one linearization character to be set to zero. Entry (L2), constructed in §0.2, is this orientation datum.

0.2 Cosection-twisted reduced d-critical orientation

The d-critical structure of the compact Hall sector on $K3 \times E$ is the second entry of the pro-object. It supplies a vanishing-cycle coefficient system, a Joyce–Upmeier–type orientation line, and a quotient-descent datum for the reduced E-quotient. These are the finite-stage operands of the Hall product. Before any Pfaffian wall or Weyl transport can be discussed, the orientation line must descend through the cosection-reduced moduli.

The compact Hall heart is determined by the chosen stability datum (σ, S) of Chapter 2; its K3-surface component lies in the Bridgeland–Bayer–Macri lineage [4, 12]. The PTVV (-1)-shifted symplectic structures [89] restrict to the finite-type stacks selected by the HN quotient and give Joyce d-critical structures [11] after classical truncation. The Joyce–Upmeier orientation line is taken on the cyclic A_∞ structure induced by this (-1)-shifted Calabi–Yau moduli geometry. The holomorphic symplectic two-form on the K3 fibre makes the unreduced Donaldson–Thomas theory of $K3 \times E$ vanish identically: it generates trivial first-order deformations that obstruct the virtual class. The standard remedy, as in reduced K3 enumerative geometry [75], is to quotient these out by a semiregularity cosection. The K3 fibre direction accordingly supplies a semiregularity cosection CS_C, surjective on reduced strata, whose kernel is the reduced obstruction theory. On the extension stack $\mathfrak{E}_{A,B}$ of short exact sequences $0 \to A \to C \to B \to 0$ — with projections p_1, p_2 to the substacks $\mathfrak{M}_A, \mathfrak{M}_B$ of sub- and quotient-object and q to $\mathfrak{M}_C$, the correspondence set up in §0.3 — the cosections of the three charge strata satisfy filtered Hall additivity

$$q^* \mathrm{CS}_C = p_1^* \mathrm{CS}_A + p_2^* \mathrm{CS}_B,$$

the compatibility hypothesis in Kiem–Li cosection localization [50, 52].

THE REDUCED DETERMINANT COMPLEX

For an object C in the chosen finite stage, write

$$K_C^{\mathrm{red}} = \mathrm{RHom}_{\mathrm{red}}(C, C),$$

the cosection-reduced self-Ext complex. The reduced orientation line is equipped with a chosen square root,

$$o_C \in \mathrm{Pic}(\mathfrak{M}_X^{\mathrm{or}}), \qquad o_C^{\otimes 2} \simeq \det K_C^{\mathrm{red}},$$

in the sense of [52], and is multiplicative under the reduced extension correspondences: $o_{C \oplus C'} \simeq o_C \otimes o_{C'}$ compatibly with the Hall additivity above. This is the data on which Thom–Sebastiani transport is supplied for the vanishing-cycle coefficient sheaves.

(O1) Strong reduced orientation through the quotient-orientation tower

The compact Hall product on $X = K3 \times E$ exists only after the reduced E-quotient. The orientation line must descend through this quotient, and descent is detected by a tower of mod-2 Borel-equivariant gerbes. For a charge stratum C:

(i) the reduced gerbe class

$$\alpha_C^{\mathrm{red}} = w_2(\det K_C^{\mathrm{red}}) + \mathrm{CS}_C^* \, w_2(\det \operatorname{coker} \mathrm{CS}_C) \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2);$$

(ii) the free E-class

$$\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2);$$

(iii) for each finite-stabilizer stratum S with stabilizer $H \subset E_{\mathrm{tors}}$, the equivariant class $\widetilde{\beta}_{C,S}^H \in H_H^2(S; \mathbb{F}_2)$, whose point-stabilizer restriction $\beta_{C,S}^H \in H^2(BH; \mathbb{F}_2)$ is read only after the mixed Borel terms, the stabilizer action on $H^\bullet(S; \mathbb{F}_2)$, and the spectral-sequence differentials have vanished.

Definition 0.8 *((O1) Strong multiplicative reduced orientation).* The (O1) datum at finite HN height R consists of: the reduced orientation line o_C with chosen square root; its multiplicativity under reduced extension correspondences with chosen Thom–Sebastiani transport; null-trivializations of α_C^{red}, $\alpha_C^{E,\mathrm{free}}$, $\widetilde{\beta}_{C,S}^H$ and $\beta_{C,S}^H$ on every charge stratum used by the Hall correspondences; and a choice of zero $H^1(BH; \mathbb{F}_2)$-linearization character $\lambda_{C,S}^H = 0$ for each finite stabilizer.

The full-level specialization at $H = E[2]$ is the Klein-four condition

$$\beta_{C,S}^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2, \quad (b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2),$$

$$\mathrm{res}_{\langle e_i \rangle} \beta_{C,S}^{E,2} = r_i \tau_i^2,$$

with vanishing $r_1 = r_2 = r_3 = 0$ on every retained two-torsion stratum. For full-level $H = E[N]$ with $2^a \parallel N$, $a \geq 2$, the two-primary residual takes value in $H^2(B(\mathbb{Z}/2^a)^2; \mathbb{F}_2) = \mathbb{F}_2\langle y_1, x_1x_2, y_2\rangle$ and contains the mixed coefficient $A_{12}^{(N)}$ *not* detected by cyclic restrictions.

(O1) is therefore not a single Picard-line assertion: it is the ordinary square root, the null-trivialization of the reduced and equivariant degree-two gerbes, and the choice of zero degree-one linearization characters for the actual finite stabilizers. At finite HN height this is the cocycle system of [52] restricted to $K3 \times E$.

$(\mathrm{O1})^+$ Weyl-equivariant transport along $W^{(2)}(\Lambda_{II}^{2,1})$

The orientation gerbe descends through the reduced E-quotient, but the Igusa structure also requires it to be compatible with the Weyl group of the BKM target. For each of the three type-II simple reflections s_{δ_i} ($i = 1, 2, 3$), the $K3 \times E$ side carries a wall transport

$$\tau_i : o_C \longrightarrow s_{\delta_i}^* o_C.$$

The word *coherent* is a hypothesis. At finite height, the first object is the finite partial action groupoid $\mathcal{G}_R$ whose objects are retained oriented quotient strata and whose arrows are the retained type-II wall transports; a choice of determinant-line lifts gives a 2-cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$. Only on a complete $W^{(2)}(\Lambda_{II}^{2,1})$-orbit with constant coefficients does this reduce to the group cohomology class

$$[c_o] \in H^2\big(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2\big).$$

Definition 0.9 *((O1)$^+$ Weyl-equivariant determinant-line transport).* The (O1)$^+$ datum is the chosen set of lifts $\{\tau_i\}$ satisfying $[c_o] = 0$ with chosen 1-cochain killing c_o, the normalization $\tau_i^2 = 1$, and the transport of the quotient-orientation cocycle

$$\mathfrak{q}_C = \left(\alpha_C^{\mathrm{red}}, \alpha_C^{E,\mathrm{free}}, \{(\beta_C^H, \lambda_C^H)\}_{H \in \mathcal{H}(C)}\right)$$

along τ_i between charge strata over Weyl-related Gram degrees. The torsor defect of this transport is

$$\omega_{i,C} \in H^1(\mathfrak{M}^{\mathrm{red}}_{s_{\vartheta_i}C}; \mathbb{F}_2) \oplus \bigoplus_{H \in \mathcal{H}(C)} H^1(BH; \mathbb{F}_2),$$

required to vanish. The Coxeter presentation of $W^{(2)}(\Lambda_{II}^{2,1})$ then assembles these lifts into an honest action on the oriented Pfaffian line over the wall-complement of the reduced compact Hall sector.

(O1)$^+$ is a determinant-line monodromy datum. No scalar formula enters; the transports must move the entire null-trivialization datum, not only the underlying line. The (O1)$^+$ data comprise $[c_o] = 0$, the chosen cochain, the unitarity $\tau_i^2 = 1$, and the torsor vanishing $\omega_{i,C} = 0$ on every retained orbit.

Cosection localization and the cosection-reduced heart

The K3 semiregularity cosection CS_C selects a reduced obstruction theory; the corresponding Behrend constructible function ν^{red} produces the reduced vanishing-cycle complex Φ^{red} used in the Hall product [5, 50]. The filtered Hall additivity $q^* \mathrm{CS}_C = p_1^* \mathrm{CS}_A + p_2^* \mathrm{CS}_B$ on extension stacks is the cosection-localization compatibility required by the Hall correspondence; the Joyce–Upmeier orientation lines are required to be Thom–Sebastiani-coherent on the resulting two-step and four-input flag stacks.

PROPOSITION 0.10 *(Finite d-critical Hall product; conditional on the (O1) and atlas data).* Fix R and closed-configuration substacks $\mathfrak{M}^{\mathrm{loc,cl}}_{c,R,I}$ in a noetherian abelian Hall heart, and assume:

(i) finite-type quasi-smooth derived Artin enhancements of the selected object, extension, and two-step flag stacks, with perfect obstruction theories;
(ii) finite residual inertia after scalar, E-translation, and wrapped rigidifications;
(iii) retained closed flag-Quot/filtration extension and flag stacks, or an explicitly specified compact-support six-functor model, with admissible p^* and $q_!$;
(iv) PTVV (-1)-shifted symplectic structures restricting to these substacks, classical truncations carrying Joyce d-critical structures [11, 89];
(v) surjective K3 semiregularity cosections with filtered Hall additivity;
(vi) Joyce–Upmeier orientation lines and orientation transports [52] with all finite-stabilizer characters trivialized;
(vii) Thom–Sebastiani transport for the reduced vanishing-cycle sheaves satisfying two-step flag pentagons and higher-coloured configuration coherences.

Then the finite reduced local-local product

$$m_R^{\mathrm{red}} = q_! \circ \mathrm{TS}_R^{\mathrm{red}} \circ p^*$$

is an honest morphism on $H_c^\bullet(\mathfrak{M}^{\mathrm{loc,cl}}_{c,R,I}, \Phi^{\mathrm{red}}_{c,R,I} \otimes o^{\mathrm{red}}_{c,R,I})$ and is associative on the finite stage.

The first obstruction to removing these hypotheses is finite inertia: proper finite-type correspondences do not by themselves eliminate residual $\mathbb{G}_m$, direct-sum, or parabolic automorphisms, and a residual $\mathbb{G}_m$

already contributes $H^\bullet(B\mathbb{G}_m) = \mathbb{C}[u]$ to protected cohomology. The phrase *finite stage* here means finite charge support, finite-type stacks, and finite-dimensional protected cohomology after the stated quotients and admissibility hypotheses; it does not mean that the Hall correspondence morphisms are themselves finite morphisms. The raw exact-triangle stack is not a proper Hall correspondence; the proper object is the retained compactified filtration stack, or else the compact-support functor $q_!$ is an extra datum.

Quotient-after-correspondence ordering

The E-quotient is taken *after* the extension correspondences, not before. A pseudofunctor $\mathsf{Corr}_R^{E,\mathrm{hyb}} \to \mathsf{Corr}_R^{\mathrm{red,hyb}}$ supplies the descent; applying the vanishing-cycle Borel–Moore functor with orientation lines included gives $Q_{E,R}$ between the corresponding finite correspondence module categories. For each operation $\mu_R^\bullet = q_!\,\mathrm{TS}\,p^*$ a chosen comparison isomorphism

$$\theta_{\mu,R}^{Q} : Q_{E,R}\, q_!\,\mathrm{TS}\, p^* \xrightarrow{\sim} \overline{q}_!\overline{\mathrm{TS}}\,\overline{p}^*(Q_{E,R}^{\boxtimes k})$$

is part of the datum. Its compatibility with the eight-word flag 2-morphisms and HN transitions is the (O1) descent calculus. A quotient-first construction (object-stack quotient followed by a new Hall product) is not the same datum.

The orientation obstruction class

The orientation lane carries the residual

$$\mathfrak{O}_{\mathrm{or},R} = \left(\mathfrak{o}_C^{\mathrm{red}}, \mathfrak{o}_C^{E,\mathrm{free}}, \{\mathfrak{o}_{C,S}^{H}\}_{H,S}, \{\mathfrak{o}_{C,S}^{\lambda^H}\}_{H,S}, [c_o]_R,\ \xi_{c_o,R},\ \{\mathfrak{o}_{i,R}^{\tau,2}\}_i,\ \{\omega_{i,C,R}\}_i\right)$$

required to vanish at every finite HN height, with chosen null-trivializations. Theorems 7.12 and 7.13 of Appendix 7 are the quotient-orientation and Weyl-transport criteria relative to that retained datum. The Klein-four and two-primary finite-stabilizer lemmas there give the edge coordinates of (O1), and the wrapped-stratum Weyl-equivariant transport along $W^{(2)}(\Lambda_{II}^{2,1})$ is reduced to the no-braid type-II Coxeter presentation and the Maass-character match.

Comparison with the Calabi–Yau quantum-group stratum

Vol III's Calabi–Yau quantum-groups stratification observes that $K3 \times E$ is not presented by a global toric/quiver-variety NCCR atlas of the Schiffmann–Vasserot type, so the orientation lane cannot be supplied by such an NCCR-induced heart; the cosection-twisted reduced d-critical structure is forced on the chosen stability heart instead. The displayed Borel calculation at $E[2]$ is the smallest finite stabilizer detected by the cosection-reduced determinant; even $N \geq 4$ stabilizers carry the mixed coefficient $A_{12}^{(N)}$, which the displayed cyclic restrictions miss, and which (O1) requires to be null-trivialized as part of the full degree-two gerbe.

The orientation o_R trivializes the Joyce–Upmeier obstruction to a compatible Hall product on the cosection-twisted heart. Once o_R is in place, the Hall product, the collision coproduct, and the protected primitive extraction are unambiguous. Entry (L3) is the Hall apparatus of §0.3.

0.3 *Hall correspondences: product, coproduct, primitives*

The third entry of the Dirac–Igusa pro-object is the Hall structure on the cosection-reduced d-critical heart. It is the algebraic content that the orientation lane equips and the hybrid carrier indexes. Three operations live here: the Hall product m_R^{red}, the collision coproduct Δ_R^{ch}, and the primitive projection

P_R^{Prim}. After normal-ordered Gram descent these form a graded Hopf datum on the protected primitive layer; recognition of this Hopf datum with the Borcherds–Kac–Moody target is the §5.6 obligation.

The Kontsevich–Soibelman cohomological Hall algebra of a 3-Calabi–Yau category [59] is the prototype; on $K3 \times E$ the cosection reduction [50, 52] replaces the unreduced critical CoHA, and the wrapped sector is the residual the hybrid carrier of §0.1 retains. The Maulik–Okounkov stable-envelope construction and the Schiffmann–Vasserot theorems on Drinfeld doubles of CoHA operate on the same input; their integral form is the Hall side of the present construction.

The compact Hall product

Definition 0.11 *(Compact Hall product on the cosection-reduced heart).* Let $C_{\sigma,S,\leq R}$ be the finite HN quotient of §0.1. Assume the retained Liu–Hilbert compactified correspondence and the compact-support pushforward of Proposition 0.10, including finite residual inertia, base-change, projection-formula, Thom–Sebastiani, and quotient-after-correspondence coherences. The compact Hall product at height R is the operation

$$m_R^{\mathrm{red}} = q_{c,c'!} \circ \mathrm{TS}_{c,c'}^{\mathrm{red}} \circ p_{c,c'}^* : H_*^{\mathrm{BM}}\big(\mathfrak{M}_c^\sigma \times \mathfrak{M}_{c'}^\sigma, \Phi_c^{\mathrm{red}} \boxtimes \Phi_{c'}^{\mathrm{red}} \otimes o_c \boxtimes o_{c'}\big) \to H_*^{\mathrm{BM}}\big(\mathfrak{M}_{c+c'}^\sigma, \Phi_{c+c'}^{\mathrm{red}} \otimes o_{c+c'}\big)$$

attached to the retained compactified extension correspondence

$$\mathfrak{M}_c^\sigma \times \mathfrak{M}_{c'}^\sigma \xleftarrow{p_{c,c'}} \overline{\mathfrak{E}}_{c,c'}^{\mathrm{ret},\sigma} \xrightarrow{q_{c,c'}} \mathfrak{M}_{c+c'}^\sigma,$$

in the cosection-reduced d-critical heart, with vanishing-cycle coefficients Φ^{red}, Joyce–Upmeier orientation lines o, and Thom–Sebastiani transport $\mathrm{TS}_{c,c'}^{\mathrm{red}}$. Associativity on the finite stage is the eight-word two-step binary flag pentagon together with the quotient-after-correspondence and HN-transition coherences of §0.1.

For local-only inputs and ordered mixed and wrapped inputs, the same construction produces the three operation families $\mu_R^{\mathrm{loc}}, \mu_R^{\mathrm{hyb}}, \mu_R^{\mathrm{wr}}$ of §0.1. Existence of the product as a genuine morphism on Borel–Moore chains requires the admissibility/properness residuals of that section to vanish.

The collision coproduct on the source coalgebra

The collision coproduct is defined on the same correspondence geometry, read in the opposite direction: instead of contracting an extension to a single middle term, one expands an object into the pairs of its sub/quotient extensions. At finite stage, this is the 2-step flag stack already supplied as part of the hybrid datum.

Definition 0.12 *(Collision coproduct).* The finite collision coproduct

$$\Delta_R^{\mathrm{ch}} : C_{X,R} \longrightarrow C_{X,R} \otimes C_{X,R}$$

is the chain-level pull-push along the 2-step flag stack $\mathfrak{Flag}_{c,c',R}^{(2)}$ of subobjects of charge c and quotients of charge c' in $\mathfrak{M}_{c+c',R}^\sigma$, computed in the cosection-reduced d-critical heart, before the reduced E-quotient. If the higher-coloured residual, unit maps, refinement coherences, and quotient-after-correspondence residuals of §0.1 vanish, then after $Q_{E,R}$ it satisfies the coalgebra identities

$$(\Delta_R^{\mathrm{ch}} \otimes 1)\Delta_R^{\mathrm{ch}} = (1 \otimes \Delta_R^{\mathrm{ch}})\Delta_R^{\mathrm{ch}}, \qquad (\varepsilon_R^C \otimes 1)\Delta_R^{\mathrm{ch}} = (1 \otimes \varepsilon_R^C)\Delta_R^{\mathrm{ch}} = \mathrm{id},$$

which are the dual of the Hall pentagon on the same flag stack. For $R' \geq R$ the transition map $\rho_{R'R}^C$ commutes with Δ^{ch}.

This is the source coalgebra of §0.5; its identification with the bar construction $\bar{B}_E^{\mathrm{ch}}(\mathcal{F}_X^{\mathrm{hyb}})$ is the bar comparison there.

Primitive projection and the Hopf datum

The primitive projection on a graded conilpotent Hopf object is unique once the augmentation, product, and coproduct are supplied.

Definition 0.13 *(Primitive Hall sub-bracket).* Write

$$\widetilde{P}_X = H^0 \operatorname{Prim}_{\mathrm{prot}}(\mathcal{F}_X) = \ker\bigl(\widetilde{\Delta} : \mathcal{F}_X \to \mathcal{F}_X \otimes \mathcal{F}_X\bigr)$$

for the primitive layer of the protected Hall Hopf object, where $\widetilde{\Delta} = \Delta - (\eta\varepsilon \otimes \mathrm{id} + \mathrm{id} \otimes \eta\varepsilon)$. The Hall product restricts to a $\widehat{\Gamma}_X$-graded bracket

$$[\,\cdot\,,\,\cdot\,]_X : \widetilde{P}_{X,\widehat{c}} \otimes \widetilde{P}_{X,\widehat{c}'} \longrightarrow \widetilde{P}_{X,\widehat{c}\star\widehat{c}'},$$

the protected primitive Hall bracket. After normal-ordered Gram descent (§0.1) and the Hopf-radical quotient, the descended bracket is Γ_{gram}-graded and the descended primitive layer is the candidate $P_X^{\Pi,\mathrm{red}}$ to be identified with $\mathfrak{g}_{\Delta_5}$.

Lemma 0.14 *(Primitive bracket lifting; conditional on the Frobenius defect $\mathfrak{o}_F = 0$).* Let $\langle\cdot,\cdot\rangle_X$ be the protected Hopf pairing on $\widetilde{P}_X$, non-degenerate on the radical quotient. The trilinear Frobenius defect

$$\mathfrak{o}_F(x,y,z) := \langle [x,y],z\rangle + (-1)^{|x|\cdot|y|}\langle y,[x,z]\rangle, \quad x,y,z \in \widetilde{P}_X, \quad c_x + c_y + c_z = 0,$$

viewed as a class in $C^3_{\mathrm{cyc}}(\widetilde{P}_X;k)$ after the reduced E-quotient and radical quotient, vanishes precisely when the bracket is graded-symmetric for the Hopf pairing. Vanishing is the cyclic-CY-3 Serre-functor identity of Theorem 4.1. Non-degeneracy of the Hopf pairing alone does *not* force $\mathfrak{o}_F = 0$; it only detects the vanishing once supplied.

Lemma 0.15 *(Hopf radical is a Lie ideal and a coideal; conditional on $\mathfrak{o}_F = 0$ and Hopf adjointness).* Let P_R^{Π} be a finite normal-ordered primitive stage with transported product m_R^{Θ}, coproduct Δ_R^{Θ}, graded symmetric Hopf pairing $\langle\cdot,\cdot\rangle_R$, and radical Rad_R^{Π}. Assume Hopf adjointness $\langle m_R^{\Theta}(x,y),z\rangle_R = \langle x\otimes y, \Delta_R^{\Theta} z\rangle_R$ and non-degeneracy on $P_R^{\Pi}/\mathrm{Rad}_R^{\Pi} \otimes P_R^{\Pi}/\mathrm{Rad}_R^{\Pi}$. Then

$$(\bar{q}_{\Pi,R} \otimes \bar{q}_{\Pi,R})\Delta_R^{\Theta}(r) = 0 \quad (r \in \mathrm{Rad}_R^{\Pi}),$$

i.e. Rad_R^{Π} is a coideal. If, in addition, $\mathfrak{o}_{F,R} = 0$, then Rad_R^{Π} is a Lie ideal.

Proof. For $r \in \mathrm{Rad}_R^{\Pi}$ and $x, y \in P_R^{\Pi}$, Hopf adjointness gives $\langle x\otimes y, \Delta_R^{\Theta}(r)\rangle_R = \langle m_R^{\Theta}(x,y), r\rangle_R = 0$; non-degeneracy of the quotient tensor pairing then kills the class. The Lie-ideal half is direct: if $\mathfrak{o}_{F,R} = 0$ then $\langle [x,r]_\Theta, z\rangle_R = -(-1)^{|x||r|}\langle r,[x,z]_\Theta\rangle_R = 0$ for every z, so $[x,r]_\Theta \in \mathrm{Rad}_R^{\Pi}$. □

Integral form and stable-envelope transport

Two parallel constructions act on the same compact Hall heart:

(i) the integral form of the Schiffmann–Vasserot elliptic Hall algebra, whose $K3 \times E$-specialization carries the Maulik–Okounkov stable envelope;

(ii) the Drinfeld double $\mathcal{D}_\hbar(\mathrm{CoHA}_{K3\times E})$ of Vol III, whose positive half is the protected primitive Hopf algebra above and whose Drinfeld pairing is the Hopf pairing.

The compatibility datum between these two constructions is a comparison with Calabi–Yau quantum groups. A different constant or sign would be a distinct mathematical claim; the Drinfeld double is not asserted here as an input theorem.

CONILPOTENT SOURCE COALGEBRA AND HOPF RADICAL QUOTIENT

Definition 0.16 *(Conilpotent source coalgebra).* At finite stage, $C_{X,R}$ is the conilpotent chiral coalgebra on E obtained from the collision coproduct on $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}$ after the reduced E-quotient: it is filtered by a finite increasing source charge filtration

$$0 = F^{\mathrm{co}}_{R,-1} \subset F^{\mathrm{co}}_{R,0} \subset \cdots \subset F^{\mathrm{co}}_{R,N_R} = C_{X,R}$$

whose associated graded pieces are finite dimensional in each Γ_R^{Π}-degree — where $\Gamma_R^{\Pi} := \overline{\Pi}_X(\widehat{\Gamma}_R)$ is the image, under the *normal-ordered* Gram homomorphism $\overline{\Pi}_X$ of Chapter 2, of the retained normal-ordered charge window $\widehat{\Gamma}_R$ at Harder–Narasimhan height R (§0.1); the additive $\overline{\Pi}_X$, not the raw quadratic Π_X, is forced as the grading by the no-go theorem of Chapter 2 — and whose reduced iterated coproduct is nilpotent on each finite stage. Conilpotence is recorded by the defect $(\overline{\Delta}^{\mathrm{ch}}_R)^{N_R+1}$, required to vanish. Inverse-limit completion gives the source coalgebra $C_X = \varprojlim_R C_{X,R}$.

PROPOSITION 0.17 *(Sectorial Hall truncation and inverse limit; conditional on* [K3×E-fin], [K3×E-atlas], *and* [ML]*).* Under the three standing hypotheses [K3×E-fin] (finite-type semistable substacks at bounded HN height), [K3×E-atlas] (finite d-critical/cosection Hall atlas with Joyce–Upmeier orientation lines), and [ML] (Mittag–Leffler descent for the HN inverse system), the finite Hall product, coproduct, primitive projection, and Hopf pairing are compatible with the transition maps $\rho_{R'R}$, and their inverse limits define the completed Hall Hopf datum $\widetilde{P}_X, C_X$. The Mittag–Leffler hypothesis controls the $R^1 \varprojlim$ obstruction on every finite-window slice; without it, only a compatible pro-object is obtained.

The hypothesis [K3×E-fin] is not a consequence of fibrewise K3 boundedness alone. A precise sufficient input is product-stability boundedness in the Abramovich–Polishchuk/Liu sense [1, 61]: for a rational K3 stability condition $\sigma_S = (\mathcal{A}_S, Z_S)$ and Liu's product stability $\sigma_{s,t}$ on $D^b(\mathrm{Coh}(S \times E))$, fixed-class boundedness

$$\mathfrak{M}^{ss}_{\sigma_{s,t}}(\gamma) \text{ is finite type over } \mathbb{C}$$

together with Liu's support property would imply [K3×E-fin] for bounded h_S-windows. Piyaratne–Toda boundedness for threefold Bridgeland/BMT [90] does not automatically supply this theorem; fibrewise Bayer–Macri boundedness alone does not either.

WALL-CROSSING AND CHAMBER DEPENDENCE

The lattice cocycle $B(c, c')$ of the dyonic Mukai construction is chamber-independent. The compact Hall side $D_0(X; \sigma, S)$ is chamber-dependent: finite-type semistable substacks change under wall-crossing, and the Kontsevich–Soibelman wall-crossing formula [59] relates Hall counts in adjacent chambers. If the datum $D_0(X; \sigma, S)$ is supplied in chamber σ, then $\overline{\Pi}_X$ remains a homomorphism and the recognition target is $\mathfrak{g}_{\Delta_5}$ provided the chamber identifications hold for the OP/DT scalar import; that the chosen σ lies in the OP/DT chamber, or that the wall-crossing transport is compatible with $\overline{\Pi}_X$, is a separate conjectural input. No chamber-independence of the recognition target is asserted.

THE HALL OBSTRUCTION CLASS

The Hall lane carries the residuals

$$\mathfrak{O}_{\mathrm{Hall},R} = (\mathfrak{o}^{\mathrm{stage}}_R,\ \mathfrak{o}^{\mathrm{conf}}_R,\ \mathfrak{o}^{\mathrm{prop}}_R,\ \mathfrak{o}^{\mathrm{TS}}_R,\ \mathfrak{o}^{\mathrm{assoc}}_R,\ \mathfrak{o}_{F,R},\ \mathfrak{o}^{\mathrm{coid}}_R,\ \mathfrak{o}^{\mathrm{ML}}_{R'R}).$$

On the projection-finite stratum the wrapped product is the classical d-critical CoHA product after cosection reduction; on the wrapped stratum the construction is open and governed by the (D0) obstruction. Hopf adjointness, Frobenius vanishing, and PBW compatibility are independent constructions on the descended primitive layer.

Comparison with Drinfeld doubling

Vol III's Drinfeld doubling $Y^+(X) \to G(X)$ is the comparison target for the compact source side; the integral form is the chiral lift of the Schiffmann–Vasserot elliptic Hall algebra after the stable-envelope and completion data are fixed. Maulik–Okounkov stable-envelope transport supplies the integral form on the projection-finite Ran stratum; the wrapped stratum is the corresponding stratum-completion. No black-box equality CoHA $= W_\infty$ is asserted: $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\mathfrak{gl}_1)$ in Vol III, and $W_{1+\infty}$ appears only after Drinfeld doubling, centre, vacuum evaluation. The same discipline applies here: the protected primitive Hopf algebra of $K3 \times E$ becomes the BKM target only after the Hopf radical quotient, the Frobenius identity, and the no-extra-relations theorem of §0.6.

Under $\mathfrak{O}_{\mathrm{Hall},R} = 0$ and the Mittag–Leffler transition datum, the Hall correspondences give a finite-stage protected primitive Hopf datum on the cosection-twisted heart. This datum lives at the operator level and has no automatic comparison to the scalar denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ imported in Chapter 4. Closing the cross-level identity ②↔③ requires a chain-level scalar incarnation of the primitive structure: a Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$ on the reduced moduli, a Dirac block decomposition, and a first-order Pfaffian section whose $\mathbf{Sp}_4(\mathbb{Z})$-pullback recovers Δ_5. This is the fourth entry (L4), constructed in §0.4.

0.4 First-order Dirac block and protected Pfaffian

The fourth entry of the Dirac–Igusa pro-object is the protected Pfaffian line and its first-order Dirac block decomposition. The Pfaffian, not the determinant, is the object of interest for a precise reason: the automorphic target Δ_5 is the square root of the scalar weight-ten form $\Delta_5^2 = \chi_{10}$, and the square root carries the order-two Maass multiplier ν_{Δ_5} that the square sends to the trivial character. Geometrically the determinant line $\mathscr{L}_{\det}$ is canonically trivialisable and forgets orientation; its square root, the Pfaffian line, retains the orientation character ϵ_o, and it is precisely the match $\epsilon_o = \nu_{\Delta_5}$ that distinguishes the first-order object Δ_5 from its orientation-blind square Δ_5^2. Two definitions of this Pfaffian meet. The geometric definition starts from the cosection-reduced d-critical heart, carries a Joyce–Upmeier orientation, descends to the reduced quotient, and associates a Pfaffian line to the determinant of the reduced obstruction theory in the Brav–Bussi–Dupont–Joyce–Szendrői framework [11]. The algebraic construction starts from the imported Borcherds–Kac–Moody denominator algebra, fixes a positive window in Γ_{gram}, and writes the algebraic super-Pfaffian

$$\mathrm{sPf}_R^{\mathrm{alg}}(D) = \prod_{\gamma \in A_R} (1 - x^\gamma)^{\mathrm{sdim}\, V_\gamma}$$

of the trivial first-order block decomposition D_γ on the target side. The Pfaffian–Dirac theorem of Chapter 6 is the identification of these two Pfaffian sections after the chosen line-comparison ι_{aut} and the four-obstruction datum of Appendix 7.

Pfaffian line and squaring isomorphism

Definition 0.18 *(Pfaffian line on the cosection-reduced quotient).* Let K_C^{red} denote the cosection-reduced self-Ext complex on a charge stratum, with reduced determinant $\det K_C^{\mathrm{red}}$ and Joyce–Upmeier orientation o_C. The Pfaffian line at finite stage is

$$\mathscr{L}_{\mathrm{Pf},R} = \bigotimes_C \mathrm{Pf}(K_{C,R}^{\mathrm{red}})$$

on the reduced moduli, with squaring isomorphism

$$\mathscr{L}_{\mathrm{Pf},R}^{\otimes 2} \simeq \mathscr{L}_{\det,R}$$

in the Brav–Bussi–Dupont–Joyce–Szendrői sense [11]. The successor maps $\rho_{R^+R}^{\mathrm{Pf}}$ commute with these squaring isomorphisms, identifying the inverse limit with a Pfaffian line $\mathscr{L}_{\mathrm{Pf},X}$ on the inverse-limit object. A normalized section $\mathrm{pf}_{X,R}$ at finite stage and $\mathrm{pf}_X = \varprojlim_R \mathrm{pf}_{X,R}$ are part of the datum.

Pfaffian-to-automorphic comparison

The recognition statement is read on the automorphic line. A section-level identification $\mathrm{pf}_X = \Delta_5$ requires a comparison of $\mathscr{L}_{\mathrm{Pf},X}$ with the pullback of the automorphic weight-five line to the moduli base.

Definition 0.19 *(Pfaffian-to-automorphic line comparison).* A section-level Pfaffian-to-automorphic comparison is a chosen isomorphism of line bundles

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$$

to the pullback of the automorphic weight-five line, twisted by the multiplier ν_{Δ_5}, compatible with cusp trivialization at $\mathfrak{c}_\infty$ and with the squaring map $\mathscr{L}_{\mathrm{Pf},X}^{\otimes 2} \simeq \mathscr{L}_{\det,X}$.

First-order Dirac block decomposition

Definition 0.20 *(First-order Dirac block).* For $\gamma \in \Gamma_R^{\Pi,+}$, write $P_{R,\gamma}^{\Pi,+}$ for the cusp-positive primitive piece in degree γ and put $x^\gamma = q^n r^l s^m$ for $\gamma = (n, l, m)$. The two-term Pfaffian/Koszul block on $P_{R,\gamma}^{\Pi,+} \oplus (P_{R,\gamma}^{\Pi,+})^\vee[1]$ is the skew operator

$$J_\gamma = \begin{pmatrix} 0 & 1 - x^\gamma \\ -(1 - x^\gamma) & 0 \end{pmatrix}, \qquad \mathrm{Pf}(J_\gamma) = (1 - x^\gamma)^{\operatorname{sdim} P_{R,\gamma}^{\Pi,+}}.$$

For $\gamma \notin \Gamma_R^{\Pi,+}$, set $P_{R,\gamma}^{\Pi,+} = 0$. The first-order block of $\mathfrak{D}_X^{D_0}$ at height R is the direct sum

$$\mathfrak{D}_{X,R}^{D_0} = \bigoplus_{\gamma \in \Gamma_R^{\Pi,+} \setminus \{0\}} J_\gamma,$$

acting on $\bigoplus_\gamma P_{R,\gamma}^{\Pi,+} \oplus (P_{R,\gamma}^{\Pi,+})^\vee[1]$.

The principal symbol of $\mathfrak{D}_{X,R}^{D_0}$ with respect to the finite Chevalley–Eilenberg/Koszul filtration is the normal-ordered Hall bracket of §0.3: the Pfaffian line is the determinant detector, and the symbol carries the Lie data. The phrase *first-order* means the Pfaffian/primitive object before scalar squaring; no analytic Dirac operator on a constructed state space is asserted.

The Pfaffian product at finite stage

Proposition 0.21 *(Pfaffian product at finite stage; conditional on (D0-Pf)).* Suppose the orientation lane (O1, O1$^+$) has descended to the reduced quotient, the first-order block decomposition is supplied, and the support and parity residuals

$$\mathfrak{d}^{\mathrm{supp}}_{R,\gamma} = \begin{cases} P^{\Pi,+}_{R,\gamma}, & a_\Delta(\gamma) = 0 \\ 0, & a_\Delta(\gamma) \neq 0 \end{cases}, \qquad \mathfrak{d}^{\mathrm{par}}_{R,\gamma} = p^{\bar 0}_{R,\gamma} - p^{\bar 1}_{R,\gamma} - a_\Delta(\gamma) \in \mathbb{Z},$$

together with the leading-coefficient residual $\mathfrak{d}^{\mathrm{lead}}_R = \mathrm{lc}_{\mathfrak{c}_\infty}(\mathrm{pf}_{X,R}) - 64$, all vanish on the test window, where $a_\Delta(\gamma)$ is the Borcherds exponent and $p^{\bar p}_{R,\gamma} = \dim P^{\Pi,+}_{R,\gamma,\bar p}$. Then the Pfaffian section at finite stage has expansion

$$\mathrm{pf}_{X,R}\,|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\substack{\gamma \in \Gamma^{\Pi,+}_R \\ \gamma \neq 0}} (1 - x^\gamma)^{\operatorname{sdim} P^{\Pi,+}_{R,\gamma}}.$$

The cusp-positive Pfaffian half is recorded; negative root representatives and the Cartan/imaginary completion enter the Lie recognition statement of §0.6.

The support residual is a super-vector-space residual, not a K_0 class: its vanishing excludes invisible even–odd cancelling primitive pieces in target-zero degrees. The parity residual is a signed superdimension residual. These finite-stage residuals on the test window do not by themselves prove compact source coverage; the inclusion $A_R \subset \Gamma^{\Pi,+}_R$ of active target degrees in the source image is itself a residual condition, not a consequence of the Borcherds product.

Local Pfaffian wall on type-II walls

The (O2) obstruction concerns the local Pfaffian normal form on the three simple type-II walls $\mathbb{H}_{\partial_i}$, $i = 1, 2, 3$, of the Borcherds–Cartan datum $\mathscr{B}_{\Delta_5}$.

Definition 0.22 *(Rank-one type-II Pfaffian crossing).* For a simple type-II root ∂, let T_∂ be a real analytic tangent chart with trivial s_∂-action, set $U_\partial = T_\partial \times (-\varepsilon, \varepsilon)$ with $s_\partial(t, u) = (t, -u)$, and let $\mathscr{L}^{\tan}_\partial$ be an oriented tangent Pfaffian line with s_∂-invariant unit v_∂. The rank-one odd Pfaffian crossing is the skew family

$$D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad K^{\mathrm{cross}}_\partial = [\mathbb{R} \xrightarrow{u} \mathbb{R}],$$

in odd Pfaffian parity. The total local Pfaffian section is

$$\mathrm{pf}_\partial = v_\partial\, u, \qquad s_\partial^* \,\mathrm{pf}_\partial = -\,\mathrm{pf}_\partial, \qquad s_\partial^*(\mathrm{pf}_\partial^2) = \mathrm{pf}_\partial^2.$$

Proposition 0.23 *(Rank-one crossing gives the type-II local sign; proved).* For $i = 1, 2, 3$, the rank-one crossing of Definition 0.22 forms the finite generic type-II local-sign datum with $N^{\mathrm{Pf}}_{\partial_i} = 1$, and at finite HN height R, with successor maps preserving the tangent charts, the wall coordinate $u_{\partial_i,R}$, and the invariant units, the (O2) residual entries

$$\mathfrak{d}^{\mathrm{chart}}_{\partial_i,R},\ \mathfrak{d}^{\mathrm{wall}}_{\partial_i,R},\ \mathfrak{d}^{\mathrm{split}}_{\partial_i,R},\ \mathfrak{d}^{\mathrm{unit}}_{\partial_i,R},\ \mathfrak{d}^{\mathrm{rank}}_{\partial_i,R},\ \mathfrak{d}^{\mathrm{tr}}_{\partial_i,R'R}$$

vanish for these local models, and $\varepsilon^{\mathrm{loc}}_{\partial_i,R} = -1$.

Proof. The fixed locus of $s_\partial(t,u) = (t,-u)$ is $u = 0$, so the wall and the tangent fixed locus agree. The determinant factor is split: the tangent line is $\mathscr{L}_\partial^{\tan}$, and the normal factor is $K_\partial^{\mathrm{cross}}$. The unit v_∂ is invariant by construction. The skew matrix has Pfaffian u, so the normal-mode rank is one. Reflection sends $u \mapsto -u$ and fixes v_∂; hence the local Pfaffian section changes sign and its square is fixed. The transition hypothesis is exactly preservation of these data under successor maps. □

Half-Hilbert orbit and the type-II sign sum

The (O2) obstruction also requires a wall *atlas*: at finite HN height R, each retained component and boundary stratum of $\mathbb{H}_{\partial_i}$ needs a chart of the rank-one form, and the normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps must agree on chart overlaps. A single generic chart computes only a local sign.

The half-Hilbert orbit of the type-II reflection group on the relevant retained stratum carries a sign sum of size $4096 = 2^{12}$, counting the supplied finite-window normal Pfaffian signs across the twelve retained components and boundary strata of the type-II wall configuration; this sum is the signature of the (O2) wall normal-form datum. Local rank-one charts contribute the local factors; the chosen Weyl lifts convert the retained global system into the orientation character on the $W^{(2)}(\Lambda_{II}^{2,1})$-orbit. The full normal-form theorem with overlap, boundary, and component coherence is relative to the retained $(O2_{\mathrm{atlas}})$ datum and Theorem 7.18 of Appendix 7.5; the local model is the finite rank-one wall factor.

PROPOSITION 0.24 *(Generic local sign and the orientation character).* Suppose $(\mathrm{O1})^+$ supplies a normalized lift $\tau_{i,R,S} : o_{R,S} \to s_{\partial_i}^* o_{R,S}$ of the simple type-II reflection on every retained orbit, and the (O2)/hybrid residual for ∂_i vanishes. In a local chart with $K^{\mathrm{nor}}_{\partial_i,R} \simeq [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$, with normal Pfaffian unit v_i and $s_{\partial_i}(v_i) = v_i$, let $\ell_{\partial_i,R,S}$ be the path in the reduced quotient wall-complement carrying $u_i = \epsilon$ to $u_i = -\epsilon$, closed by $\tau_{i,R,S}$. Then

$$M_{\ell_{\partial_i,R,S}}(v_i u_i) = v_i(-u_i) = -v_i u_i, \qquad \epsilon_{o,R}(s_{\partial_i}) = (-1)^{N^{\mathrm{Pf}}_{\partial_i,R}}.$$

For the rank-one type-II model $\epsilon_{o,R}(s_{\partial_i}) = -1 = \nu_{\Delta_5}(s_{\partial_i})$. The global statement $\epsilon_o|_{W^{(2)}(\Lambda_{II}^{2,1})} = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}$ is the chamber-permuting half of the Pfaffian–Dirac orientation identity, conditional on the (O2) wall atlas.

Pfaffian normalization and trace

The supplied Pfaffian-to-automorphic comparison ι_{aut} then gives, on the imported Borcherds product side,

$$\iota_{\mathrm{aut}}\big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)\big) = \mathcal{D}_X = \Delta_5,$$

with cusp-side expansion

$$\mathrm{pf}_X|_{\mathfrak{c}_\infty} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

matching the imported Borcherds–Gritsenko–Nikulin product. The scalar branch obtained after taking the protected trace of the inverse squared Pfaffian section is the Oberdieck–Pixton normalization

$$Z^X_{\mathrm{OP}} = -4096\, \Delta_5^{-2},$$

where the negative sign is the OP scalar branch and $4096 = 2^{12} = 64^2$ is the scalar equality between the retained half-Hilbert sign count and the squared theta-leading coefficient. Neither number defines the orientation character.

Comparison with the Pfaffian identity

The Vol II primitive-to-shadow comparison singles out the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X) = \Delta_5$ as the target identity for the retained Pfaffian datum; the $K3 \times E$ side supplies the Pfaffian line and scalar branch data. BBDJS [11] provides the orientation-line and Pfaffian calculus on the cosection-reduced d-critical heart; Joyce–Upmeier [52] provides multiplicativity under reduced extension correspondences; Joyce–Tanaka–Upmeier [51] extends the framework to 4-fold analogues but is not used directly here. The constant $4096 = 2^{12}$ is fixed at the orientation level by the retained half-Hilbert wall atlas and at the scalar level by 64^2; the local sign -1 is fixed by the rank-one Pfaffian wall computation. The Oberdieck–Pixton minus sign is a scalar branch convention, not the source of the orientation character. No identification of the Pfaffian Δ_5 with a compact microscopic state space is asserted: Δ_5 is the protected Pfaffian section, the scalar trace is Δ_5^{-2}, and the operator-level datum is the pro-object $\mathfrak{D}_X^{\mathrm{DI}}$, defined at the Pfaffian and orientation level relative to the retained data of Appendix 7.

The Pfaffian recovers Δ_5 at the target scalar level, but the cross-level recognition that identifies the primitive Hopf algebra of §0.3 with the Borcherds–Kac–Moody target $\mathfrak{g}_{\Delta_5}$ requires a separately constructed source coalgebra and an explicit chiral-Koszul cobar comparison. The source coalgebra C_X is the bar of the augmented binary/two-step hybrid correspondence algebra; it becomes the bar of a hybrid factorization algebra only after the higher-coloured residual vanishes. The cobar map $\Theta_{\mathrm{Kos},R}$ is the quasi-isomorphism to the imported current envelope $\mathsf{Den}_{\Delta_5,E}$. This Koszul lane (L5) is the content of §0.5.

0.5 Koszul source coalgebra and chiral MC twisting

The fifth entry of the Dirac–Igusa pro-object is the Koszul lane: the source coalgebra C_X, the bar comparison b_X, and the cobar quasi-isomorphism Θ_{Kos} to the imported current envelope $\mathsf{Den}_{\Delta_5,E} = U_E^{\mathrm{ch}}(\mathrm{Cur}_E(\mathfrak{g}_{\Delta_5}))$, the chiral enveloping algebra of the level-zero current algebra of the BKM denominator algebra over E (Definition 16.1). Source–target separation is structural. The target counit

$$\varepsilon_{\Delta,R} : \Omega_E^{\mathrm{ch}} \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\le R}) \longrightarrow \mathsf{Den}_{\Delta_5,E,\le R}$$

is internal to the imported denominator current envelope and produces no source data. The compact Koszul comparison $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}$ is a source-to-target statement on the supplied hybrid source, conditional on the chiral Koszul source datum and Koszul Mittag–Leffler exactness.

The Beilinson cut applies in its purest form on this lane. $\bar{B}^{\mathrm{ch}}$ is MC twisting / coupling coalgebra; it is not the level-Z bulk. The target of the lane is $\mathsf{Den}_{\Delta_5,E}$, the chiral shadow of the BKM denominator algebra; the source coalgebra C_X is the supplied compact Hall data after collision coproduct, conilpotent filtration, and bar-length truncation, and the cobar map is the comparison.

Source–target separation

Lemma 0.25 *(Source–target separation for the Koszul lane; proved).* Fix a finite height R. The target counit $\varepsilon_{\Delta,R}$ above is a morphism internal to the imported current envelope. A source Koszul datum is different data: it must first construct

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \qquad C_{X,R}, \qquad F^{\mathrm{co}}_{R,\bullet}, \qquad \Delta^{\mathrm{ch}}_R$$

from the finite hybrid correspondence category $\mathsf{Corr}_R^{E,\mathrm{hyb}}$, the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$, orientations, the eight-word two-step binary flag atlas, and the separate higher-coloured configuration residual; only then may it compare

$$b_{X,R} : C_{X,R} \simeq \bar{B}_E^{\mathrm{ch}}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R}), \quad \Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \simeq \mathsf{Den}_{\Delta_5,E,\le R}.$$

Setting $C_{X,R} = \bar{B}_E^{\mathrm{ch}}(\mathsf{Den}_{\Delta_5,E,\le R})$, or choosing a homotopy inverse to $\varepsilon_{\Delta,R}$, does not produce a source datum.

Proof. The domain of $\varepsilon_{\Delta,R}$ contains only target augmentation, target charge filtration, and target chiral operations; it does not contain rigidified wrapped prequotients, anchors, mixed local/wrapped correspondences, quotient-after-correspondence pseudofunctor, orientation transport, protected integration, or primitive representatives. These are the data from which the source coalgebra and collision coproduct are defined. □

Chiral Koszul source datum

Definition 0.26 (Chiral Koszul source datum). Fix (σ, S, R) and the Gram image Γ_R^{Π}. A finite-stage chiral Koszul source datum is

$$\mathfrak{K}_{X,R}^{\mathrm{Kos}} = (\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}, C_{X,R}, F_{R,\bullet}^{\mathrm{co}}, \Delta_R^{\mathrm{ch}}, b_{X,R}, \Theta_{\mathrm{Kos},R})$$

with the following clauses; every entry is source-side finite data except for the terminal comparison with the imported target.

(i) *Source chiral object.* $\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}$ is the finite hybrid object of §0.1, with $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and with a supplied wrapped bar-cobar domain datum. The wrapped sector is not made Koszul by the projection-finite $b = 0$ Ran sector.

(ii) *Conilpotent chiral coalgebra.* $C_{X,R}$ is a coaugmented conilpotent chiral coalgebra on E, filtered by a finite increasing source charge filtration $F_{R,\bullet}^{\mathrm{co}}$; reduced iterated coproduct nilpotent on each finite stage.

(iii) *Collision coproduct.* Δ_R^{ch} is the collision coproduct from the finite Hall correspondences (local-local, ordered local-wrapped, wrapped-wrapped, and eight-word flag stacks), defined before $Q_{E,R}$; the quotient coalgebra identities are the vanishing of the higher-coloured, unit, refinement, and quotient residuals. Transition maps $\rho_{R'R}^{C}$ commute with Δ^{ch}.

(iv) *Bar comparison.* $b_{X,R} : C_{X,R} \simeq \bar{B}_E^{\mathrm{ch}}(\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}, \varepsilon_{X,R}^{\mathrm{hyb}})$ is a charge-preserving quasi-isomorphism of conilpotent chiral coalgebras compatible with primitive projection, the completed Hopf pairing, the reduced E-quotient, and the normal-ordered descent $\overline{\Pi}_{X,*}^{\Theta}$, separately on the projection-finite and wrapped summands and on their mixed words.

(v) *Cobar comparison with the target envelope.* $\Theta_{\mathrm{Kos},R} : \Omega_E^{\mathrm{ch}} C_{X,R} \simeq \mathsf{Den}_{\Delta_5,E,\le R}$ is a charge-preserving quasi-isomorphism of augmented chiral algebras. Restricted to constant-current primitives after the Hopf-radical quotient it gives the primitive recognition morphism

$$\iota_R^{\mathrm{Prim}} : H^0\, \mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}_{X,*}^{\Theta} \mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}) \longrightarrow \mathfrak{g}_{\Delta_5,\le R}.$$

The exact finite Koszul obstruction tuple is

$$\mathfrak{O}_{\mathrm{Kos},R} = (\mathfrak{o}_R^{C}, \mathfrak{o}_R^{\mathrm{fil}}, \mathfrak{o}_R^{\mathrm{conil}}, \mathfrak{o}_R^{\Delta,\mathrm{ch}}, \{\mathfrak{o}_{R'R}^{C,\mathrm{tr}}\}_{R'\ge R}, \mathfrak{o}_R^{b}, \mathfrak{o}_R^{\Omega}, \mathfrak{o}_R^{\mathrm{wrKos}}, \mathfrak{o}_R^{\mathrm{hyb}}, \mathfrak{o}_R^{\Pi}, \mathfrak{o}_R^{\mathrm{sep}}, \mathfrak{o}_R^{\mathrm{Prim}}).$$

Canonical finite source-bar coalgebra

For a hybrid source with bounded-length augmentation ideal, a canonical model exists.

Proposition 0.27 *(Canonical finite source-bar coalgebra; conditional on $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and bounded length).* Suppose $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and there is an integer N_R such that every nonzero ordered word of non-vacuum hybrid inputs of total HN height $\le R$ has length at most N_R, with HN height positive on every non-vacuum colour surviving in the reduced bar construction. Suppose an augmentation $\varepsilon^{\mathrm{hyb}}_{X,R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \to k\mathbf{1}$ is supplied. Then the canonical source-bar coalgebra

$$C^{\mathrm{bar}}_{X,R} = \bar{B}^{\mathrm{ch}}_{E}(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R}) = k\mathbf{1} \oplus \bigoplus_{1\le p\le N_R} \bar{B}^{\mathrm{ch},p}_{E}(\overline{\mathcal{F}}_{X,R}),$$

filtered by bar-length, with deconcatenation collision coproduct

$$\Delta^{\mathrm{bar}}_{R}([x_1|\cdots|x_p]) = \mathbf{1}\otimes[x_1|\cdots|x_p] + [x_1|\cdots|x_p]\otimes\mathbf{1} + \sum_{i=1}^{p-1}[x_1|\cdots|x_i]\otimes[x_{i+1}|\cdots|x_p],$$

realizes the coalgebra and source-bar part of a chiral Koszul source datum with $\mathfrak{o}^{C}_{R} = \mathfrak{o}^{\mathrm{fil}}_{R} = \mathfrak{o}^{\mathrm{conil}}_{R} = \mathfrak{o}^{\Delta,\mathrm{ch}}_{R} = 0$ and $b_{X,R} = \mathrm{id}$.

Proof. Apply the Beilinson–Drinfeld chiral bar construction [6, §3.7] to the supplied hybrid object. Since the finite HN quotient kills charges of height $> R$ and non-vacuum words have length $\le N_R$, only finitely many bar lengths occur. The deconcatenation coproduct on each ordered word is realized by cutting the corresponding ordered collision flag in $\mathsf{Corr}^{E,\mathrm{hyb}}_{R}$; admissibility of pull-push and Thom–Sebastiani transport are part of $\mathfrak{O}_{\mathrm{hyb},R} = 0$. The eight-word flag pentagon gives binary coassociativity; the higher-coloured, unit, refinement, and quotient coherences contained in $\mathfrak{O}_{\mathrm{hyb},R} = 0$ give the chiral coalgebra identities. After at most $p \le N_R$ reduced iterations, every branch has length one and the next reduced coproduct is zero, so $(\overline{\Delta}^{\mathrm{bar}}_{R})^{N_R+1} = 0$. □

Cobar obstruction

The canonical source gives a conilpotent chiral bar coalgebra. A cobar quasi-isomorphism to the denominator chiral algebra is an additional datum, consisting of a source-internal bar-cobar counit and a source-to-target chiral algebra quasi-isomorphism.

Corollary 0.28 *(Source-to-target cobar obstruction; proved).* Under the hypotheses of Proposition 0.27, a cobar quasi-isomorphism $\Theta_{\mathrm{Kos},R}$ requires:

(i) *Source-internal bar-cobar counit.* The supplied hybrid source, including the wrapped summands and mixed local/wrapped words, must lie in the Beilinson–Drinfeld/Francis–Gaitsgory completed bar-cobar domain [33, Theorem 6.2.2], with the source-internal counit

$$\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_{E} C^{\mathrm{bar}}_{X,R} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$$

a quasi-isomorphism in the chosen completed/coderived conilpotent category. Defects: $\mathfrak{o}^{\varepsilon,\mathrm{src}}_{R} = (\mathfrak{o}^{\mathrm{aug}}_{R}, \mathfrak{o}^{\mathrm{FG}}_{R}, \mathfrak{o}^{\mathrm{wrKos}}_{R}, H^\bullet\,\mathrm{Cone}\,\varepsilon^{\mathrm{src}}_{\mathrm{bar},R})$.

(ii) *Source-to-target chiral algebra quasi-isomorphism.* $\vartheta_R : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R}$ preserving differential, product, unit, transition maps, and primitive restriction. Defects:

$$\mathfrak{o}^{\vartheta}_{R} = (d_\Delta\vartheta_R - \vartheta_R d_{\mathrm{hyb}}, \vartheta_R\mu^{\mathrm{hyb}}_{R} - \mu_{\Delta,R}(\vartheta_R\otimes\vartheta_R), \vartheta_R\eta^{\mathrm{hyb}}_{R} - \eta_{\Delta,R}, \vartheta_R\circ\tau^{E,\mathrm{hyb}}_{R'R} - \pi^{\Delta}_{R'R}\circ\vartheta_{R'}, H^\bullet\,\mathrm{Cone}\,\vartheta_R, \vartheta_R|_{\mathrm{Prim}} - \iota$$

Then $\Theta_{\mathrm{Kos},R} = \vartheta_R\circ\varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$, and the surviving cobar residual is $\mathfrak{o}^{\Omega}_{R} = (\mathfrak{o}^{\varepsilon,\mathrm{src}}_{R}, \mathfrak{o}^{\vartheta}_{R})$. No homotopy inverse to the target counit $\varepsilon_{\Delta,R}$ enters this construction.

Class M completion and chiral Koszulness

The bar/cobar comparison takes place in the completed ambient category. Class M holds in the completed ambient: analytic HS-sewing, coderived BV = bar, weight-completed, pro-, J-adic. On chain-level ordinary complexes, the chiral bar/cobar pair is the comparison input, not the closed Koszulness theorem; chiral Koszulness in the homotopy setting is a separate theorem of Francis–Gaitsgory [33].

Koszul Mittag–Leffler exactness

The completed bar/cobar inverse system is controlled by a finite-slice Mittag–Leffler condition on every relevant tower.

Proposition 0.29 *(Consequence of the chiral Koszul source datum; conditional on Koszul Mittag–Leffler).* Assume $\mathfrak{K}^{\mathrm{Kos}}_{X,R}$ is supplied for all R with compatible transition maps and $\mathfrak{o}^{C,\mathrm{tr}}_{R'R} = 0$, and that the Koszul Mittag–Leffler condition holds: $R^1 \varprojlim = 0$ for the inverse systems of source coalgebras, completed chiral bar/cobar constructions, target truncations, and the cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ [97, §3.5]. Then $C_X = \varprojlim_R C_{X,R}$ is a completed pro-conilpotent chiral coalgebra attached to the hybrid compact source, conilpotent on every finite HN quotient, and the maps $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ induce

$$b_X : C_X \xrightarrow{\simeq} \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X), \qquad \Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

This identifies the supplied source cobar algebra with the formal target current envelope; it does not construct the E_3-source, the hybrid wrapped Hall correspondences, the orientation line, the primitive recognition data, or C_X itself from the target-internal counit.

Proof. Inverse-limit existence follows from transition compatibility, the bounded exhaustive finite charge filtration, finite coassociative and counital collision coproduct, and continuity hypotheses. The coradical filtration is finite on every Γ^{Π}_R-truncation; the Koszul Mittag–Leffler hypothesis kills $R^1 \varprojlim$ on the cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$, giving completed quasi-isomorphisms. Without that exactness one obtains only a compatible pro-quasi-isomorphism. □

Chiral MC twisting / coupling coalgebra

Bar is twisting; bar is not the level-Z bulk. The reduced bar coalgebra $\bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_X)$ is the Maurer–Cartan coupling coalgebra of the supplied hybrid chiral algebra: its primitive elements are the chain-level inputs of MC twisting in the Beilinson–Drinfeld/Francis–Gaitsgory framework, and its grouplike completion records the homotopy-coherent twisting data. The level-Z bulk in Vol II is $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ for the relevant boundary algebra A; the visible target is its BKM chiral shadow $\mathsf{Den}_{\Delta_5,E}$. The Koszul comparison maps C_X, an instance of MC coupling, into that target chiral algebra by the cobar construction.

The Koszul obstruction class

The Koszul lane carries the residuals

$$\mathfrak{O}_{\mathrm{Kos},R} \quad \text{and} \quad \mathfrak{o}^{\mathrm{ML,Kos}}_{R'R}.$$

With $\mathfrak{O}_{\mathrm{hyb},R} = 0$ and the canonical source-bar construction, the coalgebra defects vanish; only the cobar residual $\mathfrak{o}^{\Omega}_R = (\mathfrak{o}^{\varepsilon,\mathrm{src}}_R, \mathfrak{o}^{\vartheta}_R)$ remains, with the wrapped Koszul defect $\mathfrak{o}^{\mathrm{wrKos}}_R$ included in $\mathfrak{o}^{\varepsilon,\mathrm{src}}_R$. Vanishing of the source-internal counit defect and construction of the source-to-target chiral algebra quasi-isomorphism operates in the completed Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar domain; the Koszul Mittag–Leffler hypothesis controls the inverse-system passage.

Source-target separation

The source–target separation is the $K3 \times E$-side instance of the Vol II Beilinson cut: bar is twisting, bar is not bulk; the boundary algebra is $A = \mathcal{F}_X^{\mathrm{hyb}}$ at finite stage, and the bulk is $Z_{\mathrm{ch}}^{\mathrm{der}}(A)$ which receives the cobar of the bar coalgebra after MC coupling. Vol I master patterns prohibit the identification of the chiral bar with the bulk chiral algebra and require the Koszul comparison to be a separate theorem in the homotopy setting; the same separation is enforced above. The compact Koszul comparison and the chiral Koszulness theorem are not identical; the present construction lives on the comparison side, with chiral Koszulness in the Francis–Gaitsgory sense as the abstract input.

Under the Koszul source datum, $\Theta_{\mathrm{Kos},R}$ identifies the cobar of the source coalgebra with the imported current envelope of $\mathfrak{g}_{\Delta_5}$. Under the recognition datum $(S1)$–$(S10)$, the source-side protected primitive $P_X^{\Pi,\mathrm{red}}$, extracted from C_X via the Hopf radical quotient, the Frobenius identity, and the no-extra-relations criterion, is identified with the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$. Entry (L6) is this recognition apparatus, in §0.6.

0.6 Primitive recognition apparatus

The sixth entry of the Dirac–Igusa pro-object is the primitive recognition criterion. Its conclusion, after the compact source data are supplied, is an isomorphism of Gram-graded Lie superalgebras

$$P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$$

between the cosection-twisted reduced Hall primitive layer of $K3 \times E$ and the imported Borcherds–Kac–Moody denominator algebra. Signed superdimensions alone do not suffice for this theorem. The Borcherds product fixes sdim $P_{X,\gamma}^{\Pi,\mathrm{red}} = f(nm, l)$ for $\gamma = (n, l, m)$; this is one integer in each degree. A source–target identification fixes two integers in every degree (the parity dimensions), the relation ideal, the radical equality, the no-extra-relations theorem, the completed PBW filtration, and the Hopf-radical coideal stability.

The determinant admits the zero-bracket graded super vector space with the same signed dimensions: the denominator is identical, while the Chevalley diagonal, the non-orthogonal brackets, and the Hopf pairing datum have disappeared. Recognition therefore requires representatives, relations, pairings, kernels, and PBW data, not numerics.

Borcherds–Cartan target datum

The target side imports the Borcherds–Kac superdatum $\mathscr{B}_{\Delta_5} = (H, I, p, \alpha_i, (\,,\,))$ of Gritsenko–Nikulin [43–45] after the [7, 54] presentation conventions. The Cartan is $H = \Lambda_{II}^{2,1} \otimes \mathbb{C}$, the three real simple roots $\delta_1, \delta_2, \delta_3$ form a hyperbolic triangle with Gram matrix

$$((\delta_i, \delta_j))_{i,j} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix},$$

and the imaginary fibres of $I \to H^*$ carry the Gritsenko–Nikulin parity split $\mu_{\bar{0}}(a), \mu_{\bar{1}}(a)$ for imaginary simple roots. The degree map is

$$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2} = n\delta_1 + m\delta_2 + (n + m - l)\delta_3,$$

with $\widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})$ the completed free Lie superalgebra on the parity-homogeneous generators e_i, f_i, $i \in I$, and J_{BK} the closed Borcherds–Kac ideal generated by Cartan, Chevalley, real Serre, orthogonality, and super sign relations. The target denominator algebra is

$$G(\mathscr{B}_{\Delta_5}) = \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})/\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}} = \mathscr{A}_{\mathrm{den}} = \mathfrak{g}_{\Delta_5}.$$

SOURCE-SIDE DATA

THEOREM 0.30 *(Primitive recognition from finite source representatives; conditional on (S1)–(S10)).* Let a compact Hall/factorization construction give a completed $\widehat{\Gamma}_X$-graded primitive Hall Lie superalgebra $\widetilde{P}_X$ and a completed Hopf pairing. Its normal-ordered Gram descent

$$P_X^{\Pi,\mathrm{desc}} = H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X) = \overline{\Pi}^{\Theta}_{X,*}\widetilde{P}_X, \quad P_X^{\Pi,\mathrm{red}} = P_X^{\Pi,\mathrm{desc}}/\overline{\mathrm{Rad}\langle\,,\,\rangle_X}.$$

Suppose the following ten source-side data are constructed.

(S1) *Charge and normal-ordered descent.* The chain-level descent $\overline{\Pi}^{\Theta}_{X,*}$ categorifies the homomorphism $\overline{\Pi}_X(c,T) = \Pi_X(c) - T$, so that the descended bracket is Γ_{gram}-graded. Each descended homogeneous parity piece is finite dimensional.

(S2) *Cartan and root-degree matching.* An even Cartan identification $\iota_H : P^{\Pi,\mathrm{red}}_{X,0} \xrightarrow{\sim} \Lambda^{2,1}_{II} \otimes \mathbb{C}$ aligning the descended Gram grading with the GN root grading via $\alpha(n,l,m)$.

(S3) *Simple primitive representatives and parities.* For every simple-root index $i \in I$, choose $\gamma_i \in \Gamma_{\mathrm{gram}}$ with $\alpha(\gamma_i) = \alpha_i$. The source supplies parity-homogeneous representatives e_i^X, f_i^X, h_i^X in $P^{\Pi,\mathrm{red}}_{X,\gamma_i}$, $P^{\Pi,\mathrm{red}}_{X,-\gamma_i}$, and $P^{\Pi,\mathrm{red}}_{X,0}$, with $|e_i^X| = |f_i^X| = p(i)$, $h_i^X = \iota_H^{-1}(\alpha_i)$, and the GN multiplicity fibres $\mu_{\bar 0}(a)$, $\mu_{\bar 1}(a)$ for imaginary simple roots. Real representatives for $\delta_1, \delta_2, \delta_3$ are even.

(S4) *Borcherds–Kac relations in the Hall bracket.* In the protected Hall/factorization bracket the representatives satisfy:

$$[h,h'] = 0, \quad [h,e_i^X] = (h,\alpha_i)e_i^X, \quad [h,f_i^X] = -(h,\alpha_i)f_i^X, \quad [e_i^X,f_j^X] = \delta_{ij}h_i^X,$$

the real Serre relations

$$(\mathrm{ad}\,e_i^X)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)}e_j^X = 0, \quad (\mathrm{ad}\,f_i^X)^{1-2(\alpha_i,\alpha_j)/(\alpha_i,\alpha_i)}f_j^X = 0 \qquad (\text{real } i,\ i \neq j),$$

the Borcherds orthogonality relations $(\alpha_i,\alpha_j) = 0 \Rightarrow [e_i^X,e_j^X] = [f_i^X,f_j^X] = 0$, and the super sign rule. In the rank-three real subsystem, the Serre exponent is 3 ($(\delta_i,\delta_j) = -2$); first brackets $[e_i^X,e_j^X]$ are not forced to vanish. For the primitive isotropic degrees $a_{ij} = \delta_i + \delta_j$ ($i < j$), choose $\gamma_{ij} \in \Gamma_{\mathrm{gram}}$ with $\alpha(\gamma_{ij}) = a_{ij}$. The source supplies the nine Gritsenko–Nikulin isotropic simple representatives $u^{\pm,X}_{ij,r} \in P^{\Pi,\mathrm{red}}_{X,\pm\gamma_{ij}}$, $1 \le r \le 9$, satisfying orthogonality and Chevalley zero with adjacent real roots, with real-string relations $(\mathrm{ad}\,e_k^X)^5 u^{+,X}_{ij,r} = 0$ for the complementary $\{i,j,k\} = \{1,2,3\}$. The tenth even vector in $\mathfrak{g}_{a_{ij}}$ is the real bracket $[e_i^X,e_j^X]$.

(S5) *Completed Hopf pairing and radical quotient.* Continuous, homogeneous, invariant for bracket and coproduct, with closed homogeneous radical $\mathrm{Rad}_X = \{x \mid \langle x,y\rangle_X = 0\ \forall y\}$ a Lie ideal and coideal. After quotienting by $\overline{\mathrm{Rad}_X}$, the source radical quotient is identified with the GN/Kac quotient $\overline{\mathrm{Rad}_{\mathrm{GN}}}$. At every finite stage the Lie-ideal and coideal stability hold and the inverse-limit kernel is exactly Rad_X.

(S6) *No extra relations.* If $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$ is the continuous map determined by the representatives, $\ker \pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$. Equivalently, $G(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$ is injective. This is checked finite stage by finite stage on saturated windows, with kernel identifications commuting with HN transitions.

(S7) *Generation.* $P_X^{\Pi,\mathrm{red}}$ is topologically generated by $P_{X,0}^{\Pi,\mathrm{red}}$ and the simple primitive representatives.

(S8) *Completed PBW.* PBW filtration compatibility:

$$\mathrm{gr}_{\mathrm{PBW}}\, U(P_X^{\Pi,\mathrm{red}})_W \xrightarrow{\sim} \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_W$$

on every saturated finite window W, with strict transition maps and inverse limit in the HN topology of §0.3.

(S9) *Exact completion.* Inverse limit exact on the saturated finite root windows: transition maps strict, images closed, and passage to the completed radical quotient commuting with finite-window kernels, cokernels, PBW associated gradeds, and parity pieces. Equivalently, no new relation, generator, radical element, or parity-cancelling summand appears only after completion.

(S10) *Full parity dimensions.* For every nonzero γ with $\alpha(\gamma) \in \mathcal{R}$,

$$\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\overline{p}} = \mathrm{mult}_{\overline{p}}(\alpha(\gamma)), \qquad p = 0, 1.$$

The same equality is required in degree $-\gamma$, with the target parity dimensions transported by the Borcherds–Kac Chevalley anti-involution; it is not inferred from the positive half or from the signed denominator. On the first timelike channel $\gamma = (1,1,1)$,

$$d^{\mathrm{GN}}_{(1,1,1),\overline{0}} = 29, \qquad d^{\mathrm{GN}}_{(1,1,1),\overline{1}} = 93.$$

These are presentation-level target dimensions, not an inference from $29 - 93 = -64$.

Then $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ as completed Lie superalgebras graded by Γ_{gram} through α, with denominator $64^{-1}\Delta_5(2Z)$. If in addition the chiral Koszul source datum of §0.5 is constructed (including Koszul Mittag–Leffler exactness), then $\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$.

Proof. The simple primitive representatives define a continuous homomorphism $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$. (S1) and (S2) make π a homomorphism of the descended Gram-graded completions. (S3) fixes the real and imaginary simple generators with their parities and multiplicity fibres. (S4) puts J_{BK} in the kernel. (S5) identifies the compact Hopf radical with the GN/Kac radical, so π descends to $G(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$. (S7) gives surjectivity; (S6) gives injectivity after the completed radical quotient. (S8) gives PBW-compatible finite-height identifications; (S9) gives exactness in the inverse limit; (S10) rules out hidden zero-superdimension root spaces and cancelling ideals invisible to the determinant. The chiral statement uses C_X, b_X, Θ_{Kos}, and the Koszul Mittag–Leffler exactness of §0.5. □

Cofinal finite-window primitive recognition

The Borcherds product fixes signed target superdimensions; the Gritsenko–Nikulin/Kac presentation fixes full target parity dimensions only in the degrees where the presentation has been computed. A compact $K3 \times E$ source is identified with the denominator algebra only by source representatives, pairings, radicals, relation ideals, PBW, and exact transition maps.

The imported target on the first window $W_{\leq 3}$ has parity table

$$\delta_i : 1|0, \quad a_{ij} : 10|0, \quad 2\delta_i + \delta_j : 1|0, \quad \delta_{123} : 29|93,$$

so $29 - 93 = -64 = f(1,1)$. The row a_{ij} : 10|0 is the real bracket $[e_i, e_j]$ together with the nine Gritsenko–Nikulin isotropic simple directions. These are target data; they do not construct compact source primitives, source pairings, radical kernels, or source bracket constants.

Definition 0.31 *(Cofinal finite-window primitive-recognition datum).* A finite set $W \subset Q_+ \setminus \{0\}$ is *downward saturated* if $0 < \beta' \leq \beta$, $\beta \in W$, $\beta - \beta' \in Q_+$ imply $\beta' \in W$. A cofinal finite-window primitive-recognition datum for $P_X^{\Pi,\mathrm{red}}$ is an increasing sequence of finite downward-saturated target-root windows $W_1 \subset W_2 \subset \cdots$ with $\bigcup_\nu W_\nu = \mathcal{R}_+$, together with the following finite data on every W_ν, compatible under $W_\nu \subset W_{\nu+1}$. Put

$$\Gamma(W_\nu) := \{\gamma \in \Gamma_{\mathrm{gram}} \mid \alpha(\gamma) \in W_\nu\}.$$

(R0) *Compact source labels.* Every source basis vector of degree in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$ satisfies the compact source admissibility condition. Target labels from $G(\mathscr{B}_{\Delta_5})$ appear only in target rows or as codomains of comparison maps.

(R1) *Representatives and parities.* Simple-root representatives e_i^X, f_i^X, h_i^X whose target roots lie in W_ν are supplied in source degrees $\pm\gamma_i$ with $\alpha(\gamma_i) = \alpha_i$, with the GN parity fibres for imaginary simple roots and even real representatives for $\delta_1, \delta_2, \delta_3$.

(R2) *Relations.* Cartan, Chevalley, real Serre, orthogonality, and super sign relations checked for every relation whose displayed target roots lie in $W_\nu \cup (-W_\nu) \cup \{0\}$ and whose source lifts lie in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$.

(R3) *Full finite parity dimensions.* For every $\gamma \in \Gamma(W_\nu)$ and for its negative $-\gamma$, $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\bar{p}} = \mathrm{mult}^{\mathrm{GN}}_{\bar{p}}(\alpha(\gamma))$ $(p = 0, 1)$, with the negative-root parity dimensions carried by the Borcherds–Kac Chevalley anti-involution, and with zero on source degrees whose image is not a target root. Two-integer equality, not signed-superdimension.

(R4) *Hopf pairing and radical.* Positive-negative pairing matrices written in parity blocks; kernels agree with GN/Kac invariant-pairing kernels; stable under finite brackets in the window; coideal for finite coproduct; carried to the next window by transitions; inverse-limit radical the closure of finite kernels.

(R5) *No extra relations.* Finite kernel equality $\ker \pi_\nu = (\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}})_{\leq W_\nu}$.

(R6) *Generation.* Source primitive spaces over $\Gamma(W_\nu) \cup (-\Gamma(W_\nu))$ are generated by Cartan and supplied simple representatives through brackets whose intermediate target roots stay in W_ν.

(R7) *Completed PBW compatibility.* Strict PBW filtration preservation under transitions, inverse-limit PBW compatibility.

(R8) *Exact triangular completion.* Words whose total degree lies in the finite test window only through positive-negative cancellation are not added as extra finite generators; transition maps strict, images closed, $\lim^1$ vanishing on the triangular finite-window systems.

The datum is finite at each ν, but not numerical: it contains representatives, relation verifications, pairing matrices, kernel equalities, and PBW transition data. The Borcherds product and the OP scalar square provide none of these source-side objects.

PROPOSITION 0.32 *(Global recognition is exactly the cofinal datum; proved).* Assume (S1)–(S4) of Theorem 0.30. Then the global source-side assertions (S5)–(S10) are equivalent to a cofinal finite-window primitive-recognition datum $\{W_\nu, \text{(R0)–(R8)}\}$ in the sense of Definition 0.31, and conversely.

Proof. The two sides record the same data: each W_ν is a saturated finite window in (S5)–(S9), and (R3) is exactly the two-integer parity equality of (S10) on W_ν. The Hopf pairing matrices, kernel equalities, PBW filtrations, no-extra-relations relations, and transition strictness are exactly the cofinal contents of (S5)–(S9), and the cusp-positive arithmetic side of $f(nm, l)$ is recovered from the parity equalities (R3) on the active target degrees. Conversely, given (R0)–(R8) on a cofinal system, inverse limit exactness (R8) gives the global statements (S5)–(S9), and (R3) on a cofinal exhaustion gives (S10). □

Recognition criterion at the window $W_{\le 3}$

For a finite window W with $S_W = W \cup (-W) \cup \{0\}$, a compact source datum consists of compact-source bases $b^X_{R,\gamma_\beta,\overline{p},a}$, parity pieces, product, coproduct, bracket, pairing, radical, and quotient matrices M, D, B, G, K, Q, comparison maps $A_{\beta,\overline{p}}$, relation identities, the no-extra kernel equality, PBW associated gradeds, and strict Mittag–Leffler transition maps. The target labels $e_i, E_{ij}, u_{ij,r}, T_i, M_{ij,r}, w_s$ occur only in target blocks or as codomain coordinates of $A_{\beta,\overline{p}}$.

The window $W_{\le 3}$ is the first arithmetic window, not a relation-closed degree window. A relation-closed degree test must include real-Serre terminal rows, doubled isotropic rows, δ_{123}-real strings, odd–odd δ_{123} rows, and complementary height-seven strings, or pass to the downward-saturated closure. In that larger target degree window, Remark 6.9 supplies full parity only for $2a_{ij}$, $C_{k,3}$, $C_{k,4}$, and $C_{k,5}$. The rows $C_{k,2}$ and $2\delta_{123}$ remain signed target rows; they cannot serve as source comparison codomains until the finite target presentation is computed in those degrees. The relation-closed degree set is therefore not yet a primitive-recognition window in the sense of (R0)–(R8).

What signed dimensions miss

Lemma 0.33 *(Signed dimensions are not recognition; proved).* Let $V_\bullet$ be a $\widehat{\Gamma}_X$-graded super vector space with zero bracket and zero coproduct, with the same signed superdimensions $\operatorname{sdim} V_\gamma = f(nm, l)$ as $P_X^{\Pi,\mathrm{red}}$. Then $V_\bullet$ has the same Borcherds denominator $64^{-1}\Delta_5(2Z)$ as $\mathfrak{g}_{\Delta_5}$ but is not isomorphic to it as a Lie superalgebra.

Proof. The denominator depends only on signed superdimensions; the bracket data, no-extra-relations theorem, Hopf-radical coideal stability, and PBW filtration are independent. If the Cartan is mapped nontrivially, the Chevalley relation $[e_i, f_i] = h_i$ fails in $V_\bullet$. If the Cartan is killed, the root-degree Cartan identification fails. In either case the kernel of the free presentation map contains all non-Cartan brackets, hence is strictly larger than $J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}$ in degrees where the BKM bracket is nonzero. With the zero Hopf pairing the radical is all of $V_\bullet$, so the radical quotient is zero. □

This lemma is the counterexample that prohibits the determinant route to recognition. It is also the structural reason the ten clauses (S1)–(S10) do not collapse to their signed-dimension shadow: the signed dimensions are necessary and not sufficient; (S1)–(S9) supply the source presentation, and (S10) supplies the missing parity split.

Obstruction classes

The recognition lane carries the residuals

$$\mathfrak{O}_{\mathrm{Rec},\nu} = \big(\mathfrak{o}_\nu^{(\mathrm{R1})}, \ldots, \mathfrak{o}_\nu^{(\mathrm{R8})}, \mathfrak{o}^{\mathrm{tr}}_{\nu+1,\nu}\big),$$

required to vanish on every cofinal window. The first arithmetic window $W_{\le 3}$ is achievable from the GN target side; the first relation-closed degree window includes the real-Serre terminal rows and the

∂_{123} odd–odd rows, but some added target rows are still only signed target rows by Remark 6.9. The finite-stage components of the (R0)–(R8) criterion, including source-radical/Hopf-coideal stability, remain source-recognition data. Appendix 7 supplies the orientation lane; it does not supply the source matrices M, D, B, G, K, Q, the comparison maps $A_{\beta,\bar{p}}$, the kernel equality, or the PBW associated-graded comparison. Construction of the source-internal recognition criterion at any one relation-closed degree window beyond $W_{\leq 3}$ is the unresolved compact-source datum of Chapter 10.

Comparison with the denominator algebra

The recognition theorem is the source side of the Vol III Drinfeld double construction. The target presentation uses the generalized Kac–Moody convention of Borcherds [7] and Kac [54]; the automorphic correction algebra is the Gritsenko–Nikulin algebra of [43, Sections 3–4, Proposition 3.1], and the product-lift data are those of [45, Theorem 2.1, §5.1.1] together with Borcherds' automorphic product [8, 9]. The identity $\mathrm{CoHA}(\mathbb{C}^3) = W_{1+\infty}$ belongs to the local toric case; the present recognition criterion is at the level of the protected primitive Hall algebra, before Drinfeld doubling; after the compact source datum is supplied, it identifies that algebra with $\mathfrak{g}_{\Delta_5}$, not with the centred or vacuum-evaluated double.

Entries (L1)–(L6) name the constructed objects. The finite HN system in §0.7 supplies the charge windows, Hall correspondences, orientation transports, Pfaffian lines, Koszul source coalgebras, primitive-recognition data, and their local obstruction classes.

0.7 *Eight finite constructions*

The seventh entry of the Dirac–Igusa pro-object is the finite HN system of charge windows, Hall correspondences, orientation transports, Pfaffian lines, Koszul source coalgebras, and primitive-recognition data. At HN height R these data are finite-type or finite-dimensional, and their obstruction classes vanish exactly when the transition system defines the inverse limit $\mathfrak{D}_X^{\mathrm{DI}}$.

At fixed HN height all objects lie on finite-type stacks and finite-dimensional Borel–Moore chain spaces; the inverse limit is taken after the finite stages have compatible transition maps. For $R \in \mathcal{R}_{\mathrm{HN}}$, the constructions assume: [K3×E-fin] (finite-type semistable substacks at bounded HN height), [K3×E-atlas] (finite d-critical/cosection Hall atlas with Joyce–Upmeier orientation lines), and [ML] (Mittag–Leffler descent).

Charge window

Hypotheses. Stability datum (σ, S), finite charge window $\Gamma_R^{\mathrm{test}} \subset \Gamma_X^{\mathrm{form}}$.

Finite datum. Finite saturated downward window $F_{\sigma,S}(R) \subset \Gamma_X^{\mathrm{HN}}(R)$, the dyonic formal Mukai/effective-charge support, the Gram map $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$, the cocycle B, and the normal-ordered extension $\widehat{\Gamma}_R = \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}}$.

Construction.

(i) Form the Liu charge $L_F(n) = Z_S(Rp_{S,*}(F \otimes p_E^* H^n))$ [61] for σ_S a rational K3 stability condition and a degree-one line bundle H on E. Restrict to charges with bounded h_S-height $\leq R$; the resulting set is $F_{\sigma,S}(R)$.

(ii) Compute the Gram map and cocycle $B(c, c')$ on $F_{\sigma,S}(R) \times F_{\sigma,S}(R)$ via the Mukai dictionary of §3.3; check the symmetric bilinear 2-cocycle condition.

(iii) Form $\widehat{\Gamma}_R$ as the central extension by B-cocycle.

Obstruction classes. [K3×E-fin] on the Liu charge bound; B cocycle property.

Finite Hall stage and atlas

Hypotheses. The charge window, plus the cosection-twisted reduced d-critical heart of §0.2.

Finite datum. The finite Hall stage $C_{\sigma,S,\le R}$, finite-type derived Artin enhancements $\mathfrak{M}_{\widehat{c},R}, \mathfrak{E}_{\widehat{c},\widehat{c}',R}, \mathfrak{F}^{(2)}_{\widehat{c},\widehat{c}',\widehat{c}'',R}$ of object, extension, and two-step flag stacks; reduced cosections; reduced vanishing cycles Φ^{red}; orientation lines o^{red}.

Construction.

(i) Restrict the chosen stability heart to charges in $F_{\sigma,S}(R)$; form the finite-type derived enhancements with PTVV (-1)-shifted symplectic structures [89] on the truncations; descend to Joyce d-critical structures [11].

(ii) Compute the K3 semiregularity cosection CS_C on each retained charge stratum; verify surjectivity on reduced strata and filtered Hall additivity on extension stacks.

(iii) Choose Joyce–Upmeier orientation lines [52] with chosen square roots, multiplicative under reduced extension correspondences.

Obstruction classes. [K3×E-atlas]; finite residual inertia after scalar, E-translation, and wrapped rigidifications.

Hybrid wrapped factorization datum

Hypotheses. The finite Hall stage.

Finite datum. The hybrid factorization carrier $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$ of §0.1: rigidified wrapped prequotients $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ with anchors $\lambda_{\eta,R}$; closed-configuration local sectors $\mathcal{F}^{\mathrm{loc}}_R$; wrapped sectors $\mathcal{G}^{\mathrm{wr}}_R$; the eight-word two-step binary flag atlas; the higher-coloured residual $\mathfrak{o}^{\mathrm{col}}_R$; the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$ and comparison isomorphisms $\theta^Q_{\mu,R}$; protected integration I^{prot}_R.

Construction.

(i) Split Γ_R into Γ^{loc}_R ($b^{\mathrm{geom}}_R = 0$) and Γ^{wr}_R ($b^{\mathrm{geom}}_R > 0$).

(ii) For $\alpha \in \Gamma^{\mathrm{loc}}_R$: form the closed-configuration incidence stacks $\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I}$ and the configuration maps $\pi_{\alpha,R,I} : \mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} \to E^I$, with vanishing-cycle/orientation pushforwards $\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I}$. Define $\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U)$ by symmetric-group descent on E^I.

(iii) For $\eta \in \Gamma^{\mathrm{wr}}_R$: form the rigidified prequotient $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \to \mathcal{M}^{\mathrm{wr}}_{\eta,R}$; choose anchor $\lambda_{\eta,R}$ with unit translation weight after the retained finite-cover normalisation; verify the typed finite anchor residual $\mathfrak{o}^{\lambda}_R = 0$ on each component.

(iv) Retain the local-local, ordered local-wrapped, and wrapped-wrapped extension correspondences with closed-support finite-type witnesses, exceptional-pushforward admissibility, and Thom–Sebastiani identifications.

(v) Retain the eight-word two-step binary flag stacks $\mathfrak{Flag}^w_{\xi_1,\xi_2,\xi_3,R}$ for $w \in \{\mathrm{L},\mathrm{W}\}^3$ and verify $\mathfrak{o}^{\mathrm{assoc}}_{w,R} = 0$ on the four-input pentagon; the unit, refinement, symmetry, descent, and higher-coloured tree data remain in $\mathfrak{o}^{\mathrm{col}}_R$.

(vi) Use the supplied quotient-after-correspondence pseudofunctor $\mathsf{Corr}^{E,\mathrm{hyb}}_R \to \mathsf{Corr}^{\mathrm{red,hyb}}_R$ and the comparison isomorphisms $\theta^Q_{\mu,R}$; verify the four quotient residuals.

(vii) Define I^{prot}_R on the reduced functor; verify the five integration defects vanish and the Gram-trace exponent law $I^{\mathrm{prot}}_R(Q_{E,R}x) \in s^{m^{\mathrm{Bch}}_R(\gamma)}\mathbb{C}[q^{\pm 1}, r^{\pm 1}]$ with $m^{\mathrm{Bch}}_R(\gamma) = \mathrm{pr}_3\,\overline{\Pi}_X(\gamma)$.

Obstruction classes. $\mathfrak{O}_{\mathrm{hyb},R}$ (§0.1).

Orientation descent and Weyl transport

Hypotheses. The hybrid wrapped factorization datum.

Finite datum. The (O1) and (O1)$^+$ data: null-trivializations of the reduced and equivariant orientation gerbes, $E[N]$-linearizations with vanishing degree-one characters, and Weyl-equivariant lifts τ_i ($i = 1, 2, 3$) compatible with the quotient-orientation cocycle $\mathfrak{q}_C$.

Construction.

(i) For each charge stratum C: compute the reduced gerbe α_C^{red}, free E-class $\alpha_C^{E,\mathrm{free}}$, and equivariant finite-stabilizer classes $\widetilde{\beta}^H_{C,S}$ for the actual stabilizers $H \subset E_{\mathrm{tors}}$ on finite-stabilizer strata.

(ii) Verify the Klein-four condition at $H = E[2]$: $(b_{20}, b_{11}, b_{02}) = (r_1, r_1 + r_2 + r_3, r_2)$ with $r_1 = r_2 = r_3 = 0$; for full-level even $N \geq 4$, verify the two-primary residual on $E[N]_{(2)}$ including the mixed coefficient $A_{12}^{(N)}$.

(iii) Choose zero $H^1(BH; \mathbb{F}_2)$-linearization characters $\lambda^H_{C,S} = 0$.

(iv) For each simple type-II reflection s_{∂_i}, supply the lift $\tau_i : o_C \to s^*_{\partial_i} o_C$ on the finite partial action groupoid $\mathcal{G}_R$; compute the 2-cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$ and verify $[c_{o,R}] = 0$ with chosen 1-cochain.

(v) Verify $\tau_i^2 = 1$ and the Coxeter relations on the $W^{(2)}(\Lambda^{2,1}_{II})$-orbit; verify $\omega_{i,C} = 0$ for each torsor defect of the quotient-orientation transport.

Obstruction classes. $\mathfrak{O}_{\mathrm{or},R}$ (§0.2); the (O2) wall-atlas enters the Pfaffian wall normal form.

Hall product, coproduct, and primitive projection

Hypotheses. The finite Hall atlas, hybrid carrier, and orientation transport.

Finite datum. The compact Hall product m_R^{red}, collision coproduct Δ_R^{ch}, primitive projection P_R^{Prim}, and Hopf pairing $\langle\, , \,\rangle_R$ on the cosection-reduced d-critical heart, with normal-ordered Gram descent $\overline{\Pi}^{\Theta}_{X,*}$.

Construction.

(i) Form $m_R^{\mathrm{red}} = q_! \circ \mathrm{TS}_R^{\mathrm{red}} \circ p^*$ on each binary correspondence (LL, LW, WL, WW); verify two-step pentagon coherence on the eight-word atlas. This operation is defined only in the retained compactified correspondence model, or in an explicitly supplied exceptional compact-support model.

(ii) Form Δ_R^{ch} by pull-push on the 2-step flag stack; verify coassociativity and counitality after $Q_{E,R}$ using the higher-coloured, unit, refinement, quotient, and transition coherences.

(iii) Form $\langle\, , \,\rangle_R$ on the primitive layer; verify Hopf adjointness with the coproduct on the radical quotient.

(iv) Compute the Frobenius defect $\mathfrak{o}_{F,R}$ via the CY-3 Serre-functor identity [59]; verify $\mathfrak{o}_{F,R} = 0$.

(v) Compute the radical Rad_R^{Π}; verify Hopf-coideal and Lie-ideal stability.

(vi) Form the chain-level normal-ordered Gram descent $\overline{\Pi}^{\Theta}_{X,*}$ via the negative-cyclic Loday–Quillen–Tsygan route or the PTVV–Joyce orientation route [62, 63, 89, 95]; verify the triple homotopy compatibility $d_{\mathrm{Hoch}}\Theta_{\Pi} = B_{\mathrm{ch}}$ and the $\widehat{\Gamma}_R$-grading absorption.

Obstruction classes. $\mathfrak{O}_{\mathrm{Hall},R}$, including $\mathfrak{o}_{F,R}$, the Hopf-coideal residual, and the Θ-pentagon residual.

Dirac block, Pfaffian line, and type-II wall atlas

Hypotheses. The orientation transport and Hall product.

Finite datum. The first-order Dirac block decomposition $\mathfrak{D}^{D_0}_{X,R} = \bigoplus_\gamma J_\gamma$, the Pfaffian line $\mathscr{L}_{\mathrm{Pf},R}$ with squaring isomorphism, the Pfaffian-to-automorphic comparison $\iota_{\mathrm{aut},R}$, the normalized section $\mathrm{pf}_{X,R}$, and the type-II wall atlas (rank-one normal model on every retained component, boundary stratum, and overlap).

Construction.

- (i) Form the Pfaffian line $\mathscr{L}_{\mathrm{Pf},R} = \bigotimes_C \mathrm{Pf}(K^{\mathrm{red}}_{C,R})$ on the cosection-reduced quotient [11]; verify the squaring isomorphism.
- (ii) Form J_γ on $P^{\Pi,+}_{R,\gamma} \oplus (P^{\Pi,+}_{R,\gamma})^\vee[1]$ for each $\gamma \in \Gamma^{\Pi,+}_R \setminus \{0\}$; verify Pfaffian contribution $(1 - x^\gamma)^{\mathrm{sdim}\, P^{\Pi,+}_{R,\gamma}}$.
- (iii) Verify the support, parity, and leading-coefficient residuals $\mathfrak{o}^{\mathrm{supp}}_{R,\gamma} = \mathfrak{o}^{\mathrm{par}}_{R,\gamma} = \mathfrak{o}^{\mathrm{lead}}_R = 0$ on the test window.
- (iv) For each simple type-II wall $\mathbb{H}_{\delta_i}$ ($i = 1, 2, 3$): take the rank-one Pfaffian crossing chart on every retained component and boundary stratum; verify $N^{\mathrm{Pf}}_{\delta_i} = 1$ and the local sign $\varepsilon^{\mathrm{loc}}_{\delta_i,R} = -1$; verify chart-overlap agreement of the normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps; verify the half-Hilbert orbit sign sum (size $4096 = 2^{12}$).
- (v) Supply the Pfaffian-to-automorphic comparison $\iota_{\mathrm{aut},R} : \mathscr{L}_{\mathrm{Pf},R} \to \mathcal{L}^5 \otimes \nu_{\Delta_5}$ compatible with cusp trivialization and squaring.

Obstruction classes. (D0-Pf), (O2)/hybrid compatibility residuals, and the Pfaffian-to-automorphic comparison residuals.

Koszul source coalgebra and cobar comparison

Hypotheses. The hybrid carrier and Hall product, an augmentation of the hybrid source, and a wrapped bar-cobar domain datum for the wrapped summands and mixed local/wrapped words.

Finite datum. The conilpotent chiral source coalgebra $C_{X,R}$, finite filtration $F^{\mathrm{co}}_{R,\bullet}$, collision coproduct Δ^{ch}_R, bar comparison $b_{X,R}$, and cobar comparison $\Theta_{\mathrm{Kos},R}$.

Construction.

- (i) Choose an augmentation $\varepsilon^{\mathrm{hyb}}_{X,R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \to k\mathbf{1}$ and verify bounded length N_R.
- (ii) Form the canonical source-bar coalgebra $C^{\mathrm{bar}}_{X,R} = \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R})$ [6, §3.7] with bar-length filtration and deconcatenation collision coproduct.
- (iii) Verify binary coassociativity on the eight-word atlas, the higher-coloured and unit coherences, bar-length conilpotence, and transition compatibility; set $b_{X,R} = \mathrm{id}$.
- (iv) Verify that the projection-finite summands, wrapped summands, and mixed local/wrapped words lie in the chosen completed Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar domain. Under that wrapped Koszul datum, verify the source-internal counit $\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_E C^{\mathrm{bar}}_{X,R} \to \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$ is a quasi-isomorphism in the chosen completed/coderived category [33, Theorem 6.2.2].
- (v) Construct the source-to-target chiral algebra quasi-isomorphism $\vartheta_R : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R}$ preserving differential, product, unit, transition maps, and primitive restriction.
- (vi) Set $\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$; verify the Koszul Mittag–Leffler condition on the cones of b and Θ_{Kos}.

Obstruction classes. $\mathfrak{O}_{\mathrm{Kos},R}$, including $\mathfrak{o}^{\Omega}_R = (\mathfrak{o}^{\varepsilon,\mathrm{src}}_R, \mathfrak{o}^{\vartheta}_R)$ and the wrapped Koszul defect $\mathfrak{o}^{\mathrm{wrKos}}_R$.

PRIMITIVE RECOGNITION IN COFINAL FINITE WINDOWS

Hypotheses. The Hall product, Pfaffian datum, and Koszul source coalgebra, plus the imported Borcherds–Kac target datum $\mathscr{B}_{\Delta_5}$.

Conditional datum. For every cofinal saturated finite window W_ν, the recognition data (R0)–(R8); after these data are supplied on a cofinal relation-closed system, the inverse-limit isomorphism $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$, and after the Koszul source datum is also supplied, the chiral identification $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}$.

Construction.

(i) For each ν, identify the saturated downward target-root window W_ν in $\mathcal{R}_+$ and the source window $\Gamma(W_\nu) = \{\gamma \in \Gamma_{\mathrm{gram}} \mid \alpha(\gamma) \in W_\nu\}$.

(ii) Choose the simple-root representatives e_i^X, f_i^X, h_i^X in source degrees $\pm\gamma_i$, with $\alpha(\gamma_i) = \alpha_i \in W_\nu$; verify parity assignments via the GN multiplicity fibres.

(iii) Verify the Borcherds–Kac relations (Cartan, Chevalley, real Serre, orthogonality, super sign) on every relation whose source and target degrees lie in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$ and $W_\nu \cup (-W_\nu) \cup \{0\}$, respectively.

(iv) Compute the full finite parity dimensions $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\bar{p}}$ $(p = 0, 1)$ for $\gamma \in \Gamma(W_\nu)$; verify two-integer equality with $\mathrm{mult}^{\mathrm{GN}}_{\bar{p}}(\alpha(\gamma))$.

(v) Compute the Hopf pairing matrices in parity blocks; verify kernel agreement with GN/Kac, coideal stability, and transition compatibility.

(vi) Verify the no-extra-relations equality $\ker \pi_\nu = (\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}})_{\le W_\nu}$.

(vii) Verify generation by Cartan and simple representatives; compute PBW associated gradeds and verify strict transition preservation.

(viii) Verify exact triangular completion: $\lim^1 = 0$ on the triangular finite-window systems.

Obstruction classes. $\mathfrak{O}_{\mathrm{Rec},\nu}$ (§0.6); for the chiral identification, the inverse-limit Koszul Mittag–Leffler condition of §0.5.

FINITE HN ORDER

At HN height R the finite datum has the following order:

$$
\begin{aligned}
(\sigma, S, R) &\longmapsto F_{\sigma,S}(R), \widehat{\Gamma}_R \\
&\longmapsto C_{\sigma,S,\le R}, \Phi^{\mathrm{red}}, o^{\mathrm{red}} \\
&\longmapsto \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \\
&\longmapsto (\mathrm{O}1), (\mathrm{O}1)^+ \\
&\longmapsto m_R^{\mathrm{red}}, \Delta_R^{\mathrm{ch}}, \langle\, ,\, \rangle_R, \overline{\Pi}^{\Theta}_{X,*} \\
&\longmapsto \mathfrak{D}^{D_0}_{X,R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, (\mathrm{O}2) \\
&\longmapsto C_{X,R}, b_{X,R}, \Theta_{\mathrm{Kos},R} \\
&\longmapsto (\mathrm{R}0)\text{–}(\mathrm{R}8) \text{ on } W_\nu
\end{aligned}
$$

The pro-object

$$
\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R \left(\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{0,R}, \mathrm{Rec}_R\right)
$$

is the inverse limit when every finite obstruction class vanishes compatibly.

Finite-stage theorems

Theorem 0.34 *(Finite-stage Dirac–Igusa pro-object; conditional on the finite obstruction classes).* Suppose the finite charge window, Hall atlas, hybrid carrier, orientation transport, Hall product, Pfaffian datum, Koszul source coalgebra, and primitive-recognition datum are defined at heights $R \in \mathcal{R}_{\mathrm{HN}}$, all finite obstruction classes vanish, and the HN transitions are strict. Then these data determine a finite-stage Dirac–Igusa datum $\mathfrak{D}^{\mathrm{DI}}_{X,R}$ at every R, with strict transition maps to $\mathfrak{D}^{\mathrm{DI}}_{X,R'}$ for $R' \leq R$, and the inverse limit $\mathfrak{D}^{\mathrm{DI}}_X = \varprojlim_R \mathfrak{D}^{\mathrm{DI}}_{X,R}$ exists in the completed chiral/Hall framework.

Theorem 0.35 *(Pfaffian–Dirac identification at finite stage; with local orientation and wall data at R).* Suppose the finite-stage Pfaffian datum, orientation transport, type-II wall atlas, and primitive-recognition datum have vanishing obstruction classes at height R. Then on the cusp expansion

$$\mathrm{pf}_{X,R}|_{\mathfrak{c}_\infty} = 64q^{1/2}r^{1/2}s^{1/2} \prod_{\gamma \in \Gamma_R^{\Pi,+}\setminus\{0\}} (1 - x^\gamma)^{\mathrm{sdim}\, P^{\Pi,+}_{R,\gamma}},$$

the Pfaffian section matches the Borcherds product on the test window, with the orientation character $\varepsilon_{0,R}|_{W^{(2),\leq R}} = \nu_{\Delta_5}|_{W^{(2),\leq R}}$ on the type-II reflection truncation. The recognition map ι_R^{Prim} : $H^0\,\mathrm{Prim}_{\mathrm{prot}}(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}) \to \mathfrak{g}_{\Delta_5,\leq R}$ is an isomorphism on the saturated finite window.

Relative theorem on $K3 \times E$

The finite Hall atlas, hybrid carrier, and orientation transport at the test window use the retained D0-HN datum through Theorem 7.6; the orientation descent at (O1) by the retained quotient-orientation datum and Theorem 7.12; these data at $(\mathrm{O1})^+$ by the chosen Weyl lifts and Theorem 7.13; the Pfaffian wall atlas at (O2) by the retained $(O2_{\mathrm{atlas}})$ datum and Theorem 7.18; the cofinal recognition criterion at the first relation-closed degree window beyond $W_{\leq 3}$ is the unresolved compact-source recognition datum of Chapter 10.

The eight finite constructions give the Dirac–Igusa pro-object only when the obstruction classes vanish. Entry (L8) names $(D0)$, $(O1)$, $(O1)^+$, and $(O2_{\mathrm{atlas}})$, locates each in the construction, and carries the retained data and the wall-atlas transport through Appendix 7 in §0.8.

0.8 Four named obstructions

The eighth entry of the Dirac–Igusa pro-object is the obstruction theory. The cross-level identity ②↔③ between the imported Borcherds–Kac–Moody denominator algebra and the compact protected Pfaffian passes through four obstruction classes and the finite Pfaffian section datum. The first obstruction is reduced on $K3 \times E$ to the retained D0-HN datum in Appendix 7 (Theorem 7.6); $(O1)$ and $(O1)^+$ are reduced there to the retained quotient-orientation and Weyl-transport data (Theorems 7.12, 7.13); the fourth is reduced to the retained type-II wall-atlas datum $(O2_{\mathrm{atlas}})$ by Theorem 7.18. The finite Pfaffian section datum supplies support, parity, active-support coverage, leading coefficient, and Pfaffian-line Mittag–Leffler compatibility for the cofinal HN product. The principal Pfaffian and orientation theorem

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) = \Delta_5, \qquad \varepsilon_0 = \nu_{\Delta_5}$$

holds relative to the retained D0-HN datum, the retained $(O1_{\mathrm{quot}})$ datum, the chosen Weyl lifts, and $(O2_{\mathrm{atlas}})$, with the finite Pfaffian section datum for the Pfaffian equality. The Lie-superalgebra recognition $P_X \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source recognition datum $(S1)$–$(S10)$ of §0.6. On $K3 \times E$ the D0 criterion, the quotient-orientation criteria, and the conditional wall-atlas transport are stated in Appendix 7; the compact-source recognition datum is a separate hypothesis.

The fourth obstruction is the type-II wall-atlas datum; the Appendix G theorem transports that datum along the Weyl orbit and compares its character with ν_{Δ_5}.

(D0) D0-DEGENERATION LIMIT OF THE COSECTION-REDUCED D-CRITICAL ORIENTATION THEORY

Obstruction. At every finite HN height R and every retained charge stratum $C \in C_{\sigma,S,\leq R}$, the cosection-reduced d-critical structure on the K3-fibre direction extends to the zero-dimensional D0 sector on $K3 \times E$. Equivalently, the chosen finite reduced obstruction theory $E_R^{\mathrm{red},\bullet}$, the K3 semiregularity cosection CS_C, the reduced vanishing-cycle complex Φ_R^{red}, and the Joyce–Upmeier orientation line o_R are compatible across the HN degeneration to zero-dimensional support, with strict HN transition maps along the degenerating family and preservation of the filtered Hall additivity

$$q^* \mathrm{CS}_C = p_1^* \mathrm{CS}_A + p_2^* \mathrm{CS}_B$$

on extension stacks throughout the limit. The local/wrapped boundary $b_R^{\mathrm{geom}} \to 0$ belongs to the hybrid Ran carrier; it enters $(D0)$ only through compatibility of the retained Hall correspondences with the zero-dimensional limit.

Finite site. The finite Hall atlas and hybrid carrier of §0.7. Failure of (D0) prevents construction of the reduced Hall product m_R^{red}, the orientation gerbe descent of (O1), and the Pfaffian line of (D0-Pf) on the wrapped sector.

Criterion. Criterion on $K3 \times E$ in Theorem 7.6, relative to the retained D0-HN datum. The fibrewise K3 framework of [50, 52] supplies the orientation technology on the projection-finite stratum; the wrapped stratum limit is part of the retained datum of Appendix 7.2.

(O1) STRONG REDUCED ORIENTATION ON THE $K3 \times E$-QUOTIENT SELF-EXT FRAME BUNDLE

Obstruction. On the cosection-reduced moduli $\mathfrak{M}_{C,R}^{\mathrm{red}}$ at every finite HN height R, the orientation line o_C has chosen square root $o_C^{\otimes 2} \simeq \det K_C^{\mathrm{red}}$ [52], multiplicative under reduced extension correspondences, with the quotient-descent equations: vanishing of the reduced gerbe class

$$\alpha_C^{\mathrm{red}} = w_2(\det K_C^{\mathrm{red}}) + \mathrm{cs}^* w_2(\det \operatorname{coker} \mathrm{cs}) \in H^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2),$$

the free E-class $\alpha_C^{E,\mathrm{free}} = a_1(C)u_1 + a_2(C)u_2 \in H^2(BE; \mathbb{F}_2)$, and, for each finite-stabilizer stratum S with stabilizer $H \subset E_{\mathrm{tors}}$, the equivariant gerbe class $\widetilde{\beta}_{C,S}^H \in H_H^2(S; \mathbb{F}_2)$ together with the choice of zero $H^1(BH; \mathbb{F}_2)$-linearization character $\lambda_{C,S}^H = 0$. The statement applies on every charge stratum used by the Hall correspondences and is compatible with HN transitions.

Finite site. The orientation descent datum of §0.7; (O1) datum of §0.2. Failure of (O1) prevents quotient-after-correspondence descent and obstructs the reduced Hall product, the Pfaffian line on the reduced quotient, and the orientation character.

Criterion. Theorem 7.12 is a quotient-orientation criterion relative to the retained $(O1_{\mathrm{quot}})$ datum. The Klein-four condition at $H = E[2]$ reduces the edge class to $r_1 = r_2 = r_3 = 0$, but quotient orientation also requires the zero degree-one linearization character. For even $N \geq 4$ the full two-primary residual includes the mixed coefficient $A_{12}^{(N)}$, which cyclic restrictions do not detect.

$(O1)^+$ WEYL-EQUIVARIANT TRANSPORT ALONG $W^{(2)}(\Lambda_{II}^{2,1})$

Obstruction. The orientation line of (O1) carries Weyl-equivariant lifts of the three type-II wall transports $\tau_i : o_C \to s_{\partial_i}^* o_C$ $(i = 1, 2, 3)$ such that:

(i) the 2-cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$ on the finite partial action groupoid $\mathcal{G}_R$ has vanishing cohomology class $[c_{o,R}] = 0$ in $H^2(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2)$ on each complete orbit, with chosen 1-cochain killing $c_{o,R}$;
(ii) $\tau_i^2 = 1$ for $i = 1, 2, 3$ and the Coxeter relations of $W^{(2)}(\Lambda^{2,1}_{II})$ hold on the lifts;
(iii) the quotient-orientation cocycle $\mathfrak{q}_C$ transports along τ_i between charge strata over Weyl-related Gram degrees, with vanishing torsor defect $\omega_{i,C} = 0$.

Finite site. The Weyl-transport datum of §0.7; (O1)$^+$ datum of §0.2. Failure of (O1)$^+$ obstructs the orientation character ε_o and prevents recognition of the Weyl orientation pattern as $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$.

Conditional theorem. The equivariance part of Theorem 7.13 is conditional on the retained quotient-orientation datum and chosen Weyl lifts. The local rank-one Pfaffian crossing supplies the local sign $\varepsilon^{\mathrm{loc}}_{\delta_i,R} = -1$ on a generic chart; the character comparison also uses the retained $(O2_{\mathrm{atlas}})$ datum. The global Coxeter coherence on $W^{(2)}(\Lambda^{2,1}_{II})$ orbits is established by the no-braid type-II Coxeter presentation and Maass-character match of Appendix 7.4.

(O2) Local Pfaffian wall normal form on type-II walls

Obstruction. At every finite HN height R and every retained component, boundary stratum, and chart overlap of the three simple type-II walls $\mathbb{H}_{\delta_i}$, $i = 1, 2, 3$, the cosection-reduced self-Ext complex splits as

$$K^{\mathrm{red}}_{\delta_i,R} \simeq K^{\mathrm{tan}}_{\delta_i,R} \oplus K^{\mathrm{nor}}_{\delta_i,R},$$

with s_{δ_i}-equivariant tangent factor and the rank-one normal form

$$K^{\mathrm{nor}}_{\delta_i,R} \simeq [\mathbb{R} \xrightarrow{u_{\delta_i,R}} \mathbb{R}], \qquad N^{\mathrm{Pf}}_{\delta_i,R} = 1,$$

with normal Pfaffian unit $\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R}) = v_{\delta_i,R} u_{\delta_i,R}$, $s_{\delta_i}(v_{\delta_i,R}) = v_{\delta_i,R}$, $s_{\delta_i}(u_{\delta_i,R}) = -u_{\delta_i,R}$. The normal Pfaffian units, tangent/normal splittings, quotient orientations, and protected integration maps agree on chart overlaps. The half-Hilbert orbit sign sum of size $4096 = 2^{12}$ on the retained components and boundary strata of the $W^{(2)}(\Lambda^{2,1}_{II})$-orbit reproduces the reflection character $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$.

Finite site. The Pfaffian wall-atlas datum of §0.7; rank-one Pfaffian crossing of §0.4. Failure of (O2) prevents the local Pfaffian section $\mathrm{pf}_{\delta_i,R} = v_{\delta_i,R} u_{\delta_i,R}$ from descending to the orientation character and obstructs the section-level identification $\mathrm{pf}_X = \Delta_5$ on the wall complement.

Conditional theorem. Conditional on $(O2_{\mathrm{atlas}})$ in Definition 7.1, Theorem 7.18. The local rank-one crossing model is established (Proposition 0.23); component, boundary, and overlap compatibility on the full half-Hilbert orbit is part of the retained wall-atlas datum and is transported by Theorem 7.18. The $4096 = 2^{12}$ sign count is the scalar shadow of that retained datum, as separated in Appendix 7.5. The full normal-form theorem with this sign count is relative to Appendix 5.

Joint theorem with recognition datum

Theorem 0.36 *(Pfaffian–Dirac identification with primitive recognition datum).* On $K3 \times E$, suppose (D0), the retained $(O1_{\mathrm{quot}})$ datum, the $(O1)^+$ Weyl lifts, and $(O2_{\mathrm{atlas}})$ hold at every finite HN height R with strict HN transitions as in Appendix 7. Suppose the recognition data (R0)–(R8) of §0.6 are supplied on a cofinal sequence of saturated downward finite windows. Suppose the chiral Koszul source datum of §0.5 is supplied, together with the Koszul Mittag–Leffler exactness hypothesis. Then

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) = \Delta_5, \qquad \varepsilon_o = \nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}, \qquad P_X \cong \mathfrak{g}_{\Delta_5},$$

and $\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$.

The eight finite constructions of §0.7, the recognition theorem of §0.6, and the Koszul comparison of §0.5 give the displayed implication. The Pfaffian and orientation conclusion holds by the retained D0-HN datum, the retained $(O1_{\mathrm{quot}})$ datum, the chosen Weyl lifts, and the retained $(O2_{\mathrm{atlas}})$ datum; the primitive Lie-superalgebra and current-envelope conclusions require the cofinal recognition datum and the Koszul source datum.

The 4096 sign sum and the (O2) wall atlas

The constant $4096 = 2^{12}$ is the size of the half-Hilbert orbit of the type-II reflection group on the retained finite-window components and boundary strata of the type-II wall configuration. At the local rank-one chart, each retained component contributes a sign $\varepsilon^{\mathrm{loc}}_{\partial_i,R} = -1$; the global sign system is the retained (O2) wall atlas of Appendix E transported along the chosen Weyl lifts, and its character is $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$. The Oberdieck–Pixton branch $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is a scalar convention and is not the source of the orientation character. The equality $\varepsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$ is relative to the retained quotient orientation, the chosen Weyl lifts, and the (O2) wall atlas; the local rank-one Pfaffian computation gives only the seed sign.

Compatibility with the companion volumes

The four obstruction classes are source-side geometric data. Their automorphic shadow is smaller: the Gritsenko–Cléry catalogue fixes the Borcherds products, weights, characters, and denominator rows; it does not record the $K3 \times E$ quotient orientation, the Weyl-equivariant determinant-line lifts, or the Pfaffian wall atlas. Thus (D0) is the cosection-reduced $K3$-fibre degeneration datum, (O1) is the reduced orientation gerbe with finite-stabilizer descent, $(\mathrm{O1})^+$ is the Weyl-equivariant determinant-line transport, and (O2) is the retained type-II Pfaffian wall atlas. The relation-closed window tests compatibility of these source data with the target BKM relations; it is not an equality theorem between geometric obstruction classes and automorphic catalogue rows.

The inverse limit on the HN system $\mathcal{R}_{\mathrm{HN}}$ contains the hybrid carrier, the cosection-twisted orientation, the Hall correspondences, the protected Pfaffian, the Koszul source coalgebra, the primitive recognition apparatus, the finite constructions, and the retained orientation and wall-atlas data into

$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R \left(\mathcal{A}^{E_3}_{X,R}, F^{\mathrm{hyb}}_{X,R}, \ldots, \mathrm{Rec}_R\right).$$

The Pfaffian and orientation part of ② $\leftrightarrow$ ③ is then the retained-data theorem of Chapter 7; the Lie-superalgebra recognition remains conditional on $(S1)$–$(S10)$.

Part III

Pfaffian–Dirac Identity

Chapter 7

The cross-level identity ② ↔ ③: Pfaffian–Dirac, orientation character, primitive recognition

The Igusa cusp form Δ_5 of weight five on $\mathbf{Sp}(2, \mathbb{Z})\backslash\mathcal{H}_2$ carries three names, each rooted in a distinct categorical-dimensional level:

- $\mathcal{D}_X = \Delta_5$ — the Borcherds–Gritsenko–Nikulin section of the weight-5 automorphic line over the genus-two Siegel space (View ①);
- $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ — the denominator of the generalized Borcherds–Kac–Moody (BKM) superalgebra $\mathfrak{g}_{\Delta_5}$ on $\Lambda^{2,1}_{II}$ (View ②);
- $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ — the protected Pfaffian of the Dirac–Igusa pro-object on $X = K3 \times E$ (View ③).

The cross-level identity ①↔② is Borcherds 1995 and Gritsenko–Nikulin 1998. The edge ②↔③ has two pieces. The four obstruction classes and the finite Pfaffian section datum give the Pfaffian; the orientation classes give the orientation character on $K3 \times E$. The Lie-superalgebra recognition has the additional finite compact-source datum $(S1)$–$(S10)$. The three statements have separate hypotheses:

(i) the *Pfaffian–Dirac theorem*: under (D0), (O1), $(\mathrm{O1})^+$, (O2), and the finite Pfaffian section datum on a cofinal HN system,
$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) \;=\; \Delta_5$$
as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$;

(ii) the *orientation character identity*: under (O1)+$(\mathrm{O1})^+$+(O2),
$$\epsilon_o \;=\; \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II});$$

(iii) the *primitive recognition theorem*: under the finite compact-source recognition datum (Cartan, simple representatives, Chevalley, real Serre, imaginary orthogonality, generation, completed PBW, Hopf radical equality, no-extra-relations, full parity dimensions),
$$H^0\,\mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big)\big/\overline{\mathrm{Rad}\langle\,\cdot\,,\,\cdot\,\rangle_X} \;\cong\; \mathfrak{g}_{\Delta_5}$$
after passage to the inverse limit over the cofinal root windows, with the chiral cobar $\Omega^{\mathrm{ch}}_E C_X$ identified as the formal current envelope $\mathsf{Den}_{\Delta_5,E}$ only after the chiral Koszul source datum is also fixed.

The eight finite constructions of §5.7 define, at every finite stage R in the Harder–Narasimhan inverse system, the data $\mathcal{A}^{E_3}_{X,R}$, $F^{\mathrm{hyb}}_{X,R}$, $\Gamma_{X,R}$, Π_X, Φ_R, o_R, H_R, P^{Π}_R, $C_{X,R}$, $\Theta_{\mathrm{Kos},R}$, $\mathscr{L}_{\mathrm{Pf},R}$, $\mathrm{pf}_{X,R}$, $\varepsilon_{o,R}$, and Rec_R of the pro-object $\mathfrak{D}^{\mathrm{DI}}_X$; the four obstruction classes of §5.8 are exactly (D0), (O1), $(\mathrm{O1})^+$, (O2). These four classes act on the Pfaffian line and the orientation character; the assertion that Rec_R is eventually an isomorphism is the separate compact-source recognition datum. The relevant geometric data are the open factorization chart of the vacuum b, the protected Pfaffian functor, the Hall–Drinfeld correspondence, the chosen $\mathrm{Sp}^{\mathrm{ch}}$ specialization, the pro-object ambient, the Do-degeneration endpoint, the Bussi–Brav–Dupont–Joyce–Szendrői orientation, the non-degeneracy of the Pfaffian line, and the type-II wall normal form.

The Pfaffian and orientation theorems are retained-data theorems on $K3 \times E$, as in Appendix 7; the primitive recognition theorem is conditional on the compact-source datum $(S1)$–$(S10)$. The BKM denominator identity (View ②) is *proved*, imported from [7, 8, 43, 45, 54]; the automorphic identification (View ①) is *proved*, imported from [48, 49]; the operator-level Pfaffian datum $\mathfrak{D}_X^{\mathrm{DI}}$ is *proved on* $K3 \times E$ *relative to the retained Do-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts,* $(O2_{\mathrm{atlas}})$*, and the retained finite Pfaffian section datum* of Appendix 7; the chiral cobar identification is *conditional* on the Koszul source datum of Definition 0.26. The κ-tuple of the Calabi–Yau quantum-groups companion, $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$, fixes the level of Δ_5: the cusp-form weight is $\kappa_{\mathrm{BKM}} = 5$ and the elliptic genus weight is $\chi_{\mathrm{top}}(K3) = \kappa_{\mathrm{fiber}} = 24$; the additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ has no invariant meaning on the κ-tuple.

1 Data and Obstructions

1.1 The pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ and its eight finite-stage components

Three names of Δ_5: the section $\mathcal{D}_X = \Delta_5$ (View ①); the BKM denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View ②); the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ (View ③, relative to retained data). The handle "Δ_5" is reserved for the working symbol; theorem-level sentences specify the named view.

The Dirac–Igusa pro-object is

$$\mathfrak{D}_X^{\mathrm{DI}} = \lim_R \big(\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \varepsilon_{o,R}, \mathrm{Rec}_R\big).$$

The components are the values of the eight finite constructions of §5.7, with the ambient category the cofinal HN inverse system of Definition 3.14; R ranges over a chosen cofinal subsystem $\mathcal{R}_{\mathrm{HN}} \subset \Lambda_{\mathrm{ample}}$. The label γ is the pro-object ambient.

Remark 1.1 *(The three Δ_5-names).* The equality $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X$ is a cross-level identity, not a tautology. The right-hand side is an automorphic section; the left-hand side is the inverse limit of finite Pfaffian sections $\mathrm{pf}_{X,R}$ of the line bundles $\mathscr{L}_{\mathrm{Pf},R}$ on the reduced quotient stacks $\mathfrak{Q}_R$; the two are related by the fixed comparison $\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$. The denominator name $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is the same global section $\mathcal{D}_X$ written in the type-II chamber polarization that exposes the Weyl–Kac–Borcherds product expansion; the prefactor 64^{-1} and the doubling $2Z$ of the modular variable enter from the theta normalization of the Borcherds product on the index-4 sublattice $\Lambda_{II}^{2,1} \subset \Lambda^{2,1}$. The source side $P_X^{\Pi,\mathrm{red}}$ identifies with $\mathfrak{g}_{\Delta_5}$ only after the recognition criterion of §4 is established.

1.2 The four obstruction classes: (Do), (O1), (O1)$^+$, (O2)

The four classes match the eight finite constructions one-to-one only on the orientation lane. They are the obstruction data of Theorem 0.35. On $K3 \times E$, Appendix 7 reduces $(D0)$ to the retained Do-HN datum by Theorem 7.6; reduces $(O1)$ to the retained quotient-orientation datum by Theorem 7.12; reduces $(O1)^+$ to the chosen Weyl lifts by Theorem 7.13; and reduces $(O2)$ to the retained type-II wall-atlas datum $(O2_{\mathrm{atlas}})$ by Theorem 7.18. The $(R1)$–$(R2)$ scalar separation for the type-II middle-wall constant $4096 = 2^{12} = 64^2$ inside $(O2)$ is Theorem 7.16.

1.2.0.1 (D0) — D0-degeneration endpoint of the cosection-reduced d-critical orientation theory. At every R, the data $D_{0,R}$ on the finite support $\widehat{\Gamma}_R$ are fixed: the finite Hall stage, the finite target test window Γ_R^{test}, successor transition maps whose composites give the smaller-stage maps, source/observable comparison maps, and a vanishing stage residual tuple $\mathfrak{O}_{D_0,R} = 0$; compatibly in R, the inverse limit gives the HN-completed state object and the finite first-order block $\mathfrak{D}_X$. The finite first-order block also carries the exponent-matching clause

$$\operatorname{sdim} P^{\Pi}_{X,(n,l,m)} = f(nm,l) \qquad ((n,l,m) \in \Gamma_{\mathrm{eff}}),$$

where $f(n,l)$ is the Fourier coefficient of the weak Jacobi form $\phi_{0,1}$ of weight zero and index one. This clause is not a consequence of the reduced orientation alone; it is part of the retained finite-stage Dirac block. The cosection contraction of the Pantev–Toen–Vaquie–Vezzosi (PTVV) (-1)-shifted symplectic form by the K3 semiregularity cosection σ_{K3} is the Maulik–Toda / Oberdieck reduced obstruction theory [76, 84]; this absorbs the parity flip $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$ into the reduced orientation line as the $\mathbb{Z}/2$-grading shift $o_C^{\mathrm{red}} = o_C^{\mathrm{std}} \otimes \Pi$.

1.2.0.2 (O1) — strong multiplicative reduced orientation datum on the reduced $K3 \times E$-quotient self-Ext frame bundle. A Joyce–Upmeier-type orientation line $o_C \in \operatorname{Pic}(\mathfrak{M}_X^{or})$ with $o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C,C)$, extension-multiplicative under $o_{C\oplus C'} \simeq o_C \otimes o_{C'}$, and equipped with the full quotient-orientation datum

$$\theta_C = \left(\theta_C^{\mathrm{red}}, \theta_C^{E,\mathrm{free}}, \{(\theta_C^H, \lambda_C^H = 0)\}_{H\in\mathcal{H}(C)}\right)$$

on every retained charge stratum. The vanishings $\alpha_C^{\mathrm{red}} = 0$, $\alpha_C^{E,\mathrm{free}} = 0$, $\widetilde{\beta}_C^H = 0$, $\lambda_C^H = 0$ are the orientation-line, the Borel-equivariant free-action, the residual finite-stabilizer Borel, and the residual finite-stabilizer linearization conditions; for $N = 2$, the Klein-four calculation $\beta_C^{E,E[2]} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2$ specializes the residual. The orientation lift technology is BBDJS [11] on the unreduced side and Joyce–Upmeier [51, 52] on the multiplicative side; the reduced $K3 \times E$-quotient theorem is separate.

1.2.0.3 (O1)$^+$ — Weyl-equivariant determinant-line transport along $W^{(2)}(\Lambda_{II}^{2,1})$ reflections. The chosen lifts $\tau_i : o_C \to s_{\delta_i}^* o_C$, $i = 1,2,3$, of the three type-II wall transports satisfy $[c_o] = 0$ in $H^2(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2)$ (equivalently, in the finite action-groupoid cohomology $H^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$ at every R), $\widetilde{\tau}_i^2 = 1$, and the quotient-orientation transport defects $\omega_{i,C} = 0$ for every simple reflection. Then $\widetilde{\tau}_i$ define an honest Weyl action on the reduced quotient orientation; the passage from the three generator signs to a character on $W^{(2)}(\Lambda_{II}^{2,1})$ is the group-theoretic part of Definition 0.9.

1.2.0.4 (O2) — Pfaffian local model and wall atlas on type-II walls. At a generic point of each $\mathbb{H}_{\delta_i}$, the chosen real local equation $u_i = u_{\delta_i}$ with $s_{\delta_i}(u_i) = -u_i$, the normal determinant-complex splitting

$$K^{\mathrm{nor}}_{\delta_i,R} \simeq \left[\mathbb{R} \xrightarrow{u_i} \mathbb{R}\right], \qquad \mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i,R}) = v_{\delta_i,R}u_i, \qquad s_{\delta_i}(v_{\delta_i,R}) = v_{\delta_i,R},$$

and the rank-one normal-form rank $N^{\mathrm{Pf}}_{\delta_i} = 1$ hold. The wall charts cover every retained component and boundary stratum of $\mathbb{H}_{\delta_i}$; on overlaps the tangent/normal splittings, normal Pfaffian units, quotient orientations, and protected integration maps agree after a zero Čech sign class. The retained atlas contains, for the middle wall $\delta_2 \leftrightarrow (0,1,1)$, a wrapped wall object $x_{\delta_2,R} \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ with $b_R^{\mathrm{geom}}(\eta) > 0$ and

$\overline{\Pi}_X(\eta) = (0, 1, 1)$. Proposition 5.2 supplies only the rank-one local Pfaffian sign on such a chart; the component, boundary, overlap, quotient, and protected-integration compatibilities are part of $(O2_{\mathrm{atlas}})$.

1.3 The OP minus sign is a scalar branch convention

The Oberdieck–Pixton (OP) signed scalar is $C_{\mathrm{OP},X} = -4096$. Its absolute square-normalization is $C_{\Box,X} = 4096 = 64^2$, and its sign is the OP scalar branch recorded by the fermionic shift $(-p_{\mathrm{DT}})^n$ of the OP/DT chamber. It is not the source of the orientation character ϵ_o; orientation lines are killed by squaring. The orientation character is computed by monodromy of the Pfaffian section around the type-II walls (Lemma 5.2); equivalently, by the $(-1)^{N^{\mathrm{Pf}}_{\delta_i}}$ values from (O2).

2 The Pfaffian–Dirac theorem

The protected Pfaffian of the Dirac–Igusa pro-object is compared with the automorphic section.

Theorem 2.1 *(Pfaffian–Dirac section identity).* Granted the four obstruction classes of Section 1.2, with $(O2)$ understood as the retained wall-atlas datum $(O2_{\mathrm{atlas}})$, and granted the finite Pfaffian section datum of Proposition 0.21 on a cofinal HN system, including support, parity, active-support coverage, leading-coefficient, and Pfaffian-line Mittag–Leffler compatibilities, the protected Pfaffian of the Dirac–Igusa pro-object $\mathfrak{D}^{\mathrm{DI}}_X$ equals the automorphic section $\mathcal{D}_X = \Delta_5$:

$$\boxed{\iota_{\mathrm{aut}}\Big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)\Big) \;=\; \mathcal{D}_X \;=\; \Delta_5}$$

as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ over the genus-two Siegel space $\mathbf{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2$, where

$$\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$$

is the fixed Pfaffian-to-automorphic line comparison, compatible with the cusp trivialization at $\mathfrak{c}_\infty$ and with squaring to $\mathcal{L}^{10}$. Equivalently, in the type-II Borcherds chamber,

$$\iota_{\mathrm{aut}}\Big(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)\Big)\Big|_{\mathfrak{c}_\infty} = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

After squaring, the orientation forgets and the OP scalar branch identifies the level-n protected scalar shadow with the inverse squared Pfaffian:

$$Z^X_{\mathrm{OP}} \;=\; -64^2\,(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X))^{-2} \;=\; -4096\,\Delta_5^{-2},$$

in agreement with [86], where $4096 = 64^2 = ([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ is the squared theta-normalization and the leading minus sign is the OP scalar branch convention. The BPS normalization rescales by -4096^{-1}: $Z^{K3\times E}_{\mathrm{BPS}} = -4096^{-1} Z^X_{\mathrm{OP}} = \Delta_5^{-2}$.

Proof. The inverse-limit assembly consists of a Pfaffian product, a leading coefficient identification with a known weight-five Siegel form, and a q-expansion principle on a punctured neighbourhood of the type-II cusp $\mathfrak{c}_\infty$.

Finite-stage Pfaffian product $(\delta + \varepsilon)$. The retained Do-HN datum supplies the cosection-reduced d-critical pro-limit on the retained HN system. The finite Pfaffian section datum supplies, at every cofinal R, the first-order block $\mathfrak{D}_{X,R}$ on the reduced compact quotient $\mathfrak{Q}_R = (\mathfrak{M}_R/E)^{\mathrm{red}}$, the Pfaffian line

$\mathscr{L}_{\mathrm{Pf},R}$, active-support coverage, the support and parity residual vanishings, and the normalized leading coefficient $\mathfrak{o}_R^{\mathrm{lead}} = 0$. Hypothesis (O1) supplies the orientation datum whose square-root line o_C satisfies $o_C^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(C,C)$ and extension-multiplicativity $o_{C\oplus C'} \simeq o_C \otimes o_{C'}$, descended under the reduced E-quotient. The protected Pfaffian functor of (β) is then the graded Pfaffian product of the determinant of $\mathrm{RHom}_{\mathrm{red}}$ over the retained charge strata. The exponent-matching clause of the retained finite-stage Dirac block gives

$$\operatorname{sdim} P^{\Pi}_{X,(n,l,m)} = f(nm,l) \quad ((n,l,m) \in \Gamma_{\mathrm{eff}}),$$

so the graded Pfaffian product, expanded in the type-II cusp polarization $(q,r,s) = (e^{2\pi i z_1}, e^{2\pi i z_2}, e^{2\pi i z_3})$, is

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_{X,R}) = v_0 \prod_{(n,l,m)\in\Gamma_{\mathrm{eff,R}}} (1 - q^n r^l s^m)^{f(nm,l)}, \qquad v_0 = 64 q^{1/2} r^{1/2} s^{1/2},$$

where $\Gamma_{\mathrm{eff,R}}$ is the active support cut by HN height R. The Weyl monomial v_0 is the theta normalization $[q^{1/2} r^{1/2} s^{1/2}]\Delta_5 = 64$ of [43, Theorem 4.1].

Inverse limit (γ). The pro-object ambient is the cofinal HN system; the retained D0-HN datum and finite Pfaffian section datum supply strict transition maps with vanishing successor residuals $\mathfrak{o}^{\mathrm{atlas}}_{R^+R} = \mathfrak{o}^{\mathrm{tr}}_{R^+R} = 0$; the finite Pfaffian sections $\mathrm{pf}_{X,R}$ form a Mittag-Leffler system on the active support, and the inverse limit is the formal Borcherds product

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})\big|_{\mathfrak{c}_\infty} = \lim_R \mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_{X,R}) = 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)}.$$

Class M completion (chain-level after weight-completion, in the sense of Vol I's class M global triangle) is exactly the ambient in which $\lim_R$ commutes with the graded Pfaffian product on the active support; this is the γ clause.

Comparison with the imported Borcherds product (β). By Proposition 7.2, the formal product $64\, q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ equals the Fourier expansion of the automorphic section $\mathcal{D}_X = \Delta_5$ at $\mathfrak{c}_\infty$; this is the Borcherds–Gritsenko–Nikulin theorem [8, 43, 45]. The fixed $\iota_{\mathrm{aut}} : \mathscr{L}_{\mathrm{Pf},X} \xrightarrow{\sim} \mathcal{L}^5 \otimes \nu_{\Delta_5}$ of (β), compatible with cusp trivialization and squaring, transports the inverse-limit Pfaffian to a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

q-expansion principle (α). The chart α is the open factorization category C on (X, D, τ) with boundary vacuum b: on the punctured neighbourhood of $\mathfrak{c}_\infty$ inside $\mathrm{Sp}(2,\mathbb{Z})\backslash\mathcal{H}_2$ the chart algebra $A_b = \mathrm{End}_C(b)$ supplies the local factorization-algebra structure within which the Pfaffian section is holomorphic. Two holomorphic genus-two Siegel modular forms with the same Maass character ν_{Δ_5} and the same Fourier expansion at $\mathfrak{c}_\infty$ are equal: this is the q-expansion principle for holomorphic Siegel modular forms [48, 49]. Both sides have weight 5 and Maass character ν_{Δ_5}: the right-hand side by the Borcherds–Gritsenko–Nikulin theorem; the left-hand side because the orientation character of the retained $\mathfrak{D}_X^{\mathrm{DI}}$, under (O1)+(O1)$^+$+(O2), is exactly ν_{Δ_5} on $W^{(2)}(\Lambda^{2,1}_{II})$ (this is Theorem 3.1). Hence $\iota_{\mathrm{aut}}(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})) = \Delta_5$ as sections of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$.

Squared scalar (β). Squaring forgets orientation: the squared line $\left(\mathcal{L}^5 \otimes \nu_{\Delta_5}\right)^{\otimes 2} = \mathcal{L}^{10}$ is trivially charactered, and $\Delta_5^2 = \Delta_{10}$ is scalar. The OP scalar normalization of Convention 1.1 fixes the partition function $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1} = -4096\,\Delta_5^{-2}$, with $\chi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2$ and $D_5 = 64^{-1}\Delta_5$. Composition of the squared Pfaffian section with the OP scalar branch gives the OP chamber scalar representative

$$Z^X_{\mathrm{OP}} \;=\; -\,(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}))^{-2} \cdot C_{\square,X}, \qquad C_{\square,X} = 4096,$$

identifying $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$; the minus sign is the OP scalar branch convention, recorded by $(-p_{\mathrm{DT}})^n$, and is not a feature of the orientation character. The protected scalar trace is $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$. □

Remark 2.2 *(Why (D0)+(O1)+(O1)$^+$+(O2)+(P_{fin})).* Each named datum is essential. (D0) gives the existence and the finite first-order block; without it there is no $\mathfrak{D}_{X,R}$ and no Pfaffian product to take. (O1) gives the orientation line and the reduced E-quotient descent; without the line the Pfaffian section has no domain. (O1)$^+$ gives the honest Weyl action on the reduced quotient orientation; without it the type-II reflection signs are projective and the orientation character is projective, not a homomorphism. (O2) gives the local Pfaffian normal form; without it the rank $N^{\mathrm{Pf}}_{\delta_i}$ is undetermined and the local sign cannot be computed. The finite Pfaffian section datum gives support, parity, active-support coverage, leading coefficient, and Pfaffian-line Mittag–Leffler compatibility; without it the inverse-limit product has no section-level domain. If one of these data is omitted, one of the proof steps loses its object.

3 The orientation character identity

The second theorem identifies the geometric orientation monodromy character ϵ_o of the Dirac–Igusa pro-object with the automorphic Maass character ν_{Δ_5} on the type-II Weyl group. The proof splits into a local Pfaffian wall computation, a group-theoretic extension, and an independent automorphic check.

Theorem 3.1 *(Orientation character).* Recall the type-II simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ on $\Lambda^{2,1}_{II} \subset \Lambda^{2,1} = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$ (Equation (4.4)). Granted the orientation classes (O1), (O1)$^+$, (O2), the geometric orientation monodromy character of the Dirac–Igusa pro-object equals the automorphic Maass character on the type-II Weyl group:

$$\boxed{\epsilon_o \;=\; \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II})}\,.$$

The character takes the value -1 on each of the three type-II simple reflections $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$, and the identity is reduced to these three values by the abelianization $W^{(2)}(\Lambda^{2,1}_{II})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^3$ (Definition 0.9).

The identity is a comparison of two characters defined independently: ϵ_o is computed by transport of the Pfaffian section under the chosen Weyl action of (O1)$^+$; ν_{Δ_5} is the Maass multiplier of the holomorphic Siegel form $\mathcal{D}_X = \Delta_5$ computed from the multiplier system of the corresponding theta lift [42, 43, 71]. No scalar branch convention enters this sign comparison: replacing Δ_5 by $c\Delta_5$, $c \in \mathbb{C}^\times$, changes neither the Maass character nor the divisor orders. The factor 64 names the monic Borcherds product $D_5 = 64^{-1}\Delta_5$; the OP/DT scalar -4096 is unused.

Proof. Three inputs combine: Proposition 5.2 computes the local Pfaffian signs on the three simple type-II walls; Definition 0.9 extends a generator sign-tuple $(\epsilon_1, \epsilon_2, \epsilon_3) \in \{\pm 1\}^3$ to a unique character on $W^{(2)}(\Lambda^{2,1}_{II})$; and Lemma 7.2 computes the automorphic target values $\nu_{\Delta_5}(s_{\delta_i}) = -1$ directly from the Maass multiplier and the Borcherds divisor order $d^{\mathrm{aut}}_{\delta_i} = 1$.

Local Pfaffian signs from (*O2*). By (O2), at a generic point of each $\mathbb{H}_{\delta_i}$, $i = 1, 2, 3$, the chosen local equation u_i satisfies $s_{\delta_i}(u_i) = -u_i$ and the normal determinant-complex splits as $K^{\mathrm{nor}}_{\delta_i} \simeq [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$ with $\mathrm{Pf} = v_{\delta_i} u_i$ and $s_{\delta_i}(v_{\delta_i}) = v_{\delta_i}$. Transport along the closed path ℓ_{δ_i} (the path in the reduced quotient wall-complement carrying $u_i = \epsilon$ to $u_i = -\epsilon$, closed by the (O1)$^+$ lift $\widetilde{\tau}_i$) sends $v_{\delta_i} u_i$ to $-v_{\delta_i} u_i$, giving

$$\epsilon_o(s_{\delta_i}) = (-1)^{N^{\mathrm{Pf}}_{\delta_i}} = -1, \qquad i = 1, 2, 3.$$

This is Proposition 5.2. Squaring eliminates the sign and hence eliminates the OP scalar branch ambiguity: the squared Pfaffian $\mathrm{Pf}(K^{\mathrm{nor}}_{\delta_i})^2 = u_i^2$ is fixed by s_{δ_i}.

Group-theoretic extension via $(O1)^+$. The type-II Weyl group has Coxeter presentation

$$W^{(2)}(\Lambda^{2,1}_{II}) \simeq \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s^2_{\delta_1} = s^2_{\delta_2} = s^2_{\delta_3} = 1\rangle,$$

with no braid relations — the off-diagonal entries $(\delta_i, \delta_j) = -2$ for $i \neq j$ give Coxeter exponents $m_{ij} = \infty$. Hence $W^{(2)}(\Lambda^{2,1}_{II})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^3$ with one generator per s_{δ_i}, and any $\{\pm 1\}$-valued assignment on the generators extends uniquely to a homomorphism. By $(\mathrm{O1})^+$, $[c_o] = 0$, $\widetilde{\tau}_i^{\,2} = 1$, and $\omega_{i,C} = 0$; the lifts $\widetilde{\tau}_i$ define an honest action on the reduced quotient orientation, and the induced character has $\epsilon_o(s_{\delta_i}) = -1$ for $i = 1, 2, 3$. This is Definition 0.9.

Independent automorphic computation of ν_{Δ_5}. The Maass multiplier ν_{Δ_5} of the singular theta lift of $\phi^{\mathrm{Weil}}_{0,1}$ is computed from the multiplier system of the Borcherds product (Borcherds [8], Gritsenko–Nikulin [43, Section 3],[42], Maass [71]). On the type-II simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$, the Borcherds divisor orders are

$$d^{\mathrm{aut}}_{\delta_i} := \mathrm{ord}_{\mathbb{H}_{\delta_i}}(\Delta_5) = f(0, \pm 1) = 1, \qquad i = 1, 2, 3,$$

so $\nu_{\Delta_5}(s_{\delta_i}) = (-1)^{d^{\mathrm{aut}}_{\delta_i}} = -1$ for $i = 1, 2, 3$. This is Lemma 7.2. The chamber-permuting roots $f_{-2} - f_2, f_2 - f_3, f_{-2} - f_3$ lie in $\Lambda^{2,1} \setminus \Lambda^{2,1}_{II}$ and are not type-II walls; their Maass values are +1 but no Hall orientation sign is asserted on them unless a semidirect-product Hall lift is constructed (Remark 3.4).

The comparison $\epsilon_o = \nu_{\Delta_5}$. Two $\{\pm 1\}$-valued characters of $W^{(2)}(\Lambda^{2,1}_{II})$ that agree on $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ agree everywhere by the Coxeter presentation above. The local Pfaffian computation and the automorphic computation give that both ϵ_o and ν_{Δ_5} take the value -1 on each of the three simple reflections. Therefore $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$. □

Remark 3.2 *(The OP minus sign is not the source of* ϵ_o*).* The local Pfaffian computation and the automorphic multiplier computation are independent. The former uses no automorphic data; the latter uses no Pfaffian local model. In particular, the OP scalar $C_{\mathrm{OP},X} = -4096 = (-1) \cdot C_{\square,X}$ does not source the orientation character: squaring forgets orientation, so $C_{\mathrm{OP},X}$ carries no information about ϵ_o. The minus sign $\mathrm{sign}(C_{\mathrm{OP},X}) = -1$ is the OP scalar branch convention of [86], recorded by the fermionic shift $(-p_{\mathrm{DT}})^n$ of the OP/DT chamber. The orientation character is a *geometric* monodromy datum, computed by transport of the Pfaffian line around type-II walls; it is fixed by the data of $(\mathrm{O1})$+$(\mathrm{O1})^+$+$(\mathrm{O2})$, and is independently checked against the automorphic target by the Maass multiplier computation.

Remark 3.3 *(Imaginary roots are not Weyl reflections).* The Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ of the type-II BKM datum is generated by real simple reflections only. The imaginary simple roots contribute to the Weyl–Kac–Borcherds denominator through the multiplicity coefficients $m(a)$ and $\tau(a)$ of Gritsenko–Nikulin [45, Theorem 2.1],[43, Section 3]; they do not define reflections in $W^{(2)}(\Lambda^{2,1}_{II})$. Hence $\epsilon_o : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ is a real-root reflection datum. Local divisor monodromy of a compact Hall theory around imaginary components of the Borcherds divisor is not a Weyl character and is not asserted here.

Remark 3.4 *(Extension to* $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$*).* The chamber-permuting symmetry group $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ acts on the Coxeter generators $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$ by permutation; its three reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$ have $\nu_{\Delta_5} = +1$. An extension of ϵ_o to $W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II})$ would require a separate Hall semidirect-product lift of the chamber-permuting reflections.

4 The primitive recognition theorem

The third theorem is a criterion for identifying the source-side primitive Lie superalgebra of the Dirac–Igusa pro-object with the BKM denominator algebra $\mathfrak{g}_{\Delta_5}$. The conditions form the *recognition criterion*: ten finite-stage data carrying compact geometric provenance, none of which is inferable from the Borcherds product or signed determinant alone.

Theorem 4.1 *(Primitive recognition).* Granted the recognition criterion (S1)–(S10) of Theorem 0.30, the cofinal finite-window datum of Definition 0.31, the chiral Koszul source datum of Definition 0.26, and the retained orientation data ($O1_{\mathrm{quot}}$), the chosen Weyl lifts, and ($O2_{\mathrm{atlas}}$), suppose the compact $K3 \times E$ Hall/factorization construction supplies, on a cofinal sequence of downward-saturated finite root windows $W_1 \subset W_2 \subset \cdots$ with $\bigcup_\nu W_\nu = \mathcal{R}_+$:

- *(S1) Chain-level descent endpoint.* The (D0)-degenerated source coalgebra $(C_{X,W_\nu}, Q_{X,W_\nu})$ of Definition 0.26, descended chain-level to its primitive part on every finite window W_ν, carrying both the orientation lane and the Frobenius–Serre primitive structure feeding the recognition window.
- *(S2) Cartan and root-degree matching.* An even Cartan identification $\iota_H : P^{\Pi,\mathrm{red}}_{X,0} \xrightarrow{\sim} \Lambda^{2,1}_{II} \otimes \mathbb{C}$, with Gram-grading $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$.
- *(S3) Simple primitive representatives.* Parity-homogeneous e_i^X, f_i^X, h_i^X for every simple-root index $i \in I$ of $\mathscr{B}_{\Delta_5}$, with Gritsenko–Nikulin parity dimensions $\mu_{\bar{0}}(a)$, $\mu_{\bar{1}}(a)$ on imaginary fibres.
- *(S4) Borcherds–Kac relations.* Cartan, Chevalley ($[e_i^X, f_j^X] = \delta_{ij} h_i^X$), real Serre, Borcherds orthogonality, super sign, and isotropic-string relations satisfied in the protected Hall bracket.
- *(S5) Completed Hopf radical equality.* The completed Hopf pairing has closed homogeneous radical Rad_X which is a Lie ideal and a coideal, and identifies under the radical quotient with $\mathrm{Rad}_{\mathrm{GN}}$.
- *(S6) No-extra-relations.* The presentation map $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P^{\Pi,\mathrm{red}}_X$ has $\ker \pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$, on every finite window separately.
- *(S7) Generation.* $P^{\Pi,\mathrm{red}}_X$ is topologically generated by $P^{\Pi,\mathrm{red}}_{X,0}$ and $\{e_i^X, f_i^X\}_{i \in I}$.
- *(S8) Completed PBW.* The associated graded $\mathrm{gr}_{\mathrm{PBW}}\, U(P^{\Pi,\mathrm{red}}_X)_W \xrightarrow{\sim} \mathrm{gr}_{\mathrm{PBW}}\, U(\mathfrak{g}_{\Delta_5})_W$ is an isomorphism on every finite window, with strict transitions.
- *(S9) Exact triangular completion.* The HN inverse limit is exact on the saturated finite root windows: $\lim^1$ vanishes for the kernels, radicals, PBW associated gradeds, and root-space parity pieces.
- *(S10) Full parity dimensions.* Two integers per nonzero root degree, in both signs: $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\bar{p}} = \mathrm{mult}^{\mathrm{GN}}_{\bar{p}}(\alpha(\gamma))$, $p = 0, 1$, and the same equality in degree $-\gamma$ through the Borcherds–Kac Chevalley anti-involution; not only their signed difference.

Then the recognition datum determines an isomorphism of completed Γ_{gram}-graded Lie superalgebras

$$\boxed{H^0\, \mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big) \big/ \overline{\mathrm{Rad}\langle\cdot,\cdot\rangle_X} \;\cong\; \mathfrak{g}_{\Delta_5}}$$

and the Weyl–Kac–Borcherds denominator of $P^{\Pi,\mathrm{red}}_X$ is $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$. If in addition the chiral Koszul source datum of Definition 0.26 is fixed, including the Koszul Mittag–Leffler exactness hypothesis of Proposition 0.29, then the chiral cobar of the source coalgebra C_X is identified with the formal current envelope:

$$\Omega^{\mathrm{ch}}_E C_X \;\simeq\; \mathsf{Den}_{\Delta_5,E}.$$

Proof. The argument is the inverse-limit assembly of (S1)–(S10) on each finite window, together with the chiral cobar identification. The window-wise theorem is Theorem 0.30; the additional input is the chiral-cobar comparison $\Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}$.

Presentation map and kernel ($\alpha + \beta$). (S2) and (S3) define the continuous Lie-superalgebra homomorphism

$$\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \longrightarrow P^{\Pi,\mathrm{red}}_X$$

on the descended Gram-graded completion. (S4) puts the Borcherds–Kac ideal J_{BK} in the kernel. (S6) gives the kernel equality $\ker \pi = \overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$ on every finite window, hence injectivity of the descended map $G(\mathscr{B}_{\Delta_5}) \to P^{\Pi,\mathrm{red}}_X$. (S7) gives surjectivity. Thus the descended map is an isomorphism on every finite window.

Radical descent (ε). (S5) identifies the source Hopf radical $\overline{\mathrm{Rad}_X}$ with the target radical $\overline{\mathrm{Rad}_{\mathrm{GN}}}$. The required Hopf adjointness, ideal/coideal property, quotient-tensor non-degeneracy, and closedness on the inverse system are part of (S5), equivalently the finite matrix conditions (R4)–(R5). The orientation lane supplies the trace pairing and Frobenius formalism on which those equations are written; it does not prove the source matrices M, D, B, G, K, Q, the kernel equality, or the PBW comparison.

Full parity dimensions (δ). (S10) is the two-integer equality $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\overline{p}} = \mathrm{mult}^{\mathrm{GN}}_{\overline{p}}(\alpha(\gamma))$ for $p = 0, 1$, not just the signed difference. This rules out hidden zero-superdimension source summands, wrong parity lifts invisible to the determinant, and parity-cancelling ideals. A putative compact source matching every Hall bracket template of $\mathfrak{g}_{\Delta_5}$ on a finite window can still fail (S5): a kernel element $k \in K_{\gamma_\rho}$ with $(Q_{\gamma_\mu} \otimes Q_{\gamma_\nu}) D^{\mu,\nu}_\rho(k) \neq 0$ is invisible to the quotient target table yet breaks the radical coideal property, so the Hopf quotient is not defined (Theorem 0.30). This is the finite tensor-Hopf obstruction hidden by signed determinant data; (S5)+(S10) together block it.

PBW comparison and exact completion (γ). (S8) is the associated-graded isomorphism on every finite window, with strict transitions. (S9) is the exact-completion clause: $\lim^1$ vanishes for kernels, radicals, PBW associated gradeds, and root-space parity pieces. Equivalently, no new generator, relation, radical element, or parity-cancelling summand appears only after completion. Class M completion is the ambient: chain-level identifications hold after weight-completion; this is the γ clause.

Passage to the inverse limit. Since the windows are cofinal ($\bigcup_\nu W_\nu = \mathcal{R}_+$) and the radical in (S5) is the closure of the finite kernels, the finite isomorphisms of the finite-window clauses determine the inverse-limit isomorphism

$$H^0\,\mathrm{Prim}_{\mathrm{prot}}\big(\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X\big)\big/\overline{\mathrm{Rad}\langle\,\cdot\,,\,\cdot\,\rangle_X} = P^{\Pi,\mathrm{red}}_X \xrightarrow{\sim} \mathfrak{g}_{\Delta_5}.$$

The Weyl–Kac–Borcherds denominator of $P^{\Pi,\mathrm{red}}_X$ is therefore $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, by Theorem 9.1.

Chiral cobar identification. Suppose the chiral Koszul source datum of Definition 0.26 is fixed: $(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R}, C_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_R, b_{X,R}, \Theta_{\mathrm{Kos},R})$ with vanishing residual $\mathfrak{O}_{\mathrm{Kos},R} = 0$ compatibly in R and with the Mittag–Leffler exactness hypothesis of Proposition 0.29. The target side is the formal current envelope $\mathsf{Den}_{\Delta_5,E} = U^{\mathrm{ch}}_E\big(\mathrm{Cur}_E(\mathcal{A}_{\mathrm{den}})\big)$ of Proposition 16.2. The cobar comparison of Proposition 0.29 identifies

$$\Omega^{\mathrm{ch}}_E C_X \;\simeq\; \mathsf{Den}_{\Delta_5,E},$$

through the source-internal bar/cobar admissibility (the residual $\mathfrak{o}^{\varepsilon,\mathrm{src}}_R = 0$) and the source-to-target quasi-isomorphism ($\mathfrak{o}^{\vartheta}_R = 0$), avoiding the target-internal bar/cobar counit; this is the β-separation between observables and states. $\square$

Remark 4.2 *(The zero-bracket graded super-vector-space counterexample).* The criterion (S1)–(S10) blocks the trivial counterexample of a graded super vector space $V_\bullet$ with all brackets zero and parity dimensions matching $\mathrm{mult}_p^{\mathrm{GN}}$ on every window. Such a $V_\bullet$ may satisfy the target parity table and the signed-superdimension residual on the active support. It fails (S4): the Chevalley relation $[e_i^X, f_j^X] = \delta_{ij} h_i^X$ fails on the diagonal. It fails (S6): the kernel of the presentation map $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to V_\bullet$ is not $\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}}$ but contains every commutator $[e_i, e_j]$ with $(\alpha_i, \alpha_j) \neq 0$, so the induced map $G(\mathscr{B}_{\Delta_5}) \to V_\bullet$ is not injective. It fails (S5) for the zero pairing because the radical is all of $V_\bullet$, so the radical quotient is zero and not the BKM target. If one installs a nonzero pairing by hand, the denominator still does not supply its invariant Hopf adjointness, coideal radical, or equality with $\mathrm{Rad}_{\mathrm{GN}}$. The counterexample shows that signed-superdimension matching alone is not recognition; the bracket-level data of (S4)+(S5)+(S6) is the mathematical content.

Remark 4.3 *(Recognition without (D0)+(O1)+(O1)$^+$+(O2)).* The recognition criterion of Theorem 4.1 nominally lists no orientation hypothesis among (S1)–(S10). The orientation classes enter through (S1) — the chain-level descent $\overline{\Pi}^{\Theta}_{X,*}\mathcal{F}_X$ is conditional on the orientation lane via Definition 0.8 and the Frobenius identity from the CY-3 Serre functor — and through (S5), where the Hopf radical's coideal property is fixed by the same Frobenius identity (Theorem 4.1). In the chapter's combined statement (Theorem 2.1+Theorem 3.1+Theorem 4.1) the orientation classes enter all three theorems.

5 Compatibility with chiral Koszul duality and Calabi–Yau quantum groups

The three theorems sit inside chiral Koszul duality and Calabi–Yau quantum groups as the View ② / View ③ face of the Calabi–Yau-to-chiral comparison. Three compatibilities are required.

5.1 Primitive-to-shadow comparison

The primitive-to-shadow comparison is the cross-level identity ②↔③ applied to the primitive-to-shadow part of the chain. The Vol II primitive datum on (X, D, τ) is the bicoloured nine-tuple

$$\bigl(C\big|_{(X,D,\tau)},\ \mathcal{D}^{\mathrm{cl}}\big|_X;\ b,\ A_b,\ F^{\mathrm{cl}};\ \sigma_{\mathrm{half}},\ \mathrm{tr}_X^{\mathrm{cl}};\ \mathrm{Tr}_C\bigr).$$

The chart of vacuum b is the α-datum; the chart algebra $A_b = \mathrm{End}_C(b)$ is the open primitive algebra; the closed-colour vacuum factorization algebra F^{cl} gives the closed-side trace $\mathrm{tr}_X^{\mathrm{cl}}$ which evaluates at the type-II cusp $\mathfrak{c}_\infty$ to the weight-5 Borcherds–Gritsenko–Nikulin section $\mathcal{D}_X = \Delta_5$ (View ①). The comparison $A_b = \Phi_d^{(\Sigma_{d-1}, C)}(C_X)$ is the chain-fusion datum of chiral Koszul duality and Calabi–Yau quantum groups at level 2: under that datum the Calabi–Yau-side chiral algebra A_X is compared with the open-side chart algebra A_b on the same curve C with boundary vacuum induced by (C_X, Σ_{d-1}, C). In the present setting, the comparison concerns the chiral algebra on E attached to the Hall source of $K3 \times E$; conditional on the chiral Koszul source datum, the chiral cobar $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5, E}$ of Theorem 4.1 is the source-side chiral expression at level 2, and its constant-current Lie superalgebra is $\mathfrak{g}_{\Delta_5}$ (View ②). The protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ is the pre-trace Pfaffian section. After squaring and applying the retained level-Z datum, Vol I Theorem H (chiral-Hochschild concentration on the Koszul locus) and Vol II's cyclic-Hochschild Beilinson stratification (i)+(ii) identify the derived-centre trace pairing of degree $-\dim_{\mathbb{C}}(K3 \times E)$ whose scalar is Δ_5^{-2} (Appendix 7, Theorem 7.22). The level-A gravity-line operator algebra acting on the boundary is the open Hall–Borcherds residual of Vol II and is not constructed by the level-Z trace criterion.

5.2 Vol III chain fusion at level 2

The Vol III κ-stratification supplies the κ-tuple $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ at $K3 \times E$; the row $\kappa_{\mathrm{BKM}} = 5$ is the cusp-form weight realized by Δ_5, and the row $\kappa_{\mathrm{fiber}} = 24$ is $\chi_{\mathrm{top}}(K3) = 24$. The additive expression $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ has no invariant meaning: the Hodge reading gives $0 + 2 = 2$ at $N = 1$, and the Heisenberg reading gives the accidental equality $3 + 2 = 5$ only on that row. The CHL frame has Borcherds weights $(5, 4, 3, 2, 1)$, fixed by $c_N(0)/2$, not by a fixed additive formula. Thus κ_{fiber} is the Euler characteristic of the topological fibre and not of $\mathcal{O}_{\mathrm{fiber}}$; for K3, $\chi_{\mathrm{top}}(K3) = 24$ but $\chi(\mathcal{O}_{K3}) = 2$. Under the Vol III chain-fusion datum at level 2, the chiral algebra A_X on the CY face is compared with the chart algebra A_b on the Vol II open face. In the $K3 \times E$ specialization this comparison becomes $A_X = A_b = \Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5, E}$ only under the hypotheses of Theorem 4.1.

Remark 5.1 *(Coefficient projection is not a Hall comparison).* A Hall–Borcherds comparison map cannot be defined by coefficient projection alone, and a coefficient projection respecting charge addition because both products use the same cone is not a recognition theorem. A coefficient projection is not a Hall product/bracket matrix comparison, and does not prove Borcherds–Serre rows, radical descent, no-extra-relations, or PBW. The recognition criterion (S4)–(S10) of Theorem 4.1 is the mathematical statement: the source-side bracket matrices, Hopf radical, kernel equality, and PBW associated-graded comparison are independent data, none of which is inferable from a coefficient projection.

5.3 BCOV / mixed holomorphic-topological strings

The mixed holomorphic-topological strings companion provides the local model $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ with brane completion as one realization of the factorization-algebra datum Φ_d^{FA}; the global obstruction is the holomorphic de Rham obstruction class. The level-n scalar shadow $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ on the closed $K3 \times E \mapsto$ point cobordism is the BPS partition function (Chapter 9). A three-dimensional gravity-line interpretation with internal compact factor $K3 \times E$ requires the Hall–Borcherds residual of Vol II and is not claimed by Theorem 2.1. The identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$ is a protected Pfaffian identity before trace; its inverse square is the level-n protected scalar trace. The Pfaffian and orientation parts of the operator-level datum $\mathfrak{D}_X^{\mathrm{DI}}$ are defined on $K3 \times E$ relative to the retained data of Appendix 7; the primitive Lie-superalgebra part is the compact-source recognition problem.

6 Residual obstructions and open subroutes

The Pfaffian and orientation theorems are retained-data theorems on $K3 \times E$, relative to Appendix 7. The residuals in this chapter are the compact-source recognition datum $(S1)$–$(S10)$ and the gravity-line Hall–Borcherds operator algebra.

6.0.0.1 (D0). Theorem 7.6 is the criterion saying that the retained D0-HN datum supplies the cosection-twisted reduced d-critical orientation through the D0-degeneration pro-limit and the cofinal HN inverse system. The D_0-state, D_0-quotient, D_0-observable, D_0-descent, and D_0-Pfaffian rows are therefore part of that datum.

6.0.0.2 (O1). Theorem 7.12 supplies the strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle relative to the retained quotient-orientation datum. The finite-stabilizer linearization characters and the connected BE-edge class are part of that datum, not consequences of the primitive recognition theorem.

6.0.0.3 (O1)$^+$. The equivariance part of Theorem 7.13 is relative to the chosen Weyl lifts, the vanishing projective cocycle, and the zero torsor defects on the retained groupoid. It gives the honest $W^{(2)}(\Lambda^{2,1}_{II})$-transport of the orientation character. The comparison with ν_{Δ_5} also uses the rank-one wall signs from $(O2_{\text{atlas}})$. Neither statement supplies the relation-closed source matrices of the recognition datum.

6.0.0.4 (O2). Relative to the retained type-II wall-atlas datum $(O2_{\text{atlas}})$, the rank-one normal-form rank $N^{\text{Pf}}_{\delta_i} = 1$ holds for $i = 1, 2, 3$ (Proposition 5.2). The type-II middle-wall constant $4096 = 2^{12} = 64^2$ admits two readings: as the squared theta-leading on the type-II Borcherds product (where the $64 = 2^6$ prefactor of $D_5 = 64^{-1}\Delta_5$ doubles by $\Delta_5^2 = \Delta_{10}$) and as the 2^{12} sign assignments over the twelve binary wall coordinates of the half-Hilbert orbit. The equality of these two readings is scalar only, by Theorem 7.16. The OP scalar branch sign -1 is recorded by $(-p_{\text{DT}})^n$ of the OP/DT chamber, not by the orientation line. The middle wall $\delta_2 \leftrightarrow (0, 1, 1)$ is represented by a wrapped wall object with positive geometric b-degree only inside the retained $(O2_{\text{atlas}})$ datum. Proposition 5.2 is the rank-one local sign calculation; the component, boundary, overlap, quotient, and protected-integration compatibilities are the global content of $(O2_{\text{atlas}})$. Higher-rank wall atlases are not used in the proof of Theorem 2.1.

6.0.0.5 Recognition criterion (S1)–(S10). The window $W_{\leq 3}$ is the first arithmetic window but is not relation-closed. Put $a_{ij} = \delta_i + \delta_j$, $\delta_{123} = \delta_1 + \delta_2 + \delta_3$, and $C_{k,s} = a_{ij} + s\delta_k$ for $\{i, j, k\} = \{1, 2, 3\}$. The first relation-closed degree window is the height-7 closure

$$W^+_{\text{rel}} = W^{\text{act}}_{\leq 3} \cup R_4 \cup I_4 \cup C_{\leq 7} \cup \{2\delta_{123}\},$$

where $R_4 = \{3\delta_i + \delta_j : i \neq j\}$, $I_4 = \{2a_{ij} : i < j\}$, and $C_{\leq 7} = \{C_{k,s} : 2 \leq s \leq 5\}$. It adds the real-Serre terminal rows, doubled isotropic rows, δ_{123}-real strings, odd–odd δ_{123} rows, and complementary height-seven strings. In this enlarged window, Remark 6.9 supplies full target parity for $I_4 = \{2a_{ij}\}$, $C_{k,3}$, $C_{k,4}$, and $C_{k,5}$, while $C_{k,2}$ and $2\delta_{123}$ remain signed target rows until the finite target presentation is computed. Thus W^+_{rel} is a relation-closed degree set, not yet a primitive-recognition window satisfying (R0)–(R8). At $W_{\leq 3}$ the target parity table $\delta_i : 1|0$, $a_{ij} : 10|0$, $2\delta_i + \delta_j : 1|0$, $\delta_{123} : 29|93$ is computed; the row $a_{ij} : 10|0$ is the real bracket $[e_i, e_j]$ together with the nine Gritsenko–Nikulin isotropic simple directions; a compact source implementing (S3)–(S10) on $W_{\leq 3}$ is open, and the source identification at W^+_{rel} requires a finite source-and-target presentation computation.

6.0.0.6 Vol III Hall–Drinfeld–Borcherds spectral sequence on compact non-toric CY3. The Hall–Drinfeld–Borcherds spectral sequence abutting from Vol III's CoHA-type algebra to the BKM denominator on $K3 \times E$ is open on the compact non-toric trivial-canonical threefold, even after (D0), (O1), (O1)$^+$, and (O2). The finite-stage Hall source kernel of Theorem 0.34 is the finite-stage substitute.

6.0.0.7 Protected determinant and character level. The Pfaffian and orientation theorems identify the 2↔3 edge at the protected determinant and character level on $K3 \times E$. The primitive Lie-superalgebra recognition remains the finite compact-source datum $(S1)$–$(S10)$. The cross-level identities 1↔2 and 2↔5 are imported from Borcherds–Gritsenko–Nikulin and the regularized theta lift. The operator-level datum $\mathfrak{D}^{\text{DI}}_X$ is defined at the Pfaffian and orientation level relative to the retained data of Appendix 7; the scalar shadow $Z^{K3\times E}_{\text{BPS}} = \Delta_5^{-2}$ is the level-n protected trace. The compact microscopic state-space interpretation requires the compact-source recognition datum; the $3d$ gravitational path integral requires the Hall–Borcherds residual; the Calabi–Yau realization requires a construction external to the Pfaffian theorem.

Part IV

Mirror Discriminant and BPS Trace

Chapter 8

The mirror discriminant

1 The mirror discriminant

The automorphic, BKM, Pfaffian, and theta-lift constructions place Δ_5 on chains that begin with explicit modular or representation-theoretic data: the Borcherds–Gritsenko–Nikulin section of the weight-5 automorphic line over $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathcal{H}_2$ (View ①), the denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ of the Borcherds–Kac–Moody Lie superalgebra on $\Lambda^{2,1}_{II}$ (View ②), the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ of the Dirac–Igusa pro-object (View ③), and the singular theta lift of the weak Jacobi form $\phi_{0,1}$, namely the Borcherds regularized theta integral on the $\mathrm{O}(3,2)$ type-IV domain, with $\mathbf{Mp}_2$-Weil input and the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ exterior-square bridge (View ⑤). The period-side assertion is geometric. Let $\mathcal{P}^{(3)}_{K3\times E}$ be the rank-3 Mukai-invariant period quotient transported to $\mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z})$. Let $\mathfrak{H}_{\mathrm{II}}$ denote the type-II Heegner divisor on this quotient. The automorphic calculation gives the divisor equality

$$[\,\Delta_5^{-2}\,] \;=\; 2[\mathfrak{H}_{\mathrm{II}}] \qquad \text{(equality of divisor classes on the } (K3\times E)\text{-component)}, \tag{8.1}$$

where $[\Delta_5^{-2}]$ is the polar divisor of the meromorphic section Δ_5^{-2} of the automorphic line. The conjectural content is the existence of a rank-3 Mukai–Hodge quotient of a mirror period family whose discriminant support is $\mathfrak{H}_{\mathrm{II}}$, together with the comparison of the automorphic line with the period Hodge line.

This identification is conjectural. The cross-level identities ④↔①, ④↔②, ④↔⑤ are open. The automorphic identity, the BKM denominator identity, the conditional Pfaffian identity, and the singular theta-lift identity do not use the period-discriminant conjecture.

The handle Δ_5 splits into three names. The automorphic section over $\mathbf{Sp}_4(\mathbb{Z})\backslash\mathcal{H}_2$ is denoted $\mathcal{D}_X = \Delta_5$ (View ①); the BKM denominator is $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View ②); the protected Pfaffian is $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ (View ③). The mirror conjecture concerns the first: the divisor identification (8.1) is a statement about $\mathcal{D}_X$ as an automorphic section, transported to the period side.

1.1 *The K3 mirror-period family*

The lattice-polarised K3 mirror is fixed by Dolgachev [30] and tracked, throughout, through the Mukai lattice. Let

$$\Lambda_{K3} \;=\; 3U \oplus 2E_8(-1), \qquad \widetilde{\Lambda}_{K3} \;=\; U \oplus \Lambda_{K3} \;\simeq\; 4U \oplus 2E_8(-1),$$

the K3 lattice of signature $(3,19)$ and the Mukai lattice of signature $(4,20)$ (Mukai [77, 78]). A *lattice polarisation* of a complex K3 surface S is a primitive embedding $\iota\colon M \hookrightarrow \mathrm{Pic}(S)$ of an even hyperbolic lattice M, of signature $(1,t)$, such that $M^\perp \subset \Lambda_{K3}$ admits a primitive embedding of the hyperbolic plane U; the pair (S,ι) is then *Nikulin-admissible*, in the convention of Gritsenko–Nikulin [45] § 3 and the rank-3 sublattice constructions of Borcherds [8] § 15. Here

$$t = \mathrm{rk}\, M - 1$$

is the negative index of M, so that $\operatorname{rk} M^\perp = 21 - t$ and $\operatorname{sign}(M^\perp) = (2, 19 - t)$. The convention is $\operatorname{sign}(L) = (\#\text{positive}, \#\text{negative})$.

The associated period domain

$$\Omega_M \;=\; \big\{[\omega] \in \P(M^\perp \otimes \mathbb{C}) \;:\; \langle \omega, \omega\rangle = 0,\ \langle \omega, \bar\omega\rangle > 0\big\}^+$$

is the type-IV symmetric domain of signature $(2, 19 - t)$ and dimension $19 - t$, with arithmetic group $\Gamma_M = \mathrm{O}^+(M^\perp)$ acting properly discontinuously. The *Dolgachev mirror lattice* is the sub-lattice

$$\check{M} \;=\; M^\perp \cap (U)^\perp \;\subset\; \Lambda_{K3}, \qquad M^\perp \;=\; U \oplus \check{M},$$

so that $\check{M}$ carries signature $(1, 18 - t)$ and is hyperbolic. The mirror K3 surface $S^\vee$ is lattice-polarised by $\check{M}$. Dolgachev mirror symmetry gives a correspondence between the M-polarised and $\check{M}$-polarised moduli quotients; it is an involution on a single quotient only in the self-mirror cases.

The Aspinwall–Morrison form of the same construction [3] uses the Mukai lattice to compare complex and complexified Kähler directions at large-volume and large-complex-structure cusps. The full K3 sigma-model moduli space is larger than Ω_M/Γ_M; the lattice correspondence is the only input needed for the Δ_5 discriminant statement. The type-IV lattice of the mirror quotient is

$$\widetilde{M} \;=\; U_{\text{mir}} \oplus M \;\simeq\; \check{M}^\perp \;\subset\; \Lambda_{K3}, \qquad \operatorname{sign}(\widetilde{M}) = (2, t+1).$$

Here U_{mir} is the hyperbolic plane split from $M^\perp = U_{\text{mir}} \oplus \check{M}$. In the full Mukai lattice $\widetilde{\Lambda}_{K3} = U_{\text{Muk}} \oplus \Lambda_{K3}$, the orthogonal complement of $U_{\text{Muk}} \oplus M$ is $M^\perp = U_{\text{mir}} \oplus \check{M}$; the extra Mukai plane is contained in the augmented sublattice. No canonical action on $\widetilde{\Lambda}_{K3}$ is needed here; only the primitive decomposition and the induced period domains enter the argument.

1.1.0.1 The rank-three case. The polarisation relevant to View ④ is

$$M \;=\; U \oplus \langle -2\rangle, \qquad \operatorname{rank}(M) = 3, \qquad \operatorname{sign}(M) = (1, 2),$$

with Dolgachev mirror lattice

$$\check{M} \;=\; \big(M^\perp \cap U^\perp\big)_{\Lambda_{K3}} \;\simeq\; U \oplus E_7(-1) \oplus E_8(-1),$$

of rank 17 and signature $(1, 16)$ inside Λ_{K3}; the convention is $\operatorname{sign} = (\#\text{positive}, \#\text{negative})$ on the bilinear form of the lattice. The decomposition follows from the embedding $\langle -2\rangle \hookrightarrow E_8(-1)$ on a single root, after which the orthogonal complement of the root inside $E_8(-1)$ is $E_7(-1)$. The rank-five mirror-domain lattice is

$$\widetilde{M} \;=\; U \oplus M \;=\; 2U \oplus \langle -2\rangle, \qquad \operatorname{rank}(\widetilde{M}) = 5, \qquad \operatorname{sign}(\widetilde{M}) = (2, 3).$$

Under the exterior-square convention (the model of $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3, 2)_+$ at § 1.4) the same lattice is written $\Lambda^{3,2} = U \oplus U \oplus \langle 2\rangle$, with the hyperbolic-plane summand $\Lambda^{(1,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ carrying signature $(1, 1)$ and the rank-1 summand carrying form value $+2$ on a generator. The two writings are the same lattice expressed in opposite sign conventions on the bilinear form: the convention $(p, q) = (3, 2)$ is signature "three positive, two negative" for the fixed quadratic form, while the Dolgachev convention writes the same signature "two positive, three negative" on the form with reversed sign. The $(+2)$-vectors in the fixed convention are the (-2)-vectors of the Dolgachev convention; the Heegner divisors and the period domain are independent of the sign choice.

In the Dolgachev sign on $\widetilde{M}$, the period domain attached to $\widetilde{M}$ is the type-IV symmetric domain

$$\Omega_{\widetilde{M}} \;=\; \left\{[Z] \in \mathbb{P}\big(\widetilde{M} \otimes \mathbb{C}\big) \;:\; (Z, Z) = 0,\ (Z, \bar{Z}) > 0\right\}^{+}, \qquad \dim_{\mathbb{C}} \Omega_{\widetilde{M}} = 3,$$

whereas the exterior-square sign of $\Lambda^{3,2}$ writes the same domain with $(Z, \bar{Z}) < 0$. The two inequalities define the same oriented two-plane after multiplying the bilinear form by -1. The arithmetic group is

$$\Gamma_{\widetilde{M}} := \wedge^2 \mathbf{Sp}_4(\mathbb{Z})/\{\pm \mathbf{I}_4\} \subset \mathrm{PO}^+(\widetilde{M})$$

acting properly discontinuously, where the component is fixed by the positive cone $C(\Lambda^{2,1})_+$ (§ 1.4). The exceptional isogeny

$$\mathbf{PSp}_4(\mathbb{R}) \;\simeq\; \mathbf{SO}(3,2)_+, \qquad \mathbf{Sp}_4(\mathbb{Z})/\{\pm 1\} \;\simeq\; \Gamma_{\widetilde{M}},$$

imported as View 5 (the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ bridge), gives the identification

$$\Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}} \;\simeq\; \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}). \tag{8.2}$$

The form $\Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ of View 1, viewed as a section on the right-hand side of (8.2), is the Borcherds lift of the weak Jacobi form $\phi_{0,1}$ under the data $(\widetilde{M}, \phi_{0,1})$ on the left-hand side (Borcherds [9, Thm. 13.3] applied to signature $(2,3)$; Gritsenko–Nikulin [45, Thm. 1.1] for the identification with the additive Gritsenko lift; the rank-3 row of the Vol III lattice-polarised $\mathfrak{g}_L$ catalogue at $L = U \oplus \langle -2 \rangle$).

The K3-side period domain on the rank-3 polarisation is a different, larger object:

$$\Omega_M \;=\; \left\{[\omega] \in \mathbb{P}\big(M^{\perp} \otimes \mathbb{C}\big) \;:\; \langle \omega, \omega \rangle = 0,\ \langle \omega, \bar{\omega} \rangle > 0\right\}^{+}, \qquad \dim_{\mathbb{C}} \Omega_M = 17,$$

where $M^{\perp} \subset \Lambda_{K3}$ has signature $(2,17)$ and parametrises the *transcendental* Hodge filtration of a lattice-polarised K3 surface. The Borcherds lift used in View 4 is on $\Omega_{\widetilde{M}}$, not on Ω_M: the difference is the adjoining of one Mukai hyperbolic plane to the rank-3 K3 lattice, not an identification of the even elliptic cohomology with the elliptic period. The two period spaces are not interchangeable: $\Omega_{\widetilde{M}}$ of dimension 3 is the genus-two Siegel space and supports Δ_5 directly via the exceptional isogeny, while Ω_M of dimension 17 is the K3-only period space, on which the rank-3 lattice-polarised lift of $\phi_{0,1}$ lands only after the additional Mukai-U augmentation.

1.2 The K3 × E Mukai–Hodge quotient

The full Calabi–Yau-threefold variation of Hodge structure on $X = S \times E_\tau$ is

$$H^3(X, \mathbb{Z}) \;=\; H^2(S, \mathbb{Z}) \otimes H^1(E_\tau, \mathbb{Z}),$$

of rank 44. The elliptic modulus τ belongs to $H^1(E_\tau)$. It does not belong to the even elliptic lattice $H^0(E, \mathbb{Z}) \oplus H^2(E, \mathbb{Z})$, whose Hodge type is constant.

The Mukai lattice $\widetilde{H}(S, \mathbb{Z}) = H^*(S, \mathbb{Z})$ of signature $(4, 20)$ carries the Mukai pairing $\langle (r, c, s), (r', c', s') \rangle_{\mathrm{Muk}} = c \cdot c' - rs' - r's$ (Definition 1.3). The even cohomology of the elliptic curve is the hyperbolic plane $U_E = H^{\mathrm{even}}(E, \mathbb{Z}) = H^0(E, \mathbb{Z}) \oplus H^2(E, \mathbb{Z})$ with the standard hyperbolic form. Since $H^{\mathrm{odd}}(S) = 0$, the even integral cohomology is, as an integral group with its Mukai tensor pairing,

$$H^{\mathrm{even}}(K3 \times E, \mathbb{Z}) \;=\; \widetilde{H}(S, \mathbb{Z}) \otimes U_E \;=\; \widetilde{H}(S, \mathbb{Z}) \otimes U,$$

a lattice of rank 48. This even Mukai lattice is not the Calabi–Yau-threefold period lattice; it is the lattice from which the rank-3 Mukai-invariant quotient is cut.

For $(K3 \times E)$-pairs (S, E_τ) with S lattice-polarised by $M = U \oplus \langle -2 \rangle$, the retained rank-3 Mukai–Hodge quotient is

$$\mathcal{P}^{(3)}_{K3\times E} \;:=\; \Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}} \;\overset{(8.2)}{\simeq}\; \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}), \tag{8.3}$$

where the augmented lattice $\widetilde{M} = U_{\mathrm{mir}} \oplus M$ is a rank-5 type-IV lattice. The summand U_{mir} is an abstract Mukai hyperbolic plane in the period quotient. It is not the constant even elliptic lattice U_E. The elliptic period enters through $H^1(E_\tau)$ and then through the exterior-square $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ coordinate system on $\Omega_{\widetilde{M}}$. Equation (8.3) is the period quotient on which View 4 is stated. The quotient is the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ model; the assertion that it is the base of the compact mirror discriminant is Conjecture 1.1.

The K3-only period domain Ω_M of dimension 17 does not enter (8.3): its role is to parametrise the transcendental Hodge filtration of S alone. The rank-3 $(K3 \times E)$ Mukai–Hodge quotient is on $\Omega_{\widetilde{M}}$, of dimension 3. The relation $\widetilde{M} = U_{\mathrm{mir}} \oplus M$ is the rank-five type-IV bridge, not an equality with the full $H^3(K3 \times E)$ period domain.

1.3 The discriminant locus

The candidate period quotient carries a natural Heegner divisor. It becomes the support of the discriminant divisor of a mirror period family only after the comparison asserted in Conjecture 1.1. On the type-IV symmetric domain $\Omega_{\widetilde{M}}$, the Heegner divisors are

$$\mathfrak{H}_{(\delta)} \;=\; \big\{[Z] \in \Omega_{\widetilde{M}} \;:\; (Z,\delta) = 0\big\}, \qquad \delta \in \widetilde{M},\;\; (\delta,\delta) = 2,$$

indexed by primitive square-2 vectors of the fixed form on $\widetilde{M} = \Lambda^{3,2}$ (equivalently, square-(-2) vectors of the Dolgachev form after the sign flip recorded above). Such a hyperplane records an extra algebraic Mukai vector; on a K3 slice its H^2-projection is the ordinary nodal (-2)-class when the projection is nonzero. The type-II Heegner divisor is

$$\mathfrak{H}_{\mathrm{II}} \;=\; \bigcup_{\substack{\delta\in\widetilde{M}\\ (\delta,\delta)=2\\ (\delta,\widetilde{M})=2\mathbb{Z}}} \mathfrak{H}_{(\delta)}/\Gamma_{\widetilde{M}} \;\subset\; \Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}}. \tag{8.4}$$

The $\Gamma_{\widetilde{M}}$-orbit structure on primitive square-2 vectors of $\widetilde{M}$ breaks into type-I and type-II classes ($\delta \cdot \widetilde{M} = \mathbb{Z}$ versus $\delta \cdot \widetilde{M} = 2\mathbb{Z}$, the dichotomy already imported as the type-II versus type-I distinction at the $W^{(2)}(\Lambda^{2,1}_{II})$-orbit level); the union in (8.4) runs over the type-II orbit, which is the orbit relevant to the $\mathbf{Sp}_4(\mathbb{Z})$-image of the exceptional isogeny. The image in the quotient is one irreducible Heegner-type divisor (Bruinier–Funke [13, § 2] for the general lattice-Heegner formulation in signature $(2,3)$; the index-1 case is implicit in Borcherds [9, Thm. 10.1]). The same divisor, transported across the bridge (8.2), is the *Humbert surface*

$$\mathcal{H}_2^{(\mathrm{Hum})} \;=\; \Big\{[Z] \in \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}) \;:\; Z = \begin{pmatrix} \tau_1 & 0 \\ 0 & \tau_2 \end{pmatrix} \text{ up to } \mathbf{Sp}_4(\mathbb{Z})\Big\},$$

the diagonal locus on which the principally polarised abelian surface $\mathbb{C}^2/(\mathbb{Z}^2 + Z\mathbb{Z}^2)$ decomposes as a product of two elliptic curves. Under the lattice identification (8.2), this Humbert surface is the Heegner support of a norm-(-2) class in the rank-3 Mukai-invariant period domain. The assertion that it is the full discriminant support of a compact mirror family is part of Conjecture 1.1.

The form Δ_5 of View ① vanishes precisely along this divisor: $\Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ has a simple zero on $\mathcal{H}_2^{(\mathrm{Hum})}$ [44, Thm. 3.1][45, § 3], with multiplier ν_{Δ_5} accounting for the half-integral shift. Concretely, the Borcherds product expansion of View ① gives

$$\Delta_5(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

and the automorphic zero divisor is exactly $\mathcal{H}_2^{(\mathrm{Hum})}$, counted with multiplicity 1. Thus the theorem-level divisor identity is

$$[\, \mathrm{div}(\Delta_5)\,] \quad = \quad [\, \mathcal{H}_2^{(\mathrm{Hum})}\,] \quad (\text{divisor classes on } \mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z})). \tag{8.5}$$

The additional equality with the mirror discriminant $\mathfrak{D}_{\mathrm{mir}}$ is not contained in the automorphic theorem; it is the support part of the mirror-discriminant conjecture.

1.4 *The conjecture*

The discriminant of a period quotient carries multiplicities, not just a support. Let $\mathfrak{D}_{\mathrm{mir}}$ denote the conjectural reduced discriminant divisor of that quotient, and let $\lambda_{\mathrm{per}} = \mathcal{F}^2$ denote the descended K3 period Hodge line on the same quotient. In the rank-3 Mukai-invariant component the expected section of $\lambda_{\mathrm{per}}^{-10}$ has polar order 2 along the type-II Heegner divisor: locally this is the Picard–Lefschetz contribution of a nodal K3 vanishing cycle to the Hodge filtration, and automorphically it is the square of the simple zero of Δ_5. This multiplicity comparison is part of the conjecture until the rank-3 quotient and the comparison line have been constructed.

Conjecture 1.1 *(The mirror discriminant identity).* On the rank-3 $(K3 \times E)$ Mukai–Hodge quotient parametrised by (8.3), the meromorphic section

$$\Delta_5^{-2} \;\in\; \Gamma\Big(\mathcal{H}_2/\mathbf{Sp}_4(\mathbb{Z}),\;\; \mathcal{L}_{\mathrm{aut}}^{-10}\Big)$$

of the inverse 10th tensor power of the automorphic line bundle $\mathcal{L}_{\mathrm{aut}}$ coincides, as a meromorphic section of $\lambda_{\mathrm{per}}^{-10}$ on $\mathcal{P}^{(3)}_{K3\times E}$, with the discriminant section of the rank-3 quotient. Equivalently, the polar divisor of Δ_5^{-2} is twice the quotient discriminant $\mathfrak{D}_{\mathrm{mir}}$, the support of $\mathfrak{D}_{\mathrm{mir}}$ is $\mathfrak{H}_{\mathrm{II}}$, and the comparison map

$$\mathcal{L}_{\mathrm{aut}}^{-10} \;\xrightarrow{\;\sim\;}\; \lambda_{\mathrm{per}}^{-10}$$

on the open complement $\mathcal{P}^{\circ}_{K3\times E} = \mathcal{P}^{(3)}_{K3\times E} \setminus \mathfrak{D}_{\mathrm{mir}}$ extends to a global identification with Δ_5^{-2} as a section of $\lambda_{\mathrm{per}}^{-10}$.

The conjecture is, at the level of divisor classes,

$$[\, \Delta_5^{-2}\,] \quad = \quad 2\,[\, \mathfrak{D}_{\mathrm{mir}}\,] \quad = \quad 2\,[\, \mathfrak{H}_{\mathrm{II}}\,] \quad = \quad 2\,[\, \mathcal{H}_2^{(\mathrm{Hum})}\,];$$

the identity (8.5) fixes the right-hand side, and the conjectural content is the matching of multiplicities and the trivialisation of the comparison line bundle.

1.4.0.1 Evidence. The genus-two Bryan–Pandharipande generating function for disconnected genus-two BPS states on $K3 \times E$ predicts the cusp branch governed by the reciprocal Igusa form [14, Conj. 1]; Oberdieck–Pixton prove the normalized OP/DT/PT scalar representative

$$Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$$

[86, Thm. 1]. In the trace convention of this manuscript

$$Z^{K3\times E}_{\mathrm{BPS}} := -4096^{-1} Z^X_{\mathrm{OP}} = \Delta_5^{-2}.$$

This is the input to View 1 on the curve-counting side. The local Picard–Lefschetz model and Saito's Hodge-filtration formalism predict the same order-two degeneration as the automorphic divisor $[\Delta_5^{-2}]$. The unproved point is the global comparison: the line $\mathcal{L}^{-10}_{\mathrm{aut}} \otimes \lambda^{10}_{\mathrm{per}}$ must be trivialised and the identification of the two sides must extend across the discriminant divisor.

A separate piece of evidence is the rank-4 precedent set by Gritsenko–Nikulin [45, Thm. 4.1] in the lattice-polarised setting at the polarisation L producing a Borcherds product identified with the quasi-pull-back of the discriminant section of the universal lattice-polarised Hodge bundle on Ω_L/Γ_L. Conjecture 1.1 is the rank-3 analogue of the same identification, evaluated at the type-II rank-3 Mukai sublattice $M = U \oplus \langle -2 \rangle$ of $\Lambda^{2,1}$ (see Appendix 3); the Fake-Monster Borcherds product Φ_{12} on the Leech-extension lattice $\mathrm{II}_{25,1}$ of [7, 8] is a distinct, signature-$(26, 2)$ object and is not invoked here.

1.4.0.2 Epistemic strength. The identification claim of Conjecture 1.1 is conjectured. The comparison line bundle is a standard Hodge-theoretic object; the automorphic Heegner identity (8.5) is a theorem; the equality $\mathrm{Supp}(\mathfrak{D}_{\mathrm{mir}}) = \mathfrak{H}_{\mathrm{II}}$, the multiplicity 2, and the global extension across the discriminant divisor are conjectural.

1.4.0.3 Consequences. If proved, View 4 embeds the Δ_5 identity inside mirror symmetry on $K3 \times E$ at the level of period-side discriminants. The identification would give a period-side derivation of Δ_5 independent of the Borcherds product (View 1), the BKM denominator (View 2), the Pfaffian of the Dirac–Igusa pro-object (View 3), and the singular theta lift (View 5). It is a conjectural fifth chain.

A proof of Conjecture 1.1 would identify Δ_5^{-2} with the section of $\lambda^{-10}_{\mathrm{per}}$ on the rank-3 $(K3 \times E)$ Mukai–Hodge quotient; together with the Bryan–Oberdieck–Pixton theorem [86, Thm. 1] that $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$, this would give a period-side interpretation of the unscaled trace $Z^{K3\times E}_{\mathrm{BPS}} := -4096^{-1} Z^X_{\mathrm{OP}}$.

1.5 Range of the period conjecture

The Beilinson cut is in force: a statement is not primitive if it holds only after passing to a trace, choosing a chamber, completing a category, or installing descent data. The conjectural identification is on the period side: it concerns the section Δ_5^{-2} of an automorphic line bundle and the discriminant divisor of the rank-3 $K3 \times E$ mirror period component. It is not a claim about the protected operator-level datum $\mathfrak{D}^{\mathrm{DI}}_X$ of View 3; that datum is established at the Pfaffian and orientation level only relative to the retained data in the View 2↔3 chapters and Appendix 7. View 4 does not enter the operator chain.

In the trace convention of Chapter 9, the level-n scalar partition function $Z^{K3\times E}_{\mathrm{BPS}} := \Delta_5^{-2}$ is the orientation-forgetful trace of the Pfaffian datum; its three-dimensional gravity-line realization requires the Hall–Borcherds residual (Vol II, [68]). The mirror conjecture above says nothing about the gravitational lift; it addresses only the geometric origin of the section Δ_5^{-2} on the rank-3 mirror period component.

The Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is on the algebraic side; it is a theorem of Gritsenko–Nikulin [45, § 3, Thm. 3.1]. The mirror conjecture is on the period side and does not enter the algebraic chain.

1.6 Lattice-convention check

The lattice convention adopted above is the Dolgachev convention $\Lambda_{K3} = 3U \oplus 2E_8(-1)$, $M = U \oplus \langle -2\rangle$, mirror lattice $\check{M} \simeq U \oplus E_7(-1) \oplus E_8(-1)$ (rank 17, signature (1, 16)). Aspinwall–Morrison's original mirror-pair construction [3] adopts the same lattice up to the choice of which hyperbolic plane in Λ_{K3} is split off; the two conventions agree on the K3 mirror lattice. The Mukai-lattice form $\widetilde{M} = U \oplus M = 2U \oplus \langle -2\rangle$ of rank 5 and signature (2, 3) is the convention adopted in Vol III [65, Prop. 6.4]; it is the convention relevant to the period-domain bridge (8.2). The lattice $\widetilde{M}$ is identified with $\Lambda^{3,2}$ of § 1.4 by the sign-flip on the rank-1 summand recorded in § 1.1. Disagreement on the lattice convention appears only as a sign convention on the Mukai pairing; it does not affect the discriminant divisor.

The K3 vs. $(K3 \times E)$ period-space matching (8.3) adjoins the distinguished hyperbolic summand U_{mir} to the rank-3 K3 lattice and lets the elliptic variation enter through $H^1(E_\tau)$. It does not identify U_{mir} with the constant even elliptic lattice U_E, and it is not an abstract equality between the K3-mirror period family and the $(K3 \times E)$-mirror period family. The possible discrepancy is exactly the comparison between the $\mathbf{Sp}_4$-side and the $\mathrm{O}^+(2,3)$-side; the isomorphism (8.2) is exactly the $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ bridge of View 5.

The word discriminant names a section, not only a support, once multiplicities are present. There are two natural divisors on the period side: the Heegner support $\mathfrak{H}_{\mathrm{II}}$, and the polar divisor of the section of $\lambda_{\mathrm{per}}^{-10}$. The conjecture concerns the second: Δ_5^{-2} is conjectured to be that section, with polar divisor twice the support divisor $\mathfrak{H}_{\mathrm{II}}$. The factor of 2 is predicted by the local Picard–Lefschetz model and is matched, on the automorphic side, by the square of Δ_5; a mismatch at the level of multiplicity would falsify the conjecture by a factor that the Borcherds product does not allow.

The relation to the Borcherds–Gritsenko–Nikulin quasi-pull-back theorems is the cleanest comparison. There the Borcherds form Φ_{12} on the even unimodular lattice of signature (2, 26) is pulled back to lattice-polarised type-IV domains, and its divisor is read as a discriminant section of the corresponding Hodge bundle [45, Thm. 4.1]. The rank-3 lattice $L = U \oplus \langle -2\rangle$ requires a separate comparison line and a separate Heegner-divisor analysis; the Φ_{12} theorem is evidence, not a proof.

1.7 Components of the period conjecture

(M1) *Trivialisation of the comparison line.* The comparison line bundle $\mathcal{L}_{\mathrm{aut}}^{-10} \otimes \lambda_{\mathrm{per}}^{10}$ on $\mathcal{P}^\circ_{K3\times E}$ must be trivialised. Since $\nu_{\Delta_5}^2 = 1$, the inverse-square section has no residual character; the obstruction is an ordinary Hodge line class. A trivialisation of this line, together with the divisor and boundary conditions below, strengthens the divisor comparison to an equality of meromorphic sections.

(M2) *Extension and multiplicity.* The section comparison is first defined on $\mathcal{P}^\circ_{K3\times E}$, where the Heegner divisor has been removed. The period condition is the extension across $\mathfrak{H}_{\mathrm{II}}$ with polar order 2, together with the extension to the rank-2 cusp with boundary exponent governed by the Borcherds singular weight $c_1(0)/2 = 5$ ([8, § 15]). Thus the divisor of the extended section, not the open part itself, is compared with $[\Delta_5^{-2}]$.

(M3) *The mirror correspondence on the section.* Dolgachev mirror symmetry gives a correspondence between the M-polarised and $\check{M}$-polarised period quotients, not a canonical involution of $\Omega_{\widetilde{M}}/\Gamma_{\widetilde{M}}$. The period condition is that a chosen correspondence identifies the period Hodge line with the automorphic line and carries the type-II Heegner divisor to the type-II Heegner divisor. After this comparison, Δ_5^{-2} is required to be the same meromorphic section of the compared inverse Hodge line.

(M4) *Rank ≥ 5 lattice compatibility.* The Vol III rank-≥ 5 lattice-polarised $\mathfrak{g}_L$ family [64] is the conjectural extension of Conjecture 1.1 to higher Picard rank. At rank ≥ 5 the discriminant identity is conditional on the Gritsenko–Cléry 2018 admissibility conjecture (Conj. 5.1 of [41]); the rank-3 statement is the corresponding conjectural statement for the K3-Mukai-invariant lattice and is the rank relevant to the Δ_5 row.

The four period conditions (M1)–(M4) are distinct from the four obstruction classes (D0), (O1), (O1$^+$), (O2) of View ③; the latter govern the operator-level chain and are the conditioning for ②↔③, while the former govern the period-side chain and are the conditioning for ④↔①. The two systems lie on different chains: (M1)–(M4) are period-line and Heegner-divisor conditions, whereas (D0), (O1), (O1$^+$), and (O2) are Pfaffian-line and orientation conditions.

1.8 Compatibility with the Vol III lattice convention

The lattice convention $M = U \oplus \langle -2 \rangle$, $\check{M} \simeq U \oplus E_7(-1) \oplus E_8(-1)$, $\widetilde{M} = U \oplus M$ of § 1.1 is the convention of Vol III [65, § 6.4 + Prop. 6.4 + Thm. 6.7]; on the $(K3 \times E)$-component the Vol III κ-tuple $\left(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}\right) = (0, 0, 3, 5, 24)$ is recorded with each entry sourced from a distinct construction. The mirror discriminant of Conjecture 1.1 is on the $\kappa_{\mathrm{BKM}} = 5$ row of the κ-tuple; the period-side datum is the discriminant divisor on the lattice-polarised period stratum of the Vol III chart classification (Vol III [65, § V]: the "lattice-polarised period domain" equivariance stratum). The $(K3 \times E)$-fiber rank $\kappa_{\mathrm{fiber}} = 24$ is the Mukai-lattice rank of $\widetilde{\Lambda}_{K3}$; it is the rank of the Heisenberg algebra of the Mukai lattice, not a discriminant invariant.

The mixed-holomorphic-topological-strings companion volume [66] records that the modular line bundle on $\overline{\mathcal{A}}_g$ at level $g = 2$ is $\Omega_{\mathrm{central}}$, and that $\Phi_{10}^{\mathrm{un}} = \chi_{10} = \Delta_5^2$ is the genus-2 section. The mirror-discriminant conjecture is consistent with that statement: the polar divisor of $\Delta_5^{-2} = (\Phi_{10}^{\mathrm{un}})^{-1}$ on $\overline{\mathcal{A}}_2$ is twice the divisor of Δ_5, and the mirror discriminant identification is precisely the geometric origin of that section.

The Vol II Hall–Borcherds residual is silent on View ④: the period-side discriminant identification is independent of the gravitational-lift bridge, and the conjecture above does not require the residual. Solving the Hall–Borcherds residual on the $(K3 \times E)$-component would not by itself prove Conjecture 1.1; conversely, a proof of Conjecture 1.1 would not automatically construct the Hall–Borcherds residual.

View ④ remains conjectural. The automorphic, BKM, Pfaffian, and theta-lift views form the theorem-level quadrilateral at the stated categorical dimensions. The mirror-discriminant identity is the remaining period-side equality.

Chapter 9

The level-n protected scalar shadow and the $K3 \times E$ realization question

The denominator algebra and the protected Pfaffian sit at the primitive and chart levels of the Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z} \to \mathsf{A}$. After taking the protected trace, the same Δ_5 survives, squared, as the partition function on the closed $K3 \times E \to \mathrm{pt}$ cobordism, equivalently the level-n modular-functor scalar trace of the realization formalism of Vol IV [69]. The scalar shadow is exactly the object allowed by the Beilinson cut: an automorphic identity, not an algebra; a trace, not an operator; an orientation-forgetful invariant, not an orientation lift. The orientation-level object is $\mathfrak{D}_X^{\mathrm{DI}}$ of Chapter 5; the scalar identity is the level-n shadow it casts.

Three distinct meanings of Δ_5 coexist and are tagged on first occurrence: the automorphic section $\mathcal{D}_X = \Delta_5$ (View I –- Borcherds–Gritsenko–Nikulin); the Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View II –- Gritsenko–Nikulin); the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ (View III –- Chapter 5). The handle "Δ_5" alone never substitutes for any of the three.

1 The level-n protected scalar shadow on $K3 \times E \to \mathrm{pt}$

1.1 The Oberdieck–Pixton scalar identity

Let $X = S \times E$, with S a smooth complex K3 surface and E a smooth elliptic curve. The reduced Donaldson–Thomas / stable-pair / Gromov–Witten correspondence on X, in the primitive K3-class chamber, gives the normalized scalar representative

$$Z^X_{\mathrm{OP/DT/PT}} = -4096\,\Delta_5^{-2}$$

on the genus-2 Siegel upper half-space $\mathfrak{H}_2$. The unscaled level-n trace retained below is $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$.

Convention 1.1 *(OP chamber, sign branch, primitive class).* The variables $(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}})$ are the Oberdieck–Pixton variables of [86, §0]: T_{OP} is the K3-surface degree, Q_{OP} the K3-curve class, and P_{OP} the elliptic-genus variable (equivalently $P_{\mathrm{OP}} = -p_{\mathrm{DT}}$, giving the $(-p_{\mathrm{DT}})^n$ DT-chamber). The K3 curve class β_h is taken *primitive* of self-intersection $\beta_h \cdot \beta_h = 2h - 2$ ([75]); the identity (9.1) is the primitive-class branch. The leading -1 of $Z^X_{\mathrm{OP}} = -(\chi_{10}^{\mathrm{OP}})^{-1}$ is the OP scalar convention; the squared theta-leading $4096 = 64^2$ is the normalization conversion $\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2$. None of these is the orientation character.

The scalar identity (9.1) is an identity in the OP chamber. If the retained compact Hall source lies in a different stability chamber, comparison with (9.1) requires the chamber-compatible Hall descent datum of Definition 4.15; the normal-ordered lattice map alone does not transport Hall brackets.

Under Convention 1.1 and [86, Theorem 1], [85, Proposition 5], [75],

$$Z^X_{\mathrm{DT}} \;=\; Z^X_{\mathrm{OP}}(P_{\mathrm{OP}}, Q_{\mathrm{OP}}, T_{\mathrm{OP}}) \;=\; -\,4096\,\Delta_5(Z)^{-2}, \qquad Z = \Omega \in \mathfrak{H}_2. \tag{9.1}$$

The constant $4096 = 64^2$ is the squared theta-leading coefficient $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5$ of the automorphic section, and the leading sign is the Oberdieck–Pixton scalar convention $Z^X_{\mathrm{OP}} = -(\chi^{\mathrm{OP}}_{10})^{-1}$. Both are normalization data of the imported reduced theory.

Theorem 1.2 *(Oberdieck–Pixton scalar branch and unscaled Igusa shadow).* With variables $(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}})$ normalized in the Oberdieck–Pixton chamber of the Oberdieck–Pixton normalization [86, Theorem 1], the proved reduced DT/PT scalar branch is

$$Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = Z^X_{\mathrm{PT}} = -4096\,\Delta_5(Z)^{-2}.$$

In the unscaled trace convention of this manuscript, the level-n protected scalar shadow on the closed $K3 \times E \to \mathrm{pt}$ cobordism is defined by

$$Z^{K3\times E}_{\mathrm{BPS}}(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}}) \;=\; \big(\Phi^{\mathrm{un}}_{10}(Z)\big)^{-1} \;=\; \Delta_5(Z)^{-2}. \tag{9.2}$$

Equivalently, in Oberdieck–Pixton normalization $\chi^{\mathrm{OP}}_{10} = 2^{-12}\Delta_5^2$, $\Phi^{\mathrm{un}}_{10} = \chi_{10} = \Delta_5^2$, and the unscaled form $(\Phi^{\mathrm{un}}_{10})^{-1}$ is the retained level-n trace convention on each primitive closed K3-class fibre in the OP chamber. The Behrend-weighted reduced quotient integration of Oberdieck's reduced PT/DT scalar-quotient integration [84] gives the OP chamber branch

$$Z^X_{\mathrm{DT}} = Z^X_{\mathrm{OP}} = Z^X_{\mathrm{PT}} = -4096\, Z^{K3\times E}_{\mathrm{BPS}}.$$

Thus the OP branch is a normalized scalar representative of the same unscaled trace, not the definition of $Z^{K3\times E}_{\mathrm{BPS}}$.

Proof. Equation (9.2) is the dictionary between the Oberdieck–Pixton primitive series and the unscaled Igusa form Φ^{un}_{10}: the Oberdieck–Pixton normalization $\chi^{\mathrm{OP}}_{10} = D_5^2 = 64^{-2}\Delta_5^2 = 2^{-12}\Delta_5^2$ of [86, §0] together with the Igusa identification $\Phi^{\mathrm{un}}_{10} = \chi_{10} = \Delta_5^2$ of [48, 49] gives $(\Phi^{\mathrm{un}}_{10})^{-1} = \Delta_5^{-2}$. Multiplying by $2^{12} = 4096$ and the OP scalar sign restores (9.1). The reduced DT/PT comparison is [84, Theorem 3(i)], the reduced PT/GW comparison on primitive K3 classes is [85, Proposition 5], and the OP product identity is [86, Theorem 1]; their composition gives the displayed equality in the OP chamber. □

1.2 *Three readings of the same exponent*

The integers $f(n,l)$ of the weight-zero index-one weak Jacobi form $\phi_{0,1}$ drive both sides of the OP/Borcherds dictionary. At the primitive datum they are virtual K_0-superdimensions of $\mathbb{U}_{n,l}$, and after taking scalar shadows they appear twice: once as exponents of the Borcherds product on the automorphic side, once as multiplicities in the Mukai dimension count of the Hilbert-scheme stratification on the geometric side. In $\mathfrak{g}_{\Delta_5}$, the same $f(nm,l)$ is the signed root supermultiplicity at $\alpha(n,l,m) = 2nf_2 - lf_3 + 2mf_{-2}$.

The variable map of the Oberdieck–Pixton normalization [86, Theorem 1] encodes this triple reading. A factor $(1 - q^n r^l s^m)^{f(nm,l)}$ of the Borcherds product on the automorphic side is, on the Oberdieck–Pixton side, a factor $(1 - p^k_{\mathrm{OP}} q^h_{\mathrm{OP}} \tilde{q}^d_{\mathrm{OP}})^{c(4hd-k^2)}$ of χ^{OP}_{10}, the K3-elliptic-genus exponents $c(4hd-k^2) = 2f(hd,k)$ recording the squared automorphic line. The multiplicities $2f$ are exactly the K3 elliptic-genus coefficients of $Z_{\mathrm{K3}} = 2\phi_{0,1}$; the squaring is part of the identification $\chi^{\mathrm{OP}}_{10} = D_5^2$ of the Oberdieck–Pixton/Pandharipande scalar identity [86, Theorem 1].

Remark 1.3 *(Constants are normalization, not orientation).* Three constants intervene in (9.1). The leading $64 = [q^{1/2}r^{1/2}s^{1/2}]\Delta_5$ is the theta-constant normalization of the Igusa section; the monic Borcherds product is $D_5 = 64^{-1}\Delta_5$. The factor $4096 = 64^2$ is the squared theta-leading coefficient produced by the OP convention $\chi_{10}^{\mathrm{OP}} = D_5^2$. The leading minus sign is the OP scalar branch $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$. None of the three is the orientation character. The protected orientation character ϵ_o and the automorphic reflection character ν_{Δ_5} are determined elsewhere: ν_{Δ_5} by the Maass transformation law of Δ_5, ϵ_o by the rank-one type-II Pfaffian signs together with the retained half-Hilbert wall atlas. The number $4096 = 2^{12}$ occurs in two different structures, not as one orientation datum: the Pfaffian-side 2^{12} is the cardinality of the retained sign orbit, while the OP-side 2^{12} is the squared theta-leading coefficient in the imported scalar identity.

1.3 *Worked Fourier-coefficient check*

The leading Fourier coefficients of Δ_5 and Δ_5^2 provide the elementary verification that the constants in (9.1) are normalization, not enumerative content. Recall $\phi_{0,1}$ has leading expansion

$$\phi_{0,1}(\tau, z) \;=\; (r^{-1} + 10 + r) + q\big(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2\big) + O(q^2),$$

with coefficients $f(0,-1) = f(0,1) = 1$, $f(0,0) = 10$, $f(1,1) = f(1,-1) = -64$, $f(1,0) = 108$, $f(1,2) = f(1,-2) = 10$, $f(1,3) = f(1,-3) = 0$. The Borcherds–Gritsenko–Nikulin product

$$\Delta_5(Z) \;=\; 64\, q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)} \tag{9.3}$$

opens with the theta-leading term $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$; the next non-trivial timelike coefficient belongs to the monic product after the leading monomial is removed:

$$[qrs]\, D_5/(qrs)^{1/2} = 93, \qquad D_5 = 64^{-1}\Delta_5.$$

The isolated factor $(1 - qrs)^{f(1,1)} = (1 - qrs)^{-64}$ contributes $64qrs$, and the remaining root factors contribute the other $29qrs$, as in Appendix 2. Squaring (9.3),

$$\Delta_5^2(Z) \;=\; 4096\, q\, r\, s \prod (1 - q^n r^l s^m)^{2f(nm,l)}, \tag{9.4}$$

with theta-leading $64^2 = 4096$. The OP product $\chi_{10}^{\mathrm{OP}} = p\, q\, \tilde{q} \prod (1 - p^k q^h \tilde{q}^d)^{c(4hd-k^2)}$, with K3-elliptic exponents $c(4hd - k^2) = 2f(hd, k)$, is exactly the right-hand side of (9.4) divided by 4096. Thus

$$\chi_{10}^{\mathrm{OP}} = 4096^{-1}\,\Delta_5^2, \qquad [pq\tilde{q}]\,\chi_{10}^{\mathrm{OP}} = 1, \qquad [q^{1/2}r^{1/2}s^{1/2}]\,(64^{-1}\Delta_5) = 1.$$

The squared theta-leading coefficient 4096 is therefore the single explicit normalization constant in the OP/DT scalar identity $Z_{\mathrm{OP}}^X = -(\chi_{10}^{\mathrm{OP}})^{-1}$; the sign -1 is OP convention; together they produce the constant -4096 of (9.1). The Fourier coefficient at $q^{1/2}r^{1/2}s^{1/2}$ identifies the scalar 4096 as $\big([q^{1/2}r^{1/2}s^{1/2}]\Delta_5\big)^2$, a leading-coefficient square, not an enumerative invariant.

1.4 *Cobordism positioning and the closed* $K3 \times E \to \mathrm{pt}$ *cobordism*

The cobordism $K3 \times E \to \mathrm{pt}$ is the closed Calabi–Yau-threefold target viewed in the retained trace convention. The trace is the Laurent expansion of the meromorphic automorphic section Δ_5^{-2}; evaluation at a point of $\mathfrak{H}_2$ outside the polar divisor gives a complex number. The "level-n" label is the curve-counting integer

$$n = \chi(O_Y)$$

for a one-dimensional subscheme $Y \subset K3 \times E$ of class (β_h, d). The target has

$$\chi(O_{K3\times E}) = \chi(O_{K3})\chi(O_E) = 2 \cdot 0 = 0,$$

and this target Euler characteristic is not the exponent n. The Laurent variables record n, the elliptic degree d, and the K3 norm $h - 1$. The dual notation $Z^{K3\times E}_{\mathrm{BPS}}(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}})$ stores the elliptic and K3 gradings as a Laurent expansion of the inverse Igusa cusp form on $\mathfrak{H}_2$.

The $K3 \times E$ target is a threefold; the perfect obstruction theories and reduced virtual fundamental classes live on the stable-pair and DT moduli spaces $P_n(X, (\beta_h, d))$ and $I_n(X, (\beta_h, d))$, not on the target itself and not on any auxiliary fourfold. The closed K3-class fibre at primitive curve class β_h of norm $2h - 2$ supplies the K3 component of the coefficient indexed by (h, n, d), not a new target dimension.

2 Spin L-factor normalization for Δ_5^2

2.1 The Saito–Kurokawa eigenline

The unscaled Igusa form $\Phi^{\mathrm{un}}_{10} = \Delta_5^2$ is the weight-10 Saito–Kurokawa lift of the unique normalized cusp form $g = \Delta E_6 \in S_{18}(\mathbf{SL}_2(\mathbb{Z}))$; the Hecke eigenline is the one-dimensional $\mathbb{C}\Delta^{\theta}_{10} = \mathbb{C}\chi^{\mathrm{OP}}_{10}$, independent of the scalar normalization of its generator.

Proposition 2.1 *(Hecke eigenvalues on the Saito–Kurokawa eigenline).* For each prime p, let $T_1(p)$ and $T_2(p^2)$ denote the degree-one and degree-two genus-two Hecke operators on $S_{10}(\mathbf{Sp}_4(\mathbb{Z}))$ of [2, §4]. The Saito–Kurokawa form $\Delta^{\theta}_{10} = \Delta_5^2$ is a simultaneous eigenform with eigenvalues

$$T_1(p)\,\Delta^{\theta}_{10} \;=\; \lambda^{\mathrm{SK}}_1(p)\,\Delta^{\theta}_{10}, \qquad T_2(p^2)\,\Delta^{\theta}_{10} \;=\; \lambda^{\mathrm{SK}}_2(p^2)\,\Delta^{\theta}_{10},$$

where the eigenvalues are determined by the Saito–Kurokawa correspondence from the elliptic Hecke eigenvalues of $g = \Delta E_6$:

$$\lambda^{\mathrm{SK}}_1(p) \;=\; \lambda_g(p) + p^8 + p^9, \qquad \lambda^{\mathrm{SK}}_2(p^2) \;=\; \lambda_g(p)^2 + \lambda_g(p)\big(p^8 + p^9\big) + p^{18},$$

read off the X- and X^2-coefficients of the local Euler factor associated with (9.6), with $\lambda_g(p)$ the normalized p-th Hecke eigenvalue of $g = \Delta E_6$, the unique normalized cusp form in $S_{18}(\mathbf{SL}_2(\mathbb{Z}))$: from $\Delta E_6 = q - 528q^2 - 4284q^3 + \cdots$, $\lambda_g(2) = -528$, $\lambda_g(3) = -4284$ [98, Table 1]. Concretely $\lambda^{\mathrm{SK}}_1(2) = -528 + 2^8 + 2^9 = 240$ and $\lambda^{\mathrm{SK}}_1(3) = -4284 + 3^8 + 3^9 = 21960$, while $\lambda^{\mathrm{SK}}_2(4) = 135424$ and $\lambda^{\mathrm{SK}}_2(9) = 293343849$; the Ramanujan bound is violated ($|\lambda^{\mathrm{SK}}_1(p)| \sim p^9$, whereas tempered classical eigenvalues at weight 10 have size $p^{17/2}$), the signature of the Saito–Kurokawa CAP form.

Proof. The Saito–Kurokawa correspondence [98, §1] produces, for each weight $2k$ elliptic newform $g \in S_{2k}(\mathbf{SL}_2(\mathbb{Z}))$, a weight-$k + 1$ Siegel cusp form $F_g \in S_{k+1}(\mathbf{Sp}_4(\mathbb{Z}))$ with spinor L-function $L_{\mathrm{spin}}(s, F_g) = \zeta(s - k + 1)\zeta(s - k)L(s, g)$ (which is exactly (9.6) at $k = 9$). The Euler product factorisation determines the Hecke eigenvalues at each prime. In Andrianov's polynomial the Siegel weight is $K = k + 1$, so $p^{2K-4} = p^{2k-2}$, $p^{2K-3} = p^{2k-1}$, and $p^{4K-6} = p^{4k-2}$. Writing $L_{\mathrm{spin}}(s, F_g) = \prod_p L_p(p^{-s}, F_g)^{-1}$ with

$$L_p(X, F_g) \;=\; (1 - p^{k-1}X)(1 - p^kX)(1 - \lambda_g(p)X + p^{2k-1}X^2)$$

and comparing with the Andrianov factorisation [2, §4.5]

$$L_p(X, F_g) = 1 - \lambda^{\mathrm{SK}}_1(p)X + (\lambda^{\mathrm{SK}}_1(p)^2 - \lambda^{\mathrm{SK}}_2(p^2) - p^{2k-2})X^2 - \lambda^{\mathrm{SK}}_1(p)p^{2k-1}X^3 + p^{4k-2}X^4,$$

the X-coefficient gives $\lambda_1^{\mathrm{SK}}(p) = \lambda_g(p) + p^{k-1} + p^k$. The X^2-coefficient of the Euler product is

$$2p^{2k-1} + (p^{k-1} + p^k)\lambda_g(p).$$

Since the Andrianov coefficient of X^2 is $\lambda_1^{\mathrm{SK}}(p)^2 - \lambda_2^{\mathrm{SK}}(p^2) - p^{2k-2}$, one obtains

$$\begin{aligned}\lambda_2^{\mathrm{SK}}(p^2) &= (\lambda_g(p) + p^{k-1} + p^k)^2 - p^{2k-2} - 2p^{2k-1} - (p^{k-1} + p^k)\lambda_g(p)\\ &= \lambda_g(p)^2 + \lambda_g(p)(p^{k-1} + p^k) + p^{2k}.\end{aligned}$$

At $k = 9$ this is the displayed formula. The dimension formula $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$ forces $g = \Delta E_6$ at $k = 9$, and the eigenvalues at $p = 2, 3$ are tabulated in [98, Table 1]. □

Remark 2.2 *(The SK-correspondence onto the full-level Maaß line).* The SK-correspondence $g \mapsto F_g$ is a Hecke-equivariant injection $S_{2k}(\mathbf{SL}_2(\mathbb{Z})) \hookrightarrow S_{k+1}(\mathbf{Sp}_4(\mathbb{Z}))$ whose image is the Maaß subspace $\mathrm{Maass}_{k+1} \subset S_{k+1}(\mathbf{Sp}_4(\mathbb{Z}))$. At $k = 9$, $\dim \mathrm{Maass}_{10} = \dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$ and $\mathrm{Maass}_{10} = \mathbb{C}\Delta_5^2$; the Hecke eigenvalues λ_i^{SK} of Proposition 2.1 record the explicit arithmetic translation of the full-level elliptic Hecke action on $g = \Delta E_6$ under the SK-correspondence. No new modular forms or arithmetic data enter; the Saito–Kurokawa eigenline of Δ_5^2 is fully determined by the full-level arithmetic of ΔE_6 and the SK-correspondence.

The OP monic normalization is

$$\chi_{10}^{\mathrm{OP}} \;=\; 2^{-12}\,\Delta_{10}^{\theta} \;=\; 4096^{-1}\,\Delta_5^2, \tag{9.5}$$

and the level-n shadow (9.2) is the inverse of the *theta-normalized* Hecke eigenline element $\Delta_{10}^{\theta} = \Delta_5^2$.

THEOREM 2.3 *(Andrianov factorization of the spinor L-function).* For the Saito–Kurokawa representation generated by $\Delta_{10}^{\theta} = \Delta_5^2$,

$$L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) \;=\; \zeta(s-8)\,\zeta(s-9)\,L(s,g), \qquad g = \Delta E_6. \tag{9.6}$$

Equivalently, for Andrianov's local polynomial

$$Q_{p,F}(t) = 1 - \lambda_F(p)t + \big(\lambda_F(p)^2 - \lambda_F(p^2) - p^{16}\big)t^2 - \lambda_F(p)p^{17}t^3 + p^{34}t^4$$

at weight 10, the Euler product $\prod_p Q_{p,F}(p^{-s})^{-1}$ is the left-hand side of (9.6). Andrianov's completed function is

$$\Psi_F(s) = (2\pi)^{-s}\Gamma(s)\Gamma(s-8)L_{\mathrm{spin}}(s,F).$$

This is Andrianov's spin L-factor factorization [2, Theorem 3.1.1]; the polynomial $Q_{p,F}$ is Andrianov's genus-two Hecke polynomial in weight 10, with coefficient $\lambda_F(p)^2 - \lambda_F(p^2) - p^{2\cdot 10-4}$ at t^2. The proof uses Andrianov's spinor L-factor formula for the degree-2 Saito–Kurokawa lift attached to the weight-18 elliptic cusp form g [2], together with the dimension count $\dim S_{18}(\mathbf{SL}_2(\mathbb{Z})) = 1$ which forces the source form to be $g = \Delta E_6$ [98].

PROPOSITION 2.4 *(Polar coefficient at $s = 10$).* With $\Lambda(g,s) = (2\pi)^{-s}\Gamma(s)L(s,g)$, the functional equation $\Lambda(g,s) = -\Lambda(g, 18-s)$ gives $L(9,g) = 0$, so the elliptic central zero cancels the possible $\zeta(s-8)$-pole in the spin product at $s = 9$. At $s = 10$ the polar coefficient is

$$\operatorname*{Res}_{s=10} L_{\mathrm{spin}}(s, \Delta_{10}^{\theta}) \;=\; \zeta(2)\,L(10,g) \;=\; \frac{\pi^2}{6}\,L(10,g),$$

with

$$L(10, g)/(2\pi i)^{10}\Omega^{+}(g) \in \mathbb{Q}$$

by Manin–Deligne periods. The period theorem fixes the normalisation of the critical value; it is not a non-vanishing theorem. Thus $L_{\mathrm{spin}}(s, \Delta_{10}^{\theta})$ has a simple pole at $s = 10$ exactly when $L(10, g) \neq 0$.

This is the Saito–Kurokawa critical-value identity [98] together with the Manin–Deligne period decomposition [27, 72]; the proof uses the Eichler–Shimura–Manin period theorem for the rational newform g [57, 72], in Deligne's period normalization [27].

Remark 2.5 *(Hecke eigenline is normalization-independent).* The spinor L-function depends on the Hecke eigenline $\mathbb{C}\Delta_{10}^{\theta} = \mathbb{C}\chi_{10}^{\mathrm{OP}}$, not on the scalar used to label its generator. Theorem 2.3 and Proposition 2.4 are statements about the eigenline. The OP scalar constant 2^{-12} of (9.5) relabels the generator from Δ_5^2 to χ_{10}^{OP}; it does not change L_{spin}, the residue, or the cancellation of the elliptic central zero against the zeta pole at $s = 9$.

2.2 Independence from the Dirac–Igusa construction

The Saito–Kurokawa factorization of $L_{\mathrm{spin}}(s, \Delta_5^2)$ and the OP/DT scalar identity (9.1) are imported results: the first from Andrianov [2], the second from Oberdieck–Pixton [86] and Oberdieck [84]. Neither uses the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}$ of Chapter 5, the orientation character ϵ_o of (O1)/(O1$^+$), the type-II wall normal form of (O2), or the primitive recognition $P_X \cong \mathfrak{g}_{\Delta_5}$ problem of Chapter 6. Conversely, the construction of $\mathfrak{D}_X^{\mathrm{DI}}$ and the Pfaffian–Dirac theorem $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5$ do not use the spinor L-factor or the scalar inverse Igusa form. The two routes meet at the squared section $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$ and at no earlier point.

3 What the scalar shadow does not supply

3.1 Z_{BPS} is the level-n protected trace, not a path integral

The squared section $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$ is the determinant section, and its inverse $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ is the protected scalar trace in the Beilinson chain, after the primitive object at level P (the open factorization dg-category, the boundary vacuum b, and $A_b = \mathrm{End}_{\mathcal{C}}(b)$) and after the chart object at level C (the disk algebra and its bar / cobar / chiral Hochschild data). The scalar partition function is the *trace* of the protected data; it does not by itself construct the level-A acting algebra.

Scope 3.1 *(No scalar trace is a 3D gravitational path integral).* The identity (9.1), proved through OP/DT/PT primary literature, is a closed automorphic equation between scalar power series. Neither this scalar identity, nor the automorphic section of Chapter 3, nor the protected Pfaffian of Chapter 5 identifies $Z_{\mathrm{BPS}}^{K3\times E}$ to a 3-dimensional gravitational path integral on $\mathrm{AdS}_3 \times S^3 \times K3$, to a metric Euclidean path-integral over the full conformal class of the boundary, or to a holographic dual partition function in any sense beyond the boundary-CFT reading recorded by Vol II. The equality $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ gives the determinant section of a candidate gravity-line theory; the partition character would be the reciprocal section Δ_5^{-2}. The construction of the gravity-line operator algebra is open.

Remark 3.2 *(Protected trace versus path integral).* The assertion that Z_{BPS} is a $3d$ gravitational path integral identifies a protected scalar trace with a level-A acting theory. The precise statement is that $Z_{\mathrm{BPS}}^{K3\times E}$ is the level-n protected trace of the retained $K3 \times E$ data; the gravitational path integral, if such a thing exists at this boundary, requires the Hall–Borcherds residual of Vol II (§3.2). The residual is open;

the path-integral realization is not claimed. In particular, no microscopic state space $\mathcal{H}_{K3\times E}$ is asserted to support $\dim \mathcal{H}_{K3\times E,\gamma} = f(\gamma)$ outside the charge-graded virtual reading recorded in Chapter 2.

3.2 The Hall–Borcherds residual

The passage from a scalar partition function to a chain-level gravity-line operator algebra requires a comparison datum that lives neither in the automorphic / OP reduced theory nor in the compact Pfaffian datum. Vol II names this comparison.

Conjecture 3.3 *(Hall–Borcherds residual; gravity-line comparison).* Let $X = K3 \times E$ and assume the oriented critical Hall datum used for the Vol III positive-half comparison

$$\Theta_{\mathrm{Hall}\to\mathrm{Borch}} : \widehat{\mathrm{CoHA}}^{\mathrm{red}}_{\Lambda^{2,1}_{II}}(X) \longrightarrow U^{\mathrm{ch}}\big(\widehat{\mathfrak{n}_+(\mathfrak{g}_{\Delta_5})}\big). \tag{9.7}$$

The Hall–Borcherds residual is the conjectural extension of (9.7), after Drinfeld doubling, current enveloping along E, and weight-completion, to a filtered $\mathsf{SC}^{\mathrm{ch,top}}$ morphism from the $K3 \times E$ Hall boundary object to the gravitational boundary line algebra, whose derived-centre trace has scalar character $(\Phi^{\mathrm{un}}_{10})^{-1} = \Delta_5^{-2}$.

The residual is the Vol II gravity-line Hall–Borcherds residual [68]: it asks for the gravity-line operator algebra acting on the boundary. It is not claimed here; its existence and the construction of its witnessing morphism are open.

Remark 3.4 *(Compatibility with the κ-stratum).* The κ-tuple of the K3×E witness consists of the structural data behind the scalar shadow:

$$\big(\kappa_{\mathrm{cat}},\ \kappa^{\mathrm{Hodge}}_{\mathrm{ch}},\ \kappa^{\mathrm{Heis}}_{\mathrm{ch}},\ \kappa_{\mathrm{BKM}},\ \kappa_{\mathrm{fiber}}\big)_{K3\times E} = (0,\ 0,\ 3,\ 5,\ 24).$$

The fourth entry $\kappa_{\mathrm{BKM}} = 5$ is the cusp-form weight $\mathrm{wt}(\Delta_5)$, the Δ_5 row; the fifth entry $\kappa_{\mathrm{fiber}} = \chi_{\mathrm{top}}(K3) = 24$ is the topological Euler characteristic of the K3 fibre, distinct from $\chi(\mathcal{O}_{K3}) = 2$. The additive formula $\kappa_{\mathrm{BKM}} = \kappa^{\mathrm{Hodge}}_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ confuses these two: with $\kappa^{\mathrm{Hodge}}_{\mathrm{ch}} = 0$ and $\chi(\mathcal{O}_{K3}) = 2$, it predicts $\kappa_{\mathrm{BKM}} = 2$, which fails because the universal Borcherds weight at $N = 1$ is $\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 10/2 = 5$ (Borcherds 1995, Gritsenko 1999). The five entries of the tuple are five *distinct* constructions; they cannot be combined additively. The Vol III counterexample at $N = 1$ is the controlling instance.

3.3 Local-to-global in mixed holomorphic–topological language

A second route from local protected data to a global compact factorization algebra runs through the mixed holomorphic–topological model of Costello and the holomorphic de Rham obstruction class. Formal local Hamiltonian BF data imply such a global object only after the global holomorphic de Rham obstruction class $[\mathrm{obs}_{HT,\mathrm{glob}}] \in H^1(X, \mathcal{O}_X)$, plus descent, quantum master equation, anomaly, and locality data, with the Hamiltonian condition supplied by the local model. For $X = K3 \times E$ the global obstruction group is $H^1(K3 \times E, \mathcal{O}_{K3\times E}) \cong \mathbb{C}$ (non-zero), so the formal-local-Hamiltonian-BF route does not by itself identify $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ to a global compact factorization algebra. Both bridges –- Hall–Borcherds for the Vol II gravity line, mixed-HT for the Costello local-to-global ascent –- are open; the only established object here is the scalar shadow.

4 The $K3 \times E$ threefold realization question

The square root in the title is meant in Dirac's sense. The OP/DT scalar branch $Z^X_{\mathrm{OP/DT}} = -4096\,\Delta_5^{-2}$ is orientation-forgetful: the sign character is lost under squaring, and the inverse-square scalar cannot recover the sign by which a Pfaffian-line section distinguishes Δ_5 from $-\Delta_5$. The missing first-order datum is the protected Pfaffian, whose square is the inverse of the protected scalar shadow.

4.1 From the scalar trace to the retained operator datum

The Dirac square-root pattern compares the chiral-shadow Pfaffian with the scalar trace. Before taking the trace one has the protected Pfaffian of View 3; after the trace one has the inverse square as scalar shadow. The Pfaffian carries the orientation data; inverting its square forgets it and gives the orientation-forgetful scalar invariant.

Theorem 4.1 *(Orientation-forgetting scalar consequence of the Pfaffian).* On $K3 \times E$, after the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum, together with the retained finite Pfaffian section datum (P_{fin}), of Appendix 7, the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)$ of Chapter 5 is the automorphic section $\mathcal{D}_X = \Delta_5$ of Chapter 3, and its inverse square is the level-n protected scalar shadow:

$$\left(\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)\right)^{-2} \;=\; \Delta_5(Z)^{-2} \;=\; \left(\Phi^{\mathrm{un}}_{10}(Z)\right)^{-1} \;=\; Z^{K3\times E}_{\mathrm{BPS}}(T_{\mathrm{OP}}, Q_{\mathrm{OP}}, P_{\mathrm{OP}}), \tag{9.8}$$

and the OP/DT scalar branch is $-4096\,Z^{K3\times E}_{\mathrm{BPS}}$.

Proof. The Pfaffian–Dirac theorem of Chapter 6, with (D0), (O1), (O1$^+$), and $(O2_{\mathrm{atlas}})$ represented by the retained data of Theorem 7.4, and with the retained finite Pfaffian section datum (P_{fin}), asserts $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X) = \mathcal{D}_X = \Delta_5$. Squaring, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)^2 = \Delta_5^2 = \Phi^{\mathrm{un}}_{10}$; inverting, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}^{\mathrm{DI}}_X)^{-2} = (\Phi^{\mathrm{un}}_{10})^{-1}$. By Theorem 1.2 the right-hand side equals the level-n protected scalar shadow. The Oberdieck–Pixton/Pandharipande scalar identity [86, Theorem 1] says that the OP chamber representative is -4096 times this trace; the factor is the OP convention, the squared theta-leading coefficient, and the sign branch of (9.1). □

4.2 The orientation datum the inverse-square scalar cannot recover

The square-root sign is invisible to the scalar. An orientation-line sign $\varepsilon \in \{\pm 1\}$ acting by $\Delta_5 \mapsto \varepsilon\Delta_5$ satisfies $\varepsilon^2 = 1$; the scalar Δ_5^{-2} is therefore ε-invariant. Selecting the sign requires a pre-trace Pfaffian-orientation datum of View 3: the protected orientation character ϵ_o of the type-II Pfaffian wall normal form (O2) and its restriction to the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$, matched against the automorphic reflection character ν_{Δ_5}. Equality

$$\epsilon_o\,\big|_{W^{(2)}(\Lambda^{2,1}_{II})} \;=\; \nu_{\Delta_5}\,\big|_{W^{(2)}(\Lambda^{2,1}_{II})}$$

is the orientation-monodromy-character identity of Chapter 7; Appendix 7 states the retained-data criteria for (D0), (O1), (O1$^+$), and (O2) on $K3 \times E$. No constant appearing in (9.1) –- not the leading 64, not the squared 4096, not the OP scalar minus sign –- supplies ϵ_o. The local sign is checked on a rank-one Pfaffian wall; the global character requires the retained quotient orientation and the chosen Weyl transport, not a normalization constant.

4.3 The four obstructions and the realization question

Let $\mathfrak{D}_X^{\mathrm{DI}}$ denote the Dirac–Igusa pro-object

$$\mathfrak{D}_X^{\mathrm{DI}} = \lim_{R}\big(\mathcal{A}^{E_3}_{X,R},\ F^{\mathrm{hyb}}_{X,R},\ \Gamma_{X,R},\ \Pi_X,\ \Phi_R,\ o_R,\ H_R,\ P^{\Pi}_R,\ C_{X,R},\ \Theta_{\mathrm{Kos},R},\ \mathcal{L}_{\mathrm{Pf},R},\ \mathrm{pf}_{X,R},\ \varepsilon_{o,R},\ \mathrm{Rec}_R\big)$$

of Chapter 5: the inverse limit, over finite root windows R, of fourteen pieces of data including the chosen finite E_3-prefactorization algebra $\mathcal{A}^{E_3}_{X,R}$, the hybrid $\mathrm{Ran}^{\mathrm{hyb}}(E)$ carrier $F^{\mathrm{hyb}}_{X,R}$, the cosection-twisted reduced d-critical orientation o_R, the Hall correspondence datum H_R, the Pfaffian line $\mathcal{L}_{\mathrm{Pf},R}$, the Pfaffian section $\mathrm{pf}_{X,R}$, and the recognition map Rec_R. The retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ datum, and retained finite Pfaffian section datum (P_{fin}) are the Pfaffian and orientation data of Appendix 7 for Theorem 4.1.

Realization question (the K3×E threefold) — retained scalar-trace level. The square-root identity (9.8) is a theorem on $K3 \times E$ at the Pfaffian and scalar-trace level relative to the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ wall datum, and retained finite Pfaffian section datum (P_{fin}) in Appendix 7:

$(D0)$ *Do-degeneration limit of the cosection-reduced d-critical orientation theory.* **Criterion**: Theorem 7.6, relative to the retained Do-HN datum.

$(O1)$ *Strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle.* **Relative to** $(O1_{\mathrm{quot}})$: Theorem 7.12.

$(O1^{+})$ *Weyl-equivariant transport of the orientation along $W^{(2)}(\Lambda^{2,1}_{II})$ reflections.* **Relative to the chosen Weyl lifts**: Theorem 7.13.

$(O2)$ *Local Pfaffian wall normal form on type-II walls; the $2^{12} = 4096$ sign sum on the half-Hilbert orbit, with the $(R1)$–$(R2)$ scalar separation.* **Relative to** $(O2_{\mathrm{atlas}})$: Theorems 7.18, 7.16.

(P_{fin}) *Finite Pfaffian sections on a cofinal HN system, with support, parity, active-support coverage, leading coefficient, and Pfaffian-line Mittag–Leffler compatibility.* **Relative to** (P_{fin}): Definition 7.3.

The level-Z Hall–Borcherds trace comparison is relative to the retained (Z_{HB}) datum and Theorem 7.22: the retained trace datum places the compact source in the level-Z chiral-Hochschild domain, and Vol I Theorem H [67] together with Vol II [68, chiral-Hochschild Beilinson stratification] clauses (i) and (ii) give the chain-level chiral-Hochschild trace $\mathrm{Tr}_n^{\mathrm{bulk}} = \Delta_5^{-2}$ on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$. The Z_{BPS} square-root identity is therefore a retained-datum theorem on $K3 \times E$ at level Z, granted (Z_{HB}) and those two cited theorems; the level-A gravity-line operator algebra remains open in Vol II.

4.4 The threefold target, not a fourfold

The K3×E target is a complex threefold: $\dim_{\mathbb{C}} K3 \times E = 3$. The integer n in the "level-n shadow" is

$$n = \chi(\mathcal{O}_Y)$$

for the one-dimensional subscheme $Y \subset K3 \times E$, whereas $\chi(\mathcal{O}_{K3\times E}) = 2 \cdot 0 = 0$ is the target holomorphic Euler characteristic. The perfect obstruction theories in Chapter 5 are threefold theories; their virtual classes are reduced virtual classes on the moduli of objects on the threefold; the Pfaffian $\mathrm{Pf}_{\mathrm{prot}}$ of Chapter 5 is the Pfaffian of the threefold's protected operator data. The Borcherds product formula and the Gritsenko–Nikulin denominator identity give the Mukai-vector dictionary on the threefold by projection along the elliptic factor; the OP/DT/PT comparison gives the scalar shadow on the threefold; the protected Pfaffian gives the square root. No fourfold target intervenes anywhere in the chain.

4.5 Comparison with the determinant-only construction

The virtual K_0-determinant identity $\mathcal{D}_X(Z) = \Delta_5(Z)$ is a Grothendieck-class identity: it sees only the signed superdimensions of the underlying virtual super vector spaces. It does not see brackets, not parities of representatives, not relation constants, not completed PBW data, not Hopf-pairing radicals, not orientation gerbes, not the Pfaffian line $\mathcal{L}_{\mathrm{Pf}}$. Squaring the determinant gives Δ_5^2; inverting gives the scalar shadow. All the information about the retained operator datum $\mathfrak{D}_X^{\mathrm{DI}}$ that distinguishes the retained-data identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \Delta_5$ from a non-Pfaffian square-root choice $\Delta_5 \mapsto -\Delta_5$ lives in the Pfaffian datum and the orientation character. The inverse-square scalar shadow (9.2), whose OP representative is (9.1), is the maximal information visible after the trace; the Pfaffian and the orientation are the stronger pre-trace data.

5 Open structures

5.1 The Hall–Borcherds residual revisited

Conjecture 3.3 asks for the filtered $\mathsf{SC}^{\mathrm{ch,top}}$ morphism from the $K3 \times E$ Hall boundary object to the gravitational boundary line algebra of Vol II, whose derived-centre trace has scalar character Δ_5^{-2}. Its trace part is the bridge from the level-S scalar identity $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ to a level-Z chiral-Hochschild trace datum. Its acting part is the further level-A gravity-line operator algebra.

The residual consists of three completions. The Drinfeld doubling $\widehat{\mathrm{CoHA}}^{\mathrm{red}} \to D(\widehat{\mathrm{CoHA}}^{\mathrm{red}})$ of the Vol III positive-half pairing: the pairing, completion, integral form, and stable-envelope transport that convert the Hall positive half $\mathsf{Y}^+(K3 \times E)$ into the full quantum group $G(K3 \times E)$. The current enveloping along E and weight-completion: the chiral specialization $\mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C}$ along the reference curve $C = E$, together with the weight-completed ambient required to host the formal generators of $\mathfrak{g}_{\Delta_5}$. The filtered $\mathsf{SC}^{\mathrm{ch,top}}$ morphism property: the resulting boundary object must be a filtered morphism in the two-coloured Swiss-cheese chiral-topological operad, not merely a graded-vector-space comparison. Their existence is the Vol II Hall–Borcherds residual conjecture.

5.2 The gravity-line operator algebra and the Pentagon-face determinant

The Vol II gravity-line problem asks for a chain-level operator algebra $\mathfrak{Op}_{\mathrm{grav}}^{K3\times E}$ whose Pentagon-face determinant section is $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$. If such an algebra is constructed, its partition character is the reciprocal section $(\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$. The Brown–Henneaux dictionary $c = 3\ell/(2G_N)$ fixes the Virasoro boundary datum at Vir_c; it does not construct the $K3 \times E$ gravity-line algebra and does not evaluate the Pentagon-face determinant. A K3-Heisenberg witness would have to specialize the Virasoro λ-bracket $\{T_\lambda T\} = \partial T + 2T\lambda + (c/12)\lambda^3$ to the closed K3 boundary at $\kappa_{\mathrm{BKM}}(\Delta_5) = 5$ and identify the resulting determinant with $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$. Existence of the gravity-line operator algebra, agreement of its reciprocal partition character with the unscaled trace (9.2), equivalently with the OP representative (9.1) after OP normalization on each primitive closed K3-class fibre, and factorisation through the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$ of Chapter 5 are open level-A conditions.

5.3 Mathieu and umbral moonshine adjacency

The K3 elliptic genus $Z_{\mathrm{K3}} = 2\phi_{0,1}$ admits the $N = 4$ superconformal-character decomposition of Eguchi–Ooguri–Tachikawa [31]: after the massless characters are removed, the holomorphic mock-modular series

$$\Sigma(\tau) = -2q^{-1/8} + \sum_{n\geq 1} A_n q^{n-1/8}$$

has coefficients A_n which are dimensions of, and after twining characters of, M_{24}-modules [35, 37]. The decomposition is the infinite massive-sector decomposition of the $N = 4$ algebra. The coefficients $f(n,l)$ of $\phi_{0,1}$ which drive the Borcherds product are the untwined Jacobi coefficients; an M_{24}-twining would replace $\phi_{0,1}$ by the corresponding twined K3 elliptic genus before the singular theta lift. The umbral moonshine extension of Cheng–Duncan–Harvey [18] is indexed by the 23 Niemeier root systems with non-empty root system, with M_{24} the A_1^{24} case. The interaction between this sporadic-group structure on the K3 elliptic genus and the automorphic / BKM denominator structure on the $K3\times E$ Igusa cusp form is the Mathieu / umbral moonshine adjacency.

5.4 The closed-cobordism partition reads as one

The scalar shadow is the trace, the Pfaffian is the pre-trace operator-level section, and the primitive input is the retained E_3-prefactorization input whose Pfaffian line is presented by $\mathfrak{D}_X^{\mathrm{DI}}$. The closed $K3\times E \to \mathrm{pt}$ cobordism sees only the orientation-forgetful scalar trace. At that level the inverse of the squared Pfaffian section is the Laurent expansion $\Delta_5^{-2} = (\Phi_{10}^{\mathrm{un}})^{-1}$ of the inverse Igusa cusp form, and the orientation information that distinguishes Δ_5 from $-\Delta_5$ lives in the Pfaffian datum at level Z. The closed-cobordism partition function reads as one because Δ_5 is the same section throughout: the Borcherds–Gritsenko–Nikulin section of the weight-5 automorphic line, the Borcherds–Kac–Moody denominator on the root lattice $\Lambda_{II}^{2,1}$, and the protected Pfaffian of the Dirac–Igusa pro-object on $K3\times E$, relative to the retained orientation and wall-atlas data. The squared shadow of this same section is the protected partition function on the closed cobordism $K3\times E \to \mathrm{pt}$; the Hall–Borcherds residual of Vol II, when constructed, will lift it to the gravity-line operator algebra at level A; the threefold realization question of §4 is the retained-data construction of the Dirac–Igusa pro-object and the remaining compact-source recognition problem.

Bibliographic notes

The OP/DT scalar identity is Oberdieck–Pixton [86], Oberdieck–Pandharipande [85], and Oberdieck [84]; the variable conventions follow Bryan [14]. The Oberdieck–Pixton/Pandharipande scalar identity [86, Theorem 1] is the consolidated form used here. The Andrianov factorization of the spinor L-function is [2]; the Saito–Kurokawa correspondence at weight 10 is [98]; the Manin–Deligne period theorem is [72], [57], [27]. The dyon-counting interpretation in CHL backgrounds is Dijkgraaf–Verlinde–Verlinde [29] and Borcherds [9]; these inputs are the boundary-row physical hosts and are kept separate from the $K3\times E$ protected scalar shadow throughout.

The Hall–Borcherds residual conjecture in §3.2 is Vol II's Construction Problem 2 [68], the gravity-line Hall–Borcherds comparison. The protected D-brane index conjecture $\Omega_X(h,d,n) \overset{\mathrm{phys}}{=} \mathbf{DT}^X_{n,(\beta_h,d)}$ is recorded as the protected D-brane-index conjecture of Chapter 5; it is not used in the OP/DT scalar proof. The Mukai dictionary, the K3-class chamber convention, and the $\Lambda_{II}^{2,1}$ lattice data are imported from Chapter 2. The protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}$ and the Dirac–Igusa pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ are specified in Chapter 5; their Pfaffian and orientation parts are established relative to the retained data of Appendix 7;

the Pfaffian–Dirac theorem is Chapter 6; the four obstruction classes (D0), (O1), (O1$^+$), (O2) are defined and tracked in §5.8 of Chapter 5.

The κ-tuple $(0, 0, 3, 5, 24)$ and its non-additivity at $N = 1$ (the controlling instance for Δ_5) are recorded in Vol III; the Hall positive half $\mathsf{Y}^+(K3 \times E)$ and its Drinfeld doubling datum toward $G(K3 \times E)$, as well as the 6d holomorphic Chern–Simons quartic obstruction $\int_X \mathrm{Tr}_{\mathrm{ad}}(A(F_A)^3)$, are also Vol III. The local model $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ and the holomorphic de Rham obstruction class are mixed-HT-strings. None of these supply a 3D gravitational path integral.

CHAPTER 10

OPEN OBSTRUCTION CLASSES

1 OPEN OBSTRUCTION CLASSES

The cross-level identity ② ↔ ③ has two logically distinct parts. The Pfaffian and orientation part, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$ and $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$, holds on $K3 \times E$ relative to the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum, together with the retained finite Pfaffian section datum (P_{fin}), of Appendix 7. The primitive recognition $P_X \cong \mathfrak{g}_{\Delta_5}$ is conditional on the finite compact-source recognition datum $(S1)$–$(S10)$. The automorphic identity ① ↔ ② is theorem [8, 43]; the singular-theta-lift identity ② ↔ ⑤ is theorem [8]; the BKM denominator computation $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is theorem; the Lie-superalgebra recognition remains conditional from the finite-stage compact $K3 \times E$ source to the closed denominator algebra.

Each of the four obstructions sits at a definite categorical-dimensional level on the universal chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z} \to \mathsf{A}$. The remaining structures beyond the theorem are the Hall–Borcherds residual comparing the level-n protected scalar shadow with a chain-level trace datum, the chain-fusion conjecture for compact non-toric $K3 \times E$, and the gravity-line operator algebra required to have Pentagon-face determinant section Δ_5^2. The Mathieu / umbral moonshine adjacency stands between the compact source and these structures: its connection to $\mathfrak{g}_{\Delta_5}$ is conjectural. The conjecture asks for the twined denominator algebra and its reciprocal partition character. The Beilinson cut separates the retained-data Pfaffian theorem from the primitive source recognition and the gravity-line operator algebra.

The three names of Δ_5 are kept distinct on first occurrence: the automorphic section $\mathcal{D}_X = \Delta_5 \in H^0(\mathcal{H}_2, \mathcal{L}^5 \otimes \nu_{\Delta_5})$ of view ①; the BKM denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ of view ②; and the compact protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of view ③, relative to the retained data of Appendix 7.

1.1 *The four named obstructions*

The four obstructions act on distinct mathematical objects before any scalar trace is taken. The compact target is the Dirac–Igusa pro-object $\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R (\mathcal{A}_{X,R}^{E_3}, F_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, H_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathcal{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}$ The level discipline is *primitive object first, cross-level identity second, scalar shadow last*. The four obstructions and (P_{fin}) are pre-trace Pfaffian-orientation data of View ③; the level-Z object is the derived-centre trace pairing. They do not identify the compact primitive Lie superalgebra; that is the separate datum $(S1)$–$(S10)$.

1.1.1 (Do): Do-degeneration limit of the cosection-reduced d-critical orientation theory

1.1.1.1 Do-degeneration. The cosection-reduced d-critical orientation theory on the retained Hall stack $\mathcal{H}_R^{\mathrm{or}}$ of finite HN height R extends to the Do (zero-dimensional brane) degeneration locus, in the sense that the orientation line of the reduced determinant complex

$$\mathscr{D}_{R,C}^{\mathrm{red}} = \det \mathrm{RHom}_{\mathrm{red}}(C, C)$$

admits a Brav–Bussi–Dupont–Joyce–Szendrői [11] square-root gerbe section over the closure $\overline{\mathfrak{M}^{\mathrm{st}}_{R,C}} \subset \overline{\mathcal{H}^{\mathrm{or}}_R}$ including the D0-degeneration stratum, compatible with the cosection contraction inherited from $K3 \times E$. The retained Hall datum supplies an oriented critical finite-stage Pandharipande–Thomas/Gholampour–Sheshmani inverse system $\widehat{\mathcal{H}}^{\mathrm{or}}_{\sigma,S} = \varprojlim_R \mathsf{F}_R \mathcal{H}^{\mathrm{or}}_{\sigma,S}$ on the finite HN quotients of Proposition 0.17; (D0) asks that this system carry a coherent orientation through the D0-degeneration limit.

1.1.1.2 Criterion. The retained D0-HN datum supplies the D0-degeneration orientation through the retained inverse system via Theorem 7.6, and Theorems 7.12, 7.13, and 7.18 transport it to the type-II Weyl character. The Maass character ν_{Δ_5} and the geometric orientation ϵ_0 then agree on $W^{(2)}(\Lambda^{2,1}_{II})$. BBDJS [11] vanishing-cycle data require an oriented d-critical locus, not boundedness alone, and Joyce–Upmeier [52] square-root data on CY_3 moduli do not automatically extend to the reduced $K3 \times E$ quotient: the cosection-induced shift in the determinant complex must be shown compatible with descent. The connected BE Borel/edge class $\alpha^{E,\mathrm{free}} = a_1u_1 + a_2u_2 \in H^2(BE;\mathbb{F}_2)$ is separated from finite $BE[N]$ Borel terms in the Borel spectral sequence; the quotient-orientation descent across object, extension, mixed, wrapped, two-step flag, overlap, Thom–Sebastiani, and $R' \to R$ transition strata is conditional, with overlap/correspondence/flag/transition compatibilities named separately.

1.1.1.3 Argument. The retained Hall boundedness datum [K3×E-fin], motivated by [61, 90], supplies a finite-type retained slice; the cited boundedness results do not by themselves prove this slice for the compact $K3 \times E$ Hall sector. The datum also does not give quasi-smoothness, a d-critical chart, BBDJS coefficients, an orientation line, or compact-support six-functor admissibility through the D0 limit. Pantev–Toën–Vaquié–Vezzosi [89] shifted symplectic structures cover ambient derived moduli under Calabi–Yau hypotheses; the retained atlas is not produced by PTVV alone. Joyce–Upmeier [52] square-root orientation on Calabi–Yau_3 moduli requires exact-sequence/direct-sum compatibility; the reduced $K3 \times E$-quotient inherits this only after the cosection-induced shift is shown compatible with descent.

1.1.1.4 Quotient-orientation classes. The retained quotient-orientation datum consists of $\alpha^{\mathrm{red}}_{R,C} = w_2(\mathscr{D}^{\mathrm{red}}_{R,C}) \in H^2(S;\mathbb{F}_2)$, $\alpha^{E,\mathrm{free}}_{R,C}, \widetilde{\beta}^H_{R,C} \in H^2_H(S;\mathbb{F}_2)$, and $\lambda^H_{R,C} \in H^1(BH;\mathbb{F}_2)$ on every retained object, extension, mixed, wrapped, and flag stratum C, with killing of the Borel mixed terms and edge-spectral-sequence differentials at every node of the descent diagram. On the rank-one D6–D2–D0 stratum $C = (1, \beta, n)$, with β the K3-fibre primitive class, the BBDJS line carries the cosection contraction explicitly, and the four obstruction classes $(\alpha^{\mathrm{red}}, \alpha^{E,\mathrm{free}}, \widetilde{\beta}^H, \lambda^H)$ are the retained quotient-orientation datum of Appendix 7.3. The remaining source-side problems are this finite-stabilizer quotient orientation and the finite compact-source recognition datum for the Hall primitive algebra.

1.1.2 (O1): Strong reduced orientation on the K3×E-quotient self-Ext frame bundle

1.1.2.1 Reduced quotient orientation. The BBDJS [11] orientation gerbe on the moduli of stable sheaves on $K3 \times E$ descends to the $K3 \times E$-quotient self-Ext frame bundle, with reduced orientation gauge: on the stable stratum $\mathfrak{M}^{\mathrm{st}}_{R,C} \subset \mathcal{H}^{\mathrm{or}}_R$ the obstruction tuple

$$\mathfrak{o}^{\mathrm{or}}_{R,C} = \left(\alpha^{\mathrm{red}}_{R,C},\ \alpha^{E,\mathrm{free}}_{R,C},\ \{(\widetilde{\beta}^H_{R,C}, \lambda^H_{R,C})\}_H,\ \mu^{\mathrm{or}}_{R,e},\ \dots\right)$$

vanishes simultaneously, with all overlap, correspondence, flag, and $R' \to R$ transition data trivialized coherently. (O1) demands the strong reduced descent: the connected-BE class $\alpha^{E,\mathrm{free}}$, the finite-$BE[N]$ gerbe classes $\widetilde{\beta}^H$, and the residual finite linearization characters λ^H all vanish, separately, on every retained stratum, and the spectral-sequence differentials and Borel mixed terms are killed.

1.1.2.2 Criterion. The primitive recognition criterion reads: target arithmetic on $W_{\leq 3}$ is target Borcherds–Gritsenko–Nikulin–Kac data only, and recognition requires an actual compact source datum supplying $M, D, B, G, K, Q, A_\beta$, relation rows, Hopf radical/coideal kernel equality, PBW associated-graded comparison, and strict transition/Mittag–Leffler rows. Theorem 7.12 of Appendix 7 states what the retained (O1) datum carries to the recognition theorem. The connected E and finite $E[N]$ cohomologies are *distinct* obstruction spaces: $H^*(BE;\mathbb{F}_2) = \mathbb{F}_2[u_1, u_2]$ with $|u_i| = 2$, so $H^1(BE;\mathbb{F}_2) = 0$; for $E[2] \cong (\mathbb{Z}/2)^2$, $H^*(BE[2];\mathbb{F}_2) = \mathbb{F}_2[x_1, x_2]$ with $|x_i| = 1$, and the residual linearization $\lambda = \lambda_1 x_1 + \lambda_2 x_2$ is independent data after the degree-two gerbe class is killed.

1.1.2.3 Argument. Translation invariance under E-action does not produce orientation descent: a fixed underlying line may carry a nontrivial finite character, and ordinary translation invariance is *not* a substitute for vanishing of λ^H on full two-primary generators of $H^*(B(\mathbb{Z}/2^a)^2;\mathbb{F}_2) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2]$ at $N = 2^a$. The quotient-orientation criterion is finite Čech–Borel descent on the retained groupoid, not a vanishing theorem.

1.1.2.4 Quotient-orientation theorem. Theorem 7.12 is the quotient-orientation criterion relative to the retained $(O1_{\mathrm{quot}})$ datum. That datum contains the BBDJS gerbe sections on the self-Ext frame bundle, their null classes on the rank-one D6–D2–D0 stratum and its extension, mixed, wrapped, and two-step flag completions, and the overlap and correspondence transitions on the cofinal HN system. The data $M, D, B, G, K, Q, A_\beta$, the relation rows, the Hopf radical equality, and the PBW comparison required by primitive recognition are still separate compact-source data.

1.1.3 $(\mathrm{O1})^+$: Weyl-equivariant transport of the orientation along $\mathrm{W}^{(2)}(\Lambda^{2,1}_{II})$ reflections

1.1.3.1 Weyl transport. The orientation character on the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ is Weyl-equivariant: the orientation line of (O1) is equipped with coherent lifts $\tau_i : o_C \to s^*_{\delta_i} o_C$ of the three type-II simple reflections, each acting with sign -1, and the projective determinant-line cocycle $c_o \in Z^2(W^{(2)}(\Lambda^{2,1}_{II});\mathbb{F}_2)$ of the lift is a coboundary, with quotient torsor defects $\omega_{i,C} = 0$ on every retained groupoid component. At finite height R, the partial action groupoid $\mathcal{G}_R$ carries $c_{o,R} \in Z^2(\mathcal{G}_R;\underline{\mathbb{F}}_2)$, and $(\mathrm{O1})^+$ asks that $[c_{o,R}] = 0$, $\tau_i^2 = 1$, and $\omega_{i,C} = 0$ hold coherently on every stratum and at every transition height.

1.1.3.2 Criterion. The local sign-extraction criterion reads: only after the unit character is killed and $N^{\mathrm{Pf}}_{\delta} = 1$ is computed from the reduced normal self-Ext chart does the local sign equal -1, and Maass values $\nu_{\Delta_5}(s_{\delta_i}) = -1$ plus divisor orders $d^{\mathrm{aut}}_{\delta_i} = f(0, \pm 1) = 1$ supply target data only –- they do not compute N^{Pf}_{δ}, χ_v, or ϵ_o. The Coxeter presentation

$$W^{(2)}(\Lambda^{2,1}_{II}) = \langle s_{\delta_1}, s_{\delta_2}, s_{\delta_3} \mid s^2_{\delta_i} = 1\rangle$$

implies that any orientation character on $W^{(2)}(\Lambda^{2,1}_{II})$ is determined by three generator signs once $(\mathrm{O1})^+$ gives the honest Weyl lift; the three rank-one signs established by $(O2_{\mathrm{atlas}})$ are -1, and then propagate to the full character. The Weyl-equivariant transport is a separate identity, distinct from the no-extra-relations and PBW identities, established relative to the Weyl lifts in Theorem 7.13.

1.1.3.3 Argument. Three local signs on the simple reflections imply a character on the full type-II Weyl group only after $[c_{o,R}] = 0$ and $\omega_{i,C} = 0$: the Weyl determinant-line cocycle measures whether the lift is honest, and the quotient torsor defects measure whether the descent commutes with the reflection action on the retained groupoid. The cocycle reduces to the classical group-cohomology class only on a complete $W^{(2)}(\Lambda^{2,1}_{II})$-orbit with constant coefficients; finite height R is a partial action groupoid whose component data must be identified on intersections.

1.1.3.4 Conditional theorem. Theorem 7.13 is the criterion for finite-height lifts $\tau_{i,R}$ whose squares are one, whose projective cocycle class $[c_{o,R}]$ is zero, and whose torsor defects $\omega_{i,C}$ vanish on the retained groupoid components. These data give Weyl-equivariant orientation transport. They do not supply the height-seven source matrices required for the relation-closed primitive recognition window.

1.1.4 (O2): local Pfaffian wall normal form on type-II walls

1.1.4.1 Type-II wall normal form. On the three type-II walls $\mathbb{H}_{\partial_i}$ for $\partial_1 = 2f_2 - f_3$, $\partial_2 = 2f_{-2} - f_3$, $\partial_3 = f_3$, the retained wall-atlas datum has half-Hilbert sign support of size $4096 = 2^{12}$. The local computation in Appendix E is only the rank-one crossing. Concretely, in an s_∂-equivariant real-analytic d-critical/Kuranishi chart $(U_\partial, x_\partial, u_\partial, K^{\mathrm{red}}_\partial)$, with $K^{\mathrm{red}}_\partial$ the real form of the cosection-reduced self-Ext complex $\mathrm{RHom}_{\mathrm{red}}(C_\partial, C_\partial)$, the splitting $K^{\mathrm{red}}_\partial \simeq K^{\mathrm{tan}}_\partial \oplus K^{\mathrm{nor}}_\partial$ is equipped with the extension class $\eta_\partial \in \mathrm{Ext}^1_{D^{s_\partial}(U_\partial)}(K^{\mathrm{nor}}_\partial, K^{\mathrm{tan}}_\partial)$ trivial after the reduced E-quotient and quotient-orientation trivializations, and the normal Pfaffian admits the rank-one form $\mathrm{Pf}(K^{\mathrm{nor}}_\partial) = v_\partial\, u_\partial^{N^{\mathrm{Pf}}_\partial}$ with $v_\partial(x_\partial) \neq 0$, $s_\partial(v_\partial) = v_\partial$, and $N^{\mathrm{Pf}}_\partial = 1$.

1.1.4.2 The 4096 sign sum. The integer $4096 = 2^{12}$ is the half-Hilbert sign count retained in $(O2_{\mathrm{atlas}})$: twelve sign degrees of freedom, each $\mathbb{Z}/2$, give $2^{12} = 4096$ signed states. The type-II wall locus realizes this count only after the component, boundary, overlap, quotient, and protected-integration compatibilities of the retained atlas hold. The arithmetic match $-4096 = -(64)^2$ with the OP/DT scalar $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is the squared-theta normalization, fixed by Borcherds normalization and the OP scalar sign convention; it is *not* a derivation of ϵ_o.

1.1.4.3 Criterion. The local model and the exact separation from OP and Maass scalars are the following: scalar constants $D_5 = 64^{-1}\Delta_5$, $\chi_{10}^{\mathrm{OP}} = 4096^{-1}\Delta_5^2$, $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ are scalar normalizations and do not define ϵ_o; the Pfaffian normal form is the content. Theorem 7.18 is relative to $(O2_{\mathrm{atlas}})$, whose matrices over $\mathbb{Z}$ or $\mathbb{Q}$ record the Pfaffian-rank identity $N^{\mathrm{Pf}}_\partial = 1$, the splitting $\eta_\partial = 0$, the unit v_∂ with s_∂-character trivial, and overlap/transition compatibility across the cofinal HN system.

1.1.4.4 Argument. The condition $(\partial, \partial) = 2$, the automorphic divisor order $d^{\mathrm{aut}}_{\partial_i} = 1$, and the Maass character $\nu_{\Delta_5}(s_{\partial_i}) = -1$ are target data that do not imply the Pfaffian rank assertion $N^{\mathrm{Pf}}_\partial = 1$. Graph-isogeny atom candidates $B_2 = i_{\Gamma_\varphi,*}L$ are Koszul-Ext-local; they are *candidates*, not constructions, until semistability, full $\widehat{\Gamma}_X$ charge match, reduced normal Ext splitting, invariant unit, and atlas overlap compatibility are established. The "higher terms do not alter" phrase in the rank-one Ext criterion requires the explicit equivariant real/parametric Morse lemma and unit check before it operates as a theorem.

1.1.4.5 Wall-atlas transport. Definition 7.1 gives the retained type-II wall-atlas datum. Theorem 7.18 transports that datum over the half-Hilbert orbit; it does not construct the graph-isogeny or reducible-curve atoms. Theorem 7.16 identifies the 2^{12} sign count with the squared theta-leading normalization only after scalarization. The remaining relation-closed source comparison is the recognition datum $(S1)$–$(S10)$, not a type-II wall normal-form problem.

1.2 Mathieu and umbral moonshine adjacency

The K3 elliptic genus has a precise representation-theoretic decomposition. Eguchi, Ooguri, and Tachikawa [31] observed in 2010 that the elliptic genus $Z_{\mathrm{K3}}(\tau, z) = 2\phi_{0,1}(\tau, z)$ admits an M_{24}-graded character expansion: the mock-modular completion $\Sigma(\tau)$ of the elliptic genus is a graded M_{24}-module, and the multiplicities of irreducible M_{24}-representations in the first several Fourier coefficients of the holomorphic part match the dimensions of irreducible M_{24}-representations on the nose. Gannon [37] proves the arithmetic identity; the geometric origin via a sigma-model action of M_{24} on the K3 elliptic-genus state space remains open outside the classes accounted for by known K3 sigma-model symmetries. The M_{24} action visible at the level of the elliptic genus does not lift to a global geometric symmetry of $D^b(\mathrm{Coh}(K3))$; the work of Gaberdiel–Hohenegger–Volpato [35] on Mathieu twining characters identifies the precise refinement at the level of supersymmetric sigma-model conformal field theory data.

The umbral moonshine of Cheng–Duncan–Harvey [18] extends Mathieu moonshine to a family of 23 cases indexed by Niemeier root systems, one case per Niemeier lattice having a non-empty root system. For a Niemeier lattice N^X with root system X, the finite group is the umbral group

$$G^X = \mathrm{Aut}(N^X)/W^X,$$

not the discriminant group. Mathieu moonshine is the $X = A_1^{24}$ case. The Leech lattice is the 24th Niemeier lattice and has empty root system; it belongs to Conway moonshine, not to the 23-case umbral list. The umbral moonshine conjecture of Cheng–Duncan–Harvey [18], in the module-existence form proved by Gannon [37], asserts that each pair (X, G^X) carries a graded module whose graded character is the prescribed vector-valued mock modular form.

1.2.0.1 The conjectural connection to $\mathfrak{g}_{\Delta_5}$. The Mathieu moonshine module lies on the $N = 4$ superconformal character side of the K3 elliptic genus. The Eguchi–Ooguri–Tachikawa module is graded by the massive level n; the z-dependence and the l-index are carried by the $N = 4$ character, not by a primitive (n, l)-grading on the M_{24}-module. The Beilinson-cut statement is therefore a conjectural extension of the twined K3 elliptic genus to the Borcherds–Kac–Moody datum on $\Lambda_{II}^{2,1}$, producing a g-twined refinement of $\mathfrak{g}_{\Delta_5}$ with denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)})$ for each M_{24}-conjugacy class g. The umbral cases of [18, 19] extend the conjecture to a family of 23 Borcherds–Kac–Moody-style algebras associated to the 23 Niemeier root systems with non-empty root sets.

1.2.0.2 Precise conjecture. For each conjugacy class $g \in M_{24}$, let $\phi_{0,1}^{(g)}(\tau, z)$ denote the g-twined K3 elliptic genus of [35]. The Cheng–Duncan–Harvey–Gannon umbral moonshine conjecture, refined to the BKM stratum of $\Lambda_{II}^{2,1}$, asserts the existence of a g-twined Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}^{(g)}$ with denominator

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)}) = \Theta(\phi_{0,1}^{(g),\mathrm{Weil}})\big|_{\mathbb{H}_2},$$

the singular theta lift of the g-twined Weil-representation extension of $\phi_{0,1}^{(g)}$, with the chamber and Weyl-vector data determined by the relevant umbral pair (X, G^X), when such a pair is part of the twining datum. No map from all M_{24}-classes to Niemeier root systems is asserted here.

1.2.0.3 The order-seven and order-fifteen classes. The four classes $\{7A, 7B, 15A, 15B\}$ lie outside the diagonal Gritsenko–Cléry/Borcherds frame used here, namely the rows selected by $\varphi(N) \mid 2$; this does not exclude their occurrence in other CHL dyon towers. Their cyclotomic fields have degrees 6 and 8. Their cycle shapes on the standard 24-dimensional permutation representation are

$$7A : 1^3 \cdot 7^3, \qquad 7B : 1^3 \cdot 7^3, \qquad 15A : 1 \cdot 3 \cdot 5 \cdot 15, \qquad 15B : 1 \cdot 3 \cdot 5 \cdot 15,$$

the $7A/7B$ pair distinguished by Galois conjugation under $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ and the $15A/15B$ pair by $\mathrm{Gal}(\mathbb{Q}(\zeta_{15})/\mathbb{Q})$; the rational character on each pair is integer-valued, but individual class characters take values in the cyclotomic field of order equal to the class order. These cycle shapes have different geometric status: $7A$ and $7B$ have six orbits and pass the Mukai–Kondo orbit test, while $15A$ and $15B$ have four orbits and are non-geometric in that test [36, §2]. The orders $|g| \in \{7, 15\}$ satisfy $\varphi(|g|) \nmid 2$ ($\varphi(7) = 6$, $\varphi(15) = 8$), placing them outside the Gritsenko–Cléry frame $N \in \{1, 2, 3, 4, 6\}$ of κ_{BKM}-universality selected by $\varphi(N) \mid 2$ of Theorem 8.1. A Borcherds–Kac–Moody construction for these twined genera therefore requires data beyond that denominator frame.

Conjecture 1.1 *(Order-seven and order-fifteen twined denominators).* For each $g \in \{7A, 7B, 15A, 15B\}$:

- (M★7A) The Gaberdiel–Hohenegger–Volpato $7A$-twined K3 elliptic genus $\phi_{0,1}^{(7A)}$ is a weight-0, index-1 weak Jacobi form with multiplier for the appropriate congruence subgroup attached to the $7A$ class. Its singular theta lift, if the Weil extension and Weyl chamber data are fixed, has Borcherds weight $c_{7A}(0)/2$ and multiplier, and equals $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(7A)})$ for a conjectural $7A$-twined Borcherds–Kac–Moody algebra.
- (M★7B) Identical statement for the Galois conjugate class $7B$, with $\phi_{0,1}^{(7B)} = \phi_{0,1}^{(7A)} \circ \sigma_{\zeta_7 \to \zeta_7^3}$ the Galois-conjugate character.
- (M★15A) The $15A$-twined character $\phi_{0,1}^{(15A)}$ is a weight-0, index-1 weak Jacobi form with multiplier for the appropriate congruence subgroup attached to the $15A$ class. Its singular theta lift, if the Weil extension and Weyl chamber data are fixed, has Borcherds weight $c_{15A}(0)/2$ and multiplier, and equals $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(15A)})$.
- (M★15B) Identical statement for the Galois conjugate class $15B$.

The four clauses have finite necessary conditions in their first Fourier coefficients, but no finite truncation proves the denominator identity.

Remark 1.2 *(Fourier conditions and the parity obstruction).* Three Fourier conditions constrain each clause (M★g) at finite truncation depth.

(i) The leading Fourier coefficient $c_g(0) = [q^0 r^0]\phi_{0,1}^{(g)}$ is the Gaberdiel–Hohenegger–Volpato McKay–Thompson value of g at the $q^0 r^0$ position of the K3 elliptic genus, tabulated in [35, Table 1] for all 26 conjugacy classes. Outside the CHL frame $\{1A, 2A, 3A, 4A, 6A\}$, this leading coefficient need not be an even integer in the Borcherds lattice normalisation; for the classes $\{7A, 7B, 15A, 15B\}$ it is not one of the even values $c_N(0) \in \{10, 8, 6, 4, 2\}$ of Theorem 8.1.

(ii) The Borcherds weight formula gives $\kappa_{\mathrm{BKM}}(\Phi^{(g)}) = c_g(0)/2$ only after the twined Borcherds datum and its lattice have been fixed. For the order-seven and order-fifteen classes this weight need not be an integer; the lift must therefore be treated as a multiplier-bearing automorphic product, not as a weight-5 paramodular form. This is the arithmetic obstruction to extending the level-Z denominator construction beyond the CHL frame.

(iii) Mathieu-moonshine consistency with the EOT M_{24}-multiplicity predictions through the first eight Fourier coefficients of $\Sigma(\tau)$ imposes a finite necessary condition on each clause; proving $(M^{\star}g)$ requires identifying the geometric origin of the M_{24}-action on the K3 elliptic-genus state space and its umbral refinement in the sense of Cheng–Harrison [19].

1.2.0.4 Independence from the four Pfaffian–Dirac obstructions. The Pfaffian theorem 2.1 and the orientation-character theorem 3.1 are retained-data theorems on $K3 \times E$, after the retained D0-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\text{atlas}})$ wall datum, and retained finite Pfaffian section datum (P_{fin}) of Theorem 7.4. The primitive recognition theorem remains conditional on $(S1)$–$(S10)$, and the level-A gravity-line operator algebra remains conditional on the Vol II Hall–Borcherds residual. The moonshine adjacency is independent of both: it lives at level Z (centre/denominator) and at level A (acting object on the K3 elliptic-genus state space). A proven identification of the moonshine module as a g-twined character of $\mathfrak{g}_{\Delta_5}$ does not by itself solve the Vol II Hall–Borcherds residual; conversely the Hall–Borcherds residual does not prove the moonshine adjacency.

1.2.0.5 Twined denominator problem. The g-twining operation would produce $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)})$ from $\mathrm{den}(\mathfrak{g}_{\Delta_5})$ and an M_{24}-action on a denominator-compatible lift of the Mathieu module. The ordinary massive-level grading of the M_{24}-module supplies the mock modular coefficients; the Borcherds product requires the additional Jacobi variable and $\Lambda_{II}^{2,1}$-root grading. That lift, and the denominator-compatible action on it, are not part of the EOT theorem.

1.2.0.6 Compatibility with Vol III. The Vol III dimension-stratified BKM catalogue [65] places $\mathfrak{g}_{\Delta_5}$ as the $d = 3$ sibling, with companions at $d = 3$ (Borcherds Monster $V^{\natural}$ on $\mathrm{II}_{1,1}$) and $d = 5$ (Fake Monster on $\mathrm{II}_{25,1}$ via $K3 \times K3 \times E$). The Mathieu/umbral moonshine adjacency to $\mathfrak{g}_{\Delta_5}$ is a level-Z cross-stratum conjecture: the denominator-compatible M_{24} action on the K3 elliptic-genus state space for $\{7A, 7B, 15A, 15B\}$ is the missing piece on the K3-fibre side; the g-twined denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)})$ is the missing piece on the BKM side.

1.2.0.7 Twined denominator identity. For each of the 26 conjugacy classes of M_{24}, the expected object is a g-twined Borcherds product on $\Lambda_{II}^{2,1}$. Its specialization to the $N = 4$ character expansion must recover the twined McKay–Thompson coefficients of EOT–Gannon. A coefficientwise refinement of the EOT multiplicities by BKM denominator coefficients is therefore a consequence only after the denominator-compatible Jacobi lift and the $\Lambda_{II}^{2,1}$-root grading have been fixed. The required identity is the consistency of the g-twined denominator with the OP scalar $Z_{\mathrm{OP}}^{X} = -4096\,\Delta_5^{-2}$ under the g-twining construction, including the squared sign convention.

1.3 Residual operator structures

The Beilinson cut separates the remaining operator data by the object on which the missing structure must live: the Hall–Borcherds operator lift, the compact non-toric chain-fusion equivalence, and the gravity-line algebra.

1.3.1 The Hall–Borcherds residual

1.3.1.1 Hall–Borcherds residual. The level-n protected scalar shadow $Z_{\mathrm{BPS}}^{K3\times E} = (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$ on the closed $K3 \times E \mapsto$ point cobordism is the BPS partition function. A 3d gravitational path-integral interpretation with a genuine operator-level lift requires the Hall–Borcherds residual: an operator-valued completion of the scalar trace whose trace recovers the level-n shadow and whose algebraic content carries the metric degrees of freedom. This residual remains open in Vol II.

1.3.1.2 Comparison with Vol II. Vol II [68] names the gravity-line operator algebra required to have Pentagon-face determinant section $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$, whose partition character is $(\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$, and identifies the Hall–Borcherds residual as the comparison arrow from the bulk $Z_{\mathrm{ch}}^{\mathrm{der}}(A_b)$ to the line-operator algebra. The Vol II residual asks for the construction of the gravity-line operator algebra with Pentagon-face determinant $\det_{\mathrm{Pentagon}} = \Phi_{10}^{\mathrm{un}} = \Delta_5^2$ on the closed cobordism, whose primitive square-root Borcherds weight is $\omega_{\mathrm{Borcherds}}(\Delta_5) = c_1(0)/2 = 5$ at the K3-Heisenberg witness. The Igusa-side datum is the same automorphic determinant identity $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$, with the Appendix G Pfaffian datum $\mathfrak{D}_X^{\mathrm{DI}}$ carrying the retained orientation and Weyl-transport data on the chiral side.

1.3.1.3 Argument. The protected graded trace

$$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F q^{L_0} r^{J_0} s^{\widetilde{L}_0}\big) = \Delta_5^{-2}$$

forgets orientation; the OP chamber representative is -4096 times this trace. The sign-character ϵ_o is invisible in the squared form and is part of the Pfaffian–Dirac orientation theorem, not of the gravity-line residual. The Hall–Borcherds residual asks instead for an acting operator algebra whose trace recovers the scalar shadow. The realization as a three-dimensional gravitational path integral is conditional on Brown–Henneaux, modular invariance, vacuum dominance, and saddle dominance [46]; no such realization is asserted.

1.3.1.4 Scalar non-faithfulness. The structural reason Δ_5^{-2} does not by itself recover the operator-level gravity-line algebra is the scalar non-faithfulness of the graded Euler character on filtered $SC^{\mathrm{ch,top}}$-objects, proved as a lemma in Vol III [65]: an equality $\chi(C_1) = \chi(C_2)$ between filtered $SC^{\mathrm{ch,top}}$-objects does not determine a quasi-isomorphism, an $SC^{\mathrm{ch,top}}$-morphism, a Hall product, a Drinfeld pairing, a current-envelope locality structure, or a derived-centre trace –- adding a filtered acyclic summand K with $\chi(K) = 0$ leaves the scalar unchanged while admitting non-trivial extensions of the operadic action, and a filtered differential may be perturbed within the associated graded without altering χ. The missing chain-level content underneath Δ_5^{-2} sits under Vol III's Hall–Borcherds datum: a positive-half Hall–Borcherds bialgebra morphism on E, a completed Drinfeld-double extension, a current-envelope morphism along E, and trace compatibility with Δ_5^{-2} [65]. The Igusa scalar is only the trace shadow of this datum. The Hall–Borcherds residual is the missing chain-level lift together with its finite-window identities.

1.3.1.5 Pentagon-face cyclic trace. The required cyclic trace is $\mathrm{Tr}^{\mathrm{cl}}_{\mathrm{Pentagon}}$ on the closed-colour factorization algebra F^{cl} at the $K3 \times E$-Heisenberg witness. Its evaluation must recover $c_1(0)/2 = 5$ on the universal Borcherds-weight identity, and its operator-level lift to the open-colour boundary algebra $A_b = \mathrm{End}_C(b)$ must factor through the chain-fusion datum below. The compatibility condition is the consistency of the Pentagon-face scalar with the unscaled trace Δ_5^{-2} under the squared-trace operation. The OP chamber representative is -4096 times this trace.

1.3.2 Chain fusion for compact non-toric K3×E

1.3.2.1 Chain-fusion datum. For a Calabi–Yau$_d$ category $\mathcal{C}_X$ and a chart datum (Σ_{d-1}, C), a chain-fusion datum consists of a boundary vacuum $b(X, \Sigma, C)$ in an open factorization dg-category on (C, D_C, τ_C), a comparison morphism

$$\Phi_d^{(\Sigma_{d-1},C)}(\mathcal{C}_X) \longrightarrow \mathrm{End}_C(b(X, \Sigma, C)),$$

and the compatibility of this morphism with clutching, trace, and the Calabi–Yau orientation. The chain-fusion conjecture asserts that this morphism is a quasi-isomorphism for the admissible compact chart. The affine toric examples $\mathbb{C}^3$, local $\mathbb{P}^2$, and the conifold furnish local comparison models; they do not construct the compact non-toric boundary vacuum for $K3 \times E$. At $d = 3$, the retained Dirac–Igusa data supply the protected scalar shadow on $K3 \times E$; the boundary-vacuum construction required for chain fusion remains open.

1.3.2.2 Comparison with Vol III. Vol III's catalogue [65] names this conjecture in the two-stage factorization frame and identifies, after chain fusion, the level-Z Drinfeld double obligation $G(X) = D(Y^+(X))$ for compact non-toric CY_3, together with the conditional TCFT/framing/anomaly/completion datum needed for compact non-formal CY_3 in general. Chain fusion remains a conjecture, not a theorem; the assertion of Δ_5 as a protected Pfaffian section on $K3 \times E$ is the retained-data identity $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5$ of view ③, not an underlying microscopic state space.

1.3.2.3 Argument. The compact $K3 \times E$ case at $d = 3$ is a candidate instance of chain fusion via the rectified finite $K3$-to-E specialization (Definition 0.8). The proposed chiral specialization, if constructed from the retained finite-stage datum, has target form $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}(\mathfrak{A}_{K3\times E}) = U^{\mathrm{ch}}(\mathfrak{heis}_{\mathrm{Muk}}) \otimes U^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{BPS}}_{K3})$ on $\mathrm{Ran}(E)$. The map $\mathrm{Sp}^{\mathrm{ch}}_{K3,E}$ is the chosen $[K3{\to}E\text{-sp}]$ datum, not a canonical functor. The boundary algebra $A_b = \mathrm{End}_C(b)$ on the open factorization side can coincide with this target only after the chain-fusion boundary vacuum $b(K3 \times E, \Sigma_2, E)$ exists. Compact non-toric $K3 \times E$ generalizations to other compact non-toric CY_3-data (e.g. Borcea–Voisin threefolds, Schoen's small resolutions) require independent boundary vacua.

1.3.2.4 Boundary vacuum condition. The missing object is $b(K3 \times E, \Sigma_2, E)$, a boundary vacuum in an open factorization dg-category on (E, D_E, τ_E) with D_E at the Igusa cusp $\mathfrak{c}_\infty$ of the Siegel space. The chain-fusion isomorphism on $W_{\leq 3}$ at finite HN height $R = 3$ must identify the chiral algebra with $A_{b(K3\times E,\Sigma_2,E)}$. The necessary compatibility condition is the consistency of the chain-fusion isomorphism with the unscaled trace Δ_5^{-2}, whose OP representative is $-4096\,\Delta_5^{-2}$, and with the universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 5$; see the κ-tuple discussion below.

1.3.3 The gravity-line operator algebra

1.3.3.1 Gravity-line algebra. A gravity-line operator algebra, if constructed, is an operator algebra on the closed-colour factorization algebra F^{cl} on $K3 \times E$ whose Pentagon-face determinant section is required to be $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$, a weight-10 automorphic section. Its partition character is the reciprocal section Δ_5^{-2}. The primitive square-root row has Borcherds weight

$$\omega_{\mathrm{Borcherds}}(\Delta_5) = \kappa_{\mathrm{BKM}}(\Delta_5) = c_1(0)/2 = 5.$$

This is the same Hall–Borcherds residual and remains open in Vol II.

1.3.3.2 Comparison with Vol II. Vol II [68] names the gravity-line Hall–Borcherds comparison conjecture and the Pentagon-face determinant; Vol III [65] names $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ as the universal Borcherds-weight identity [8], with $\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 5$ at the paramodular Δ_5 datum.

1.3.3.3 The κ-tuple. The five-fold κ-tuple $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ at $K3 \times E$ records five separate invariants, distinct in construction and meaning, never additively combined. Here $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3}) \cdot \chi(\mathcal{O}_E) = 2 \cdot 0 = 0$ is the Künneth-multiplicative categorical Euler; $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = \sum_q (-1)^q h^{0,q}(K3 \times E) = 0$ by Serre duality on the compact total space; $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$ is the rank of the Heisenberg–Cartan in the chiral Hall specialization on $\mathrm{Ran}(E)$ and equals the Cartan rank of $\mathfrak{g}_{\Delta_5}$; $\kappa_{\mathrm{BKM}} = c_1(0)/2 = 5$ is the universal Borcherds-weight identity at the Gritsenko–Cléry row whose Jacobi seed is $\phi_{0,1}$ and whose Borcherds product is Δ_5, with $c_1(0) = 10$; $\kappa_{\mathrm{fiber}} = 24$ is the Mukai-lattice rank $\mathrm{rk}\,\widetilde{H}^*(K3, \mathbb{Z}) = 24$ of the K3 fibre, equal to $\chi_{\mathrm{top}}(K3)$. The additive identity $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ fails because it confuses $\chi_{\mathrm{top}}(K3) = 24$ with $\chi(\mathcal{O}_{K3}) = 2$; the universal identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is the correct frame.

1.3.3.4 Argument. The Pentagon-face determinant is the modularity-witness of the open-side cyclic trace plus clutching; modularity is a property of the open trace, not of the closed algebra. On the Δ_5 row, $c_1(0)/2 = 5$ is the weight of the primitive Borcherds factor. The determinant required by chain fusion is its square $\Delta_5^2 = \Phi_{10}^{\mathrm{un}}$, hence a form of total weight 10 on the closed cobordism; the level-n protected scalar shadow is the reciprocal Laurent expansion Δ_5^{-2}.

1.3.3.5 Gravity-line trace condition. The required object is the gravity-line operator algebra $\mathrm{GravLine}_{K3\times E}$ on the open factorization dg-category on (E, D_E, τ_E) at the Igusa cusp, with Pentagon-face determinant identity $\det^{\mathrm{cl}}_{\mathrm{Pentagon}}(\mathrm{GravLine}_{K3\times E}) = \Delta_5^2$ and compatibility with the universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_1) = 5$ and with the κ-tuple $(0, 0, 3, 5, 24)$ at the K3-Heisenberg witness. The compatibility condition is the existence of chain fusion at $b(K3 \times E, \Sigma_2, E)$ and the Hall–Borcherds residual lifting the protected trace $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ toward the acting operator-algebra level.

The finite compact-source recognition datum $(S1)$–$(S10)$ on $W_{\leq 3}$ and $W_{\leq 7}$, the chain-fusion boundary vacuum $b(K3 \times E, \Sigma_2, E)$, the Hall–Borcherds residual lifting $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$, the gravity-line operator algebra with Pentagon-face determinant Δ_5^2, and the Mathieu-twined denominators $\mathrm{den}(\mathfrak{g}_{\Delta_5}^{(g)})$ on the 26 M_{24}-conjugacy classes are the unresolved data. The four obstructions concern, respectively, the D0-degeneration of the reduced orientation theory, quotient orientation, Weyl transport of the orientation line, and the type-II Pfaffian wall atlas. The remaining operator obstructions concern the Hall–Borcherds operator lift, compact non-toric chain fusion, and the gravity-line algebra. The Beilinson cut separates these objects before any scalar trace is taken.

1 Lattice computations

The seven integral lattices are recorded here in one place: the hyperbolic plane $U = \mathrm{II}_{1,1}$, the Lorentzian root lattice $\Lambda_{II}^{2,1}$ of the Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$, the K3 cohomology lattice $\Lambda_{K3} = \mathrm{II}_{3,19}$, the lattice-polarized K3 sublattice Λ_{pol}, the Mukai lattice $\widetilde{H}(S, \mathbb{Z}) \simeq \mathrm{II}_{4,20}$, the formal dyonic charge lattice $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z})$, and the signature-(3, 2) lattice $\Lambda^{3,2}$ that bridges $\mathbf{Sp}_4(\mathbb{Z})$ to the type-IV orthogonal group via the projective exterior-square image. Each is given by its Gram matrix or root data, signature, discriminant, and automorphism group, with primary citations (Conway–Sloane 1999 §2.5, §4, §16; Nikulin 1980 Theorem 1.1.2 and Theorem 1.14.4; Mukai 1984; Gritsenko–Nikulin 1998 Part II §2, §5). Ranks, signatures, and the Cartan matrix of the type-II simple roots are checked by the rational computations displayed below.

The identities about Δ_5, $\mathfrak{g}_{\Delta_5}$, the orientation character ν_{Δ_5}, and the Pfaffian wall normal form on $W^{(2)}(\Lambda_{II}^{2,1})$ use the lattice data fixed here, not a sign or scaling chosen later.

Convention 1.1 *(Sign of the bilinear form; orientation of the time-like cone).* The standard mathematical convention for the hyperbolic plane U is $U = (\mathbb{Z}^2, ((0,1),(1,0)))$, even unimodular of signature (1, 1) and discriminant -1 (Conway–Sloane 1999 §2.5). This convention is primary.

The Borcherds–Gritsenko–Nikulin packaging used in §1.4 of Chapter 5 and the exterior-square model of $\mathbf{Sp}_4(\mathbb{Z})$ employs the negated form

$$\Lambda^{(1,1)} = \left(\mathbb{Z}^2, \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\right),$$

so that the time-like cone $C(\Lambda^{2,1})_+ \subset \Lambda^{2,1} \otimes \mathbb{R}$ has *negative* square $(x, x) < 0$. The relation is $\Lambda^{(1,1)} = U(-1)$. Every Gram matrix below is given in the standard convention; a sign flip transports it to the BKM convention without changing rank, discriminant absolute value, or automorphism group.

1.1 The hyperbolic plane $U = \mathrm{II}_{1,1}$

The hyperbolic plane is the unique even unimodular indefinite lattice of signature (1, 1) (Conway–Sloane 1999 §2.5; Nikulin 1980 Theorem 1.1.2).

Definition 1.2 *($U = \mathrm{II}_{1,1}$).* [definitional] Let $U := \mathbb{Z}e \oplus \mathbb{Z}f$ with Gram matrix

$$G_U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad e^2 = f^2 = 0, \quad e \cdot f = 1.$$

Verification 1.3 *(Rank, signature, discriminant of U).* [computed] $\operatorname{rk} U = 2$. The eigenvalues of G_U are ± 1, so the signature is (1, 1). The discriminant is

$$\operatorname{disc} U = \det G_U = -1.$$

U is even (every G_U-diagonal entry is $0 \equiv 0 \pmod 2$) and unimodular ($|\det G_U| = 1$). These are the defining invariants of $\mathrm{II}_{1,1}$ in the Conway–Sloane classification (Conway–Sloane 1999 §2.5, Theorem 2.5.1).

Proposition 1.4 *($\mathrm{Aut}(U)$).* [proved] The orthogonal group of U is

$$\mathrm{O}(U) = \{\pm \mathbf{I}\} \times \langle \sigma \rangle \simeq (\mathbb{Z}/2)^2, \qquad \sigma : e \leftrightarrow f.$$

The proper subgroup $\mathrm{O}^+(U) \simeq \mathbb{Z}/2$ swaps e and f but fixes the spinor norm.

Proof. An isometry $\phi \in \mathrm{O}(U)$ preserves the isotropic vectors of U. In rank 2 over $\mathbb{Z}$, the isotropic primitive vectors are exactly $\pm e, \pm f$, so ϕ permutes these four vectors. The permutations preserving the bilinear form give the four-element group above (Conway–Sloane 1999 §4.1). □

Remark 1.5 *(Standard sign convention versus BKM convention).* The BKM convention uses $\Lambda^{(1,1)} = U(-1)$ in the cusp polarization of §1.4 and in the exterior-square model of the automorphic group, where the time-like cone of $\Lambda^{2,1}$ has negative square. The map $U \to U(-1)$, $(e, f) \mapsto (e, f)$ is a bijection of underlying $\mathbb{Z}$-modules; only the bilinear form flips sign. The lattices $\Lambda^{2,1}$ and $\Lambda^{3,2}$ are presented in BKM coordinates; even-unimodular ambient lattices $(\Lambda_{K3}, \widetilde{H}(S, \mathbb{Z}))$ are presented in the standard Conway–Sloane convention.

1.2 The Lorentzian root lattice $\Lambda^{2,1}_{II}$

The Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$ of [45, Theorem 2.1, Section 5.1.1, case $(t = 1, II, \bar{0})$] is graded by an integral lattice of signature $(2, 1)$. This is the lattice on which the type-II simple-root Cartan matrix is $2(2\mathbf{I}_3 - \mathbf{J}_3)$ where $\mathbf{J}_3$ is the all-ones matrix.

Definition 1.6 *(Ambient $\Lambda^{2,1} = U(-1) \oplus \langle 2 \rangle$).* [definitional] Set

$$\Lambda^{2,1} := U(-1) \oplus \langle 2 \rangle = \mathbb{Z}e \oplus \mathbb{Z}f \oplus \mathbb{Z}h, \qquad e \cdot f = -1,\ e^2 = f^2 = 0,\ h^2 = 2.$$

The Gram matrix in the basis (e, f, h) is

$$G_{2,1} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad \det G_{2,1} = -2, \qquad \operatorname{disc} \Lambda^{2,1} = -2.$$

The BKM lattice $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2]$ in (f_2, f_3, f_{-2})-coordinates uses exactly this Gram, with $\Lambda^{(1,1)} = U(-1)$ on the (f_2, f_{-2})-block and $[2] = \langle 2 \rangle$ on the f_3-axis.

Definition 1.7 *($\Lambda^{2,1}_{II}$ as the index-four type-II sublattice of $\Lambda^{2,1}$).* [definitional] The type-II sublattice

$$\Lambda^{2,1}_{II} \ := \ \{ m f_2 + l f_3 + n f_{-2} \in \Lambda^{2,1} \mid m \equiv 0\,(2),\ n \equiv 0\,(2) \} \ \subset \ \Lambda^{2,1}$$

is the index-4 sublattice on which the type-II Cartan matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$ is the Gram of the simple roots $(\delta_1, \delta_2, \delta_3)$ of Proposition 1.10. Abstractly it is the lattice

$$\Lambda^{2,1}_{II} \ \simeq \ U(4) \oplus \langle 2 \rangle, \qquad \operatorname{disc} \Lambda^{2,1}_{II} = -32,$$

matching the terminology of Chapter 1 and the type-II sublattice convention of Chapter 4.

VERIFICATION 1.8 *(Rank, signature, discriminant of $\Lambda^{2,1}_{II}$).* [computed] $\operatorname{rk} \Lambda^{2,1}_{II} = 3$. The Gram of $\Lambda^{2,1}_{II}$ in $(\delta_1, \delta_2, \delta_3)$ of Proposition 1.10 is the Cartan matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$ of equation (4.6); its determinant is

$$\det\big(2(2\mathbf{I}_3 - \mathbf{J}_3)\big) = -32.$$

The hyperbolic-plus-rank-one block of the ambient $\Lambda^{2,1}$ has signature $(2, 1)$; the index-4 sublattice $\Lambda^{2,1}_{II}$ inherits this signature, so

$$\sigma(\Lambda^{2,1}_{II}) = (2, 1).$$

The discriminant group $(\Lambda^{2,1}_{II})^* / \Lambda^{2,1}_{II}$ has order 32 as required. The lattice $\Lambda^{2,1}_{II}$ is even because every $\delta_i^2 = 2$ and the off-diagonal couplings $(\delta_i, \delta_j) = -2$ are integral. Changing the sign of one hyperbolic basis vector gives the standard hyperbolic-plane convention $U = ((0,1),(1,0))$ and presents $\Lambda^{2,1} \simeq U \oplus \langle 2\rangle$ and $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2\rangle$; rank, signature, discriminant group order, automorphism group, and Cartan datum are unchanged.

Remark 1.9 *(Identification with $\Lambda^{2,1}$).* The BKM convention uses $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$ with Gram in (f_2, f_3, f_{-2})

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

matching $G_{2,1}$ of Definition 1.6 up to the basis swap $e \mapsto f_2, f \mapsto f_{-2}, h \mapsto f_3$. The type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$ of Definition 1.7 is the index-4 sublattice on which the simple roots $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$ of Proposition 1.10 live; it is the even Lorentzian root lattice of $\mathfrak{g}_{\Delta_5}$.

Proposition 1.10 *(Type-II simple-root Cartan).* [proved] In the coordinates (f_2, f_3, f_{-2}) fixed by the Weyl chamber, set

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Then $\delta_i^2 = 2$ for $i = 1, 2, 3$ and

$$\big((\delta_i, \delta_j)\big)_{i,j=1,2,3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix} = 2(2\mathbf{I}_3 - \mathbf{J}_3),$$

where $\mathbf{J}_3$ is the 3×3 all-ones matrix. Each δ_i is type-II in the sense that $(\delta_i, \Lambda^{2,1}) = 2\mathbb{Z}$.

Proof. Direct computation in the BKM Gram gives the matrix $2(2\mathbf{I}_3 - \mathbf{J}_3)$, equal to the displayed Cartan matrix. The type-II divisibility statement follows because each δ_i pairs against $\Lambda^{2,1}$ with even values: $(\delta_i, f_2), (\delta_i, f_3), (\delta_i, f_{-2}) \in 2\mathbb{Z}$ for each i. This is the GN root datum of [45, Section 5.1.1, case $(1, II, \overline{0})$]. $\square$

Proposition 1.11 *(Weyl vector ρ).* [proved] The Weyl vector

$$\rho := \tfrac{1}{2}(\delta_1 + \delta_2 + \delta_3) = f_2 - \tfrac{1}{2}f_3 + f_{-2} \in \Lambda^{2,1} \otimes \mathbb{Q}$$

satisfies $(\rho, \delta_i) = -1$ for $i = 1, 2, 3$ and

$$(\rho, \rho) = -\tfrac{3}{2}, \qquad \rho \in C(\Lambda^{2,1})_+.$$

In particular ρ lies in the time-like cone, so the chamber $\mathcal{P}_{II} = \{x \mid (x, \delta_i) \le 0,\ i = 1, 2, 3\}$ has finite hyperbolic volume.

Proof. The simple-root condition $(\rho, \delta_i) = -(\delta_i^2)/2 = -1$ follows from Kac [54, §2.5, Lemma 2.5] for any choice of simple roots of square 2; for the GN type-II datum see [45, (2.7)]. The rational computation gives $\rho = (1, -1/2, 1)$ in (f_2, f_3, f_{-2}) coordinates and $(\rho, \delta_i) = -1$. Direct computation in the BKM Gram of Definition 1.7 gives

$$(\rho, \rho) = 2(f_2, f_{-2}) + \tfrac{1}{4}(f_3, f_3) = 2(-1) + \tfrac{1}{4}(2) = -\tfrac{3}{2}.$$

Since $(\rho, \rho) < 0$, $\rho \in C(\Lambda^{2,1})_+$. $\square$

PROPOSITION 1.12 *(Reflection group; Nikulin's theorem).* [proved; Nikulin 1980 Theorem 1.4.1] Let $W^{(2)}(\Lambda^{2,1})$ be the subgroup of $\mathbf{O}(\Lambda^{2,1})$ generated by reflections in primitive roots of square 2. Then

$$\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1}),$$

where the subscript + denotes the index-two subgroup preserving the time-like cone $C(\Lambda^{2,1})_+$. Equivalently, every isometry of $\Lambda^{2,1}$ preserving the chosen positive cone is a product of square-2 reflections.

Proof. Nikulin [82, Theorem 1.4.1] establishes for the rank-three hyperbolic lattice $\Lambda^{(1,1)} \oplus [2]$ the equality $\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1})$; see also Gritsenko–Nikulin [43, Proposition 1.5]. □

VERIFICATION 1.13 *(Type-II reflection orbit).* [computed] For each $\delta \in \{\delta_1, \delta_2, \delta_3\}$, the reflection $s_\delta : x \mapsto x - (x, \delta)\delta$ satisfies $s_\delta(\delta) = -\delta$ and preserves the Cartan matrix on the δ-orbit. Concretely, $s_{\delta_1}(\delta_1) = -\delta_1$, $s_{\delta_1}(\delta_2) = \delta_2 + 2\delta_1$, etc.; the bilinear form on $\{\delta_1, \delta_2, \delta_3\}$ is reflection-invariant by direct matrix computation.

Remark 1.14 *(The two type-2 divisibility classes).* Each $\delta \in \Lambda^{2,1}$ with $\delta^2 = 2$ has divisibility $(\delta, \Lambda^{2,1}) \in \{\mathbb{Z}, 2\mathbb{Z}\}$. The two cases are recorded as *type I* ($(\delta, \Lambda^{2,1}) = \mathbb{Z}$) and *type II* ($(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$); both classes generate index-finite normal subgroups of $W^{(2)}(\Lambda^{2,1})$:

$$[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_I)] = 2, \qquad [W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda^{2,1}_{II})] = 6,$$

following [43, Sections 4–5]. The denominator algebra $\mathfrak{g}_{\Delta_5}$ uses the type-II reflection chamber.

1.3 The K3 lattice $\Lambda_{K3} = \mathrm{II}_{3,19}$

For S a complex K3 surface, the integral cohomology $H^2(S, \mathbb{Z})$ with cup-product pairing is the unique even unimodular indefinite lattice of signature $(3, 19)$ (Conway–Sloane 1999 §16.4; Nikulin 1980 Theorem 1.1.4).

Definition 1.15 *($\Lambda_{K3} = \mathrm{II}_{3,19}$).* [definitional] The K3 cohomology lattice is

$$\Lambda_{K3} := H^2(S, \mathbb{Z}) \simeq -E_8 \oplus -E_8 \oplus U \oplus U \oplus U \equiv -2E_8 \oplus 3U,$$

where E_8 is the rank-8 root lattice with the standard positive- definite Cartan pairing (Conway–Sloane 1999 §4.8) and $-E_8$ denotes E_8 with bilinear form negated.

VERIFICATION 1.16 *(Rank, signature, discriminant of Λ_{K3}).* [computed]

$$\operatorname{rk} \Lambda_{K3} = 8 + 8 + 2 + 2 + 2 = 22, \qquad \sigma(\Lambda_{K3}) = (3, 19).$$

The three U-summands contribute signature $(3, 3)$ and the two $-E_8$-summands contribute signature $(0, 16)$; total $(3, 19)$. The discriminant is

$$\operatorname{disc} \Lambda_{K3} = (\det -E_8)^2 \cdot (\det U)^3 = 1 \cdot (-1)^3 = -1,$$

so $|\operatorname{disc}| = 1$ and the lattice is unimodular (Conway–Sloane 1999 §16.4, Theorem 16.4.1). Even unimodularity follows because each summand is even unimodular.

Proposition 1.17 *(Aut and Hodge data on Λ_{K3}).* [proved] The orthogonal group $\mathbf{O}(\Lambda_{K3})$ acts transitively on the set of primitive vectors of any fixed square (Eichler 1952; Nikulin [81, Theorem 1.14.4]). For a polarized K3 with period $\omega \in H^{2,0}(S) \subset \Lambda_{K3} \otimes \mathbb{C}$, the transcendental and algebraic sublattices fit into the orthogonal splitting

$$\Lambda_{K3} = \mathrm{NS}(S) \oplus T(S),$$

where $\mathrm{NS}(S) \subset H^{1,1}(S, \mathbb{Z})$ is the Néron–Severi (algebraic) sublattice of signature $(1, \rho - 1)$ and $T(S)$ is the transcendental lattice of signature $(2, 20 - \rho)$, with $\rho = \mathrm{rk}\,\mathrm{NS}(S)$ the Picard rank (Beauville–Bourguignon–Demazure 1985 §6; [81, §2]).

Proof. The transitivity statement is Eichler's theorem on indefinite unimodular lattices, sharpened by Nikulin [81, Theorem 1.14.4]: for an even unimodular indefinite lattice L of rank ≥ 3, $\mathbf{O}(L)$ acts transitively on primitive vectors of fixed square in each squared-class. Decomposition by Hodge type partitions $\Lambda_{K3} \otimes \mathbb{C}$ into $H^{2,0} \oplus H^{1,1} \oplus H^{0,2}$; intersecting with Λ_{K3} gives the transcendental–algebraic splitting at the integral level. □

1.4 The lattice-polarized K3 sublattice Λ_{pol}

A lattice polarization is a primitive embedding of an integral lattice Λ_{pol} into Λ_{K3} whose orthogonal complement carries period structure.

Definition 1.18 *(Lattice polarization $\Lambda_{\mathrm{pol}} \hookrightarrow \Lambda_{K3}$).* [definitional] A *lattice polarization* of signature $(1, \rho - 1)$ on a K3 surface S is a primitive embedding

$$\Lambda_{\mathrm{pol}} \hookrightarrow \Lambda_{K3} = \mathrm{II}_{3,19}, \qquad \sigma(\Lambda_{\mathrm{pol}}) = (1, \rho - 1), \quad 1 \leq \rho \leq 19,$$

where $\Lambda_{\mathrm{pol}} \subset H^{1,1}(S, \mathbb{Z})$ contains an ample class. The orthogonal complement

$$\Lambda_{\mathrm{pol}}^{\perp} \subset \Lambda_{K3}, \qquad \sigma(\Lambda_{\mathrm{pol}}^{\perp}) = (2, 20 - \rho),$$

is the lattice on which the period map of the family $\mathcal{F}_{\Lambda_{\mathrm{pol}}} \to \mathcal{D}_{\Lambda_{\mathrm{pol}}^{\perp}}$ is defined (Dolgachev 1996; Nikulin 1980 Proposition 1.6.1).

Verification 1.19 *(Numerics of Λ_{pol} and $\Lambda_{\mathrm{pol}}^{\perp}$).* [computed]

$$\mathrm{rk}\,\Lambda_{\mathrm{pol}} = \rho, \quad \sigma(\Lambda_{\mathrm{pol}}) = (1, \rho - 1), \quad \mathrm{rk}\,\Lambda_{\mathrm{pol}}^{\perp} = 22 - \rho, \quad \sigma(\Lambda_{\mathrm{pol}}^{\perp}) = (2, 20 - \rho).$$

Since Λ_{K3} is unimodular and $\Lambda_{\mathrm{pol}} \subset \Lambda_{K3}$ is primitive, Nikulin's gluing theorem gives an anti-isometry of finite quadratic forms

$$(A_{\Lambda_{\mathrm{pol}}}, q_{\Lambda_{\mathrm{pol}}}) \simeq (A_{\Lambda_{\mathrm{pol}}^{\perp}}, -q_{\Lambda_{\mathrm{pol}}^{\perp}}),$$

hence

$$|A_{\Lambda_{\mathrm{pol}}}| = |A_{\Lambda_{\mathrm{pol}}^{\perp}}|, \qquad |\det \Lambda_{\mathrm{pol}}| = |\det \Lambda_{\mathrm{pol}}^{\perp}|.$$

The quotient $\Lambda_{K3}/(\Lambda_{\mathrm{pol}} \oplus \Lambda_{\mathrm{pol}}^{\perp})$ is the isotropic gluing graph inside $A_{\Lambda_{\mathrm{pol}}} \oplus A_{\Lambda_{\mathrm{pol}}^{\perp}}$; there is no quotient $\Lambda_{\mathrm{pol}}^{\perp}/\Lambda_{\mathrm{pol}}^{*}$.

Proposition 1.20 *(Nikulin embedding criterion).* [proved; Nikulin 1980 Theorem 1.14.4] A primitive embedding $\Lambda_{\mathrm{pol}} \hookrightarrow \Lambda_{K3}$ exists if and only if there is an isomorphism

$$(A_{\Lambda_{\mathrm{pol}}}, q_{\Lambda_{\mathrm{pol}}}) \simeq (A_{\Lambda_{\mathrm{pol}}^{\perp}}, -q_{\Lambda_{\mathrm{pol}}^{\perp}})$$

of finite quadratic forms, where $A_L := L^*/L$ is the discriminant group and q_L its quadratic form (Nikulin 1980 Definition 1.3.1 and Theorem 1.14.4; Conway–Sloane 1999 §15.5). The embedding is unique up to $\mathbf{O}(\Lambda_{K3})$ when

$$\operatorname{rk}\Lambda_{\mathrm{pol}} + \ell(A_{\Lambda_{\mathrm{pol}}}) \le 22 - \operatorname{rk}\Lambda_{\mathrm{pol}}$$

and the discriminant form satisfies the Nikulin uniqueness criterion (Nikulin 1980 Theorem 1.14.4).

Proof. This is the gluing-via-discriminant-form theorem ([81, §1.14]; [20, §15.5]). An embedding into Λ_{K3} exists iff the discriminant form on $\Lambda_{\mathrm{pol}}^*/\Lambda_{\mathrm{pol}}$ is realized by the orthogonal complement; uniqueness is the Nikulin orbit-uniqueness statement under the corank inequality. □

Remark 1.21 *(Mirror lattice convention).* Choose a primitive hyperbolic plane $U_{\mathrm{mir}} \subset \Lambda_{\mathrm{pol}}^\perp$. The Aspinwall–Morrison–Dolgachev mirror to a Λ_{pol}-polarized K3 family is polarized by the orthogonal complement

$$\Lambda_{\mathrm{pol}}^\perp = U_{\mathrm{mir}} \oplus \check{\Lambda}_{\mathrm{pol}}, \qquad \check{\Lambda}_{\mathrm{pol}} = U_{\mathrm{mir}}^\perp \cap \Lambda_{\mathrm{pol}}^\perp \simeq \Lambda_{\mathrm{pol}}^\perp \ominus U_{\mathrm{mir}},$$

the splitting depending on the cusp $U_{\mathrm{mir}} \hookrightarrow \Lambda_{\mathrm{pol}}^\perp$ (Dolgachev 1996 §5; Aspinwall 1997). Thus $\operatorname{rk}\check{\Lambda}_{\mathrm{pol}} = 20 - \operatorname{rk}\Lambda_{\mathrm{pol}}$. For $\Lambda_{\mathrm{pol}} = U \oplus \langle -2\rangle$ one obtains $\check{\Lambda}_{\mathrm{pol}} \simeq U \oplus E_7(-1) \oplus E_8(-1)$, of signature $(1, 16)$. The hyperbolic plane U_{mir} re-enters only in the Mukai/type-IV comparison, where the period Hodge line is compared after adjoining the standard Mukai U-summand. The assertion that Δ_5^{-2} is the corresponding mirror section of the inverse Hodge line with reduced type-II Heegner support (View 4) remains conjectural and is independent of Chapter 6.

1.5 The Mukai lattice $\widetilde{H}(S,\mathbb{Z}) = \mathrm{II}_{4,20}$

For S a complex K3 surface, the full integral cohomology $H^*(S,\mathbb{Z})$ with the Mukai pairing is an even unimodular lattice of signature $(4, 20)$ (Mukai 1984 Theorem 1.5; Mukai 1987).

Definition 1.22 *(Mukai lattice $\widetilde{H}(S,\mathbb{Z})$).* [definitional] The Mukai lattice of S is

$$\widetilde{H}(S,\mathbb{Z}) := H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z}), \qquad \widetilde{H}(S,\mathbb{Z}) \simeq U \oplus U \oplus U \oplus U \oplus -2E_8 \equiv 4U \oplus 2(-E_8),$$

graded by even cohomological degree. The Mukai pairing is

$$\langle (r, D, s), (r', D', s')\rangle_{\mathrm{Muk}} := D \cdot D' - rs' - r's$$

for $(r, D, s), (r', D', s') \in \mathbb{Z} \oplus H^2(S,\mathbb{Z}) \oplus \mathbb{Z}$ (Mukai 1984 Theorem 1.5). The pairing of an H^0-class with an H^4-class carries a minus sign relative to the cup product, so that the rank-4 component of $H^*(S,\mathbb{Z}) \otimes H^*(S,\mathbb{Z})$ contributes $-rs' - r's$.

Verification 1.23 *(Rank, signature, discriminant of $\widetilde{H}(S,\mathbb{Z})$).* [computed]

$$\operatorname{rk}\widetilde{H}(S,\mathbb{Z}) = 1 + 22 + 1 = 24, \qquad \sigma(\widetilde{H}(S,\mathbb{Z})) = (4, 20).$$

The four U-summands give signature $(4, 4)$ and the $-2E_8$ pieces give signature $(0, 16)$, total $(4, 20)$. Discriminant

$$\operatorname{disc}\widetilde{H}(S,\mathbb{Z}) = (\det U)^4 \cdot (\det -E_8)^2 = (-1)^4 \cdot 1 = 1,$$

so $\widetilde{H}(S,\mathbb{Z})$ is even unimodular, denoted $\mathrm{II}_{4,20}$ (Conway–Sloane 1999 §16.4). The lattice-polarization symbol is Λ_{pol}; the full Mukai lattice is written $\widetilde{H}(S,\mathbb{Z})$.

Proposition 1.24 *(Mukai vector).* [proved; Mukai 1987] For F a coherent sheaf on S, the Mukai vector is

$$v(F) := \mathrm{ch}(F)\sqrt{\mathrm{td}(S)} = (\mathrm{rk}\,F,\ c_1(F),\ \mathrm{ch}_2(F) + \mathrm{rk}\,F \cdot [\mathrm{pt}]) \in \widetilde{H}(S,\mathbb{Z}),$$

where $\sqrt{\mathrm{td}(S)} = (1,0,1)$ in Mukai notation on a K3 surface ([78, Theorem 3]). The Mukai pairing on $\widetilde{H}(S,\mathbb{Z})$ recovers the Euler characteristic of Hom:

$$\langle v(F), v(G)\rangle_{\mathrm{Muk}} = -\chi(F,G) = -\sum_i (-1)^i \dim \mathrm{Ext}^i(F,G).$$

Proof. Mukai [78, Theorem 3] computes $\sqrt{\mathrm{td}(S)}$ for K3 surfaces. The Hirzebruch–Riemann–Roch formula gives $\chi(F,G) = \int_S \mathrm{ch}(F^\vee)\mathrm{ch}(G)\,\mathrm{td}(S) = -\langle v(F), v(G)\rangle_{\mathrm{Muk}}$ once the sign of the H^0–H^4 pairing is taken into account; see Mukai 1984 §2. □

Remark 1.25 *(Sign of the Mukai pairing on rank-0 components).* The Mukai pairing $\langle (r,D,s),(r',D',s')\rangle_{\mathrm{Muk}} = D\cdot D' - rs' - r's$ restricts to the cup-product pairing on the middle H^2-component

$$\langle (0,D,0),(0,D',0)\rangle_{\mathrm{Muk}} = D\cdot D',$$

and to the negative cup-product on the rank–Euler component

$$\langle (r,0,s),(r',0,s')\rangle_{\mathrm{Muk}} = -rs' - r's.$$

The relative minus sign is forced by Hirzebruch–Riemann–Roch [47, Theorem 5.1.1]: $-\chi(F,G)$ contains a $-\,\mathrm{rk}\,F\cdot\chi(\mathcal{O}_S,G) - \mathrm{rk}\,G\cdot\chi(\mathcal{O}_S,F)$ term coming from $\sqrt{\mathrm{td}(S)} = (1,0,1)$. Negating the H^0–H^4 pairing or dropping the minus sign converts the Mukai lattice to an isotropic-rank-2 unimodular form not of signature $(4,20)$, in disagreement with the Mukai 1984 classification ([77, Theorem 1.5]) and the unique even unimodular type $\mathrm{II}_{4,20}$. The sign as stated is forced by these requirements.

Proposition 1.26 *(Aut of the Mukai lattice).* [proved] $\mathbf{O}(\widetilde{H}(S,\mathbb{Z}))$ acts transitively on the set of primitive vectors of any fixed square (Eichler 1952; Nikulin 1980 Theorem 1.14.4). In particular, the Mukai vector $v = (1,0,1)$ (corresponding to the structure sheaf $\mathcal{O}_S$) and the generator $(0,0,1)$ of $H^4(S,\mathbb{Z}) \cap \widetilde{H}(S,\mathbb{Z})$ are $\mathbf{O}(\widetilde{H}(S,\mathbb{Z}))$-conjugate to any other primitive vector of the same square. The derived autoequivalence group of $D^b(S)$ maps to $\mathbf{O}(\widetilde{H}(S,\mathbb{Z}))$ via the action on Mukai vectors (Mukai 1987 §2; Orlov 1997).

Proof. Eichler's theorem ([81, Theorem 1.14.4]) gives the transitivity statement. Mukai [78, §2] constructs the representation of derived autoequivalences on $\widetilde{H}(S,\mathbb{Z})$ via Fourier–Mukai kernels. □

1.6 The formal dyonic charge lattice $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z})$

The formal additive dyonic lattice on $X = S \times E$ encodes the electric–magnetic charge data on which the Gram-shadow projection Π_X is defined. Its two factors are full Mukai lattices, not lattice-polarization sublattices of $H^2(S,\mathbb{Z})$.

Definition 1.27 *(Γ_X^{form} and its Gram pairing).* [definitional; from Definition 1.3] For $X = S \times E$ with S a K3 surface and E an elliptic curve,

$$\Gamma_X^{\mathrm{form}} := \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z}), \qquad c = (Q,P) \in \Gamma_X^{\mathrm{form}}, \quad Q,P \in \widetilde{H}(S,\mathbb{Z}).$$

The Gram pairing $\langle\ ,\ \rangle_\Gamma : \Gamma_X^{\mathrm{form}} \times \Gamma_X^{\mathrm{form}} \to \mathbb{Z}$ is inherited from the Mukai pairing on each factor:

$$\langle (Q,P),(Q',P')\rangle_\Gamma := \langle Q,Q'\rangle_{\mathrm{Muk}} + \langle P,P'\rangle_{\mathrm{Muk}}.$$

The Gram-index projection

$$\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}} = \mathbb{Z}^3, \qquad (Q,P) \mapsto \left(\tfrac{Q^2}{2},\ Q\cdot P,\ \tfrac{P^2}{2}\right),$$

records the genus-two Gram triple of the dyonic class. This map is quadratic, not additive: the obstruction is the symmetric bilinear 2-cocycle B of Lemma 2.4.

Verification 1.28 *(Rank, signature of* Γ_X^{form}*).* [computed]

$$\mathrm{rk}\,\Gamma_X^{\mathrm{form}} = 2\cdot \mathrm{rk}\,\widetilde{H}(S,\mathbb{Z}) = 48, \qquad \sigma(\Gamma_X^{\mathrm{form}}) = 2\cdot(4,20) = (8,40).$$

The discriminant is

$$\mathrm{disc}\,\Gamma_X^{\mathrm{form}} = (\mathrm{disc}\,\widetilde{H}(S,\mathbb{Z}))^2 = 1,$$

so Γ_X^{form} is even unimodular. In Conway–Sloane notation it is $\mathrm{II}_{8,40}$ (Conway–Sloane 1999 §16.4).

Remark 1.29 *(Relation to the K3×E Mukai geometry).* Under the Künneth decomposition for $X = S\times E$, the integral even cohomology decomposes as

$$H^{\mathrm{ev}}(X,\mathbb{Z}) = H^*(S,\mathbb{Z})\otimes H^{\mathrm{even}}(E,\mathbb{Z}) = H^*(S,\mathbb{Z})\otimes(\mathbb{Z}[1]\oplus\mathbb{Z}[\omega_E]),$$

where $\omega_E \in H^2(E,\mathbb{Z})$ is the cohomological point class with $\int_E \omega_E = 1$. Every Mukai vector on X in the formal even sector decomposes as $v_X = P\otimes 1_E + Q\otimes\omega_E$ with $Q,P\in\widetilde{H}(S,\mathbb{Z})$, and $\Gamma_X^{\mathrm{form}} = \widetilde{H}(S,\mathbb{Z})\oplus\widetilde{H}(S,\mathbb{Z})$ is the additive lattice on which the (P,Q)-grading lives (Definition 1.5). The tensor Künneth pairing on the formal even sector pairs against the elliptic U-summand $H^0(E,\mathbb{Z})\oplus H^2(E,\mathbb{Z})\simeq U$, giving

$$\langle v_X, v'_X\rangle_{\mathrm{Kun}} = Q\cdot P' + Q'\cdot P,$$

where each summand is the Mukai pairing on $\widetilde{H}(S,\mathbb{Z})$. The direct-sum Gram pairing on Γ_X^{form} keeps Q^2, $Q\cdot P$, and P^2 separately; the resulting genus-two Gram triple $(Q^2/2, Q\cdot P, P^2/2)$ is the Gram-index data $\Pi_X(Q,P)$ used in the Borcherds product (Theorem 3.9).

Lemma 1.30 *(Two orthogonal hyperbolic planes inside* $\widetilde{H}(S,\mathbb{Z})$*).* [proved] $\widetilde{H}(S,\mathbb{Z})\simeq 4U\oplus 2(-E_8)$ contains four mutually orthogonal copies of U. In particular, two of them give

$$U_1\oplus U_2 \subset \widetilde{H}(S,\mathbb{Z}), \qquad U_i\simeq U,\ U_1\perp U_2,$$

which is the hypothesis of the formal primitive lift Lemma 2.8: every Gram triple $(n,l,m)\in\mathbb{Z}^3$ has a primitive saturated formal lift $c=(Q,P)\in\Gamma_X^{\mathrm{form}}$ with $\Pi_X(Q,P)=(n,l,m)$.

Proof. By Definition 1.3, $\widetilde{H}(S,\mathbb{Z})\simeq 4U\oplus 2(-E_8)$ contains four mutually orthogonal U-summands. Selecting any two of them gives the displayed sublattice, which is itself isomorphic to $U\oplus U\simeq \mathrm{II}_{2,2}$ as a sublattice of $\widetilde{H}(S,\mathbb{Z})$. The Gram-triple lift then follows from Lemma 2.8: take $Q = e_1 + nf_1$, $P = lf_1 + e_2 + mf_2$ in $U_1\oplus U_2$ to obtain $\Pi_X(Q,P) = (n,l,m)$. □

1.7 *The signature-*$(3,2)$ *lattice* $\Lambda^{3,2}$ *and the* $\mathbf{PSp}_4 \simeq \mathbf{SO}(3,2)_+$ *bridge*

The exceptional isomorphism $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}_+(3,2)$ identifies the Siegel domain with the type-IV orthogonal domain of an integral lattice $\Lambda^{3,2}$ of signature $(3,2)$, and it identifies $\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}\}$ with the exterior-square arithmetic group $\Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+$ ([43, Section 3]; Borcherds 1995 §5). The Igusa cusp form Δ_5 lives on the Siegel side; the Borcherds product expansion lives on the orthogonal side.

Definition 1.31 *($\Lambda^{3,2}$ in the exterior-square model).* [definitional] Let $\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4$ and consider the exterior square $\wedge^2\Lambda^4$ with the symmetric pairing

$$u \wedge v \cdot u' \wedge v' = [\det(u,v,u',v')]_{e_1\wedge e_2\wedge e_3\wedge e_4} \in \mathbb{Z}.$$

Then $(\wedge^2\Lambda^4, (\ ,\ )) \simeq \mathrm{II}_{3,3}$ as an even unimodular lattice of signature $(3,3)$. Fix the primitive vector $q_0 := e_1 \wedge e_3 + e_2 \wedge e_4 \in \wedge^2\Lambda^4$, for which $q_0^2 = -2$, and define

$$\Lambda^{3,2} := q_0^{\perp} \subset \wedge^2\Lambda^4.$$

Verification 1.32 *(Rank, signature, discriminant of* $\Lambda^{3,2}$*).* [computed] $\operatorname{rk}\wedge^2\Lambda^4 = \binom{4}{2} = 6$. Taking the orthogonal complement of the primitive vector q_0 gives $\operatorname{rk}\Lambda^{3,2} = 5$. The explicit Gram matrix of $\Lambda^{3,2}$ in the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ is

$$G_{3,2} = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where $f_1 = e_1 \wedge e_2$, $f_2 = e_2 \wedge e_3$, $f_3 = e_1 \wedge e_3 - e_2 \wedge e_4$, $f_{-2} = e_4 \wedge e_1$, $f_{-1} = e_4 \wedge e_3$. Direct computation: the matrix is antidiagonal except for the central $h^2 = 2$ entry, with off-diagonal entries $-1, -1, -1, -1$. Expansion gives

$$\det G_{3,2} = 2 \cdot (-1)^4 \cdot \operatorname{sgn}(1,2,3,4 \mapsto 4,3,2,1) = 2 \cdot 1 \cdot (+1) = +2,$$

since the permutation $(1,2,3,4) \mapsto (4,3,2,1)$ has six inversions and even parity. Therefore

$$\sigma(\Lambda^{3,2}) = (3,2), \qquad \operatorname{disc}\Lambda^{3,2} = 2.$$

This is a rational computation in the exterior-square lattice.

Proposition 1.33 *($\Lambda^{3,2} \simeq U \oplus U \oplus \langle 2\rangle$ in the fixed convention; $\Lambda^{3,2} \simeq U(-1) \oplus U(-1) \oplus \langle 2\rangle$ in the standard convention).* [proved] In the BKM convention of the exterior-square model,

$$\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2],$$

with $\Lambda^{(1,1)} = U(-1)$ and $[2]$ the rank-1 lattice with Gram (2). The Gram presentation displayed in Verification 1.32 arises from the f-basis: (f_1, f_{-1}) pairs as a copy of $\Lambda^{(1,1)}$ ($(f_1, f_{-1}) = -1$), (f_2, f_{-2}) pairs as a second copy of $\Lambda^{(1,1)}$ ($(f_2, f_{-2}) = -1$), and f_3 contributes $(f_3, f_3) = 2$.

Proof. The pairings $(f_1, f_{-1}) = -1$, $(f_2, f_{-2}) = -1$, $(f_3, f_3) = 2$ are read off from $G_{3,2}$. The orthogonal decomposition is direct: $\mathbb{Z}f_1 + \mathbb{Z}f_{-1} \perp \mathbb{Z}f_2 + \mathbb{Z}f_{-2} \perp \mathbb{Z}f_3$ inside $\Lambda^{3,2}$ because every cross-pair vanishes in $G_{3,2}$. The identification with $\Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ follows because each rank-2 block has Gram $((0,-1),(-1,0)) = \Lambda^{(1,1)}$ and the rank-1 block has Gram $(2) = [2]$. $\square$

Proposition 1.34 *(The projective exterior-square arithmetic group).* [proved; exterior-square model of $\mathbf{Sp}_4(\mathbb{Z})$] The exterior-square representation $\wedge^2$ induces an isomorphism

$$\wedge^2 : \mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\} \xrightarrow{\sim} \Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+, \qquad \Gamma_{3,2} := \wedge^2\mathbf{Sp}_4(\mathbb{Z})/\{\pm\mathbf{I}_4\}.$$

Concretely, the integral generators

$$g_0 = \begin{pmatrix} 0 & \mathbf{I}_2 \\ -\mathbf{I}_2 & 0 \end{pmatrix}, \quad g_M = \begin{pmatrix} \mathbf{I}_2 & M \\ 0 & \mathbf{I}_2 \end{pmatrix}, \quad g_A = \begin{pmatrix} {}^tA^{-1} & 0 \\ 0 & A \end{pmatrix} \quad (A \in \mathbf{GL}_2(\mathbb{Z}))$$

of $\mathbf{Sp}_4(\mathbb{Z})$ map under $\wedge^2$ to integral isometries of $\Lambda^{3,2}$, displayed as the three explicit 5×5 matrices $\wedge^2(g_0)$, $\wedge^2(g_M)$, $\wedge^2(g_A)$ in the exterior-square model. The image is the arithmetic group through which the Siegel modular group acts on the type-IV model. The Siegel space $\mathbb{H}_2$ embeds into the type-IV upper half-space $\mathbb{H}_+^{\mathbf{IV}} \subset \mathbb{P}(\Lambda^{3,2} \otimes \mathbb{C})$ via

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \mapsto {}^t(z_2^2 - z_1z_3,\ z_3,\ z_2,\ z_1,\ 1)z_0,$$

and the action of $\mathbf{Sp}_4(\mathbb{Z})$ on $\mathbb{H}_2$ corresponds to the action of $\Gamma_{3,2}$ on $\mathbb{H}_+^{\mathbf{IV}}$.

Proof. The Pfaffian pairing on $\wedge^2\Lambda^4$ is preserved by $\wedge^2\mathbf{SL}_4$. The subgroup fixing q_0 is $\mathbf{Sp}_4(\mathbb{Z})$ by definition of the symplectic form $B_{q_0}(x,y)e_1 \wedge e_2 \wedge e_3 \wedge e_4 = -x \wedge y \wedge q_0$. The kernel on $q_0^\perp = \Lambda^{3,2}$ is $\{\pm\mathbf{I}_4\}$. The exceptional Lie-algebra isomorphism $\mathfrak{sp}_4(\mathbb{R}) \simeq \mathfrak{so}(3,2)$ extends to the integral form on the displayed generators. The discriminant group of $\Lambda^{3,2}$ has one non-zero element; hence the stable and full projective components coincide in the Gritsenko–Nikulin normalization. This is the content of [43, Section 3] and the exterior-square model of the automorphic group. $\square$

Remark 1.35 *(Embedding $\Lambda_{II}^{2,1} \subset \Lambda^{3,2}$).* The primitive sublattice

$$\Lambda^{2,1} = \Lambda^{(1,1)} \oplus [2] = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}$$

of the Weyl-chamber section is the BKM root lattice: discarding the (f_1, f_{-1})-block of $\Lambda^{3,2} \simeq \Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$ leaves $\Lambda^{(1,1)} \oplus [2] \simeq \Lambda^{2,1}$. The type-II sublattice $\Lambda_{II}^{2,1} = \{mf_2 + lf_3 + nf_{-2} \mid m, n \equiv 0 \pmod 2\}$ is the lattice of even integral type-II classes, on which the simple-root Cartan

$$\big((\delta_i, \delta_j)\big) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}$$

of Proposition 1.10 is realized via $\delta_1 = 2f_2 - f_3$, $\delta_2 = 2f_{-2} - f_3$, $\delta_3 = f_3$. Every isometry of $\Lambda^{2,1}$ extends to an isometry of $\Lambda^{3,2}$ by the identity on $(\Lambda^{2,1})^\perp \subset \Lambda^{3,2}$.

1.8 Cross-references and consistency check

The lattices recorded above enter as follows.

- $U = \mathrm{II}_{1,1}$: every U-summand of $\Lambda^{3,2}$, Λ_{K3}, and $\widetilde{H}(S, \mathbb{Z})$. The relation $\Lambda^{(1,1)} = U(-1)$ records the BKM negated convention.

- $\Lambda^{2,1}_{II} \simeq U(4) \oplus \langle 2\rangle$ (with $U(4)$ written as $U(-4)$ in the BKM sign convention): the BKM root lattice on which the type-II simple-root Cartan $2(2\mathbf{I}_3 - \mathbf{J}_3)$ and the Weyl vector $\rho = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3)$ are realized. The denominator algebra $\mathfrak{g}_{\Delta_5} = \mathfrak{g}_{\Delta_5}$ is graded by $\Lambda^{2,1}_{II}$ ([45, Theorem 2.1]).
- $\Lambda_{K3} = \mathrm{II}_{3,19}$: K3 cohomology with intersection pairing. Polarized K3 sublattices Λ_{pol} embed primitively into Λ_{K3}.
- Λ_{pol}: signature $(1, \rho - 1)$ for $\rho \leq 19$ (the algebraic Picard rank). It is the lattice-polarization sublattice of Λ_{K3}, not the full Mukai lattice.
- $\widetilde{H}(S, \mathbb{Z}) = \mathrm{II}_{4,20}$: the Mukai lattice on which the D6–D2–D0 Mukai vectors $v_X(\mathcal{I}_Y)$ live; the Gram-index map Π_X is computed via this pairing (Theorem 3.9).
- $\Gamma^{\mathrm{form}}_X = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z})$: the formal additive dyonic charge lattice of $X = S \times E$, on which the quadratic Gram-shadow projection Π_X and the symmetric bilinear 2-cocycle B are defined (Definition 1.3).
- $\Lambda^{3,2} \simeq U(-1) \oplus U(-1) \oplus \langle 2\rangle$: the signature-$(3, 2)$ orthogonal lattice that bridges $\mathbf{PSp}_4(\mathbb{Z})$ to the type-IV orthogonal group via the exterior-square representation; on this lattice the Borcherds product expansion of Δ_5 is written.

Verification 1.36 *(End-to-end computation check).* [computed] The lattice data satisfy, in one self-contained $\mathbb{Q}$-arithmetic computation: (i) the rank-5 Gram matrix $G_{3,2}$ in the $(f_1, f_2, f_3, f_{-2}, f_{-1})$-basis; (ii) the orthogonality of the symplectic invariant $q_0 = e_1 \wedge e_3 + e_2 \wedge e_4$ to the basis; (iii) the type-II simple-root Cartan $2(2\mathbf{I}_3 - \mathbf{J}_3)$; (iv) the Weyl vector $\rho = (1, -1/2, 1)$ in (f_2, f_3, f_{-2}) coordinates and the $(\rho, \delta_i) = -1$ relation; (v) the type-II reflection action and Cartan invariance under each s_{δ_i}. Thus the Gram, Cartan, and Weyl-vector data of §1.2 and §1.7 agree in the common rational lattice.

Remark 1.37 *(Discriminant convention; sign of $\Lambda^{3,2}$).* In the BKM convention $\Lambda^{(1,1)} \oplus \Lambda^{(1,1)} \oplus [2]$, both hyperbolic-plane summands have Gram determinant -1 and the $[2]$ summand contributes $+2$, so $\operatorname{disc}\Lambda^{3,2} = (-1)^2 \cdot 2 = +2$. In the standard convention $U \oplus U \oplus \langle -2\rangle$, the two U-summands contribute determinant -1 each and $\langle -2\rangle$ contributes -2, so $\operatorname{disc}\Lambda^{3,2} = (-1)^2 \cdot (-2) = -2$. The two presentations differ by an overall sign on the rank-2 hyperbolic blocks; both carry an order-2 discriminant group with single nontrivial generator $f_3/2 \in (\Lambda^{3,2})^*/\Lambda^{3,2}$. The same analysis applies to the ambient $\Lambda^{2,1}$: discriminant -2 in the BKM convention, discriminant $+2$ in the standard convention, and discriminant group $\mathbb{Z}/2$ generated by $h/2$. The type-II sublattice $\Lambda^{2,1}_{II} \subset \Lambda^{2,1}$ of index four has $|\operatorname{disc}\Lambda^{2,1}_{II}| = 32$, with discriminant group $(\Lambda^{2,1}_{II})^*/\Lambda^{2,1}_{II}$ of order 32 (Verification 1.8); the type-II divisibility statement $(\delta, \Lambda^{2,1}) \in \{\mathbb{Z}, 2\mathbb{Z}\}$ depends on the order 2 of the ambient $\Lambda^{2,1}$ discriminant group, not on the sign of any determinant.

Remark 1.38 *(Cited primary literature).* The primary references behind every lattice claim above are: Conway–Sloane 1999 §2.5, §4.8, §15.5, §16.4 (classification of even unimodular indefinite lattices, E_8 root system, discriminant-form gluing, K3 lattice as $\mathrm{II}_{3,19}$); Nikulin 1980 Theorem 1.1.2, Theorem 1.4.1 (= [82] Theorem 1.4.1 in the K3-paper version), Theorem 1.6.1, Theorem 1.14.4 (signature classification, reflection group of square-2 vectors, primitive embeddings, discriminant-form uniqueness); Mukai 1984 Theorem 1.5 (Mukai pairing, signature $(4, 20)$), Mukai 1987 Theorem 3 ($\sqrt{\mathrm{td}(S)} = (1, 0, 1)$ on K3); Gritsenko–Nikulin 1998 Part II Theorem 2.1, Section 5.1.1, case $(t = 1, II, \overline{0})$ (BKM root datum for $\mathfrak{g}_{\Delta_5}$, Weyl vector, type-II simple-root Cartan).

2 Borcherds-product expansion: $\phi_{0,1}$, the Δ_5 infinite product, and the Gritsenko–Cléry eight-row catalogue

The Fourier expansion of the weight-zero index-one weak Jacobi form $\phi_{0,1}$, the Borcherds–Gritsenko–Nikulin product expansion of the Igusa cusp form Δ_5 in the type-II cusp chamber, the theta-constant normalization constant $64 = 2^6$, the doubling identity that turns the monic Borcherds product into the denominator of the Lorentzian Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$, and the full eight-row Gritsenko–Cléry catalogue of diagonal-divisor Siegel modular forms are the automorphic data at the type-II cusp. Every displayed numerical Fourier coefficient is produced by exact rational expansion of the theta quotient; every identity is attributed to author, year, and theorem of the primary literature. Orientation data, Pfaffian wall normal forms, and compact Hall input do not enter these automorphic identities; they live entirely on the automorphic / target side, prior to the geometric content of (D0), (O1), $(\mathrm{O1})^+$, and (O2).

Three names for Δ_5 appear and are kept distinct: the theta-constant automorphic section $\mathcal{D}_X = \Delta_5$ (View 1); the Borcherds–Kac–Moody denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ (View 2); and the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ of the pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ (View 3, after the retained D0-HN datum, retained quotient-orientation datum, chosen Weyl lifts, retained $(O2_{\mathrm{atlas}})$ wall datum, and finite Pfaffian section datum of Appendix 7). The first and third sections coincide as weight-five automorphic sections; the second coincides with their Borcherds denominator after the normalization $D_5 = 64^{-1}\Delta_5$ and the pullback $Z \mapsto 2Z$. The orientation character that distinguishes Δ_5 from $-\Delta_5$ enters only at View 3.

B.1 The weak Jacobi form $\phi_{0,1}$

The K3 elliptic genus is the modular input. Eichler–Zagier [32, §2.2 and Theorem 9.3] fix the weight-zero index-one weak-Jacobi normalization

$$\phi_{0,1}(\tau,z) \in J^{\mathrm{w}}_{0,1}, \qquad \phi_{0,1}(\tau,z) = \sum_{n\geq 0,\, l\in\mathbb{Z}} f(n,l)\, q^n r^l, \qquad q = e^{2\pi i\tau},\ r = e^{2\pi i z}.$$

The relation to the K3 right-moving elliptic genus is $Z_{\mathrm{K3}}(\tau,z) = 2\phi_{0,1}(\tau,z)$ [32, §2.2]; the half is the Borcherds-lift representative for the Borcherds product. The Jacobi weak-form support is $f(n,l) = 0$ whenever $4n - l^2 < -1$, with isolated extreme entries $f(0,\pm 1) = 1$ on the discriminant hyperbola $4n - l^2 = -1$. *(proved;* [32, §2.2, Theorem 9.3]*;* [45, Example 2.4]*.)*

2.0.0.1 Verified Fourier table. The exact coefficients $f(n,l)$ for $n \leq 2$ are the rational integers computed by the theta-quotient identity $\phi_{0,1} = 4\sum_{i=2,3,4} \theta_i(\tau,z)^2/\theta_i(\tau,0)^2$ of [32, §2.2] by rational arithmetic:

	$l=-3$	$l=-2$	$l=-1$	$l=0$	$l=1$	$l=2$	$l=3$
$n=0$	0	0	1	10	1	0	0
$n=1$	0	10	-64	108	-64	10	0
$n=2$	1	108	-513	808	-513	108	1

(computed; cross-checked with [45, Example 2.4] *and* [43, Theorem 4.1]*.)* The negative entries reflect the index-one Jacobi-form theta-decomposition $\phi_{0,1} = \theta_{1,0}h_0 + \theta_{1,1}h_1$ with theta-coefficient expansions $h_0 = 2(45E_4E_{4,1} - 50E_6E_{6,1})/\eta^{24} + \cdots$ in [32, §2.2]; the integer values are not target signs but signed multiplicities of the Borcherds source.

2.0.0.2 Fourier identities.

$$\begin{array}{lll} f(0,0)=10, & f(0,\pm1)=1, & f(0,l)=0\ \ (|l|\geq 2), \\ f(1,0)=108, & f(1,\pm1)=-64, & f(1,\pm2)=10, \\ f(2,0)=808, & f(2,\pm1)=-513, & f(2,\pm2)=108, \\ & f(2,\pm3)=1, & f(1,\pm3)=0. \end{array}$$

The diagonal entry $f(0,0)=10$ controls the Borcherds-lift weight $\mathrm{wt}(\Delta_5)=f(0,0)/2=5$; the entries $f(0,\pm1)=1$ control the simple type-II divisor orders $d^{\mathrm{aut}}_{\delta_i}=1$; the entry $f(1,1)=-64$ controls the multiplicity of the first timelike imaginary root, computed and used below.

B.2 The cusp chamber and the effective semigroup

Set $Z=\left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right)\in\mathbb{H}_2$, $q=e^{2\pi i z_1}$, $r=e^{2\pi i z_2}$, $s=e^{2\pi i z_3}$. The standard type-II cusp polarization is the ordered limit

$$0<|s|\ll|q|\ll|r|^{-1}\ll 1,$$

which determines the effective semigroup

$$\Gamma_{\mathrm{eff}}=\{(n,l,m)\in\mathbb{Z}^3\mid m>0,\ n\geq 0\}\cup\{(n,l,0)\mid n>0\}\cup\{(0,l,0)\mid l<0\}.$$

Inside the Gram-index lattice $\Gamma_{\mathrm{ind}}=\mathbb{Z}^3$, the *active support*

$$\Gamma_{\mathrm{act}}:=\{(n,l,m)\in\Gamma_{\mathrm{eff}}\mid f(nm,l)\neq 0\}$$

is the set of triples that contribute a visible product factor; the remaining triples in Γ_{eff} have zero exponent and fix ordering only. The signed Gram pairing $(\alpha(n,l,m),\alpha(n,l,m))=-2(4nm-l^2)$ under

$$\alpha:(n,l,m)\longmapsto 2nf_2-lf_3+2mf_{-2}\in\Lambda^{2,1}_{II}$$

restricts to the Lorentzian root lattice of $\mathfrak{g}_{\Delta_5}$.

B.3 The Borcherds product formula for Δ_5

Theorem 2.1 *(Borcherds–Gritsenko–Nikulin product expansion).* On the type-II cusp polarization $0<|s|\ll|q|\ll|r|^{-1}\ll 1$, the weight-five Igusa cusp form admits the absolutely convergent infinite product

$$\Delta_5(Z)=64\,q^{1/2}r^{1/2}s^{1/2}\prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}}(1-q^n r^l s^m)^{f(nm,l)},$$

with theta-leading coefficient

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5\ =\ 64\ =\ 2^6.$$

The monic product is $D_5:=64^{-1}\Delta_5$; the constant 64 is the theta-constant Igusa normalization, not a separately chosen scalar. *(proved;* [8, Theorem 10.1], [45, Theorem 2.1, Example 2.4, equations (2.15)–(2.16)], [43, Theorem 4.1].*)*

Proof. The general Borcherds singular theta-lift sends a weakly holomorphic vector-valued modular form of negative weight for the Weil representation of a lattice of signature $(2,n)$ to a meromorphic orthogonal automorphic form realized as an infinite product [8, Theorem 10.1]; the regularisation and convergence on divisors with singularities in higher rank is in [9, §§6–13]. The genus-two specialization at the Siegel cusp uses the exterior-square isogeny $\mathbf{PSp}_4(\mathbb{R}) \simeq \mathbf{SO}(3,2)_+$ and the Eichler–Zagier dictionary $J^{\mathrm{w}}_{0,1} \leftrightarrow$ vector-valued Mp_2-modular forms [32, §5]; the ordered limit $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ selects the type-II cusp [45, §2, Example 2.4]. Gritsenko–Nikulin [45, (2.15)–(2.16)] write the resulting product in the unnormalized form

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod_{(n,l,m)\in\Gamma_{\mathrm{eff}}} (1 - q^n r^l s^m)^{f(nm,l)},$$

and identify D_5 with 64^{-1} times the theta-constant cusp form Δ_5 of Igusa [43, Theorem 4.1], [48], [49]; the leading constant $2^6 = 64$ is the explicit theta-product coefficient. □

2.0.0.3 The constant 2^6 as a theta-product leading coefficient. Igusa's classical product formula displays Δ_5 as the product of the ten even theta constants of genus two,

$$\Delta_5(Z) = \prod_{\substack{a,b\in(\mathbb{Z}/2\mathbb{Z})^2 \\ {}^t ab\equiv 0 \pmod 2}} \nu_{a,b}(Z), \qquad \nu_{a,b}(Z) = \sum_{l\in\mathbb{Z}^2} \exp\bigl(\pi i (Z[l + \tfrac{1}{2}a] + {}^t bl)\bigr),$$

[48], [43, Theorem 4.1], [34, §6]. In the type-II cusp polarization, with z_2 the off-diagonal modulus, the four characteristics $a = (0,0)$ start with 1, the two characteristics $a = (1,0)$ start with $2q^{1/8}$, the two characteristics $a = (0,1)$ start with $2s^{1/8}$, and the two characteristics $a = (1,1)$ start with $2q^{1/8} r^{1/4} s^{1/8}$; the total leading monomial is $q^{1/2} r^{1/2} s^{1/2}$ with constant $1 \cdot 2^2 \cdot 2^2 \cdot 2^2 = 2^6 = 64$ *(proved;* [48], [43, Theorem 4.1].*)* This is not a normalization choice: the constant 64 is read off the theta product, with no scalar freedom. Replacing Δ_5 by $c\Delta_5$, $c \in \mathbb{C}^\times$, would change neither the Maass character ν_{Δ_5} nor the divisor orders $d^{\mathrm{aut}}_{\partial}$ of [43, Theorem 4.1].

B.4 Sample exponent verifications

The exponent $f(nm, l)$ in the product formula is read directly from the Jacobi-coefficient table of B.1. The following sample is verified exactly by the displayed coefficient table:

factor (n,l,m)	Lorentzian Gram label $\alpha = 2nf_2 - lf_3 + 2mf_{-2}$	exponent $f(nm,l)$
$(1,1,0)$	$\delta_1 = 2f_2 - f_3$	$f(0,1) = 1$
$(0,1,1)$	$\delta_2 = 2f_{-2} - f_3$	$f(0,1) = 1$
$(0,-1,0)$	$\delta_3 = f_3$	$f(0,-1) = 1$
$(0,0,m)$ $(m \geq 1)$	cusp ray $2mf_{-2}$	$f(0,0) = 10$
$(1,0,1)$	$2f_2 + 2f_{-2}$ (timelike, $(\alpha,\alpha) = -8$)	$f(1,0) = 108$
$(1,1,1)$	$2f_2 - f_3 + 2f_{-2}$ (timelike, $(\alpha,\alpha) = -6$)	$f(1,1) = -64$
$(1,2,1)$	$2f_2 - 2f_3 + 2f_{-2}$ (isotropic, $(\alpha,\alpha) = 0$)	$f(1,2) = 10$
$(1,3,1)$	$2f_2 - 3f_3 + 2f_{-2}$ (spacelike, $(\alpha,\alpha) = 10$)	$f(1,3) = 0$
$(2,1,1)$	$4f_2 - f_3 + 2f_{-2}$ ($(\alpha,\alpha) = -14$)	$f(2,1) = -513$
$(1,2,2)$	$2f_2 - 2f_3 + 4f_{-2}$ ($(\alpha,\alpha) = -12$)	$f(2,2) = 108$

The triple $(1,1,1)$ gives the first timelike imaginary root: its exponent $f(1,1) = -64$ is the signed superdimension of the root space $\mathfrak{g}_{\Delta_5,\alpha(1,1,1)}$ on the active support, and is the content of the lattice cross-check $f(1,1) = 29 - 93 = -64$ from the free-Lie/imaginary-simple split computed by the Gritsenko–Nikulin target presentation. The triple $(1,3,1)$ has exponent zero because $f(1,3) = 0$: this triple lies in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ and contributes ordering, not a visible factor. These are signed superdimensions, not positive dimensions; negative entries record an odd excess in the Borcherds source, and only after the (O1), (O1)$^+$, (O2) orientation data become available do they translate to operator parities.

2.0.0.4 Theta-leading coefficient. The coefficient 64 is the coefficient of the leading monomial of Δ_5:

$$\Delta_5(Z) = 64\, q^{1/2} r^{1/2} s^{1/2}\big(1 + \cdots\big).$$

After the leading monomial is divided out, the first qrs-coefficient of $D_5 = 64^{-1}\Delta_5$ is 93, not 64. The factor $(1 - qrs)^{-64}$ contributes $64\, qrs$, and the remaining root factors contribute the other $29\, qrs$; this is the identity

$$[qrs]\, D_5/(qrs)^{1/2} = 93, \qquad 29 - 93 = -64 = f(1,1).$$

(computed by exact rational expansion of the Borcherds product.)

B.5 The doubling identity $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$

Lemma 2.2 *(Two exponential normalizations).* The product chamber convention writes monomials as $q^n r^l s^m$ with $q^n r^l s^m = \exp(-\pi i(\alpha(n,l,m), z))$, while the standard denominator-side convention of a Borcherds–Kac–Moody algebra writes formal characters as $\exp(-2\pi i(\alpha, z))$. Replacing Z by $2Z$ converts the first character to the second, so

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) \;=\; D_5(2Z) \;=\; 64^{-1}\Delta_5(2Z).$$

(proved; [43, Proposition 3.1], [45, Section 5.1.1] *together with Theorem 2.1.)*

Proof. The product side of Theorem 2.1 gives $D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \prod (1 - q^n r^l s^m)^{f(nm,l)}$ with leading monomial $q^{1/2} r^{1/2} s^{1/2}$. Doubling $Z \mapsto 2Z$ replaces $q^n r^l s^m$ by $q^{2n} r^{2l} s^{2m}$ and hence $\exp(-\pi i(\alpha, z))$ by $\exp(-2\pi i(\alpha, z))$. The Lorentzian Borcherds–Kac–Moody algebra $\mathfrak{g}_{\Delta_5}$ of Gritsenko–Nikulin [43, Sections 3–4, Proposition 3.1] carries the denominator identity

$$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = \sum_{w \in W^{(2)}(\Lambda_{II}^{2,1})} \det(w) \left(e^{-2\pi i (w\rho, z)} - \sum_a m(a)\, e^{-2\pi i (w(\rho + a), z)} \right)$$

on the imaginary-positive cone, where $m(a)$ is determined by the negative-norm additive coefficients and by $1 - \sum_{t\ge 1} m(ta) q^t = \prod_{t \ge 1}(1 - q^t)^{\tau(ta)}$ on isotropic rays. The right-hand side is equal to $D_5(2Z)$ in the doubled exponential. Equivalently $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$. □

The factor 64 in $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is the same theta-constant 2^6 from the leading-coefficient calculation, transported through $Z \mapsto 2Z$; it is not a separate normalization, an integration constant, or a free scalar. The three identifications $\mathcal{D}_X = \Delta_5$, $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$, $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{DI})$ are kept distinct; the doubling lemma supplies the denominator-character normalization.

B.6 The eight-row Gritsenko–Cléry catalogue

Gritsenko and Cléry [41, Theorem 1.4] classify all Siegel modular forms of genus two, with character or multiplier system on a Hecke-type congruence subgroup $\Gamma_t(N) \subset \mathrm{Sp}(4, \mathbb{Q})$, whose divisor is supported on the $\Gamma_t(N)$-translates of the diagonal Humbert surface $\mathcal{H}_{\mathrm{diag}} = \{Z = \mathrm{diag}(a, b) : a, b \in \mathbb{H}_1\}$ with multiplicity one. The classification produces exactly eight rows. *(proved;* [41, Theorem 1.4 and Theorem 3.1]*.)*

The rows are:

j	F_j	(t, N)	Γ_j	$\mathrm{wt}(F_j)$	multiplier ν_j	$c_j(0)$
1	Δ_5	$(1, 1)$	$\Gamma_1 = \mathrm{Sp}(4, \mathbb{Z})$	5	$v_\eta^{12} \times v_H$	10
2	Δ_2	$(2, 1)$	Γ_2^+	2	$v_\eta^{6} \times v_H$	4
3	Δ_1	$(3, 1)$	Γ_3^+	1	$v_\eta^{4} \times v_H$	2
4	$\Delta_{1/2}$	$(4, 1)$	Γ_4^+	$\frac{1}{2}$	$v_\eta^{3} \times v_H$ (degree 8)	1
5	∇_3	$(1, 2)$	$\Gamma_0^{(2)}(2)$	3	$\chi_2^{(2)} \times v_H$	6
6	∇_2	$(1, 3)$	$\Gamma_0^{(2)}(3)$	2	$\chi_2^{(3)} \times v_H$	4
7	$\nabla_{3/2}$	$(1, 4)$	$\Gamma_0^{(2)}(4)$	$\frac{3}{2}$	$\chi_4^{(4)} \times v_H$ (order 4)	3
8	Q_1	$(2, 2)$	$\Gamma_2(2)$	1	$\chi_4^{(2)} \times v_H$	2

In the displayed F_j-order the weights and constant Fourier coefficients are

$$(5, 2, 1, \tfrac{1}{2}, 3, 2, \tfrac{3}{2}, 1), \qquad (10, 4, 2, 1, 6, 4, 3, 2).$$

In the input-pair order

$$(1, 1), (2, 1), (1, 2), (3, 1), (1, 3), (4, 1), (1, 4), (2, 2)$$

the same data read

$$(5, 2, 3, 1, 2, \tfrac{1}{2}, \tfrac{3}{2}, 1), \qquad (10, 4, 6, 2, 4, 1, 3, 2).$$

Each row satisfies the Borcherds–Gritsenko universal weight identity

$$\mathrm{wt}(F_j) \;=\; \frac{c_j(0)}{2} \;=\; \kappa_{\mathrm{BKM}}(F_j), \qquad j = 1, \dots, 8.$$

(proved; [8], [41, Theorem 3.1]*; the κ_{BKM} relabelling is the universal Borcherds-weight identity used in the Calabi–Yau quantum-groups κ-stratification.)*

The half-integer weights $\frac{1}{2}$ (row 4) and $\frac{3}{2}$ (row 7) are realized by multiplier systems $v_\eta^k \times v_H$ on the ambient paramodular cover, not by passing to a metaplectic double cover. The multiplier v_η is the Dedekind eta multiplier on $\mathrm{SL}_2(\mathbb{Z})$; the multiplier v_H is the Heisenberg multiplier on the Heisenberg lift of $\mathrm{Sp}(4, \mathbb{Z})$; and $v_\eta^3 \times v_H$ has order 8 on the row-4 cover, while $\chi_4^{(4)} \times v_H$ has order 4 on the row-7 cover. *(proved;* [41, Theorem 3.1]*.)*

2.0.0.5 Denominator content of the rows. Rows 1–4 carry Gritsenko–Nikulin denominator theorems [45, Theorem 2.1, equations (2.11), (2.20), (2.21)]; in the $e_t^{(n,l,m)} = q^n r^l s^{tm}$ exponential normalization of [45, §5.1.1], the products are

$$\Delta_2(Z) = q^{1/4} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_2^{\mathrm{prod}}} (1 - q^n r^l s^{2m})^{f_2(nm,l)},$$

$$\Delta_1(Z) = q^{1/6} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_3^{\mathrm{prod}}} (1 - q^n r^l s^{3m})^{f_3(nm,l)},$$

$$\Delta_{1/2}(Z) = q^{1/8} r^{1/2} s^{1/2} \prod_{(n,l,m) \in \Gamma_4^{\mathrm{prod}}} (1 - q^n r^l s^{4m})^{f_4(nm,l)},$$

with cusp Jacobi inputs $\phi_{0,2}, \phi_{0,3}, \phi_{0,4}$ supplied explicitly by [45, §5]. Rows 5–8 carry Cléry–Gritsenko products on the corresponding congruence covers, with cusp Jacobi inputs ϕ_j as in [41, Theorem 3.1], and the additional denominator data (Weyl chamber, simple real roots, reflection character, Weyl alternating sum) supplied by Govindarajan's twisted-CHL theorem [38], [39], [17] for $F_5 = \nabla_3$, $F_6 = \nabla_2$, $F_7 = \nabla_{3/2}$, $F_8 = Q_1$. For F_7 the denominator theorem lives on the order-four multiplier cover $\Gamma_0^{(2)}(4)\langle\chi_4^{(4)} \times v_H\rangle$. *(proved;* [38], [45, Theorem 2.1], [41, Theorem 3.1].*)*

2.0.0.6 Level separation. The eight-row catalogue is a list of *automorphic products plus denominator theorems on the rows where the data have been supplied*. It is not a list of compact Hall sources, reduced curve-counting backgrounds, or chiral hosts. For a row j, the automorphic product F_j acquires a compact physical host only after separate choices of an automorphic line, a denominator algebra, a scalar trace, a chiral source, and a Hall realization; these choices are row-dependent. The row $j = 1$ ($F_1 = \Delta_5$) is the only row with a proved reduced CY/DT scalar host (Oberdieck–Pixton, Oberdieck–Pandharipande: $Z^X_{\mathrm{OP}} = -4096\Delta_5^{-2}$); even row 1 has open compact Hall/chiral host at the primitive-recognition level, governed by $(S1)$–$(S10)$.

B.7 The Borcherds-weight coordinate $\kappa_{\mathrm{BKM}} = 5$

The universal Borcherds-weight identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ is the row-by-row specialization of the Borcherds singular-theta-lift weight formula [8, Theorem 10.1] together with the Gritsenko–Nikulin unification [43, §2], [42]. The relevant row is $j = 1$, with

$$\kappa_{\mathrm{BKM}}(\Delta_5) = \mathrm{wt}(\Delta_5) = 5 = \frac{c_1(0)}{2} = \frac{f(0,0)}{2} = \frac{10}{2}.$$

(proved; [8, Theorem 10.1], [32, §2.2], [41, Theorem 3.1].*)*

This is the Borcherds-weight invariant: the same integer $\kappa_{\mathrm{BKM}} = 5$ identifies the Δ_5 row inside the companion Calabi–Yau quantum-groups κ-stratification, and it is the entry that distinguishes $\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Delta_5)$ (the Igusa cusp-form weight) from the unrelated quantities $\kappa_{\mathrm{ch}}^{\mathrm{Hodge}} = 0$, $\kappa_{\mathrm{ch}}^{\mathrm{Heis}} = 3$, $\kappa_{\mathrm{cat}} = 0$, $\kappa_{\mathrm{fiber}} = 24 = \chi_{\mathrm{top}}(\mathrm{K3})$. The additive formula $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ would identify $24 = \chi_{\mathrm{top}}(\mathrm{K3})$ with $2 = \chi(\mathcal{O}_{\mathrm{K3}})$. The Heisenberg reading gives an accidental equality at $N = 1$, but it does not extend to the CHL weights $c_N(0)/2 = (5, 4, 3, 2, 1)$; the appendix asserts $\kappa_{\mathrm{BKM}} = 5$ as the row entry of the Borcherds-weight identity, not as the value of any additive formula.

B.8 Normalizations and residual obstructions

The following normalizations fix the Borcherds product without residual scalar, chamber, multiplier, or coefficient ambiguity.

- *Theta-constant normalization.* The constant 64 is not a free scalar on Δ_5. The identity $64 = 2^6$ is the explicit theta-product leading coefficient [48], [43, Theorem 4.1]; with the theta normalization fixed, no residual scalar freedom remains. Cross-checked against the consistency $f(1,1) = -64$ and the qrs-coefficient identity $c_\Delta(1,1,1) = 64$ [43, Theorem 4.1].
- *Cusp chamber definition Γ_{eff}.* The semigroup Γ_{eff} is the ordered-limit semigroup of the type-II cusp polarization $0 < |s| \ll |q| \ll |r|^{-1} \ll 1$ [45, Example 2.4]; not a free choice. Triples in $\Gamma_{\mathrm{eff}} \setminus \Gamma_{\mathrm{act}}$ have zero exponent and therefore contribute no visible product factor.
- *Multiplier system for half-integer weights.* Half-integer weights $\frac{1}{2}$ and $\frac{3}{2}$ are realized by $v_\eta^k \times v_H$, not by the Weil-representation metaplectic double cover; the integer weight 5 of Δ_5 uses the abelian multiplier $v_\eta^{12} \times v_H$ (which is automorphic-trivial modulo the Maass character ν_{Δ_5} on the type-II Weyl generators). No metaplectic factor enters the Borcherds weight identity. See [41, Theorem 3.1] for the full classification.

- *Doubling factor* $64^{-1}\Delta_5(2Z)$. The factor 64^{-1} in $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ is the same theta-constant 2^6 from Theorem 2.1, transported through $Z \mapsto 2Z$; no new constant. See Lemma 2.2.
- *The* $\kappa_{\mathrm{BKM}} = 5$ *row.* The integer $\kappa_{\mathrm{BKM}} = 5$ is the row entry of the Borcherds-weight identity $\mathrm{wt}(F_j) = c_N(0)/2$ in the catalogue of [41, Theorem 3.1]; it agrees with $\mathrm{wt}(\Delta_5) = f(0,0)/2 = 5$ from [32, §2.2]; it agrees with the entry $(c_1(0), \mathrm{wt}) = (10, 5)$ in the companion Calabi–Yau quantum-groups κ-stratification.
- *Eichler–Zagier vs Gritsenko–Nikulin convention.* The present appendix uses the Eichler–Zagier convention $\phi_{0,1} = Z_{\mathrm{K3}}/2$ [32, §2.2]; this matches the Gritsenko–Nikulin Jacobi convention in [45, Example 2.4] and [45, §5.1.1] on the (n, l)-coefficient table given in B.1, and matches the singular theta-lift normalization [8, Theorem 10.1] after the Mp_2-vector-valued bridge. No residual factor of 2 enters the exponent $f(nm, l)$ of the Borcherds product.

2.0.0.7 Residual obstructions. The four obstruction classes (D0), (O1), $(\mathrm{O1})^+$, (O2) are not part of the Borcherds product calculation; they are geometric content on the source side of Δ_5. Within the automorphic calculation, no residual Borcherds-product obstruction remains. The universal identity $\mathrm{wt}(F_j) = c_N(0)/2$ fixes the row $j = 1$, $\kappa_{\mathrm{BKM}} = 5$, and any disagreement with the Vol III catalogue, the Borcherds singular theta-lift weight formula, or the Gritsenko–Cléry classification would be a contradiction among the stated comparison theorems.

2.0.0.8 Computations. Fourier coefficients are obtained from the rational theta-quotient expansion; lattice and Gram-pairing identities are the computations of Appendix 1.2; the type-II cusp chamber, active support, and Borcherds product are in Proposition 7.2; the doubling identity is Remark 5.5; the catalogue tables are in Appendix 6.

3 Mukai–Gram dictionary on $K3 \times E$

Let $X = S \times E$ be the smooth trivial-canonical threefold, with S a smooth complex K3 surface and E a smooth complex elliptic curve. The Mukai vector and Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$, together with the Künneth decomposition on $H^*(X, \mathbb{Z})$, produce the Gram-degree projection $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ and its normal-ordered companion $\overline{\Pi}_X$ on the central extension $\widehat{\Gamma}_X$. The Borcherds expansion variables $q^n r^l s^m$ of an Igusa cusp expansion record Gram-degree shadows in Γ_{gram}. When the shadow is carried by an actual one-dimensional subscheme $Y \subset X$, its dyonic charge $c = (Q_Y, P_Y)$ satisfies the formula below. The existence of such a subscheme is the algebraic-effective Hall-support condition, not a formal consequence of the Fourier monomial. For Y of class $[Y] = (\beta_h, d)$ and $\chi(\mathcal{O}_Y) = n$,

$$\Pi_X(Q_Y, P_Y) = (h - 1,\ n,\ d - 1).$$

The threefold arithmetic Euler characteristic $n = \chi(\mathcal{O}_Y)$ is the direct structure-sheaf invariant; no "Todd correction" enters, and no ambient fourfold appears.

The Mukai–Gram construction sits at Beilinson chart layer α (intrinsic charge data) and γ (ambient Mukai/Gram lattice). The κ-tuple, orientation character ϵ_o, Pfaffian section, and recognition theorem live at higher Beilinson levels and depend on this chart datum.

3.1 Mukai vector and Mukai pairing on a K3 surface

Definition 3.1 *(Mukai lattice and Mukai vector).* Let S be a smooth complex K3 surface. The Mukai lattice is the total cohomology

$$\widetilde{H}(S,\mathbb{Z}) := H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z}),$$

graded by even cohomological degree, an even unimodular lattice of signature $(4,20)$ and rank 24 when equipped with the Mukai pairing below. For a coherent sheaf $F \in \mathrm{Coh}(S)$, respectively for an object $F \in D^b(S)$, the Mukai vector is

$$v_S(F) := \mathrm{ch}(F)\cdot\sqrt{\mathrm{td}(S)} = \big(\mathrm{rk}\,F,\ c_1(F),\ \mathrm{ch}_2(F) + \mathrm{rk}\,F\cdot[\mathrm{pt}]\big) \ \in\ \widetilde{H}(S,\mathbb{Z}).$$

The square root $\sqrt{\mathrm{td}(S)} = (1,0,1)$ on the K3 surface uses the Todd class $\mathrm{td}(S) = (1,0,2[\mathrm{pt}])$, so $\sqrt{\mathrm{td}(S)} = (1,0,[\mathrm{pt}])$; identifying $[\mathrm{pt}]$ with the integer 1 in the displayed coordinates gives the displayed vector. The triple $v_S(F) = (r,D,s)$ records the rank $r \in H^0$, the first Chern class $D \in H^2$, and the Mukai-corrected second component $s = \mathrm{ch}_2(F) + r \in H^4$. Mukai introduced this lattice and pairing in [78, Theorem 1.5], with the symplectic structure of the moduli space of sheaves established in [77].

Proposition 3.2 *(Mukai pairing on the K3 surface).* The Mukai pairing on $\widetilde{H}(S,\mathbb{Z})$ is the symmetric $\mathbb{Z}$-bilinear form

$$\big\langle (r,D,s),(r',D',s')\big\rangle_{\mathrm{Muk}} = D\cdot D' - rs' - r's,$$

equivalently

$$\big\langle v,w\big\rangle_{\mathrm{Muk}} = -\int_S v^\vee \cup w, \qquad v^\vee := (r,-D,s),$$

where $v_2 \cup w_2$ is the K3 cup product on $H^2(S,\mathbb{Z})$ and $v_0 \cup w_4, v_4 \cup w_0$ record the cross-degree pairings. The form is symmetric, even, unimodular, of signature $(4,20)$; its hyperbolic-plane summands are

$$\widetilde{H}(S,\mathbb{Z}) \simeq U^{\oplus 4} \oplus E_8(-1)^{\oplus 2}$$

as integral lattices. For a polarized K3 of Picard rank ρ, the Néron–Severi lattice $\mathrm{NS}(S)$ has signature $(1,\rho-1)$, while the algebraic Mukai lattice

$$\widetilde{H}^{1,1}_{\mathrm{alg}}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus \mathrm{NS}(S) \oplus H^4(S,\mathbb{Z})$$

has signature $(2,\rho)$. This is Mukai [78, Theorem 1.5(ii)]; see also Huybrechts–Lehn [47, Chapter 6].

Proof. The bilinear identity is the definition; its symmetry is immediate from the symmetry of the K3 cup product and the identification of degree-zero and degree-four components. Even unimodularity follows from the K3 intersection lattice on H^2 (even unimodular of signature $(3,19)$) and the unimodular pairing $H^0 \times H^4 \to \mathbb{Z}$ under $rs' \mapsto rs'$. The signature $(4,20)$ follows by Hodge index plus the hyperbolic factor on $H^0 \oplus H^4$. The same hyperbolic factor changes $\mathrm{NS}(S)$ of signature $(1,\rho-1)$ into $\widetilde{H}^{1,1}_{\mathrm{alg}}(S,\mathbb{Z})$ of signature $(2,\rho)$. The lattice isomorphism is [78, Theorem 1.5(ii)]. □

Remark 3.3 *(Mukai pairing is symmetric on both parts).* The Mukai pairing on $\widetilde{H}(S,\mathbb{Z})$ is symmetric in both the algebraic and transcendental parts. There is no orthosymplectic $\mathfrak{osp}$ refinement at the level of the lattice. The Hodge $\mathbb{Z}/2$-super-extension $(\mathbb{C},\omega_S) \mapsto \mathbb{C}\oplus\omega_S$ used in Mukai self-mirror language imposes a separate $\mathbb{Z}/2$-grading on a Yangian-type quantum group constructed from this lattice; that grading does not change the underlying lattice pairing, which remains symmetric. The classical Lie limit of the Mukai-form Yangian is the orthogonal Lie algebra $\mathfrak{so}(4, 20)$, not $\mathfrak{osp}(4 \mid 20)$; see [92, §2] for the Mukai-side Yangian classical limit and [59] for the cohomological Hall algebra realization.

Example 3.4 *(Mukai vector of a point and of an ideal sheaf).* $v_S(\mathcal{O}_p) = (0, 0, 1)$ for the structure sheaf of a point $p \in S$; $v_S(\mathcal{O}_S) = (1, 0, 1)$ for the trivial line bundle; $v_S(L) = (1, c_1(L), c_1(L)^2/2 + 1)$ for a line bundle L. For an ideal sheaf $\mathcal{I}_C$ of a smooth curve $C \subset S$ of genus g, the adjunction formula on the K3 surface gives $C^2 = c_1(\mathcal{O}_S(C))^2 = 2g - 2$, and

$$v_S(\mathcal{I}_C) = (1, -c_1(\mathcal{O}_S(C)), \mathrm{ch}_2(\mathcal{I}_C) + 1).$$

The Mukai pairing $\langle v_S(\mathcal{O}_p), v_S(\mathcal{O}_p)\rangle_{\mathrm{Muk}} = 0 - 0 \cdot 1 - 0 \cdot 1 = 0$, recording that points have zero self-pairing in the Mukai form. The pairing $\langle v_S(\mathcal{O}_S), v_S(\mathcal{O}_p)\rangle_{\mathrm{Muk}} = 0 - 1 \cdot 1 - 0 \cdot 0 = -1$ records the Euler characteristic $\chi(\mathcal{O}_S, \mathcal{O}_p) = 1$ up to the Mukai sign convention $\chi(F, G) = -\langle v_S(F), v_S(G)\rangle_{\mathrm{Muk}}$ of [78, Proposition 2.2].

3.2 Künneth on $K3 \times E$ and the rank-3 Gram lattice

PROPOSITION 3.5 *(Künneth for $X = K3 \times E$).* Let S be a smooth complex K3 surface and E a smooth complex elliptic curve. The even integral cohomology of $X = S \times E$ splits as

$$H^{\mathrm{even}}(X, \mathbb{Z}) \simeq H^*(S, \mathbb{Z}) \otimes_{\mathbb{Z}} H^{\mathrm{even}}(E, \mathbb{Z}),$$

the odd–odd Künneth term $H^{\mathrm{odd}}(S) \otimes H^{\mathrm{odd}}(E)$ absent because S carries no odd cohomology, both factors free $\mathbb{Z}$-modules. Fixing a generator $\omega_E \in H^2(E, \mathbb{Z})$ with $\int_E \omega_E = 1$, $H^{\mathrm{even}}(E, \mathbb{Z}) = \mathbb{Z} \cdot 1_E \oplus \mathbb{Z} \cdot \omega_E$, and every even class on X decomposes uniquely as

$$v_X = v^{(0)} \otimes 1_E + v^{(\omega)} \otimes \omega_E, \qquad v^{(0)}, v^{(\omega)} \in H^*(S, \mathbb{Z}).$$

The formal Künneth lattice

$$H^{\mathrm{even}}(X, \mathbb{Z}) \simeq \widetilde{H}(S, \mathbb{Z}) \otimes U_E, \qquad U_E := H^0(E, \mathbb{Z}) \oplus H^2(E, \mathbb{Z}),$$

carries the tensor Künneth pairing of rank 48 and signature (24, 24), obtained from the (4, 20)-Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$ and the (1, 1)-hyperbolic pairing on U_E. This cross-pairing is distinct from the direct-sum componentwise Gram pairing on Γ_X^{form} used by Π_X. The odd Künneth term $H^{\mathrm{odd}}(S) \otimes H^{\mathrm{odd}}(E)$ is absent because the K3 surface S carries no odd cohomology. See Huybrechts [47, Chapter 3].

Proof. E and S have torsion-free integral cohomology, so Künneth holds integrally and produces the displayed tensor decomposition. The tensor-pairing factorization is bilinearity of the cup product over the Künneth isomorphism. The lattice signature comes from (4, 20) on $\widetilde{H}(S, \mathbb{Z})$ (Proposition 3.2) tensored with (1, 1) on U_E. □

Definition 3.6 *(Gram-degree projection on $X = K3 \times E$).* The formal additive even-Mukai sector is

$$\Gamma_X^{\mathrm{form}} := \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z}), \qquad c = (Q, P) \in \Gamma_X^{\mathrm{form}},$$

with $Q, P \in \widetilde{H}(S, \mathbb{Z})$. The Gram-index lattice is

$$\Gamma_{\mathrm{gram}} := \mathbb{Z}^3,$$

and the Gram-degree projection is the quadratic map

$$\Pi_X : \Gamma_X^{\mathrm{form}} \longrightarrow \Gamma_{\mathrm{gram}}, \qquad (Q, P) \longmapsto \left(\tfrac{(Q,Q)_{\mathrm{Muk}}}{2},\ (Q, P)_{\mathrm{Muk}},\ \tfrac{(P,P)_{\mathrm{Muk}}}{2}\right).$$

The triple $(n, l, m) = \Pi_X(Q, P)$ is the Gram-degree shadow; it is the target of the Igusa Fourier expansion at the standard cusp. It is *not* an additive Hall grading: the additivity defect is the symmetric bilinear cocycle B of §3.4 below.

3.3 D6–D2–D0 dictionary

The Künneth decomposition of $v_X(\mathcal{I}_Y)$ on the threefold $X = K3 \times E$ produces the D6–D2–D0 dictionary $\Pi_X(Q_Y, P_Y) = (h-1, n, d-1)$ of Theorem 3.9 as a chart-data identity. The Mukai-vector calculation isolates the K3-fibre rank-one component $P_Y = (1, 0, 1-d)$ on the 1_E-summand and the curve-class component $Q_Y = (0, -\beta^\vee, -n)$ on the ω_E-summand, with $\beta^\vee \in H^2(S, \mathbb{Z})$ the Poincaré dual of β; the Mukai pairing on the K3 components delivers the Gram-degree shadow.

THEOREM 3.7 *(Mukai vector of an ideal sheaf on the threefold $K3 \times E$).* Let $X = S \times E$ with S a smooth complex K3 surface and E a smooth complex elliptic curve, so $K_X \simeq \mathcal{O}_X$ and X is a smooth trivial-canonical threefold. Let $Y \subset X$ be a one-dimensional closed subscheme of class $[Y] = (\beta, d) \in H_2(S, \mathbb{Z}) \oplus H_2(E, \mathbb{Z})$, write $\beta^\vee \in H^2(S, \mathbb{Z})$ for the Poincaré dual of β, and let $n = \chi(\mathcal{O}_Y) \in \mathbb{Z}$. The Mukai vector of the ideal sheaf $\mathcal{I}_Y \in D^b(X)$ decomposes against the Künneth basis as

$$v_X(\mathcal{I}_Y) = (1, 0, 1-d) \otimes 1_E + (0, -\beta^\vee, -n) \otimes \omega_E.$$

Proof. The trivial-canonical threefold condition gives $\sqrt{\mathrm{td}(X)} = \pi_S^*\sqrt{\mathrm{td}(S)}\,\pi_E^*\sqrt{\mathrm{td}(E)} = (1, 0, 1) \otimes 1$, since the elliptic curve is one-dimensional with trivial Todd class $\mathrm{td}(E) = 1$. The ideal-sheaf identity is $\mathrm{ch}(\mathcal{I}_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y)$. For the structure sheaf of a one-dimensional subscheme of a trivial-canonical threefold, $\mathrm{ch}_0(\mathcal{O}_Y) = 0$, $\mathrm{ch}_1(\mathcal{O}_Y) = 0$, $\mathrm{ch}_2(\mathcal{O}_Y) = \mathrm{PD}[Y]$, and $\int_X \mathrm{ch}_3(\mathcal{O}_Y) = \chi(\mathcal{O}_Y) = n$. Decomposing $[Y] = (\beta, d)$ on the Künneth basis, $\mathrm{PD}[Y] = \beta^\vee \otimes \omega_E + d[\mathrm{pt}_S] \otimes 1_E$, where $[\mathrm{pt}_S]$ is the K3 point class; the fibrewise elliptic-degree component $d[\mathrm{pt}_S] \otimes 1_E$ records the elliptic degree. Hence

$$\mathrm{ch}(\mathcal{O}_Y) = (0, 0, d) \otimes 1_E + (0, \beta^\vee, n) \otimes \omega_E,$$

and $1 - \mathrm{ch}(\mathcal{O}_Y) = (1, 0, -d) \otimes 1_E + (0, -\beta^\vee, -n) \otimes \omega_E$. Multiplying by $\sqrt{\mathrm{td}(X)} = (1, 0, 1) \otimes 1_E$ using the K3 cup product on each Künneth summand, $(1, 0, 1) \cup (1, 0, -d) = (1, 0, 1-d)$ since $[\mathrm{pt}_S] \cup 1 = [\mathrm{pt}_S]$, and $(1, 0, 1) \cup (0, -\beta^\vee, -n) = (0, -\beta^\vee, -n)$ since $\beta^\vee \cup [\mathrm{pt}_S] = 0$ in $H^*(S, \mathbb{Z})$, yields the displayed Mukai vector $v_X(\mathcal{I}_Y) = (1, 0, 1-d) \otimes 1_E + (0, -\beta^\vee, -n) \otimes \omega_E$. □

Remark 3.8 *(Threefold Euler characteristic, no Todd correction).* The integer $n = \chi(\mathcal{O}_Y) \in \mathbb{Z}$ is the holomorphic Euler characteristic of the structure sheaf $\mathcal{O}_Y$ on the threefold $X = S \times E$; the trivial-canonical threefold condition makes $\sqrt{\mathrm{td}(X)}$ act as identity on the third Chern character so that $\int_X \mathrm{ch}_3(\mathcal{O}_Y) = n$ directly. There is no auxiliary "Todd correction" to install on top of n; the Mukai vector $(0, -\beta^\vee, -n) \otimes \omega_E$ records n in the K3 H^4-coordinate on the elliptic ω_E-summand. The sign is fixed by $\mathrm{ch}(\mathcal{I}_Y) = 1 - \mathrm{ch}(\mathcal{O}_Y)$ and the Künneth orientation $\int_E \omega_E = +1$.

THEOREM 3.9 *(D6–D2–D0 Mukai–Gram dictionary, restated as chart data).* With the standing data of Theorem 3.7, set

$$P_Y := (1, 0, 1-d), \qquad Q_Y := (0, -\beta^\vee, -n) \ \in\ \widetilde{H}(S, \mathbb{Z}),$$

the Künneth components of $v_X(\mathcal{I}_Y)$. Then the Gram-degree projection of Definition 3.6 reads

$$\Pi_X(Q_Y, P_Y) = \left(\tfrac{(Q_Y,Q_Y)_{\mathrm{Muk}}}{2},\ (Q_Y, P_Y)_{\mathrm{Muk}},\ \tfrac{(P_Y,P_Y)_{\mathrm{Muk}}}{2}\right) = \left(\tfrac{(\beta^\vee)^2}{2},\ n,\ d-1\right).$$

For the elliptically fibered K3 surface $S \longrightarrow \mathbb{P}^1$ with section s_1 and curve class $\beta = \beta_h = s_1 + hF$ ($h \in \mathbb{Z}_{\geq 0}$, F the fibre class), the self-intersection $\beta_h^2 = 2h - 2$ gives

$$\Pi_X(Q_Y, P_Y) = (h-1,\ n,\ d-1).$$

Proof. Apply the Mukai pairing of Proposition 3.2 component by component:

$$\begin{aligned}
(Q_Y, Q_Y)_{\mathrm{Muk}} &= (-\beta^\vee)\cdot(-\beta^\vee) - 0\cdot(-n) - 0\cdot(-n) = (\beta^\vee)^2,\\
(Q_Y, P_Y)_{\mathrm{Muk}} &= (-\beta^\vee)\cdot 0 - 0\cdot(1-d) - 1\cdot(-n) = n,\\
(P_Y, P_Y)_{\mathrm{Muk}} &= 0\cdot 0 - 1\cdot(1-d) - 1\cdot(1-d) = 2(d-1),
\end{aligned}$$

giving the half-pairings $((\beta^\vee)^2/2, n, d-1)$. For the elliptic class $\beta_h = s_1 + hF$, identified with its Poincaré dual, the K3 intersection $s_1^2 = -2$ on a section of an elliptic K3 surface (genus formula $2g(s_1) - 2 = s_1^2 + s_1\cdot K_S = -2 + 0$, giving $g(s_1) = 0$), $s_1 \cdot F = 1$, $F^2 = 0$, so $\beta_h^2 = s_1^2 + 2h(s_1\cdot F) + h^2F^2 = -2 + 2h = 2h - 2$. The Gram-degree shadow follows. The cross-level identity with the OP/DT scalar normalization is the $Q_Y^2/2 = h - 1$ line of Theorem 3.9. □

Remark 3.10 *(Algebraic vs primitive D2 charges).* The K3 curve class $\beta \in H_2(S,\mathbb{Z})$ appearing through its Poincaré dual in $Q_Y = (0, -\beta^\vee, -n)$ is algebraic: $\beta^\vee \in \mathrm{NS}(S)$, since $Y \subset X$ is a closed subscheme. Primitivity of β — that β is not a non-trivial integer multiple of another effective class — is a separate condition; the dictionary holds for any algebraic β, not only primitive ones. The Mukai-pairing formula $(Q_Y, Q_Y)_{\mathrm{Muk}} = (\beta^\vee)^2$ records the $\mathrm{NS}(S)$-self-intersection regardless of primitivity. The Bryan–Oberdieck–Pandharipande–Yin/Maulik–Pandharipande conventions [14, 76, 85] take β algebraic; the elliptic degree $d \in \mathbb{Z}_{\geq 0}$ is unconstrained beyond non-negativity in the effective semigroup.

Remark 3.11 *(Chart-data layer; not the additive Hall grading).* The triple $(n, l, m) = \Pi_X(Q_Y, P_Y)$ records the Gram-degree invariants of the dyonic charge $c = (Q, P)$. It is the target of the Igusa Fourier expansion and the Borcherds product variables $q^n r^l s^m$. It is *not* the additive Hall grading on a hypothetical compact $K3 \times E$ Hall sector; the additive grading is the upstairs charge $c \in \Gamma_X^{\mathrm{form}}$ (or its normal-ordered lift $\widehat{\Gamma}_X$ below), and Π_X is a quadratic map onto Gram invariants. The polarization defect that obstructs the upstairs additive grading from descending directly is the symmetric bilinear cocycle B of §3.4.

3.4 Symmetric bilinear Gram cocycle

Lemma 3.12 *(Gram additivity defect, restated as chart structure).* For $c_i = (Q_i, P_i) \in \Gamma_X^{\mathrm{form}}$, $i = 1, 2$,

$$\Pi_X(c_1 + c_2) = \Pi_X(c_1) + \Pi_X(c_2) + B(c_1, c_2),$$

where the symmetric bilinear cocycle B is

$$B(c, c') := \big((Q, Q')_{\mathrm{Muk}},\ (Q, P')_{\mathrm{Muk}} + (Q', P)_{\mathrm{Muk}},\ (P, P')_{\mathrm{Muk}}\big).$$

The map B is symmetric and $\mathbb{Z}$-bilinear with values in $\Gamma_{\mathrm{gram}} = \mathbb{Z}^3$, satisfies the polarization identity

$$B(c, c) = 2\Pi_X(c),$$

and equals $-\delta\Pi_X$ as an additive coboundary, $(\delta\Pi_X)(c,c') := \Pi_X(c) + \Pi_X(c') - \Pi_X(c+c')$. Thus the ordinary additive cohomology class $[B] = 0$; the obstruction recorded here is the impossibility of an *additive linear* trivialization, not a non-zero additive cohomology class.

Proof. This is Lemma 2.4. Symmetry, bilinearity, integrality, polarization, and δ-cocycle property are Lemma 3.12(i)–(iii). The splitting $B = -\delta\Pi_X$ by the quadratic refinement Π_X is clause (vi) of the same lemma; clause (vii) shows no additive linear $L : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ can satisfy $B = \delta L$, since $B(c,c) = 2\Pi_X(c)$ is non-zero on charges with non-trivial Mukai self-pairings. □

Remark 3.13 *(Cleanly stated polarization identity).* The polarization identity $B(c,c) = 2\Pi_X(c)$ is the symmetric companion to $\Pi_X(c+c') = \Pi_X(c) + \Pi_X(c') + B(c,c')$; evaluated at $c = c'$ the latter gives $\Pi_X(2c) = 2\Pi_X(c) + B(c,c)$, and $\Pi_X(2c) = 4\Pi_X(c)$ by quadraticity of Π_X, hence $B(c,c) = 2\Pi_X(c)$. This is purely chart-data structure; no Heisenberg or central extension hypothesis on the source is invoked.

3.5 Normal-ordered Gram map

Definition 3.14 *(Normal-ordered charge lattice).* Set

$$\widehat{\Gamma}_X := \Gamma_X^{\mathrm{form}} \oplus_B \Gamma_{\mathrm{gram}},$$

the underlying set $\Gamma_X^{\mathrm{form}} \times \Gamma_{\mathrm{gram}}$ equipped with the abelian-group law

$$(c,T) \star (c',T') := \big(c+c',\ T+T'+B(c,c')\big).$$

The identity is $(0,0)$; the inverse of (c,T) is $(-c,\ -T+B(c,c)) = (-c,\ -T+2\Pi_X(c))$. The split section

$$s : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \qquad s(c) := (c, \Pi_X(c)),$$

is a homomorphism of abelian groups; the raw placement

$$i_0 : \Gamma_X^{\mathrm{form}} \longrightarrow \widehat{\Gamma}_X, \qquad i_0(c) := (c, 0),$$

is not a homomorphism unless $B \equiv 0$.

Proposition 3.15 *(Normal-ordered Gram homomorphism).* The map

$$\overline{\Pi}_X : \widehat{\Gamma}_X \longrightarrow \Gamma_{\mathrm{gram}}, \qquad \overline{\Pi}_X(c,T) := \Pi_X(c) - T,$$

is a homomorphism of abelian groups, $\overline{\Pi}_X : (\widehat{\Gamma}_X, \star) \to (\Gamma_{\mathrm{gram}}, +)$; equivalently,

$$\overline{\Pi}_X\big((c,T) \star (c',T')\big) = \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T').$$

On the split section, $\overline{\Pi}_X(s(c)) = 0$; on the raw placement, $\overline{\Pi}_X(i_0(c)) = \Pi_X(c)$.

Proof. Direct calculation,

$$\begin{aligned}
\overline{\Pi}_X((c,T) \star (c',T')) &= \overline{\Pi}_X(c+c', T+T'+B(c,c')) \\
&= \Pi_X(c+c') - T - T' - B(c,c') \\
&= \big(\Pi_X(c) + \Pi_X(c') + B(c,c')\big) - T - T' - B(c,c') \\
&= \big(\Pi_X(c) - T\big) + \big(\Pi_X(c') - T'\big) \\
&= \overline{\Pi}_X(c,T) + \overline{\Pi}_X(c',T'),
\end{aligned}$$

using Lemma 3.12. The split-section and raw-placement identities are the definitions. □

Remark 3.16 *(Polarization defect).* The composite $\overline{\Pi}_X \circ i_0 = \Pi_X$ records the quadratic Gram-degree shadow of the bare upstairs charge; the composite $\overline{\Pi}_X \circ s \equiv 0$ records the split-section convention that absorbs the entire Gram-degree into the central coordinate T. The choice of section is the choice of normal ordering: bare placements collect cross-pairing $B(c_i, c_j)$ in the central coordinate (Lemma 3.12), while s-placements record $\overline{\Pi}_X = 0$ on every component and the Gram-degree information lives entirely in the upstairs $\Pi_X(c)$. The two presentations agree on the homomorphism $\overline{\Pi}_X$. This is the additive group law on which a hypothetical compact Hall product can descend without a chain-level associator at the lattice level.

Remark 3.17 *(Chart data, not Hall support).* The lattice $\widehat{\Gamma}_X$ is the formal additive group on which charges and their Gram-degree shadows are tracked. It is not the algebraic effective semigroup Γ_X^{eff} of curve classes representable by closed subschemes $Y \subset X$; membership in the effective sub-semigroup is a separate algebraic-effective condition (Definition 1.6). The Beilinson chart layer α (charge data) and γ (Mukai lattice) license $\widehat{\Gamma}_X$ as ambient; the Hall-support restriction is part of the categorical-input layer at higher Beilinson level, not the Mukai–Gram chart datum.

3.6 Three Igusa names and the Mukai–Gram chart

Three Igusa names are distinct mathematical objects and remain distinct under the Mukai–Gram chart datum:

- the automorphic section $\mathcal{D}_X = \Delta_5$, a section of the weight-5 automorphic line over the genus-2 Siegel space with Maass character ν_{Δ_5};
- the BKM denominator $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$ on the root lattice $\Lambda_{II}^{2,1}$; and
- the compact protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$, a section of $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on the conditional pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ of §5.

The Mukai–Gram dictionary supplies the Gram-degree shadow $\Pi_X : \Gamma_X^{\mathrm{form}} \to \Gamma_{\mathrm{gram}}$ on which the cusp-Fourier expansion of Δ_5, the root-degree map of $\mathrm{den}(\mathfrak{g}_{\Delta_5})$, and the wall data of $\mathrm{Pf}_{\mathrm{prot}}$ all read. The Mukai pairing on $\widetilde{H}(S, \mathbb{Z})$ is symmetric in both algebraic and transcendental parts; the ungraded Mukai-form classical limit of an associated Yangian-type construction is the orthogonal Lie algebra $\mathfrak{so}(4, 20)$, not the orthosymplectic $\mathfrak{osp}(4 \mid 20)$. The Hodge $\mathbb{Z}/2$-super-extension that appears in Yangian conventions imposes a separate $\mathbb{Z}/2$-grading on the algebraic side; it does not enter the lattice pairing above. The Bryan/Oberdieck–Pandharipande/ Maulik–Toda conventions [14, 76, 85] take β algebraic and d elliptic-degree, agreeing with Theorem 3.9.

The Mukai–Gram chart datum does not touch the orientation character ϵ_o of the Weyl-transport condition 0.9, the Pfaffian section $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X)$, or the recognition theorem of Theorem 0.35; those structures live at higher Beilinson levels and depend on the chart datum.

4 Hall correspondences on $\mathfrak{M}^+_{\mathrm{eff}}(K3 \times E)$

The Hall side of the Dirac–Igusa construction rests on the following imported theorems and on their $K3 \times E$ specialization. The Hall correspondences are the convolution correspondences through which the Beilinson–Drinfeld factorization structure on the elliptic carrier acts on the retained $K3 \times E$ Hall stack [6]. Equivalently, they comprise the chart-augmented A_b data for the open factorization dg-category (X, D, τ) at a chosen boundary vacuum b, with chiral shadow on the elliptic carrier supplied separately by § 9.

The cited theorems fix the exact hypotheses needed for the $K3 \times E$ specialization: finite retained substacks, proper extension correspondences, compact-support functoriality, vanishing cycles, Thom–Sebastiani maps, and orientation coefficients.

The standing data are: S is a smooth complex projective $K3$ surface, E is a smooth elliptic curve, $X = S \times E$ is the trivial-canonical threefold, $\sigma_S \in H^0(S, K_S)$ is the holomorphic two-form trivializing K_S, and $\sigma = \mathrm{pr}_S^* \sigma_S \in H^0(X, p_S^* K_S)$ is the inflated cosection of §9. The algebraic Mukai lattice used by the Hall support is

$$\Lambda_{S,\mathrm{alg}} = H^0(S, \mathbb{Z}) \oplus \mathrm{NS}(S) \oplus H^4(S, \mathbb{Z}),$$

the algebraic sublattice of the full Mukai lattice $\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$, endowed with the restricted Mukai pairing of [77]. Γ_{eff} is the algebraic/effective Hall support of Definition 1.6; and $\mathfrak{M}^+_{\mathrm{eff}}(X)$ is the moduli stack of perfect complexes of effective dyonic class semistable for the supplied stability datum on X.

D.1 *The Hall product on the moduli of stable sheaves*

The first imported theorem is the existence and associativity of a Hall product on the equivariant cohomology of the moduli stack of semistable sheaves, with vanishing-cycle coefficients, after Joyce and Song.

Theorem 4.1 *(Joyce–Song generalised* DT *Hall product; γ, proved on* CY_3 *projective; conditional on (O1), (O1)$^+$ on $K3 \times E$).* Let Y be a smooth complex projective Calabi–Yau threefold. The moduli stack $\mathfrak{M}(Y)$ of compactly supported coherent sheaves on Y carries a critical *Hall product*

$$m \;:\; H^\bullet_c\big(\mathfrak{M}(Y), \Phi_W \otimes o\big) \otimes H^\bullet_c\big(\mathfrak{M}(Y), \Phi_W \otimes o\big) \longrightarrow H^\bullet_c\big(\mathfrak{M}(Y), \Phi_W \otimes o\big),$$

with Φ_W the perverse sheaf of vanishing cycles of the holomorphic Chern–Simons potential and o the orientation line supplied by Joyce–Upmeier orientation data. The product is defined by pull-push along the universal extension stack

$$\mathfrak{M}(Y) \xleftarrow{\;p\;} \mathfrak{Ext}^1(\mathfrak{M}(Y), \mathfrak{M}(Y)) \xrightarrow{\;q\;} \mathfrak{M}(Y),$$

where $\mathfrak{Ext}^1$ is the moduli stack of short exact sequences $0 \to A \to C \to B \to 0$, $p(0 \to A \to C \to B \to 0) = (A, B)$, and $q(0 \to A \to C \to B \to 0) = C$. The product $m = q_! \circ \mathrm{TS} \circ p^*$ is associative, where TS is the Massey–Saito–Joyce Thom–Sebastiani isomorphism on the universal extension stack [50, Theorem 5.4], [73], [11]. Its unit is the class of the empty object.

Proof. The product is the pull-push form of three standard structures. The moduli stack $\mathfrak{M}(Y)$ is locally the critical locus of the Joyce–Song holomorphic Chern–Simons potential $W : \mathrm{Crit}(W) \hookrightarrow Y_W$ where Y_W is a Darboux chart of the (-1)-shifted symplectic structure on $\mathfrak{M}(Y)$ supplied by Pantev–Toën–Vaquié–Vezzosi [89, Theorem 0.3]; the perverse sheaf Φ_W of vanishing cycles is well defined on the oriented d-critical locus by Brav–Bussi–Dupont–Joyce–Szendrői [11, Theorem 6.9]. The universal extension stack is locally a critical locus of $W_A \boxplus W_B$ on $Y_A \times Y_B$, and the Massey–Saito Thom–Sebastiani isomorphism [73, Theorem 1.1], [11, Theorem 5.10] identifies $\Phi_{W_A \boxplus W_B}$ with $\Phi_{W_A} \boxtimes \Phi_{W_B}$ on $\mathrm{Crit}(W_A) \times \mathrm{Crit}(W_B)$. The multiplicative orientation cocycle of Joyce–Upmeier [52, Theorem 4.1], [51, Theorem 1.11] trivializes the orientation gerbe on extensions, supplying the line bundle factor. When the exceptional pushforward is admissible, pulling $\Phi_W \otimes o$ along p, tensoring by Thom–Sebastiani, and pushing along

$q_!$ defines m. Associativity reduces, via the two-step flag stack $\mathfrak{Flag}(\mathfrak{M}(Y), \mathfrak{M}(Y), \mathfrak{M}(Y))$, to the pentagon coherence of Thom–Sebastiani triple stratification [11, Theorem 5.10(iv)] together with the Joyce–Upmeier multiplicative cocycle on triple extensions [52, §4]. See Joyce–Song [50, §5.2, Theorem 5.5] for the original construction and Davison [25, Theorem A] for the cohomological amplification. □

$K3 \times E$ specialization. On $Y = K3 \times E$, the cosection $\sigma = \mathrm{pr}_S^* \sigma_S$ of the holomorphic two-form on the K3 factor reduces the obstruction theory through Kiem–Li (Appendix 4); the supplied moduli is $\mathfrak{M}^{\mathrm{red}}_{\mathrm{eff}}(X) \subset \mathfrak{M}(X)$, and the product is the reduced critical product

$$m^{\mathrm{red}} \;=\; q_!^{\mathrm{red}} \circ \mathrm{TS}^{\mathrm{red}} \circ p^*$$

of Proposition 0.10. Vanishing of the seven obstruction classes $\mathfrak{o}_R^{\mathrm{conf}}$, $\mathfrak{o}_R^{\mathrm{prop,LL}}$, $\mathfrak{o}_R^{\mathrm{prop,mix}}$, $\mathfrak{o}_R^{\mathrm{prop,WW}}$, $\mathfrak{o}_R^{\mathrm{conf\text{-}prop}}$, $\mathfrak{o}_{w,R}^{\mathrm{assoc}}$, and $\mathfrak{o}_{w,R}^{I,\mathrm{prod}}$, together with finite residual inertia, retained compactified flag-Quot correspondences or an explicit compact-support six-functor model, base-change, projection-formula, and quotient-after-correspondence coherences, is the data $(D0)$+$(O1)$, absorbed into Bnd, dCrit, RedOr, and Six of Definition 0.11. *The existence of a chain-level Hall product on $K3 \times E$ without those four data is open.* The resulting invariant $Y^+(X) := H_c^\bullet(\mathfrak{M}^+_{\mathrm{eff}}(X), \Phi_W \otimes o)$ is the positive-half cohomological Hall algebra (CoHA) of Kontsevich–Soibelman [59] restricted to effective dyonic class; its Drinfeld double, regulated tensor product completion, and integral form constitute the level-3 object $G(X)$, conditional on the Hall pairing and stable-envelope transport. Thus $G(X)$ arises from $Y^+(X)$ by Hall pairing, integral-form completion, and stable-envelope transport, as in the Vol III Δ_5 row [65].

D.2 The Hall coproduct via collisions

The Hall product does not by itself supply a coproduct. A collision coproduct is additional six-functor data on the reversed extension correspondence: properness or compact-support replacement for the left leg, inverse Thom–Sebastiani transport, quotient descent, and a Hopf-pairing twist. When these data are supplied, $Y^+(X)$ becomes a graded bialgebra object; after Drinfeld doubling, completion, integral form, and stable-envelope transport, it is a candidate for the level-3 quantum group $G(X) = D_\hbar(Y^+(X))$ of Schiffmann–Vasserot type [92].

Theorem 4.2 *(Collision coproduct datum, formal conditional).* Let Y be a smooth projective Calabi–Yau threefold and suppose the Joyce–Song Hall product of Theorem 4.1 is given. Assume, in addition, that the reversed extension correspondence carries admissible compact-support six-functor data, inverse Thom–Sebastiani transport, and a completed Hall pairing for which the bialgebra compatibility residual vanishes. Then the critical Hall positive half $Y^+(Y) = H_c^\bullet(\mathfrak{M}^+_{\mathrm{eff}}(Y), \Phi_W \otimes o)$ carries a collision coproduct

$$\Delta \;:\; Y^+(Y) \longrightarrow Y^+(Y) \otimes Y^+(Y), \qquad \Delta \;=\; p_!^{\mathrm{coll}} \circ \mathrm{TS}^{-1} \circ q^{\mathrm{coll},*},$$

defined by the collision/extension correspondence

$$\mathfrak{M}^+_{\mathrm{eff}}(Y) \xleftarrow{\;q^{\mathrm{coll}}\;} \mathfrak{Ext}^{1,\mathrm{coll}}(\mathfrak{M}^+_{\mathrm{eff}}(Y)) \xrightarrow{\;p^{\mathrm{coll}}\;} \mathfrak{M}^+_{\mathrm{eff}}(Y) \times \mathfrak{M}^+_{\mathrm{eff}}(Y),$$

with $\mathfrak{Ext}^{1,\mathrm{coll}}$ the moduli of short exact sequences $0 \to A \to C \to B \to 0$ read with C on the left and (A, B) on the right, $q^{\mathrm{coll}}(0 \to A \to C \to B \to 0) = C$, $p^{\mathrm{coll}}(0 \to A \to C \to B \to 0) = (A, B)$. The coproduct Δ is graded coassociative. Together with the Hall product m of Theorem 4.1, $(Y^+(Y), m, \Delta)$ is a graded bialgebra.

Proof. The formula is the formal reverse of the Hall product. Its existence requires the extra six-functor hypotheses because the reversed leg is not automatically proper in the compact $K3 \times E$ problem. Given those hypotheses, coassociativity is the inverse Thom–Sebastiani pentagon on the three-step flag stack. The compatibility of m and Δ is not a consequence of the product theorem alone; it is exactly the vanishing of the Hall-pairing bialgebra residual assumed in the statement. On quivers with potential this is the Davison–Meinhardt bialgebra theorem [26, Theorem A]; on the present compact threefold it is retained as part of the Hall datum. □

Drinfeld double. The Hall pairing $\langle \cdot, \cdot \rangle_{\mathrm{Hall}}$, when supplied and non-degenerate after the completed radical quotient, pairs Y^+ with itself; the Drinfeld double $D_\hbar(Y^+(X))$ is the quantum double in the sense of Drinfeld [54, §13], supplying the negative half $Y^-(X)$ and the Cartan generators $K_\gamma, \gamma \in \Gamma_{\mathrm{eff}}$, with relations encoding the Hall pairing. The *level-3 $K3 \times E$ quantum group* is then

$$G(X) \;:=\; D_\hbar(Y^+(X))^{\mathrm{int}}_{\mathrm{compl}}$$

where $(-)^{\mathrm{int}}_{\mathrm{compl}}$ is the integral-form completion under the Hall pairing, conditional on the stable-envelope transport of § D.6. In Schiffmann–Vasserot [92] the analogous identification on toric CY_3 ambient is established; $K3 \times E$ is not equipped here with a global toric or quiver-variety NCCR model of that kind: there is no finite fixed-point McKay presentation, no compatible product-polarized HPD self-dual model, and no Schiffmann–Vasserot stable-envelope atlas on the compact non-toric ambient. The cosection-twisted reduced obstruction theory of § D.4 is the replacement; the global $G(X)$ on the compact non-toric $K3 \times E$ is not constructed from these data without the Hall pairing, integral form, completion, and stable-envelope transport.

D.3 The Hall primitives and the formal recognition criterion

For a connected graded bialgebra $B = \bigoplus_c B_c$, the primitive elements at charge c are

$$\mathrm{Prim}_c(B) \;=\; \{x \in B_c \mid \Delta(x) = x \otimes 1 + 1 \otimes x\},$$

equivalently the kernel of the reduced coproduct $\bar{\Delta}$. On a graded connected cocommutative Hopf algebra in characteristic zero, the Milnor–Moore theorem [70, Theorem 5.18] identifies $B \simeq U(\mathrm{Prim}_\bullet(B))$ as Hopf algebras, where $\mathrm{Prim}_\bullet(B)$ is the Lie algebra of primitives under the bracket $[x, y] = xy - (-1)^{|x|\,|y|}yx$.

Theorem 4.3 *(Hall primitives, formal level; β, formal).* Let $\mathfrak{M}$ carry a critical Hall positive half $Y^+(Y) = (B, m, \Delta)$ of Theorem 4.2. The graded subspace

$$\mathrm{Prim}_\bullet(Y^+(Y)) \;=\; \bigoplus_{c \in \Gamma_{\mathrm{eff}}} \mathrm{Prim}_c(Y^+(Y))$$

is a graded Lie subalgebra of $Y^+(Y)$ under the antisymmetrised product $[\cdot, \cdot]$. At a charge c such that the moduli $\mathfrak{M}^+_c(Y)$ is connected and the C-component of Δ on $\mathfrak{M}^+_c(Y)$ factors through extensions whose graded pieces lie in strictly lower charges, the inclusion $\mathrm{Prim}_c \hookrightarrow Y^+_c$ is the kernel of $\bar{\Delta}_c$. When $Y^+(Y)$ is graded connected and cocommutative, Milnor–Moore identifies $Y^+(Y) \simeq U(\mathrm{Prim}_\bullet(Y^+(Y)))$ as graded Hopf algebras.

Proof. Primitivity of $\mathrm{Prim}_\bullet$ under the antisymmetrised product is verified by direct computation of $\Delta([x, y])$ using the cocommutative-up-to-twist axiom and the multiplicativity of Δ; the resulting expression collapses to $[x, y] \otimes 1 + 1 \otimes [x, y]$. On a connected graded cocommutative Hopf algebra in characteristic zero, the Milnor–Moore identification is [70, §21.1]; on graded connected non-cocommutative

Hopf algebras, the Cartier–Quillen–Milnor–Moore theorem [70, Theorem 21.2.1] requires either cocommutativity or a chain-level lift through bar / cobar, which is exactly the chiral Koszul separation theorem of Vol I [33]. □

Recognition. The recognition theorem of Theorem 0.30 identifies the completed formal primitive Lie superalgebra with the imported Borcherds–Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ of Theorem 9.1, conditional on the finite compact-source datum $(S1)$–$(S10)$: the completed PBW filtration, the Hopf pairing radical equality, and the parity dimension recovery. In particular, on $K3 \times E$ the formal primitive subspace $\mathrm{Prim}_\bullet(Y^+(X))$ is the *candidate* primitive Lie superalgebra; its identification with the imported BKM denominator algebra $\mathfrak{g}_{\Delta_5}$ is the primitive recognition part of View ③. The formal Hopf algebra structure alone gives no recognition theorem; recognition acts on it.

D.4 The cosection-twist on $K3 \times E$: Kiem–Li localisation

On $K3 \times E$, the holomorphic two-form on the K3 factor supplies the semiregularity cosection on the retained nonzero K3-class branches of the obstruction theory of $\mathfrak{M}(X)$. The Kiem–Li [55] localisation theorem then localises the virtual fundamental class to the zero-locus of the cosection when the surjectivity hypothesis holds. The localisation is the source of the *reduced* virtual class on which the Hall product of § D.1 carries chain-level coherence.

Definition 4.4 *(Cosection).* Let $\mathfrak{M}$ be a Deligne–Mumford stack equipped with a perfect obstruction theory $\phi : \mathbb{E} \to \mathbb{L}_{\mathfrak{M}}$ of amplitude $[-1, 0]$, and let $\mathrm{Ob}(\mathfrak{M}) = h^1(\mathbb{E}^\vee)$ be the coherent obstruction sheaf. A *cosection* of the obstruction theory is an $\mathcal{O}_{\mathfrak{M}}$-linear morphism

$$\sigma \;:\; \mathrm{Ob}(\mathfrak{M}) \longrightarrow \mathcal{O}_{\mathfrak{M}};$$

its *vanishing locus* is the closed substack $\mathfrak{M}^{\mathrm{red}} = (\sigma = 0)$ on which the obstruction sheaf restricts to the kernel $\ker \sigma \subset \mathrm{Ob}(\mathfrak{M})$.

THEOREM 4.5 *(Kiem–Li cosection localisation;* [55] *Theorem 5.0.1; γ, proved).* Let $\mathfrak{M}$ be a Deligne–Mumford stack with perfect obstruction theory ϕ and surjective cosection $\sigma : \mathrm{Ob}(\mathfrak{M}) \twoheadrightarrow \mathcal{O}_{\mathfrak{M}}$ on $\mathfrak{M} \setminus \mathfrak{M}^{\mathrm{red}}$, where $\mathfrak{M}^{\mathrm{red}} = (\sigma = 0)$. The virtual fundamental class $[\mathfrak{M}]^{\mathrm{vir}} \in A_{\mathrm{vd}}(\mathfrak{M})$ lifts to a cosection-localised class

$$[\mathfrak{M}]^{\mathrm{vir}}_{\mathrm{loc}} \;\in\; A_{\mathrm{vd}}(\mathfrak{M}^{\mathrm{red}})$$

satisfying $\iota_*[\mathfrak{M}]^{\mathrm{vir}}_{\mathrm{loc}} = [\mathfrak{M}]^{\mathrm{vir}}$, where $\iota : \mathfrak{M}^{\mathrm{red}} \hookrightarrow \mathfrak{M}$ is the inclusion. If the cosection is used separately to quotient a trivial obstruction direction and form a reduced obstruction theory, that reduced theory has virtual dimension $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd} + 1$; this dimension shift is not the Kiem–Li localized cycle itself.

Proof. The cosection-kernel cone $C(\sigma) = \ker(\sigma : \mathrm{Ob} \to \mathcal{O})$ is a perfect obstruction cone of amplitude $[-1, 0]$ on $\mathfrak{M}^{\mathrm{red}}$ with $\mathrm{rk}\, C(\sigma) = \mathrm{rk}\,\mathrm{Ob} - 1$. The intrinsic normal cone $\mathfrak{C}$ of $\mathfrak{M}$ sits in the total obstruction bundle on $\mathfrak{M} \setminus \mathfrak{M}^{\mathrm{red}}$, where σ is surjective and the cone has codimension one in Ob; on $\mathfrak{M}^{\mathrm{red}}$, the cone embeds into $C(\sigma)$ by [55, Proposition 4.3]. Intersection with the zero section of $C(\sigma)$ on $\mathfrak{M}^{\mathrm{red}}$, pushed forward to $\mathfrak{M}$ through ι, yields $[\mathfrak{M}]^{\mathrm{vir}}$ by [55, Theorem 5.0.1]. The localized Gysin map preserves the original virtual dimension. The shift $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd} + 1$ enters only after a separate reduced obstruction theory is formed by removing the trivial obstruction summand. □

$K3 \times E$ cosection. On $X = K3 \times E$, the cosection of the obstruction theory of $\mathfrak{M}(X)$ is induced by the pullback $\sigma = \mathrm{pr}_S^* \sigma_S$ of the K3 holomorphic two-form. Concretely, for a perfect complex F on X, the chain-level cosection acts on $\mathrm{Ext}^2(F, F)$ by the semiregularity map of Buchweitz–Flenner / Bloch [76],

$$\mathrm{cs}_F \;:\; \mathrm{Ext}^2(F, F) \longrightarrow \mathbb{C}, \qquad \mathrm{cs}_F(\xi) \;=\; \int_X \sigma \wedge \mathrm{Tr}\big((\xi \otimes 1_{\Omega^1_X}) \circ \mathrm{At}(F)\big),$$

where $\mathrm{At}(F) \in \mathrm{Ext}^1(F, F \otimes \Omega^1_X)$ is the Atiyah class. The cosection vanishes precisely on the *reduced* moduli $\mathfrak{M}^{\mathrm{red}}(X) \subset \mathfrak{M}(X)$ consisting of sheaves whose Atiyah class is annihilated by σ; equivalently, sheaves whose deformation space along the K3 direction is constrained. See Maulik–Toda [76, Lemma 2.6, Theorem 2.7] and Oberdieck [84, Theorem 1, Theorem 3] for the explicit identification on the DT/PT side. The cosection is surjective on the retained nonzero K3-class branch. On the $s = 0$ Hall cusp the retained charge can lose the K3 component that makes the semiregularity map surjective; the holomorphic two-form σ_S does not vanish. The localisation then passes to the $D0$-degeneration limit named at obstruction $(D0)$ of § 9.

Theorem 4.6 *(Reduced virtual class on $K3 \times E$; primitive stable-pair branch).* On the stable-pair or ideal-sheaf Deligne–Mumford branch carrying a perfect obstruction theory and a surjective K3 semiregularity cosection, Kiem–Li localization gives

$$[\mathfrak{M}_c(X)]^{\mathrm{vir}}_{\mathrm{loc}} \in A_{\mathrm{vd}}(\mathfrak{M}_c(X)^{\mathrm{red}}).$$

The reduced obstruction theory obtained by quotienting the trivial obstruction direction gives a distinct reduced class

$$[\mathfrak{M}_c(X)]^{\mathrm{red}} \in A_{\mathrm{vd}+1}(\mathfrak{M}_c(X)^{\mathrm{red}}).$$

The reduced DT/PT theory of the primitive nonzero K3-class branch uses this reduced class and gives the Oberdieck–Pixton scalar $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ [84, Theorem 3]. The semistable Hall stack requires the retained compact-support six-functor and d-critical Hall datum of §0.3. On the $s = 0$ cusp the cosection degenerates; *the cosection-reduced theory there is the* $(D0)$ *degeneration limit named at § 9 and controlled by the retained D0-HN criterion of Theorem 7.6.* $[\partial, (D0)]$

D.5 BBDJS reduced d-critical orientation

The Hall product of § D.1 requires not just a virtual fundamental class but a perverse-sheaf-of-vanishing-cycles refinement and an orientation gerbe; the source is the Brav–Bussi–Dupont–Joyce–Szendrői [11] extension of the Joyce [50] d-critical-locus framework.

Theorem 4.7 *(BBDJS perverse-sheaf-of-vanishing-cycles; oriented* CY_3 *version of* [11] *Theorem 6.9; γ, proved).* Let $\mathfrak{M}$ be a derived Artin stack with a (-1)-shifted symplectic structure (PTVV [89]) whose classical truncation is a d-critical locus in the sense of Joyce [50, Definition 2.5]. Suppose $\mathfrak{M}$ carries an *orientation*, i.e. a square root o of the Joyce determinant line $K^{\mathrm{vir}}_{\mathfrak{M}} = (\det \mathbb{L}_{\mathfrak{M}})^{1/2}$, in the sense of [11, Theorem 6.9]. There is a perverse sheaf

$$\Phi_{\mathfrak{M}} \;\in\; \mathrm{Perv}(\mathfrak{M}),$$

the *perverse sheaf of vanishing cycles* of the d-critical locus; locally on a Joyce–Song Darboux chart $\mathrm{Crit}(W)$ inside a smooth ambient Y_W, $\Phi_{\mathfrak{M}}|_{\mathrm{Crit}(W)} \;\simeq\; \Phi_W[d_{Y_W}]$ is the standard perverse sheaf of vanishing cycles of the holomorphic potential $W : Y_W \to \mathbb{C}$, shifted by the ambient dimension. The chart-comparison cocycle is canonical up to the orientation choice; the orientation determines a multiplicative cocycle of Joyce–Upmeier [52, Theorem 4.1] on the moduli of objects.

Proof. Three layers stack. PTVV [89, Theorem 0.3] produces the (-1)-shifted symplectic form on the derived moduli of CY_3 coherent sheaves; Bussi–Brav–Joyce [11, §5] produce the d-critical-locus chart structure on the classical truncation; Bussi–Brav–Dupont–Joyce–Szendrői [11, Theorem 6.9] produce the perverse sheaf of vanishing cycles with its chart cocycle. The orientation is the datum needed to unify chart-local Φ_W into a global perverse sheaf $\Phi_{\mathfrak{M}}$; without it, the chart-cocycle has a sign ambiguity that obstructs the descent to a perverse sheaf [11, §6.4]. Joyce–Upmeier [52, Theorem 1.11] construct the multiplicative orientation cocycle on the universal extension stack; this is the CY_3 projective theorem for the retained Hall orientation. □

Reduced refinement on $K3 \times E$. On $X = K3 \times E$, the (-1)-shifted symplectic structure $\omega_{\mathfrak{M}}$ on $\mathfrak{M}(X)$ does not restrict to a (-1)-shifted symplectic structure on $\mathfrak{M}^{\mathrm{red}}(X)$; the cosection-twist of § D.4 contracts $\omega_{\mathfrak{M}}$ by σ, and the *reduced* (-1)-shifted form $\omega_{\mathfrak{M}}^{\mathrm{red}}$ is non-degenerate only after restriction to the cosection vanishing locus [76, §2.6]. The corresponding reduced d-critical-locus structure is supplied by the Maulik–Toda [76, Theorem 2.7] extension; the reduced perverse sheaf of vanishing cycles $\Phi_{\mathfrak{M}}^{\mathrm{red}}$ and the reduced orientation line o_C^{red} are exactly the data $(O1)$ of § 9. In the notation of (P4), the reduced orientation is twisted by a parity-flip line Π generated by the cosection cokernel,

$$o_C^{\mathrm{red}} \;=\; o_C^{\mathrm{std}} \otimes \Pi,$$

absorbing the dimension shift $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$ into a $\mathbb{Z}/2$-grading shift on the oriented BBDJS line.

PROPOSITION 4.8 *(BBDJS reduced d-critical orientation on $K3 \times E$; ∂, $(O1) + (O1)^+$).* Let $\mathfrak{M}_c^{\mathrm{red},+}(X)$ be the cosection-reduced moduli of semistable sheaves of effective dyonic class c on $X = K3 \times E$. The hypotheses *strong reduced orientation on the $K3 \times E$-quotient self-Ext frame bundle (O1) and Weyl-equivariant transport along $W^{(2)}(\Lambda_{II}^{2,1})$ reflections (O1)$^+$* of § 9 are exactly the following. There exist a multiplicative orientation cocycle of Joyce–Upmeier type extending equivariantly across the reduced E-quotient, the σ_S-cosection-induced parity flip line Π, and a connected BE-equivariant trivialisation of the small-orbit gerbe class $\overline{\beta}_C^E$ on the finite-stabilizer strata, such that

$$\Phi^{\mathrm{red}}_{\mathfrak{M}_c^{\mathrm{red},+}(X)} \otimes o_c^{\mathrm{red}}$$

descends through the free E-translation quotient and trivializes the Pin^c-lift gerbe class of the reduced derived self-Ext frame bundle. The strong reduced orientation $(O1)$ is established from the retained quotient-orientation datum. Joyce–Upmeier [52, Theorem 1.11] supplies the projective Calabi–Yau-threefold orientation technology used here; the quasi-projective and quotient descent belongs to the retained $(O1_{\mathrm{quot}})$ datum. The Cao–Kool–Monavari CY_4 theorem [16, Theorem 1.6] is a fourfold analogue, not an ingredient of the $K3 \times E$ CY_3 reduction. The free E-quotient, finite-stabilizer, and Weyl-transport theorem on $K3 \times E$ follows only relative to the retained quotient-orientation datum and the chosen Weyl lifts of Theorems 7.12 and 7.13.

Proof. The construction of the reduced perverse sheaf $\Phi^{\mathrm{red}}_{\mathfrak{M}_c^{\mathrm{red},+}(X)}$ on the cosection-reduced moduli is by Maulik–Toda [76, Theorem 2.7]; the orientation gerbe lifts to the reduced Joyce d-critical locus by the BBDJS [11, §6.4] chart-cocycle calculation. The multiplicative *equivariant* orientation cocycle across the E-quotient and across the finite-stabilizer locus $BE[N]$-strata is the joint vanishing of $(O1)$ and $(O1)^+$: on the connected BE-equivariant trivialisation of $\alpha_C^E \in H_E^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$ of Definition 0.9 it suffices to trivialize the small-orbit gerbe class $\overline{\beta}_C^E$ on $BE[N]$-strata, governed by the Klein-four detection criterion of Lemma 7.9. The Weyl-equivariant transport across $W^{(2)}(\Lambda^{2,1}_{II})$ reflections is the orientation character calculation of Theorem 0.35 on the three type-II walls. Theorems 7.12 and 7.13 supply these transports on $K3 \times E$; the Pin^c-lift theory of [51, 52] supplies the analytic technology in the Calabi–Yau-threefold setting. □

D.6 Stable-envelope transport: from Hall to Yangian

The level-3 quantum-group structure on $G(X) = D_\hbar(Y^+(X))$ of § D.2 is, on toric ambient or quiver varieties, identified by Maulik–Okounkov [74] with the geometric R-matrix of the *stable envelope*, providing the link from the Hall side to the Yangian / quantum-group side.

Definition 4.9 *(Stable envelope).* Let $\mathfrak{M}$ be a smooth symplectic variety with a torus T-action and a finite T-fixed locus $\mathfrak{M}^T$; let $C \subset \mathrm{Lie}(T)$ be a chamber of the T-equivariant cohomology of $\mathfrak{M}$. The *stable envelope* [74, §3] is the T-equivariant cohomological correspondence

$$\mathrm{Stab}_C \;:\; H^\bullet_T(\mathfrak{M}^T) \longrightarrow H^\bullet_T(\mathfrak{M}),$$

characterised by support, normalization, and degree axioms [74, Theorem 3.3.4]. The geometric R-matrix is the comparison

$$R^{\mathrm{MO}}_{C,C'} \;=\; \mathrm{Stab}^{-1}_{C'} \circ \mathrm{Stab}_C \;\in\; \mathrm{End}(H^\bullet_T(\mathfrak{M}^T))$$

across two chambers C, C'; it satisfies the Yang–Baxter equation by [74, Theorem 4.6.1].

THEOREM 4.10 *(Maulik–Okounkov;* [74] *Theorem 10.2.1; β, proved on quiver varieties; conditional on $K3 \times E$ compact non-toric).* Let $\mathfrak{M} = \mathfrak{M}_{Q,\mathbf{v},\mathbf{w}}$ be a Nakajima quiver variety [79, 80] attached to a finite quiver Q with dimension vectors $\mathbf{v}$, $\mathbf{w}$. The stable envelope Stab_C makes $H^\bullet_T(\mathfrak{M}_{Q,\mathbf{v},\mathbf{w}})$ a module over a Yangian $Y_\hbar(\mathfrak{g}_Q)$ attached to Q, with $\mathfrak{g}_Q$ the Kac–Moody Lie algebra of the quiver, and the geometric R-matrix R^{MO} is the rational R-matrix of $Y_\hbar(\mathfrak{g}_Q)$.

Proof. The Maulik–Okounkov construction proceeds through: (i) the existence and uniqueness of the stable envelope (axioms of support, polarisation, normalization), [74, §3.3]; (ii) the Yang–Baxter equation as a chamber-wall cocycle, [74, §4]; (iii) the identification of the Bethe algebra generated by the matrix elements of Stab with a Yangian, [74, §5–10]. The Schiffmann–Vasserot identification $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$ of [92, Theorem A] is the affine A_0 specialization; $\mathcal{W}_{1+\infty}$ arises *only after* Drinfeld doubling and Fock-evaluation. The three-arrow chain $\mathrm{CoHA} = Y^+ \hookrightarrow Y \xrightarrow{\mathrm{ev}_\lambda} \mathrm{End}(\mathcal{W}_{1+\infty}[\lambda]\text{-vac})$ of [59], [92] traverses the level-2 → level-3 → level-3-evaluated passage; the named identity on each arrow measures the structural distance from $\mathrm{CoHA}(\mathbb{C}^3)$ to $\mathcal{W}_{1+\infty}$. □

$K3 \times E$ specialization: open. On $X = K3 \times E$, the moduli $\mathfrak{M}^+_{\mathrm{eff}}(X)$ is not a Nakajima quiver variety, the ambient is compact non-toric, and the global stable envelope is unconstructed. The five obstructions to a global toric/quiver-variety NCCR model of the Schiffmann–Vasserot type obstruct the literal [74] construction. The cosection-twist of § D.4 supplies the *Serre-equivariant reduced substitute*: the cosection-reduced obstruction theory of Theorem 4.6 restores a CY_3-symmetric reduced obstruction theory on the $s \neq 0$ sector, and the Hall-to-Yangian comparison is then expected at the level of the reduced theory and conditional on the stable-envelope transport, after the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum of Appendix 7.

The Hall–Yangian comparison on $K3 \times E$ is therefore conjectural:

$$G(X) \;\stackrel{?}{=}\; D_\hbar(\mathrm{CoHA}^{\mathrm{red},+}_{K3\times E})^{\mathrm{int}}_{\mathrm{compl}} \;\stackrel{?}{=}\; Y_\hbar(\widehat{\mathfrak{g}}_{\Delta_5})$$

where $\widehat{\mathfrak{g}}_{\Delta_5}$ is the central extension of the imported BKM Lie superalgebra of Theorem 9.1, and the equality is conditional on a $K3 \times E$ stable-envelope transport. The Hall–Drinfeld double $\mathcal{D}_\hbar(\mathrm{CoHA}_{K3\times E})$ of Kontsevich–Soibelman [59] places this identification at the same level-3 quantum group seen from the geometric side; it is the open compact non-toric instance of the chain-fusion conjecture.

D.7 Level separations

The Beilinson cut separates the following objects.

(i) $Y^+(X)$ is the cohomological-Hall positive half $H_c^\bullet(\mathfrak{M}^+_{\mathrm{eff}}(X), \Phi_W \otimes o)$ with the Joyce–Song product/coproduct of Theorems 4.1, 4.2; $G(X) = D_\hbar(Y^+(X))^{\mathrm{int}}_{\mathrm{compl}}$ is the Drinfeld double after Hall pairing, completion, integral form, stable-envelope transport, and descent.

(ii) $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$ [92, Theorem A]; $\mathcal{W}_{1+\infty}$ appears only after Drinfeld doubling and Fock evaluation.

(iii) No global Schiffmann–Vasserot/Nakajima atlas is known for compact non-toric $K3 \times E$. The cosection-twist of § D.4 supplies the Serre-equivariant reduced substitute used here.

(iv) $\mathrm{Prim}_\bullet(Y^+(X))$ is the formal primitive Lie superalgebra of $Y^+(X)$; its identification with $\mathfrak{g}_{\Delta_5}$ is the primitive-recognition theorem of Theorem 0.30, conditional on the finite compact-source datum $(S1)$–$(S10)$.

(v) BBDJS [11, Theorem 6.9] requires an *oriented* d-critical locus; the orientation is the square root of the Joyce determinant line, and the multiplicative cocycle is Joyce–Upmeier [52, Theorem 4.1].

(vi) Kiem–Li [55, Theorem 5.0.1] requires the cosection to be surjective off the localised locus; on $K3 \times E$, the cosection degenerates on the $s = 0$ cusp, which is the $(D0)$ degeneration limit controlled by the retained D0-HN criterion of Theorem 7.6.

(vii) The cosection contracts the (-1)-shifted symplectic form; ω^{red} is non-degenerate only after restriction to the cosection vanishing locus [76, §2.6], and the orientation line carries the $\mathbb{Z}/2$-parity flip Π absorbing the dimension shift $\mathrm{vd}^{\mathrm{red}} = \mathrm{vd}^{\mathrm{std}} + 1$.

(viii) $K3 \times E$ is a trivial-canonical threefold. The scalar-trace index is $n = \chi(\mathcal{O}_Y)$ on the closed target, whereas $\chi(\mathcal{O}_{K3\times E}) = \chi(\mathcal{O}_{K3})\chi(\mathcal{O}_E) = 2 \cdot 0 = 0$, and the BBDJS / Joyce–Upmeier machinery applies in the CY_3 form [11, §6], not in the CY_4 Borisov–Joyce [10] or Oh–Thomas [87, 88] form.

D.8 Four local obstruction calculations

Four local calculations enter the Hall correspondence.

(F1) Hall associativity. The Joyce–Song associativity of Theorem 4.1 requires the pentagon coherence of Thom–Sebastiani triple stratification [11, Theorem 5.10(iv)]. The chart-cocycle of the d-critical locus carries the orientation sign whose pentagon coherence is measured by the BBDJS orientation gerbe. That gerbe [11, §6.4] trivializes exactly this ambiguity; the multiplicative cocycle of Joyce–Upmeier [52, §4] extends the trivialisation to the universal extension stack. On $K3 \times E$, the gerbe descends through the free E-quotient and trivializes the small-orbit class by Theorems 7.12 and 7.13.

(F2) Cosection surjectivity. Kiem–Li [55, Theorem 5.0.1] requires $\sigma : \mathrm{Ob} \twoheadrightarrow \mathcal{O}$ on $\mathfrak{M} \setminus \mathfrak{M}^{\mathrm{red}}$. On the $s = 0$ cusp of the dyonic charge lattice of $K3 \times E$, the K3-class component needed for semiregularity can disappear and surjectivity fails, while the K3 holomorphic two-form itself remains nonzero. The smooth primitive nonzero K3-class branch satisfies surjectivity by Maulik–Toda [76, §2]; the cusp branch is the $(D0)$ degeneration limit named at obstruction $(D0)$ of § 9, controlled by the retained D0-HN criterion of Theorem 7.6. The Oberdieck–Pixton scalar $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ is proved on the $s \neq 0$

branch [84, Theorem 3]; on the cusp the chain-level theory degenerates to the Borcherds imaginary-simple-root contribution.

(F3) BBDJS orientation existence on compact non-toric. The Joyce–Upmeier multiplicative cocycle [52, Theorem 4.1] is established on the CY_3 projective ambient; its extension across the cosection-reduced quotient ambient is part of the retained $(O1_{\mathrm{quot}})$ datum. On the $K3 \times E$ quotient by free E-translation, descent of the orientation line requires the small-orbit gerbe class on $BE[N]$-strata to trivialize. The joint vanishing of the connected BE-equivariant degree-two obstruction $\alpha_C^E \in H_E^2(\mathfrak{M}_C^{\mathrm{red}}; \mathbb{F}_2)$ the finite-stabilizer Cech–Borel gerbe classes, and the residual degree-one stabilizer characters is exactly $(O1)$; Theorem 7.12 of Appendix 7 is the quotient-orientation criterion relative to this retained datum. The remaining open data here are the retained quotient orientation and the compact-source recognition rows.

(F4) (−1)-shifted vs (−2)-shifted symplectic. $K3 \times E$ is a trivial-canonical threefold; the moduli $\mathfrak{M}(K3 \times E)$ carries a (-1)-shifted symplectic structure by PTVV [89, Theorem 0.3]. The Borisov–Joyce [10] and Oh–Thomas [87, 88] theories are CY_4 theories. The Calabi–Yau dimension of $X = K3 \times E$ is $\dim_{\mathbb{C}}(K3) + \dim_{\mathbb{C}}(E) = 2 + 1 = 3$, the holomorphic three-form is $\sigma_S \wedge dz_E$, and the shifted symplectic is $[-1]$-shifted by [89, §2.1]. The cosection of § D.4 removes one obstruction direction and increases the reduced virtual dimension by one, but does not reduce the shifted-symplectic degree; ω^{red} is still (-1)-shifted, on the cosection vanishing locus, by the Maulik–Toda [76, Theorem 2.7] extension. The cosection-twist of § D.4 is the Serre-equivariant reduced substitute for the missing global toric/quiver-variety atlas.

The four calculations above are the local Hall forms of $(D0)$, $(O1)$, $(O1)^+$, and $(O2)$. Their vanishing is the hypothesis of the conditional cross-level identity ②↔③. The Hall correspondences above supply the compact-support functorial maps; the recognition $\mathrm{Prim}_\bullet(Y^+(X)) \simeq \mathfrak{g}_{\Delta_5}$ of Theorem 0.30 remains conditional on $(S1)$–$(S10)$. The protected Pfaffian theorem compares the View ③ Pfaffian section with $\mathcal{D}_X = \Delta_5$; the level-Z scalar trace is a later trace operation.

5 Pfaffian wall normal form on type-II walls

The three local Pfaffian wall computations supply the sign data used in Chapter 7 for the orientation-character match

$$\epsilon_o = \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda_{II}^{2,1}),$$

after the global orientation transport of Theorems 7.12, 7.13, and 7.18. Every numerical identity of §5.1 and §5.6 is independently verified by exact rational lattice arithmetic and exact rational expansion of the Borcherds product.

5.1 *The three simple type-II walls*

The hyperbolic root lattice of signature $(2, 1)$ is $\Lambda^{2,1} = \Lambda^{(1,1)} \oplus \langle 2 \rangle$, with basis (f_2, f_3, f_{-2}) and bilinear form $(f_2, f_{-2}) = -1$, $(f_3, f_3) = 2$. Its index-four type-II sublattice is

$$\Lambda_{II}^{2,1} = \{ m f_2 + l f_3 + n f_{-2} \in \Lambda^{2,1} \mid m \equiv n \equiv 0 \pmod 2 \},$$

abstractly $\Lambda_{II}^{2,1} \simeq U(4) \oplus \langle 2 \rangle$ with discriminant -32; the reflection-group index $[W^{(2)}(\Lambda^{2,1}) : W^{(2)}(\Lambda_{II}^{2,1})] = 6$ tracks the orbit data on the chamber-walls and is independent of the sublattice index. The three simple type-II roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with pairwise pairings forming the type-II Cartan matrix

$$A = \big((\partial_i, \partial_j)\big)_{1\le i,j\le 3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}, \qquad \det A = -32.$$

The matrix is symmetric generalized Cartan, hyperbolic (one negative eigenvalue, two positive), and the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ generated by $s_{\partial_1}, s_{\partial_2}, s_{\partial_3}$ is the Coxeter group with relations $s^2_{\partial_i} = 1$ and no braid relation between distinct generators; the off-diagonal value -2 is the infinite Coxeter exponent. It acts on the hyperbolic plane $\mathbb{RP}_{II}$. The Weyl vector ρ determined by $(\rho, \partial_i) = -1$ for $i = 1, 2, 3$ is

$$\rho = \tfrac{1}{2}(\partial_1 + \partial_2 + \partial_3) = f_2 - \tfrac{1}{2}f_3 + f_{-2}.$$

The Cartan matrix, the Weyl vector, the reflection identity $s_{\partial_i}(\partial_i) = -\partial_i$, and the preservation of the form under s_{∂_i} are all direct computations in the BKM Gram.

Remark 5.1 *(Type-II versus type-I).* The lattice $\Lambda^{2,1}$ admits both type-I roots (divisibility $(\partial, \Lambda^{2,1}) = \mathbb{Z}$, simple system $\{f_{-2} - f_2, f_2 - f_3, f_{-2} - f_3\}$, trivial multiplier) and type-II roots (divisibility $2\mathbb{Z}$, simple system $\{\partial_1, \partial_2, \partial_3\}$ above, multiplier -1 on each generator). The Borcherds–Gritsenko–Nikulin denominator algebra $\mathfrak{g}_{\Delta_5}$ is the type-II algebra, with denominator $64^{-1}\Delta_5(2Z)$ on $\Lambda^{2,1}_{II}$; the doubling $2Z$ is the passage from the type-I to the type-II lattice, and the constant $64 = 2^6$ is the theta-normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5$. The Pfaffian wall calculation is type-II throughout.

5.2 *The rank-one type-II Pfaffian crossing*

For each simple type-II wall $\partial \in \{\partial_1, \partial_2, \partial_3\}$, the retained rank-one chart in the datum $(O2_{\text{atlas}})$ has local model

$$U_\partial = T_\partial \times (-\varepsilon, \varepsilon), \qquad s_\partial(t, u) = (t, -u),$$

with wall locus $T_\partial \times \{0\}$ coinciding with the fixed locus of s_∂. The reduced normal self-Ext factor on this retained chart is the rank-one odd Pfaffian crossing

$$K^{\text{cross}}_\partial = [\mathbb{R} \xrightarrow{u} \mathbb{R}], \qquad D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad \operatorname{Pf}(D_u) = u.$$

Choose an oriented tangent Pfaffian line $\mathscr{L}^{\tan}_\partial$ on T_∂ with s_∂-invariant unit v_∂:

$$s_\partial(v_\partial) = v_\partial.$$

The total local Pfaffian section is

$$\operatorname{pf}_\partial = v_\partial\, u.$$

PROPOSITION 5.2 *(Local Pfaffian sign; conditional on the retained rank-one wall chart).* For the local rank-one type-II Pfaffian crossing $(U_\partial, \operatorname{pf}_\partial)$ of §5.2,

$$s^*_\partial \operatorname{pf}_\partial = -\operatorname{pf}_\partial, \qquad s^*_\partial(\operatorname{pf}^2_\partial) = \operatorname{pf}^2_\partial.$$

Equivalently, the local Hall/Pfaffian sign is

$$\epsilon^{\text{loc}}_o(s_\partial) = (-1)^{N^{\text{Pf}}_\partial} = -1, \qquad N^{\text{Pf}}_\partial = 1.$$

Proof. The fixed locus of $s_\partial(t,u) = (t,-u)$ is $\{u = 0\}$, and the wall coincides with the tangent fixed locus, so the determinant factor is split: $\mathscr{L}_\partial^{\tan}$ on T_∂ is the tangent factor and $K_\partial^{\mathrm{cross}}$ is the normal factor. The unit v_∂ is invariant by construction; the skew matrix D_u has Pfaffian u; and s_∂ sends $u \mapsto -u$ while fixing v_∂. Therefore

$$s_\partial^*(v_\partial\, u) = v_\partial\,(-u) = -v_\partial\, u = -\mathrm{pf}_\partial,$$

and squaring gives the second equality. The Pfaffian rank exponent $N_\partial^{\mathrm{Pf}} = 1$ follows because the normal factor is the single odd crossing $[\mathbb{R} \to \mathbb{R}]$. The displayed sign is $(-1)^1 = -1$. □

5.3 Match with the Maass multiplier

The Maass multiplier ν_{Δ_5} of the weight-5 Igusa cusp form is the multiplier system of the section $\Delta_5 \in M_5(\mathbf{Sp}_4(\mathbb{Z}), \nu_{\Delta_5})$ of the automorphic line bundle $\mathcal{L}^5 \otimes \nu_{\Delta_5}$ on the genus-two Siegel space $\mathbf{Sp}(2,\mathbb{Z})\backslash\mathbb{H}_2$. Its values on the type-II reflections in $\mathbf{O}(\Lambda^{2,1})_+ = W^{(2)}(\Lambda^{2,1})$ are computed in [43, Proposition 2.1] (citing Maass [71, (1.3)–(1.5)]) via the exterior-square identification $\mathbf{PSp}_4(\mathbb{Z}) \simeq \Gamma_{3,2} \subset \mathrm{P}\mathbf{O}(\Lambda^{3,2})_+$:

LEMMA 5.3 *(Maass multiplier on simple type-II reflections).* The Maass character of [43, Proposition 2.1], in the normalization of [71, (1.3)–(1.5)], evaluates on the simple type-II reflections as

$$\nu_{\Delta_5}(s_{\partial_i}) = -1, \qquad i = 1,2,3,$$

and trivially on the three $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$ reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$:

$$\nu_{\Delta_5}(s_{f_{-2}-f_2}) = \nu_{\Delta_5}(s_{f_2-f_3}) = \nu_{\Delta_5}(s_{f_{-2}-f_3}) = +1.$$

The character is invariant under rescaling Δ_5 by a nonzero constant: the theta-normalization $[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64$ and the monic $D_5 = 64^{-1}\Delta_5$ give the same multiplier values on generators.

Derivation. Each s_{∂_i} lifts under the exterior-square isomorphism $\mathbf{PSp}_4(\mathbb{Z}) \xrightarrow{\sim} \Gamma_{3,2}$ of Theorem 1.5 to a paramodular involution acting on the genus-two Siegel space; Δ_5 transforms by the Maass character on these involutions, valued -1 on each type-II generator (Maass [71, (1.3)–(1.5)], Gritsenko–Nikulin [43, Proposition 2.1]). The trivial S_3-reflections $s_{f_{-2}-f_2}, s_{f_2-f_3}, s_{f_{-2}-f_3}$ lift to inner Weyl permutations on the level-one paramodular and act with multiplier $+1$.

Combining Proposition 5.2 and Lemma 5.3:

THEOREM 5.4 *(Local orientation match on type-II generators).* Composing Proposition 5.2 and Lemma 5.3, on the three simple type-II reflections,

$$\epsilon_o^{\mathrm{loc}}(s_{\partial_i}) = -1 = \nu_{\Delta_5}(s_{\partial_i}), \qquad i = 1,2,3.$$

The local Pfaffian sign and the Maass multiplier coincide on generators of the type-II reflection group, relative to the retained rank-one wall charts.

The match $\epsilon_o^{\mathrm{loc}} = \nu_{\Delta_5}$ on the three type-II simple reflections is the local seed of the global orientation-character identity. Its extension to the whole type-II Weyl group requires global Hall-stack transport of the local crossing, the $(O2)$ obstruction.

5.4 *Extension to the type-II Weyl group*

5.4.1 Generation and a presentation

The type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ is generated by $s_{\partial_1}, s_{\partial_2}, s_{\partial_3}$ (Nikulin's reflection theorem for $\Lambda^{(1,1)} \oplus \langle 2\rangle$, [43, Theorem 4.1]). The generator pairings $(\partial_i, \partial_j) = -2$ for $i \neq j$ are obtuse and equal to -2; the corresponding Coxeter exponent m_{ij} is the order of $s_{\partial_i}s_{\partial_j}$, which is infinite because the rotation $s_{\partial_i}s_{\partial_j}$ acts on the hyperbolic plane by a hyperbolic translation. Thus

$$W^{(2)}(\Lambda^{2,1}_{II}) = \langle s_{\partial_1}, s_{\partial_2}, s_{\partial_3} \mid s^2_{\partial_i} = 1\rangle$$

is the free product of three involutions, equivalently the (∞, ∞, ∞)-triangle reflection group on the hyperbolic plane.

5.4.2 The determinant character

The determinant character $\det : W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ is the unique homomorphism with $\det(s_{\partial_i}) = -1$ on each generator. The relations $s^2_{\partial_i} = 1$ are respected since $(-1)^2 = 1$, and there are no further relations, so det extends unambiguously. The Maass multiplier $\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})}$ is also a $\{\pm 1\}$-character that equals -1 on each generator (Lemma 5.3); it therefore equals det:

$$\nu_{\Delta_5}|_{W^{(2)}(\Lambda^{2,1}_{II})} = \det = (\text{sign character}).$$

This is the global Maass character on the type-II Weyl group.

5.4.3 The orientation character is a $\{\pm 1\}$-character

The orientation character ϵ_o on the Hall-stack moduli is, by construction, a homomorphism $W^{(2)}(\Lambda^{2,1}_{II}) \to \{\pm 1\}$ provided the orientation cocycle on the half-Hilbert orbit trivializes consistently with the local rank-one crossings of §5.2 across all generators. Under that trivialisation, the global orientation character is determined by its values on $s_{\partial_1}, s_{\partial_2}, s_{\partial_3}$, each equal to -1 by Theorem 5.4. Hence

$$\epsilon_o = \det = \nu_{\Delta_5} \quad \text{on} \quad W^{(2)}(\Lambda^{2,1}_{II}).$$

This is the global orientation-character match relative to the retained quotient orientation and the chosen Weyl transport. Its proof in this appendix establishes the $(O2)$ wall-atlas part: the trivialisation of the orientation cocycle on the half-Hilbert orbit.

5.5 *The (O2) obstruction: half-Hilbert orbit transport*

5.5.1 Statement of (O2)

Definition 5.5 *((O2) local Pfaffian wall normal form, global form).* The obstruction $(O2)$ is the assertion that, for every retained HN height R and every retained type-II wall transport $s_\partial \in W^{(2)}(\Lambda^{2,1}_{II})$, the local rank-one crossing of §5.2, defined on the generic wall chart $U_{\partial,R}$, extends to a strictly compatible system of charts on the half-Hilbert orbit $\mathfrak{H}^{\mathrm{half}}_R$, with the orientation cocycle on this orbit trivialized by an explicit cochain compatible with the local Pfaffian sign on each component, the boundary of each component, and the overlap of every adjacent pair of components.

5.5.2 Local computation and global transport

Theorem 5.4 proves the local sign identity on each of the three simple type-II generators by a single chart computation: *three local sign computations, one per generator*. The global statement of §5.4, $\epsilon_o = \nu_{\Delta_5}$ on the entire Weyl group, requires that these three local seeds extend to a strictly coherent system on the half-Hilbert orbit, namely the entire orbit of the half-Hilbert moduli stack under the $W^{(2)}$-action. The component, boundary, and overlap compatibility data of Definition 5.5 are *not* produced by the local rank-one crossings alone. They are the content of the retained $(O2_{\mathrm{atlas}})$ datum transported by Theorem 7.18.

5.5.3 Two appearances of the constant 4096

The integer 4096 appears in two structures, and the equality is scalar only. On the orientation side, the half-Hilbert orbit of the type-II reflection group on the cosection-reduced self-Ext stack carries 12 binary signs, so the orientation-cocycle support carries $2^{12} = 4096$ sign assignments; this is the reading on which the $(O2_{\mathrm{atlas}})$ transport of Appendix 7, Theorem 7.18, operates. On the scalar side, the Oberdieck–Pixton normalization

$$Z^X_{\mathrm{OP}}(P_{\mathrm{OP}}, Q_{\mathrm{OP}}, T_{\mathrm{OP}}) = -\big(\chi_{10}^{\mathrm{OP}}\big)^{-1} = -4096\,\Delta_5(Z)^{-2}, \qquad \chi_{10}^{\mathrm{OP}} = D_5^2 = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2,$$

exhibits $4096 = 64^2 = ([q^{1/2}r^{1/2}s^{1/2}]\Delta_5)^2$ as the squared theta-leading coefficient, with $-4096 = (-1)\cdot 64^2$ the OP scalar branch convention times the squared theta-leading ([85, 86]; arithmetic check by the product expansion).

The two values coincide because $2^{12} = 64^2$: the half-Hilbert orbit sign count and the squared theta-leading are the same integer after orientation has been forgotten. Theorem 7.16 of Appendix 7 makes the scalar separation precise: the orientation character $\epsilon_o = \nu_{\Delta_5}$ is fixed, relative to the retained orientation and Weyl-transport data, by wall transport of the sign-count, while the OP scalar magnitude 4096 enters $Z^X_{\mathrm{OP}} = -4096\,\Delta_5^{-2}$ through the squared theta-leading.

5.5.4 The retained (O2) wall-atlas route

The retained $(O2_{\mathrm{atlas}})$ datum consists of:

(O2.a) A strictly compatible system of charts $U_{\delta,R}$, $\delta \in W^{(2)}(\Lambda^{2,1}_{II})$, on the half-Hilbert orbit at height R, each presenting the rank-one type-II Pfaffian crossing of §5.2.

(O2.b) An orientation cocycle $c_o \in Z^2(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2)$ on the orbit, with explicit cochain trivialisation $\eta_o \in C^1(W^{(2)}(\Lambda^{2,1}_{II}); \mathbb{F}_2)$ satisfying $\delta\eta_o = c_o$, compatible with the chosen charts.

(O2.c) Strict compatibility of the trivialisation with the component decomposition, the boundary of each component, and the overlap of every adjacent pair of components — equivalently, the component, boundary, and overlap residual entries of the retained wall-atlas datum vanish with explicit null-trivialisations.

Once (O2.a)–(O2.c), the retained quotient orientation, and the chosen Weyl transport are fixed, Theorem 5.4 extends to the global Weyl-group character match $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$.

5.6 First-timelike parity split as a recognition obligation

The orientation-character match of §5.4 is the conditional *multiplier* comparison. The recognition theorem $\mathcal{P}_X \cong \mathfrak{g}_{\Delta_5}$ of Chapter 7 requires further data, namely full parity dimensions per root degree, not just signed dimensions. The first nontrivial test occurs at the imaginary simple root $\delta_{123} = \delta_1 + \delta_2 + \delta_3$:

Lemma 5.6 *(First-timelike parity split).* The signed root multiplicity at δ_{123} is

$$\mathrm{smult}\ \mathfrak{g}_{\Delta_5,\alpha(\delta_{123})} = f(1,1) = -64,$$

and the full target parity computed from the Gritsenko–Nikulin presentation is

$$\big(\mathrm{mult}_{\bar{0}}(\delta_{123}),\ \mathrm{mult}_{\bar{1}}(\delta_{123})\big) = (29, 93), \qquad 29 - 93 = -64.$$

The decomposition of the even part is

$$29 = 2 + 3 \cdot 9,$$

where $2 = \frac{1}{3}\,\mu_{\mathrm{Mob}}(1)\,3!$ is the Witt dimension of the free Lie algebra on three generators in multidegree $(1,1,1)$ and $3 \cdot 9$ is the three real–isotropic mixings each contributing the eta-9 multiplicity $\tau = 9$ of the doubled isotropic ray. The odd part 93 is the absolute value of the timelike Borcherds correction $m(\delta_{123})$, with

$$m(\delta_{123}) = -\big[\,q^1 r^1 s^1\,\big]\big((qrs)^{-1/2} 64^{-1}\Delta_5\big) = -93.$$

Proof. The signed multiplicity is $f(1,1) = -64$ by direct expansion of $\phi_{0,1}$ in the Eichler–Zagier normalization [32, §2.2]. The full parity decomposition is the Gritsenko–Nikulin presentation reading: the even part counts free Lie words in three real-root letters, the odd part counts simple imaginary timelike roots. Witt's dimension formula $W_{\mathrm{multidegree}}(1,1,1) = \frac{1}{n}\sum_{d|\gcd}\mu_{\mathrm{Mob}}(d)\,(n/d)!/\prod(n_i/d)!$ with $n = 3$, $\gcd(1,1,1) = 1$ gives $\frac{1}{3}\cdot 1 \cdot 6 = 2$. The three real–isotropic mixings each contribute $\tau = 9$ on the primitive isotropic rays (the eta-9 expansion $\prod_{k\geq 1}(1-q^k)^9$ of [45, Theorem 2.1, (5.1)]); summing over the three independent isotropic directions gives $3 \cdot 9 = 27$. Hence $29 = 2 + 27$. The odd part is $-m(\delta_{123}) = 93$ by the Gritsenko–Nikulin imaginary simple-root formula. □

Corollary 5.7 *(Signed dimension is not recognition).* A graded super vector space with parity dimensions $(\dim_{\bar 0}, \dim_{\bar 1}) = (29, 93)$ and zero bracket has the same signed dimension -64 as the $\mathfrak{g}_{\Delta_5,\alpha(\delta_{123})}$ component but the wrong Lie superalgebra structure. Recognition of $\mathcal{P}_X$ with $\mathfrak{g}_{\Delta_5}$ at degree δ_{123} requires both parity dimensions 29 and 93 and the Borcherds–Kac–Moody bracket; the determinant identity alone does not imply this recognition statement.

5.6.1 Higher-degree consistency

The same Borcherds–Kac arithmetic gives higher-degree signed checks and the finite rows where full target parity is known:

- *Doubled isotropic.* For $\delta = 2a_{ij}$, $i \neq j$,

  $$(\mathrm{smult},\ \tau,\ \mathrm{gap}) = (10,\ 9,\ 1),$$

  the signed multiplicity 10 splits as eta-9 multiplicity 9 plus the Kac null-root residual 1 of the rank-two affine subroot system, for each of the three pairs (i,j).
- *Height-four timelike.* For the three rows

  $$C_{k,2} = a_{ij} + 2\delta_k, \qquad \{i,j,k\} = \{1,2,3\},$$

  namely coefficient vectors $(2,1,1)$, $(1,2,1)$, $(1,1,2)$,

  $$(\mathrm{smult},\ m,\ \mathrm{gap}) = (108,\ 90,\ 18),$$

  each with the same triple of integers. These rows are not Weyl translates of δ_{123}; rather

  $$s_{\delta_k}(\delta_{123}) = a_{ij} + 3\delta_k = C_{k,3}.$$

  Thus $C_{k,2}$ is a signed target row. The integer $18 = 108 - 90$ is the signed lower-bracket residual at height four, not a full parity block.
- *Real-string exponents.* The Borcherds–Kac real-root strings through δ_{ij}-style mixed degrees give (real-real exponent) $= 3$ and (complement-isotropic exponent) $= 5$; the timelike string through δ_{123} has exponent 3. All three are pairwise consistent under Weyl reflection.

These integers are the degree-wise Borcherds–Kac arithmetic used in the recognition criterion.

5.7 Local Pfaffian sign computation

Let $\delta \in \{\delta_1, \delta_2, \delta_3\} \subset \Lambda^{2,1}$, with $(f_2, f_{-2}) = -1$ and $(f_3, f_3) = 2$. On a normal chart

$$U_\delta = T_\delta \times (-\epsilon, \epsilon), \qquad s_\delta(t,u) = (t,-u), \qquad \mathbb{H}_\delta \cap U_\delta = T_\delta \times \{0\},$$

choose an oriented tangent Pfaffian unit v_δ satisfying $s_\delta^* v_\delta = v_\delta$. The rank-one normal complex is

$$K_\delta^{\mathrm{nor}} = [\mathbb{R} \xrightarrow{u} \mathbb{R}], \qquad D_u = \begin{pmatrix} 0 & u \\ -u & 0 \end{pmatrix}, \qquad \mathrm{Pf}(D_u) = u.$$

Thus

$$\mathrm{pf}_\delta = v_\delta u, \qquad s_\delta^* \,\mathrm{pf}_\delta = -\,\mathrm{pf}_\delta, \qquad s_\delta^* \,\mathrm{pf}_\delta^2 = \mathrm{pf}_\delta^2\,.$$

The normal Pfaffian exponent is $N_\delta^{\mathrm{Pf}} = 1$, hence

$$\epsilon_o^{\mathrm{loc}}(s_\delta) = (-1)^{N_\delta^{\mathrm{Pf}}} = -1 = \nu_{\Delta_5}(s_\delta)$$

for the three simple type-II reflections, by the Maass-character calculation of Gritsenko–Nikulin [43, Proposition 2.1].

The $(O2_{\mathrm{atlas}})$-transport formulas of Appendix 6 extends this local computation to the global half-Hilbert orbit by component, boundary, and overlap compatibility.

5.8 O2 relative summary

- **(O2.a) Strictly compatible chart system on the half-Hilbert orbit.** The local rank-one crossing is fixed at each simple type-II generator (§5.2); strict compatibility across all elements of $W^{(2)}(\Lambda_{II}^{2,1})$ is the half-Hilbert orbit chart system retained in $(O2_{\mathrm{atlas}})$ and transported by Theorem 7.18.
- **(O2.b) Orientation cocycle trivialisation.** $c_o \in Z^2(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2)$ is the orientation cocycle; its trivializing cochain $\eta_o \in C^1(W^{(2)}(\Lambda_{II}^{2,1}); \mathbb{F}_2)$ is part of the chosen Weyl-transport datum of Theorem 7.13 and is transported in Theorem 7.18.
- **(O2.c) Component, boundary, overlap compatibility.** The residual entries $\mathfrak{d}_{\delta,R}^{\mathrm{chart}}, \mathfrak{d}_{\delta,R}^{\mathrm{wall}}, \mathfrak{d}_{\delta,R}^{\mathrm{split}}, \mathfrak{d}_{\delta,R}^{\mathrm{unit}}, \mathfrak{d}_{\delta,R}^{\mathrm{rank}}, \mathfrak{d}_{\delta,R^+R}^{\mathrm{tr}}$ of Proposition 0.23 are fixed at finite height R for the local rank-one models; the global compatibility across the half-Hilbert orbit is retained in $(O2_{\mathrm{atlas}})$ and transported by Theorem 7.18.
- $(R1)$–$(R2)$ **scalar separation.** Separated in Appendix 7, Theorem 7.16: the orientation-cocycle support on the half-Hilbert orbit has size 2^{12} (R1), and this integer is equal to 64^2, the theta-leading square on the scalar side (R2). This is not an identification of the two data.

Relative to the chosen Weyl lifts and $(O2_{\mathrm{atlas}})$, Theorems 7.13, 7.18, and 7.16 extend Theorem 5.4 from the three local generators to the global character match $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda_{II}^{2,1})$.

6 Eight finite constructions for the Dirac–Igusa pro-object

The Dirac–Igusa pro-object

$$\mathfrak{D}_X^{\mathrm{DI}} = \varprojlim_R \left(\mathcal{A}_{X,R}^{E_3}, \mathcal{F}_{X,R}^{\mathrm{hyb}}, \Gamma_{X,R}, \Pi_X, \Phi_R, o_R, \mathcal{H}_R, P_R^{\Pi}, C_{X,R}, \Theta_{\mathrm{Kos},R}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R}, \mathrm{Rec}_R\right)$$

of Theorem 0.34 and Definition 0.8 is the finite source datum. Its protected Pfaffian and orientation character, after the retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained ($O2_{\text{atlas}}$) wall datum, together with the finite Pfaffian section datum, of Appendix 7, give the Igusa section and Maass character,

$$\mathrm{Pf}_{\text{prot}}(\mathfrak{D}_X^{\text{DI}}) = \mathcal{D}_X = \Delta_5, \qquad \epsilon_o = \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The primitive identification $P_X^{\Pi,\text{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the separate finite compact-source recognition datum. The finite Harder–Narasimhan height R is relative to the standing hypotheses [K3×E-fin], [K3×E-atlas], [ML], [E_3-HolFA], and [K3→E-sp]. These finite data constitute the datum $\mathfrak{R}_X = (L_X, H_X, O_X, \mathsf{Desc}_X, \mathsf{Rec}_X)$ of Theorem 0.34. The charge, moduli, and hybrid data give (L_X) and (H_X); the reduced orientation data give the local content of (O_X); the Hall correspondence object gives the primitive product; the Pfaffian–Dirac line carries the orientation character; the Koszul source coalgebra and primitive-recognition data give (Desc_X) and (Rec_X).

Throughout, $X = S \times E$ with S a smooth complex K3 surface and E a smooth elliptic curve. $\sigma = \sigma_{s,t}$ denotes the chosen Liu product-stability datum on the Abramovich–Polishchuk heart in $\mathsf{D}^b(\mathrm{Coh}(X))$ [1, 61]. The HN sector (σ, S) is fixed and the cofinal subsystem $\mathcal{R}_{\text{HN}} = \{R_0 < R_1 < \cdots\}$ is the one of Definition 0.8, governed by [K3×E-fin], [K3×E-atlas], and [ML].

The ordering charge→moduli→hybrid→orientation→Hall →Dirac+Pfaffian→Koszul→recognition is forced by the mathematics: a Pfaffian line needs the orientation datum, the orientation datum needs the quotient Hall atlas, the Hall bracket needs the hybrid correspondences, and primitive recognition needs the normal-ordered Hall primitive layer.

The four obstructions are attached to the finite data: (D_0) is the cofinal-trivialization datum at every finite R, with components in the charge, moduli, hybrid, orientation, Hall, and Pfaffian data; $(O1)$ and $(O1)^+$ live in the reduced orientation data, with Coxeter coherence feeding the Pfaffian transport; $(O2)$ lives in the Pfaffian wall normal form. Primitive recognition inherits these obstruction classes and adds the exact-completion residual (R8) of Definition 0.31.

6.1 Charge window

6.1.0.1 Hypotheses. HN height $R \in \mathcal{R}_{\text{HN}}$; the formal dyonic Mukai lattice Γ_X^{form} of Definition 2.1; the Gram map $\Pi_X : \Gamma_X^{\text{form}} \to \Gamma_{\text{gram}}$ and the cocycle B of Lemma 2.4; the active support $\Gamma_{\text{act}} = \{\gamma \in \Gamma_{\text{eff}} \mid f(nm, l) \neq 0\}$ of the Igusa Jacobi form $\phi_{0,1}$; the cofinal subsystem $\mathcal{R}_{\text{HN}}$.

6.1.0.2 Charge-window datum. A finite cusp-positive active window

$$A_R \subset \Gamma_{\text{act}} \subset \Gamma_{\text{gram}}, \qquad A_R \text{ downward saturated for the cofinal target-window order,}$$

together with the normal-ordered extension

$$\widehat{\Gamma}_R = \{(c, T) \mid c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\} \subset \widehat{\Gamma}_X, \qquad \Gamma_R^{\Pi} = \overline{\Pi}_X(\widehat{\Gamma}_R) \subset \Gamma_{\text{gram}},$$

the lift $L : \Gamma_{\text{gram}} \to \Gamma_X^{\text{form}}$ of Lemma 2.8, and the finite test window Γ_R^{test} of Definition 0.8.

6.1.0.3 Construction.

(i) Compute the active support cone

$$\Gamma_{\text{act}} = \{(n, l, m) \in \Gamma_{\text{eff}} \mid f(nm, l) \neq 0\}$$

from the Fourier expansion of the index-one weak Jacobi form $\phi_{0,1}(\tau, z) = \sum_{n,l} c(n, l) q^n r^l$ of [32], with the index substitution $f(nm, l) := c(nm, l)$ tabulated in Appendix 8 for the $N = 1$ Igusa boundary.

(ii) Set

$$A_R^\circ = \{\gamma = (n, l, m) \in \Gamma_{\mathrm{act}} \mid n + |l| + m \leq R\}.$$

Form its downward saturation

$$A_R = \{\gamma' \in \Gamma_{\mathrm{act}} \mid \exists \gamma \in A_R^\circ,\ 0 < \gamma' \leq \gamma,\ \gamma - \gamma' \in Q_+\},$$

in the sense of Definition 0.31.

(iii) Normal-ordered lift. For each $c \in \Gamma_R^{HN}$ compute the B-translate orbit

$$\mathcal{T}_R(c) = \Big\{\sum_{i<j} B(c_i, c_j) \ \Big|\ c = \sum_i c_i,\ c_i \in F_{\sigma,S}(R),\ \sum_i h_S(c_i) \leq R\Big\}.$$

Set $\widehat{\Gamma}_R = \{(c, T) : c \in \Gamma_R^{HN},\ T \in \mathcal{T}_R(c)\}$.

(iv) Finite target test window: $\Gamma_R^{\mathrm{test}} = \overline{\Pi}_X(\widehat{\Gamma}_R)$, compatibly with $R \to R'$ by inclusion.

(v) Obtain $(A_R, \widehat{\Gamma}_R, \Gamma_R^{\Pi}, \Gamma_R^{\mathrm{test}}, L)$.

6.1.0.4 Identities. A_R is finite and downward saturated; $\overline{\Pi}_X(\widehat{\Gamma}_R) = \Gamma_R^{\Pi}$ lies in Γ_{gram}; $\Gamma_R^{HN} \subset F_{\sigma,S}(R)$ is finite by [K3×E-fin]; the cocycle identity $\Pi_X(c + c') = \Pi_X(c) + \Pi_X(c') + B(c, c')$ is an axiom of the cocycle.

6.1.0.5 Obstruction classes. None; this part is purely lattice combinatorics from the imported Gritsenko–Nikulin Borcherds product [7, 8, 43–45]. The downward saturation is the input requirement of (R0) in Definition 0.31.

6.2 Finite moduli

6.2.0.1 Hypotheses. The active window A_R and lift L of the charge window; Liu's product stability condition $\sigma_{s,t}$; the standing hypotheses [K3×E-fin] and [K3×E-atlas].

6.2.0.2 Finite-moduli datum. A finite retained Liu–Hilbert type set C_R, whose elements are tuples

$$c = (\widehat{c}, [a_c, b_c], (P_{c,i}), N_c),$$

and a lift $\ell_R : \Gamma_R^{\Pi} \to C_R$ with $\Pi_X(\widehat{c}_{\ell_R(\gamma)}) = \gamma$; the family of finite-type quasi-smooth derived Artin substacks

$$\mathfrak{M}_{R,c}^{ss} = \mathfrak{M}_{\sigma_{s,t}}^{ss}(X, c)_{\leq R} \subset \mathrm{Perf}(X), \qquad c \in C_R,$$

of $\sigma_{s,t}$-semistable objects of retained numerical class $\widehat{c}$, fixed cohomological amplitude $[a_c, b_c]$, fixed Hilbert polynomials $(P_{c,i})$, regularity bound N_c, and HN height $\leq R$; the scalar-rigidified truncation $\mathfrak{M}_{R,c}^{\mathrm{sc}}$; the universal perfect complex $\mathcal{A}_{R,c}$ on $X \times \mathfrak{M}_{R,c}^{\mathrm{sc}}$; the finite stratification by retained closed substacks of finite residual inertia.

6.2.0.3 Construction.

(i) Form C_R from the lifts $L\gamma$, $\gamma \in A_R$, by choosing the amplitude, Hilbert-polynomial, and regularity bounds above. Close C_R under retained extension sums, subobjects, quotients, duals, and intermediate flag classes.

(ii) For each $c \in C_R$ cut the substack of $\sigma_{s,t}$-semistable objects of retained HN type c by HN height $\leq R$:

$$\mathfrak{M}^{ss}_{R,c} = \{[A] \in \mathrm{Perf}(X) \mid \sigma_{s,t}\text{-semistable, } \mathrm{hn}(A) \leq R,\ \nu(A) = \widehat{c},\ A \text{ has HN type } ([a_c, b_c], (P_{c,i}), N_c)\}.$$

The retained substack is required to specialization-closed in the bounded Quot/Postnikov ambient stack.

(iii) Verify finite-typeness: [K3×E-fin] gives bounded Quot/Postnikov charts for the retained HN types. Liu product stability alone is not the finiteness assertion. PTVV [89] on the trivial-canonical threefold X gives the (-1)-shifted symplectic structure on $\mathrm{Perf}(X)$, restricted to the retained semistable sector.

(iv) Quasi-smoothness: the universal perfect complex $\mathcal{A}_{R,c}$ supplies a relative cotangent complex of bounded Tor-amplitude on the retained charts. Tangent at A is $T_A\mathfrak{M}^{ss}_{R,c} \simeq \mathrm{RHom}_X(A, A)[1]$.

(v) Scalar rigidification: form $\mathfrak{M}^{sc}_{R,c}$ by killing the scalar $\mathbb{G}_m$-automorphism of c-stable factors.

(vi) Two-step flag stack $\mathfrak{F}^{(2)}_{R,c,c',c''}$ and extension stack $\mathfrak{E}_{R,c,c'}$ are the retained closed flag-Quot/filtration stacks over the finite HN-type set, with induced subobject and quotient HN types. A raw stack of exact triangles is only formal until this compactified model, or an explicitly admissible compact-support $q_!$, is fixed. In the compactified model the stacks are finite type and quasi-smooth in the retained sector by Proposition 0.10.

(vii) Obtain $(C_R, \ell_R, \{\mathfrak{M}^{ss}_{R,c}\}, \{\mathfrak{E}_{R,c,c'}\}, \{\mathfrak{F}^{(2)}_{R,c,c',c''}\})$ with universal perfect complexes and the finite stratification.

6.2.0.4 Identities. $\mathfrak{M}^{ss}_{R,c}$ is a finite-type quasi-smooth derived Artin substack of $\mathrm{Perf}(X)$; the scalar rigidification leaves only finite residual inertia $H_S \subset E_{\mathrm{tors}}$ on each connected stabilizer component, by Proposition 0.10; PTVV (-1)-shifted symplectic structure restricts.

6.2.0.5 Obstruction classes. [K3×E-fin] is invoked here in its product-boundedness form and supplies the finite-type assertion not in the cited literature for the K3×E threefold; it is part of the standing datum. [K3×E-atlas] supplies the quasi-smooth d-critical charts. These enter (D_0) at the level of the finite Hall atlas of Theorem 0.34.

6.3 Hybrid wrapped factorization carrier

6.3.0.1 Hypotheses. The finite moduli stacks $\{\mathfrak{M}^{sc}_{R,c}\}$ of the finite moduli datum; the projection-finite/wrapped split by elliptic degree $b^{\mathrm{geom}}_R : \Gamma^{HN}_{\sigma,S}(R) \to \mathbb{Z}_{\geq 0}$ of Definition 0.5; a fixed origin $o_E \in E$; the Abramovich–Polishchuk/Liu wrapped anchor candidate.

6.3.0.2 Hybrid factorization datum. The finite hybrid local/wrapped factorization datum

$$\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\leq R} = (C_{\sigma,S,\leq R}, b^{\mathrm{geom}}_R, \mathcal{F}^{\mathrm{loc}}_R, \mathcal{G}^{\mathrm{wr}}_R, \mathfrak{E}^{\mathrm{loc}}_R, \mathfrak{E}^{\mathrm{hyb}}_R, \mathfrak{E}^{\mathrm{wr}}_R, \mathfrak{Flag}^{\mathrm{hyb}}_R, \mathsf{Corr}^{E,\mathrm{hyb}}_R, \lambda_R, Q_{E,R}, \theta^Q_R, o_R, I^{\mathrm{prot}}_R)$$

of Definition 0.5, together with the hybrid Ran base $\mathrm{Ran}^{\mathrm{hyb}}(E)$, the eight-word two-step binary flag atlas $\{w \in \{\mathrm{L},\mathrm{W}\}^3\}$, and the quotient-after-correspondence pseudofunctor inducing $Q_{E,R}$.

6.3.0.3 Construction.

(i) *Elliptic-degree split.* Decompose $\Gamma^{HN}_{\sigma,S}(R) = \Gamma^{\mathrm{loc}}_R \sqcup \Gamma^{\mathrm{wr}}_R$ where $\Gamma^{\mathrm{loc}}_R = \{b^{\mathrm{geom}}_R = 0\}$ and $\Gamma^{\mathrm{wr}}_R = \{b^{\mathrm{geom}}_R > 0\}$. Verify additivity $b^{\mathrm{geom}}_R(\gamma+\gamma') = b^{\mathrm{geom}}_R(\gamma) + b^{\mathrm{geom}}_R(\gamma')$ on retained extensions remaining in the quotient.

(ii) *Projection-finite local sector.* For $\alpha \in \Gamma^{\mathrm{loc}}_R$ and finite index set I, take the closed-support incidence stack over E^I:

$$\mathfrak{M}^{\mathrm{loc,cl}}_{\alpha,R,I} = \left\{(A,\mathbf{e}) \in \mathfrak{M}^{\mathrm{sc}}_{R,L\alpha} \times E^I \;\middle|\; p_E(\mathrm{Supp}\,A) \subset \bigcup_{i\in I} S \times \{e_i\}\right\}.$$

Form $\mathscr{H}^{\mathrm{loc}}_{\alpha,R,I} = (\pi_{\alpha,R,I})_!(\Phi^{\mathrm{loc}}_{\alpha,R,I} \otimes o^{\mathrm{loc}}_{\alpha,R,I})$ and the open-support sheaf $\mathcal{F}^{\mathrm{loc}}_{\alpha,R}(U)$ over $U \subset E$ by restricted Borel–Moore configuration sums.

(iii) *Wrapped prequotient sector.* For $\eta \in \Gamma^{\mathrm{wr}}_R$, form the rigidified prequotient $\mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \to \mathcal{M}^{\mathrm{wr}}_{\eta,R}$ by the Abramovich–Polishchuk / determinant anchor

$$\lambda_\eta : \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R} \longrightarrow E, \qquad \lambda_\eta(A) = \det Rp_{E,*}A \otimes O_E(-\chi(A)\cdot 0_E),$$

divided by a finite cover when $\chi(A) \neq 1$ so that $\lambda_\eta(t\cdot A) = \lambda_\eta(A) + t$ holds. Verify the losslessness residual $\mathfrak{o}^{\lambda,\mathrm{loss}}_R = 0$ by the Ext-distinguishing test of Definition 0.5, separating $W_0 = i_{s,*}(O_E \oplus O_E)$ from $W_L = i_{s,*}(L \oplus L^{-1})$.

(iv) *Mixed local/wrapped correspondences.* For an ordered hybrid input $\alpha_- \in \Gamma^{\mathrm{loc}}_R$, $\eta \in \Gamma^{\mathrm{wr}}_R$, $\beta_+ \in \Gamma^{\mathrm{loc}}_R$, form $\mathfrak{E}^{\mathrm{mix}}_{\alpha_-;\eta;\beta_+,R}$ before the reduced E-quotient.

(v) *Eight-word two-step flag atlas.* Retain flag stacks $\mathfrak{F}^{(2),w}_R$ for each $w \in \{\mathrm{L},\mathrm{W}\}^3 = \{\mathrm{LLL},\mathrm{LLW},\mathrm{LWL},\mathrm{WLL},\mathrm{LWW},\mathrm{WLW},\mathrm{WWL},\mathrm{WWW}\}$, classifying filtrations with three associated graded factors of those types. Verify the four-input pentagon coherence and the quotient-after-correspondence descent. This is binary/two-step data; unit maps, refinements, symmetry conventions, and higher-coloured trees remain in $\mathfrak{o}^{\mathrm{col}}_R$.

(vi) *Hybrid correspondence-module functor.* Let $\mathsf{Corr}^{E,\mathrm{hyb}}_R$ be the correspondence category, and define $\mathcal{B}^{E,\mathrm{hyb}}_R : \mathsf{Corr}^{E,\mathrm{hyb}}_R \to \mathsf{Ch}_{\mathbb{C}}$ by $\mathcal{B}^{E,\mathrm{hyb}}_R(e) = q_!\,\mathrm{TS}^{\mathrm{red}}_e\,p^*$ for every generating arrow $e = (Y \xleftarrow{p} \mathfrak{E}_e \xrightarrow{q} Z)$.

(vii) *Quotient pseudofunctor $Q_{E,R}$.* Use the chosen quotient-after-correspondence pseudofunctor, including coefficient, orientation, Thom–Sebastiani, base-change, flag, stabilizer, anchor, and transition coherences. Applying compactly supported vanishing-cycle chains to this datum gives the reduced object

$$\mathcal{F}^{\mathrm{geom}}_{X,\sigma,S,\le R} = Q_{E,R} \circ \mathcal{B}^{E,\mathrm{hyb}}_R.$$

(viii) Obtain the hybrid datum with all coherences and the protected integration map I^{prot}_R, with homogeneous s-degree the Borcherds trace exponent $m^{\mathrm{Bch}}_R = \mathrm{pr}_3\,\overline{\Pi}_X$.

6.3.0.4 Identities. The carrier $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$ is a hybrid correspondence-module functor on $\mathsf{Corr}^{E,\mathrm{hyb}}_R$ with values in $\mathsf{Ch}_{\mathbb{C}}$; the four-input pentagon follows from the eight-word two-step flag atlas; the anchor residual tuple $\mathfrak{o}^{\lambda}_R = (\mathfrak{o}^{\lambda,\mathrm{ex}}_R, \mathfrak{o}^{\lambda,\mathrm{unit}}_R, \mathfrak{o}^{\lambda,\mathrm{loss}}_R, \mathfrak{o}^{\lambda,\mathrm{multi}}_R, \mathfrak{o}^{\lambda,\mathrm{tr}}_{R'R})$ vanishes; the hybrid residual $\mathfrak{D}_{\mathrm{hyb},R} = 0$ at every height R, compatibly with transition maps.

6.3.0.5 Obstruction classes. The middle type-II wall lies in $\delta_2 \leftrightarrow (0,1,1)$, elliptic degree $d = m+1 = 2$ by Lemma 5.1; it cannot be represented by a projection-finite local stack and demands the wrapped prequotient. The existence of the wrapped representative $x_{\delta_2,R} \in \mathcal{M}^{\mathrm{wr,rig}}_{\eta,R}$ with $\overline{\Pi}_X(x_{\delta_2,R}) = (0,1,1)$ and its compatible anchor $\lambda_{\eta,R}$ — is precisely the hybrid input to $(O2)$ at the middle wall.

6.4 Reduced d-critical orientation and Joyce–Upmeier transport

6.4.0.1 Hypotheses. The hybrid carrier $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$ of the preceding section; the K3 holomorphic 2-form σ_S on S; BBDJS d-critical chart machinery [11]; the Joyce–Upmeier multiplicative orientation formalism [51, 52]; the cosection localization machinery of Kiem–Li and the cosection-reduced obstruction theory; the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$ generated by $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$.

6.4.0.2 Reduced orientation datum. The cosection $\sigma = p_S^*\sigma_S$; the reduced tangent complex

$$\mathrm{RHom}_{\mathrm{red}}(A,A) = \mathrm{Cone}\Big(\mathrm{RHom}_X(A,A) \xrightarrow{(\mathrm{tr,cs})} R\Gamma(X,\mathcal{O}_X) \oplus H^2(S,\mathcal{O}_S)[-1]\Big)[-1];$$

the BBDJS vanishing-cycle complex $\Phi_{R,c}$; the reduced orientation line

$$o_{R,c} = \det \mathrm{Ext}^1_{\mathrm{red}}(A,A)^{-1} \quad \text{on Darboux charts,} \qquad o^{\otimes 2}_{R,c} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A,A);$$

the Joyce–Upmeier multiplicative transport $o_{c\oplus c'} \simeq o_c \otimes o_{c'}$ compatible with the reduced extension correspondences; the quotient orientation $\mathfrak{Q}^{or}_R$ of Theorem 7.12; the Weyl-equivariant lifts $\tau_{i,R,S} : o_{R,S} \to s^*_{\delta_i} o_{R,S}$ for $i = 1,2,3$ of $(O1)^+$; the local generic wall-sign data $\epsilon^{\mathrm{loc}}_{o,R}$.

6.4.0.3 Construction.

(i) *Cosection.* Pull back the K3 semiregularity cosection cs along $p_S : X \to S$ to obtain the 2-step cosection $\sigma = p_S^*\sigma_S$ on the K3 factor; verify additivity in exact triangles by (RedOr) of Theorem 0.34.

(ii) *Reduced obstruction.* Form the reduced tangent complex by the displayed cone above (Kiem–Li cosection localization). PTVV [89] (-1)-shifted symplectic on $\mathrm{RHom}_X(A,A)$ descends, after the cosection contraction, to the reduced complex.

(iii) *BBDJS d-critical chart.* On each Darboux chart of $\mathfrak{M}^{\mathrm{sc}}_{R,c}$, take the BBDJS vanishing-cycle complex $\Phi_{R,c}$ [11]; verify reduced Thom–Sebastiani transport coherent on two-step flag stacks of the hybrid carrier.

(iv) *Reduced orientation line.* Define $o_{R,c} = \det \mathrm{Ext}^1_{\mathrm{red}}(A,A)^{-1}$; verify the squared isomorphism $o^{\otimes 2}_{R,c} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A,A)$ by 3-Calabi–Yau Serre duality $\mathrm{Ext}^2_{\mathrm{red}}(A,A) \simeq \mathrm{Ext}^1_{\mathrm{red}}(A,A)^\vee$.

(v) *Quotient-orientation descent on the K3×E-quotient self-Ext frame bundle.* Compute the four Cech–Borel obstruction classes for every retained stratum S:

$$\alpha^{\mathrm{red}}_{R,S} = w_2(\mathscr{D}_{R,S}), \quad \alpha^{E,\mathrm{free}}_{R,S} = \mathrm{edge}_{2,0}(w_2^E(\mathscr{D}_{R,S})), \quad \widetilde{\beta}^H_{R,S} = w_2^G(\mathscr{D}_{R,S}), \quad \lambda^H_{R,S} \in H^1_G(S;\mathbb{F}_2),$$

and verify simultaneous vanishing with compatible null-trivializations on overlaps, on Hall correspondence pullbacks, on two-step flag stacks, and under successor maps. The full calculation proceeds primary stabilizer by stabilizer; for cyclic $\mathbb{Z}/2^a$ the absent mixed coefficient requires the rank-two two-primary check. Obtain the descended quotient orientation system $\mathfrak{Q}^{or}_R$.

(vi) *Multiplicative Joyce–Upmeier transport.* Transport $o_{c\oplus c'} \simeq o_c \otimes o_{c'}$ coherently across two-step flag stacks [51, 52]; verify the reduced Thom–Sebastiani identification on three-input flags.

(vii) *Weyl-equivariant lifts* $(O1)^+$. For each $i \in \{1,2,3\}$ construct the lift $\tau_{i,R,S} : o_{R,S} \to s^*_{\delta_i} o_{R,S}$; verify the projective Weyl cocycle $c_{o,R} \in Z^2(\mathcal{G}_R; \underline{\mathbb{F}}_2)$ on the retained action groupoid; null-trivialize $[c_{o,R}]$ and choose a 1-cochain killing it; verify $\tau^2_{i,R,S} = 1$; verify the Coxeter relations, namely $s^2_{\delta_i} = 1$ and no braid relation for $i \neq j$, as in Appendix 5.1; verify that the torsor defects $\omega_{i,C}$ of Theorem 0.34 (O_X) vanish.

(viii) *Local wall-sign datum.* At each generic type-II wall point $\mathbb{H}_{\delta_i}$, with the rank-one local model $K^{\mathrm{cross}}_{\delta_i} = [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$, record the local generic sign $\epsilon^{\mathrm{loc}}_{o,R}(s_{\delta_i}) = -1$ of Proposition 0.23.

(ix) Obtain $(o_R, \mathfrak{Q}^{or}_R, \{\tau_{i,R,S}\}, \epsilon^{\mathrm{loc}}_{o,R})$.

6.4.0.4 Identities. o_R is a Picard-line section squaring to the reduced determinant; the four Cech–Borel classes vanish with compatible null-trivializations; the lifts $\tau_{i,R}$ satisfy $\tau^2_{i,R} = 1$ and the type-II Coxeter relations; multiplicative transport agrees on flag stacks.

6.4.0.5 Obstruction classes. This datum is the entire site of $(O1)$ (strong reduced orientation on the K3×E-quotient self-Ext frame bundle) and of $(O1)^+$ (Weyl-equivariant transport along $W^{(2)}(\Lambda^{2,1}_{II})$ reflections). Discharging $(O1)$ requires the simultaneous vanishing and compatible null-trivialization of the four Cech–Borel classes on every retained stratum; discharging $(O1)^+$ requires $[c_{o,R}] = 0$, the Coxeter coherence, and the vanishing of all torsor defects $\omega_{i,C}$. The local wall-sign datum is the input to $(O2)$ in the Pfaffian wall normal form.

6.5 Hall correspondences and primitive part

6.5.0.1 Hypotheses. The oriented hybrid carrier from the reduced orientation datum; the extension stack $\mathfrak{E}^{\mathrm{geom}}_{R,\hat{c},\hat{c}'}$; the two-step flag stack $\mathfrak{F}^{(2),\mathrm{geom}}_{R,\hat{c},\hat{c}',\hat{c}''}$; the BBDJS reduced Thom–Sebastiani transport; six-functor admissibility of p^*, $q_!$, $p_!$ from the finite moduli datum. The input includes compactified retained filtration/flag-Quot correspondences or an explicitly admissible exceptional compact-support model, finite inertia, base-change and projection-formula witnesses, quotient-after-correspondence pseudofunctoriality, and HN-transition compatibility.

6.5.0.2 Hall correspondence datum. The finite Hall object

$$\mathcal{H}^{\mathrm{geom}}_R = \bigoplus_{\hat{c}\in\hat{\Gamma}_R} H^\bullet_c(\mathfrak{M}^{\mathrm{geom}}_{R,\hat{c}}, \Phi_{R,c} \otimes o_{R,c});$$

the Hall product $m_R = q_!\, \mathrm{TS}^{\mathrm{red}}\, p^*$ and Hall coproduct $\Delta_R = p_!\, (\mathrm{TS}^{\mathrm{red}})^{-1}\, q^*$; the finite-stage Hall bialgebra object $(\mathcal{H}^{\mathrm{geom}}_R, m_R, \Delta_R, \eta_R, \epsilon_R)$; the primitive part

$$P^{\mathrm{geom}}_R = \ker(\Delta_R - 1 \otimes \mathrm{id} - \mathrm{id} \otimes 1) \cap \ker \epsilon_R;$$

the normal-ordered Gram descent $P^{\Pi,\mathrm{geom}}_R = \overline{\Pi}^{\Theta_R}_{X,*} P^{\mathrm{geom}}_R$; the Γ^{Π}_R-graded super-Lie bracket $[\,\bar{x}, \bar{y}\,]_{\mathrm{red}}$ of Theorem 4.3; the Hopf pairing matrix $G_{\gamma,ab}$ and radical K_γ.

6.5.0.3 Construction.

(i) *Pullback.* For $\widehat{c}, \widehat{c}' \in \widehat{\Gamma}_R$, pull back along $p : \mathfrak{E}^{\mathrm{geom}}_{R,\widehat{c},\widehat{c}'} \to \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}} \times \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}'}$; apply reduced Thom–Sebastiani $\mathrm{TS}^{\mathrm{red}}$ to combine vanishing cycle data [11]; pushforward exceptionally along $q : \mathfrak{E}^{\mathrm{geom}}_{R,\widehat{c},\widehat{c}'} \to \mathfrak{M}^{\mathrm{geom}}_{R,\widehat{c}\star\widehat{c}'}$ to obtain $m_R(x \otimes y) = q_! \,\mathrm{TS}^{\mathrm{red}}\, p^*(x \boxtimes y)$. This operation is defined only in the retained compactified or exceptional compact-support model specified in the input.

(ii) *Coproduct.* Dually pull back along q, apply inverse reduced Thom–Sebastiani, pushforward along p to obtain $\Delta_R(z) = p_! \,(\mathrm{TS}^{\mathrm{red}})^{-1}\, q^*(z)$. The compact geometric coproduct correspondence $\mathfrak{C}^{\mathrm{geom}}_{\rho;\mu,\nu}$ of Theorem 4.3 is the support data for splitting.

(iii) *Bialgebra check.* Verify associativity of m_R and coassociativity of Δ_R on the two-step flag stack $\mathfrak{F}^{(2),\mathrm{geom}}_{R,\widehat{c},\widehat{c}',\widehat{c}''}$ by base change and Thom–Sebastiani coherence. Verify the bialgebra identity (Hopf compatibility) via the Massey–Saito Joyce–Upmeier multiplicative orientation transport.

(iv) *Primitive extraction.* Compute $P^{\mathrm{geom}}_R = \ker(\Delta_R - 1 \otimes \mathrm{id} - \mathrm{id} \otimes 1) \cap \ker \epsilon_R$; since $\Delta_R - 1 \otimes \mathrm{id} - \mathrm{id} \otimes 1$ is degree-graded on $\widehat{\Gamma}_R$, this reduces to a finite linear computation in each charge.

(v) *Normal-ordered descent.* Apply $\overline{\Pi}^{\Theta_R}_{X,*}$ of Theorem 4.1 to obtain the Γ^{Π}_R-graded $P^{\Pi,\mathrm{geom}}_R$; verify additivity $\overline{\Pi}_X(\widehat{c} \star \widehat{c}') = \overline{\Pi}_X(\widehat{c}) + \overline{\Pi}_X(\widehat{c}')$.

(vi) *Bracket and pairing matrices.* Compute the matrices

$$M^{\alpha+\beta,c}_{\alpha,a;\beta,b} = \int_{\mathfrak{E}^{\mathrm{geom}}_{\alpha,\beta;\alpha+\beta}} q^!(e^\vee_{\alpha+\beta,c}) \cap \mathrm{TS}^{\mathrm{red}}(p^*(e_{\alpha,a} \boxtimes e_{\beta,b})), \quad B_{\alpha,\beta}(x \otimes y) = M_{\alpha,\beta}(x \otimes y) - (-1)^{|x||y|} M_{\beta,\alpha}\mathsf{sw}(x \otimes y),$$

$$G_{\gamma,ab} = \mathrm{tr}_0(q_! \,\mathrm{TS}^{\mathrm{red}}\, p^*(e_{\gamma,a} \boxtimes e_{-\gamma,b})) \quad \text{on} \quad \mathfrak{P}^{\mathrm{geom}}_{\gamma,-\gamma;0}.$$

Form the radical $K_{\gamma,\bar{p}} = \ker G_{\gamma,\bar{p}} \cap \ker G^t_{-\gamma,\bar{p}}$ and the quotient $L_{\gamma,\bar{p}}$; fix splittings.

(vii) Obtain $(\mathcal{H}^{\mathrm{geom}}_R, m_R, \Delta_R, P^{\mathrm{geom}}_R, P^{\Pi,\mathrm{geom}}_R, \{B_{\alpha,\beta}\}, \{G_{\gamma,ab}\}, \{K_\gamma\}, \{L_\gamma\})$.

6.5.0.4 Identities. $\mathcal{H}^{\mathrm{geom}}_R$ is a finite $\widehat{\Gamma}_R$-graded Hall bialgebra; m_R is associative on the two-step flag stack; Δ_R is coassociative; the supercommutator of primitives is primitive (the bialgebra identity); the descended bracket is Γ^{Π}_R-graded after $\overline{\Pi}^{\Theta_R}_{X,*}$; finite-dimensional in each homogeneous parity piece by finite-typeness.

6.5.0.5 Obstruction classes. The global Do-orientation follows from the retained Do-HN datum through Theorem 7.6. This finite construction produces the finite Hall datum $(\mathcal{H}^{\mathrm{geom}}_R, m_R, \Delta_R, P_R)$. The seven obstruction classes $\mathfrak{o}^{\mathrm{top}}_\Theta, \mathfrak{o}^{\mathrm{Hoch}}_\Theta, \mathfrak{o}^{\mathrm{tri}}_\Theta, \mathfrak{o}^{\mathrm{Jac}}_\Theta, \mathfrak{o}_F, \mathfrak{o}^{\mathrm{cyc}}_\Theta, \mathfrak{o}^{\mathrm{rad}}_\Theta$ of Theorem 4.1 are required to vanish on this finite Hall stage and compatibly in the inverse limit.

6.5.0.6 Comparison with Vol III. The datum $(\mathcal{H}^{\mathrm{geom}}_R, m_R, \Delta_R, P_R)$ is the finite protected Hall positive-half candidate attached to the K3-side Hall–Drinfeld–Borcherds problem of Vol III [65]. In the Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S}$, P^{geom}_R is the finite primitive Hall layer which can identify with the positive half $Y^+(X)$ only after the compact-source recognition datum, the Hopf pairing, completion, integral form, stable-envelope transport, and strict inverse-limit exactness have been fixed. The Drinfeld double $G(X) = D(Y^+(X))$ is therefore not obtained from $(\mathcal{H}^{\mathrm{geom}}_R, m_R)$ alone. The κ-tuple coordinate $\kappa_{\mathrm{BKM}} = 5$ is the BKM/automorphic weight of the denominator section; it is not a level-P invariant of the Hall carrier itself.

6.6 *First-order Pfaffian block and Pfaffian line*

6.6.0.1 Hypotheses. The oriented Hall datum $(\mathcal{H}_R^{\mathrm{geom}}, o_R, \{\tau_{i,R,S}\})$ of the reduced orientation and Hall correspondence data; the local wall-sign datum $\epsilon_{o,R}^{\mathrm{loc}}$ of the reduced orientation datum; the rank-one type-II Pfaffian crossing datum of Definition 0.22 and Appendix 5.2; the half-Hilbert orbit under $W^{(2)}(\Lambda_{II}^{2,1})$ of size $2^{12} = 4096$.

6.6.0.2 Pfaffian datum. The finite skew Pfaffian block on the orientation gerbe $\mathfrak{D}_R^{\mathrm{Pf}}$; the Pfaffian line $\mathscr{L}_{\mathrm{Pf},R} = \det \mathrm{Ext}^1_{\mathrm{red}}(A, A)^{-1}$ (squaring to the reduced determinant); the protected Pfaffian section $\mathrm{pf}_{X,R}$; the global wall-sign character $\epsilon_{o,R} : W^{(2)}(\Lambda_{II}^{2,1}) \to \{\pm 1\}$; the Pfaffian section asymptotic

$$\mathrm{pf}_{X,R} = 64 q^{1/2} r^{1/2} s^{1/2} \prod_{\gamma=(n,l,m)\in A_R} (1 - q^n r^l s^m)^{f(nm,l)} + \text{higher-window correction},$$

which exhausts to Δ_5 as $R \to \infty$.

6.6.0.3 Construction.

(i) *First-order Pfaffian block.* On each Darboux chart of $\mathfrak{M}_{R,c}^{\mathrm{sc}}$, with reduced normal self-Ext splitting $K_{R,c}^{\mathrm{nor}}$, form the finite skew block $\mathfrak{D}_{R,c}^{\mathrm{Pf}}$ representing the Pfaffian of the reduced determinant. At a generic type-II wall point the rank-one normal summand is

$$J_{\mathfrak{d}_i,R} = \left(\begin{smallmatrix} 0 & u_i \\ -u_i & 0 \end{smallmatrix}\right), \qquad \mathrm{Pf}(J_{\mathfrak{d}_i,R}) = u_i$$

(Definition 0.22).

(ii) *Pfaffian line.* Set $\mathscr{L}_{\mathrm{Pf},R} = \det \mathrm{Ext}^1_{\mathrm{red}}(A, A)^{-1}$; verify $\mathscr{L}_{\mathrm{Pf},R}^{\otimes 2} \simeq \det \mathrm{RHom}_{\mathrm{red}}(A, A)$ by 3-Calabi–Yau Serre duality.

(iii) *Local wall sign.* At each generic point of $\mathbb{H}_{\mathfrak{d}_i}$ with normal model $K_{\mathfrak{d}_i,R}^{\mathrm{nor}} = [\mathbb{R} \xrightarrow{u_i} \mathbb{R}]$, compute the monodromy of the Pfaffian section $\mathrm{pf}_{\mathfrak{d}_i,R} = v_i u_i$ along the wall-reflection loop $\ell_{\mathfrak{d}_i,R,S}$ closed by the $(O1)^+$ lift $\tau_{i,R,S}$:

$$M_{\ell_{\mathfrak{d}_i,R,S}}(v_i u_i) = v_i(-u_i) = -v_i u_i, \qquad \epsilon_{o,R}^{\mathrm{loc}}(s_{\mathfrak{d}_i}) = (-1)^{N_{\mathfrak{d}_i,R}^{\mathrm{Pf}}} = -1.$$

(iv) *Half-Hilbert orbit transport.* The reflection $s_{\mathfrak{d}_i}$ acts on the retained half-Hilbert orbit $\mathfrak{H}_R^{\mathrm{half}}$, whose wall-atlas support has twelve binary Pfaffian parities and cardinality $2^{12} = 4096$. The quotient $\Lambda_{II}^{2,1}/2\Lambda_{II}^{2,1}$ has only eight elements and is not the source of this count. The Weyl lift transports the rank-one sign to a character value on each component:

$$\epsilon_{o,R}(s_{\mathfrak{d}_i}) = \epsilon_{o,R}^{\mathrm{loc}}(s_{\mathfrak{d}_i}; x) = -1, \qquad x \in \mathfrak{H}_R^{\mathrm{half}}.$$

The orbit support enters the protected integration datum; it is not multiplied to define the value of a simple reflection. The transported datum is the global character $\epsilon_{o,R} : W^{(2)}(\Lambda_{II}^{2,1}) \to \{\pm 1\}$.

(v) *Compatibility datum.* Verify the component, boundary, overlap, quotient, and protected integration compatibilities of Proposition 5.2 on every retained type-II wall stratum.

(vi) *Pfaffian section.* Define $\mathrm{pf}_{X,R}$ as the protected integration of the oriented Hall product over $\mathcal{F}_{X,\sigma,S,\le R}^{\mathrm{hyb}}$:

$$\mathrm{pf}_{X,R} = 64 q^{1/2} r^{1/2} s^{1/2} I_R^{\mathrm{prot}}\left(\bigotimes_{\gamma\in A_R} \mathrm{pf}_{\gamma,R}\right), \qquad \mathrm{pf}_{\gamma,R} = (1 - q^n r^l s^m)^{f(nm,l)}.$$

(vii) Obtain $(\mathfrak{D}_R^{\mathrm{Pf}}, \mathscr{L}_{\mathrm{Pf},R}, \mathrm{pf}_{X,R}, \epsilon_{o,R})$, with the protected integration compatibility data.

6.6.0.4 Identities. $\mathscr{L}_{\mathrm{Pf},R}$ is a Picard-line whose square isomorphism follows from reduced Serre duality; $\mathrm{pf}_{X,R}$ is a section of $\mathscr{L}_{\mathrm{Pf},R}$ with the displayed cusp expansion; the character $\epsilon_{o,R}$ takes values in $\{\pm 1\}$ and satisfies the Coxeter relations of $W^{(2)}(\Lambda_{II}^{2,1})$; on each simple type-II reflection $\epsilon_{o,R}(s_{\delta_i}) = -1$ by the rank-one local model.

6.6.0.5 Obstruction classes. This datum is the entire site of $(O2)$ (local Pfaffian wall normal form on type-II walls; the $2^{12} = 4096$ sign sum on the half-Hilbert orbit transport). The local rank-one Pfaffian crossing of Appendix 5.2 supplies the generic chart sign; the global half-Hilbert orbit transport is defined only relative to $(O2_{\mathrm{atlas}})$. The character identity $\epsilon_o = \nu_{\Delta_5}|_{W^{(2)}(\Lambda_{II}^{2,1})}$ becomes a theorem only after the retained Do-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and $(O2_{\mathrm{atlas}})$ are compatible in R.

6.7 Chiral Koszul source coalgebra

6.7.0.1 Hypotheses. The oriented Hall primitives $P_R^{\Pi,\mathrm{geom}}$ of the Hall correspondence datum; the hybrid carrier $\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$; the augmentation $\varepsilon^{\mathrm{hyb}}_{X,R} : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \longrightarrow k\mathbf{1}$; the hybrid residual $\mathfrak{O}_{\mathrm{hyb},R} = 0$; the bounded length N_R (positivity of HN height on every non-vacuum colour); the Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar formalism [6, 33]; the conilpotent completed domain on the elliptic curve E; the wrapped bar-cobar domain datum for the wrapped summands and mixed local/wrapped words.

6.7.0.2 Koszul coalgebra datum. The conilpotent chiral coalgebra on E:

$$C^{\mathrm{geom}}_{X,R} = \bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R}) = k\mathbf{1} \oplus \bigoplus_{1\le p\le N_R} \bar{B}^{\mathrm{ch},p}_E(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R});$$

the bar-length filtration $F^{\mathrm{co}}_{R,p}$; the chiral collision coproduct Δ^{ch}_R; the bar comparison $b_{X,R} = \mathrm{id}_{\bar{B}^{\mathrm{ch}}_E(\cdot)}$; conditional on the wrapped bar-cobar domain and Francis–Gaitsgory hypotheses, the source cobar counit $\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_E C^{\mathrm{geom}}_{X,R} \longrightarrow \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$; after a separate source-to-target quasi-isomorphism $\vartheta_R : \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R} \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E,\le R}$, the cobar comparison $\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$.

6.7.0.3 Construction.

(i) *Augmentation kernel.* Set $\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R} = \ker(\varepsilon^{\mathrm{hyb}}_{X,R})$; verify the bounded-length hypothesis (positivity of HN height on every non-vacuum surviving colour, hence at most N_R factors in any non-zero word).

(ii) *Bar coalgebra.* Form the reduced chiral bar

$$C^{\mathrm{geom}}_{X,R} = k\mathbf{1} \oplus \bigoplus_{1\le p\le N_R} \bar{B}^{\mathrm{ch},p}_E(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R}),$$

length p summand generated by ordered p-fold inputs from the augmentation ideal of total HN height $\le R$ (height $> R$ terms zero); coaugmentation by the inclusion of the vacuum summand; counit by projection.

(iii) *Bar-length filtration.*

$$F^{\mathrm{co}}_{R,p} C^{\mathrm{geom}}_{X,R} = k\mathbf{1} \oplus \bigoplus_{1 \le j \le p} \bar{B}^{\mathrm{ch},j}_{E}(\overline{\mathcal{F}}^{\mathrm{geom}}_{X,R}), \qquad 0 \le p \le N_R.$$

(iv) *Collision coproduct.* For $x = [x_1|\cdots|x_p]$, define the cut-map coproduct

$$\Delta^{\mathrm{ch}}_R(x) = \mathbf{1} \otimes x + x \otimes \mathbf{1} + \sum_{i=1}^{p-1} [x_1|\cdots|x_i] \otimes [x_{i+1}|\cdots|x_p],$$

with each cut realized by the corresponding finite collision/flag pull-push map in $\mathsf{Corr}^{E,\mathrm{hyb}}_{R,\mathrm{geom}}$ of the hybrid carrier. Verify coassociativity by the four-input pentagon and the eight-word flag coherences; verify counit identities via vacuum collision flags.

(v) *Conilpotence.* Verify $(\overline{\Delta}^{\mathrm{bar}}_R)^{N_R+1} = 0$: each reduced cut strictly lowers bar length on each branch.

(vi) *Bar comparison.* Take $b_{X,R}$ to be the identity of $\bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}, \varepsilon^{\mathrm{hyb}}_{X,R})$, eliminating the coalgebra-side finite Koszul residuals $\mathfrak{o}^{C}_R$, $\mathfrak{o}^{\mathrm{fil}}_R$, $\mathfrak{o}^{\mathrm{conil}}_R$, $\mathfrak{o}^{\Delta,\mathrm{ch}}_R$ of Definition 0.26.

(vii) *Source cobar counit.* Verify that the projection-finite summands, wrapped summands, and mixed words all lie in the chosen conilpotent completed Beilinson–Drinfeld/Francis–Gaitsgory bar-cobar domain. Under that wrapped Koszul datum, apply [33, Theorem 6.2.2] to obtain $\varepsilon^{\mathrm{src}}_{\mathrm{bar},R} : \Omega^{\mathrm{ch}}_E C^{\mathrm{geom}}_{X,R} \xrightarrow{\simeq} \mathcal{F}^{\mathrm{hyb}}_{X,\sigma,S,\le R}$; verify it is a source-internal quasi-isomorphism of chiral algebras on E, not a homotopy inverse of the target counit.

(viii) *Source-to-target identification.* Construct ϑ_R from the geometric primitive representatives, Hall products, coproducts, pairings, and HN transition compatibility; verify the residual $\mathfrak{o}^{\vartheta}_R$ of Corollary 0.28 vanishes; set $\Theta_{\mathrm{Kos},R} = \vartheta_R \circ \varepsilon^{\mathrm{src}}_{\mathrm{bar},R}$.

(ix) *Inverse limit.* Verify the Koszul Mittag–Leffler condition $R^1 \varprojlim = 0$ for the inverse systems of source coalgebras, completed chiral bar/cobar constructions, target truncations, and the cones of $b_{X,R}$ and $\Theta_{\mathrm{Kos},R}$ [97, §3.5]. Form

$$C^{\mathrm{geom}}_X = \varprojlim_R C^{\mathrm{geom}}_{X,R}, \qquad \Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C^{\mathrm{geom}}_X \xrightarrow{\simeq} \mathsf{Den}_{\Delta_5,E}.$$

(x) Obtain $(C^{\mathrm{geom}}_{X,R}, F^{\mathrm{co}}_{R,\bullet}, \Delta^{\mathrm{ch}}_R, b_{X,R}, \Theta_{\mathrm{Kos},R})$ at every R, with successor compatibility.

6.7.0.4 Identities. $C^{\mathrm{geom}}_{X,R}$ is a coaugmented conilpotent chiral coalgebra on E with bounded bar-length filtration of length at most N_R; the coalgebra identities for Δ^{ch}_R are part of the chosen higher-coloured, unit, refinement, quotient, and transition coherences on the chosen flag stacks; the bar-cobar inversion follows from Francis–Gaitsgory chiral Koszul only after the wrapped bar-cobar domain datum is fixed; the cobar comparison $\Theta_{\mathrm{Kos},R}$ is source-internal, not the target bar-cobar counit (the source/target separation of Lemma 0.25).

6.7.0.5 Obstruction classes. The Koszul Mittag–Leffler exactness hypothesis is the separate inverse-system hypothesis for the chiral Koszul lane; it is not a consequence of the reduced-orientation (D0) endpoint. The total finite Koszul residual $\mathfrak{O}_{\mathrm{Kos},R} = (\mathfrak{o}^{C}_R, \mathfrak{o}^{\mathrm{fil}}_R, \mathfrak{o}^{\mathrm{conil}}_R, \mathfrak{o}^{\Delta,\mathrm{ch}}_R, \{\mathfrak{o}^{C,\mathrm{tr}}_{R'R}\}, \mathfrak{o}^{b}_R, \mathfrak{o}^{\Omega}_R, \mathfrak{o}^{\mathrm{wrKos}}_R, \mathfrak{o}^{\mathrm{hyb}}_R, \mathfrak{o}^{\Pi}_R, \mathfrak{o}^{\mathrm{sep}}_R, \mathfrak{o}^{\mathrm{Prim}}_R)$ has its coalgebra terms killed by the Koszul source coalgebra; the wrapped Koszul term, $\mathfrak{o}^{\Omega}_R$, the source-to-target map, and the ML exactness remain conditions until the displayed identities hold.

6.7.0.6 Source-target separation. The bar coalgebra $\bar{B}^{\mathrm{ch}}_E(\mathcal{F}^{\mathrm{hyb}}_{X,R})$ is a *twisting / coupling* coalgebra (Maurer–Cartan classification), *not* the bulk $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{F}^{\mathrm{hyb}}_X) = \mathrm{ChirHoch}^\bullet(\mathcal{F}^{\mathrm{hyb}}_X, \mathcal{F}^{\mathrm{hyb}}_X)$; cf. Vol II [68] (Beilinson chain $\mathsf{P} \to \mathsf{C} \to \mathsf{S} \to \mathsf{Z}$). The chiral Koszul comparison $\Theta_{\mathrm{Kos}} : \Omega^{\mathrm{ch}}_E C^{\mathrm{geom}}_X \longrightarrow \mathsf{Den}_{\Delta_5,E}$ is a comparison between two algebras of the same level. It is an equivalence only under the Koszul source datum. The homotopy inverse to the target-internal counit $\Omega^{\mathrm{ch}}_E \bar{B}^{\mathrm{ch}}_E(\mathsf{Den}_{\Delta_5,E}) \to \mathsf{Den}_{\Delta_5,E}$ is not used in either direction.

6.8 Primitive recognition

6.8.0.1 Hypotheses. The descended source primitive Lie superalgebra $P^{\Pi,\mathrm{red}}_X$ of the Hall correspondence datum (after the radical quotient $\overline{\mathrm{Rad}\langle\,\cdot,\cdot\,\rangle_X}$); the imported target denominator algebra $\mathfrak{g}_{\Delta_5} = G(\mathscr{B}_{\Delta_5})$ on $\Lambda^{2,1}_{II}$ of Definition 3.2 and Definition 4.2; the increasing cofinal sequence of finite downward-saturated windows $W_1 \subset W_2 \subset \cdots$ with $\bigcup_\nu W_\nu = \mathcal{R}_+$; the Borcherds–Kac–Moody data $(H, I, p, \alpha_i, (\ ,\ ))$ of Theorem 0.30.

6.8.0.2 Conditional primitive-recognition datum. The cofinal finite-window primitive-recognition datum $\{(\mathrm{R0}),(\mathrm{R1}),(\mathrm{R2}),(\mathrm{R3}),(\mathrm{R4}),(\mathrm{R5}),(\mathrm{R6}),(\mathrm{R7}),(\mathrm{R8})\}$ of Definition 0.31 for every W_ν, compatible under $W_\nu \subset W_{\nu+1}$; the source representatives e^X_i, f^X_i, h^X_i; the source presentation map $\pi_W : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5})_W \to P^{\Pi,\mathrm{red}}_X\big|_W$; the kernel equality $\ker \pi_W = \overline{(J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}})}_W$; the conditional isomorphism

$$\boxed{P^{\Pi,\mathrm{red}}_X \cong \mathfrak{g}_{\Delta_5}, \quad \Omega^{\mathrm{ch}}_E C_X \simeq \mathsf{Den}_{\Delta_5,E}.}$$

This construction is a finite criterion. The isomorphism is asserted only after the compact-source rows (R0)–(R8) are fixed on a cofinal relation-closed system.

6.8.0.3 Construction. For every cofinal W_ν:

(R0) *Compact source labels.* Verify every finite source basis vector of degree in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$, where

$$\Gamma(W_\nu) := \{\gamma \in \Gamma_{\mathrm{gram}} \mid \alpha(\gamma) \in W_\nu\},$$

satisfies the compact source-admissibility condition of Theorem 0.34.

(R1) *Representatives and parities.* Supply parity-homogeneous primitive representatives $e^X_i \in P^{\Pi,\mathrm{red}}_{X,\gamma_i}$, $f^X_i \in P^{\Pi,\mathrm{red}}_{X,-\gamma_i}$, $h^X_i = \iota^{-1}_H(\alpha_i)$ for every simple-root index i with $\alpha_i \in W_\nu$ and $\alpha(\gamma_i) = \alpha_i$, with the Gritsenko–Nikulin parity fibres $\mu_{\bar 0}(a), \mu_{\bar 1}(a)$ for imaginary simple roots and even real representatives for $\delta_1, \delta_2, \delta_3$.

(R2) *Relations.* Verify in the protected Hall super-bracket the Cartan, Chevalley, real Serre, Borcherds orthogonality, and super sign relations for every relation whose displayed source and target degrees lie in $\Gamma(W_\nu) \cup (-\Gamma(W_\nu)) \cup \{0\}$ and $W_\nu \cup (-W_\nu) \cup \{0\}$, respectively. In the real rank-three subsystem the Cartan matrix is $((\delta_i, \delta_j)) = \mathrm{diag}(2) - 2(J - \mathrm{diag}(1))$ with real Serre exponent 3.

(R3) *Full finite parity dimensions.* For every $\gamma \in \Gamma(W_\nu)$ verify $\dim P^{\Pi,\mathrm{red}}_{X,\gamma,\bar p} = \mathrm{mult}^{\mathrm{GN}}_{\bar p}(\alpha(\gamma))$ for both parities $p = 0, 1$. In the first timelike channel $\gamma = (1,1,1)$, whose target root is $\alpha(\gamma) = \delta_{123}$, Proposition 6.7 requires $(d^{\mathrm{GN}}_{(1,1,1),\bar 0}, d^{\mathrm{GN}}_{(1,1,1),\bar 1}) = (29, 93)$. Determinant information $29 - 93 = -64 = f(1,1)$ is consistent but not the recognition.

(R4) *Hopf pairing and radical.* Compute the positive-negative pairing matrices in parity blocks; verify the kernel agrees with the Gritsenko–Nikulin/Kac invariant-pairing kernel; verify Lie-ideal and coideal stability under all finite brackets; verify carrying under HN transitions to the next window.

(R5) *No extra relations.* Verify the kernel equality $\ker \pi_\nu = (\overline{J_{\mathrm{BK}} + \mathrm{Rad}_{\mathrm{GN}}})_{\le W_\nu}$ on the finite triangular span generated by $W_\nu \cup (-W_\nu) \cup \{0\}$.

(R6) *Generation.* Verify that every source primitive root space in $W_\nu \cup (-W_\nu)$, after the finite radical quotient, is generated by the Cartan and the chosen simple-root representatives through brackets whose intermediate positive degrees stay in W_ν.

(R7) *Completed PBW compatibility.* Verify the PBW-graded isomorphism $\mathrm{gr}_{\mathrm{PBW}} U(P_X^{\Pi,\mathrm{red}})_{\le W_\nu} \xrightarrow{\sim} \mathrm{gr}_{\mathrm{PBW}} U(\mathfrak{g}_{\Delta_5})_{\le W_\nu}$; verify strict preservation under $W_{\nu+1} \to W_\nu$ transition maps.

(R8) *Exact triangular completion.* Verify strictness of the transition maps and closedness of images on every finite window, so that passage to the inverse limit commutes with the finite kernels, cokernels, radicals, PBW associated gradeds, and parity pieces. Equivalently the relevant $\lim^1$ terms vanish for these triangular finite-window systems.

(i) If the preceding rows hold on a cofinal relation-closed system, obtain the global isomorphism $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ by passage to the inverse limit (compatible cofinal datum); together with the chiral Koszul comparison with the Koszul source coalgebra this gives $\Omega_E^{\mathrm{ch}} C_X \simeq \mathsf{Den}_{\Delta_5,E}$. The protected Pfaffian identification follows separately from Theorem 2.1.

6.8.0.4 Identities. The source presentation map $\pi : \widehat{\mathfrak{F}}(\mathscr{B}_{\Delta_5}) \to P_X^{\Pi,\mathrm{red}}$ is continuous and degree-preserving; (R5) gives injectivity after the completed radical quotient; (R6) gives surjectivity; (R7) gives PBW compatibility; (R8) gives exactness in the inverse-limit passage. No claim on signed superdimensions $\mathrm{sdim} = f(nm, l)$ substitutes for the two-integer parity dimensions of (R3).

6.8.0.5 Obstruction classes. Primitive recognition inherits the obstruction tower carried by the charge, moduli, hybrid, orientation, Hall, Pfaffian, and Koszul data and adds the cofinal exact-completion residual (R8). The Beilinson–Drinfeld bar-cobar separation is respected throughout: (R5) is a source kernel-equality theorem, not an inversion of the target bar-cobar counit; (R7) is a PBW graded comparison of source and target enveloping algebras, not a graph-equality argument from the scalar square.

6.8.0.6 Comparison with Vol II. The Vol II [68] operator-algebra realization problem separates into two assertions for $K3 \times E$. The retained Pfaffian and orientation data give

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5, \qquad \epsilon_o = \nu_{\Delta_5},$$

relative to (D_0), $(O1)$, $(O1)^+$, $(O2)$. The primitive Lie-superalgebra assertion $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the cofinal compact-source datum (R0)–(R8). The Drinfeld doubling $Y^+(X) \to G(X) = D(Y^+(X))$ of Vol III [65] requires the Hopf pairing of (R4), completion by (R8), integral form, and stable-envelope transport. The κ-tuple coordinate $\kappa_{\mathrm{BKM}} = 5$ is the weight of Δ_5 and lives at level 4 of the universal stage chain; it is a consequence of the construction, not a hypothesis.

6.9 Finite HN dependencies

The finite HN system has a strict ordered sequence of mathematical dependencies. The pro-object $\mathfrak{D}_X^{\mathrm{DI}}$ is determined by the following data for every $R \in \mathcal{R}_{\mathrm{HN}}$:

(i) the lattice combinatorics of the charge window;
(ii) the finite-type quasi-smooth derived stacks of the finite moduli datum;
(iii) the hybrid local/wrapped factorization datum and the eight-word two-step flag atlas of the hybrid carrier;
(iv) the four Cech–Borel orientation classes and the $(O1)^+$ Coxeter coherence of the reduced orientation datum;
(v) the finite-stage Hall bialgebra object and primitive matrices of the Hall correspondence datum;
(vi) the rank-one Pfaffian crossing and the half-Hilbert orbit transport of the Pfaffian line;
(vii) the bar coalgebra, conilpotent collision coproduct, and the Francis–Gaitsgory cobar quasi-isomorphism of the Koszul source coalgebra;
(viii) the cofinal datum (R0)–(R8) of primitive recognition;

together with the compatible vanishing of the obstruction tuples $(\mathfrak{D}_R^{E_3-\mathrm{src}}, \mathfrak{D}_{\mathrm{hyb},R}, \mathfrak{D}_{\mathrm{Kos},R}, \mathfrak{D}_R^{\mathrm{O1}}, \mathfrak{D}_R^{\mathrm{O1}^+}, \mathfrak{D}_R^{\mathrm{O2}}, \mathfrak{D}_R^{D_0})$ at every height R, and the Mittag–Leffler exactness condition [ML] of Definition 0.8 on the cofinal HN inverse system. Each obstruction is attached to the unique finite datum in which its components live.

6.9.0.1 Order rigidity. Each of the seven adjacent dependencies fails under exchange:

(i) charge window $\to$ finite moduli: Λ_{ample} parametrises the HN strata; without the window, the moduli stack has no finite cofinal subsystem.
(ii) finite moduli $\to$ hybrid carrier: $\mathrm{Ran}^{\mathrm{hyb}}(E)$ needs the finite Hall stack as factor on which to graft the wrapped/local atlas.
(iii) hybrid carrier $\to$ reduced orientation: the cosection-twisted Joyce–Upmeier line bundle is defined on the carrier, not on the raw moduli (the cosection lives on $\mathrm{Ran}^{\mathrm{hyb}}$).
(iv) reduced orientation $\to$ Hall correspondences: the Pfaffian line bundle on the Hall stack is the orientation-twisted sign data; without orientation, the Hall product is unsigned.
(v) Hall correspondences $\to$ Pfaffian line + first-order block: the primitive Hall bracket $[-,-]_X$ defines the Hall–Drinfeld correspondence whose Pfaffian section is the Dirac–Igusa Pfaffian.
(vi) Pfaffian line $\to$ chiral Koszul source coalgebra: the source C_X is the bar of the augmented binary/two-step hybrid correspondence algebra pulled back along the Pfaffian comparison ι_{aut}; it becomes the bar of a hybrid factorization algebra only after the higher-coloured residual vanishes.
(vii) chiral Koszul source $\to$ primitive recognition: the primitive part $P_X^{\Pi,\mathrm{red}}$ is the Frobenius–Serre primitive filtration on the source; the recognition theorem matches it to $\mathfrak{g}_{\Delta_5}$ only after the finite compact-source datum is fixed.

The order of the constructions is forced. If the orientation datum is placed before the hybrid correspondence datum, the middle type-II wall $\delta_2 \leftrightarrow (0,1,1)$ has no representative by Lemma 5.1. If primitive recognition is placed before the Koszul cobar comparison, the morphism $\Omega_E^{\mathrm{ch}} C_X \to \mathsf{Den}_{\Delta_5,E}$ is absent. The same dependency obstruction applies to the remaining adjacent transpositions.

6.9.0.2 Conditional theorem. With the retained D0-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained $(O2_{\mathrm{atlas}})$ wall datum of Appendix 7, Theorem 0.35 gives

$$\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}}) = \mathcal{D}_X = \Delta_5, \qquad \epsilon_0 = \nu_{\Delta_5}\big|_{W^{(2)}(\Lambda_{II}^{2,1})}.$$

The primitive isomorphism $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source recognition datum. The scalar square $Z_{\mathrm{BPS}}^{K3\times E} = (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$ is the level-Z protected scalar trace. A three-dimensional gravity-line interpretation requires the Hall–Borcherds residual of Vol II and is not asserted here.

7 The D0 criterion and the retained orientation data of the Pfaffian–Dirac theorem

The Pfaffian part of the cross-level identity ②↔③ of Chapter 7 is conditional on four obstruction classes of §1.2, $(D0)$, $(O1)$, $(O1)^+$, $(O2)$, and on the finite Pfaffian section datum (P_{fin}). The retained D0-HN orientation datum belongs to the trivial-canonical threefold $X = K3 \times E$; the retained quotient-orientation datum belongs to $(O1)$; the $(O1)^+$ transport is relative to that datum; the retained wall-atlas datum belongs to $(O2)$. The primitive-recognition datum and the level-A gravity-line problem remain separate. The retained D0 datum, the orientation data, and the finite Pfaffian sections combine Joyce–Upmeier orientation theory [52], Kiem–Li cosection localisation [55], the perverse-sheaf chart cocycle of Brav–Bussi–Dupont–Joyce–Szendrői [11], and the K3-side reduced theory of Maulik–Pandharipande [75], with $K3 \times E$-specific finite-stabilizer criteria at the Klein-four edge, the full two-primary $E[N]$-edge, half-Hilbert orbit transport, and the $(R1)$–$(R2)$ scalar separation.

7.1 $K3 \times E$ datum and criterion theorem

The standing data are: S a smooth complex projective $K3$ surface, E a smooth elliptic curve, $X = S \times E$ the trivial-canonical threefold, $\sigma_S \in H^0(S, K_S)$ the holomorphic two-form trivialising K_S, $\sigma = \mathrm{pr}_S^*\sigma_S \in H^0(X, p_S^*K_S)$ the inflated K3 two-form defining the semiregularity cosection, and $\mathfrak{M}_c^{\mathrm{red},+}(X)$ the cosection-reduced moduli of effective dyonic class $c \in \Gamma_{\mathrm{eff}}$ on X in the chosen stability heart fixed in Chapter 2. The D0 sector is the zero-dimensional sector on the threefold X; the Hilbert–Chow morphism $\nu_n : \mathrm{Hilb}^n(K3) \longrightarrow \mathrm{Sym}^n(K3)$ furnishes only the K3-fibre scalar test for the reduced theory. The Mukai pairing of Definition 1.3 is symmetric of signature $(4, 20)$.

Definition 7.1 (Retained type-II wall-atlas datum). The datum $(O2_{\mathrm{atlas}})$ consists, for the three simple type-II walls $\partial_1, \partial_2, \partial_3$, of retained wall objects with full $\widehat{\Gamma}_X$-charge match, semistability on the wall, reduced self-Ext splitting, rank-one normal Pfaffian block, invariant Pfaffian unit, quotient orientation, and overlap compatibility on the cofinal HN system. Concretely, for each retained HN height R it is a system of wall charts $(U_{\partial,R}, u_{\partial,R}, \mathscr{L}_{\partial,R}^{\tan}, v_{\partial,R})$ with $s_\partial(u_{\partial,R}) = -u_{\partial,R}$,

$$\mathrm{pf}_{\partial,R} = v_{\partial,R} u_{\partial,R},$$

transition isomorphisms on double overlaps, a Cech 2-cocycle $c_{\alpha\beta\gamma}^{\mathrm{wall}} \in \{\pm 1\}$, and a 1-cochain $b_{\alpha\beta}^{\mathrm{wall}}$ satisfying $\delta b^{\mathrm{wall}} = c^{\mathrm{wall}}$. These data are required to commute with Thom–Sebastiani products, HN transition maps, the chosen Weyl lifts, and the half-Hilbert orbit chart system carrying the twelve binary sign coordinates used in the scalar wall count.

Definition 7.2 (Retained quotient-orientation datum). The datum $(O1_{\mathrm{quot}})$ consists, on every retained object, extension, wrapped, mixed, two-step flag, overlap, and HN-transition stratum, of null-trivialisations of the ordinary reduced orientation gerbe, the connected BE-edge class, the finite-stabilizer Cech–Borel classes, the zero degree-one stabilizer linearizations, and the Thom–Sebastiani compatibilities of these choices.

Definition 7.3 (Retained finite Pfaffian section datum). The datum (P_{fin}) is a cofinal HN system carrying the finite Pfaffian section datum of Proposition 0.21: support, parity, active-support coverage, normalized leading coefficient, and Pfaffian-line Mittag–Leffler compatibilities for the sections $\mathrm{pf}_{X,R}$ of $\mathscr{L}_{\mathrm{Pf},R}$.

Theorem 7.4 *(Relative Pfaffian-orientation criterion on $K3 \times E$).* On the rank-one and finite-stabiliser strata of the moduli stack $\mathfrak{M}_c^{\mathrm{red},+}(X)$, $X = K3 \times E$, in the stability heart fixed in Chapter 2, the obstruction class $(D0)$ is represented by the retained D0-HN orientation datum $(D0_{\mathrm{HN}})$ of Definition 7.5. The class $(O1)$ is represented by $(O1_{\mathrm{quot}})$, $(O1)^+$ by $(O1_{\mathrm{quot}})$ together with the chosen Weyl lifts, and $(O2)$ by $(O2_{\mathrm{atlas}})$. If the Cech, Borel, torsor, and wall-atlas equations in these retained data vanish with the stated null-trivialisations, then the Pfaffian–Dirac theorem 2.1 and the orientation-character match 3.1 hold on $K3 \times E$ relative to $(D0_{\mathrm{HN}})$, $(O1_{\mathrm{quot}})$, the chosen Weyl lifts, $(O2_{\mathrm{atlas}})$, and (P_{fin}). The primitive recognition theorem 4.1 remains conditional on the finite compact-source recognition datum $(S1)$–$(S10)$.

The four clauses are separated below according to their retained data. The level-Z Hall–Borcherds trace comparison is distinct from the level-A gravity-line residual.

7.2 (D0) D0-degeneration limit of the cosection-reduced d-critical orientation theory

The first obstruction concerns the inverse-limit assembly of the Joyce–Upmeier orientation cocycle along the HN system $\mathcal{R}_{\mathrm{HN}}$ when the chosen stability heart degenerates to the $D0$-supported subcategory of zero-dimensional sheaves on $X = K3 \times E$. The $D0$-charge is the threefold length n. Projection to the K3 factor gives the reduced K3 scalar test; it does not replace the threefold D0 moduli. The D0-degeneration limit must commute with cosection-reduction and with the orientation extension.

Definition 7.5 (Retained D0-HN orientation datum). [definitional] For each $R \in \mathcal{R}_{\mathrm{HN}}$, let $\mathfrak{M}_R^{\mathrm{red}} \subset \mathfrak{M}_c^{\mathrm{red},+}(X)$ be the open substack of objects supported on the HN window of height $\leq R$ under the chosen stability slope. The *D0-degenerated HN direct diagram of stacks* is

$$\mathfrak{M}_R^{\mathrm{red}} \xhookrightarrow{\iota_R} \mathfrak{M}_{R+1}^{\mathrm{red}} \hookrightarrow \cdots, \qquad \mathfrak{M}_\infty^{\mathrm{red}} := \varinjlim_R \mathfrak{M}_R^{\mathrm{red}}.$$

The datum $(D0_{\mathrm{HN}})$ consists of this HN diagram together with:

(i) retained quasi-smooth d-critical atlases and reduced vanishing-cycle complexes on the object, extension, mixed, wrapped, and two-step flag strata;
(ii) surjective K3 semiregularity cosections on the retained charts, additive on extension and flag stacks;
(iii) Joyce–Upmeier orientation lines on the reduced d-critical charts with Thom–Sebastiani multiplicativity;
(iv) restriction isomorphisms along ι_R, compatible with Hall extensions, quotient-after-correspondence, protected integration, and Pfaffian lines;
(v) Mittag–Leffler exactness for the inverse systems of orientation gerbes, vanishing-cycle complexes, compact-support six-functor operations, and Pfaffian lines;
(vi) compatibility of the pro-object with the zero-dimensional $D0$-supported subcategory, where the $D0$-charge is length.

The D0-degeneration limit is the pro-object obtained from these restriction data.

Theorem 7.6 *((D0) criterion).* Granted $(D0_{\mathrm{HN}})$, the cosection-reduced d-critical orientation theory extends compatibly along the HN inclusions ι_R and has a D0-degeneration pro-limit on $\mathfrak{M}_\infty^{\mathrm{red}}$.

Proof. PTVV supplies the ambient (-1)-shifted symplectic source on the trivial-canonical threefold moduli [89, Theorem 0.3]; BBDJS constructs vanishing cycles after an oriented d-critical atlas is present [11, Theorem 6.9]; Kiem–Li gives the cosection localisation formalism [55]; Joyce–Upmeier gives the orientation-line and multiplicativity technology [51, 52]. These theorems do not by themselves produce the retained HN-compatible reduced orientation system. The clauses of Definition 7.5 supply that system at finite height, its Hall and quotient compatibilities, and the Mittag–Leffler exactness needed to pass to the pro-limit. The pro-limit therefore inherits the reduced vanishing-cycle complexes, orientation lines, Thom–Sebastiani transports, compact-support operations, and Pfaffian lines. Thus the retained HN datum gives the $(D0)$ vanishing condition on the pro-limit. □

Remark 7.7 *(The role of the K3 elliptic genus).* The K3 elliptic-genus formula $\chi_y(\mathrm{Hilb}^n(K3))$ [75, §4.1] tests the scalar D0 specialization. It is not a substitute for Mittag–Leffler exactness of the orientation gerbes, vanishing-cycle complexes, compact-support functors, or Pfaffian lines.

7.3 (O1) strong reduced orientation on the K3×E-quotient self-Ext frame bundle

The second obstruction is the strong reduced orientation: the multiplicative orientation cocycle of $(D0)$ must be equivariant under the free E-translation quotient and under the finite-stabilizer strata $BE[N]$ on which the E-action degenerates.

Definition 7.8 *(Strong reduced orientation on the K3×E-quotient frame).* [definitional] The *strong reduced orientation* on $\mathfrak{M}_c^{\mathrm{red},+}(X)$ is a multiplicative orientation cocycle o_c^{strong} on the principal Spin^c-bundle $\mathcal{F}_c^{\mathrm{red}}$ of the reduced derived self-Ext frame bundle on $\mathfrak{M}_c^{\mathrm{red},+}(X)$, satisfying:

- (O1.a) *Free E-translation equivariance.* On the generic E-translation orbit of codimension 0, o_c^{strong} descends through the free quotient $\mathfrak{M}_c^{\mathrm{red},+}(X)/E$.
- (O1.b) *Klein-four edge and linearization condition.* Across the $BE[2]$-strata, the Cech–Borel class reduces to the edge class of Lemma 7.9, that class vanishes, and the residual degree-one $E[2]$-linearization character is zero.
- (O1.c) *Full two-primary finite-stabilizer condition.* Across the $BE[N]$-strata for $N \geq 3$, the two-primary edge class and the residual stabilizer character satisfy the criterion of Lemma 7.10.

Lemma 7.9 *(Klein-four edge criterion on rank-2 E-strata).* Assume that the finite-stabilizer Cech–Borel class on a retained $BE[2]$-stratum has no ordinary, mixed, or differential term and therefore has an edge representative in $H^2(BE[2], \mathbb{F}_2)$. For $E[2] \simeq (\mathbb{Z}/2)^2$,

$$H^*(BE[2], \mathbb{F}_2) = \mathbb{F}_2[x_1, x_2], \qquad |x_i| = 1,$$

and the edge class has the form

$$\beta^{E,2} = b_{20}x_1^2 + b_{11}x_1x_2 + b_{02}x_2^2.$$

If r_i denotes its restriction to the three non-zero cyclic subgroups of $E[2]$, then

$$b_{20} = r_1, \qquad b_{02} = r_2, \qquad b_{11} = r_1 + r_2 + r_3.$$

Hence $\beta^{E,2} = 0$ is equivalent to $r_1 = r_2 = r_3 = 0$ for this edge class. After this degree-two class is killed, the remaining linearization character is

$$\lambda^{E,2} = \lambda_1 x_1 + \lambda_2 x_2 \in H^1(BE[2], \mathbb{F}_2),$$

and its restrictions satisfy

$$\rho_1 = \lambda_1, \qquad \rho_2 = \lambda_2, \qquad \rho_3 = \lambda_1 + \lambda_2.$$

The vanishing of r_i is therefore not the vanishing of ρ_i; quotient orientation requires both sets of equations.

Proof. The cohomology ring is the polynomial ring displayed above. Restricting to $\langle e_1\rangle$, $\langle e_2\rangle$, and $\langle e_1 + e_2\rangle$ sends the quadratic form respectively to $b_{20}t^2$, $b_{02}t^2$, and $(b_{20} + b_{11} + b_{02})t^2$. Over $\mathbb{F}_2$ this gives the displayed invertible linear change of coordinates. The last assertion is the standard statement that, after a degree-two gerbe is trivialised, linearizations form an $H^1(BE[2], \mathbb{F}_2)$-torsor. □

LEMMA 7.10 *(Full two-primary finite-stabilizer criterion).* For a complex elliptic curve,

$$E[N] \simeq (\mathbb{Z}/N)^2.$$

If N is odd, the mod-2 finite-stabilizer orientation class restricts trivially by transfer. If $2^a \parallel N$, then the mod-2 calculation factors through

$$E[N]_{(2)} \simeq (\mathbb{Z}/2^a)^2.$$

For $a = 1$ one obtains Lemma 7.9. For $a \geq 2$,

$$H^*\big(B(\mathbb{Z}/2^a)^2, \mathbb{F}_2\big) = \Lambda(x_1, x_2) \otimes \mathbb{F}_2[y_1, y_2], \qquad |x_i| = 1, \quad |y_i| = 2,$$

and every degree-two edge class has the form

$$\beta^{E,N} = A_1^{(N)} y_1 + A_{12}^{(N)} x_1 x_2 + A_2^{(N)} y_2.$$

The residual stabilizer character is

$$\lambda^{E,N} = \lambda_1^{(N)} x_1 + \lambda_2^{(N)} x_2.$$

Thus quotient orientation at the N-stratum requires

$$A_1^{(N)} = A_{12}^{(N)} = A_2^{(N)} = \lambda_1^{(N)} = \lambda_2^{(N)} = 0.$$

The coefficient $A_{12}^{(N)}$ is not detected by cyclic order-two restrictions.

Proof. The group of N-torsion points of a complex elliptic curve is $(\mathbb{Z}/N)^2$. For N odd the group order is invertible in $\mathbb{F}_2$, hence $H^{>0}(BE[N], \mathbb{F}_2) = 0$ by transfer. The even-primary reduction is the primary decomposition of $E[N]$. The cohomology of $B(\mathbb{Z}/2^a)$ is $\Lambda(x) \otimes \mathbb{F}_2[y]$, with $|x| = 1$ and $|y| = 2$; the Künneth formula gives the displayed degree-two basis for the square. A cyclic order-two subgroup pulls back $x_1 x_2$ to zero, so it cannot see the mixed coefficient. □

Remark 7.11 (What the finite-stabilizer criteria prove). The two preceding lemmas are detection statements inside the retained datum $(O1_{\mathrm{quot}})$. They do not construct the Cech–Borel null-trivialisations, do not kill mixed Borel terms, and do not remove the residual H^1-linearization characters. They say which coordinates must vanish after the finite-stabilizer problem has been reduced to the classifying-space edge.

Theorem 7.12 *((O1) quotient-orientation criterion).* Granted the retained quotient-orientation datum $(O1_{\text{quot}})$ of Definition 7.2, the reduced orientation of Theorem 7.6 descends to the strong reduced orientation o_c^{strong} of Definition 7.8 on $\mathfrak{M}_c^{\text{red},+}(X)$.

Proof. The reduced orientation o_∞^{red} of Theorem 7.6 and the datum $(O1_{\text{quot}})$ determine $(O1.a)$, $(O1.b)$, and $(O1.c)$.

$(O1.a)$ *Free E-equivariance.* On the free translation locus the connected BE-edge class $\alpha^{E,\text{free}} \in H^2(BE, \mathbb{F}_2)$ is one of the null-trivialised classes in $(O1_{\text{quot}})$. The chosen trivialisation, together with Joyce–Upmeier multiplicativity [52, §4], gives descent through the open quotient.

$(O1.b)$ *Klein-four extension.* Lemma 7.9 identifies the $BE[2]$-edge equations. The datum $(O1_{\text{quot}})$ supplies their vanishing and the zero $H^1(BE[2], \mathbb{F}_2)$-linearization character.

$(O1.c)$ *Higher-rank extension.* Lemma 7.10 identifies the full two-primary finite-stabilizer equations for $BE[N]$, $N \geq 3$. The same datum supplies their vanishing and the compatible zero stabilizer characters on overlaps, Hall correspondences, flags, and HN transitions.

The combined cocycle is multiplicative and Pin^c-equivariant by the multiplicativity of the supplied Thom–Sebastiani transports; this is o_c^{strong}. By the multiplicativity, every product $F_1 \to E \to F_2$ of effective dyonic objects has $o_c^{\text{strong}}(F_1 \to E \to F_2) = o_c^{\text{strong}}(F_1) \otimes o_c^{\text{strong}}(F_2)$. This is the strong reduced orientation. □

7.4 $(O1)^+$ *Weyl-equivariant transport along* $W^{(2)}(\Lambda^{2,1}_{II})$

The third obstruction is the Weyl-equivariance of the strong reduced orientation along the type-II Weyl group $W^{(2)}(\Lambda^{2,1}_{II})$. The Weyl group is the 3-generator Coxeter group generated by the three simple type-II reflections $s_{\delta_1}, s_{\delta_2}, s_{\delta_3}$. Since $(\delta_i, \delta_j) = -2$ for $i \neq j$, the Coxeter exponents are $m_{ij} = \infty$; there are no braid relations among distinct generators. Hence $(W^{(2)})_{\text{ab}} \simeq (\mathbb{Z}/2)^{\oplus 3}$.

Theorem 7.13 *((O1)$^+$ Weyl transport criterion and type-II character comparison).* The strong reduced orientation o_c^{strong} of Theorem 7.12, together with the chosen lifts of Definition 0.9 with vanishing projective cocycle and zero quotient-torsor defects, is $W^{(2)}(\Lambda^{2,1}_{II})$-equivariant. If, in addition, the rank-one type-II wall signs of $(O2_{\text{atlas}})$ are fixed on the three simple walls, then its character on $W^{(2)}$ is

$$\epsilon_o = \nu_{\Delta_5} \quad \text{on } W^{(2)}(\Lambda^{2,1}_{II}),$$

identical to the Maass multiplier system of the Igusa cusp form.

Proof. The proof compares the equivariant orientation line with the Maass character on the type-II Coxeter generators.

Equivariant determinant line. The vanishing of the projective cocycle c_o makes the chosen determinant-line lifts into an honest action of $W^{(2)}(\Lambda^{2,1}_{II})$ on o_c^{strong}. The vanishing of the torsor defects $\omega_{i,C}$ says that the quotient-orientation cocycle is transported with the line, not only after forgetting the finite-stabilizer data. This proves the $(O1)^+$ equivariance assertion.

Generator values from $(O2)$. Under $(O2_{\text{atlas}})$, Proposition 5.2 gives the local Pfaffian sign at each simple type-II wall:

$$\epsilon_o^{\text{loc}}(s_{\delta_i}) = (-1)^{N^{\text{Pf}}_{\delta_i}} = -1, \qquad i = 1, 2, 3,$$

with $N^{\text{Pf}}_{\delta_i} = 1$ the rank-one Pfaffian rank exponent. By Lemma 5.3, the Maass character on these generators is $\nu_{\Delta_5}(s_{\delta_i}) = -1$ for $i = 1, 2, 3$. Hence $\epsilon_o(s_{\delta_i}) = \nu_{\Delta_5}(s_{\delta_i}) = -1$ for the three generators.

The abelianisation $(\mathbb{Z}/2)^{\oplus 3}$. The type-II Weyl group has the Coxeter presentation

$$W^{(2)}(\Lambda^{2,1}_{II}) = \langle s_1, s_2, s_3 \mid s_1^2 = s_2^2 = s_3^2 = 1 \rangle,$$

with no relation on $s_i s_j$ for $i \neq j$. This is the Coxeter presentation attached to the symmetric matrix (δ_i, δ_j), because $(\delta_i, \delta_j) = -2$ gives $m_{ij} = \infty$. Abelianising only imposes commutativity and $s_i^2 = 1$, hence

$$(W^{(2)})_{\mathrm{ab}} \simeq (\mathbb{Z}/2)^{\oplus 3}.$$

Extension to the full Weyl group. A character of $W^{(2)}$ is determined by its values on the abelianisation, hence by its values on the three generators. The extension principle is standard: given $\chi : \{s_1, s_2, s_3\} \to \{\pm 1\}$, define $\chi(w) = \prod_{j=1}^{k} \chi(s_{i_j})$ for any reduced word $w = s_{i_1} \cdots s_{i_k}$. The only defining relations are $s_i^2 = 1$; hence the formula is well-defined.

Character comparison. The group $H^1(W^{(2)}, \{\pm 1\})$ is $\mathrm{Hom}(W^{(2)}, \{\pm 1\}) \simeq (\mathbb{Z}/2)^{\oplus 3}$, not the obstruction group. The obstruction is represented by the projective cocycle c_o and torsor defects $\omega_{i,C}$ of Definition 0.9. When these classes vanish and the three rank-one $(O2)$ signs are fixed, both ϵ_o and ν_{Δ_5} are $\{\pm 1\}$-valued characters of $W^{(2)}$ with the same values on the three generators by the local $(O2)$ computation; hence they are equal: $\epsilon_o = \nu_{\Delta_5}$ on $W^{(2)}(\Lambda^{2,1}_{II})$. □

Remark 7.14 *(Why the type-II Coxeter group reduces to its abelianisation).* The reduction to the abelianisation is forced by the order-two structure of the $\{\pm 1\}$-character group: any group homomorphism into a group of exponent two factors through the abelianisation, since exponent-two characters automatically annihilate the commutator subgroup. There is no order-three braid relation in the type-II Coxeter group; the off-diagonal value -2 is the infinite Coxeter exponent. Thus no order-three obstruction class is present in this comparison.

7.5 (O2) Local Pfaffian wall normal form and the $(R1)$–$(R2)$ *scalar separation*

The fourth obstruction is the type-II wall normal form: the local Pfaffian section on each type-II wall is the rank-one odd crossing of Appendix E, and the $2^{12} = 4096$ sign-sum on the half-Hilbert orbit remains separate from the OP normalization until the scalar shadow is taken.

THEOREM 7.15 *((O2) wall-atlas consequence: type-II wall normal form).* Assume $(O2_{\mathrm{atlas}})$ and the Weyl-transport datum of Theorem 7.13. On each type-II wall $\delta \in W^{(2)}(\Lambda^{2,1}_{II}) \cdot \{\delta_1, \delta_2, \delta_3\}$ of $\mathfrak{M}_c^{\mathrm{red},+}(X)$, the local Pfaffian section pf_δ is the rank-one odd crossing $[\mathbb{R} \xrightarrow{u} \mathbb{R}]$ with Pfaffian rank exponent $N_\delta^{\mathrm{Pf}} = 1$ and local sign $\epsilon_o^{\mathrm{loc}}(s_\delta) = -1$.

Proof. The proof reduces higher-rank type-II walls to the rank-one normal form of Appendix E by Weyl-orbit transport.

Rank-one walls. For $\delta \in \{\delta_1, \delta_2, \delta_3\}$, the rank-one normal form is Proposition 5.2, supplying $N_\delta^{\mathrm{Pf}} = 1$ and $\epsilon_o^{\mathrm{loc}}(s_\delta) = -1$.

Orbit-stabiliser theorem. For higher-rank walls $\delta = w \cdot \delta_i$ with $w \in W^{(2)}(\Lambda^{2,1}_{II})$ and $i \in \{1, 2, 3\}$, the local geometric chart $U_\delta = T_\delta \times (-\varepsilon, \varepsilon)$ is identified in the retained wall atlas with the w-transport of the rank-one chart $U_{\delta_i} = T_{\delta_i} \times (-\varepsilon, \varepsilon)$. By Theorem 7.13 (Weyl-equivariance of the strong reduced orientation), the local Pfaffian section transports as

$$\mathrm{pf}_\delta \;=\; w_* \, \mathrm{pf}_{\delta_i} \;\cdot \epsilon_o(w),$$

with $\epsilon_o(w) \in \{\pm 1\}$ the Weyl character value.

Rank exponent invariance. The Pfaffian rank exponent $N_{\hat{\partial}}^{\mathrm{Pf}} = 1$ is preserved under the supplied Weyl-orbit chart identifications, since it is the rank of the normal factor in the reduced d-critical chart.

Local sign. The local sign $\epsilon_o^{\mathrm{loc}}(s_{\hat{\partial}}) = \epsilon_o^{\mathrm{loc}}(w s_{\hat{\partial}_i} w^{-1})$. Since the orientation sign is a character after Theorem 7.13,

$$\epsilon_o(w s_{\hat{\partial}_i} w^{-1}) = \epsilon_o(w)\epsilon_o(s_{\hat{\partial}_i})\epsilon_o(w)^{-1} = \epsilon_o(s_{\hat{\partial}_i}) = -1.$$

Hence $\epsilon_o^{\mathrm{loc}}(s_{\hat{\partial}}) = -1$ for every Weyl-translate. □

Theorem 7.16 *($(R1)$–$(R2)$ scalar separation: $4096 = 2^{12} = 64^2$).* Two independent normalizations have the same integer value:

(R1) (operator-level) $N_{\mathrm{wall}} = 2^{12}$, the count of sign assignments in the half-Hilbert wall atlas of (O2);

(R2) (automorphic-level) $N_{\mathrm{OP}} = 64^2 = c_\Delta(1,1,1)^2$, the squared theta-leading constant of the type-II Borcherds product.

The equality $N_{\mathrm{wall}} = N_{\mathrm{OP}} = 4096$ is an equality of integers after passing to a scalar shadow. It is not an identification of the orientation character with the OP scalar normalization.

Proof. The (O2) atlas supplies 12 binary wall signs, hence

$$N_{\mathrm{wall}} = 2^{12} = 4096.$$

The leading Fourier coefficient of Δ_5 at the type-II vacuum is

$$c_\Delta(1,1,1) \;=\; [q^{1/2}r^{1/2}s^{1/2}]\,\Delta_5 \;=\; 64$$

by Lemma 5.2. Squaring,

$$N_{\mathrm{OP}} = c_\Delta(1,1,1)^2 = 64^2 = 4096.$$

Thus

$$N_{\mathrm{wall}} = N_{\mathrm{OP}}.$$

No step identifies the wall orientation line with the OP scalar branch. □

Remark 7.17 *($(R1)$ and $(R2)$ live on different levels).* The wall count belongs to the Pfaffian orientation atlas. The factor 64^2 belongs to the automorphic normalization $D_5 = 64^{-1}\Delta_5$ and to the OP scalar convention. Their common integer is used only after orientation has been forgotten by squaring.

Theorem 7.18 *((O2) wall-atlas transport on the half-Hilbert orbit).* Assume $(O2_{\mathrm{atlas}})$ and the Weyl-transport datum of Theorem 7.13. The orientation cocycle on the half-Hilbert orbit on $\mathfrak{M}_c^{\mathrm{red},+}(X)$ extends from the rank-one local data of Theorem 7.15 to a globally consistent $(\mathbb{Z}/2)$-valued system on the entire half-Hilbert orbit, with the total sign count 2^{12} realised by the twelve binary wall coordinates of the $(O2_{\mathrm{atlas}})$ datum. Equivalently, the type-II Pfaffian-wall normal form is reduced to $(O2_{\mathrm{atlas}})$.

Proof. Theorem 7.15 supplies the rank-one normal form on every type-II wall relative to the wall atlas. Theorem 7.13 supplies the $W^{(2)}$-equivariance. Theorem 7.16 identifies the integer $2^{12} = 4096$ with the squared theta-leading normalization, after passing to the scalar shadow. The half-Hilbert orbit transport is the following Weyl-orbit calculation. A base point on the orbit determines every other orbit point by Weyl transport, and the orientation sign is the product of generator signs along the chosen path. The result is independent of path by the chosen Weyl lifts and zero torsor defects of Theorem 7.13, and the total count is the $2^{12} = 4096$ count supplied by the wall atlas. This is the $(O2)$ transport theorem relative to $(O2_{\mathrm{atlas}})$. □

7.6 Vol II Hall–Borcherds trace comparison at level Z: chain-level chiral-Hochschild trace

The retained Do-HN datum, retained quotient-orientation datum, chosen Weyl lifts, and retained type-II wall atlas give the relative Pfaffian and orientation hypotheses of the Pfaffian–Dirac theorem on $K3 \times E$. A separate trace comparison relates the level-n protected scalar shadow $Z_{\mathrm{BPS}}^{K3\times E} = (\Phi_{10}^{\mathrm{un}})^{-1} = \Delta_5^{-2}$ of Chapter 9 to a chain-level chiral-bulk trace on the derived chiral centre $Z_{\mathrm{ch}}^{\mathrm{der}}(C_X)$ on $K3 \times E$. The level-A Hall–Borcherds residual is different: it asks for the gravity-line operator algebra acting on the boundary and remains open in Vol II.

Definition 7.19 *(Vol II Hall–Borcherds trace comparison at level* Z*).* [definitional] The *level-*Z *Hall–Borcherds trace comparison* on $K3 \times E$ is the assertion identifying the protected graded trace

$$\mathrm{Tr}_{\mathrm{prot}}\big((-1)^F q^{L_0} r^{J_0} s^{\widetilde{L}_0}\big) = Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$$

of the retained E_3-prefactorization algebra $\mathcal{A}_{K3\times E}$ to the level-Z trace pairing of the chain-level chiral derived centre $Z_{\mathrm{ch}}^{\mathrm{der}}(C_X)$ on $(X = K3 \times E, D, \tau)$ at the boundary vacuum b; the identification $Z_{\mathrm{ch}}^{\mathrm{der}}(C_X) = \mathrm{ChirHoch}^\bullet(C_X, C_X)$ is the chiral-Hochschild concentration theorem of Vol I (Theorem H, [67]) on the Koszul locus, arranged by the level-Z clause (i) of the cyclic-Hochschild Beilinson stratification of Vol II [68, chiral-Hochschild Beilinson stratification]; the trace pairing of degree -3 is the CY-orientation pairing clause (ii) of the same theorem under $\mathsf{Eff}_{\mathrm{Pf,or}}$. The comparison is a class in

$$H^*_{\mathrm{Hall}}(\mathcal{A}_{K3\times E}) \;\to\; H^*_{\mathrm{Borcherds}}\big(Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}_{K3\times E})\big),$$

governing whether the chiral-shadow Hall integral Δ_5^{-2} is the shadow of a chain-level chiral-Hochschild Borcherds integral on the target.

Remark 7.20 *(Distinguishing the level-*Z *trace from the level-*A *gravity line).* The level-Z Hall–Borcherds trace comparison of Definition 7.19 has trace-level scope, whereas level A asks for a gravity-line operator algebra acting on the boundary chiral algebra of the $K3 \times E$ holomorphic-topological theory, with Pentagon-face determinant section $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ and reciprocal partition character Δ_5^{-2}. This residual remains open in Vol II; its construction requires the operator-valued completion that the present level-Z trace identity does not supply. The level-Z facet is a trace pairing on $Z_{\mathrm{ch}}^{\mathrm{der}}(C_X)$; the level-A facet is an acting algebra on a gravity-line module. No theorem constructs that acting algebra.

7.7 The level-Z Hall–Borcherds trace comparison on $K3 \times E$

The level-Z Hall–Borcherds trace comparison of Definition 7.19 is a criterion relative to a retained trace datum on $K3 \times E$. Vol I Theorem H [67] and Vol II's chiral-Hochschild Beilinson stratification [68, chiral-Hochschild Beilinson stratification] apply only after that datum has placed the compact source in their domain. The level-A operator-algebra facet remains open in Vol II (Remark 7.20).

Definition 7.21 *(Retained level-*Z *Hall–Borcherds trace datum).* [definitional] The datum (Z_{HB}) consists of the following finite and pro-compatible data:

- (Z1) a boundary chart algebra A_b in the Vol I/Vol II chiral-Koszul domain, together with a filtered quasi-isomorphism $A_b \simeq C_X$;
- (Z2) a class-M weight completion of $\mathrm{Bar}(A_b)$, compatible with the HN inverse system and the Koszul Mittag–Leffler data;
- (Z3) the Vol I chiral-Hochschild concentration hypothesis on A_b and the Vol II stratification hypotheses at level Z;

(Z4) a degree -3 cyclic trace pairing on $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$, compatible with the Calabi–Yau three-form, the Pfaffian line, and HN transition;

(Z5) compatibility of the trace normalization with the type-II Borcherds product of Δ_5, the OP chamber scalar branch, and the protected trace convention of Chapter 9.

(Z_{HB}) has trace-level scope: it supplies a trace pairing on the centre, not an operator algebra acting on a gravity-line module.

Theorem 7.22 *(Level-Z Hall–Borcherds trace criterion on $K3 \times E$).* Granted the retained level-Z trace datum (Z_{HB}), Vol I Theorem H [67], and Vol II's chiral-Hochschild Beilinson stratification [68] as established theorems, the level-Z Hall–Borcherds trace comparison of Definition 7.19 holds on $K3 \times E$. Equivalently, the level-n protected scalar trace $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ of Chapter 9 is identified with the chain-level CY-orientation trace of the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$ on $K3 \times E$:

$$\boxed{\mathrm{Tr}^{\mathrm{bulk}}_n\big((-1)^F \cdot \mathrm{id}\big)_{Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)} \;=\; \Delta_5^{-2}}\,.$$

The OP chamber branch is the normalized scalar representative $-4096\,\mathrm{Tr}^{\mathrm{bulk}}_n$, not the trace itself. The comparison is at level Z in the Beilinson chain of Vol II; the level-A gravity-line operator algebra is not constructed here and remains open in Vol II (Remark 7.20).

Proof. The retained datum (Z_{HB}) supplies the chart algebra A_b, its comparison with the compact source C_X, the class-M completion, the cyclic trace pairing, and the normalization comparison with the type-II Borcherds product. Clause (Z3), together with Vol I Theorem H and the level-Z clause of Vol II, identifies the chiral derived centre with chiral Hochschild cohomology on the retained Koszul locus:

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \;=\; \mathrm{ChirHoch}^{\bullet}(A_b, A_b),$$

and clause (Z4) supplies the degree--3 trace pairing

$$\mathrm{Tr}_C \colon Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \otimes Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \;\longrightarrow\; \mathbb{C}[-3]$$

at level Z. Clause (Z5), the Pfaffian normalization of Theorem 2.1, and the OP/Igusa scalar normalization of Theorem 1.2 identify this trace with the protected scalar:

$$\mathrm{Tr}^{\mathrm{bulk}}_n = \Delta_5^{-2}.$$

The OP branch convention of Convention 1.1 gives the separate scalar representative $Z^X_{\mathrm{OP}} = -4096\,\mathrm{Tr}^{\mathrm{bulk}}_n$. This proves exactly the level-Z criterion: once the retained trace datum exists with the stated normalization compatibilities, the trace comparison follows. The $\mathsf{SC}^{\mathrm{ch,top}}$-brace, the E_3-topological factorisation-algebra structure on $X \times \mathbb{R}$, and the gravity-line operator algebra are not consequences of this trace datum. □

Theorem 7.23 *(Pfaffian–Dirac theorem and level-Z Z_{BPS} trace identity on $K3 \times E$, relative to retained data).* Granted Theorems 7.6, 7.12, 7.13, 7.18, the retained finite Pfaffian section datum (P_{fin}), and 7.22, the last itself granted (Z_{HB}), Vol I Theorem H [67] and Vol II's chiral-Hochschild Beilinson stratification [68] as established theorems, the Pfaffian–Dirac theorem 2.1, the orientation-character match 3.1, and the level-Z trace identity for the protected scalar $Z^{K3\times E}_{\mathrm{BPS}} = \Delta_5^{-2}$ to a chain-level CY-orientation trace on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$ all hold on $K3 \times E$. The primitive recognition theorem 4.1 is a separate conditional theorem: it requires the compact-source representatives, relations, pairings, radicals, PBW comparison, and full parity dimensions of $(S1)$–$(S10)$. The level-A gravity-line operator algebra remains open in Vol II (Remark 7.20).

Proof. $(D0)$ follows from the retained D0-HN datum through Theorem 7.6; $(O1)$ follows from the retained quotient-orientation datum through Theorem 7.12; the equivariance part of $(O1)^+$ follows from the chosen Weyl lifts through Theorem 7.13; and $(O2)$ is transported from the retained wall-atlas datum by Theorem 7.18. The finite Pfaffian sections are the datum (P_{fin}) of Definition 7.3. The level-Z Hall–Borcherds trace comparison follows from the retained (Z_{HB}) datum through Theorem 7.22. The Pfaffian and orientation theorems of Chapter 7 then hold on $K3 \times E$ relative to these retained data. The primitive-recognition theorem asks for the additional compact-source datum $(S1)$–$(S10)$, and Appendix G gives the Pfaffian-orientation lane, not the compact-source Lie-superalgebra recognition lane. The level-Z trace identity of Chapter 9 reads $\mathrm{Tr}_n^{\mathrm{bulk}} = \Delta_5^{-2}$. The OP scalar branch is $-4096\,\mathrm{Tr}_n^{\mathrm{bulk}}$, by Convention 1.1. □

Remark 7.24 *(The conditioning is the retained level-*Z *datum and Vol I/Vol II chain-level theorems).* After the $K3 \times E$ retained D0-HN datum, the retained $(O1_{\mathrm{quot}})$ datum, the chosen Weyl lifts, the retained $(O2_{\mathrm{atlas}})$ datum, and the trace comparison of §7.6, the remaining chain-level conditioning for the level-Z Z_{BPS} trace identity is the retained datum (Z_{HB}), Vol I Theorem H [67] (chiral Hochschild concentration on the Koszul locus), and Vol II [68, chiral-Hochschild Beilinson stratification] clauses (i) and (ii) (chain-level bulk identification and CY-orientation trace pairing) as correct theorems. The K3×E-specific trace obstruction is precisely (Z_{HB}). The theorem is conditional on this retained datum and on the Vol I/Vol II chain-level constructions. The level-A gravity-line operator algebra sits outside this conditioning and remains open in Vol II. Primitive recognition is independent of the retained orientation data: it is exactly the finite compact-source recognition problem of Theorem 4.1.

7.8 Compatibility of the retained data

The retained D0-HN datum, the retained quotient-orientation datum, the chosen Weyl lifts, and the retained $(O2_{\mathrm{atlas}})$ datum convert the Pfaffian and orientation parts of the Pfaffian–Dirac theorem to statements on $K3 \times E$ at the Pfaffian and orientation level. The primitive Lie-superalgebra recognition remains conditional on $(S1)$–$(S10)$. The proof uses the retained data in a fixed implication chain. The retained D0-HN datum gives Theorem 7.6. The retained quotient-orientation datum gives Theorem 7.12. The Weyl lifts and simple-wall signs give Theorem 7.13. The retained $(O2_{\mathrm{atlas}})$ datum together with these Weyl lifts gives Theorem 7.18. The scalar separation $(R1)$–$(R2)$ is Theorem 7.16. The Hall–Borcherds trace comparison is Theorem 7.22. The gravity-line algebra is a separate Vol II construction.

The criteria use the following theorems, with the fourfold analogue separated from the $K3 \times E$ argument:

- Joyce–Upmeier orientation theorem [52, Theorem 1.11];
- Joyce–Tanaka–Upmeier quasi-projective extension [51, Theorem 1.4];
- the Cao–Kool–Monavari CY_4 cosection-twist extension [16, Theorem 1.6] is a fourfold analogue, not a theorem used in the $K3 \times E$ CY_3 reduction above;
- Kiem–Li cosection localisation [55, Theorem 5.0.1];
- Brav–Bussi–Dupont–Joyce–Szendrői d-critical chart cocycle [11, Theorem 6.9];
- Pantev–Toën–Vaquié–Vezzosi (-1)-shifted symplectic structure [89, Theorem 0.3];

- Maulik–Pandharipande K3 reduced theory [75, Theorem 1];
- Oberdieck–Pandharipande $K3 \times E$ Igusa cusp form factorisation [83, Theorem 0.2], [86, Theorem 1].

The $K3 \times E$-specific finite-stabilizer criteria are Lemma 7.9 (Klein-four detection on rank-2 E-strata), Lemma 7.10 (full two-primary $E[N]$-edge detection on finite-stabilizer strata), and Theorem 7.16 ($(R1)$–$(R2)$ scalar separation). The foundational citations give the chart and orientation technology; Mittag–Leffler exactness is a condition in the retained datum $(D0_{\mathrm{HN}})$.

7.9 The relative theorem on K3×E

The Pfaffian theorem 2.1 and the orientation-character theorem 3.1 hold on $K3 \times E$ at the Pfaffian and orientation level relative to $(D0_{\mathrm{HN}})$, $(O1_{\mathrm{quot}})$, the Weyl lifts, $(O2_{\mathrm{atlas}})$, and (P_{fin}). The obstruction $(D0)$ of Chapter 6 is reduced to the retained D0-HN datum by Theorem 7.6; $(O1)$ and $(O1)^{+}$ are reduced through Theorems 7.12 and 7.13 relative to their retained data; $(O2)$ is transported from $(O2_{\mathrm{atlas}})$ by Theorem 7.18. The $(R1)$–$(R2)$ scalar separation $4096 = 2^{12} = 64^{2}$ is Theorem 7.16. The level-Z Hall–Borcherds trace identity $\mathrm{Tr}_n^{\mathrm{bulk}} = \Delta_5^{-2}$ on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$ is Theorem 7.22, granted (Z_{HB}), Vol I Theorem H [67], and Vol II's chiral-Hochschild Beilinson stratification [68] as established theorems; no appeal is made to the Vol II Universal Holography master theorem, whose standard-landscape scope (affine Kac–Moody at non-critical level, principal $W_{N,c}$, Vir_c, Schellekens $c = 24$, Monster $V^{\natural}$, VSKR+BGG-tempered irrational cosets on a smooth projective curve) does not include the BKM-derived hybrid factorisation chart C_X on $K3 \times E$. The Saito–Kurokawa eigenline arithmetic of Proposition 2.1 and the κ-universality identity $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ across the CHL frame $N \in \{1, 2, 3, 4, 6\}$ of Theorem 8.1 are inscribed by direct computation. Throughout, the cited foundational results (Joyce–Upmeier, BB-DJS, Maulik–Pandharipande, Oberdieck–Pandharipande, Kiem–Li, Pantev–Toën–Vaquié–Vezzosi) are applied in their published form. The Cao–Kool–Monavari CY_4 theorem is a separate fourfold analogue.

The theorem-level automorphic, BKM-denominator, and singular-theta readings of Δ_5 close at the automorphic level, and the protected Pfaffian $\mathrm{Pf}_{\mathrm{prot}}(\mathfrak{D}_X^{\mathrm{DI}})$ is identified with Δ_5 relative to the retained data. The Lie-superalgebra recognition $P_X^{\Pi,\mathrm{red}} \cong \mathfrak{g}_{\Delta_5}$ requires the finite compact-source datum $(S1)$–$(S10)$. The mirror discriminant of Chapter 8 is the remaining view: the automorphic calculation gives $[\Delta_5^{-2}] = 2[\mathfrak{H}_{\mathrm{II}}]$, while the identification of this Heegner divisor with the reduced mirror discriminant and the global trivialisation of the comparison line bundle are conjectural and refined into four period conditions in Conjecture 1.1.

The unresolved data are the finite compact-source recognition datum $(S1)$–$(S10)$, the mirror period comparison of Conjecture 1.1, the explicit OPE structure on $Z^{\mathrm{der}}_{\mathrm{ch}}(C_X)$, the Type II sigma-model on $K3 \times E$ with worldsheet BPS partition function Δ_5^{-2}, the level-A gravity-line operator algebra, and the Mathieu–umbral twined denominator system. For the order-seven and order-fifteen conjugacy classes $\{7A, 7B, 15A, 15B\}$ of M_{24}, Conjecture 1.1 of Chapter 10 refines the Cheng–Duncan–Harvey–Gannon umbral moonshine conjecture to four assertions on $\Theta(\phi_{0,1}^{(g),\mathrm{Weil}})\big|_{\mathbb{H}_2}$, and identifies the arithmetic obstruction in the parity of $c_g(0)$. The level-A gravity-line problem demands an operator algebra acting on the boundary chiral algebra of the $K3 \times E$ holomorphic-topological theory with Pentagon-face determinant section $\Phi_{10}^{\mathrm{un}} = \Delta_5^{2}$ and reciprocal partition character Δ_5^{-2}, an operator-valued Hall–Borcherds completion, and the 6d holomorphic Chern–Simons quartic anomaly vanishing on $K3 \times E$ of Vol III. The level-Z trace criterion of Theorem 7.22 is a trace pairing on the centre, not an acting algebra on a gravity-line module.

8 $N = 1$ Boundary-Compatibility Conditions

Let a chambered Hall-descent datum supply the compact comparison data of Theorem 0.34, so that the criterion of Theorem 0.30 applies. The smooth compact-realization standing datum is fixed as follows: S is a smooth complex K3 surface and E is a smooth elliptic curve. Singular K3 limits, nodal or cuspidal elliptic limits, imprimitive multiple-cover contributions, the $s = 0$ compact Hall cusp, and multi-component charge closures require separate boundary-degeneration data. Its $N = 1$ decategorified boundary would then be constrained by the Igusa datum

$$\begin{aligned}\Gamma_{\text{eff}} = \{&(n, l, m) \in \mathbb{Z}^3 \mid m > 0,\ n \geq 0,\ l \in \mathbb{Z}\} \\ &\cup \{(n, l, 0) \in \mathbb{Z}^3 \mid n > 0,\ l \in \mathbb{Z}\} \\ &\cup \{(0, l, 0) \in \mathbb{Z}^3 \mid l < 0\} \subset \mathbb{Z}^3.\end{aligned}$$

comparison Hall descent	Igusa boundary
Gram index lattice	$\Gamma_{\text{ind}} = \mathbb{Z}^3$
sector completion	Γ_{eff}
active denominator support	$\Gamma_{\text{act}} = \{\gamma \in \Gamma_{\text{eff}} \mid f(nm, l) \neq 0\}$
Lorentzian degree	$\alpha(n, l, m) = 2nf_2 - lf_3 + 2mf_{-2}$
orientation character	required geometric monodromy comparison $\epsilon_o\|_{W^{(2)}} = \nu_{\Delta_5}\|_{W^{(2)}}$
automorphic boundary	$\mathrm{Borch}_\theta(\phi_{0,1}^{\text{Weil}}) = \Delta_5$
BKM denominator	$\mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z)$
denominator algebra	$\mathcal{A}_{\text{den}} = \mathfrak{g}_{\Delta_5}$
formal current envelope	$\mathsf{Den}_{\Delta_5,E} = U_E^{\text{ch}}(\mathfrak{g}_{\Delta_5} \otimes \mathcal{O}_E)$
scalar square	$Z_{\text{OP}}^X = Z_{\text{DT}}^X = -4096\Delta_5^{-2}$ (proved for the smooth primitive nonzero $K3$-class branch).

The degree map satisfies

$$(\alpha(\gamma), \alpha(\eta)) = -2\langle \gamma, \eta \rangle_{\text{ind}}, \qquad (\alpha(n, l, m), \alpha(n, l, m)) = -2(4nm - l^2).$$

The real walls are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

These equations give the denominator polarisation of the same Gram-index lattice whose cusp polarisation gives the Fock determinant. The cone is generated by the active support. Triples in Γ_{eff} with zero Jacobi coefficient fix ordering and contribute no visible product factors. The product alone does not exclude invisible zero-superdimension root spaces.

The comparison is the $N = 1$ normalization

$$\mathrm{Borch}_\theta(\phi_{0,1}^{\text{Weil}}) = \Delta_5, \qquad \mathrm{den}(\mathfrak{g}_{\Delta_5}) = 64^{-1}\Delta_5(2Z), \qquad \kappa_{\text{BKM}}(\mathfrak{g}_{\Delta_5}) = \mathrm{wt}(\Delta_5) = 5.$$

It contains no denominator theorem beyond the $N = 1$ theorem and no row map

$$\Phi_j^{\text{cmp}} \longleftrightarrow F_i.$$

The protected scalar has the form

$$Z_{\square}^{X} = C_{\square}\Delta_5^{-2}.$$

The Oberdieck–Pixton and reduced Hilbert-scheme DT branches have constant -4096 in the quotient-by-E normalization of Theorem 5.3. The chiral/denominator half is Δ_5.

A finite radical-quotient Hall–Drinfeld double realizes the windowed protected bracket from the compactification only after the finite-first oriented Hall descent, Serre-dual opposite half, protected integration, charge-additive Hall primitives, Euler–Hall pairing, and radical quotient have been fixed. Its recognition as the compact Δ_5 current object still requires the comparison datum of Theorem 0.34 and Theorem 0.30: Borcherds relations inside the Hall super-commutator, PBW/no-extra-relations, parity, centre, associator, and the completed Hopf pairing with orientation character required to restrict to ν_{Δ_5}. Homotopy descent must produce the positive half, its completion data, and the Hall-side domain on which the comparison character is defined. This orientation character is independent of rescaling Δ_5; it comes from monodromy of the chosen square root of the Hall determinant line. The universal recognition envelope may be formed as the finite quotient killing the radical, Serre, coproduct, centre, and associator defects; it is the unquotiented compact Hall object only when compact source matrices satisfy those identities and the source-faithfulness theorem gives injectivity at every height. The Igusa boundary identifies the denominator algebra $\mathfrak{g}_{\Delta_5}$ and its protected decategorified multiplicities before those comparison data are proved.

For the Cléry–Gritsenko rows F_5, F_6, F_7, F_8, this table contains no nonabelian chamber, no simple real roots, no reflection character, and no Weyl-sum side. Rows F_5, F_6, F_7, F_8 use the separate Govindarajan CHL/WKB denominator theorem when a nonabelian bracket is needed. For F_7 this is a theorem on the μ_4-multiplier cover determined by the Cléry–Gritsenko order-four multiplier. None of these denominator rows supplies a comparison row map or a CY/DT/chiral boundary host.

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